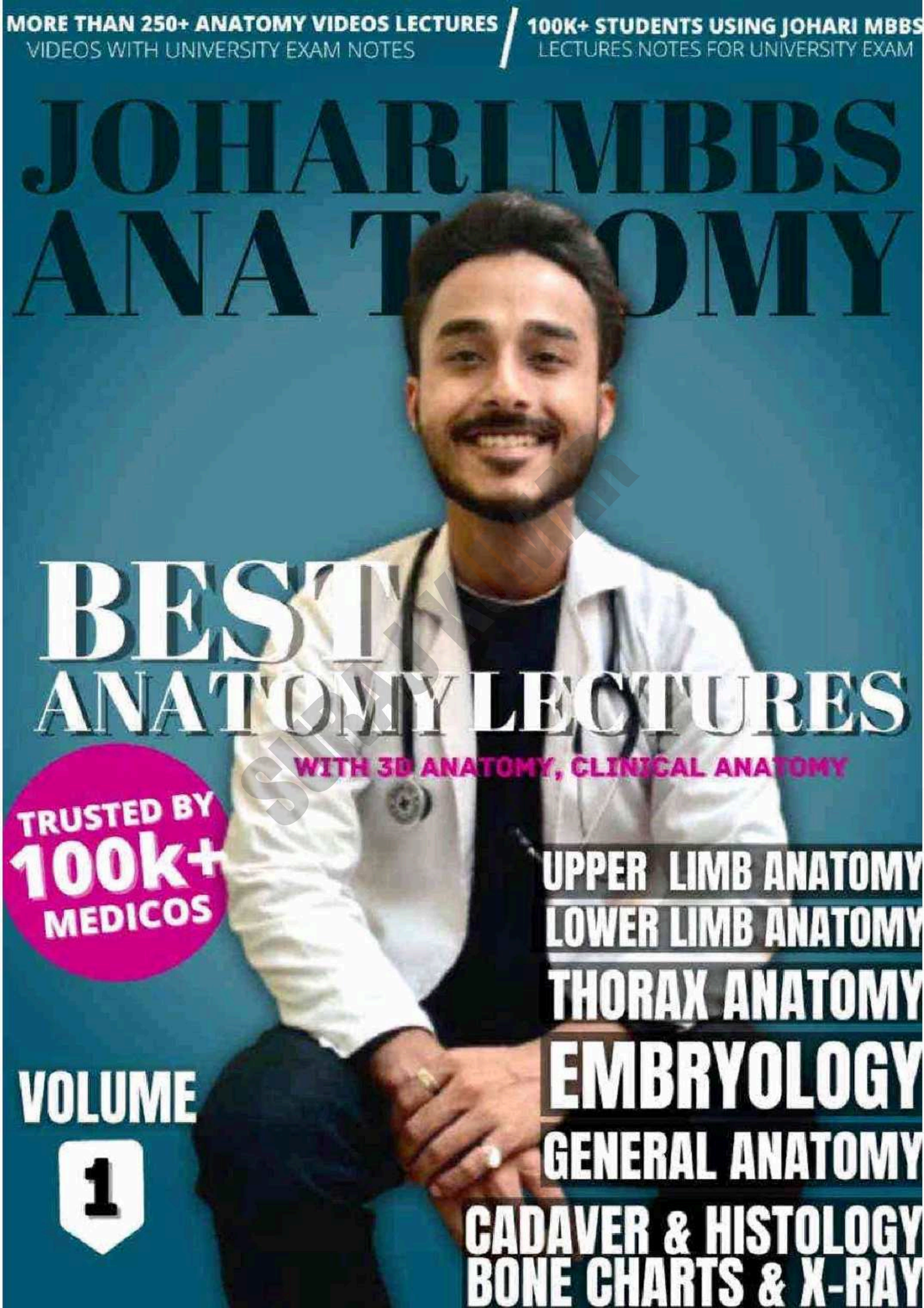
MORE THAN 250+ ANATOMY VIDEOS LECTURES
VIDEOS WITH UNIVERSITY EXAM NOTES

100K+ STUDENTS USING JOHARI MBBS
LECTURES NOTES FOR UNIVERSITY EXAM

JOHARI MBBS ANATOMY



BEST ANATOMY LECTURES

WITH 3D ANATOMY, CLINICAL ANATOMY

TRUSTED BY 100k+ MEDICOS

UPPER LIMB ANATOMY
LOWER LIMB ANATOMY
THORAX ANATOMY
EMBRYOLOGY
GENERAL ANATOMY
CADAVER & HISTOLOGY
BONE CHARTS & X-RAY

VOLUME 1

SUMIT

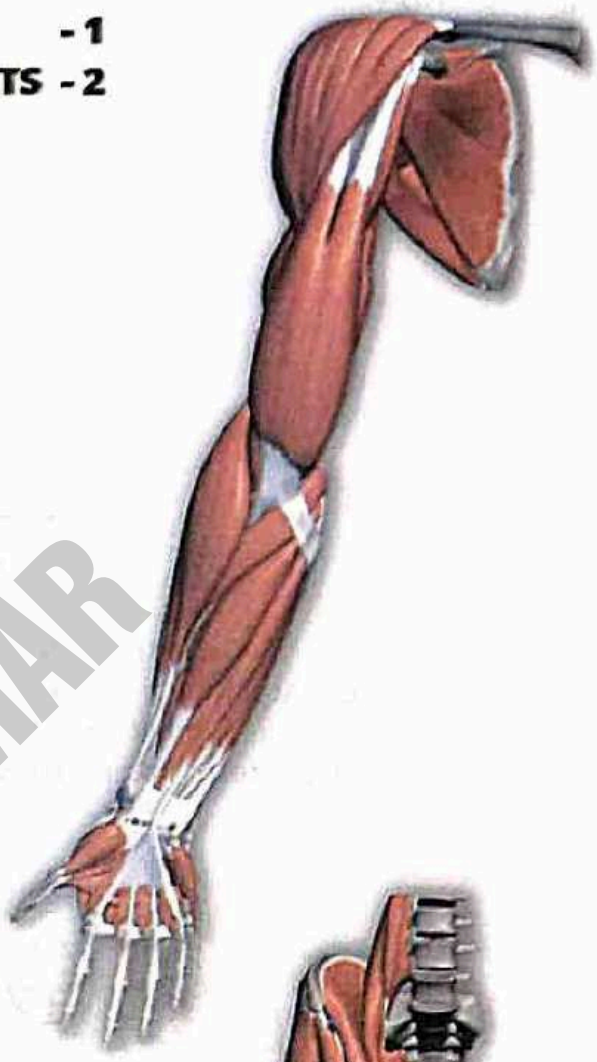
UPPER LIMB IMPS TOPICS

JOHARI MBBS | YOUTUBE

UPPER LIMB MUSCLE CHARTS - 1
BLOOD SUPPLY OF UPPER LIMB CHARTS - 2

1) BRACHIAL PLEXUS - 3
2) AXILLA - 4
3) AXILLARY ARTERY - 5
4) CUBITAL FOSSA - 6
5) MAMMARY GLAND - 8
6) RADIAL NERVE - 11
7) ULNAR NERVE - 13
8) MEDIAN NERVE - 15
9) ROTATOR CUFF - 17
10) FLEXOR RETINACULUM - 17
11) MID PALMAR SPACE - 19
12) SNUFF BOX - 19
13) LUMBRICAL - 20
14) THENAR - 20
15) SHOULDER JOINT - 21
14) ELBOW JOINT - 23
15) SUPINATION & PRONATION - 24

UNIVERSITY QUESTIONS - 26
MCQS OF UEPPR LIMB - 29



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LOWER LIMB IMPS

LOWER LIMB MUSCLE CHART - 30
NERVE SUPPLY OF LOWER LIMB CHARTS - 31

1) FEMORAL TRIANGLE - 32
2) FEMORAL SHEATH & CANAL - 34
3) FEMORAL ARTERY - 34
4) FEMORAL NERVE - 35
5) ADDUCTOR CANAL - 37
6) POPLITEAL FOSSA - 39
7) OBTURATOR NERVE - 39
8) SCIATIC NERVE - 41
9) ARCHE OF FOOT - 43
10) VENOUS DRAINAGE OF LOWER LIMB - 44
11) HAMSTRING - 47

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12) FLEXOR RETTINACULUM - 47
13) HIP JOINT - 49
14) LOCKING & UNLOCKING OF KNEE - 50
15) KNEE JOINT - 52

UNIVERSITY QUESTIONS - 55
MCQS OF LOWER LIMB - 56

THORAX IMPS

THORAX MUSCLE CHARTS - 57
ABDOMEN MUSCLE CHARTS - 58

1) PLEURA - 59
2) RECESSES OF PLEURA - 61
3) PULMONARY LIGAMENT - 61
4) FISSURE AND LOBE OF LUNG - 62
5) ROOT OF LUNG - 64
6) BRONCHOPULMONARY SEGMENTS - 66
7) SUPERIOR & INFERIOR MEDIASTINUM - 68
8) PERICARDIUM - 71
9) RIGHT ATRIUM - 73
10) ARTERY SUPPLY OF HEART - 74
11) VEINS OF HEART - 75
12) AZYGOS VEIN - 77
13) ARCH OF AORTA - 77
14) THORACIC DUCT - 79
15) INTERCOSTAL NERVE - 80
14) INTERCOSTAL SPACE - 82
15) INLET OF THORAX - 83
16) OUTLET OF THORAX - 84

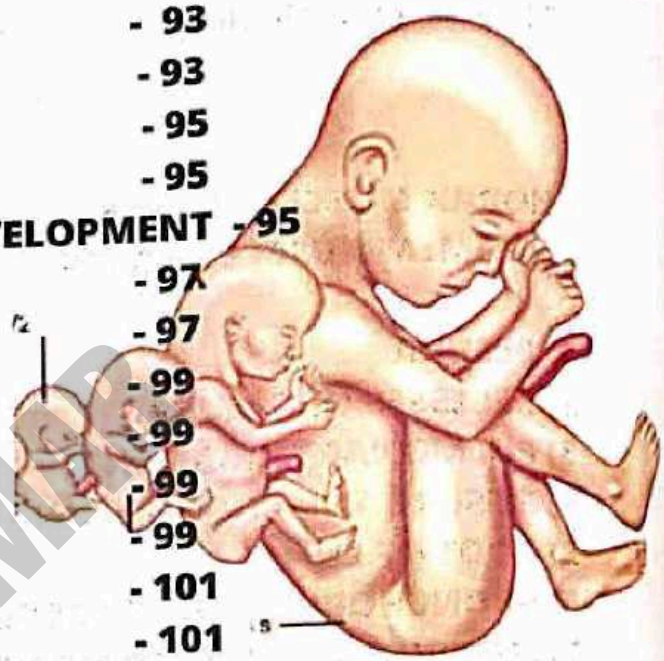
UNIVERSITY QUESTION - 85
MCQ OF THORAX LIMB - 86

SURAJ KUMAR

EMBRYOLOGY

1) FIRST WEEK OF DEVELOPMET - 87
2) SECOND WEEK OF DEVELOPMENT - 89
3) THIRD WEEK OF DEVELOPMENT - 90
4) NOTOCHORD DEVELOPMENT - 90
5) PHARYNGEAL ARCHES - 92
6) THYROID DEVELOPMENT - 93
7) TONGUE DEVELOPMENT - 93
8) HEART DEVELOPMENT - 95
9) RIGHT ATRIUM DEVELOPMENT - 95
10) INTERVENTRICULAR SEPTUM DEVELOPMENT - 95
11) DEVELOPMENT OF FACE - 97
12) DEVELOPMENT OF SOFT PALATE - 97
13) DEVELOPMENT OF PANCREAS - 99
14) DEVELOPMENT OF DUODENUM - 99
15) DEVELOPMENT OF LIVER - 99
16) ROTATION OF MIDGUT - 99
17) DEVELOPMENT OF KIDNEY - 101
18) DEVELOPMENT OF HIDGUT - 101
19) DEVELOPMENT OF GONADS - 103
20) DESCENT OF TESTIS - 103

UNIVERITY QUESTION - 104



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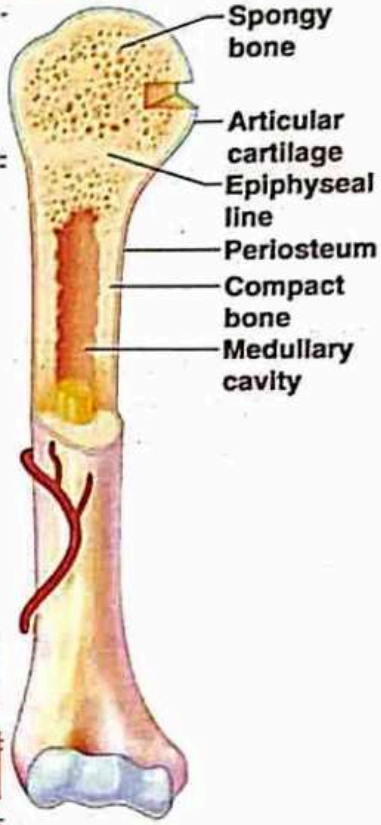
BONE CHARTS

1) NORMA VERTICALIS - 113
2) NORMA OCCIPITALIS - 113
3) NORMA FRONTALIS - 116
4) NORMA LATERALIS - 117
5) NORMA BASALIS - 118
6) CLAVICLE BONE - 119
7) SCAPULA BONE - 121
8) HUMERUS BONE - 125
9) RADIUS BONE - 128
10) ULNA BONE - 129
11) HIP BONE - 131
12) FEMUR BONE - 136

13) FIBULA BONE - 139
14) TIBIA BONE - 140

GENERAL ANATOMOY

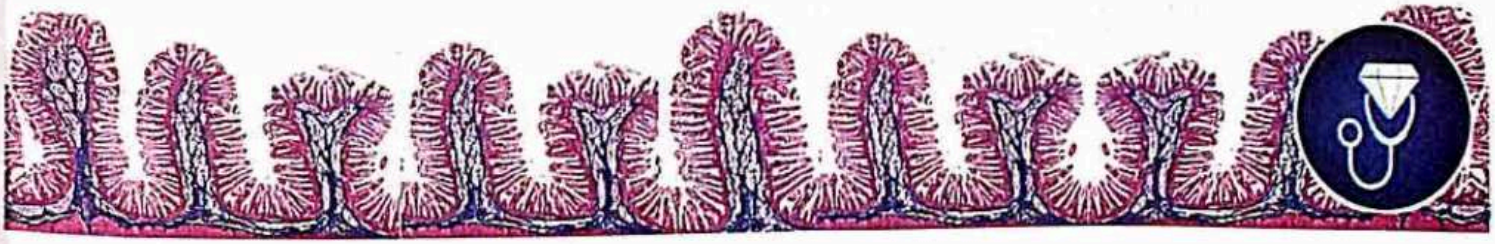
1) EPIPHYSIS - 105
2) BONE - 105
3) BLOOD SUPPLY OF BONE - 107
4) CARTILAGE - 108
5) JOINT - 109
6) SYNOVIAL - 110
7) KARYOTYPING - 111
8) DOWN SYNDROME - 112
9) KLINEFELTER SYNDROME - 112
10) TURNER SYNDROME - 112



X - RAY - 143 TO 145
CADAVER - 146 TO 148
HISTOLOGY - 149

1) HYALINE CARTILAGE , SEROUS GLAND - 149
2) SKELETON MUSCLE , CARDIAC MUSCLE - 149
3) CARDIAC MUSCLE , SPLEEN - 150
4) LIVER , PALATINE TONSIL - 150
5) PANCREAS , APPENDIX - 151
6) UROTHELIUM , RESPIRATORY EPITHELIUM - 151
7) DUODENUM , STOMACH - 152
8) KIDNEY , URINARY BALDDER - 152
11) PROSTRATE GLAND , UTERUS - 153
15) THYROID , CEREBELLUM - 153

SUNIL KUMAR



UPPER LIMB MUSCLES

- Radial Nerve
- Radial Nerve
- Posterior interosseous Nerve

- Musculocutaneous Nerve
- Median Nerve
- Ulnar Nerve

ARM

POSTERIOR COMPARTMENT
- TRICEPS BRACHI

ANTERIOR COMPARTMENT
- BICEP BRACHII
- BRACHIALIS
- CORACOBRACHIALIS

FOREARM

POSTERIOR / EXTENSOR COMPARTMENT

SUPERFICIAL
- ANCONEUS
- BRACHIORADIALIS
- EXTENSOR CARPI RADIALIS LONGUS
- EXTENSOR CARPI RADIALIS BREVIS
- EXTENSOR DIGITORUM
- EXTENSOR DIGITI MINIMI
- EXTENSOR CARPI ULNARIS

DEEP
- SUPINATOR
- ABDUCTOR POLLICIS LONGUS
- EXTENSOR POLLICIS BREVIS
- EXTENSOR POLLICIS LONGUS
- EXTENSOR INDICIS

ANTERIOR / FLEXOR COMPARTMENT

SUPERFICIAL
- PRONATOR TERES
- FLEXOR CARPI RADIALIS
- PALMARIS LONGUS
- FLEXOR DIGITORUM SUPERFICIALIS
- FLEXOR CARPI ULNARIS

DEEP
- FLEXOR DIGITORUM PROFUNDA
  - MEDIAL 1/2
  - LATERAL 1/2
- FLEXOR POLLICIS LONGUS
- PRONATOR QUADRATUS

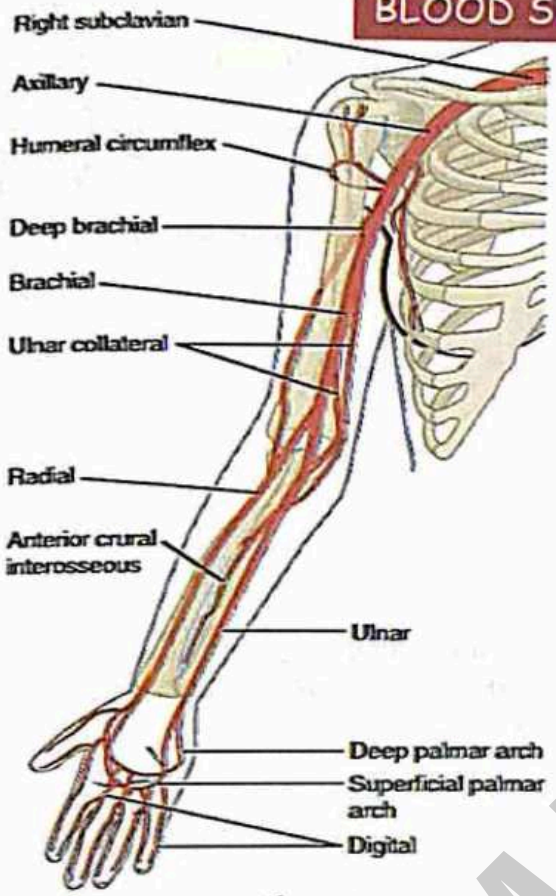
HAND

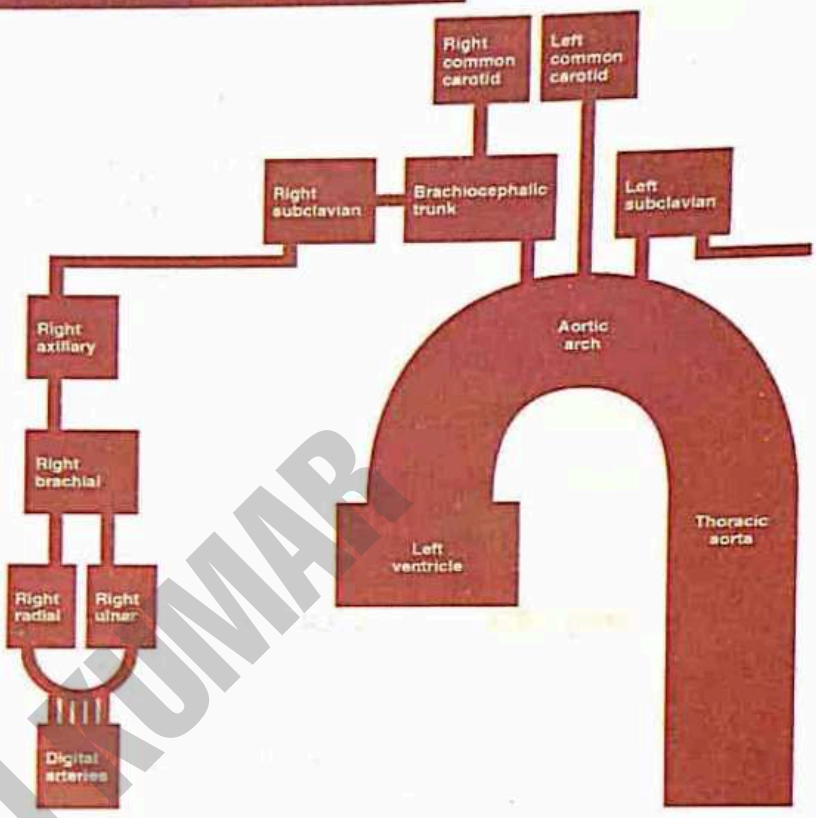
- THENAR
  - Abductor Pollicis brevis
  - Flexor Pollicis brevis
  - Opponens pollicis
  - Adductor pollicis
- HYPOTHENAR
  - Abductor digiti minimi
  - Flexor digiti minimi
  - Opponens digiti minimi
- LUMBRICALS
  - 1st & 2nd
  - 3rd & 4th
- INTEROSSEI
  - Palmar
  - Dorsal

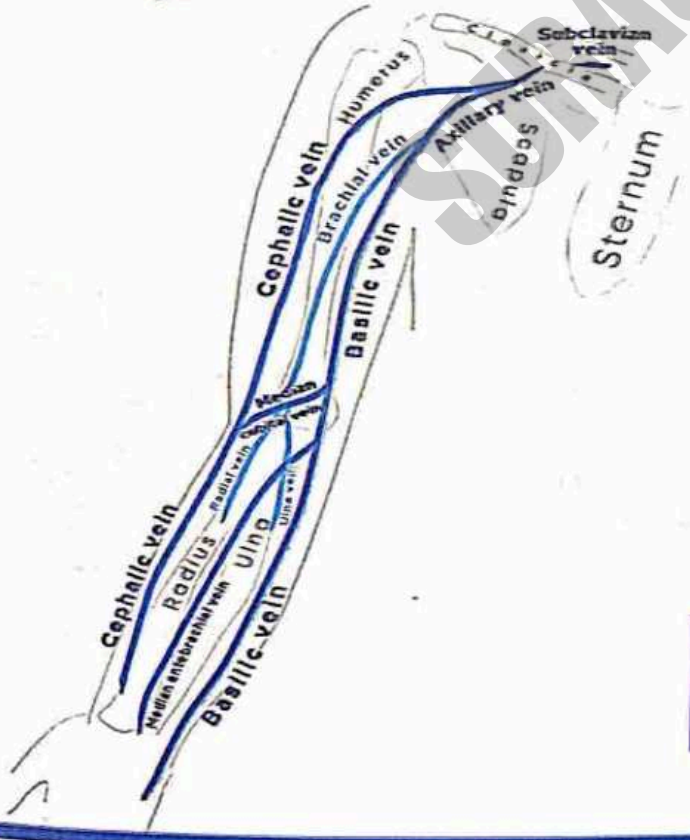
- Median Nerve
- Deep branch of Ulnar Nerve

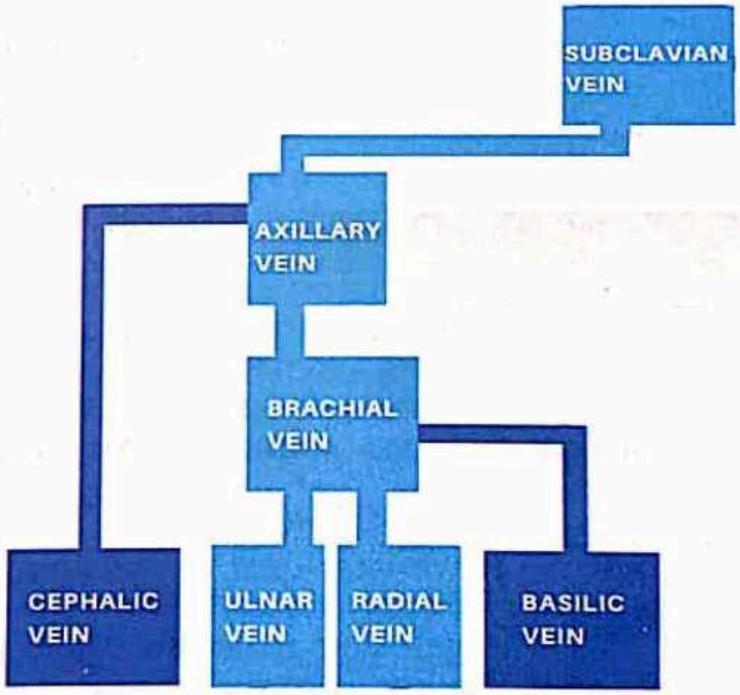
IRAJ KUMAR

BLOOD SUPPLY OF UPPER LIMB









SUBAJ KUMAR

BRACHIAL PLEXUS

- BRACHIAL PLEXUS IS A NETWORK OF NERVE WHICH SUPPLY UPPER LIMB
- CONSIST ⟶ ROOTS, TRUNKS, DIVISIONS, CORDS AND BRANCHES.
- ROOTS ⟶ CONSTITUTED BY ANTERIOR PRIMARY RAMI C5 - C8 AND T1.
  - ORIGIN OF PLEXUS MAY SHIFT UPWARD & DOWNWARD

PREFIXED PLEXUS ⟶ C4 IS LARGE AND T2 ABSENT
POSTFIXED PLEXUS ⟶ T1 LARGE, T2 PRESENT, C4 ADSENT C5 REDUCED IN SIZE

- **TRUNK** ⟶ C5 & C6 - UPPER TRUNK
  C7 - MIDDLE TRUNK
  C8 & T1 - LOWER TRUNK

- **DIVISION** ⟶ EACH TRUNK DIVIDE INTO VENTRAL & DORSAL DIVISION

- **CORD** ⟶ LATERAL CORD - UNION VENTRAL DIVISION OF UPPER & MIDDLE TRUNK
  MEDIAL CORD - VENTRAL DIVISON OF LOWER TRUNK
  POSTERIOR CORD - DORSAL DIVISION OF ALL 3 TRUNK

- **BRANCHES** ⟶ LATERAL CORD - LATERAL PECTORAL NERVE (C5-C7)
  - MUSCULOCUTANEOUS NERVE (C5-C7)
  - LATERAL ROOT OF MEDIAN (C5-C7)

⟶ MEDIAL CORD - MEDAIL PECTORAL (C8 , T1)
- MEDIAL CUTANEOUS NERVE OF ARM (C8,T1)
- MEDIAL CUTANEOUS NERVE OF FOREARM (C8,T1)
- ULNAR (C7,C8,T1)
- MEDIAL ROOT OF MEDIAN NERVE (C8-T1)

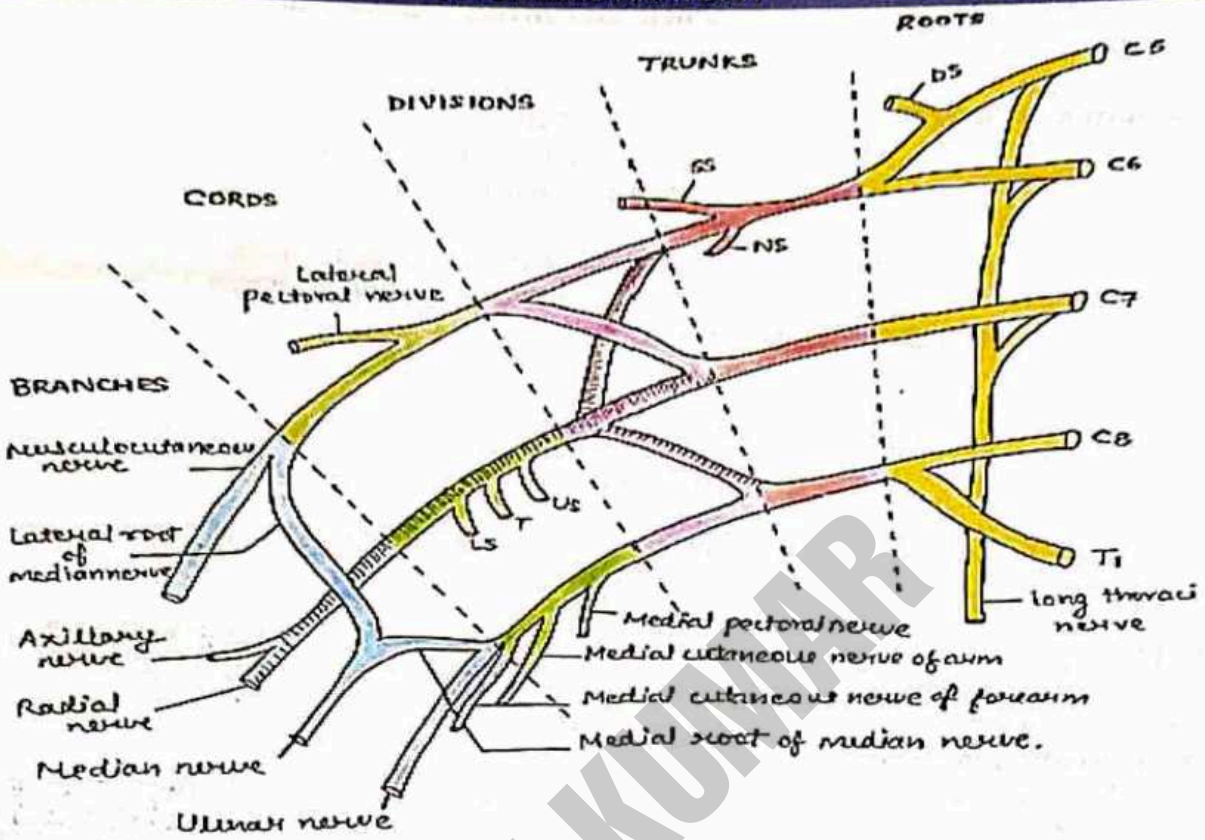
⟶ POSTERIOR CORD - UPPER SUBSCAPULAR (C5 , C6)
- NERVE TO LATISSIMUS DORSI (C6-C8)
- LOWER SUBSCAPULAR NERVE (C5-C6)
- AXILLARY NERVE (C5-C6)
- RADIAL NERVE (C5-C8 , T1)

SURAJ KUMAR

CLINICAL ANATOMY

1. <u>ERB'S PARALYSIS</u> - SITE OF INJURY - ERBS POINT IN UPPER TRUNK
   DEFORMITY - ARM - ADDUCTED AND MEDIALLY ROTATED
   FOREARM - EXTENDED & PRONATED
   THE DEFORMITY ALSO KNOWN AS POLICEMAN'S TIP HAND
   NERVE ROOT INVOLVE - MAINLY C5 & PARTIALLY C6

2. <u>KLUMPKE'S PARALYSIS</u> - SITE OF INJURY - LOWER TRUNK
   DEFORMITY - CLAW HAND - DUE TO UNOPPOSED ACTION OF LONG FLEXORS AND EXTENSORS
   CLAW HAND - HYPEREXTENSION AT METACARPOPHALANGEAL JOINT
   -FLEXION AT INTERPHALANGEAL JOINT
   NERVE ROOT INVOLVE - MAINLY T1 & PARTIALLY C8



Brachial plexus and its branches
[SS - Suprascapular nerve
US - Upper subscapular nerve
T - Thoraco dorsal nerve
NS - Nerve to subclavius
LS - Lower subscapular nerve
DS - Dorsal scapular nerve]

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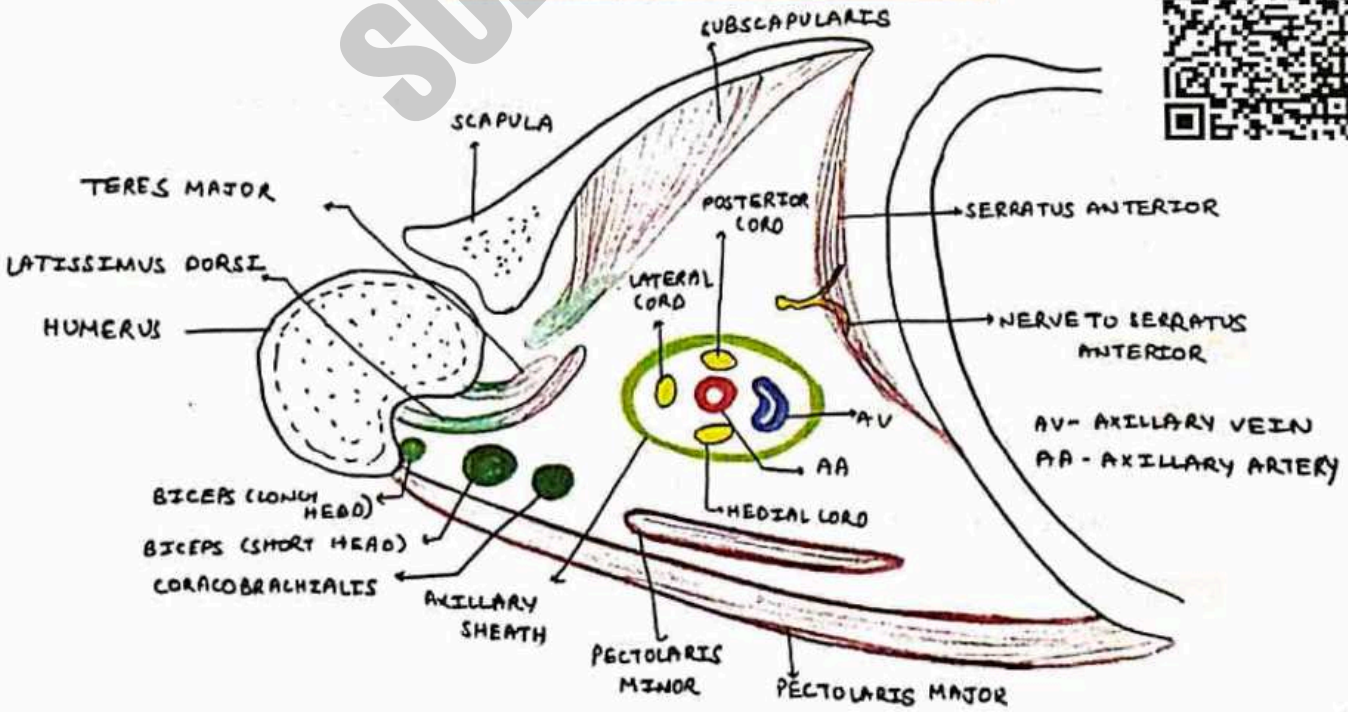
AXILLA



¶ DEFINITION :- PYRAMIDAL SPACE, SITUATED - UPPER PART OF THE ARM AND CHEST WALL

BOUNDARIES :- APEX - ANTERIOR COVER BY CLAVICLE
- POSTERIOR BY SUPERIOR BORDER OF SCAPULA
- MEDIAL BY OUTER BORDER OF FIRST RIB

BASE - SKIN , SUPERFICIAL FASCIA

ANTERIOR WALL - PECTORAL MAJOR & MINOR , CLAVIPECTORAL FASCIA

POSTERIOR WALL - SUBSCAPULARIS , TERES MAJOR , LATISSIMUS DORSI

LATERAL WALL - NARROW , CORACOBRACHIALIS & SHORT HEAD OF BICEP BRACHII

CONTENTS :-

1) AXILLARY ARTERY & VEIN BRANCHES
2) INFRACLAVICULAR PART OF BRACHIAL PLEXUS
3) AXILLARY LYMPH NODE
4) LONG THORACIC & INTERCOSTOBRACHIAL NERVE
5) AXILLARY FAT AND AREOLAR TISSUE

SCAN QR WITH QRBOT APP TO DIRECTLY OPEN VIDEO LINK

SURAJ KUMAR

AXILLARY ARTERY

INTRODUCTION - CONTINUATION OF SUB CLAVIAN ARTERY
-EXTEND - OUTER BORDER OF FIRST RIB TO LOWER BORDER OF TERES MAJOR MS.
-DEVIDED—3 PARTS

PECTORALIS MINOR MUSCLE CROSS ARTERY - DIVIDES INTO 3 PARTS

- 1ST PART - BRANCHES - SUPERIOR THORACIC ARTERY
- 2ND PART - BRANCHES - THORACOCROMIAL ARTERY
  -LATERAL THORACIC ARTERY
- * 3RD PART-BRANCHES - SUBSCAPULAR ARTERY
  -ANTERIOR CIRCUMFLEX HUMERAL ARTERY
  -POSTERIOR CIRCUMFLEX HUMERAL ARTERY

*BRANCHES :-

1.SUPERIOR THORACIS ARTERY -
BRANCH - 1ST PART OF AXILIARY ARTERY
DIRECTION - DOWN, FORWARD & MEDICAL
SUPPLY - PECTORAL MAJOR

2.THORACOCROMIAL ARTERY -
BRANCH - 2ND PART OF AXILLARY ARTERY

3.LATERAL THORACIC ARTERY -

- BRANCH - 2ND PART OF AXILLARY ARTERY
  IT EMERGES - LOWER BORDER OF PECTORALIS MINOR

4.ANTERIOR CIRCUMFLEX HUMERAL ARTERY -

- BRANCH -3RD PART OF AXILLARY ARTERY
  ARISE - LOWER PART OF SUBSCAPULARIS
  SUPPLY -ANASTOMOSIS POSTERIOR CIRCUMFLEXX HUMERAL ARTERY

5.POSTERIOR CIRCUMFLEX HUMERAL ARTERY -

- BRANCH -3RD PART OF AXILLARY ARTERY
  EXTEND - LOWER BORDER OF SUBSCAPULARIS
  SUPPLY -SHOULDER JOINT,DELTOID

6.SUBSCAPULARIS ARTERY -

- BRANCH -3RD PART OF AXILLARY ARTERY
  EXTEND -LOWER BORDER OF SUBSCAPULARIS TO TERMINATE NEAR INFERIOR ANGLE OF SCAPULA.
  SUPPLY - LATISSIMUS DORSI & SERRATUS ANTERIOR MUSCLE.

NOTE :- DIAGRAM OF AXILLARY ARTERY - PAGE NO.005

CUBITAL FOSSA

BIRAJ KUMAR

INTRODUCTION - TRIANGULAR HOLLOW , SITUATED - FRONT OF ELBOW

***BOUNDARIES :-**

1. LATERAL - MEDIAL BORDER OF BRACHIORADIALIS
2. MEDIAL - LATERAL BORDER OF PRONATOR TERES
3. BASE - IMAGINARY LINE OF EPICHONDYLES
4. APEX - AREA WHERE - BRACHIORADIALIS & PRONATOR TERES CROSSES
5. ROOF - SKIN , SUPERFICIAL FASCIA
   -BICIPITAL APONEUROSIS
6. FLOOR - BRACHIALIS

***CONTENTS :-**

1) MEDIAN NERVE - GIVE BRANCHES - FLEXOR CARPI RADIALIS
   -PALMARIS LONGUS , FLEXOR DIGITORUM SUPERFICIALIS

2) RADIAL NERVE - GIVE - POSTERIOR INTEROSSEI NERVE AND DEEP BRANCH OF RADIAL NERVE

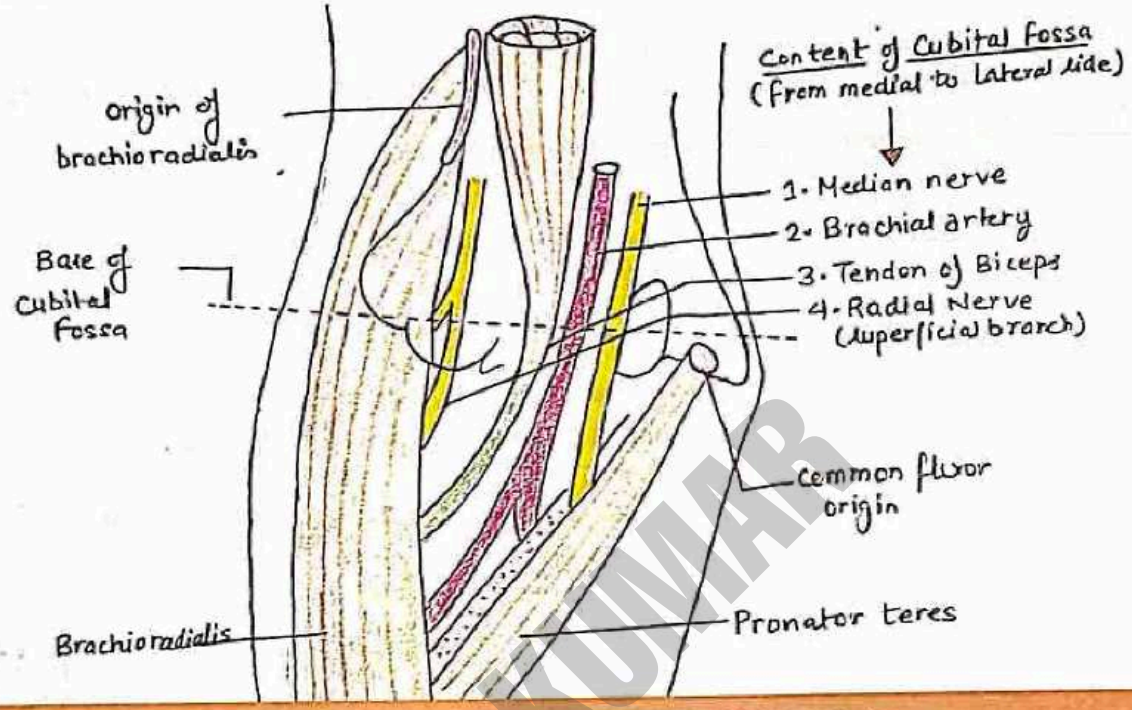
3) BRACHIAL ARTERY - GIVE - RADIAL ARTERY -> RADIAL RECURRENT BRANCH
   -ULNAR ARTERY -> ANTERIOR & POSTERIOR RECURRENT BRANCH , COMMON INTEROSSEOUS BRANCH

4) TENDON OF BICEP BRACHII

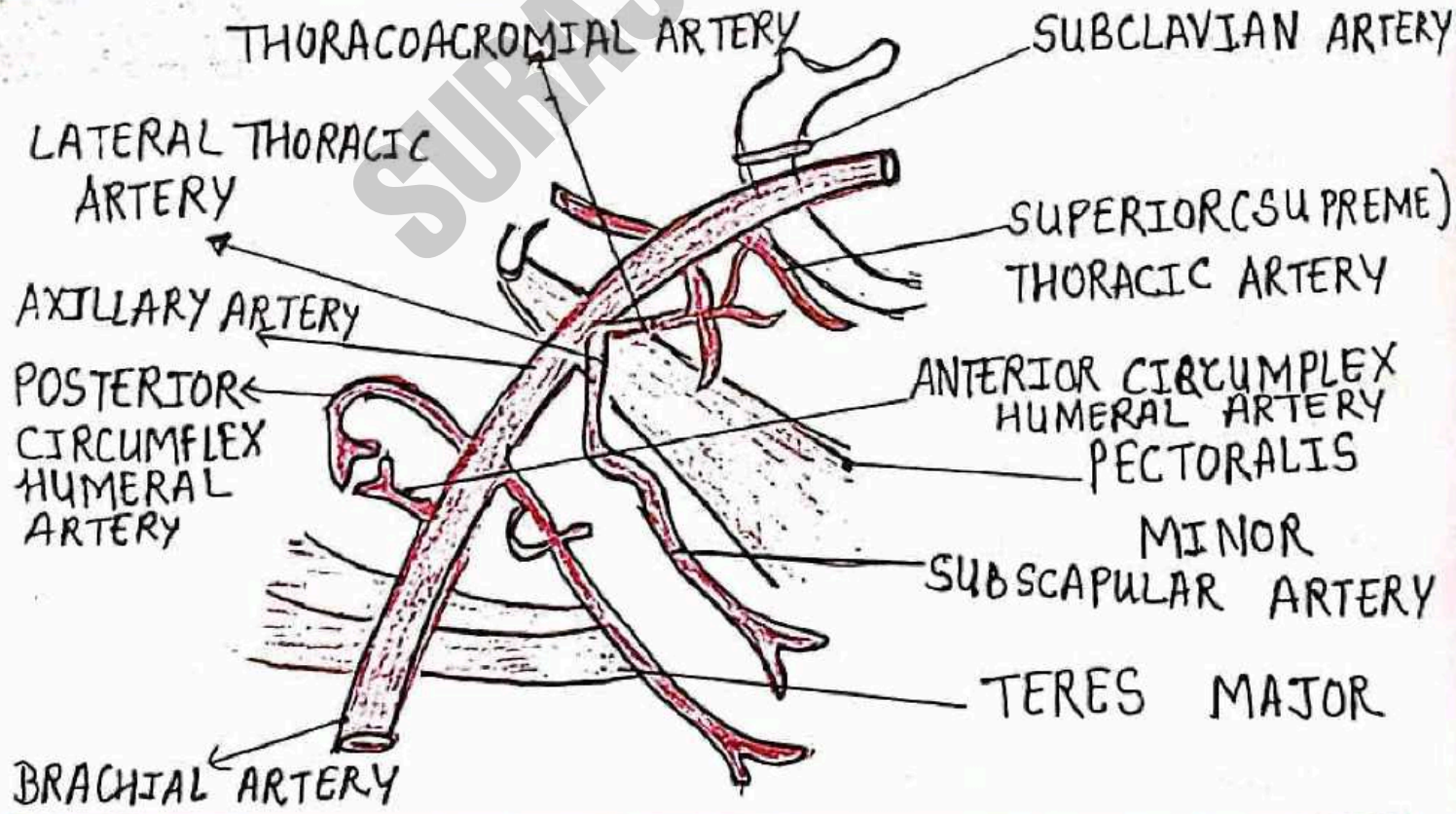
CLINICAL ANATOMY

1. MEDIAN CUBITAL VEIN - VEIN OF CHOICE FOR INTRAVENOUS INJECTION.

2. BLOOD PRESSURE UNIVERSALLY RECORDED - BRACHIAL ARTERY INFRONT OF ELBOW



CONTENT OF CUBITAL FOSSA



BRANCHES OF AXILLARY ARTERY

SURAJ KUMAR

MAMMARY GLAND

BREAST IS MOST IMPORTANT STRUCTURE PRESENT IN PECTORAL REGION.
FOUND IN BOTH SEX BUT RUDIMENTARY IN MALE.
MODIFIED SWEAT GLAND.
*SITUATION - LIE IN SUPERFICIAL FASICA.
DIVIDE INTO 4 QUADRANT - UPPER MEDIAL, UPPER LATERAL, LOWER MEDIAL, LOWER LATERAL

*EXTENSION OF UPPER QUADRANT CALLED AXILLARY TAIL OF SPENCE ->WHICH PASSES THROUGH OPENING IN DEEP FASICA.
*OPENING CALLED FORAMEN OF LANGER.

*EXTENT OF BREAST -
VERTICALLY - 2ND - 6TH RIB
HORIZONTAL - LATERAL BORDER OF STERNUM TO MID AXILLARY LINE
DEEP RELATION - BREAST LIE ON DEEP FASCIA
-DEEPER THERE ARE PART OF 3 MUSCLES - PECTORALIS MAJOR
- SERRATAUS ANTERIOR
- EXTERNAL OBLIQE MUSCLE

*STRUCTURE OF BREAST -
SKIN :- CONICAL POSITION OF NIPPLE ->AT LEVEL->4TH INTERCOASTAL SPACE.
- NIPPLE PIERCED BY 15-20 LECTIFEROUS DUCT.
- IT CONTAIN CIRCULAR & LONGITUDINAL SMOOTH MUSCLE.
- SKIN SURROUNDING BASE OF NIPPLE IS PIGMENTED & FORM CIRCULAR AREA CALLED AREOLA

PARENCHYMA :- MAMMARY GLAND COMPOUND TUBULOALVEOAR GLAND WHICH SECRETES MILK
- IT IS A MODIFIED SWEAT GLAND - GLAND CONSISTS 50-20 LOBES
- EACH LOBE -> CLUSTER OF ALVEOLI -> DRAINED BY LECTIFEROUS DUCT
- LECTIFEROUS DUCT -> CONVERGE -> TOWARD NIPPLE
- LECTIFERAL DUCT NEAR ITS TERMINATION DILATATION -> CALLED LECTIFEROUS SINUS.

STROMA :- FROM SUPPORTIVE FRAMEWORK OF GLAND
- PARTLY FIBROUS PARTLY FATTY

BLOOD SUPPLY :-
1.INTERNAL THORACIC ARTERY
2.LATERAL THORACIC, SUPERFICIAL THORACIC, ACROMIOTHORACIC
3.LATERAL BRANCH -> POSTERIOR INTERCOASTAL ARTERY

*VEINS -> FOLLOW THE ARTERY
1.SUPERFICIAL VEIN -> DRAIN INTERNAL THORACIC VEIN
2.DEEP VEIN -> DRAIN-> AXILLARY & POSTERIOR INTERCOASTAL VEIN

*NERVE SUPPLY :-
SUPPLY BY ANTERIOR AND LATERAL CUTANEOUS BRANCHES OF 4TH & 6TH INTERCOASTAL NERVE.

POINTS TO REMEMBER !

* NERVE DO NOT CONTROL SECRETIION OF MILK

*SECRETION CONTROL BY HORMONE PROLACTIN

SUNIL KUMAR

LYMPHATIC DRAINAGE OF BREAST

*LYMPHATIC DRAINAGE OF BREAST ASSUMED GREAT IMP. TO SURGEON BECAUSE CARCINOMA SPREAD THROUGH LYMPH NODE.

***LYMPH NODE -:**

1. AXILLARY LYMPH NODE - ANT., POST., LAT., CENTRAL, APCAL L.N. (75%)
2. INTERNAL MAMMARY L.N. - (20%)
3. SOME LYMPH (5%) -> FROM BREAST REACH->* SUPRACLAVICULAR NODE
   * DELTOPECTORAL NODE
   * POST.INTERCOASTAL NODE
   * SUB DIAPHRAGMATIC

***LYMPHATIC VESSELS**

1. SUPERFICAL LYMPHATICS --->DRAIN->SKIN OVER BREAST EXCEPT NIPPLE & AREOLA

2. DEEP LYMPHATIC -->DRAIN->PARENCHYMA OF BREAST

3. LYMPH FROM ANTERIOR & POSTERIOR LYMPH NODE PASS -> CENTRAL & LATERAL LYMPH NODE -->THROUGH-->APICAL GROUP-> -->SUPRACLAVICAL NODES

4. INTERNAL MAMMARY NODE DRAIN LYMPH NOT ONLY FROM INNER HALF OF BREAST BUT ALSO OUTER HALF OF BREAST.

5. PLEXUS OF LYMPHATIC VESSELS PRESENT DEEP TO AREOLA IS SUBAEROLAR PLEXUS OF SAPPEY --->DRAIN--> ANTERIOR LYMPH NODE.

6. DEEP LYMPHATIC OF BREAST -->PASS THROUGH PECTORALIS MAJOR MUSCLE & CLAVIPECTORAL FASICA->APICAL NODE.

SURAJ KUMAR

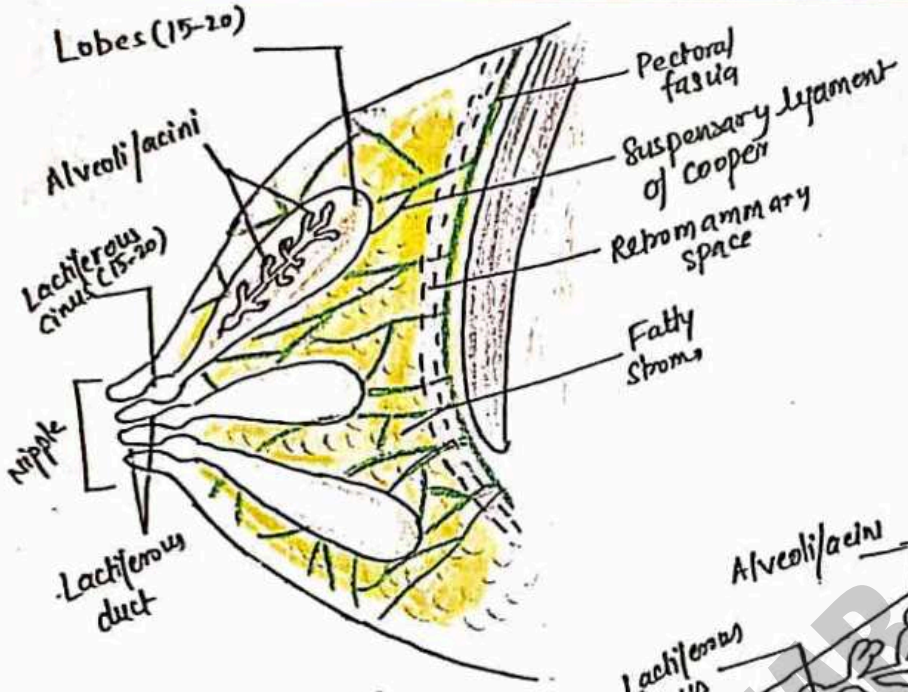
CLINICAL ANATOMY

1. UPPER AND LOWER QUADRANT OF BREAST IS FREQUENT SITE OF CARCINOMA

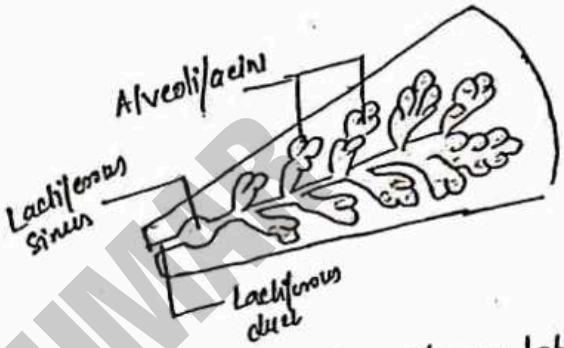
2. CANCER CELL MAY INFILTRATE THE SUSPENSARY LIGAMENT THEN BREAST BECOMES FIX CONTRACTION OF LIGAMENT->CAUSE RETRACTION OR FOLDING OF SKIN.

3. OBSTRUCTION OF SUPERFICIAL LYMPH VESSEL BY CANCER CELL->PRODUCE OEDEMA OF SKIN-->APPEAR GIVING APPEARANCE LIKE SKIN OF ORRANGE (*PEAU D'ORANGE APPEARANCE*).

4. CARCINOMA USUALLY ARISES FROM EPITHILIUM OF LARGE DUCTS.

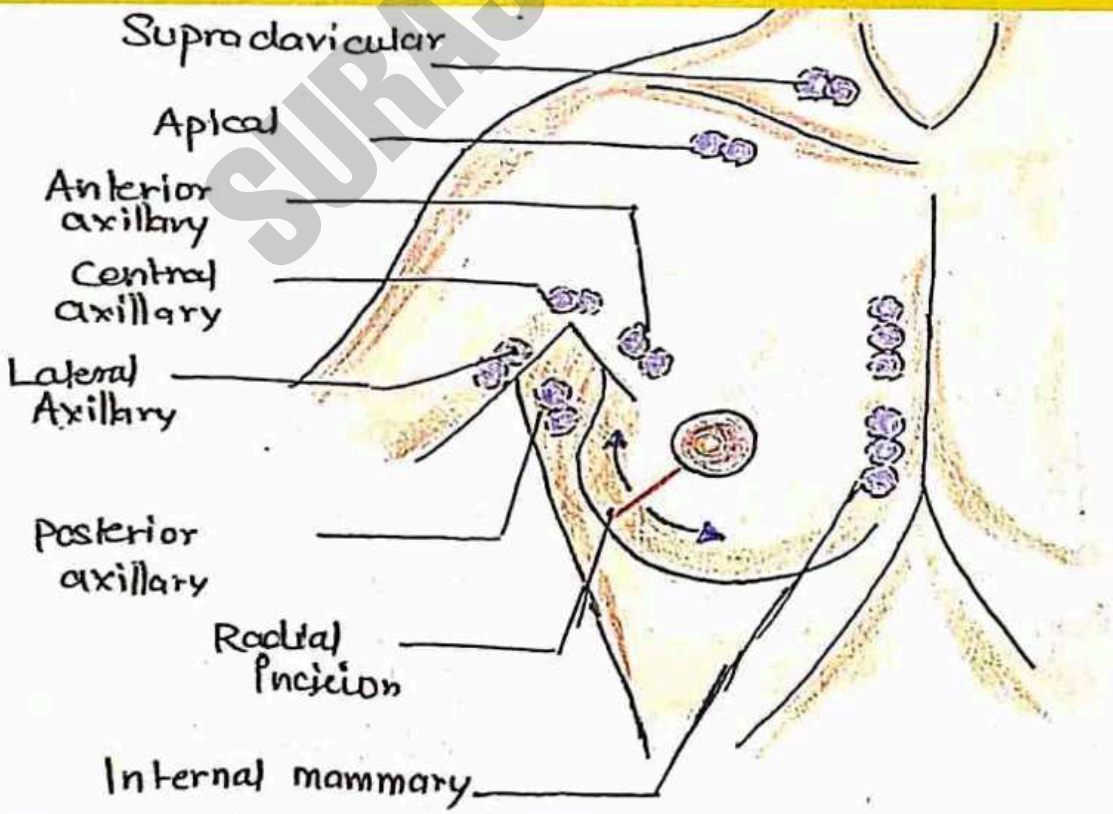


∴ Suspensory ligament of breast & its lobe



∴ Structure of one lobe of mammary gland

STRUCTURE OF BREAST



LYMPH NODE DRAINING BREAST

SURAJ KUMAR

RADIAL NERVE

*THICKEST BRANCH OF BRACHIAL PLEXUS.

➢ ROOT VALUE - VENTRAL RAMI OF C5, C8 & T1

➢ COURSE :-

AXILLA - RADIAL NERVE LIE POSTERIOR TO AXILLARY ARTERY
-THEN ENTER IN LOWER TRIANGULAR SPACE.

RADIAL SULCUS - RADIAL NERVE ENTER THROUGH LOWER TRIANGULAR SPACE INTO RADIAL SULCUS.

CUBITAL FOSSA - NERVE ENTER LATERAL SIDE OF CUBITAL FOSSA.
-TERMNATE INTO SUPERFICIAL & DEEP BRANCH.
DEEP BRANCH-->SUPPLY->EXTENSOR CARPIRADIALIS BREVIS
-->SUPINATOR.
*THEN IT COURSE BETWEEN TWO HEAD OF SUPINATOR MUSCLE -->REACH BACK OF FOREARM.
SUPERFICIAL BRANCH ->LEAVE CUBITAL FOSSA TO ENTER LATERAL SIDE OF FRONT OF FOREARM.

DORSUM OF FOREARM - SUPERFICIAL BRANCH--->DESCEND TILL SNUFF BOX
- TO REACH DORSUM OF HAND--->SUPPLY SKIN OF DORSAL HALF HAND.

➢ BRANCHES :-

1.AXILLA--->MUSCULAR--->LONG HEAD OF TRICEP BRACHI
--->MEDIAL HEAD OF TRICEP BRACHI
--->CUTANEOUS--->POSTERIOR CUTANEOUS NERVE OF ARM

2.RADIAL SULCUS--->MUSCULAR->LATERAL HEAD OF TRICEP BRACHII
-->ANCONEAUS
-->CUTANEOUS-->LATERAL CUTANEOUS NERVE OF ARM
->POSTERIOR CUTANEOUS NERVE OF FOREARM.

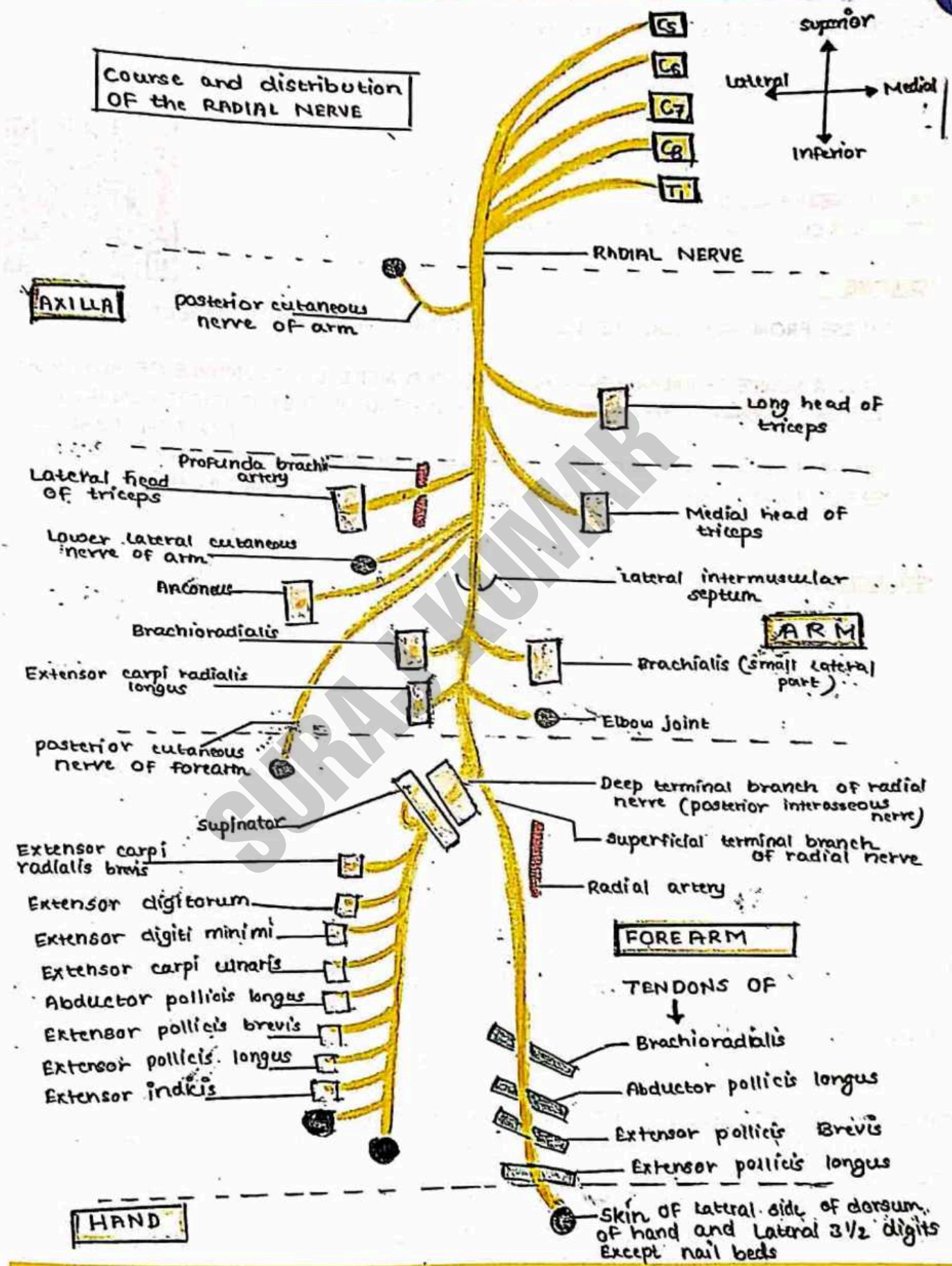
3.LATERAL SIDE OF ARM-->MUSCULAR->BRACHIORADIAL
-->EXTENSOR CARPI RADIALIS LONGUS

4.DEEP BRANCH-->DEEP BRANCH OF RADIAL NERVE
->SUPINATOR MUSCLE
-->EXTENSOR CARPI RADIALIS BREVIS

5.SUPERFICIAL BRANCH-->SUPERFICIAL BRANCH OF RADIAL NERVE
-->CUTANEOUS BRANCHES IN ANATOMICAL SNUFFBOX
(TO SUPPLY LATERAL HALF OF DORSUM OF HAND)

SURAJ KUMAR

SCAN QR WITH QRBOT APP TO DIRECTLY OPEN VIDEO LINK



BRANCHES OF RADIAL NERVE

SURAJ KUMAR

WRIST DROP-RADIAL NERVE INJURY IN RADIAL GROOVE DUE TO FRACTURE OF SHAFT OF HUMEROUS CAUSE WRIST DROP.
THIS RESULT IN WEAKNESS AND LOSS OF POWER OF EXTENSION IN WRIST.

ULNAR NERVE

*ULNAR NERVE ALSO KNOWN AS MUSICIAN NERVE BECAUSE IT CONTROLS FINE MOVEMENTS OF FINGERS.

*COURSE :-

*ARISE FROM BRACHIAL PLEXUS -->DESCEND DOWN MEDIAL ASPECT OF UPPER ARM.
*ULNAR NERVE IS PALPABLE--->LIES BEHIND MEDIAL EPICONDYLE OF HUMEROUS
*ENTER FOREARM--->PASSING BETWEEN 2 HEAD OF FLEXOR CARPI ULNARIS (CUBITAL TUNNEL)
*IN FOREARM--->LIE ALONG--->LATERAL BORDER OF FLEXOR CARPI ULNARIS.
*ENTER PALM--->PASSING SUPERFICIAL TO FLEXOR RETINACULUM

*BRANCHES:-

1.MUSCULAR--->FLEXOR CARPI ULNARIS
-->FLEXOR DIGITORUM PROFUNDUS (HALF)

2.CUTANEOUS---->PALMER CUTANEOUS BRANCH
(SUPPLIES SKIN OVER HYPOTHENAR MUSCLE)
---->DORSAL CUTANEOUS BRANCH
(SUPPLIES MEDIAL 2HALF FINGER DORSUM OF HAND)

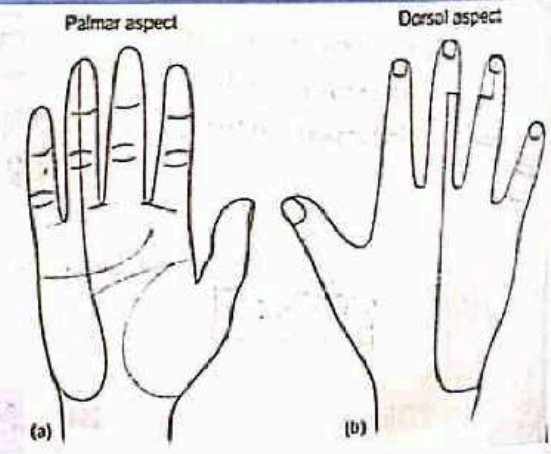
3.PALMAR---->SUPERFICIAL TERMINAL BRANCH->PALMARIS BREVIS
->DIGITAL BRANCH
----->DEEP TERMINAL BRANCH - HYPOTHENAR
- INTEROSSEOUS MUSCLE
-MEDIAL 2 LUMBRICAL

CLINICAL ANATOMY

1.ULNAR NERVE INJURY AT WRIST- PRODUCE ULNAR CLAW HAND

*CAUSE - HYPEREXTENSION AT METACARPOPHALANGEAL JOINTS AND FLEXION AT THE INTERPHALANGEAL JOINTS, INVOLVE-INDEX OR MIDDLE FINGER.

*SENSORY LOSS - MEDIAL 1/3 OF PALM AND MEDIAL 1HALF FINGER.
-MEDIAL HALF OF DORSUM OF HAND.



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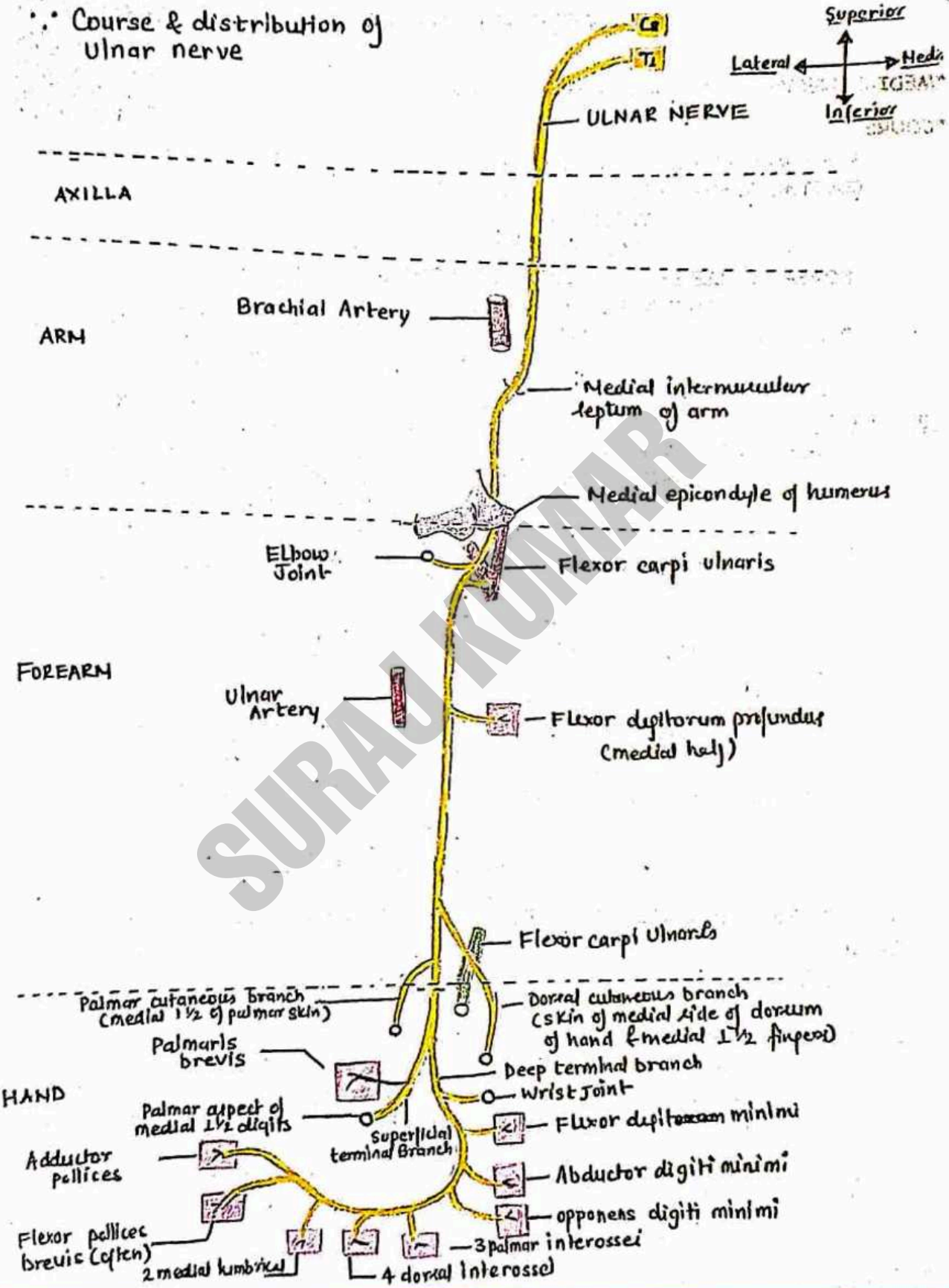
∴ Course & distribution of Ulnar nerve



BRANCHES OF ULNAR NERVE

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MEDIAN NERVE

*MEDIAN NERVE IS THE MAIN NERVE OF FOREARM.

*COURSE -

ARM - MEDIAL NERVE LIE MEDIAL TO BRACHIAL ARTERY ENTER CUBITAL FOSSA.

CUBITAL FOSSA - LIE MEDIAL IN CUBITAL FOSSA
- THEN LEAVE FOSSA BY PASSING BETWEEN TWO HEAD OF PRONATOR TERES.

FOREARM - LIE BETWEEN FLEXOR DIGITORUM SUPERFICIALIS AND FLEXOR DIGITORUM PROFUNDUS.

PALM - ENTER IN HAND LIE DEEP TO --->FLEXOR RATINOCULUM THROUGH CARPAL TUNEL.

*BRANCH :-

1.MUSCULAR-ARM-PRONATOR TERES
CUBITAL FOSSA----->FLEXOR CARPI RADIALIS
---->FLEXORDIGITORUM SUPERFICIALIS
-->PALMARIS LONGUS
FOREARM---->FLEXOR DIGITORUM PROFUNDUS
---->PRONATOR QUADRATUS
--->FLEXOR POLLICIS LONGUS
PALM ----> THENAR MUSCLE
--->LATERAL 2 LUMBRICALS

2.CUTANEOUS-PALMARIS CUTANEOUS PALM
(SUPPLY SKIN OVER THENAR MUSCLE)

3.ARTICULAR BRANCHES----->ARE GIVEN TO ELBOW JOINT

4.MUSCULAR BRANCH--->SUPPLY RADIAL AND ULNAR NERVE.

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CLINICAL ANATOMY

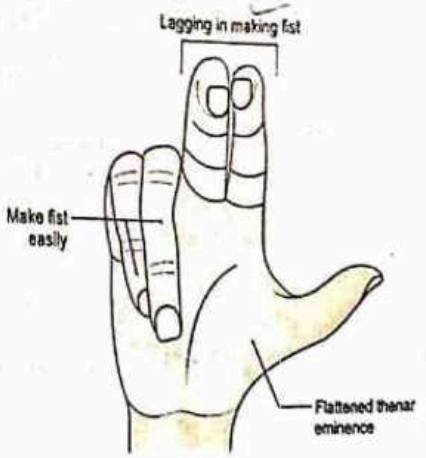
*CARPAL TUNNEL SYNDROME- CAUSED BY COMPRESSION OF MEDIAL NERVE AND CARPI TUNNEL.

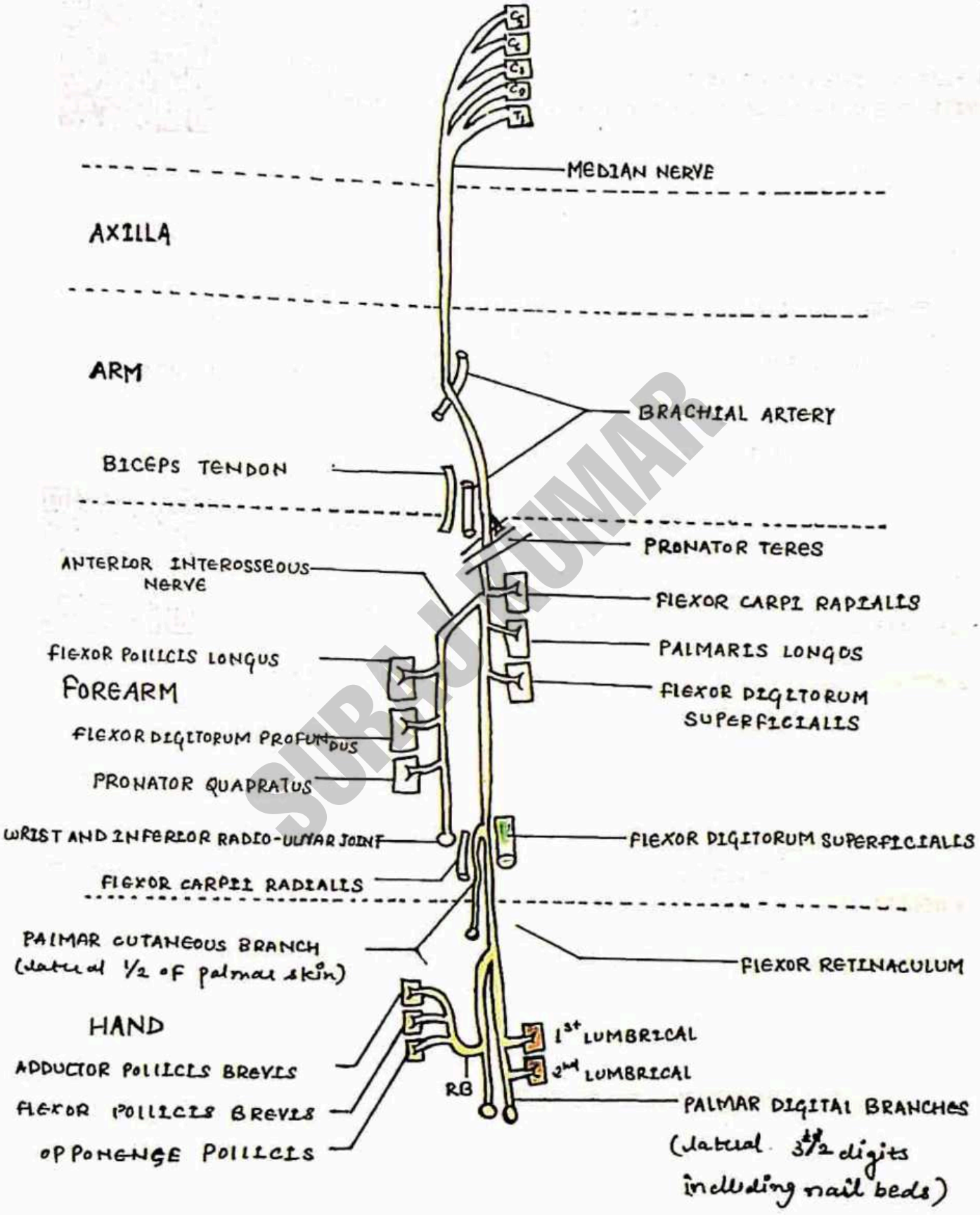
SENSORY CHANGE- LOSS OF SENSATION IN LATERAL 3 HALF DIGITS INCLUDING NAIL BEDS ON DORSUM OF HAND.

MOTOR CHANGE- APE / MONKEY LIKE THUMB DEFORMITY
LOSS OF OPPOSITION OF THUMB, INDEX AND MIDDLE FINGER LAG BEHIND MAKING FIST DUE TO PARALYSIS OF FIRST AND SECOND LUMBRICAL MUSCLE.

VASOMOTOR CHANGE- SKIN AREA WITH SENSORY LOSS WARMER DUE TO ARTERIAL DILATATION.

TROPIC CHANGE- LONG STANDING CAUSE OF PARALYSIS LEAD TO DRYER SCALY SKIN.





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BRANCHES OF MEDIAN NERVE

ROTATOR CUFF

1.FIBROUS SHEATH→FORM BY 4 FLATTENED TENDONS WHICH BLEND WITH THE CAPSULE OF SHOULDER JOINT AND STRENGTHENED IT.

2.MUSCLE WHICH FORM ROTATOR CUFF ARISE -→SCAPULA
INSERTION-→LESSER & GREATER TUBERCLE HUMEROUS

3.MUSCLE FORM ROTATOR CUFF-→SUBSCAPULARIS, SUPRASPINATUS, INFRASPINATUS, TERES MINOR.

4.CUFF GIVE STRENGTH TO SHOULDER JOINT--→EXCEPT-→INFERIORLY.

5.THUS DISLOCATION OF HUMEROUS OCCUR COMMONLY INFERIOLATERAL.

6.SUBACROMIAL BURSA---→FLUID FILLED SAC. REDUCED FRICTION
--→LARGEST BUSRA OF BODY - BLOW CORACOACROMIAL ARC.

NOTE :- DIAGRAM OF ROTATOR CUFF - PAGE NO.16

FLEXOR RATINACULUM

1.STRONGEST FIBROUS BAND WHICH BRIDGE ANTERIOR CONCAVITY OF CARPUS & CONVERT INTO TUNNEL---→CARPLE TUNNEL.

2.ATTACHMENT - MEDIAL------→PISIFORM BONE, HOOK OF HAMATE
LATERAL---→TUBERCLE OF SCAPHOID, CREST OF TRAPEZIUM.

MEDIAL SUPERFICIAL SLIP---→PISIFORM BONE
(CONVERT INTO TUNNEL-→PASS-→ULNAR VESSEL & NERVE)

LATERAL DEEP SLIP--→MEDIAL GROOVE ON TRAPEZIUM
(CONVERT INTO TUNNEL--→PASS--→FLEXOR CARPI RADIALIS TENDON)

3.RELATION:-

*STRUCTURE PASSES SUPERFICIAL TO FLEXOR RETINACULUM :-
1.PALMARIS CUTANEOUS BRANCH OF MEDIAN NERVE & ULNAR NERVE.
2.PALMARIS LONGUS TENDON.
3.ULNAR NERVE AND VESSEL.

*STRUCTURE PASSES DEEP TO F.R.:-
1.MEDIAN NERVE.
2.ULNAR & RADIAL BURSA.
3.4 TENDON OF FLEXOR DIGITORUM SUPERFICIALIS.
4.4 TENDON OF FLEXOR DIGITORUM PROFUNDUS.
5.TENDON OF FLEXOR POLLICIS LONGUS.

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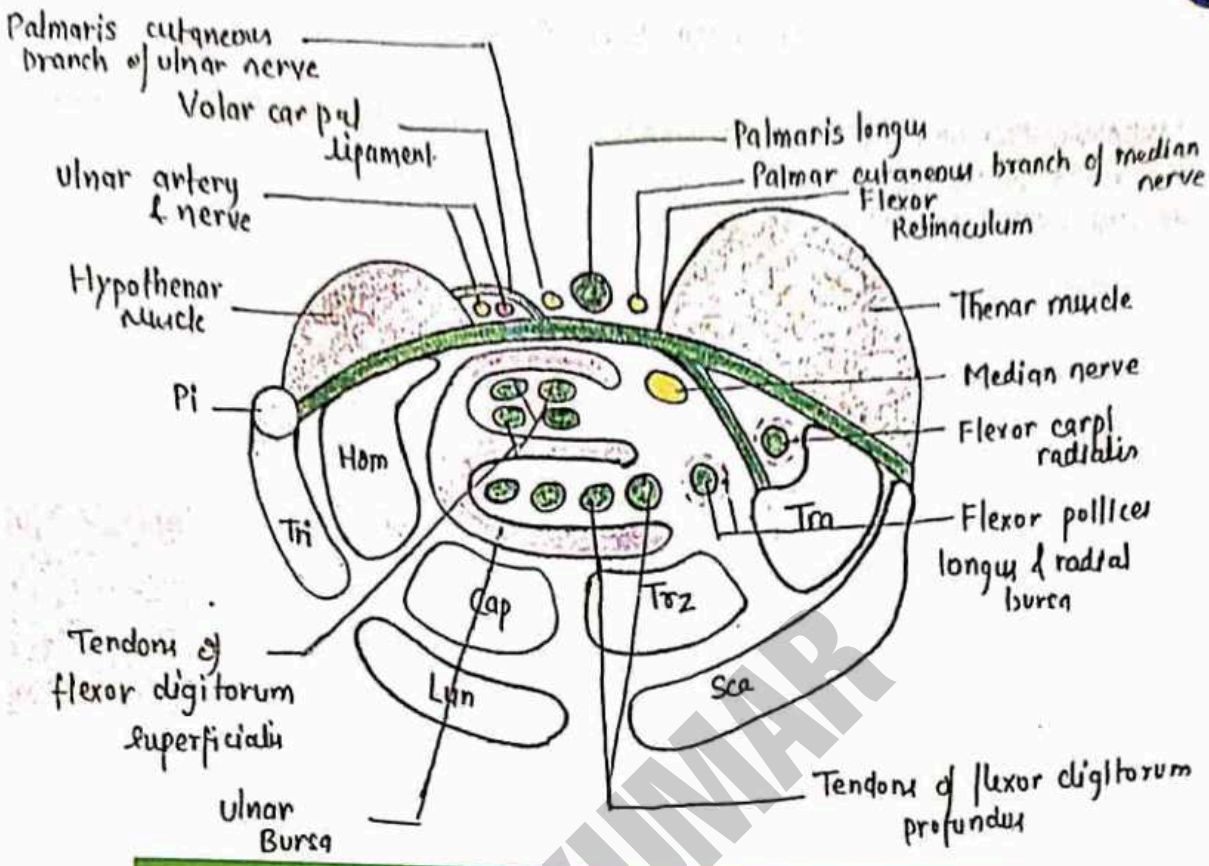
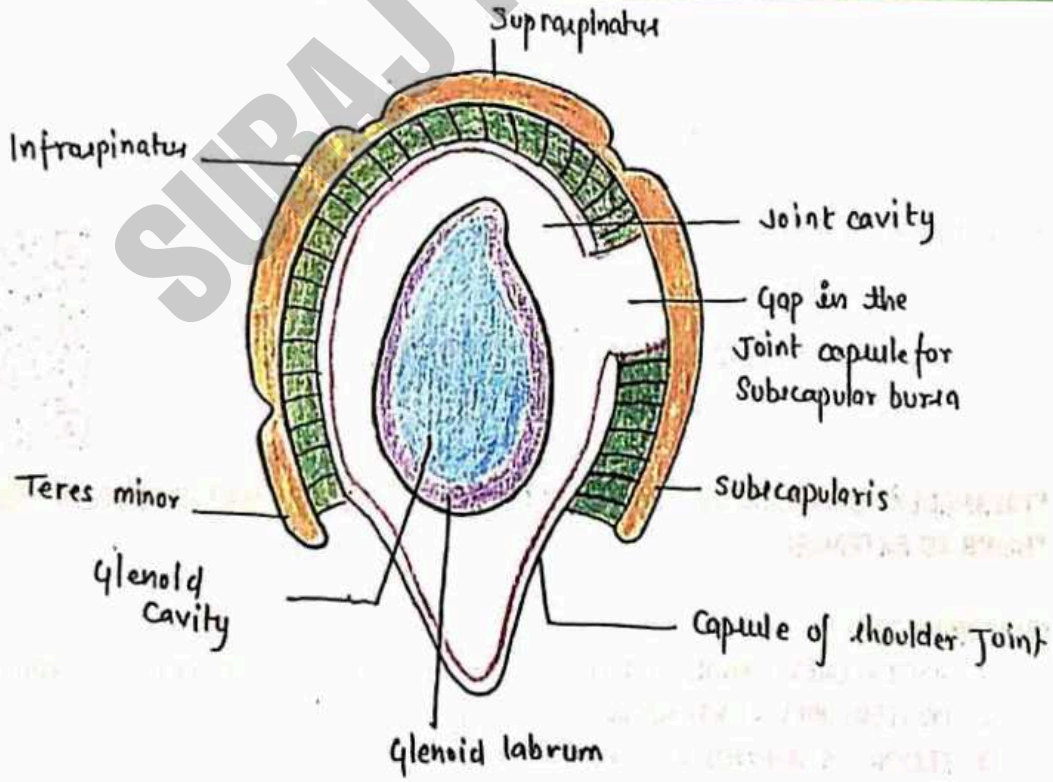


FLEXOR RETINOCULUM



∵ Rotator cuff

ROTATOR CUFF

SUBAJ KUMAR

MID PALMAR SPACE

*ARRANGEMENT OF FASCIAE AND THE FASCIAL SEPTA IN THE HAND IN SUCH A WAY MANY SPACES ARE FORMED, SPACES ARE SURGICAL IMPORTANCE BECAUSE THEY MAY BECOME INFECTED- IMPORTANT SPACES ARE AS FOLLOWS :-

A.PALMAR SPACES-->PULP SPACE OF FINGER, MID PALMAR SPACE, THENAR SPACE.

B.DORSAL SPACE----->DORSAL SUBCUTANEOUS AND DORSAL SUBAPONEUROTIC SPACE.

C.THE FOREARM SPACE OF PARONA.

MID PALMAR SPACE

1. SHAPE--->TRIANGULAR

2. SITUATION-->UNDER INNER HALF OF HOLLOW OF PALM

3. EXTEND--->PROXIMAL - FOREARM SPACE OF PARONA
   DISTAL - DISTAL PALMAR CREASE.

4.BOUNDARY-->ANTERIOR - FLEXOR TENDONS OF 3RD,4TH,& 5TH DIGITS.
-2ND, 3RD, & 4TH LUMBRICALS.
- PALMAR APONEUROSIS.

-->POSTERIOR - FASCIA COVERING INTEROSSEI & METACARPALS.
-->LATERAL - INTERMIDIATE PALMAR SEPTUM.
-->MEDIAL - MEDIAL PALMAR SEPTUM.

5.DRAINAGE-->INCISION IN EITHER THE 3RD OR 4TH WEB SPACE.

SURAJ KUMAR

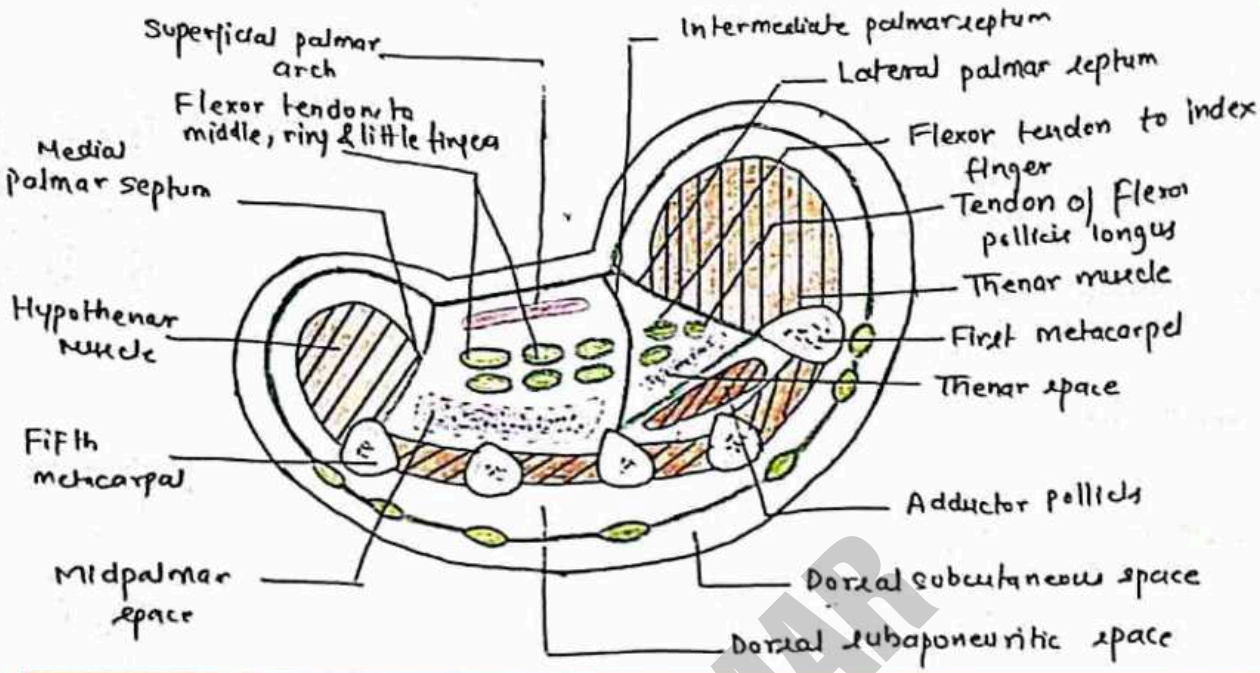
ANATOMICAL SNUFFBOX

*TRIANGULAR DEPRESSION-->POSTERIOR LATERAL SIDE OF WRIST, SEEN WHEN THUMB IS EXTENDED.

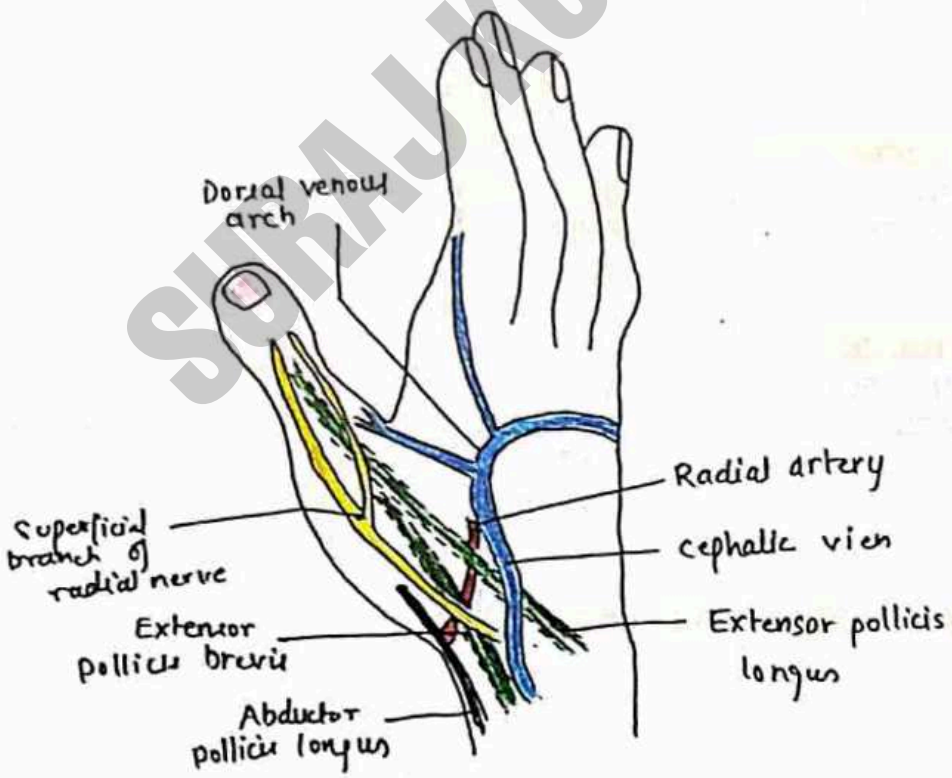
*BOUNDARIES :-

1. ANTERIORLY - ABDUCTOR POLLICIS LONGUS, EXTENSOR POLLICIS BREVIS.
2. POSTERIORLY - EXTENSOR POLLICIS LONGUS.
3. FLOOR - SCAPHOID & TRAPEZIUM.

*CONTENT :- 1.RADIAL ARTERY & RADIAL NERVE
2.CEPHALIC VEIN



MID PALMAR SPACE



∴ Anatomical snuffbox

ANATOMICAL SNUFF BOX

SUBAJ KUMAR

LUMBRICAL MUSCLE

*ORIGIN :-

1ST - LATERAL SIDE OF TENDON OF FLEXOR DIGITORUM PROFUNDUS OF 2ND DIGIT.
2ND - LATERAL SIDE OF SAME TENDON OF 3RD DIGIT.
3RD - ADJACENT SIDES OF SAME TENDONS OF 3RD & 4TH DIGIT.
4TH - ADJACENT SIDES OF SAME TENDONS OF 4TH & 5TH DIGIT.

*INSERTION -

EXTENSOR EXPANSION--->DORSUM OF BASE OF--->DISTAL PHALANGES

*NERVE SUPPLY - 1ST & 2ND -->MEDIAN NERVE
- 3RD & 4TH -->ULNAR NERVE

*ACTION - FLEX METACARPOPHALANGEAL JOINTS
- EXTEND INTERPHALANGEAL JOINT

SCAN QR WITH QRBOT APP TO DIRECTLY OPEN VIDEO LINK

THENAR EMINENCE

1.ABDUCTOR POLLICIS BREVIS :-

*ORIGIN - TUBERCLE OF SCAPHOID, CREST OF TREPAZIUM, FLEXOR RETINACULUM
*INSERTION - BASE OF PROXIMAL PHALANX OF THUMB
*ACTION - ABDUCTION OF THUMB

2.FLEXOR POLLICIS BREVIS :-

*ORIGIN - FLEXOR RETINACULUM, CREST OF TREPAZIUM & CAPITATE BONES
*INSERTION - BASE OF PROXIMAL PHALANX OF THUMB
*ACTION - FLEXES METACARPOPHALANGEAL JOINT OF THUMB

3.OPPONENS POLLICIS :-

*ORIGIN - FLEXOR RETINACULUM CREST OF TREPAZIUM
*INSERTION - LATERAL HALF OF PALMAR SURFACE OF THE SHAFT OF MATACARPAL BONE OF THUMB
*ACTION - PULLS THUMB MEDIALLY AND FORWARD ACROSS PALM (OPPOSES THUMB TOWARDS THE FINGERS)

4.NERVE SUPPLY OF THENAR MUSCLE :- MEDIAN NERVE

POINTS TO REMEMBER

1.PALMAR INTEROSSEI - ACTION-->ADDUCTION OF FINGERS TOWARDS CENTER.

2.DORSAL INTEROSSEO - ACTION-->ABDUCTION OF FINGER FROM CENTER OF THIRD DIGIT.

SUBAJ KUMAR

SHOULDER JOINT

*TYPE - SYNOVIAL JOINT OF BALL & SOCKET VARIETY.

*ARTICULATING SURFACE :- GLENOID CAVITY OF SCAPULA & HEAD OF HUMERUS.

*WEAK JOINT BECAUSE GLENOID CAVITY IS SMALL & SHALLOW
*STABILITY OF JOINT IS MAINTAIN BY -->CAROCOACROMIAL ARCH
-->ROTATOR CUFF
-->GLENOID LEBRUM (LIP.)

*LIGAMENT :-

1.CAPSULAR LIGAMENT- ATTACHMENT- VERY LOOSE AND PERMIT FREE MOVEMENT
*MEDIALLY- SUPRAGLANOID TENDON
*LATERAL- ANATOMICAL NECK OF HUMERUS
*ANTERIOR- CAPSULE IS REINFORCED BY SUPERIOR, MIDDLE, INFERIOR GLANOHUMERAL LIGAMENT

2.CORACOACROMIAL LIGAMENT :- ATTACHMENT - ROOT OF CORACOID PROCESS TO NECK OF HUMERUS.

3.TRANSVERSE HUMERAL LIGAMENT - BRIDGE UPPER PART OF BICIPITAL GROOVE OF HUMERUS.
-TENDON OF LONG HEAD OF BICEP BRACHII PASS DEEP TO THE LIGAMENT

4.GLANOID LABRUM - FIBRO CARTILAGINOUS RIM COVERS GLENOID CAVITY

*BURSAE- SUBACROMIAL BURSA, SUBSCAPULARIS BURSA, INFRASPINATOUS BURSA

*RELATION - SUPERIOR - CORACOACROMIAL ARCH, SUBACROMIAL BURSA
-SUPRASPINATOUS & DELTOID
INFERIOR - LONG HEAD OF TRICEP BRACHII
- AXILLARY NERVE
-POSTERIOR CIRCUMFLEX HUMERAL ARTERY
ANTERIOR - SUBSCAPULARIS
-CORACOBRACHIALIS & SHORT HEAD OF BICEP BRACHII
POSTERIOR - INFRASPINATOUS & TERES MINOR

*BLOOD SUPPLY - ANT. & POST. CIRCUMFLEX HMERAL ARTERY
- SUPRASCAPULAR VESSEL & SUBSCAPULAR VESSEL

*NERVE SUPPLY - AXILLARY NERVE SUPPLY
MUSCULOCUTANOUS NERVE
SUPRASCAPULAR NERVE

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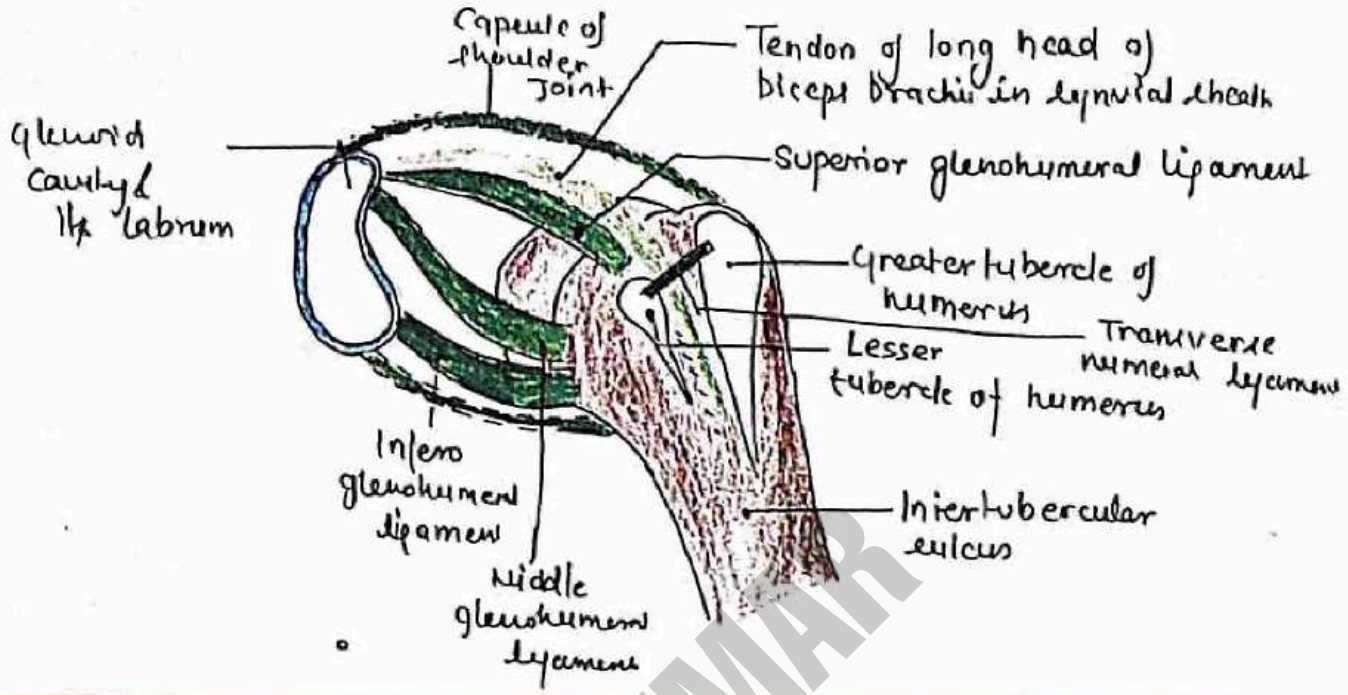
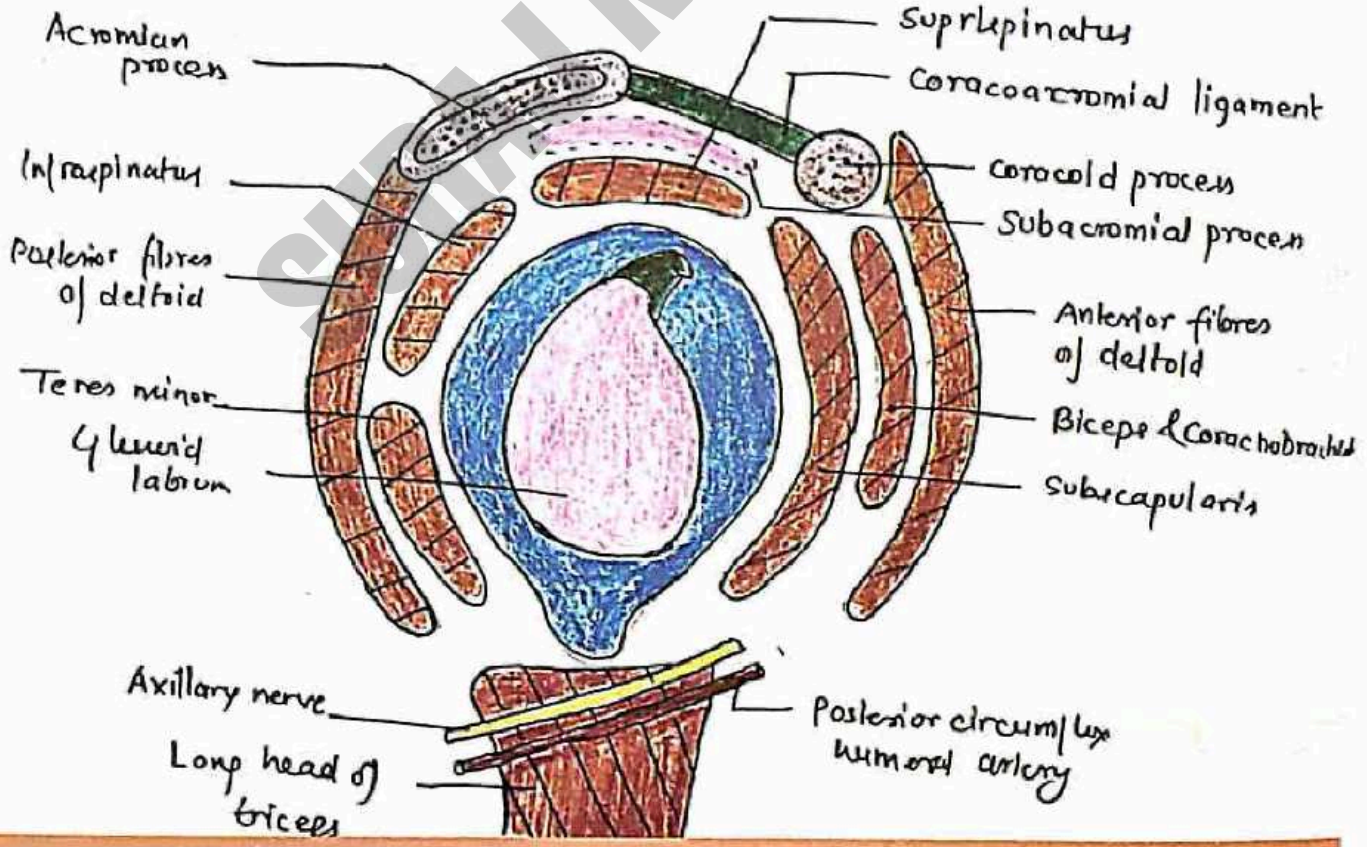


LIGAMENT OF SHOULDER JOINT



RELATION OF SHOULDER JOINT

SUNIL KUMAR

*MOVEMENT OF SHOULDER JOINT :-

1.FLEXION - ARM MOVE FORWARD & MEDIAL
2.EXTENSION - ARM MOVE LATERAL BACKWARD
3.ADDUCTION - OPPOSITE OF ABDUCTION
4.ABDUCTION - ARM MOVE ANTEROLATERALLY AWAY FROM TRUNK
5.MEDIAL ROTATION - HAND MOVE MEDIAL ACROSS THE CHEST
6.LATERALL ROTATION - HAND MOVE LATERAL OF THE SHOULDER JOINT
7.CIRCUMDUCTION - COMBINATION OF DIFFERENT MOVEMENT

ELBOW JOINT

*TYPE :- SYNOVIAL JOINT OF VARIETY OF HINGE JOINT BETWEEN LOWER END OF HUMEROUS & UPPER END OF RADIUS & ULNA.

*ARTICULATION SURFACE :-

1.UPPER-->CAPITULUM & TROCLEAR CONDYLE OF HUMERUS
-CORNOID FOSSA JUST LIE ABOVE TROCHLEAR
(CORONOID FOSSA FIT CORONOID PROCESS OF ULNA IN FLEXION)
-RADIAL FOSSA JUST ABOVE CAPITULUM
(RADIAL FOSSA FIT RADIAL HEAD OF FLEXION)

2.LOWER----->HEAD OF RADIUS WITH CAPITULUM
-TROCHLEAR NOTCH OF ULNA WITH TROCHLEAR CONDYLE OF HUMERUS

*LIGAMENT :-

1.CAPSULAR LIGAMENT-ATTACHMENT- SUPERIOR - LOWER END OF HUMERUS
INFEROMEDIAL- MARGIN OF TROCHLEAR NOTCH OF ULNA
INFERIOLATERAL- ANGULAR LIGAMENT

2.ULNAR COLLATERAL LIGAMENT- TRIANGULAR IN SHAPE
-APEX-MEDIAL EPICHONDYLE OF HUMERUS
-BASE-ULNA
-ANTERIAR BAND- BELOW CORONOID PROCESS
-POSTERIOR BAND- BELOW OLECRANON PROCESS

3.RADIAL COLLATERAL LIGAMENT-FAN SHAPE
-EXTEND LATERAL EPICHONDYLE-->ANGULAR LIGAMENT

SURAJ KUMAR

*BLOOD SUPPLY - FROM ANATOMOSES AROUND ELBOW JOINT

*NERVE SUPPLY - ULNAR, MEDIAN, RADIAL & MUSCULOCUTANEOUS NERVE

*MOVEMENT - FLEXION -->BRACHIALIS
-->BICEP BRACHII & BRACHIORADIALIS

EXTENSION-->TRICEPS BRACHII
--->ANCHONEUS

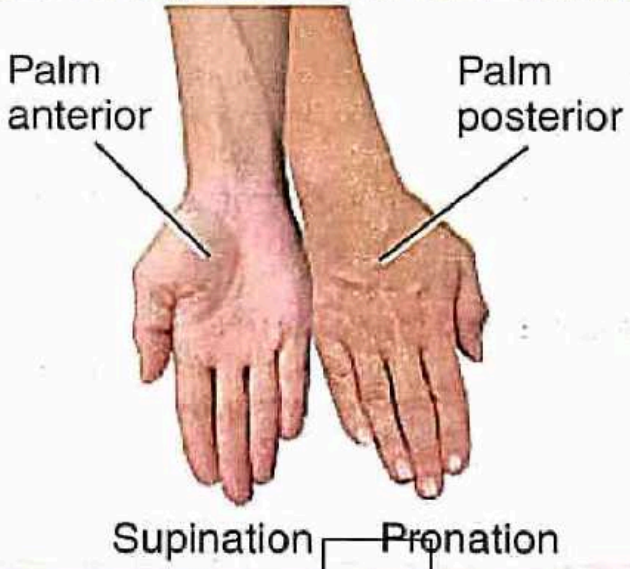
SPINATION & PRONATION

1.SUPINATION & PRONATION ARE ROTATORY MOVEMENT OF FOREARM-->VERTICAL AXIS IN SEMIFLEX ELBOW.

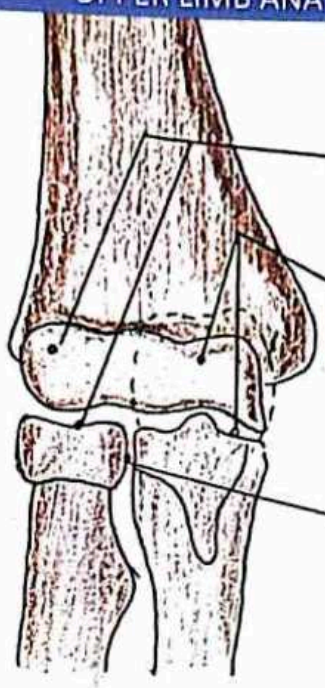
2.SUPINATION-->PALM--->TURN UPWARD.
3.PRONATION----->PALM TURN DOWNWARDS.

4.VERTICAL AXIS MOVEMENT PASS CENTER OF HEAD OF RADIUS ABOVE.
5.THIS AXIS IS NOT STATIONARY BECAUSE -->LOWER END OF ULNA IS NOT FIXED.
6.IT MOVE -->BACKWARD LATERALLY IN PRONATION
-->FORWARD MEDIALLY IN SUPINATION
7.SUPINATION IS POWERFUL BECAUSE IT IS ANTI GRAVITATIONAL
8.SUPINATION MOVEMENT IS RESPONSIBLE FOR ALL SCREWING MOVEMENT OF HAND

9.AROUND 50 DEGREE OF SUPINATIN AND PRONATION ARE GENERALLY TO PERFORM MANY ROUTINE ACTIVITIES
10.PRONATION------>CHEIFLY BY PRONATOR QUADRATUS
----->RAPIDLY MOVEMENT BY PRONATOR TERES
11.SUPINATION----->CAUSE BY SUPINATOR MUSCLE & BICEP BRACHII
---->SLOW SUPINATION---->BY SUPINATR MUSCLE
---->RAPID SUPINATION --->BY BICEP BRACHII



SIBRAS KUMAR



Elbow Joint

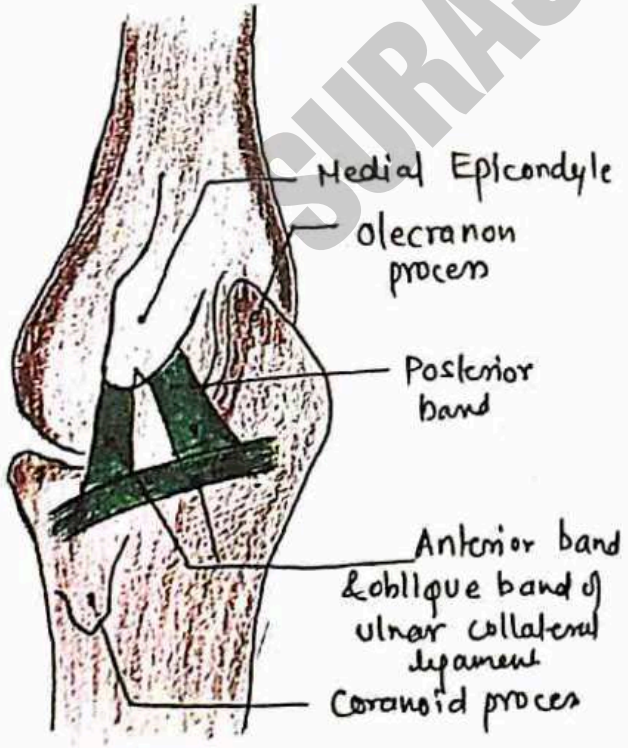
Capitulum of humerus articulates with upper surface of head of radius

Trochlea of humerus articulates with trochlear notch of ulna

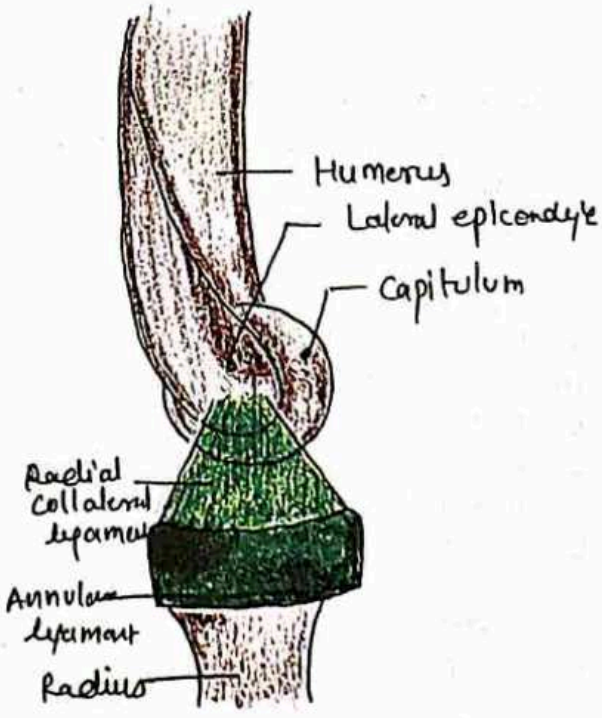
Superior radioulnar joint

The circumference of the head of radius articulates with the radial notch of ulna

∴ The cubital articulations



∴ Ulnar collateral ligament



∴ Radial collateral ligament

ELBOW JOINT

SURAJ KUMAR

PREVIOUS YEAR UNIVERSITY QUESTIONS

*LONG QUESTIONS - 20MARKS

1.DESCRIBE BRACHIAL PLEXUS WITH ITS APPLIED ANATOMY ?

STUDY TIPS FOR MAXIMUM MARKS !!!

1.IN CLINICAL ANATOMY OR APPLIED ANATOMY - YOU HAVE TO WRITE ONLY 3-4 CLINICAL ANATOMY POINTS (ONLY BDC CLINICAL ANATOMY POINTS)

2.IF IT IS MENTION TO WRITE ON SPECIFIC CLINICAL ANATOMY -:

FOR EXAMPLE : WRITE CLINICAL ANATOMY OF ERBS PALSY THEN YOU HAVE TO WRITE ONLY ABOUT ERB PALSY - IN 1-1/2 PAGE.

2.DESCRIBE MEDIAL NERVE UNDER FOLLOWING HEADS -
1.ROOT VALUE 2.COURSE & RELATION 3.DISTRIBUTION 4.CLINICAL ANATOMY

HOW TO WRITE IN UNIVERSITY EXAM (TIME MANAGEMENT)

1.IN MULTIPLE HEADING QUESTION - SPECIFIC MARKS ARE GIVEN TO EACH HEADING

FOR EXAMPLE - 2MARKS FOR ROOT VALUE, 8MARKS FOR COURSE, 5MARKS FOR DISTRIBUTION & 5MARKS FOR CLINICAL ANATOMY.

2.MARKS DISTRIBUTION OF EACH HEADING ARE GIVEN IN PAPER.

3.WRITE ACCORDING TO MARKS DISTRIBUTION.

(ZYADA LIKHNE SE ZYADA MARKS NAHI MILENGE, 2MARKS HAIN TO 2MARKS HI MILENGE - ZYADA LIKHNE KE CHAKKAR ME TIME WASTE MAT KARNA PAPER MISS HO JAAYEGA)

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3. A 30-YEAR-OLD MALE PRESENTED TO EMERGENCY DEPARTMENT AFTER A ROAD TRAFFIC ACCIDENT WITH COMPLAINING OF RIGHT ARM PAIN AND INABILITY TO MOVE HIS EXTREMITY. THE FOLLOWING MORNING, HE DEVELOPED CLAWING. NERVE CONDUCTION STUDIES ON THE PERIPHERAL NERVES OF HIS ARM CONFIRMED THE DIAGNOSIS OF INJURY TO THE ULNAR NERVE. KEEPING THIS SCENARIO IN MIND, ANSWER THE FOLLOWING -
QUESTIONS

1) ORIGIN, ROOT VALUE OF THE NERVE 2) COURSE & DISTRIBUTION 3) BRANCHES 4) EXPLAIN REASON FOR DEVELOPING THE CLAW HAND

HOW TO SOLVE CASE STUDY !!!!!!

1.MAINLY NERVES (MEDIAN, RADIAL, ULNAR NERVE) ARE GIVEN IN CASE STUDY.

2.IDENTIFICATION TRICK -
1.IF MENTION INJURY TO SPECIFIC NERVE THEN WRITE ABOUT THAT(FOR EXAMPLE - IN ABOVE CASE STUDY IT MENTION'S ULNAR NERVE INJURY)

2.IF INJURY IS NOT MENTIONED THEN SEE SENSORY LOSS

-IN RADIAL NERVE INJURY SENSORY LOSS - IN DORSAL LATERAL HALF
-IN ULNAR NERVE INJURY SENSORY LOSS IN - MEDIAL HALF OF BOTH DORSAL & PALMAR SIDE
-IN MEDIAN NERVE INJURY SENSORY LOSS IN - LATERAL HALF OF PALMAR SIDE.

(IDENTIFY AARAM SE KARNA, GALAT CASE STUDY IDENTIFY KARNE SE SEEDHE 20MARKS KA LOSS HOGA AUR FAIL.)

4.MID PALMAR SPACE IN HAND ?

HOW TO DRAW A DIAGRAM !!!!!!

1.TRY TO MAKE COLOURFUL DIAGRAM - 1.RED COLOUR - ARTERY
2.BLUE - VEIN
3.BROWN - MUSCLE
4.YELLOW - NERVE
5.GREEN - LIGAMMENT

2.IN LONG QUESTION TRY TO MAKE 2-3 DIAGRAMS ONLY.
(NUMBER OF DIAGRAM = MARKS)
(ZYAADA DIAGRAM BANANE ME TIME WASTE MAT KARNA WARNA NEXT QUESTION MISS HO JAYEGA - TIME MANAGE KARE CHALNA)

3.TRY TO MAKE CLEAR, COLOURFUL & WELL LABELLED DIAGRAM.

5.DESCRIBE MAMMARY GLAND?

HOW TO WRITE IN LONG QUESTIONS !!!!!!

1.IF IN LONG QUESTIONS HEADINGS ARE NOT GIVEN, THEM WRITE IN FOLLOWING HEADINGS
1.DEFINITION
2.EXTERNAL FEATURES
3.BLOOD SUPPLY
4.NERVE SUPPLY
5.LYMPHATIC DRAINAGE
6.CLINICAL ANATOMY

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6.DESCRIBE SHOULDER JOINT UNDER THE FOLLOWING HEADINGS -
1.TYPE OF JOINT & ARTICULAR SURFACE 2.CAPSULAR LIGAMMENT 3.RELATION
4.APPLIED ANATOMY

7. DESCRIBE ELBOW JOINT UNDER THE FOLLOWING HEADINGS -
1.TYPE OF JOINT & ARTICULAR SURFACE 2.LIGAMENT 3.APPLIED ANATOMY

*SHORT QUESTIONS - 5MARKS

HOW TO WRITE IN SHORT QUESTIONS !!!!!!

1.IN SHORT QUESTIONS WRITE 2-3 PAGES WITH ONE DIAGRAM & 1 CLINICAL ANATOMY - MAXIMUM.
2.MINIMUM 1-2 PAGES WITH DIAGRAM.

(ZYADA LIKHNE ME TIME WASTE MAT KARNA, 5 ME SE 10 MARKS NAHI MILENGE 5 ME 5 HI MILENGE, ZYADA LIKHNE KE CHAKKAR ME PAPER AUR MISS HO JAYEGA)

*SHORTS NOTES :-

1) AXILLA
2) AXILLARY ARTERY
3) CUBITAL FOSSA
4) ROTATOR CUFF
5) FLEXOR RETINACULUM
6) SNUFF BOX
7) LUMBRICAL
8) THENAR MUSCLE
9) SUPINATION & PRONATION
10) FLEXOR RETINACULUM

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UPPER LIMB ANATOMY IS COMPLETED !!

BEST WAY TO PREDICT FREE FUTURE IS TO CREATE IT

ALL THE BEST FOR YOUR EXAMS

DR.JOHARI

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MCQ UPPER LIMB

1) Brachial plexus is formed by:
a ventral ramii of C5 to T1 spinal nerves
b ventral root of C5to T1 spinal nerves
c dorsal ramii of C5 to T1 spinal nerves
d dorsal roots of C5 to T1 spinal nerves

2) All of the following are branches of posterior cord of brachial plexus :
a axillary nerve
b radial nerve
c suprascapular nerve
d thoracodorsal

3) The action of acromial fibres of deltoid are :
a adduction of arm
b abduction of arm
c extension of arm
d flexion of arm

4) all of the following are innervated by musculocutaneous nerve except:
a Coracobrachialis
b biceps brachii
c brachialis
d brachioradialis

5) Which structure does nit lie in anatomical snuff box:
a cephalic vein
b radial artery
c first metacarpal bone
d extensor pollicis longus

6) Following nerve may get injured in downward dislocation of head of humerus
a musculocutaneous
b Axillary
c Radial
d median

7) Following nerve may get injured in fracture of surgical neck of humerus:
a musculocutaneous
b subscapular
c axillary
d radial

8) Fracture of medial epicondyle of humerus may injure :
a median nerve
b ulnar nerve
c ulnar artery
d brachial artery

9) The nerve supply of serratus anterior is:
a axillary nerve
b C4
c thoracodorsal nerve
d long thoracic nerve

10) The most preffered vein for venipuncture is:
a cephalic vein
b basilic vein
c median cubital vein
d brachial vein

11) Which structure lies in the deltopectoral groove:
a basilic vein
b cephalic vein
c axillary vein
d medial pectoral nerve

12) Which nerve passes between ulnar and humeral head of pronator teres
a ulnar nerve
b radial nerve
c median nerve
d anterior interosseous nerve

13) Which nerve passes between ulnar and humeral head of fkexor carpii ulnaris:
a ulnar nerve
b radial nerve
c median nerve
d anterior interosseous nerve

14) All of the following structures pass superficial to flexor retinaculum EXCEPT :
a median nerve
b superficial cutaneous branch of median nerve
c ulnar nerve
d superficial cutaneous branch of ulnar nerve

15) All of the following muscles supplied by median nerve except:
a abductor pollicis brevis
b flexor pollicis
c opponens pollicis
d adductor pollicis

16) Which nerve is involved in carpal tunnel syndrome:
a ulnar nerve
b median nerve
c radial nerve
d anterior interosseous nerve

17 Which lymphatic does not drain the breast:
a subscapular nodes
b infraclavivular nodes
c parasternal nodes
d pectoral nodes

18 Wrist drop occurs due to injury of :
a axillary nerve
b ulnar nerve
c median nerve
d radial nerve

1) A 2)C 3)B 4)D 5)D 6) B 7)C 8)B 9)D 10)C 11)B 12)C 13)A 14)A 15)D 16)B 17)C 18)D

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LOWER LIMB MUSCLES

GLUTEAL REGION

MAJOR MUSCLES
- Gluteus maximus
- Gluteus medius
- Gluteus minimus
- Tensor fasciae latae

MINOR MUSCLES
- Piriformis
- Inferior Gemellus
- Superior Gemellus
- Obturator internus
- Quadratus femoris

THIGH

ANTERIOR COMPARTMENT
- Quadriceps femoris
  - Rectus femoris
  - Vastus lateralis
  - Vastus medialis
  - Vastus intermedius
- Articularis Genu
- Sartorius
- Tensor fasciae Latae

- Inferior Gluteal Nerve
- Superior Gluteal Nerve
- Ventral rami $S_1 S_2$
- Nerve to quadratus femoris
- Nerve to obturator internus
- Quadratus femoris

LATERAL COMPARTMENT
- Peroneus longus
- Peroneus brevis

POSTERIOR COMPARTMENT
SUPERFICIAL
- Gastrocnemius
- Soleus
- Plantaris

DEEP
- Popliteus
- Flexor digitorum longus
- Flexor hallucis longus
- Tibialis posterior

POSTERIOR COMPARTMENT
- Bicep Femoris — short head, long head
- Semitendinosus
- Semimembranosus
- Adductor magnus (ischial part)

MEDIAL COMPARTMENT
ANTERIOR / FIRST LAYER
- Pectineus
- Adductor longus
- Gracilis

MIDDLE / SECOND LAYER
- Adductor brevis

POSTERIOR / THIRD LAYER
- Adductor Magnus
  - Adductor part
  - Hamstring part

- Tibial Sciatic Nerve
- Common Peroneal Sciatic Nerve
- Obturator Nerve
- Femoral Nerve

LEG

MEDIAL COMPARTMENT
Tendons of
- Sartorius
- Gracilis
- Semi tendinosus

ANTERIOR COMPARTMENT
- Tibialis anterior
- Extensor hallucis longus
- Extensor digitorum longus
- Peroneus tertius

- Saphenous Nerve
- Deep peroneal Nerve
- Superficial peroneal Nerve
- Tibial Nerve

FOOT

MUSCLES OF SOLE

1st Layer
- Flexor digitorum brevis
- Abductor hallucis
- Abductor digiti minimi

2nd Layer
- Flexor digitorum accessorius
- 4 lumbricals — 1, 2, 3, 4
- 2 tendon — Flexor digitorum longus, Flexor hallucis longus

3rd Layer
- Flexor hallucis brevis
- Flexor digiti minimi brevis
- Adductor hallucis

4th Layer
- Interossei — 3 plantar, 4 dorsal

- Tendons of
  - tibialis posterior
  - peroneus longus

- Medial plantar
- Lateral plantar

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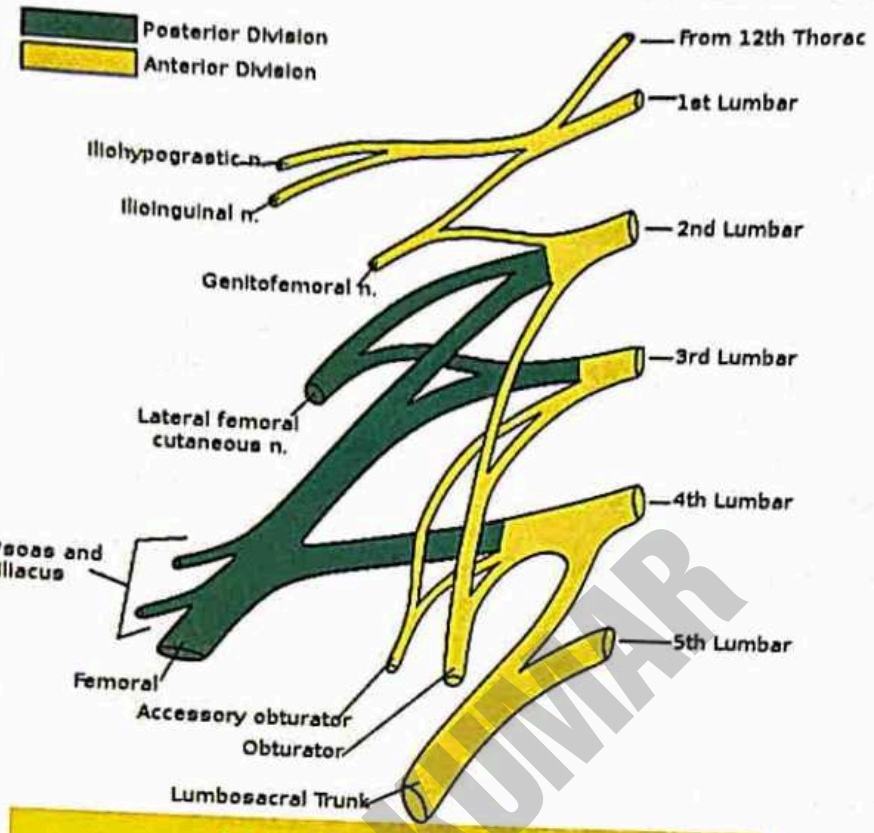
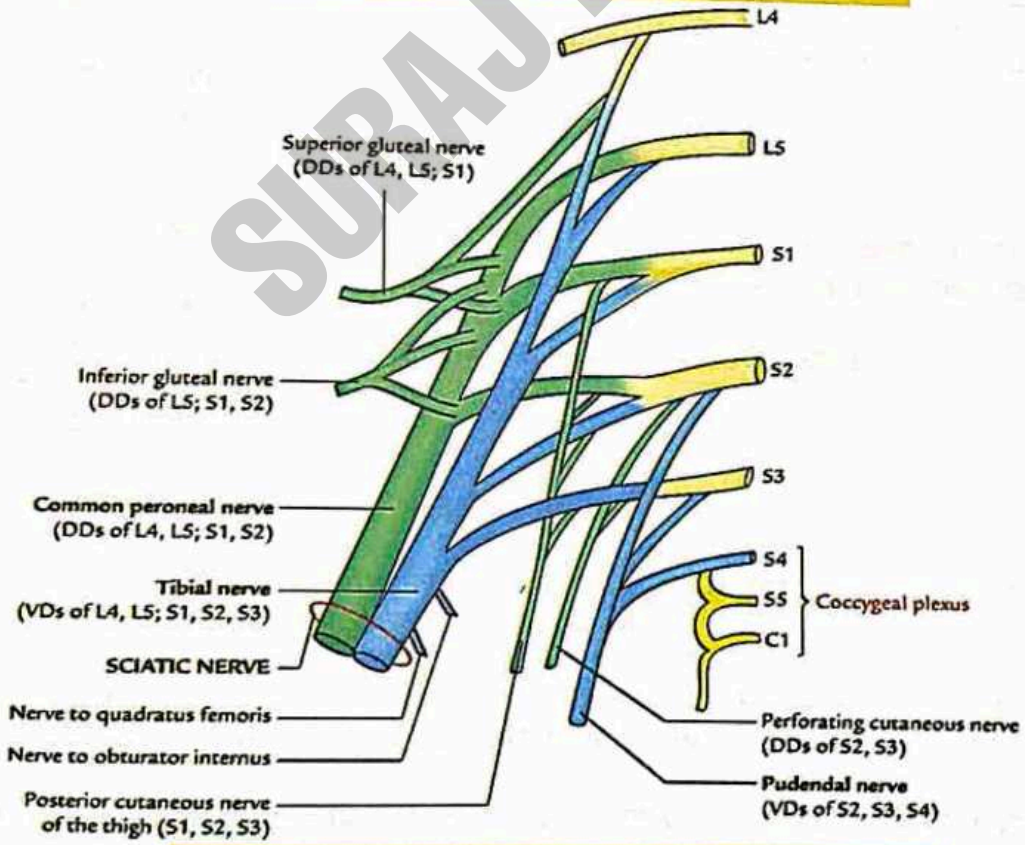


LUMBAR PLEXUS



SACRAL PLEXUS

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FEMORAL TRIANGLE

SCAN QR WITH QRBOT APP TO DIRECTLY OPEN VIDEO LINK

*TRIANGULAR DEPRESSION-->INFRONT OF UPPER 1/3RD THIGH BELOW INGUINAL LIGAMENT

*BOUNDARIES :-

1.LATERAL - MEDIAL BORDER SARTORIOUS MUSCLE.
2.MEDIAL - MEDIAL BORDER OF ADDUCTOR LONGUS.
3.BASE - INGUINAL LIGAMENT.
4.APEX - LATERAL & MEDIAL BORDER CROSS.
5.ROOF - SKIN, SUPERFICIAL FASICA & DEEP FASICA.

SUPERFICIAL FASICA---->CONTENT--> SUPERFICIAL INGUNAL L.N.
-> FEMORAL BRANCH OF GENITOFEMORAL NERVE
-> ILIOINGUINAL NERVE
-> GREAT SAPHENOUS VEIN
-> SUPERFICIAL BRANCH OF FEMORAL ARTERY

6.FLOOR - MEDIAL -> ADDUCTOR LONGUS & PECTINEOUS MUSCLE
LATERAL -> PSOAS MAJOR & ILIACUS

*CONTENT :-

1.FEMORAL ARTERY-->IT GIVE 6 BRANCHES - 3 SUPERFICIAL & 3 DEEP BRANCHES
2.FEMORAL VEIN-->MEDIAL TO ARTERY
-->FEMORAL VEIN RECEIVE--->GREAT SAPHENOUS VEIN
--->CIRCUMFLEX VEIN
--->VEIN CORRESPONDING TO BRANCH OF FEMORAL ARTERY

3.FEMORAL SHEATH - ENCLOSE UPPER 4CM OF FEMORAL VESSEL
4.DEEP INGUAL L.N.--->DEEP TO DEEP FASICA
--->MEDIAL TO UPPER PART OF FEMORAL VEIN
->RECEIVED ->LYMPH FROM - SUPERFICIAL INGUINAL L.N.
-DEEP LYMPHATIC OF LOWER LIMB
-FROM GLANS PENIS & CLOTRIES

5.NERVE -

*FEMORAL NERVE - LIE LATERAL TO FEMORAL ARTERY->OUTSIDE FEMORAL SHEATH

*NERVE TO PECTINOUS - ARISE FROM FEMORAL NERVE
-PASS BEHIND FEMORAL SHEATH

*FEMORAL BRANCH OF GENIOFEMORAL NERVE - OCCUUPY LATERAL
-COMPARTMENT TO FEMORAL SHEATH ALONG FEMORAL ARTERY

*LATERAL CUTANEOUS NERVE OF THIGH - CROSS THE LATERAL ANGLE OF

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TRIANGLE.

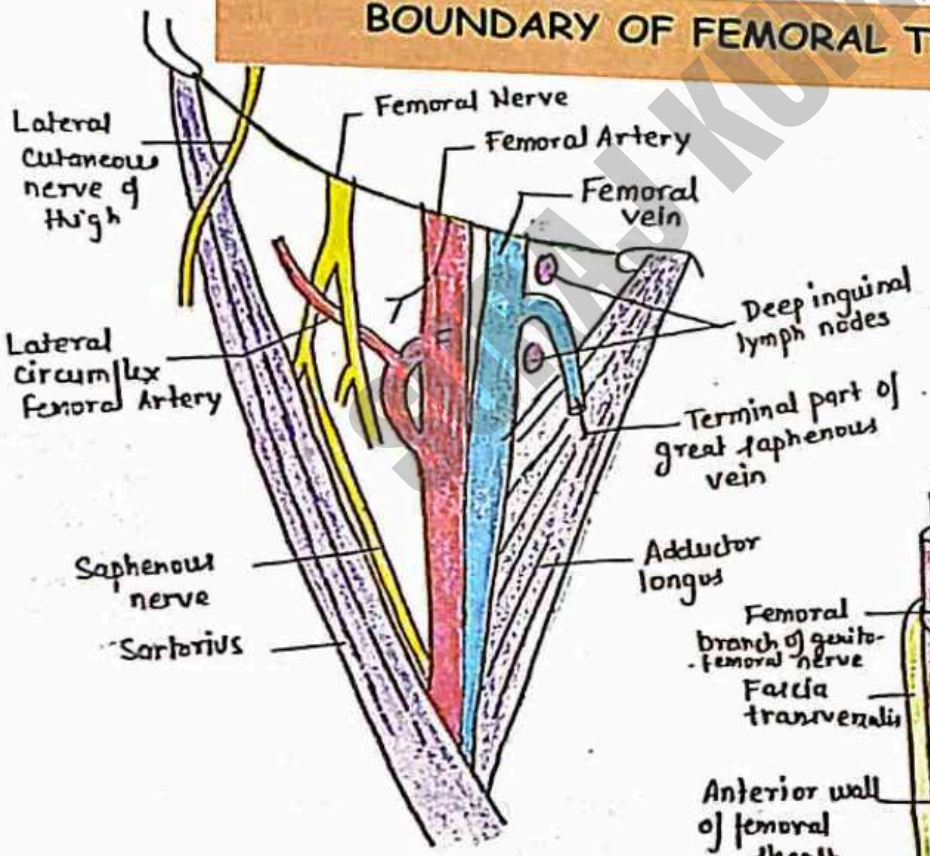
* Boundaries

1) Inguinal ligament
2) Medial border of Sartorius
3) Medial Border of Adductor longus

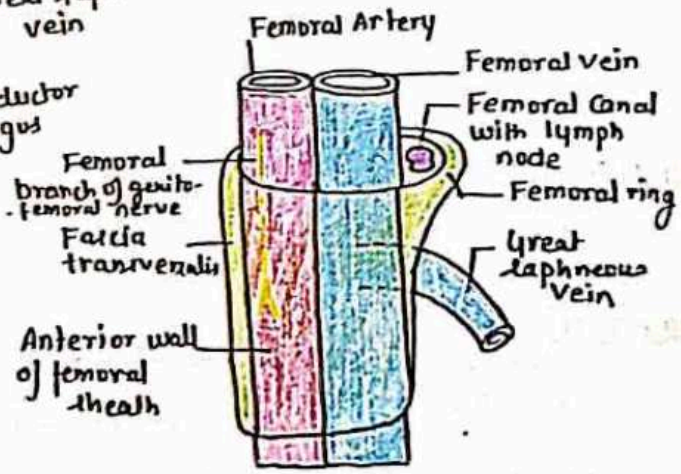
Floor of Femoral Δ

1) Iliacus
2) Psoas Major
3) Pectineus
4) Adductor longus

BOUNDARY OF FEMORAL TRIANGLE



∵ Content of femoral Δ



∵ Femoral canal

CONTENT OF FEMORAL TRIANGLE

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FEMORAL SHEATH

*FUNNEL SHAPE SLEEVE OF FASCIA - ENCLOSING UPPER 4CM OF FEMORAL VESSEL
*SHEATH IS FORM DOWNWARD EXTENSION OF - 2 LAYER OF FASCIA OF ABDOMEN

*ANTERIOR WALL --->FORM--->FASCIA TRANSVERSALIS
*POSTERRIOR WALL--->FORM --->FASCIA ILIACA
*INFERIORLY - SHEATH MERGE WITH CONNECTIVE TISSUE AROUND FEMORAL VESSEL

*SHEATH IS DIVIDED INTO 3 COMPARTMENT BY SEPTA:-
1.LATERAL COMP.-->CONTENT->FEMORAL ARTERY
-->FEMORAL BRANCH OF GENITOFEMORAL NERVE
2.INTERMEDIATE COMP. -->FEMORAL VEIN
3.MEDIAL COMP.--> ALSO KNOWN AS FEMORAL CANAL

FEMORAL CANAL

*MEDIAL COMP.OF FEMORAL SHEATH - CONICAL IN SHAPE - WIDE IN ABOVE, NARROW BLOW
*BASE OR UPPER END OF FEMORAL CANAL ALSO KNOWN AS FEMORAL RING.

CONTENT :- LYMPH NODE & SMALL AMOUNT OF AREOLAR TISSUE

CLINICAL ANATOMY

*FEMORAL HERNIA :-FEMORAL CANAL IS A AREA OF POTENTIAL WEAKNES IN ABDOMINAL WALL-->THROUGH WHICH ABDOMINAL CONTENT MAY BULGE OUT FORM FEMORAL HERNIA

FEMORAL ARTERY

*CHIEF ARTERYOF LOWER LIMB - CONTINUATION OF EXTERNAL ILIAC ARTERY

COURSE - BEGIN - BEHIND INGUINAL LIGAMENT - MID INGUINAL POINT
- PASSES DOWNWARD MEDIALY - 1ST - ENTERING IN FEMORAL TRIANGLE
-THEN ENTER IN ADDUCTOR CANAL - PASSES THROUGH OPENING OF ADDUCTOR MAGNUS ->BECOME POPLITEAL ARTERY

RELATION :-
1.ANTERIOR - SKIN, SUPERFICIAL & DEEP FASCIA, ANTERIOR WALL OF FEMORAL SHEATH
2.POSTERIOR - POSTERIOR WALL OF FEMORAL SHEATH
-PSOAS MAJOR & PECTINEUS
3.MEDIAL - FEMORAL VEIN
4.LATERAL- FEMORAL NERVE

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*BRANCHES :- 3 SUPERFICIAL & 3 DEEP

SUPERFICIAL BRANCH - 1.SUPERFICIAL EXTERNAL PUDENDAL ARTERY
2.SUPERFICIAL EPIGASTRIC ARTERY
3.SUPERFICIAL CIRCUMFLEX ILIAC

DEEP BRANCH - 1.PROFUNDA FEMORAL ARTERY
2.DEEP EXTERNAL PUDENDAL ARTERY
3.MUSCULAR BRANCH

FEMORAL NERVE

*CHIEF NERVE OF ANTERIOR COMPARTMENT OF THIGH

*ORIGIN - LARGEST BRANCH OF LUMBER PLEXUS - FORM BY DORSAL DIVISION OF ANTERIOR PRIMARY RAMI (L2, 3,4)

*COURSE - FEMORAL NERVE-→ENTER FEMORAL TRIANGLE BEHIND INGUINAL LIGAMENT LATERAL TO FEMORAL ARTERY
-IN THIGH-→LIE--→BETWEEN ILIACUS & PSOAS OUTSIDE FEMORAL SHEATH.
-AFTER SHORT COURSE--→2.5CM BELOW THE INGUINAL LIGAMENT NERVE DIVIDE INTO ANTERIOR AND POSTERIOR

*BRANCHES :-

MUSCULAR - 1.TRUNK GIVE - BRANCH -→ ILIACUS & PECTINEUS
2.ANTERIOR DIVISION -→SARTORIUS
3.POSTERIOR DIVISION-→RECTUS FEMORIS & 3 VASTI MUSCLE

CUTANEOUS - 1.ANTERIOR DIVISION - GIVE 2 CUTANEOUS BRANCH (MEDIAL & INTERMEDIATE CUTANEOUS NERVE OF THIGH)

2.POSTERIOR DIVISION - GIVE ONLY ONE CUTANEOUS BRANCH (SAPHANOUS NERVE)

ARTICULAR - ARTICULAR BRANCH TO HIP JOINT & KNEE JOINT
VASCULAR BRANCH TO FEMORAL ARTERY

CLINICAL ANATOMY

*FEMORAL NERVE :-FEMORAL NERVE INJURY CAUSE - PARALYSIS OF QUADRICEPS FEMORIS
*SENSORY LOSS - ANTERIOR MEDIAL SIDE OF THIGH &
- MEDIAL SIDE OF LEG

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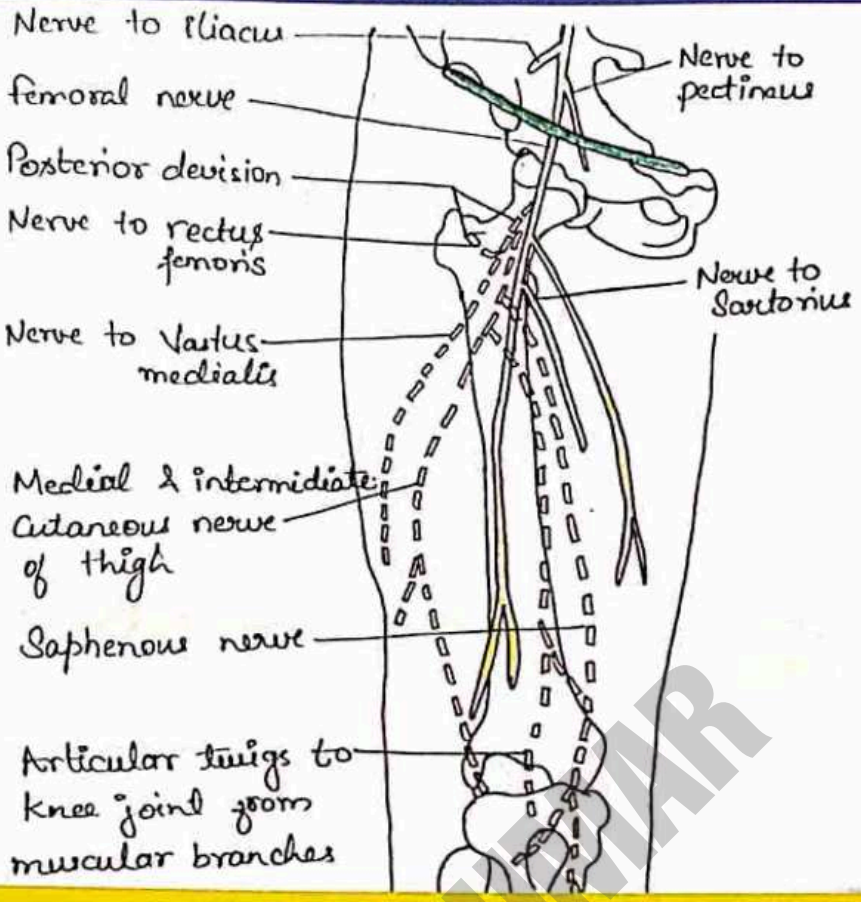
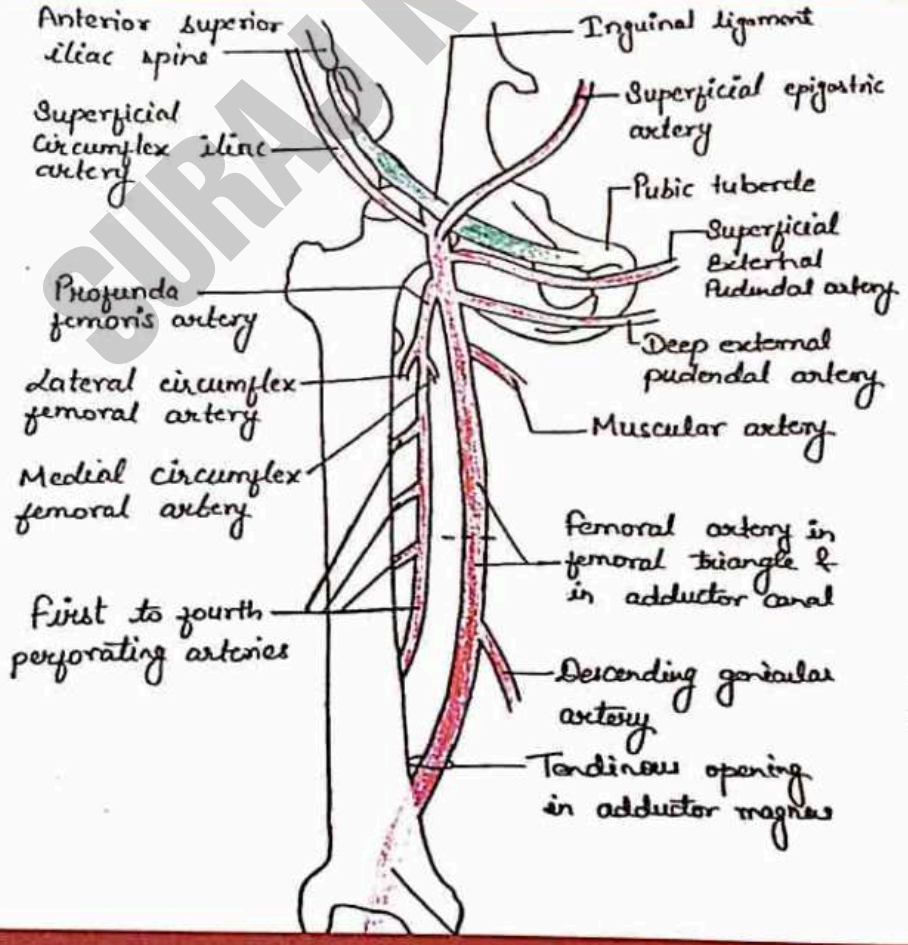


FEMORAL NERVE



FEMORAL ARTERY

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ADDUCTOR CANAL

*INTERMUSCULAR SPACE SITUATED ON MEDIAL SIDE MIDDLE 1/3RD OF THIGH
*EXTEND -APEX OF FEMORAL TRIANGLE→TENDINOUS OPENING OF ADDUCTOR MAGNUS

*BOUNDARIES:-

1.ANTEROLATERAL - VASTUS MEDIALIS

2.POSTEROMEDIAL - ADDUCTOR LONGUS ABOVE
-ADDUCTOR MAGNUS BELOW

3.MEDIAL WALL - SARTORIUS

*SUBSARTORIAL PLEXUS NERVE -→ LIE UNDER SARTORIUS
--→ FORM BY - SAPHENOUS NERVE
- ANT.DIVISION OF OBTURATOR NERVE
- MEDIAL CUTANEOUS NERVE OF THIGH

*CONTENT :-

1.FEMORAL ARTERY--→ENTER CANAL AT APEX OF FEMORAL TRIANGLE
--→GIVE -→MUSCULAR BRANCH AND DESCENDING GENICULAR BRANCH

2.FEMORAL VEIN--→LIE --→POSTERIR TO FEMORAL ARTERY IN UPPER PART
--→LATERAL TO FEMORAL ARTERY IN LOWER PART

3.SAPHENOUS NERVE-→CROSS-→FEMORAL ARTERY ANTERIORLY

4.NERVE TO VASTUS MEDIALIS--→LATERAL TO FEMORAL ARTERY

5.BRANCH OF TWO DIVISION OF OBTURATOR NERVE

-→ANTERIOR DIVISION--→EMERGE AT LOWER BORDER OF ADDUCTOR MAGNUS
-->POSTERIOR DIVISION--→ RUN ON ANT. SURFACE OF ADDUCTOR MAGNUS

CLINICAL ANATOMY

*A DISEASE OF HIP JOINT MAY CAUSE REFERRED PAIN IN KNEE AND ON THE LATERAL SIDE OF THIGH BECAUSE OF THEIR COMMON NERVE SUPPLY BY THE OBTURATOR NERVE.

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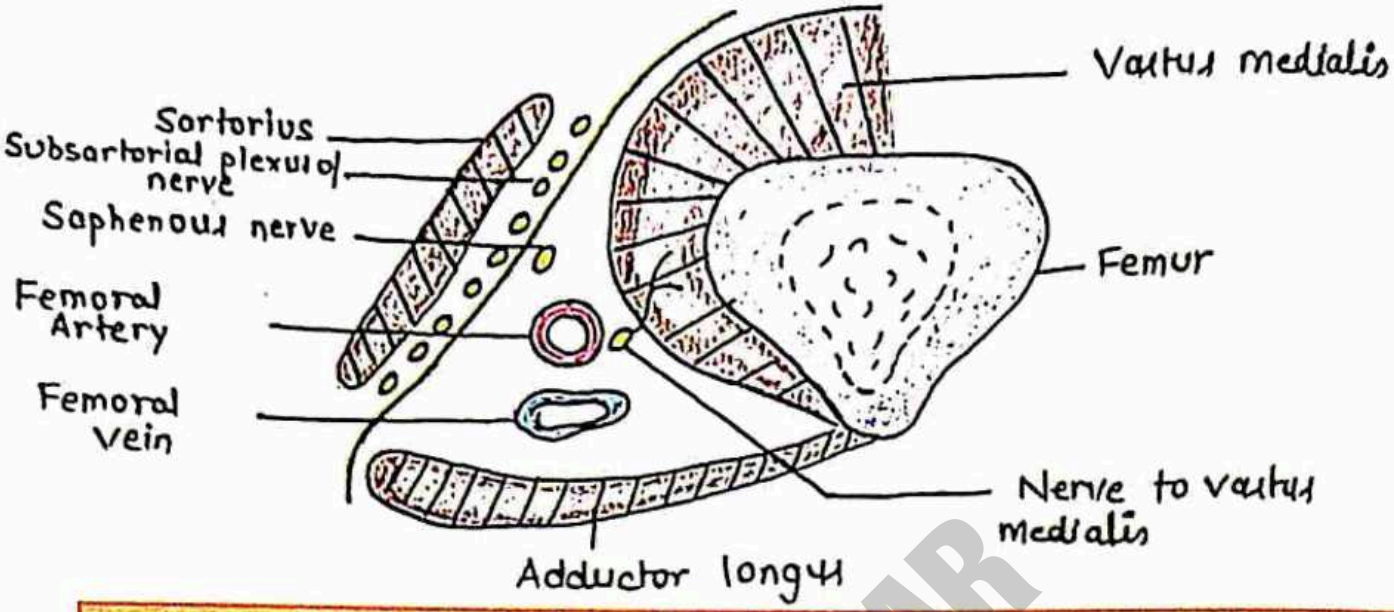
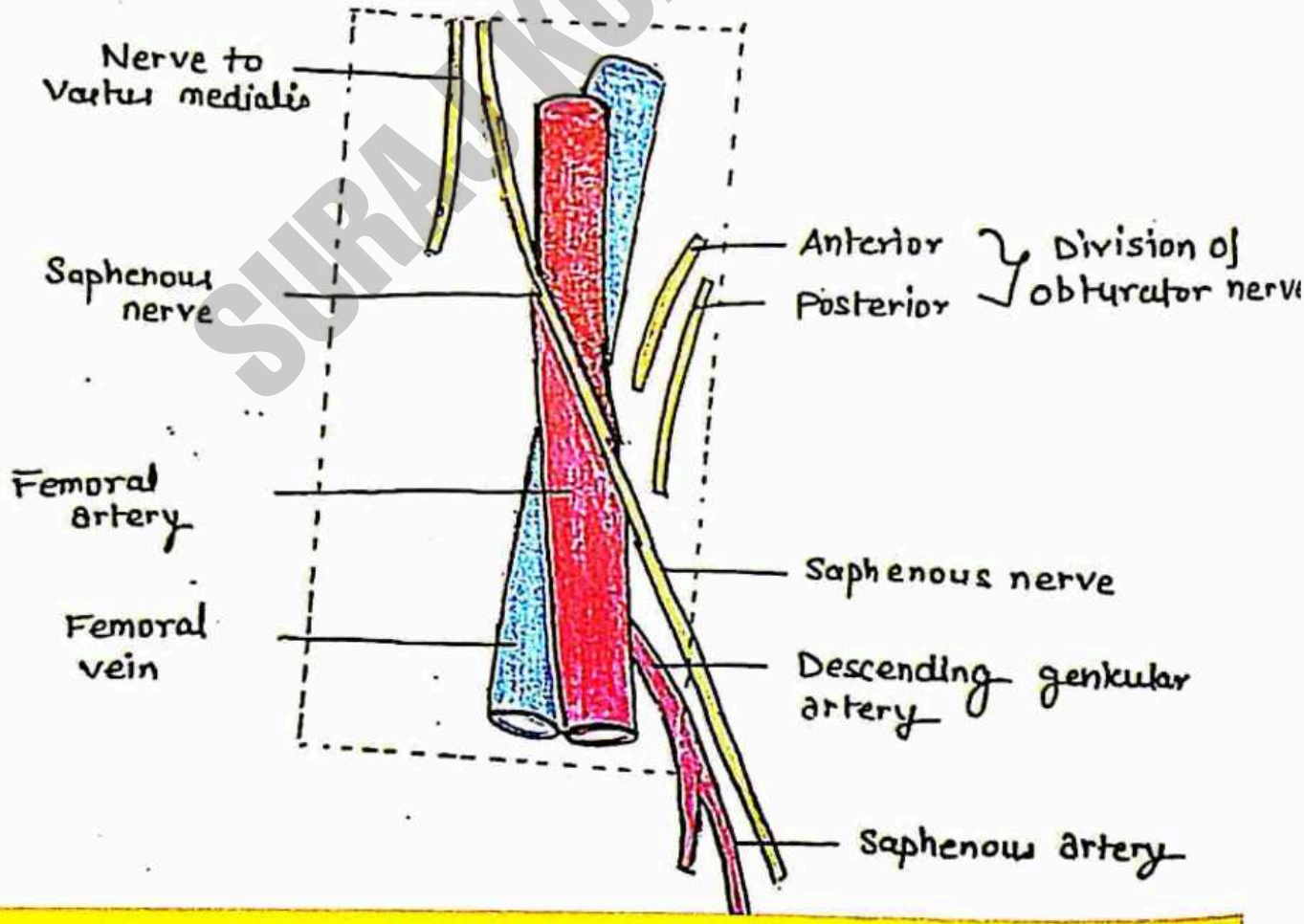


BOUNDARIES OF ADDUCTOR CANAL



CONTENT OF ADDUCTOR CANAL

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POPLITEAL FOSSA

*DIAMOND SHAPE DEPRESSION -->BEHIND KNEE JOINT, LOWER PART OF FEMUR AND UPPER PART OF TIBIA

*BOUNDARIES :-

1.SUPEROLATERAL - BICEP FEMORIS MUSCLE
2.SUPEROMEDIAL - SEMITENDINOSUS MUSCLE
3.INFEROLATERAL - LATERAL HEAD GASTROCNEMIUS
4.INFEROMEDIAL - MEDIAL HEAD OF GASTROCNEMIUS
5.ROOF - DEEP FASICA
-SUPERFICIAL FASICA -->SMALL SAPHENOUS VEIN
-->POSTERIOR CUTANEOUS NERVE OF THIGH
-->POSTERIOR DIVISION OF MEDIAL CUTANEOUS NERVE OF THIGH
-->PERONEAL NERVE
6.FLOOR - POPLITEAL SURFACE OF FEMUR
-CAPSULE OF KNEE JOINT
-POPLITEAL MUSCLE

*CONTENT :- 1.POPLITEAL ARTERY & VEIN
2.TIBIAL NERVE
3.COMMON PERONEAL NERVE
4.GENICULAR BRANCH OF OBURATOR NERVE
5.POPLITEAL LIMPH NODE & FAT SURROUNDING ALL STRUCTURE

SCAN QR WITH QRBOT APP
TO DIRECTLY OPEN VIDEO LINK

OBTURATOR NERVE

*CHIEF NERVE OF MEDIAL COMPARTMENT OF THIGH

ORIGIN -: BRANCH OF LUMBER PLEXUS -->FORM BY VENTRAL DIVISION OF L2,3,4
-ENTER THE THIGH-->PASSINGG THROUGH OBTURATOR CANAL

COURSE :-

1.WITH OBTURATOR CANAL -->NERVE DIVIDED INTO ANTERIOR & POSTERIR DIVISION
2.ANTERIOR DIVISION-->PASS DOWNWARD INTO OBTURATOR EXTERNUS
-->THEN DESCEND BEHIND ADDUCTOR LONGUS & PECTINEOUS
-->SUPPLY - PECTINEOUS & GRACILIUS
-ADDUCTOR LONGUS
-ADDUCTOR BRAVIS (IF NOT SUPPLY BY P.D.)

3.POSTERIOR DIVISION --->ENTER THIGH BY PIERCING OBTURATOR EXTERNUS
-->THEN DESCEND BEHIND ADDUCTOR BREVIS
--->SUPPLY - OBTURATOR EXTERNUS
-ADDUCTOR MAGNUS
-ADDUCTOR BREVIS(IF NOT SUPPLY BY ANT.DIVISION)

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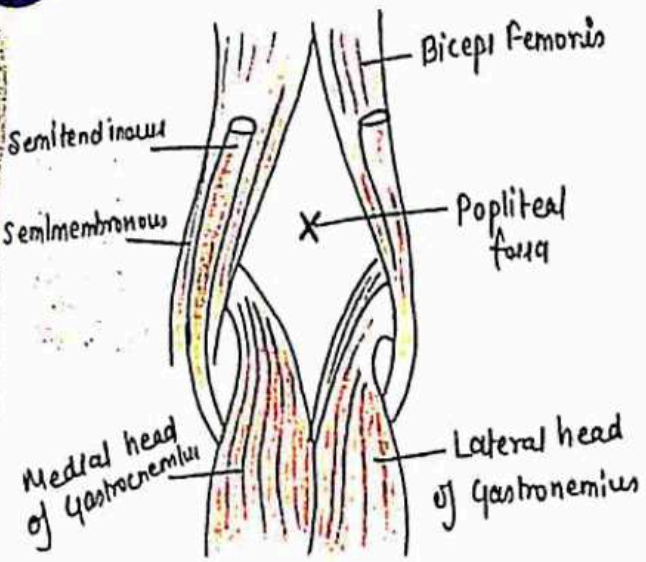
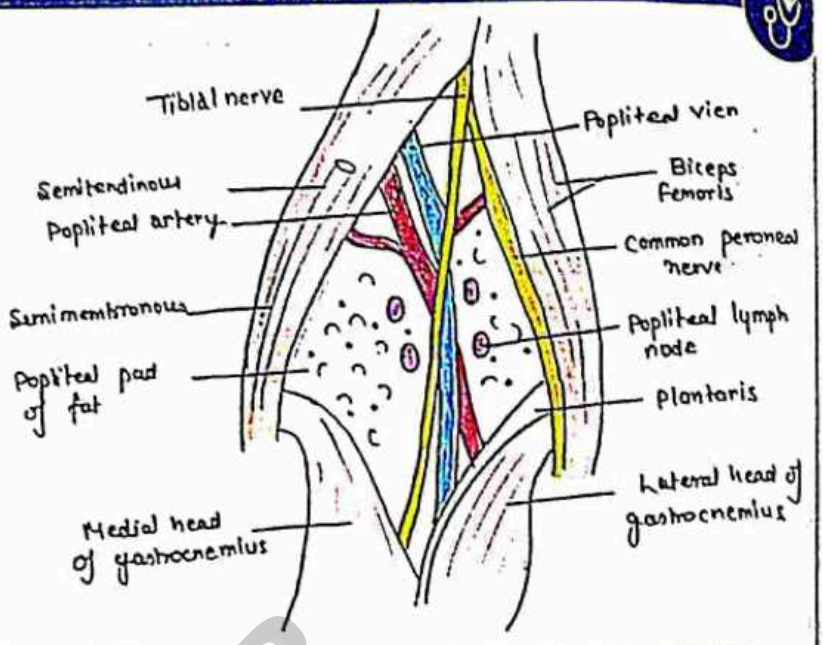
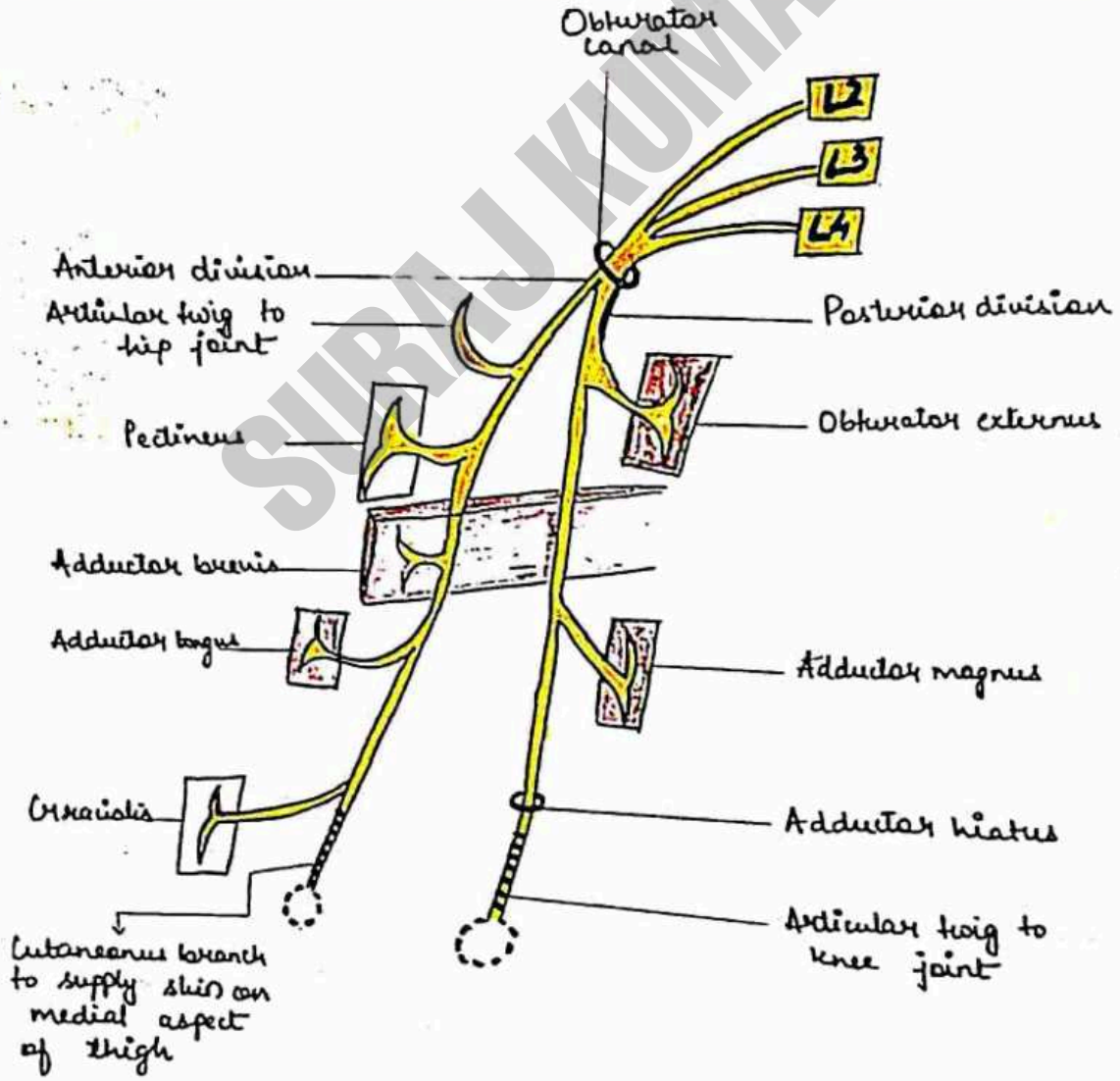




BOUNDARY & CONTENT OF POPLITEAL FOSSA



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OBTURATOR NERVE

SCIATIC NERVE

*THICKEST NERVE IN THE BODY-→BEGIN IN PELVIS -→TERMINATE AT SUPERIOR ANGLE OF POPLITEAL FOSSA BY DIVIDING - TIBIAL NERVE & COMMON PERONEAL NERVE

*ORIGIN :- LARGEST BRANCH OF SACRAL PLEXUS - ROOT VALUE L4,5 & S1,2,3
-TIBIAL PART - FORM BY - VENTRAL DIVISION OF L4,5 & S1,2,3
-COMMON PERONEAL PART - FORM BY - DORSAL DIVISION OF L4,5 & S1,2

*COURSE :-
1.IN PELVIS - NERVE LIE INFRONT OF-PIRIFORMIS - UNDER COVER OF ITS FASCICA
2.IN GLUTEAL REGION - ENTER THROUGH GREATER SCIATIC FORAMEN
-→ RELATION IN GLUTEAL REGION
1.SUPERFICIAL - GLUTEAL MAXIMUS
2.DEEP - BODY OF ISCHISM
-TENDON OF OBTURATOR INTERNUS WITH GEMELLI
-QUADRATUS FEMORIS
-OBTURATOR EXTERNUS
-ADDUCTOR MAGNUS
3.IN THIGH - ENTER BACK OF THIGH
-RUN VERTICALLY DOWNWARDS UPTO SUPERIOR ANGLE OF POPLITEAL FOSSA
--→ RELATION THIGH :-
1.SUPERFICIAL - CROSS LONG HEAD OF BICEPS FEMORIS
2.DEEP - ADDUCTOR MAGNUS
3.MEDIAL - SEMIMEMBRANOSUS & SEMITENDINOUS
4.LATERAL - BICEPS FEMORIS

BRANCHES :-
1.ARTICULATING BRANCH OF HIP JOINT
2.MUSCULAR BRANCH OF TIBIAL PART --→SEMITENDINOSUS
- SEMIMEMBRANOSUS
- LONG HEAD OF BICEPS FEMORIS
COMMON PARONEAL PART - SHORT HEAD OF BICEPS FEMORIS

POINTS TO REMEMBER

*COMMON PERONEAL NERVE BRANCH -SUPERFICIAL & DEEP PERONEAL NERVE
1.SUPERFICIAL PERONEAL NERVE-→BRANCH--→TIBIALIS ANTERIOR
--→EXTENSOR DIGITORUM LONGUS
--→EXTENSOR HALLUCIS LONGUS
--→PERONEAL TERTIS
2.DEEP PERONEAL NERVE-→BRANCH-→PERONEUS LONGUS & BREVIS

FOOT DROP :- INJURY TO DEEP PERONEAL NERVE RESULT IN LOSS OF POWER OF DORSIFLEXION OF FOOT AS A RESULT FOOT IS PLANTARFLEXED CONDITION IS KNOWN AS FOOT DROP.

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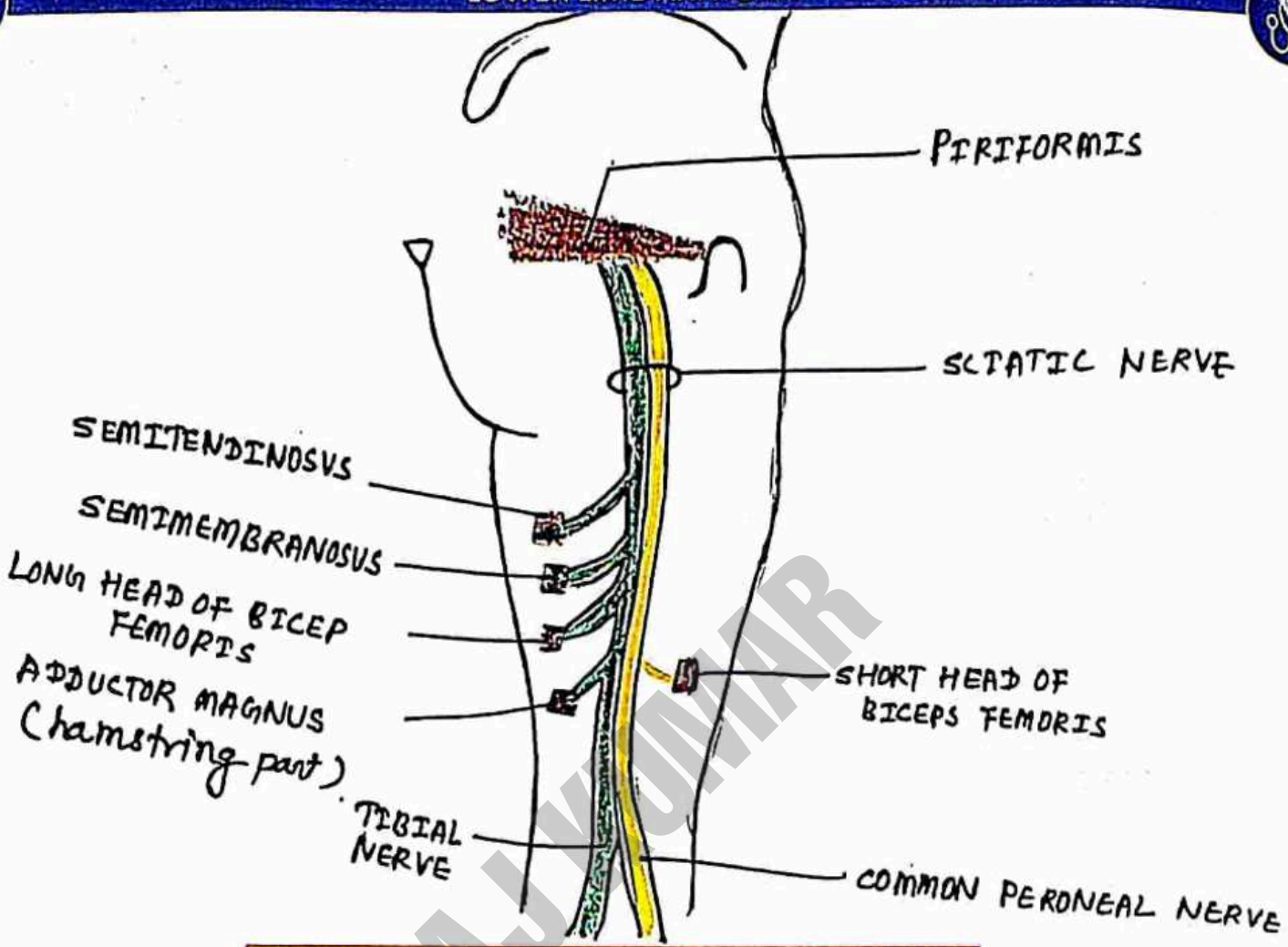
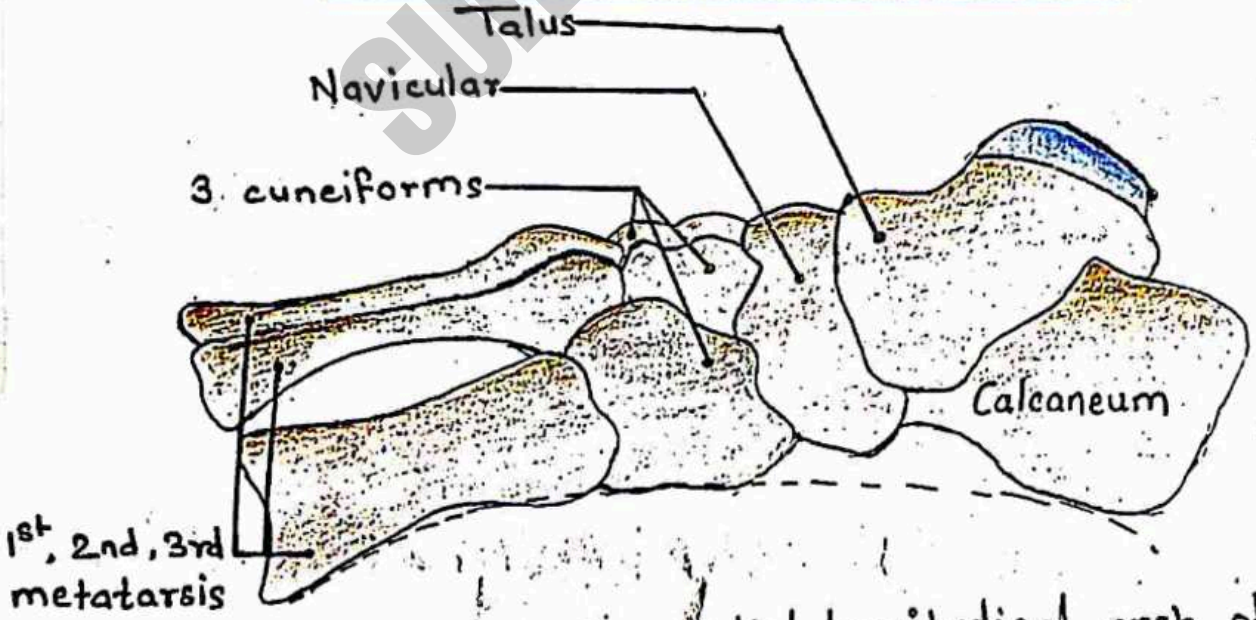


SCIATIC NERVE



Bones forming the medial longitudinal arch of foot: Medial view. Inset showing big arc of a small circle

ARCHES OF FOOT

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ARCHES OF FOOT

*ARCHES OF FOOT HELP IN FAST WALKING & JUMPING.

CLASSIFICATION OF ARCHES :-

1.LONGITUDINAL - MEDIAL & LATERAL ARCH
2.TRANSVERSE - ANTERIOR & POSTERIOR

MEDIAL LONGITUDINAL ARCH

ARCH IS CONSIDER HIGHER, MORE MOBILE-→IT CONSIDER AS BIG ARCH OF MORE BONE & MORE JOINT

*ENDS - ANT.END-→HEAD OF 1ST,2ND & 3RD METATARSAL BONE
-POST.END-→MEDIAL TUBERCLE OF CALCANEUM
*SUMMIT :- SUPERIOR ARTICULAR SURFACE OF BODY OF TALUS

*PILLARS :-
ANT.PILLAR - LONG AND WEAK
-FORM BY TALUS, THE NAVICULAR & 3 CUNEIFORM BONES
POST.PILLAR - SHORT AND STRONG
-FORM BY MEDIAL PART OF CALCANEUM
*MAIN JOINT :- TALOCALCANEONAVICULAR JOINT.

LATERAL LONGITUDINAL ARCH

CHARACTERISLY LOW WITH LESS BONE & JOINT & MOBILITY

*END :- ANT.END-→HEAD OF 5TH & 4TH METATARSLE BONE
POST.END-→LATERAL TUBERCLE OF CALCANIUM

*SUMMIT :- SUPERIOR SPACE OF CALCANIUM

*PILLAR :- ANT.PILLAR-→LONG & WEAK
-→FORM BY CUBOID BONE, 4TH & 5TH METATARSAL
POST.PILLAR-→SHORT & STRONG
-→FORM BY LATERAL HALF OF CALCANEUM
*MAIN JOINT :- CALCANEOCUBOIDAL JOINT

TRANSVERSE ARCH (T.A.)

1.ANTERIOR T.A-→FORM BY 5 METATARSAL BONE
-→COMPLETE ARCH BECAUSE HEAD OF 1ST & 2ND METATARSAL COME IN CONTACT WITH GROUND
2.POSTERIOR T.A-→FORM BY GREATER PART OF TARSAL & BASE OF METATARSAL
-→IN COMPLETE ARCH BECAUSE ONLY THE LATERAL END COME IN CONTACT WITH THE GROUND

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VENOUS DRAINAGE OF LOWER LIMB

*VEIN OF LOWER LIMB CLASSIFIED-→1.SUPERFICIAL- 2.DEEP - 3.PERFORATING VEIN

*DEEP VEIN - MEDIAL PLANTAR, LATERAL PLANTAR, DORSALIS PEDIS , ANTERIOR & POSTERIOR TIBIAL, PERONEAL, POPLITEAL & FEMORAL VEIN
- VALVE ARE MORE NUMEROUS IN DEEP VEIN
- MORE EFFICIENT CHANNEL BECAUSE DRIVING FROCE OF MUSCULAR CONTRACTION

*SUPERFICIAL VEIN - GREATER & SMALL SAPHENOUS VEIN → LIE IN SUPERFICIAL FASCIA
- VALVE ARE MORE IN DISTAL PART
- LARGE PORTION OF BLOOD-→ DRAINAGE - INTO DEEP VEIN THROUGH PERFORATING VEIN

LONG SAPHENOUS VEIN

*FORMATION - UNION OF MEDIAL END OF DORSAL VENOUS ARCH WITH MEDIAL MARGINAL VEIN

*COURSE :-
- PASS UPWARD IN FRONT OF MEDIAL MALLEOLUS
- CROSS LOWER 1/3RD OF MEDIAL SURFACE OF TIBIA TO REACH KNEE JOINT
- TN THING - INCLINE FORWARD TO REACH SAPHENOUS OPENING -→WHERE IT PIERCE CIRBRIFORM FASCIA & OPEN INTO FEMORAL VEIN

*CONNECTION -
- TO DEEP VEIN BY PERFORATING VEIN
  1) 3 MEDIAL PERFORATOR JUST ABOVE ANKLE
  2) 1 PERFORATOR JUST BELOW KEE
  3) ANOTHER ONE IN ADDUCTOR CANA

*TRIBUTARIES -

1.AT COMMENCEMNET - MEDIAL MARGINAL VEIN
2.LEG- COMMUNICATE WITH SMALL SAPHENOUS VEIN & DEEP VEIN
- POSTERIOR ARCH VEIN & VEIN FROM CALF
3.THING - ACCESSARY SAPHENOUS VEIN
- ANTERIOR CUTANEOUS VEIN
4.JUST BEFORE PIERCING THE CRIBRIFORM FASCIA-SUP. EPIIGASTRIC
-SUP.CIRCUMFLEX ILIAC
5.JUST BEFORE TERMINATION - DEEP EXTERNAL PUDENDAL VEIN

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SMALL SAPHENOUS VEIN

*FORMATION - UNION OF LATERAL END OF DORSAL VENOUS ARCH WITH LATERAL MARGINAL VEIN

*COURSE - IT ENTER THE BACK OF LEG BY PASSING BEHIND LATERAL MALLEOUS TO LOWER PART OF POPLITEAL FOSSA

*DRAIN - LATERAL BORDER OF FOOT - HEEL & BACK OF LEG

PERFORATING VEIN

INDIRECT PERFORATING VEIN -CONECT SUPERFICIAL VEIN WITH DEEP VEIN THROUGH MUSCULAR VEIN.

DIRECT PERFORATING VEIN - CONECT SUPERFICIAL VEIN WITH DIRECTLY WITH DEEP VEIN

1.IN THIGH - ADDUCTOR CANAL PERFORATOR--->CONNECT->GREAT SAPHENOUS WITH FEMORAL VEIN.

2.BELOW THE KNEE - PERFORATOR--->CONNECT--> GREAT SAPHENOUS VEIN WITH POSTERIOR TIBIAL VEIN.

3.IN LEG - LATERAL PERFORATOR---> CONNECT-->SMALL SAPHENOUS VEIN WITH PERONEAL VEIN

- MEDIAL THERE ARE 3 PERFORATORS - CONNECT-->POSTERIOR ARCH VEIN WITH POSTERIOR TIBIAL VEIN.
- MEDIAL 3 PERFORATOR ARE - UPPER MEDIAL PERFORATOR
  -MIDDLE MEDIAL PERFORATOR
  -LOWER MEDIAL PERFORATOR

CLINICAL ANATOMY

1.GREAT SAPHENOUS VEIN IS USED FOR BYPASING THE BLOCKED CORONARY ARTERY THE VEIN IS REVERSED SO THAT VALVE DO NOT BLOCK THE PASSAGE OF BLOOD.

2.VARICOSE VEIN - IF THE VALVES IN PERFORATING VEIN OR AT THE TERMINAL OF SUPERFICIAL VEIN BECOME INCOMPETENT --->RESULT IN DILATATION OF SUPERFICIAL VEIN AND GRADUAL DE-GENERATION OF WALL TO PRODUCE VARICOSE VEIN.

3.VERICOSE VEIN OFTEN OCCUR DURING THIRD TRIMESTER OF PREGNANCY,AS THE ILIAC VEINGET PRSSED DUE TO ENLARGED UTERUS.

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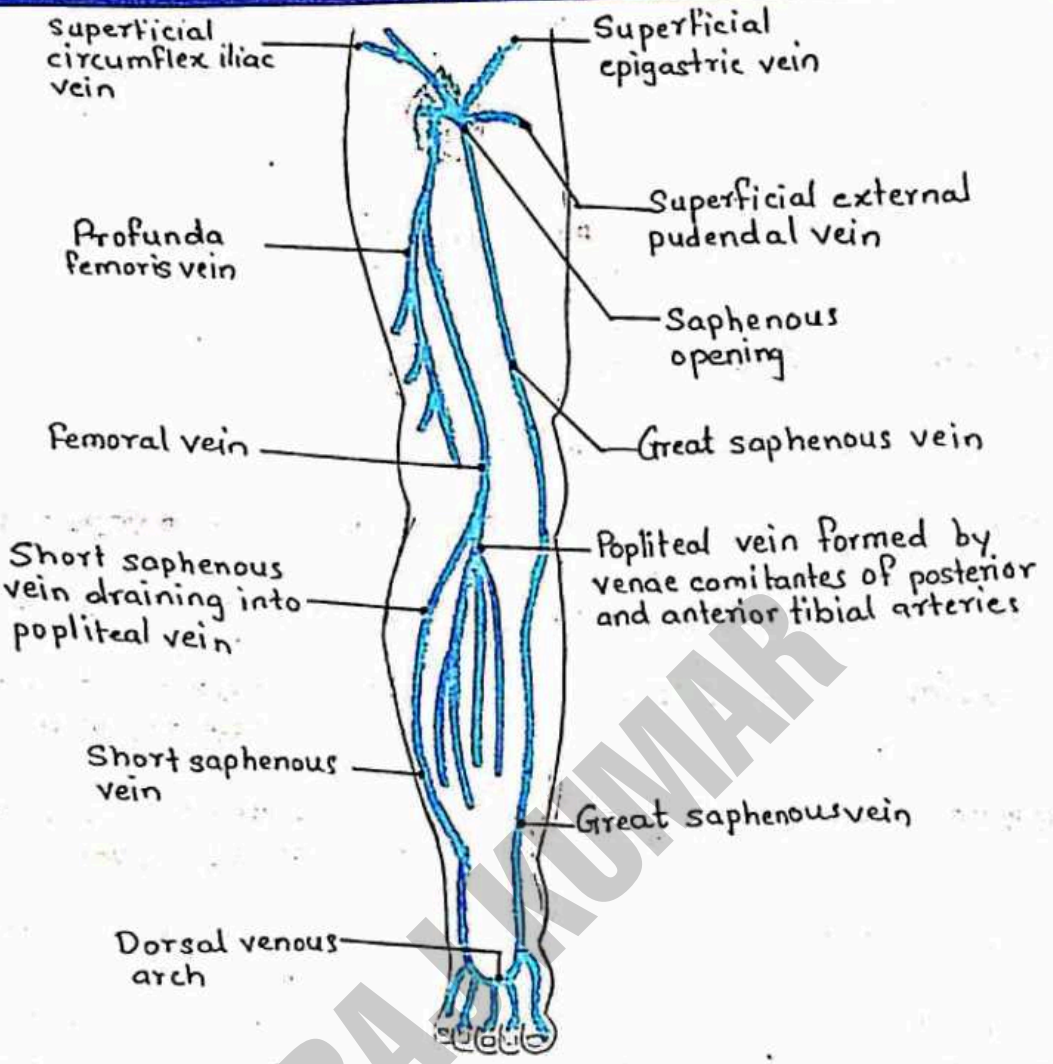

Scheme to show the arrangement of the veins of lower limb. Popliteal, short saphenous and venae comitantes of posterior tibial artery are on posterior aspect.

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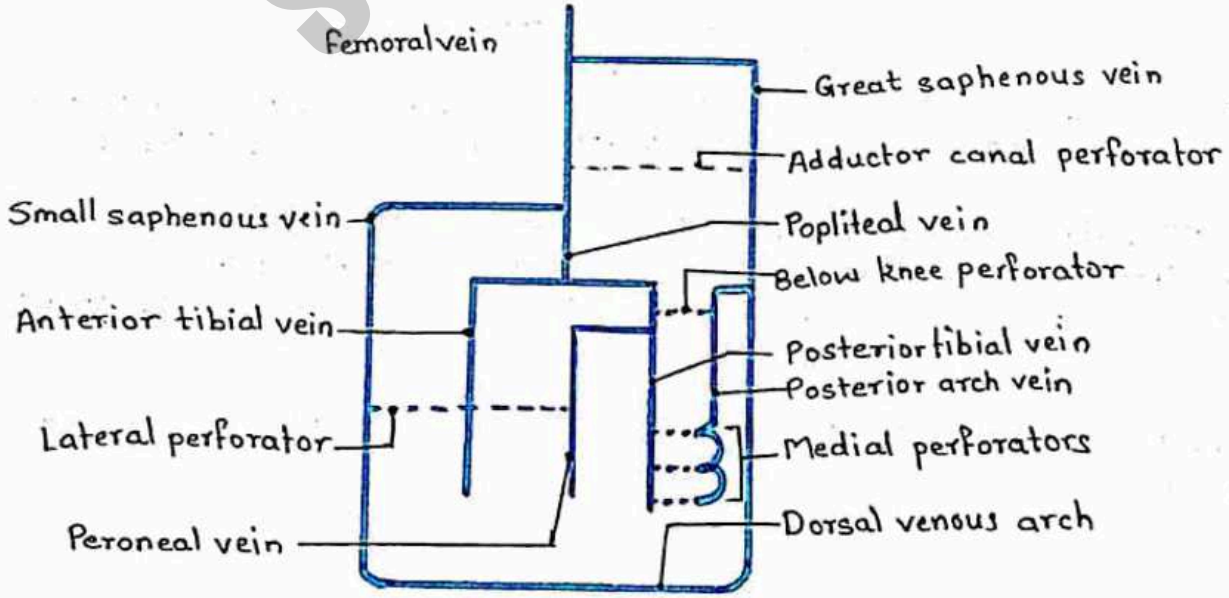

VENOUS DRAINAGE OF LOWER LIMB

HAMSTRING MUSCLE

*FEATURES - ORIGIN FROM ISCHIAL TUBEROCITY
- NERVE SUPPLY - TIBIAL PART OF SCIATIC NERVE
- MUSCLE ACT AS FLEXOR KNEE & EXTENSOR OF HIP

1.SEMITENDINOUS :-

*ORIGIN - INFEROMIDIAL IMPRESSION ON THE UPPER PART OF ISCHIAL TUBEROSITY
*INSERTION - UPPER MEDIAL SURFACE OF TIBIA
*NERVE SUPPLY - TIBIAL PART OF SCIATIC NERVE
*ACTION - FLEXION OF KNEE & MEDIAL ROTATION OF LEG, WEEK EXTENSOR OF HIP

2.SEMIMEMBRANOSUS :-

*ORIGIN - SUPEROLATERAL IMPRESSION - ISCHIAL TUBEROSITY
*INSERTION - POSTERIOR SURFACE OF MEDIAL CONDYLE OF TIBIA
*NERVE SUPPLY - TIBIAL PART OF SCIATIC NERVE
*ACTION - FLEXION OF KNEE & MEDIAL ROTATION OF LEG, WEEK EXTENSOR OF HIP

3.BICEPS FEMORIS :-

*ORIGIN - LONG HEAD - INFEROMEDIAL IMPRESSION
SHORT HEAD - ISCHIAL TUBEROSITY
*INSERTION - HEAD OF FIBULA
*NERVE SUPPLY - LONG HEAD - TIBIAL PART OF SCIATIC NERVE
- SHORT HEAD - COMMON PERONEAL NERVE
*ACTION - LATERAL ROTATION OF KNEE - WEEK EXTENSOR OF HIP

4.ADDUCTOR MAGNUS :-

*ORIGIN - LOWER LATERAL PART OF ISCHIAL TUBEROSITY
- INFERIOR RAMUS OF PUBIS
*INSERTION - MEDIAL MARGIN OF GLUTEAL TUBEROSITY
- LINEA ASPERA & MEDIAL SUPRACHONDYLAR LINE
- ADDUCTOR TUBERCLE
*NERVE SUPPLY - DOUBLE NERVE SUPPLY -TIBIAL PART OF SCIATIC NERVE
- OBTURATOR NERVE
*ACTION - EXTENSION OF HIP
- ADDUCTOR PART - CAUSE ADDUCTION OF THIGH

FLEXOR RETINOCULUM

*ATTACHMENT - ANTERIORLY--→ MEDIAL MALLEOLUS
- POSTERIOR & LATERALLY --→ MEDIAL TUBERCLE OF CALCANEUM

*STRUCTURE PASSING DEEP TO FLEXOR RETINOCULUM :-

1.TENDON OF TIBIALIS POSTERIOR & FLEXOR DIGITORUM LONGUS
2.POSTERIOR TIBIAL ARTERY & TIBIAL NERVE
3.TENDON OF FLEXOR HALLUCIS LONGUS

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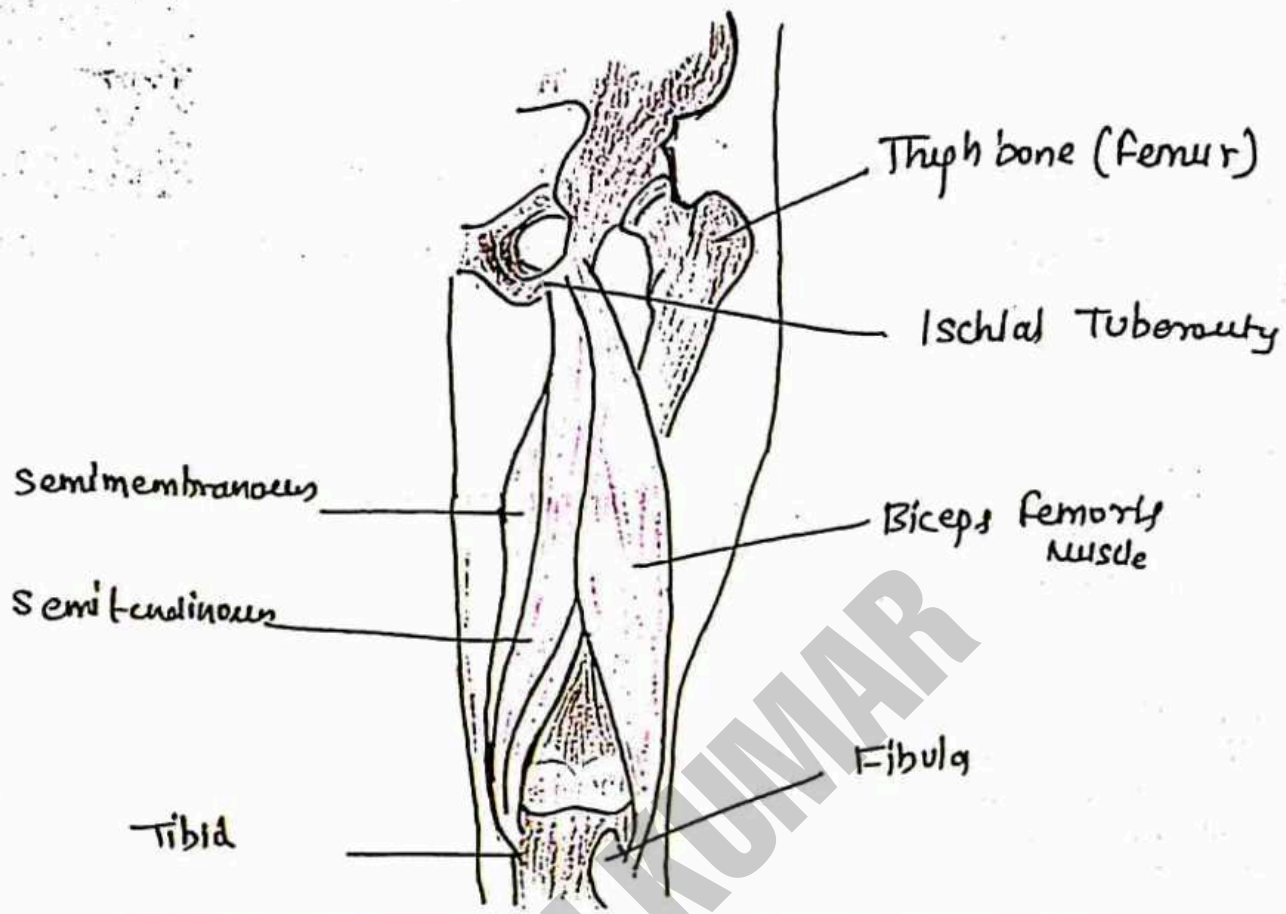

HAMSRING MUSCLE

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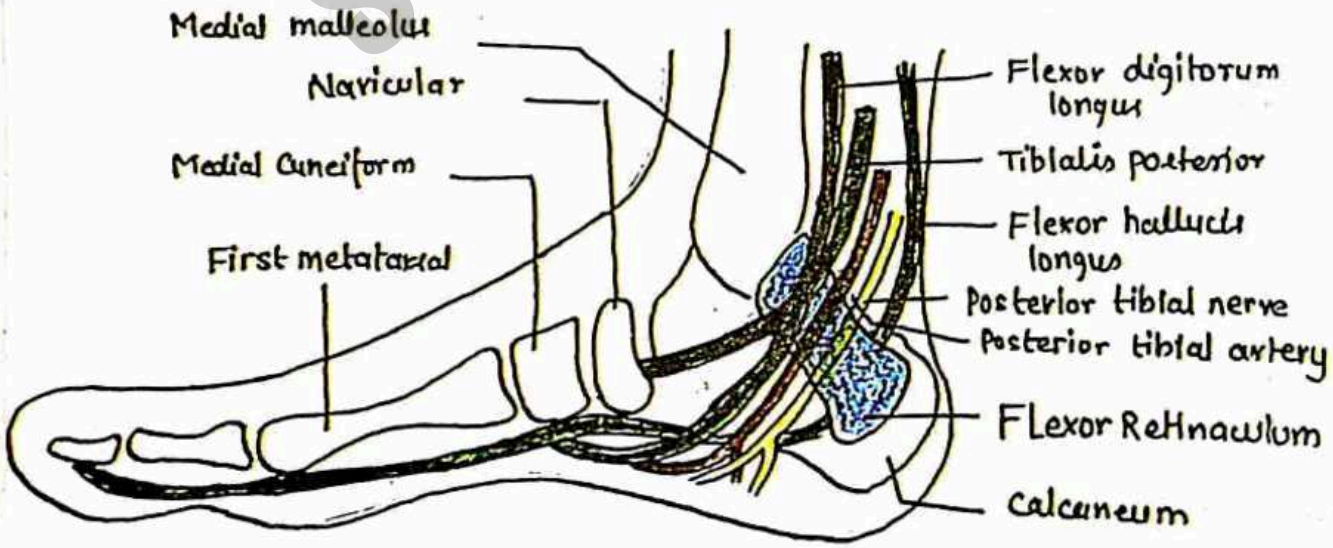

FLEXOR RETINACULUM

HIP JOINT

*TYPE - BALL & SOCKET VARIETY OF SYNOVIAL JOINT.

*ARTICULATING SURFACE - HEAD OF FEMUR WITH ACETABULUM OF HIP JOINT.

*HIP JOINT IS UNIQUE HAVING HIGH DEGREE OF STABILITY AS WELL AS MOBILITY.
*STABILITY DEPEND UPON - DEPTH OF ACETABULUM TENSION & STRENGTH OF LIGAMENT.
*STRENGTH OF SURROUNDING MUSCLE
*LENGTH AND OBLIQUITY OF NECK OF FEMUR.

*LIGMANET

1.FIBROUS CAPSULE - ATTACHED ON HIP BONE TO ACETABULUM LABRUM
- ON FEMUR TO THE INTERTROCHANTERIC LINE INFRONT

2.ILIOFEMORAL LIGAMENT - STRONGEST LIGAMENT OF BODY
- TRIANGULAR IN SHAPE
- APEX IS ATTACHED - ANT. INFERIOR ILIAC SPINE
- BASE - INTERTRCHANTERIC LINE.

3.PUBOFEMORAL LIGAMENT - SUPPORT THE JOINT INFEROMEDIALLY
- TRIANGULAR IN SHAPE
- SUPERIOR ATTACH - ILIOPUBIC EMINENCE & OBTURATOR CRUST
-INFERIORLY - ANTEROINFERIOR PART OF CAPSULE
- LOWER BORDER OF ILIOFEMORAL LIGAMENT.

4.ISCHIOFEMORAL LIGAMNET - COVER THE JOINT POSTERIORLY
- FIBRE ARE TWISTED EXTEND FROM ISCHIUM TO ACETABULUM

5.LIGAMENT OF HEAD OF FEMUR - TRIANGULAR IN SHAPE
- APEX ATTACHED - FOVEA CAPITIS
- BASE - MARGIN OF ACETABULUM NOTCH

6.ACETABULAR LABRUM - FIBROCARTILAGINOUS RIM
- ATTACHED TO MARGIN OF ACETABULUM.

*RELATION :-

1. ANTERIORLY - TENDON OF ILIOPSOAS
- FEMORAL ARTERY
- FEMORAL NERVE
- FEMORAL VEIN

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2.POSTERIORLY - PIRIFORMIS
- SCIATIC NERVE
- OBTURATOR INTERNUS & GEMILLI
- QUADRATUS FEMORIS
- GLUTEUS MAXIMUS

3.SUPERIOR - GLUTEUS MAXIMUS
- GLUTIUS MINIMUS
- GLUTIUS MEDIUS

4.INFERIOR - OBTURATOR EXTERNUS
- HAMSTRING MUSCLE

BLOOD SUPPLY :- OBTURATOR ARTERY & 2 CIRCUMFLEX FEMORAL, 2 GLUTIAL ARTERY

NERVE SUPPLY :- FEMORAL NERVE--->THROUGH NERVE TO RECTUS FEMORIS
-ANTERIOR DIVISION OF OBTURATOR NERVE
-NERVE TO QUADRATUS FEMORIS
-SUPERIOR GLUTEAL NERVE

MOVEMENT :- FLEXION & EXTENSION - TRANSVERSE AXIS
- ADDUCTION & ABDUCTION - ANTERIOR & POSTERIOR AXIS
- MEDIAL & LATERAL ROTATION - VERTICAL AXIS
- CIRCUMDUCTION

LOCKING & UNLOCKING OF KNEE JOINT

1.MECHANISM THAT ALLOW THE KNEE TO REMAIN IN THE POSITION FULL EXTENSION WITHOUT MUSCULAR EFFORT.

2.LOCKING OCCUR -->MEDIAL ROTATION OF FEMUR DURING LAST STAGE OF EXTENSION

3.DIAMETER OF LATERAL FEMORAL CONDYLE IS LESS THAN MEDIAL CONDYLE

4.AS RESULT -->LATERAL CONDYLE FULLY ARTICULATE IN EXTENSION
-->PART OF MEDIAL CONDYLE REMAIN UNUSED
-->AT THIS STAGE LATERAL CONDYLE SERVE AS AXIS & MEDIAL ROTATION OF FEMUR OCCUR
--->SO REMAINING PART OF MEDIAL CONDYLAR SURFACE IS ALSO TAKEN UP.

5.LOCKING PRODUCED BY CONTINUED ACTION OF EXTENSOR MUSCLE ->QUADRICEPS FEMORIS

6.UNLOCK IS REVERSE OF LOCKING BY LATERAL ROTATION OF FEMUR

7.UNLOCK CAUSED BY ---->POPLITEUS MUSCLE

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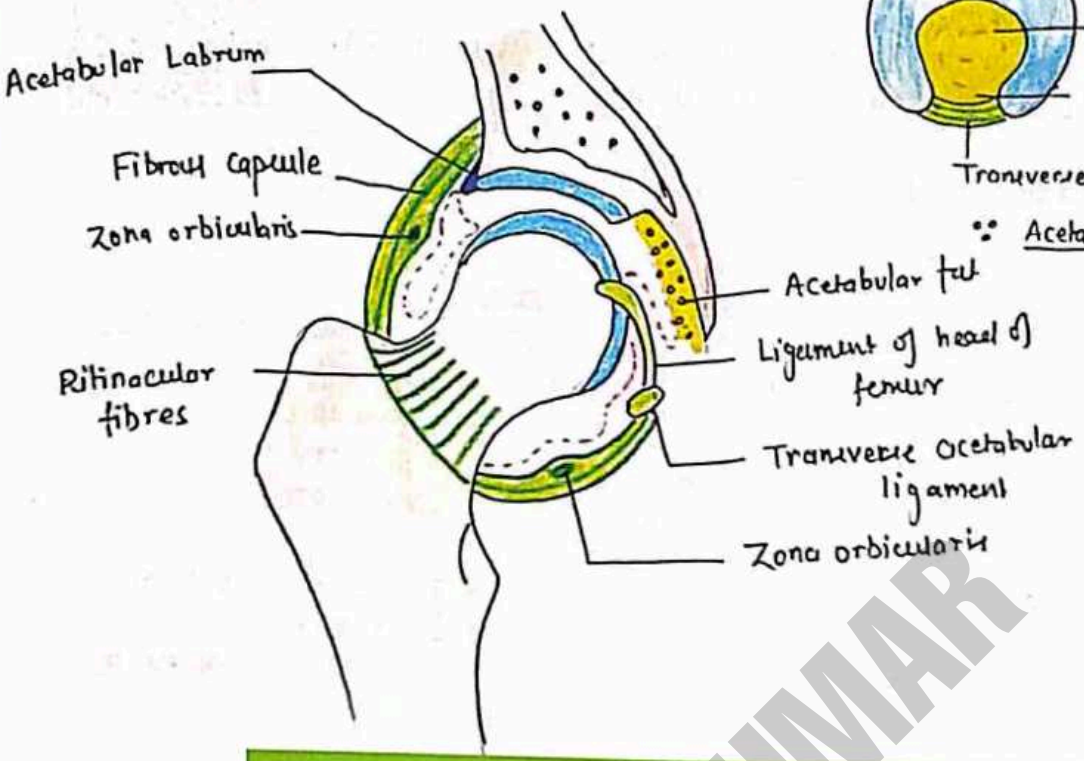
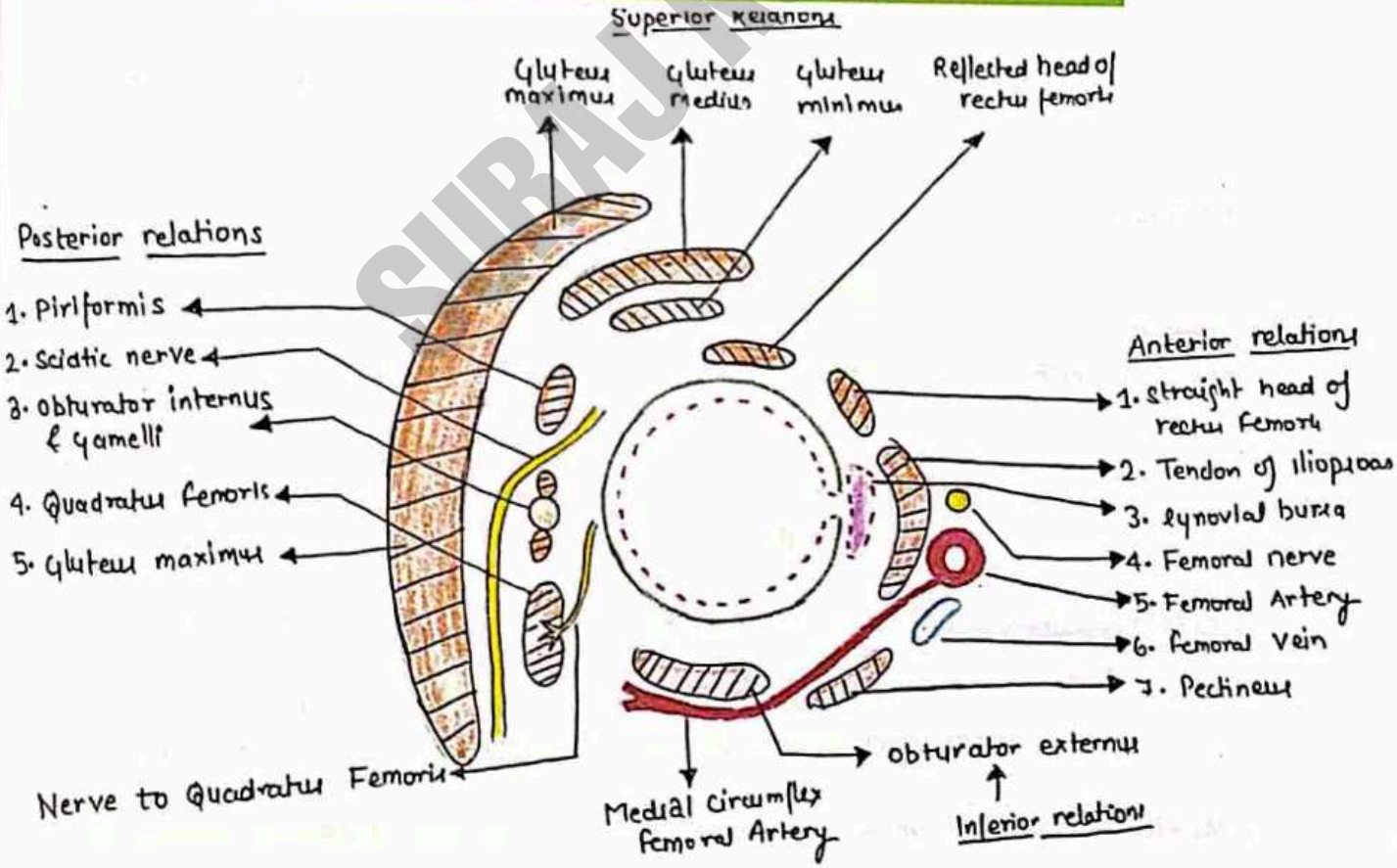
∴ Acetabulum of Hip bone



LIGAMENT OF HIP JOINT



RELATION OF HIP JOINT

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KNEE JOINT

*TYPE :- CONDYLAR SYNOVIAL JOINT
- 2 CONDYLAR JOINT BETWEEN CONDYLE OF FEMUR & TIBIA.
- 1 SADDLE JOINT BETWEEN FEMUR & PATELLA.

*ARTICULAR SURFACE :- CONDYLE OF FEMUR, PATELLA, CONDYLE OF TIBIA

*LIGAMENT :-

1.FIBROUS CAPSULE - THIN AND DEFICIENT ANTERIORLY.
- REPLACE BY QUADRICEP FEMORIS & LIGAMENTUM PATELLAE
- FEMORAL ATTACHMENT - ANT. - DEFICIENT
- POST.- INTERCONDYLAR LINE
- TIBIAL ATTACHMENT - ANT. - TIBIAL TUBEROSITY
- POST. - INTERCONDYLAR RIDGE

2.LIGAMENTUM PATELLAE - THIS IS THE CENTRAL PORTION OF COMMON TENDON OF INSERTION OF QUADRICEPS FEMORIS
-ATTACHMENT-->ABOVE-POSTERIOR SURFACE OF APEX OF PATELLA
BELOW - TIBIAL TUBEROSITY

3.TIBIAL COLLATERAL LIGAMENT - THIS IS A LONG BAND OF GREAT STRENGTH
-ATTACHMENT-->SUPERIOR-MEDIAL EPICONDYLE SHAFT OF FEMUR
-INFERIOR - DEVIDED INTO ANT. AND POST. PART
-ANT.PART-MEDIAL SURFACE OF SHAFT OF TIBIA
-POST.PART-MEDIAL CONDYLE OF TIBIA

4.FIBULAR COLLATERAL - SUPERIOR - ATTACHMENT - LATERAL EPICONDYLE OF FEMUR
-INFERIORLY - HEAD OF FEBULA

5.OBLIQUE POPLITEAL LIGAMENT - EXPANSION FROM TENDON OF SEMIMEMBRANOSUS
-RUN LATERAL WITH POST.SURFACE OF CAPSULE
-ATTACHMENT -->INRECONDYLAR LINE & LATERAL CONDYLE OF FEMUR

6.ARCUATE POPLITEAL LIGAMENT - POSTERIOR EXPANSION FROM SHORT LATERAL LIGAMENT
-EXTEND BACKWARD FROM HEAD OF FIBULA TO POSTERIOR BORDER OF INTERCONDYLAR AREA OF TIBIA.

7.MINISCUS - ARE 2 FIBROCARTILAGINOUS DISC - LATERAL & MEDIAL MINISCUS.

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8.CRUCIATE LIGAMENT - VERY THICK & STRONG FIBROUS BAND
*ANT.CRUCIATE LIGAMENT - BEGIN FROM ANT.INTRACONDYLAR AREA OF TIBIA TO POSTERIOR PART OF LATERAL CONDYLE OF FEMUR.
*POST.CRUCIATE LIGAMENT - POSTERIOR PART OF INTERCONDYLAR AREA OF TIBIA TO ANT. PART OF MEDIAL CONDYLE OF FEMUR.

9.TRANSVERSE LIGAMENT - CONNECT ANTERIOR END OF LATERAL & MEDIAL MENISCUS

*BLOOD SUPPLY :-

1.5 GENICULAR BRANCH OF POPLITEAL ARTERY
2.DESCENDING GENICULAR BRANCH OF FEMORAL ARTERY
3.DESCENDING BRANCH OF LATERAL CIRCUMFLEX ARTERY
4.2 RECURRENT BRANCH OF ANTERIOR TIBIAL ARTERY
5.CIRCUMFLEX FIBULAR BRANCH OF POSTERIOR TIBIAL ARTERY

*NERVE SUPPLY :-

1.FEMORAL NERVE →BRANCH TO VASTI ESPECIALLY VASTUS MEDIALIS
2.SCIATIC NERVE THROUGH GENICULAR BRANCH OF THE TIBIAL & COMMON PERONEAL NERVES
3.OBTURATOR NERVE---→>THROUGH POSTERIOR DIVISION

INVERSION & EVERSION OF FOOT

1.INVERSION - MOVEMENT IN WHICH MEDIAL BORDER OF FOOT IS ELEVATED
- SOLE FACE MEDIALLY.
2.EVERSION - MOVEMENT IN WHICH LATERAL BORDER OF FOOT IS ELEVATED.
- SOLE FACE LATERALLY.

*PERFORM VOLUNTARY ONLY WHEN THE FOOT IS OFF THE GROUND.
*WHEN FOOT IS ON THE GROUND THIS MOVEMENT HELP TO ADJUST THE UNEVEN GROUND.

3.INVERSION - ACCOMPANIED BY PLANTAR FLEXION OF FOOT
- ADDUCTION OF FORE FOOT.
4.EVERSION - ACCOMPANIED BY DORSI FLEXION OF FOOT
- ABDUCTION OF FORE FOOT.

DORSIFLEXION & PLANTAR FLEXION

1.DORSIFLEXION → FORE-FOOT IS RAISED
→ANGLE BETWEEN FRONT OF LEG & DORSUM OF THE FOOT IS DIMINISHED

2.PLANTAR FLEXION→ FORE-FOOT IS DEPRESSED
→ANGLE BETWEEN LEG & FOOT IS INCREASED.

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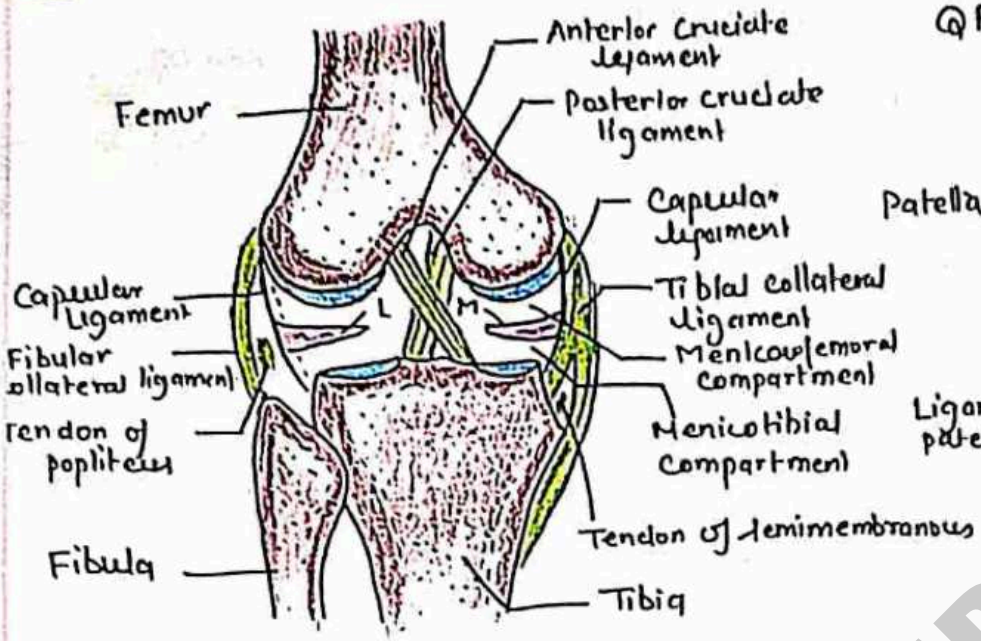
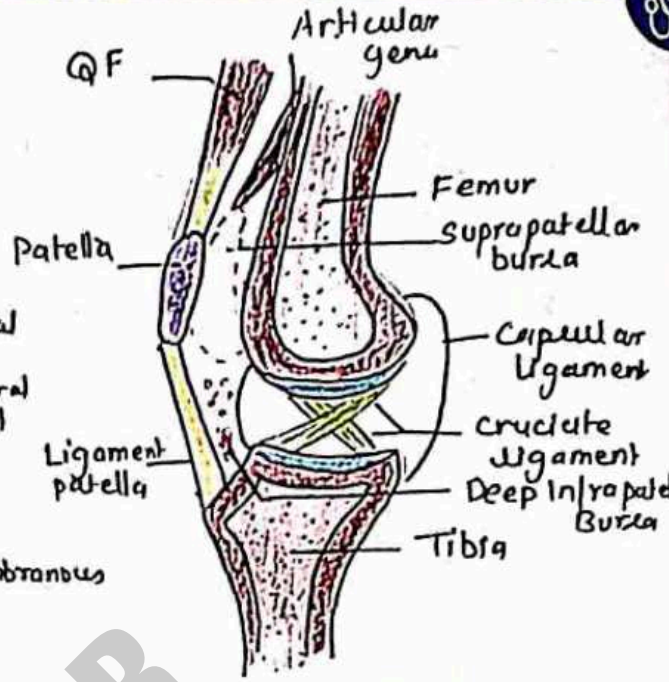
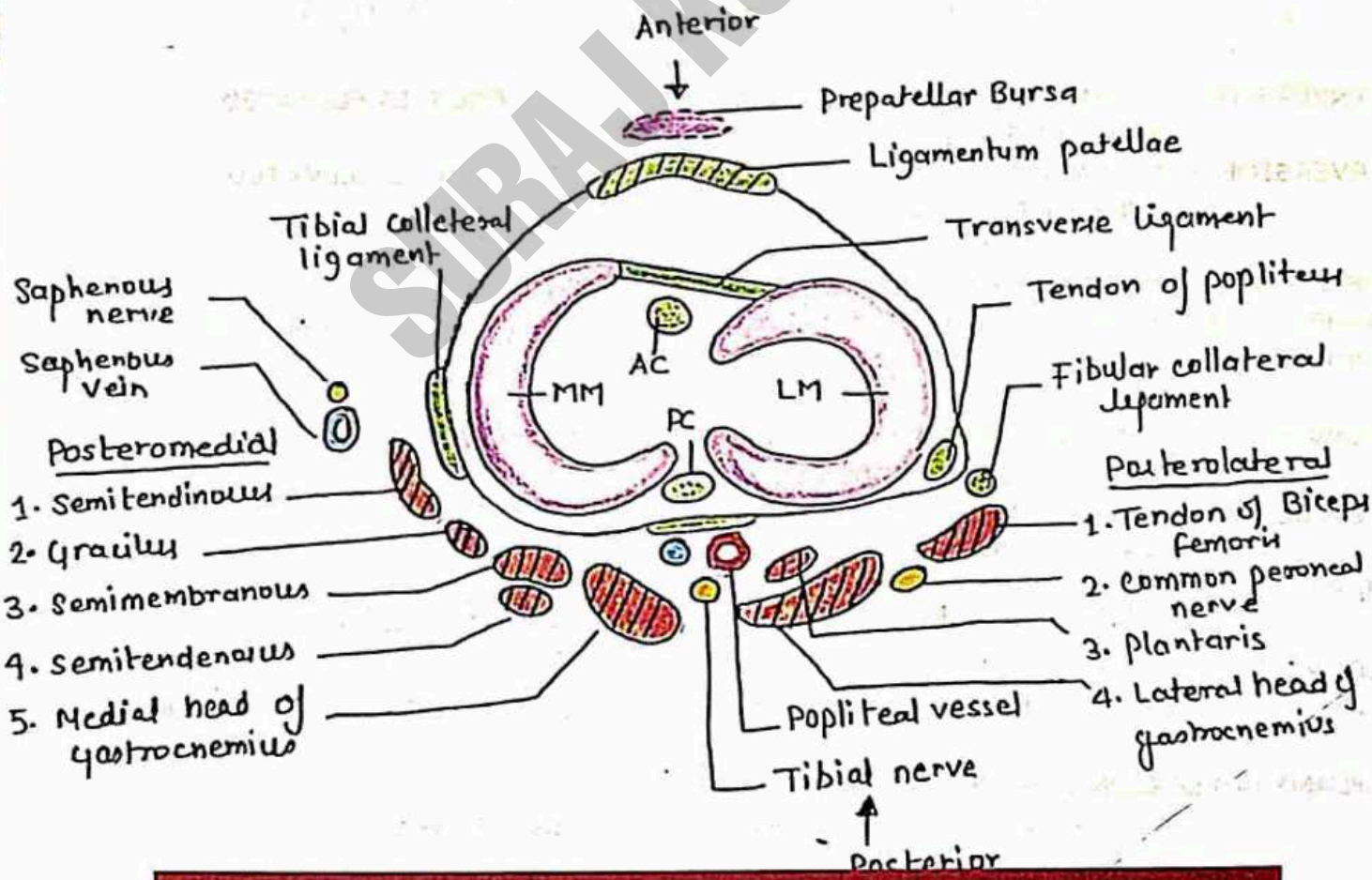


∴ Knee Joint – coronal section



∴ Knee joint – Sagittal section

TIBIAL & FIBULAR COLLATERAL LIGAMENT OF KNEE JOINT



TRANSVERSE SECTION THROUGH KNEE JOINT

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STUDY TIPS FOR MAXIMUM MARKS

1.HOW TO WRITE IN UNIVERSITY EXAM AND HOW TO DRAW DIAGRAMS ALREADY MENTION IN UPPER LIMB (PAGE NUMBER - 25)

PREVIOUS YEAR UNIVERSITY QUESTIONS

*LONG QUESTIONS - 20 MARKS

1) DESCRIBE THE ROOT VALUE. COURSE, RELATIONS BRANCHES AND DISTRIBUTION AND APPLIED ANATOMY OF OBTURATOR NERVE ?

2) DESCRIBE THE TYPE LIGAMENTS, RELATION, MUSELES PRODUCING THE MOVEMENTS AND APPLIED ANATOMY OF HIP JOINT ?

3) FEMORAL TRIANGLE AND FEMORAL SHEATH & CANAL WITH CLINICAL ANATOMY

4) DESCRIBE VENOUS DRAINAGE OF LOWER LIMBS ITS APPLIED ASPECT ?

5) DESCRIBE THE TYPE LIGAMENTS , MOVEMENTS AND MUSELES PRODUCING THE MOVEMENTS AND APPLIED ANATOMY OF HIP JOINT ?

*SHORTS QUESTION - 5 MARKS

1)FEMORAL ARTERY
2)FEMORAL NERVE
3)ADDUCTOR CANAL
4)POPLITEAL FOOSA
5)ARCHES OF FOOT
6)HAMSTRING MUSCLE
7)LOCKING & UNLOCKING OF KNEE
8) FLEXOR RETINACULUM OF FOOT

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LOWER LIMB ANATOMY IS COMPLETED !!

STOP DOUBTING YOURSELF , WORK HARD AND MAKE IT HAPPEN

ALL THE BEST FOR YOUR EXAMS

DR.JOHARI

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MCQ LOWER LIMB

1) Which of the following does not form boundary of femoral triangle :
a Inguinal ligament
b medial border of adductor longus
c lateral border of adductor longus
d medial border of sartorius

2) All of the following are content of femoral triangle except :
a femoral artery
b femoral nerve
c deep inguinal lymph nodes
d great saphenous vein

3) Femoral nerve supplies all of the following except :
a adductor longus
b pectineus
c iliacus
d sartorius

4) All of the following have dual nerve supply except :
a Pectineus
b biceps femoris
c quadriceps femoris
d adductor magnus

5) The following muscle is flexor of both hip & knee joint:
a rectus femoris
b biceps femoris
c gracilis
d sartorius

6) Anterior division of femoral nerve supplies the following muscle :
a iliacus
b pectineus
c rectus femoris
d sartorius

7) Anterior division of obturator nerve supplies all the following except :
a obturator externus
b pectineus
c gracilis
d adductor longus

8) Medial boundary of femoral ring is formed by:
a inguinal ligament
b pectineus muscle
c lacunar ligament
d femoral vein

9) Which muscle of the thigh is preffered to give intramuscular injection in children :
a sartorius
b vastus lateralis
c vastus medialis
d rectus femoris

10) All of the following muscle are inserted on greater trochanter EXCEPT :
a obturator externus
b piriformis
c gluteus medius
d gluteus maximus

11) All the following are supplied by posterior division of obturator nerve EXCEPT :
a adductor brevis
b adductor magnus
c obturator internus
d obturator externus

12) Which structure lies midway between anterior superio iliac spine and pubic symphysis :
a femoral artery
b femoral vein
c femoral nerve
d deep inguinal ring

13) Which of the following ligament prevents hypertensio hip:
a inguinal ligament
b iliofemoral ligament
c pubofemoral ligament
d ischiofemoral ligament

14) All the following are lateral rotators of thigh except :
a gluteus medius
b obturator internus
c sartorius
d quadratus femoris

15) The following are part of hamstrings EXCEPT :
a Semitendinosus
b Semimembranosus
c biceps femoris
d tensor fascia lata

16) The superficial inguinal lymph nodes drain all the following except :
a lower part of ant Abdominal wall
b thigh
c scrotum
d glans penis

17) Saphenous nerve is a branch of:
a femoral nerve
b obturator nerve
c tibial nerve
d common peroneal nerve

18) The deep inguinal lymph node of Rosenmuller drains the:
a Testis
b scrotum
c glans penis
d ductus deferens

1) C 2)D 3)A 4)B 5)D 6)D 7)A 8)C 9)B 10)D 11)C 12)A 13)B 14)A 15)D 16)D 17)A 18)C

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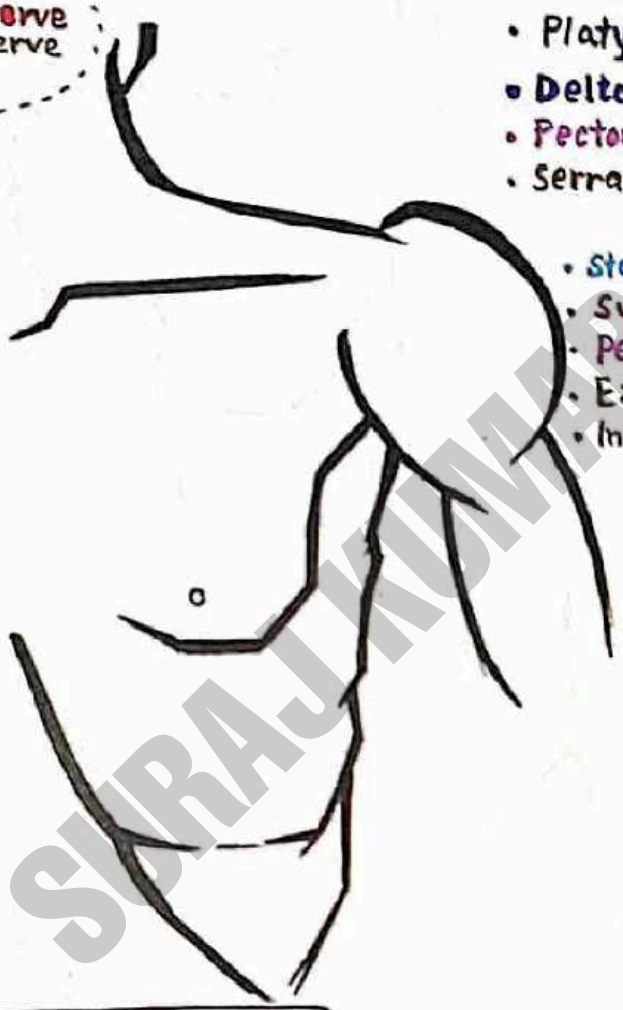
THORAX MUSCLES

- Facial Nerve
- Axillary Nerve
- Lateral & Medial Pectoral Nerve
- Long thoracic Nerve
- Accessory Nerve
- Subclavian Nerve
- Intercostal Nerve



ANTERIOR AXIO-APPENDICULAR

SUPERFICIAL

- Platysma
- Deltoid
- Pectoralis Major
- Serratus Anterior

DEEP

- Sternocleido mastoid
- Subclavius
- Pectoralis Minor
- External intercostal Muscle
- Internal intercostal Muscle

POSTERIOR AXIO-APPENDICULAR

SUPERFICIAL

- Deltoid
- Trapezius
- Rhomboid Major
- Rhomboid Minor
- Latissimus dorsi

DEEP

- Supraspinatus
- Infraspinatus
- Teres minor
- Teres major
- Levator scapulae

- Dorsal scapular Nerve
- Thoracodorsal Nerve
- Suprascapular Nerve
- Lower subscapular Nerve

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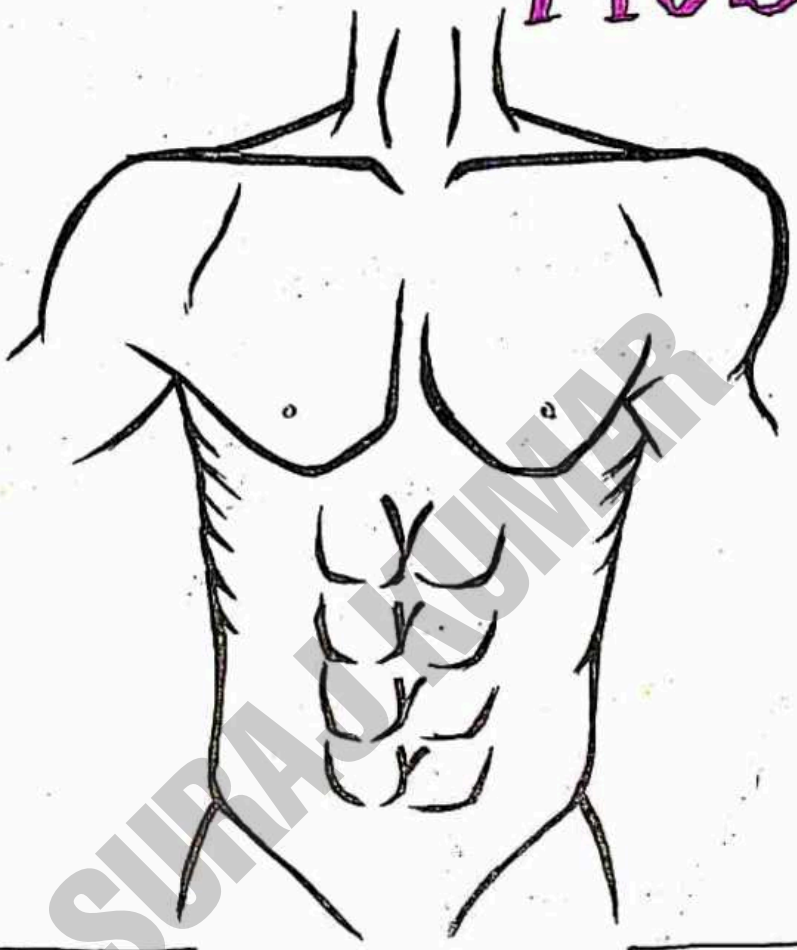
ABDOMINAL MUSCLES



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ANTERIOR WALL

- External Oblique
- Internal Oblique
- Trans-verse Abdominis
- Rectus Abdominis

- Anterior primary rami of lower 6 thoracic nerves (t_7-12)
- Lower 5 intercostal & subcostal nerves (t_7-12)
- Anterior primary rami of lower 6 thoracic nerves (t_7-12) & 1st lumbrical (L_1)

POSTERIOR WALL

- Quadratus lumborum
- Psoas minor
- Psoas major
- Iliacus

- Ventral rami of T_{12} & L_{1-4} lumbar spinal nerves
- Ventral rami of L_{1-4} spinal nerves
- Branch of L_1 spinal nerve
- Femoral nerve (L_{2-4})

PLEURA

*PLEURA IS A SEROUS MEMBRANE - LINED BY MESOTHELIUM.
*2 PLEURAL SAC, 1 ON EITHER SIDE OF THE MEDIASTINUM.
*OUTER LAYER IS PARIETAL PLEURA.
*INNER LAYER IS VISCERAL OR PULMONARY PLEURA.
*2 PLEURA CONTINUE WITH EACH OTHER AROUND HILIUM OF LUNGS BETWEEN PLEURAL CAVITY.

COMPARISON OF VISCERAL & PARIETAL PLEURA

DEVELOPMENT	SPLANOPLEURIC MESODERM	SOMATOPLEURIC MESODERM
POSITION	LINES SURFACE OF LUNG INCLUDING TISSUES	LINE THORACIC WALL, MEDIASTINUM & DIAPHRAM.
NERVE SUPPLY	SYMPATHETIC NERVES FROM T2-T5 GANGLIA, PARASYMPATHETIC FROM VAGUS NERVE	THORACIC NERVES & PHRENIC NERVES
SENSITIVITY	INSENSITIVE TO PAIN	SENSITIVE TO PAIN
BLOOD SUPPLY	BRONCHIAL VESSELS	INTERCOASTAL & PERICARDIACOPHRENIC VESSELS
LYMPHATIC DRAINAGE	TRACHEOBRONCHIAL LYMPH NODE	INTERCOASTAL LYMPH NODE

VISCERAL PLEURA :- COVER SURFACE & FISSURES OF LUNGS EXCEPT HILUM WHERE IT CONTINUE WITH PARIETAL PLEURA.
-FIRMLY ADHERENT TO THE LUNGS CAN NOT BE SEPARATED.

PARIETAL PLEURA :- THICKER THAN VISERAL PLEURA SUB DIVIDED INTO 4 PARTS - 1.COSTAL, 2.DIAPHRAGMATIC, 3.MEDIASTINAL, 4.CERVICAL

SURFACE MARKING OF PLEURA :- (PARIETAL)

1.CERVICAL PLEURA - REPRESENT BY CURVED LINE. - ABOUT 2.5CM ABOVE CLAVICLE.

2.ANTERIOR MARGIN OF COSTOMEDIASTINAL -
- RIGHT SIDE - EXTEND - STERNOCLAVICULAR JOINT TO STERNUM ANGLE.
- VERTICALLY DOWNWARD TO XIPHISTERNAL POINT
- LEFT SIDE - FOLLOW THE SAME COURSE UP TO 4TH COSTOCARTILAGE LEVEL.
- THEN ARCH DOWN TO 6TH COSTOCARTILAGE.

3.POSTERIOR MARGIN - PASS 2CM LATERAL TO 12 THORACIC SPINE TO 2CM LATERAL TO 4TH CERVICLE SPINE.

4.INFERIOR MARGIN - CROSS 8TH RIB IN MID CLAVICULAR LINE - 10TH RIB IN MID AXILLARY LINE

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SURFACE MAKING OF PLEURA :-(VISCERAL)

*APEX - 2.5CM ABOVE MEDIAL 2/3 CLAVICLE.

1.ANTERIOR BORDE OF VISCERAL PLEURA -

-RIGHT SIDE :- CORRESPOUND CLOSELY WITH COASTO MEDIASTINAL LINE.

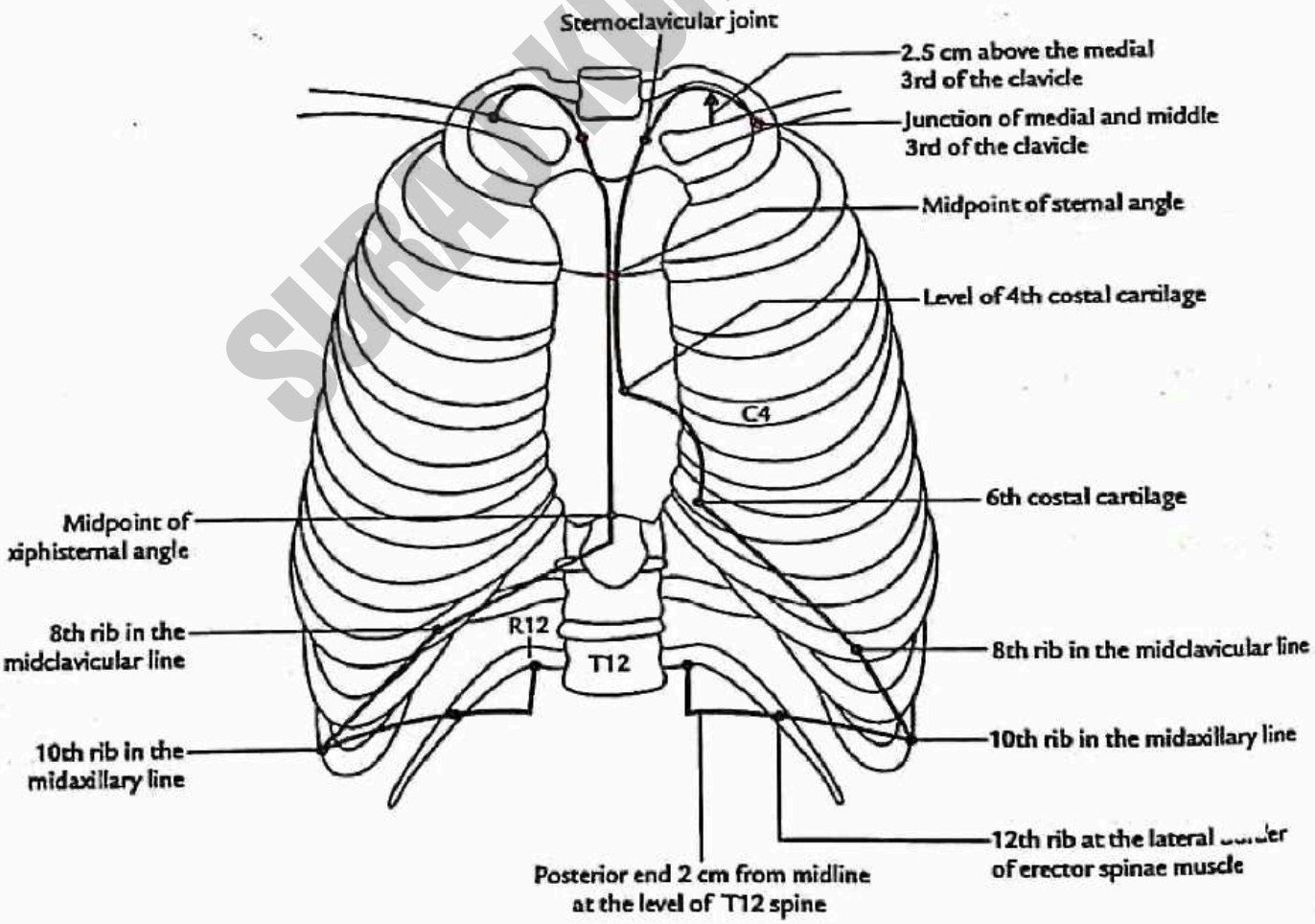
-OBTAIN BY JOINING - STERNOCLAVICULO JOINT & STERNAL ANGLE

-MEDIAL PLANE JUST ABOVE XIPHISTERNAL JOINT.

-LEFT SIDE :- CORRESPOUND TO COASTOMEDIASTINAL LINE UP TO LEVEL OF 4TH COSTOCARTILAGE

- LOWER PART REPRESENT A CARDIAC NOTCH
- FROM 4TH COASTOCARTILAGE IT PASS - LATERAL 3.5CM FROM STERNAL MARGIN
- THEN CURVE DOWN MEDIALLY TO REACH 6TH COASTOCATILAGE
- AREA OF CARDIAC NOTCH IS DULL ON PERCUSSION CALLED AREA OF SUPERFICIAL CARDIAC DULLNESS.

2.LOWER BORDER- CROSS 6TH RIB IN MIDCLAVICULAR LINE, 8TH RIB IN MID AXILLARY LINE.



SURFACE MARKING OF PARIETAL PLEURA

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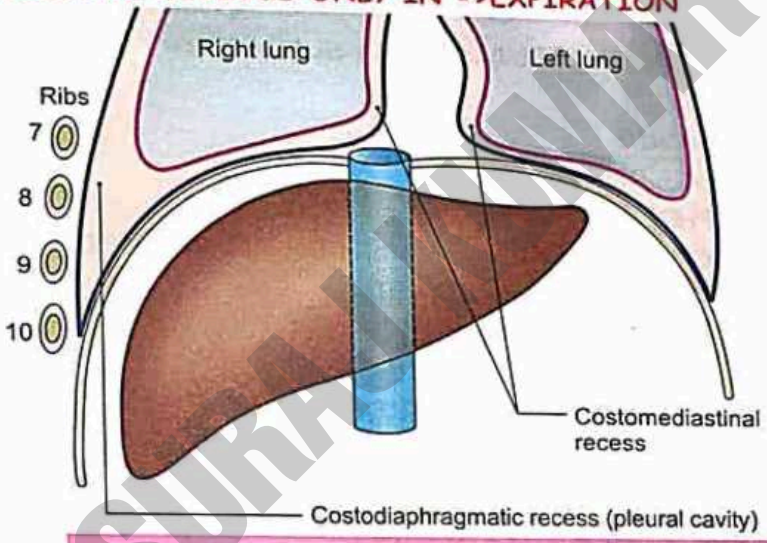
RECESSES OF PLEURA

*THERE ARE 2 RECESSES OF PARIETAL PLEURA ACT →RESERVE SPACE FOR LUNGS TO EXPAND DURING INSPIRATION

1) COSTOMEDIASTINAL RECESS-
- *LIE ANTERIORLY ,BEHIND STERNUM AND COSTAL CARTILAGE
- *BETWEEN COSTAL AMD MEDIASTINAL PLEURA
- *PARTICULARLY IN RELATION →CARDIAC NOTCH OF LEFT LUNG
- *FILLED UP BY -→ANTERIOR MARGIN OF LUNGS EVEN DURING QUIET BREATHING
- *ONLY OBVIOUS-→REGION OF CARDIAC NOTCH OF LUNG

2) COSTODIAPHRAGMATIC RRECESS-
- *LIE INFERIORLY BETWEEN COSTAL AND DIAPHRAGMATIC PLEURA
- *VERTICALLY 5CM AND EXTEND FROM 8TH →TO 10TH RIB IN MIDAXILLARY LINE
- *FILLED BY -→LUNGS EXPEND INTO RECESSES DURING INSPIRATION
- *SO THIS RECESS ARE OBVIOUS ONLY IN ->EXPIRATION



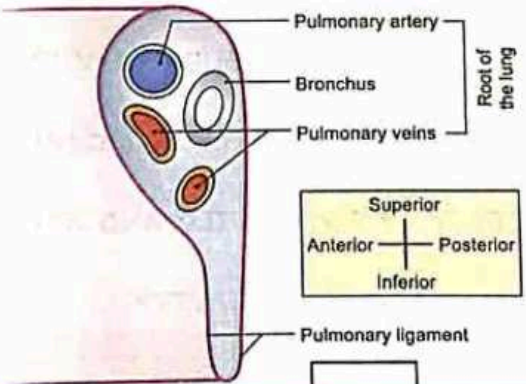
SCAN QR WITH QRBOT APP
TO DIRECTLY OPEN VIDEO LINK

PULMONARY LIGAMENT

1.PARITEAL PLEURA- SURROUND ROOT OF LUNGS-→EXTEND DOWNWARD BEYOUND ROOT AS FOLD CALLED PULMONARY LIGAMENT

2.FOLD CATAIN -→THIN LAYER OF LOOSE AREOLAR TISSUE WITH FEW LYMPHATICS

3.PROVIDE - DEAD SPACE--→INTO PULMONARY VEIN EXPAND DURING -→INCREASE IN VENOUS RETURN IN EXCERCISE



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FISSURES & LOBES OF LUNGS

*RIGHT LUNG-→ DIVIDED INTO 3 LOBES - UPPER, MIDDLE, & LOWER BY OBLIQUE FISSURE AND HORIONTAL FISSURE

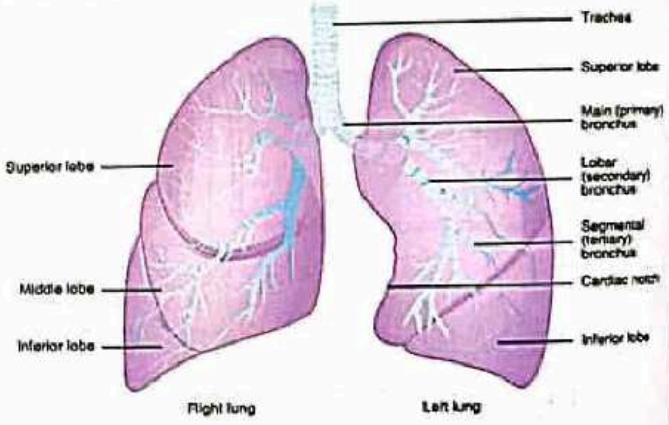
*LEFT LUNG-→ DIVIDED INTO 2 LOBES - UPPER & LOWER LOBE BY OBLIQUE FISSURE

*OBLIQUE FISSSURE - CUT INTO WHOLE THICKNESS OF LUNGS, EXPECT AT HILUM
-DUE TO OBLIQUE PLANE OF FISSURE -→LOWER LOBE IS MORE POSTERIOR
-→UPPER & MIDDLE LOBE MORE ANTERIOR

*HORIZONTAL FISSURE - PASSES FROM ANTERIOR BORDER UP TO OBLIQUE FISSURE
-SEPARATE WEDGE SHAPED MIDDLE LOBE FROM UPPER LOBE
-RUN HORIZONTALLY AT LEVEL OF FOURTH COSTAL CARTILAGE
-TONGUE SHAPE PROJECTION OF LEFT LING BELOW CARDIAC NOTCH CALLED LINGULA

IMPRESSION ON MEDIASTINAL SURFACE :-

1.RIGHT LUNGS - OESOPHAGUS
- TRACHEA - RIGHT VAGUS NERVE
- RIGHT PHRENIC NERVE
- SUPERIOR VENA CAVA_-→ ARCH OF AZYGOS VEIN
- RIGHT BRACHIOCEPHALIC VEIN
- INFERIOR VENA CANA
- HIULM
- IMPRESSION OF RIGHT ATRIUM



2.LEFT LUNGS - OESPHAGUS
- THORACIC DUCT
-TRACHEA
- LEFT PHRENIC NERVE
- LEFT VAGUS MERVE
- PULMONARY TRUNK
- HILUM
- IMPRESSION OF LEFT VENTICLE

CLINICALANATOMY

1.PNEUMOTHORAX- PRESENCE OF AIR IN THE PLEURAL CAVITY

2.HAEMOTHORAX- PRESENCE OF BLOOD IN THE PLEURAL CAVITY

3.HYDROPNEUMOTHORAX- PRESENCE OF BOTH FUILD AND AIR CAVITY IN PLEURAL CAVITY

4.EMPYEMA- PRESENCE OF PUS IN THE PLEURAL CAVITY

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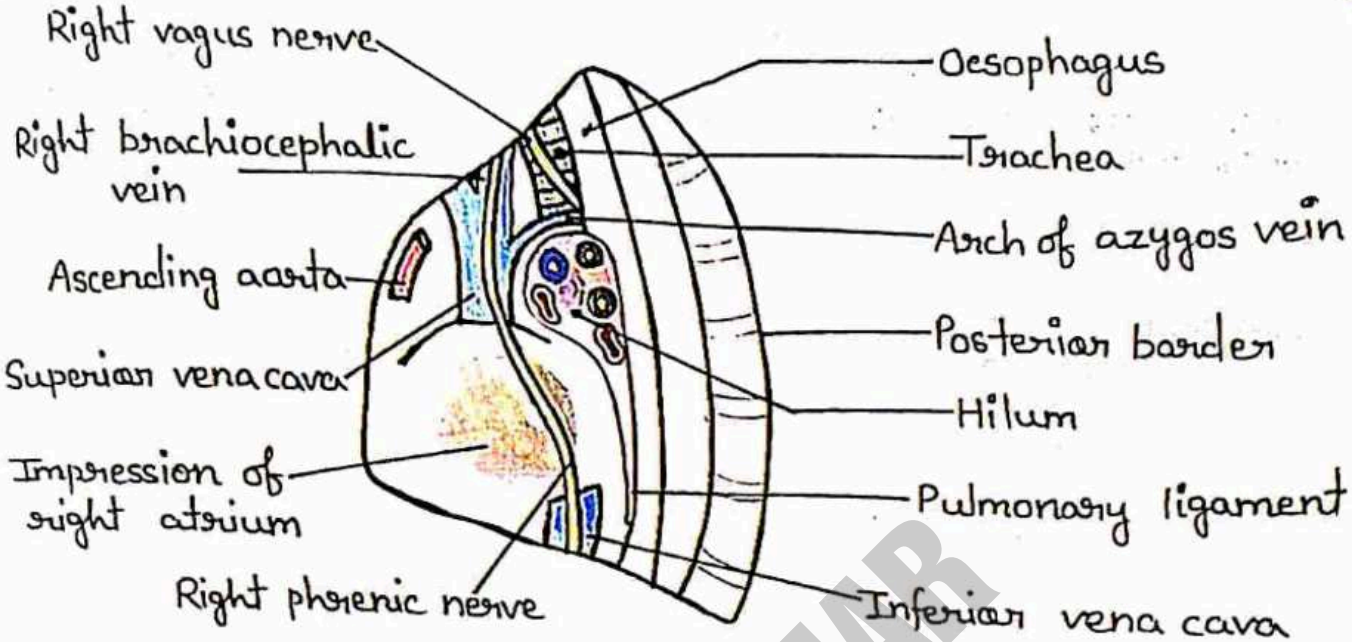
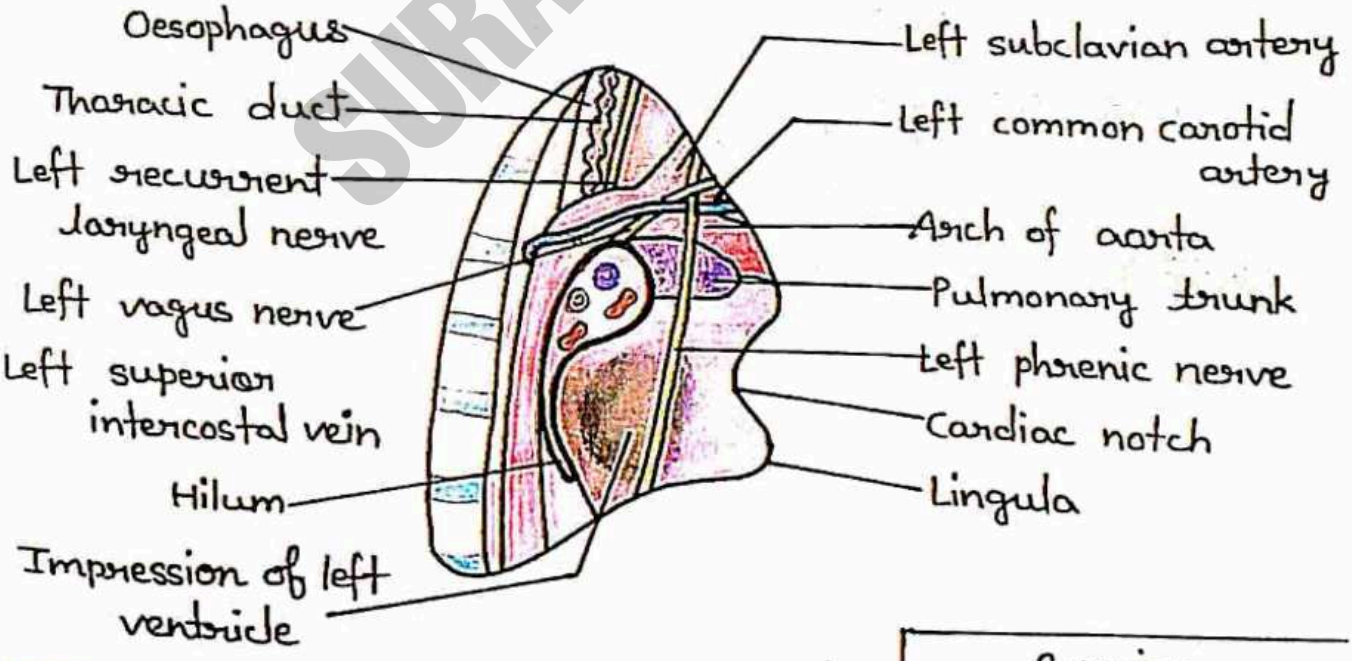


IMPRESSION IN MEDIASTINUM SURFACE OF RIGHT LUNGS

Superior
Anterior — Posterior
Inferior



IMPRESSION IN MEDIASTINUM SURFACE OF LEFT LUNGS

Superior
Posterior — Anterior
Inferior

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ROOTS OF LUNGS

*SHORT, BROAD PEDICLE WHICH CONNECT MEDIAL SURFACE OF LUNGS TO MEDIASTINUM

*FROM BY STRUCTURE WHICH EITHER ENTER OR COME OUT OF LUNGS AT HILUM.

*LIE OPPOSITE THE BODY OF 5TH 6TH & 7TH THOARCIC VERTEBRAE

*CONTENTS - PRINCIPAL BRONCHUS ON LEFT SIDE
- EPARTERIAL & HYPARTERIAL BRONCHI ON RIGHT SIDE
- 1 PULMONARY ARTERY
- 2 PULOMONARY VEIN - SUPERIOR & INFERIOR
- BRONCHIAL ATERIES
- BRONCHIAL VEINS
- ANTERIOR & POSTERIOR PULMONARY PLEXUSES OF NERVE
- LYMPHATICS OF LUNGS

*ARRANGEMENT OF STRUCTURE IN ROOT :-

* ANT. TO POST. - SIMILAR IN BOTH SIDES - SUPERIOR PULMONARY VEIN
- PULMONARY ARTERY
- BRONCHUS

*ABOVE DOWNWARD - DIFFERENT IN 2 SIDES

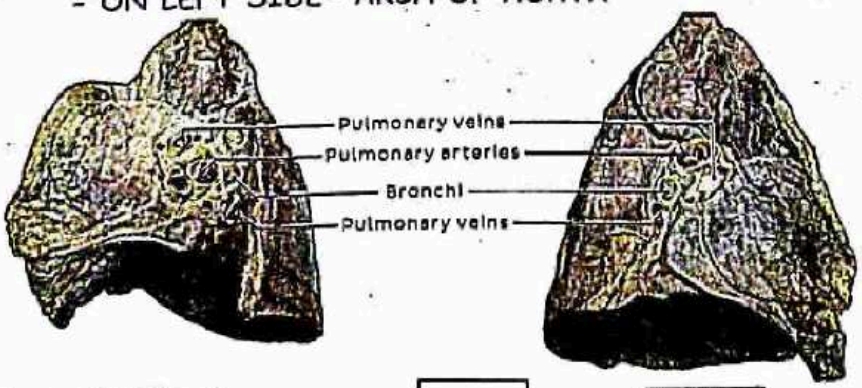
1) RIGHT SIDE - EPARTERIAL BRONCHUS
-PULMONARY ARTERY
-HYPARTERIAL BRONCHUS
-INFERIOR PULMONARY VEIN

2) LEFT SIDE - PULMONARY ARTERY
- BRONCHUS
- INFERIOR PULMONARY VEIN

*RELATONS :- ANTERIOR - COMMON ON 2 SIDES - PHRENIC NERVE
- PERICARDIACOPHRENIC VESSELS
- ON RIGHT SIDE - SUPERIOR VENA CAVA

POSTERIOR - COMMON ON 2 SIDES - VAGUS NERVE
- ON LEFT SIDE - DESCENDING THORACIC AORTA

SUPERIOR - ON RIGHT SIDE - TERMINAL PART OF AZYGOS VEIN
- ON LEFT SIDE- ARCH OF AORTA



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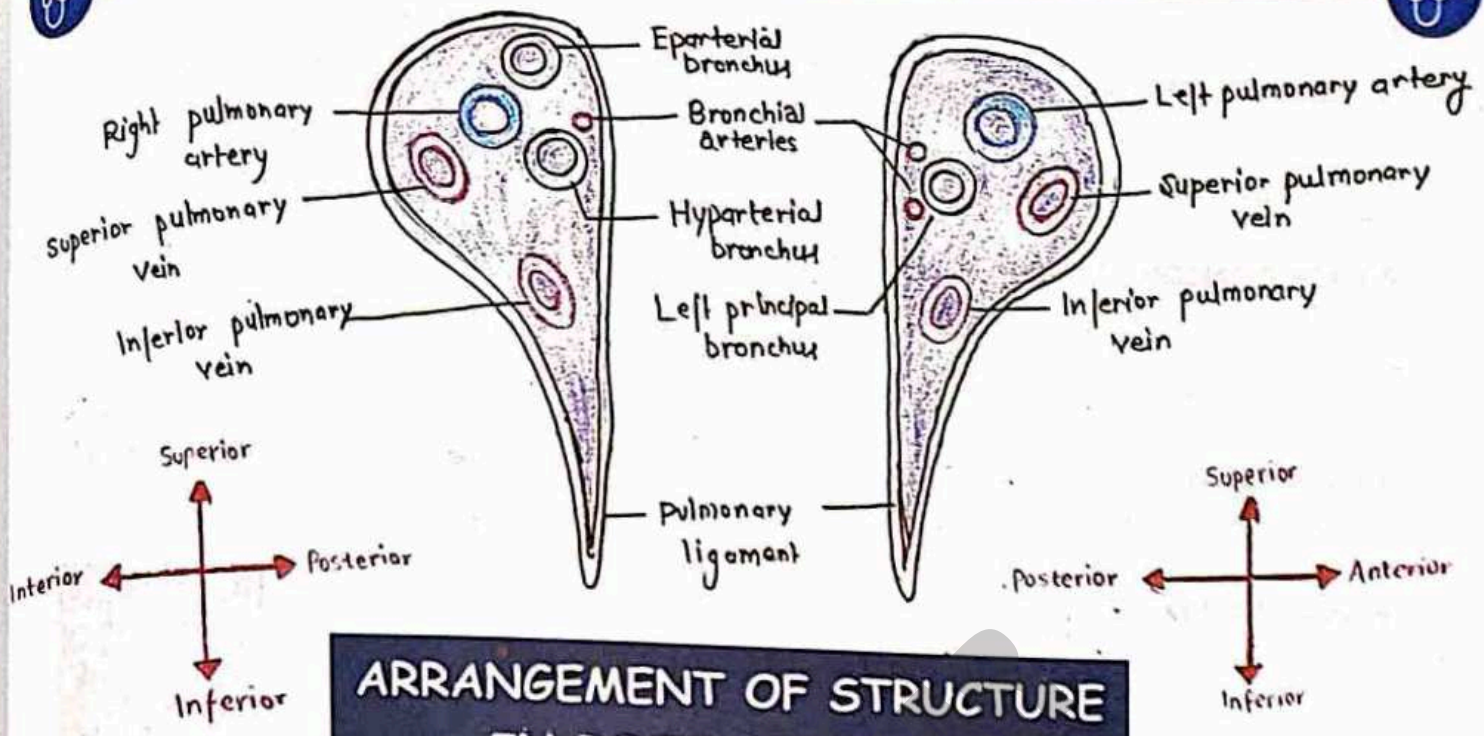
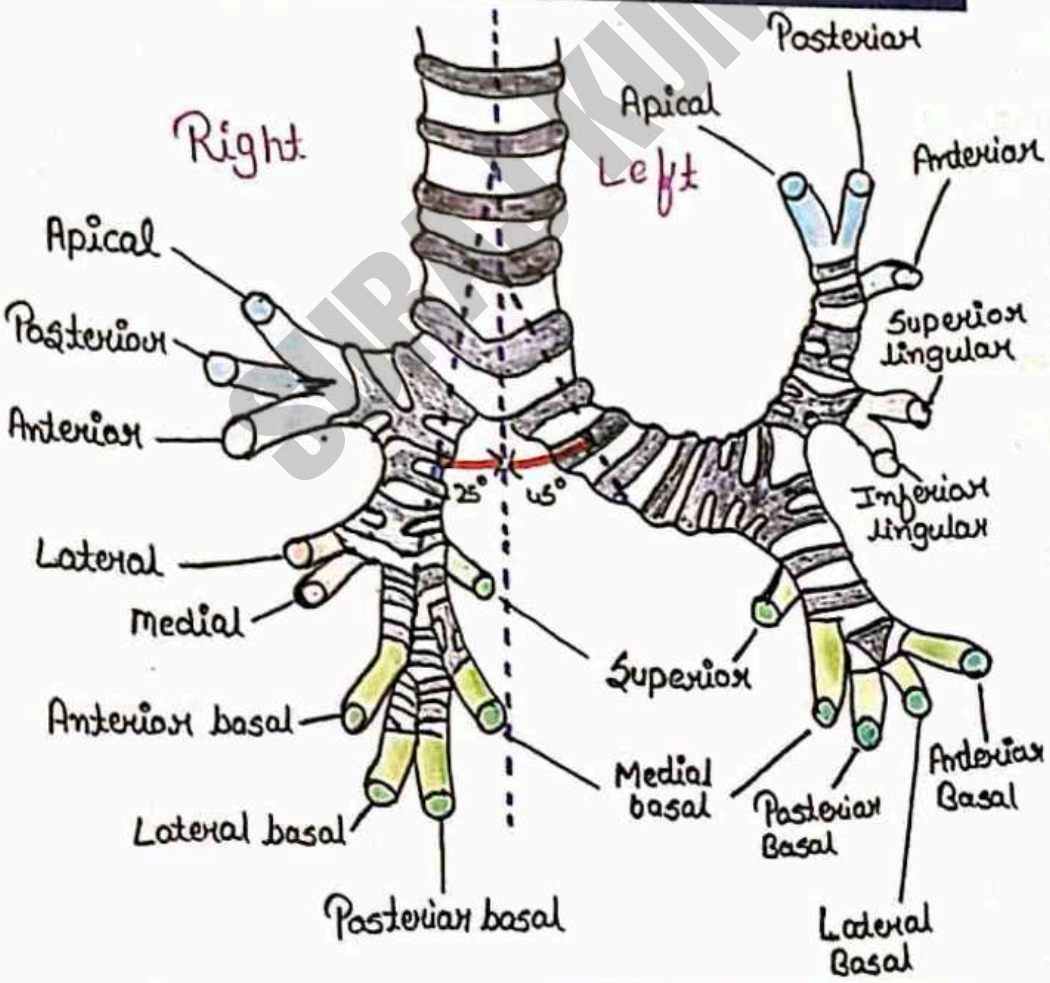


ARRANGEMENT OF STRUCTURE IN ROOT OF LUNGS



BRONCHOPULMONARY SEGMENTS

BRONCHIAL TREE & BRONCHOPULMONARY SEGMENTS

1.TRACHEA DIVIDES INTO 2 PRIMARY PRINCIPAL BRONCHI -→RIGHT & LFT BRONCHUS

2.EACH PRINCIPAL BRONCHUS ENTER LUNGS -→THROUGH HILIUM

3.DIVIDES INTO SECONDARY LOBAR BRONCH-→ONE FOR EACH LOEBS -→ 3 LOBAR BRONCHI - RIGHT SIDE 2 LOBAR BRONCHI - LEFT SIDE

4.EACH LOBAR BRONCHUS DIVIDE INTO TERTIARY -→ 1 FOR EACH BRONCHOPULMONARY SEGMENTS --→ WHICH ARE 10 ON RIGHT AND LEFT SIDE

5.SMALLER BRANCHES ARE CALLED RESPIRATORY BRONCHIOLES

BRONCHOPULMONARY SEGMENTS -

*10 SEGEMNTS IN RIGHT SIDE AND 10 ON LEFT SIDE

***FEATURES :-**

1. WELL DEFINED ANATOMIC , FUNCTIONAL AND SURGICAL SECTOR OF LUNGS
2. EACH ONE IS AERATED BY TERTIARY BRONCHUS
3. EACH SEGMENTS IS PYRAMIDAL IN SPACE -→APEX -→TOWARD ROOT OF LUNGS
4. EACH SEGMENTS HAS SEGMENTAL BRONCHUS & ARTERY AND AUTONOMIC NERVE
5. SEGMENTAL VENULES LIE-→IN CONNECTIVE TISSUE BETWEEN ADJACENT PULMONARY UNITS
6. DURING SEGMENTAL RESCENTION, THE SURGEON WORK ALONG SEGMENTAL VEIN TO ISOLATE A PARTICULAR SEGMEMT

***RELATION TO PULMONARY ARTERY** - BRANCHES OF PULMONARY ARTERY ACCOMPANY THE BRONCHI

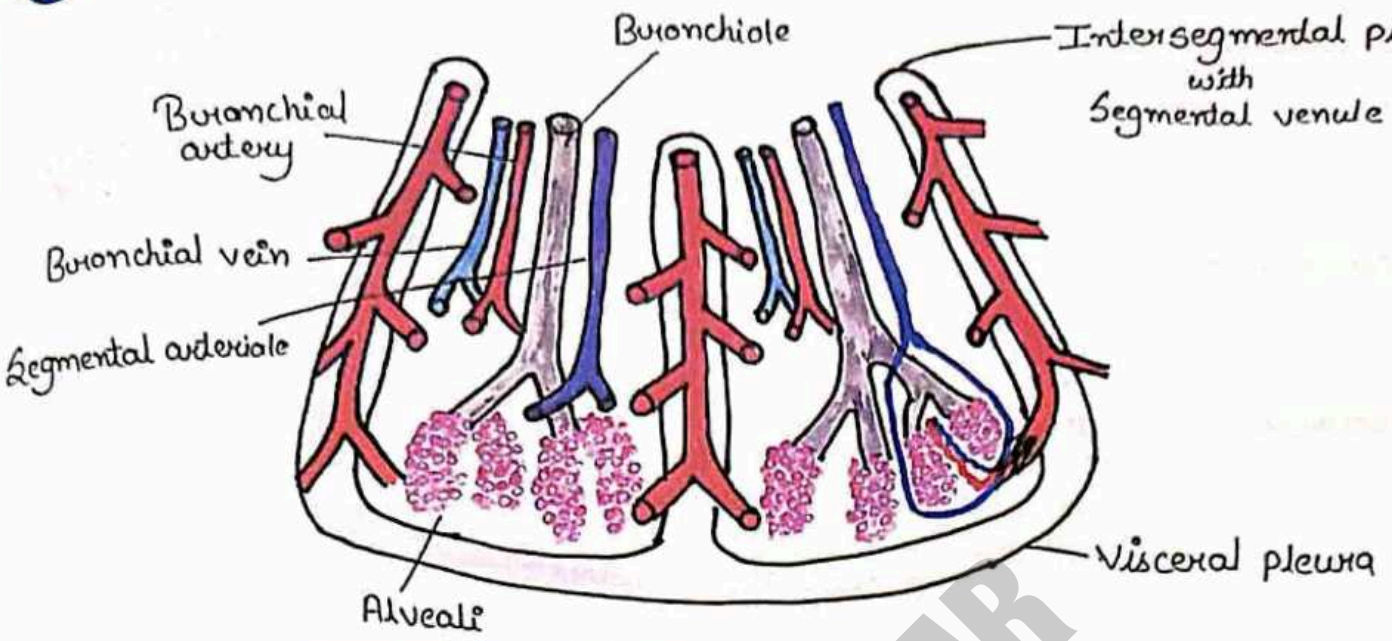
***RELATION TO PULMONARY VEIN** - PULMONARY VEIN DO NOT ACCOMPANY THE BRONCHI NOR PULMONARY ARTERIES
-RUN IN THE INTERSEGMENTAL PLANES

CLINICAL ANATOMY

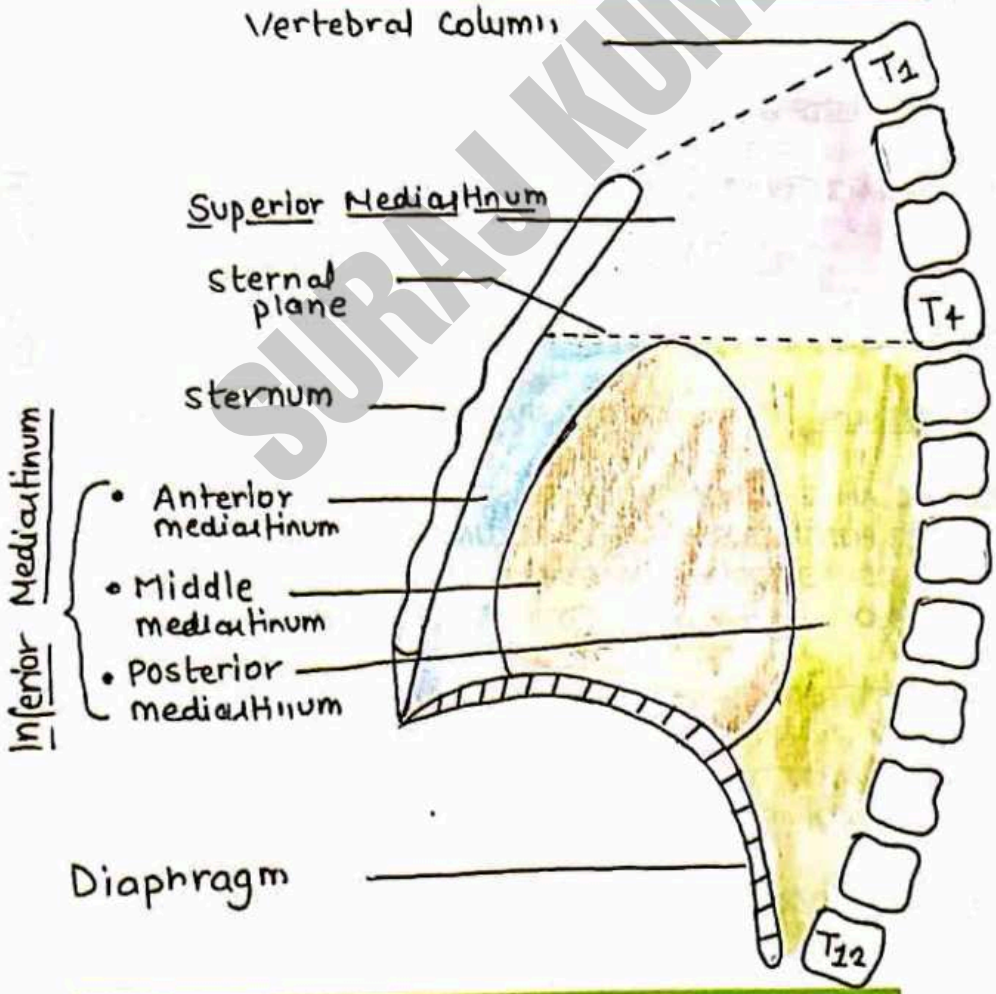
1.USUALLY, THE INFECTION OF A BRONCHOPULMONARY SEGMENT REMAINS RESTRICTED TO IT, ALTHOUGH TUBERCULOSIS & BRONCHOGENIC CARCINOMA MAY SPREAD FROM ONE SEGMENT TO ANOTHER.

2.KNOWLEDGE OF THE DETAILED ANATOMYOF THE BRONCHIAL TREE HELPS CONSIDERABLY IN :- SEGMENTARY RESECTION.

SUBAL KUMAR



BRONCHOPULMONARY SEGMENTS



MEDIASTINUM

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SUPERIOR MEDIASTINUM

*MEDIASTINUM - MIDDLE SPACE IN THORACIC CAVITY BETWEEN LUNGS
*MOST IMP CONTENT IS HEART ENCLOSE IN PERCARDIUM

*DIVISION :-SUPERIOR & INFERIOR MEDIASTINUM

*SUPERIOR MEDIASTINUM - SEPARATED FROM INFERIORLY BY IMAGINARY PLANE PASSING THROUGH STERNAL ANGLE ANTERIORLY & THORACIC VERTEBRAE POSTERIORLY

*BOUNDARIES :
- ANTERIORLY - MANUBRIUM STERNI
- POSTERIORLY - UPPER THORACIC 4 VERTEBRAE
- SUPERIORLY - PLANE OF THORACIC INLET
- INFERIORLY - AN IMAGINARY LINE
- ON EACH SIDE - MEDIASTINAL PLEURA

*CONTENTS :-
1. TRACHEA & OESOPHAGUS
2. MUSCLE - STERNOHYOID & STERNOTHYROID
3. ARTERIES - ARCH OF AORTA
   - BRACHIOCEPHALIC ARTERY
   - LEFT COMMON CAROTID ARTERY & SUBCLAVIAN ARTERY
4. VEIN - LETF & RIGHT BRACHIOCEPHALIC VEIN
5. NERVE - VAGUS , PHRENIC & CARDIAC NERVE OF BOTH SIDE
6. THORACIC DUCT & LYMPH NODES

INFERIOR MEDIASTINUM

*DIVIDED INTO - ANTERIOR, MIDDLE , POSTERIOR MEDIASTINUM

*ANTERIOR MEDIASTINUM - VERY NARROW SPACE INFRONT OF PERICARDIUM

*BOUNDARIES :-
1. ANTERIORLY - BODY OF STERNUM
2. POSTERIORLY - PERCARDIUM
3. SUPERIORLY - IMAGINARY LINE
4. ON EACH SIDE - MEDIASTINAL PLEURA

*CONTENTS :-
1. STERNOPERICARDIAL LIGAMENT
2. LYMPH NODE
3. LOWEST PART OF THYMUS
4. AREOLAR TISSUE

*MIDDLE MEDIASTINUM :- OCCUPIED BY PERICARDIUM & ITS CONTENTS.

SUNIL KUMAR

*BOUNDARIES :- 1.ANTERIORLY - STERNOPERICARDIAL LIGAMENT
2.POSTERIORLY - OESOPHAGUS
-DESCENDING THORACIC AORTA.
3.ON EACH SIDE - MEDIASTINAL PLEURA

*CONTENT :- 1.HEART ENCLOSED IN PERICARDIUM
2.ASCENDING AORTA
3.PULMONARY TRUNK
4.2 PULMONARY ARTERIES
5.VEINS - LOWER PART OF SUPERIOR VENACAVA
- TERMINAL PART OF AZYGUS VEIN
- RIGHT AND LEFT PULMONARY VEIN
6.NERVES - PHRENIC
- DEEP CARDIAC PLEXUS
7.LYMPH NODES
8.BIFURCATION OF TRACHEA

*POSTERIOR MEDIASTINUM :-

*BOUNDARIES :- ANTERIORLY - PERICARDIUM
- BIFURCATION OF TRACHEA
- PULMONARY VESSELS
- POSTERIOR PART OF THE UPPER SURFACE O THE DIAPHRAM.
-POSTERIORLY - LOWER 8 THORACIC VERTEBRAE & INTERVENING DISCS.
-ON EACH SIDE - MEDIASTINAL PLEURA.

*CONTENT :- 1.OESOPHAGUS
2.ARTERIES - DESCENDING THORACIC AORTA & ITS BRANCHES
3.VEINS - 1.AZYGOS VEIN
2.HEMIAZYGOS VEIN
3.ACCESSORY HEMIAZYGOS VEIN
4.NERVES - 1.VAGI
2.SPLANCHNIC NERVES (GREATER, LESSER)
3.8 THORACIC GANGLIA OF THE SYMPATHETIC CHAIN
5.LYMPH NODE

CLINICAL ANATOMY

1.MEDIASTINAL SYNDROME:-COMPRESSION OF MEDIASTINAL STRUCTURES BY ANY TUMOUR GIVES RISE TO A GROUP OF SYMPTOMS KNOWN AS MEDIASTINAL SYNDROME.

THE COMMON SYMPTOMS ARE AS FOLLOWS -

A. OBSTRUCTION OF SUPERIOR VENA CAVA GIVES RISE TO ENGORGEMENT OF VEINS IN THE UPPER HALF OF THE BODY.
B. PRESSURE OVER THE TRACHEA CAUSES DYSPNOEA, AND COUGH.
C. PRESSURE ON OESOPHAGUS CAUSES DYSPHAGIA.

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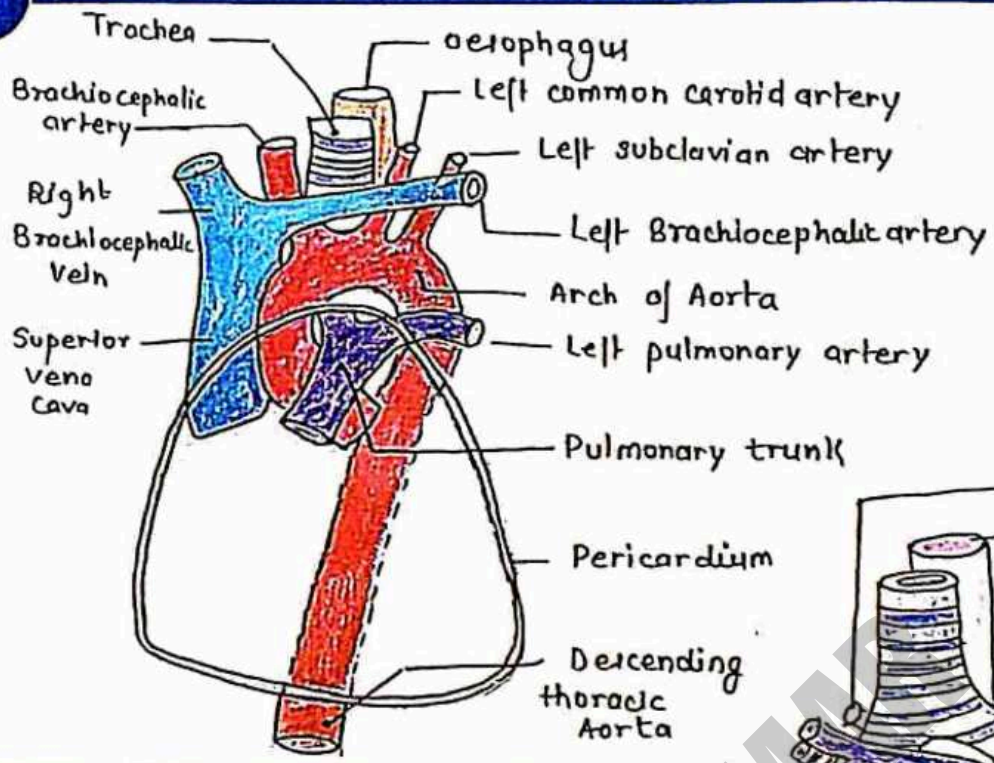
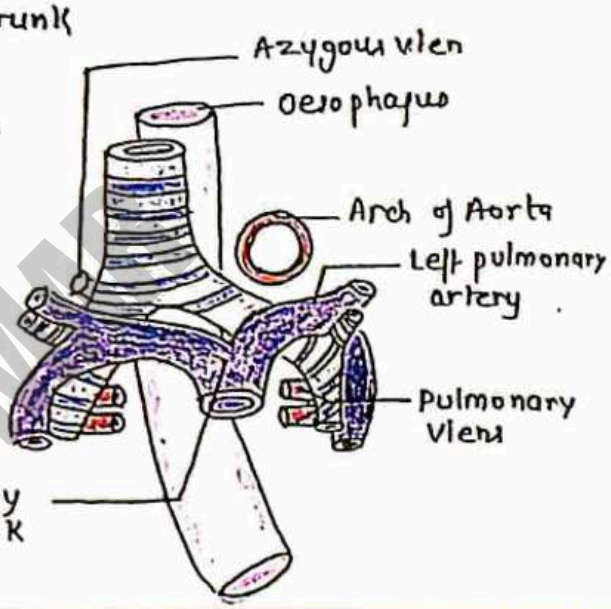
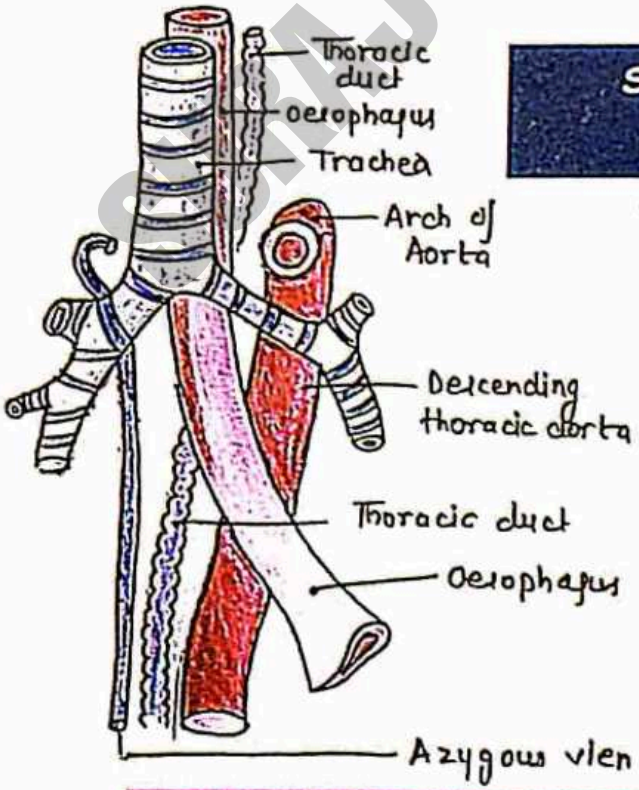


STRUCTURE IN SUPERIOR MEDIASTINUM



STRUCTURE IN MIDDLE MEDIASTINUM



STRUCTURE IN POSTERIOR MEDIASTINUM

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PERICARDIUM

*FIBROSEROUS SAC WHICH ENCLOSED HEART & ROOT OF GREAT VESSEL
*CONSISTS OF FIBROUS OF PERICARDIUM & SEROUS PERICARDIUM

1) FIBROUS PERICARDIUM :-
- CONICAL SAC MADE OF FIBROUS TISSUE
- PARIETAL LAYER OF SEROUS PERICARDIUM ATTACHED TO ITS DEEP SURFACE

*FEATURE :-
1. APEX IS BLUNT AND LIE AT LEVEL OF STERNAL ANGLE - FUSED WITH GREAT VESSEL.
2. BASE IS BROAD AND INSEPARABLY BLENDED WITH CENTRAL TENDON OF DIAPHRAGM
3. ANTERIOR CONNECTED TO STERNUM BY SUP. & INF. STERNOPERICARDIAL LIGAMENT
4. POSTERIOR RELATED TO PRINCIPAL BRONCHI, OESOPAGUS WITH NERVE PLEXUS
5. EACH SIDE BY →MEDIASTINAL PLEURA.

BLOOD SUPPLY :- FIBROUS & PARIETAL PARICARDIA-->SUPPLY BY INTERNAL THORACIC, MUSCULOPHRENIC ARTERY
VEIN DRAIN INTO CORRESPONDING VEIN.

NERVE SUPPLY :- FIBROUS AND PARIETAL ARE SUPPLIED BY -->PHRENIC NERVE
-SENSITIVE TO PAIN

SEROUS PERICARDIUM :-
-SEROUS PERICARDIUM IS THIN, DOUBLE LAYERED SEROUS MEMBRANE
-OUTER LAYER - PARIETAL PERICARDIUM FUSED - WITH FIBROUS PERICARDIUM
-INNER LAYER - UISCERAL PERICARDIUM FUSED TO HEART

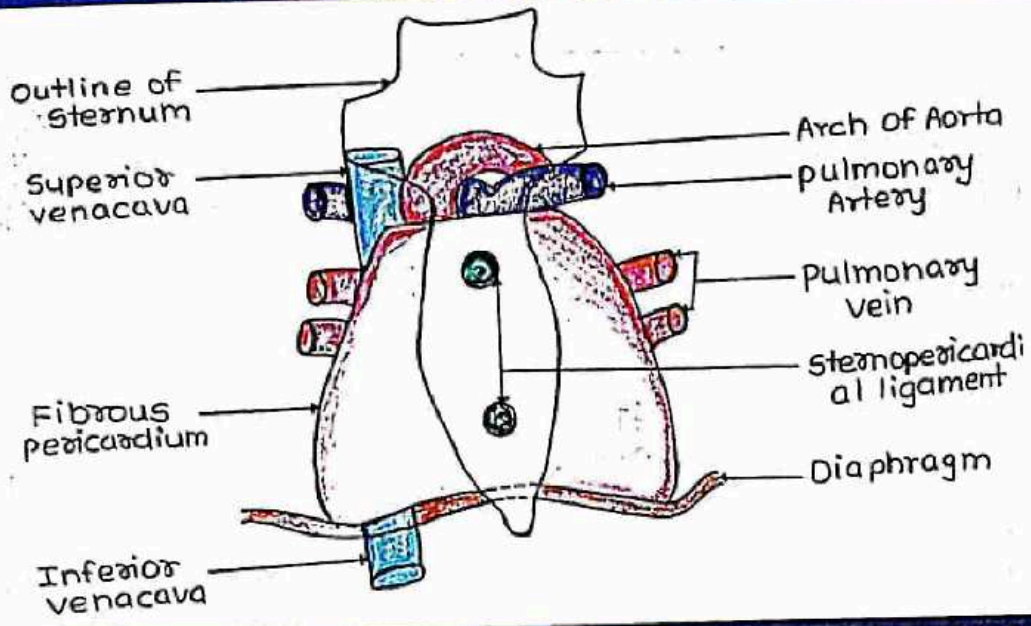
PERICARDIAL CAVITY-POTENTIAL SPACE - BETWEEN PARIETAL & VISCERAL PERCARDIUM
- CONTAIN ONLY THIN FILM OF SEROUS FLUID

2) SINUSES OF PERICARIUM :-
1. EPICARDIUM AT ROOT GREAT VESSEL ARRANGED IN FORM OF TWO TYPE -→
   -->ARTERIAL TUBE ENCLOSE -->ASCENDING AORTA & PULMONARY TRUNK
   --→ VENOUS TUBE ENCLOSE-->VENACAVA & PULMONARY TRUNK
2. PASSAGE BETWEEN TWO TUBES--->TRANSVERSE SINUS
3. DURING DEVELOPMENT TO BEGIN-→VEIN OF HEART CROWDED TOGETHER
4. HEART INCREASE IN SIZE & VEIN SEPARATE OUT
5. PERICARDIAL REFLECTION SURROUND ALL OF THEM & FORM OBLIOUS PERICARDIAL SINUS.

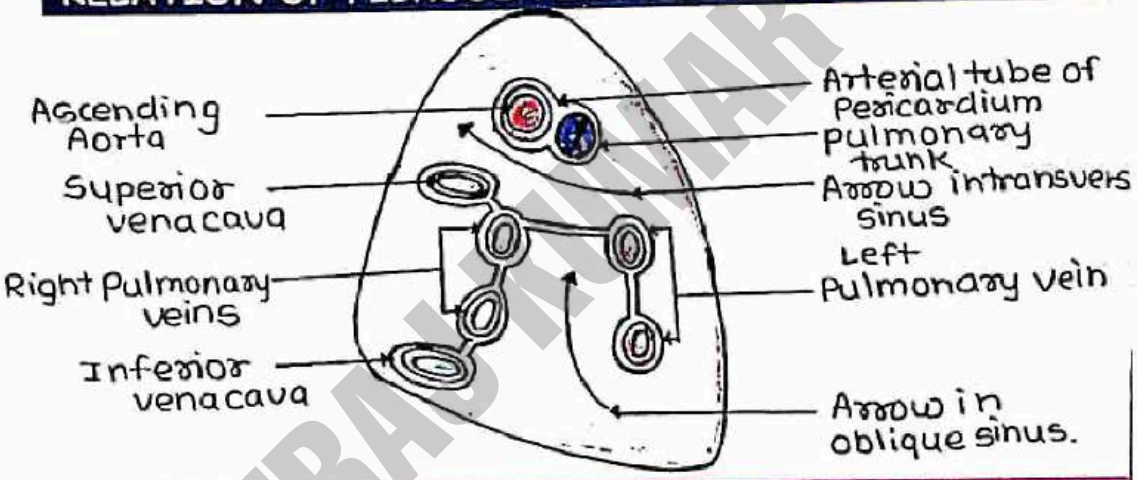
TRANSVERE SINUS :-BOUNDED - ANTERIORLY - ASCENDING AORTA,PULMONARY TRUNK
-POSTERIORLY - SUPERIOR VENACAVA

OBLIQUE SINUS :- BOUNDED - ANTERIORLY-LEFT ATRIUM
POSTERIORLY-PARIETAL PERICARDIUM OESOPHAGUS

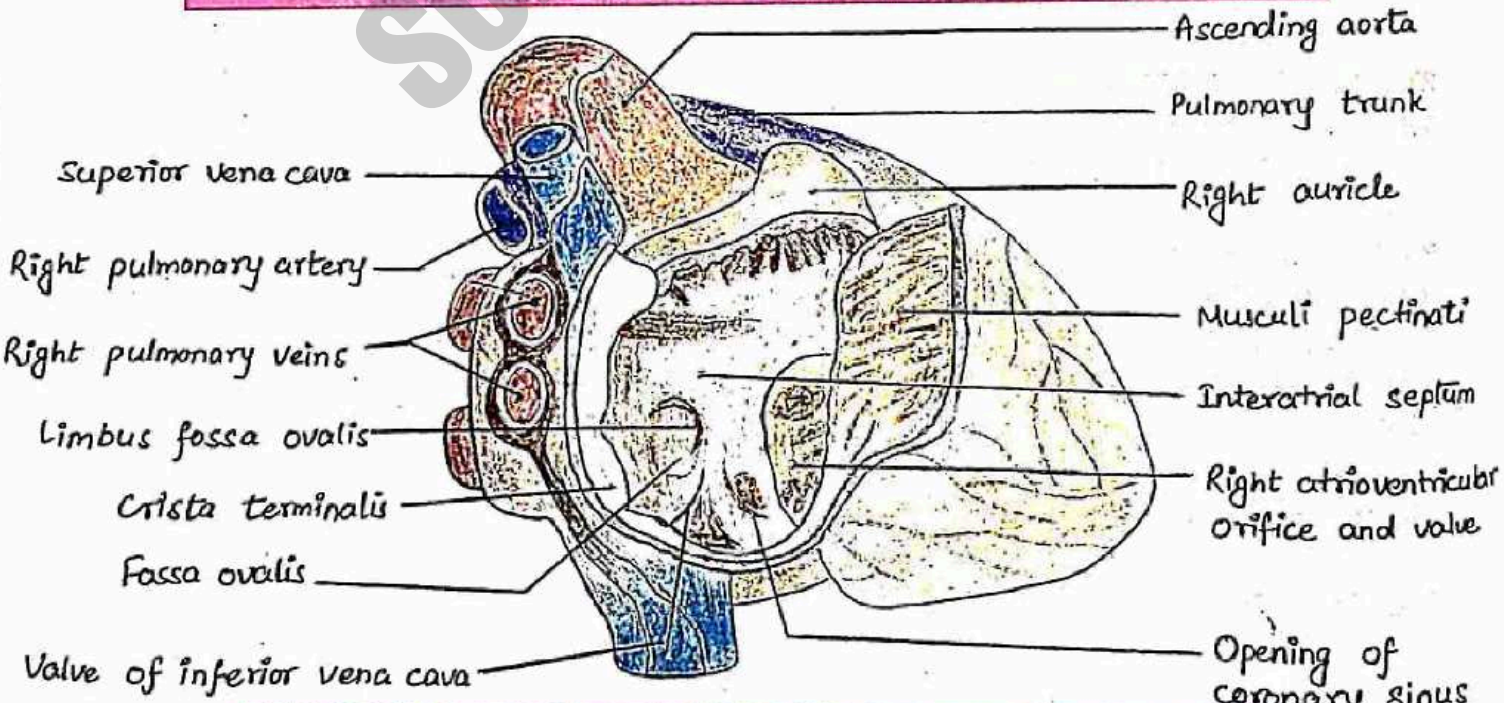
SUBAL KUMAR



RELATION OF FIBROUS PERCARDIUM TO STURNUM



SINUS OF PERICARDIUM



INTERIOR OF RIGHT ATRIUM

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RIGHT ATRIUM

*RIGHT UPPER CHAMBER OF HEART RECEIVE VENOUS BLOOD FROM BODY
*PUMP INTO RIGHT VENTRICLE THROUGH TRICUSPID VALVE

EXTERNAL FEATURE -

1. CHAMNER IS ELONGATED VERTICALLY, RECEIVING HE SUP. VENA CAVA AT THE UPPER END & INF. VENA CAVA AT THE LOWER END.

2. ALONG RIGHT BORDER OF ATRIUM--> SHALLOW VENTRICAL GROOVE WHICH PASSES FROM ---> SUPERIOR VENA CAVA TO INFERIOR VENA CAVA IS CALLED SULCUS TERMINALIS

3. IT PRODUCED BY INTERNAL MUSCULAR RIDGE CALLE CRISTA TERMINALIS

4. UPPER PART OF SULCUS CONTAINED S.A NODE

5. ANTRIOVENTRICULAR GROOVE SEPARATED RIGHT ATRIUM FROM RIGHT VENTRICLE
INLET OF RIGHT ATRIUM - 1. SUPERIOR VENA CAVA
2. INFERIOR VENA CAVA
3. CORONARY SINUS
4. ANTERIOR CARDIAC VEINS

INTERNAL FEATURES :- DIVIDED INTO 3 PARTS

1.SMOOTH PART - DEVELOPMENT - DRIVED FROM RIGHT HORN SINUS VENOSUS
- MOST OF TRIBUTARIES EXCEPT ANT. CARDIAC VEINS OPEN INTO IT
1.SUPERIOR VENA CAVA
2.INFERIOR VENA CAVA - EUSTACHIAN VALVE
3.CORONARY SINUS - THEBESIAN VALVE
4. VENAE CORDIS MINIMAE

2.ROUGH ANT.PART - DEVELOPMENT - DERIVED FROM PRIMITIVE ATRIAL CHAMBER
- IT PRESENT A SERIES OF MUSCULAR RIDGE CALLED MUSCULI PECTINATI

3.INTERATRIAL SEPTUM - DEVELOPMENT - DRIVED FROM PRIMITIVE ATRIAL CHAMBER
- IT PRESENT THE FOSSA OVALIS A SHALLOW DEPRESSION IN LOWER PART
- ANNULUS OVALIS & FOSSA OVALIS THE PROMINENT MARGIN FOSSA OVALIS

NOTE :- DIAGRAM OF RIGHT ATRIUM - PAGE NO.65

SURAJ KUMAR

ARTERIAL SUPPLY OF HEART

*HEART IS SUPPLY BY 2 CORONARY ARTERIRES -→ARISING FROM ASCENDING AORTA

RIGHT CORONARY ARTERY

POSITION - SMALLER THAN LEFT CORONARY ARTERY
- ARISE FROM ANT. AORTIC SINUS

COURSE - FIRST PASS FORWARD BETWEEN ROOT OF PULMONARY TRUNK & RIGHT AURICLE
- THEN RUN DOWNWARD - RIGHT ANTERIOR CORONARY SULCUS
- WINDS AROUND THE INFERIOR BORDER TO REACH - DIAPHRAGMATIC SURFACE OF HEART
- TERMINATE BY ANASTOMOSIS WITH LEFT CORONARY ARTERY

BRANCHES :-

1. ARTERIAL BRANCH - ANT., POST. & LATERAL
2. RIGHT CONUS ARTERY - FROM CIRCLE AROUND PULMONARY TRUNK CALLED 'ANNULUS OF VIEUSSENS'
3. VENTRICLAR BRANCH- ANT. & POST GROUP
4. RIGHT MARGINAL ARTERY
5. POSTERIOR INTERVENTRICULAR BRANCH - LIE IN POST. INTERVENTRICULAR GROOVE

AREA OF DISTRIBUTION :- RIGHT ATRIUM
- VENTRICLE - GREAT PART OF RIGHT VENTRICLE - EXCEPT AREA ADJOING INTERVENTRICULAR GROOVE
- SMALL PART OF LEFT VENTRICLE
- WHOLE CONDUCTING SYSTEM OF HEART EXCEPT A PART OF LEFT BRANCH OF AV BUNDLE.

LEFT CORONARY ARTERY

***POSITION** - ARISE FROM LEFT AORTIC SINUS

***COURSE** - FIRST RUN FORWARD TO LEFT END EMERGE BETWEEN PULMONARY TRUNK
-HERE IT GIVE THE INTERVENTRICULAR BRANCH WHICH RUN DOWNWARD IN GROOVE OF SAME NAME
-FUTHER CONTINUATION OF LCA IS CALLED CIRCUMFLEX ARTERY
-IT WIND AROUND LEFT BORDER OF HEART & CONTINUE IN LEFT CORONARY SULCUS
-TERMINATE BY ANASTOMOSIS WITH RCA

SURAJ KUMAR

BRANCHES :- 1.ANT. INTERVENTRICULAR BRANCH - LARGEST BRANCH
2.CIRCUMFLEX BRANCH IS TERMINATE PART OF LCA
3.ANT.POST VENTRICULAR BRANCH
4.ARTERIAL BRANCH - ANT.POST. LAT. GROOVE

AREA OF DISTRIBUTION :- 1.LEFT ATRIUM
2.VENTRICLE - GREATER PART OF LEFT VENTRICLE EXCEPT AREA ADJOINING POSTERIOR INTERVENTRICULAR GROOVE & SMALL PART OF RIGHT VENTRICLE OR ANT.PART OF INTERVENTRICULAR SEPTUM.
3.ANT.PART OF INTERVENTRICULAR SEPTUM.
4.LEFT BRANCH OF AV BUNDLE.

CARDIAC DOMINANCE :-
1.ARTERY GIVING POSTERIOR INTERVENTRICULAR BRANCH IS DOMINANT ARTERY
2.IN 90% RIGHT CORONARY ARTERY GIVE POSTERIOR INTERVENTRICULAR ARTERY
-→SUCH HEART CALLED RIGHT DOMINANCE
3.IN 10% OF CIRCUMFLEX ARTERY CONTINUATION OF LEFT CORONARY ARTERY
4.PROVIDE POSTERIOR INTERVENTRICULAR BRANCH SUCH HEART CALLED LEFT DOMINANCE HEART.

VEINS OF HEART

*THESE ARE GREAT CARDIAC VEIN, MIDDLE CARDIAC, RIGHT MARGINAL VEIN POSTERIOR VEIN OF LEFT VENTRICLE, ANTERIOR CARDIAC VEIN, VENAE CORDIS MINIMAE.

*ALL VEIN EXCEPT ---→ANT.CARDIAC VEIN OPEN INTO -→CORONARY SINUS

*CORONARY SINUS OPENS INTO --→RIGHT ATRIUM.

1.CORONARY SINUS :- LARGEST VEIN OF HEART
-SITUATED INTO LEFT POSTERIOR CORONARY SULCUS END INTO RIGHT ATRIUM

-TRIBUTRIES OF CORONARY SINUS -
1.GREAT CARDIAC VEIN - ACCOMPANY FIRST ANT.VENTRICULAR ARTERY THAN LEFT CORONARY ARTERY
2.MIDDLE CARDIAC VEIN - ACCOMPANY - POSTERIOR INTERVENTRICULAR ARTERY
3.SMALL CARDIAC VVEIN - ACCOMPANY - RIGHT CORONARY ARTERY
4.RIGHT MARGINAL VEIN - ACCOMPANY - MARGINAL BRANCH OF RIGHT CORONARY ARTERY
5.POST.VEIN OF LEFT VENTRICLE - RUN ON DIAPHRAGMATIC SURFACE ON LEFT VENTRICLE
6.OBLIQUE VEIN OF LEFT ATRIUM - RUNNING OF POST.SURFACE LEFT ATRIUM

2.ANT.CARDIAC VEIN :- 3 TO 4 SMALL VEIN WHICH RUN PARALLEL TO EACH OTHER ON THE ANT.WALL OF RIGHT VENTRICLE.
-OPEN DIRECTLY INTO RIGHT ATRIUM.

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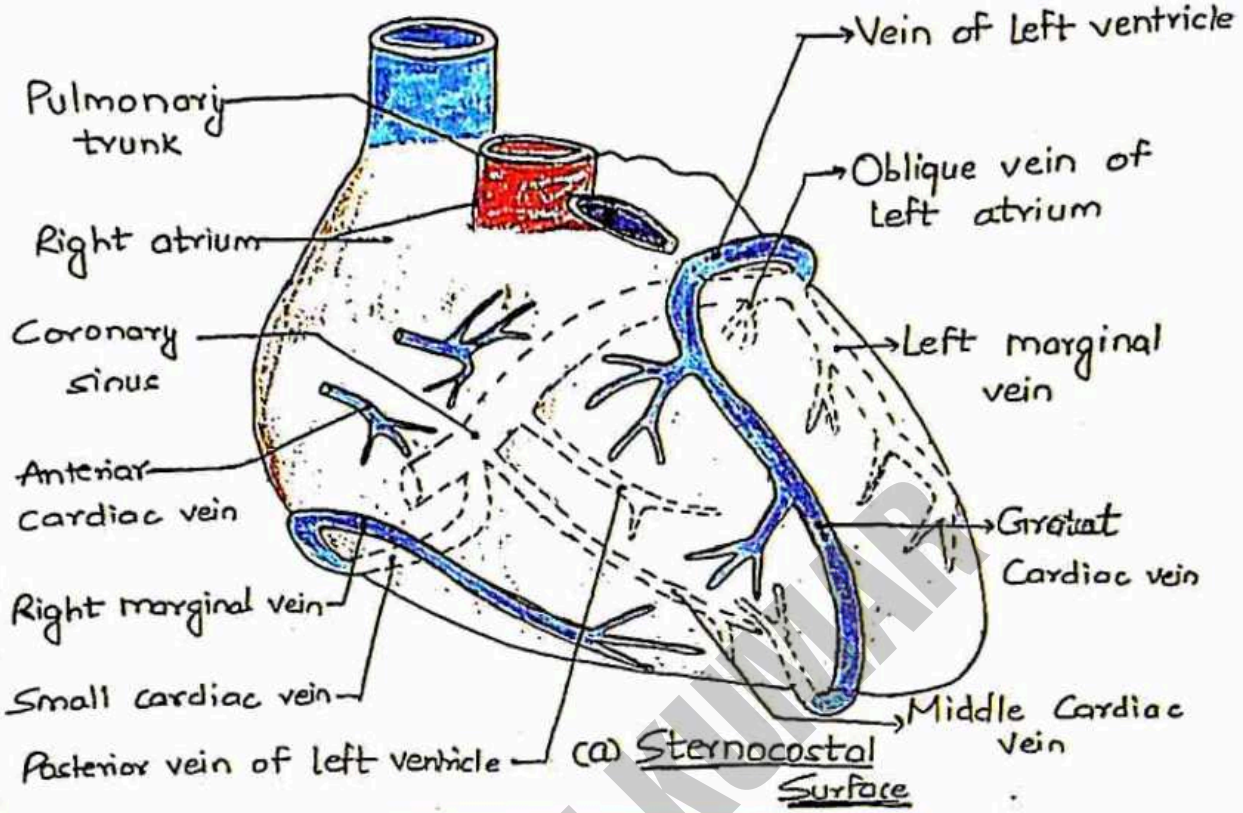
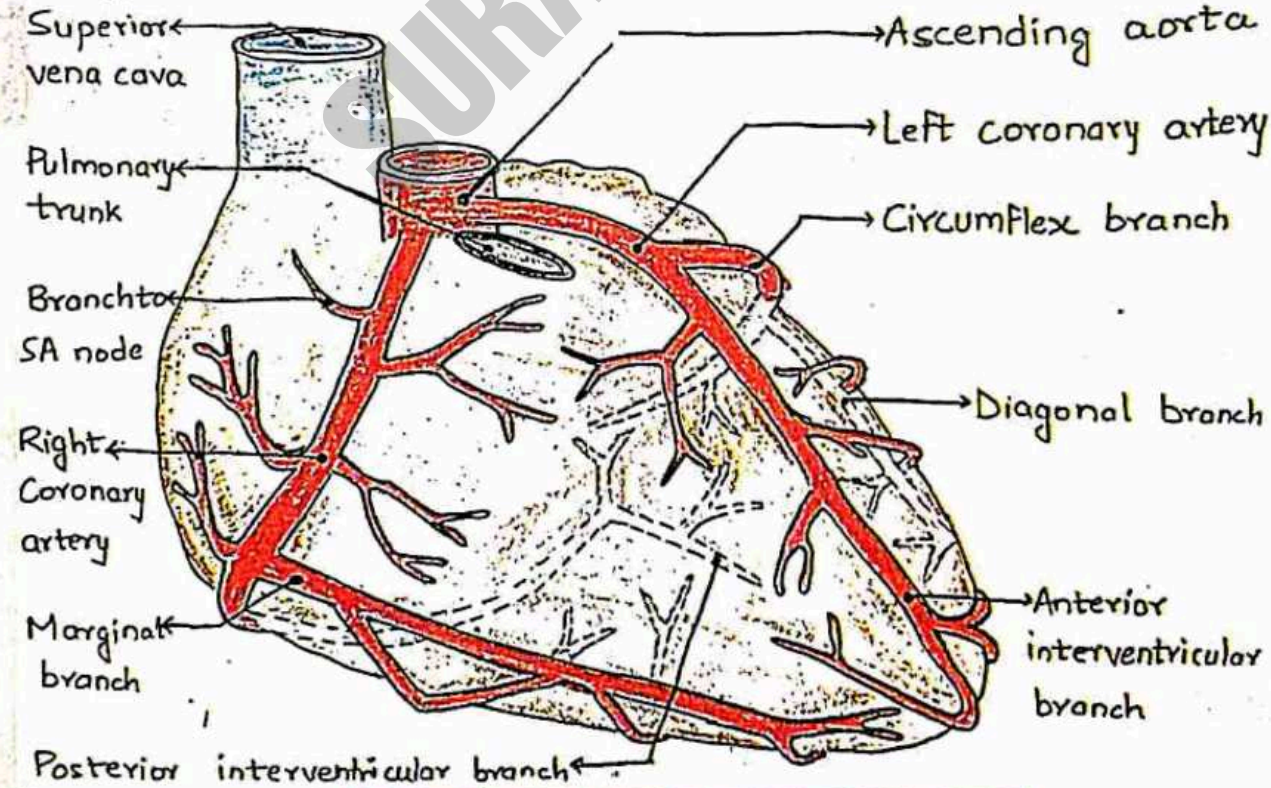
3. VENA CORDIS MINIMAE :- NUMEROUS SMALL VALVE LESS VEIN PRESENT IN ALL FOUR CHAMBER OF HEART --→ DIRECTLY OPEN RIGHT ATRIUM.



VEINS OF HEART



ARTERY OF HEART

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AZYGOS VEIN

*THE AZYGOS TERM MEANS UNPAIR.
*AZYGOS VEIN DRAIN THORACIC WALL & THE UPPER LUMBAR REGION.
*FROM IMPORTANT CHANNEL CONNECTING IVC.
*VEIN OCCUPIES UPPER PART OF POST. WALL OF ABDOMINAL WALL & THE POST.MEDIASTINUM.

*FORMATION :- FORM BY UNION OF LUMBAR AZYGOS
- RIGHT SUB COASTAL
- ASCENDING LUMBAR VEIN.

*TRIBUTARIES :-

1.RIGHT SUPERIOR INTERCOASTAL VEIN-→ FORM BY UNION OF 2ND, 3RD,4TH POST.INTERCOASTAL VEIN.
2.5TH TO 8TH POST.INTERCOASTAL VEIN.
3.HEMIAZYGOS---→AT LEVEL OF LOWER BORDER OF 8 THORACIC VERTEBRAE.
4.ACCESSORY HEMIAZYGOS VEIN - AT THE LEVEL OF UPPER BORDER OF 8TH THORACIC VERTEBRA
5.RIGHT BRONCHIAL VEIN, NEAR THE TERMINAL END OF AZYGOS VEIN.
6.SEVERAL OESOPHAGEAL, MEDIASTINAL, PERICARDIAL VEINS.

ARCH OF AORTA

*CONTINUATION OF ASCENDING AORTA
*SITUATED IN MEDIASTINUM.

*COURSE :-
1.BEGIN BEHIND THE UPPER BORDER OF THE 2ND RIGHT STERNOCHONDRAL JOINT.
2.RUN UPWARD, BACKWARD & BIFURACTION OF TRACHEA.
3.IT THUS ARCHES OVER ROOT OF LUNG.
4.ENDS AT LOWER BORDER OF BODY OF 4TH THORACIC VERTEBRAE.

*BRANCHES :-
- BRACHIOCEPHALIC ARTERY ---→DIVIDES RIGHT COMMON CAROTID RIGHT SUBCLAVIAN ARTERIES
- LEFT COMMON CAROTID ARTERY
- LEFT SUBCLAVIAN ARTERY

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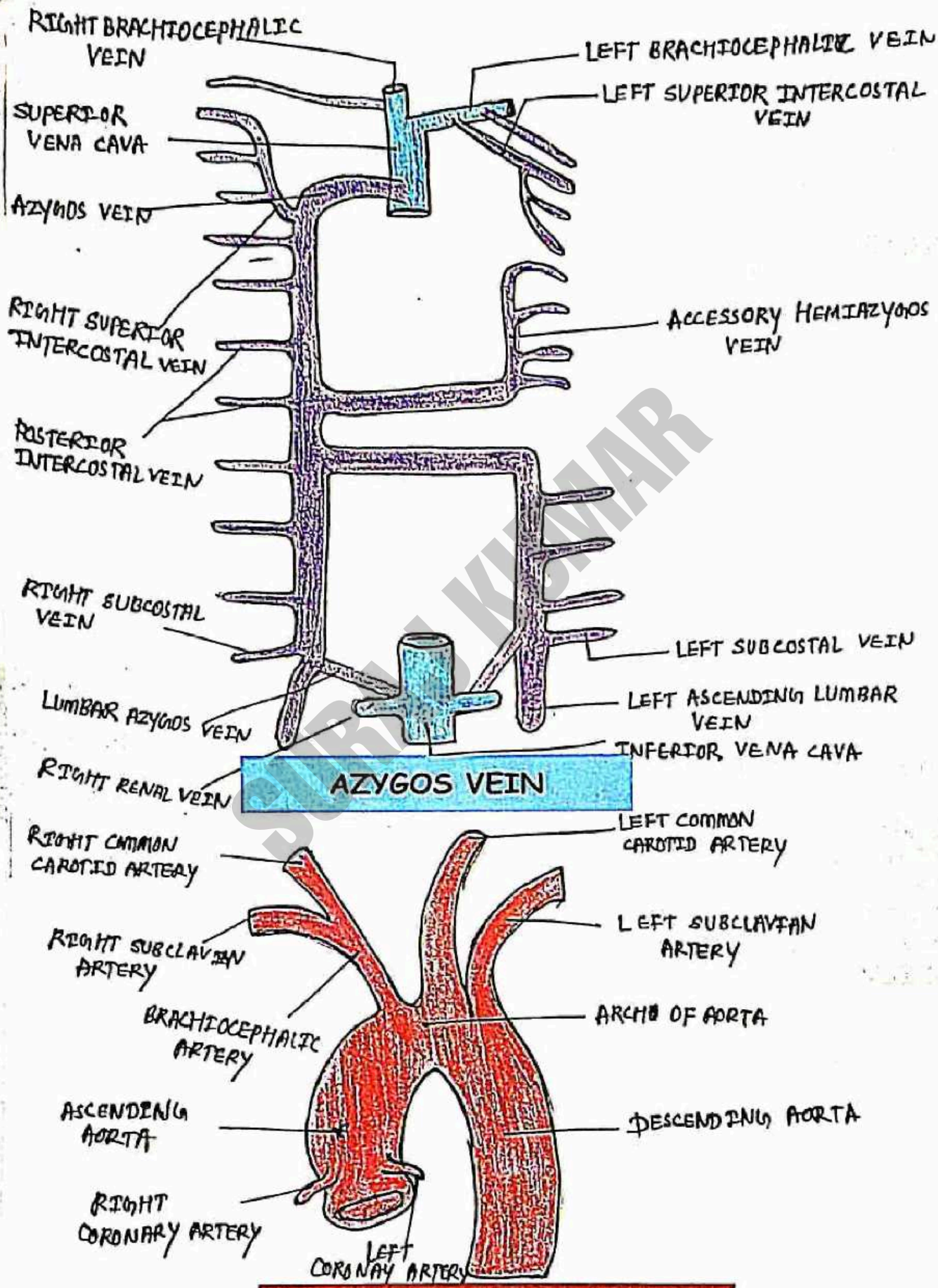
RIGHT BRACHIOCEPHALIC VEIN
LEFT BRACHIOCEPHALIC VEIN
LEFT SUPERIOR INTERCOSTAL VEIN
SUPERIOR VENA CAVA
AZYGOS VEIN
RIGHT SUPERIOR INTERCOSTAL VEIN
ACCESSORY HEMIAZYGOS VEIN
POSTERIOR INTERCOSTAL VEIN
RIGHT SUBCOSTAL VEIN
LEFT SUBCOSTAL VEIN
LUMBAR AZYGOS VEIN
LEFT ASCENDING LUMBAR VEIN
INFERIOR VENA CAVA
RIGHT RENAL VEIN
AZYGOS VEIN
LEFT COMMON CAROTID ARTERY
RIGHT COMMON CAROTID ARTERY
LEFT SUBCLAVIAN ARTERY
RIGHT SUBCLAVIAN ARTERY
ARCH OF AORTA
BRACHIOCEPHALIC ARTERY
ASCENDING AORTA
DESCENDING AORTA
RIGHT CORONARY ARTERY
LEFT CORONARY ARTERY
ARCH OF AORTA

SURYA KUMAR

THORACIC DUCT

*LARGEST LYMPHATIC VESSEL OF THE BODY
*EXTEND FROM UPPER PART OF ABDOMEN TO LOWER PART OF NECK
*CROSS POST.BORDER SUP.PART OF MEDIASTINUM

*COURSE :-

1. BEGINS AS A CONTINUATION OF THE UPPER END OF THE CISTERNA CHYLI NEAR LOWER BORDER -→12 THORACIC VERTEBRAE
2. ENTER THORAX THROUGH AORTIC OPENING OF DIAPHRAM
3. THEN ASCEND THROUGH POST.MEDIASTINUM FROM LEVEL OF 12^{TH} - 5^{TH} THORACIC VERTEBRAE.
4. THEN IT COURSE THROUGH SUP.MEDIASTINUM
5. IN NECK - ARCH LAT. AT LEVEL OF 7^{TH} CERVICAL VERTEBRAE
6. AND BY OPENING INTO ANGLE OF JUNCTION BETWEEN LEFT SUBCLAVIAN & LEFT INTERNAL JUGULAR VEIN.

*RELATION :-

-AT AORTIC OPENING OF DIAPHRAM - ANTERIORLY-DIAPHRAM
-POSTERIORLY-VERTEBRAL COLUMN
-TO THE RIGHT - AZYGOS VEIN
-TO THE LEFT - AORTA

-IN THE POST.MEDIASTINUM - ANTERIORLY - DIAPHRAGM
-OESOPHAGUS
-RIGHT PLEURAL RECESS
-POSTERIORLY - VERTEBRAL COLUMN
-RIGHT POST.INTERCOASTAL ARTERIES
TO THE RIGHT - AZYGOS VEIN
TO THE LEFT - DESCENDING AORTA

-IN SUP.MEDIASTINUM - ANTERIORLY - ARCH OF AORTA
-POSTERIORLY - VERTEBRAL COLUMN
-TO THE RIGHT - OESOPHAGUS
-TO THE LEFT - PLEURA

-IN THE NECK - THE THORACIC DUCT FORMS AN ARCH RISING ABOUT 3-4CM ABOVE THE CLAVICLE
-ANTERIORLY - LEFT COMMON CAROTID ARTERY
- LEFT VAGUS
- INTERNAL JUGULAR VEIN
-POSTERIORLY - VERTEBRAL ARTERY
- SYMPATHETIC TRUNK
- LEFT PHRENIC NERVE

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*TRIBUTARIES :- THORACIC DUCT RECEIVE LYMPH FROM BOTH HALF OF BODY BELOW DIAPHRAM & LEFT HALF ABOVE ABDOMEN.
- IN THORAX THORACIC DUCT RECEIVE - LYMPH VESSEL FROM POST.MEDIASTINAL NODE & SMALL INTERCOASTAL NODE.
- LEFT JUGULAR TRUNK, LEFT SUBCLAVIAN TRUNK END IN THORACIC DUCT
- LEFT BRONCHOMEDIASTINAL TRUNK - DRAIN LYMPH FROM LEFT HALF OF THORAX & END IN THORACIC DUCT .

RIGHT HALF ABOVE DIAPHRAGM :-
-RIGHT SIDE THERE IS RIGHT LYMPHATIC DUCT.-→INTO WHICH RIGHT BRONCHIOMEDIASTINAL, RIGHT JUGULAR, RIGHT SUBCLAVIAN LYMPH DRAIN.
-RIGHT LYMPHATIC TRUNK END IN RIGHT BRONCHIOCEPHALIC VEIN.

NOTE :- DIAGRAM OF THORACIC DUCT - PAGE NO.74

INTERCOSTAL NERVE

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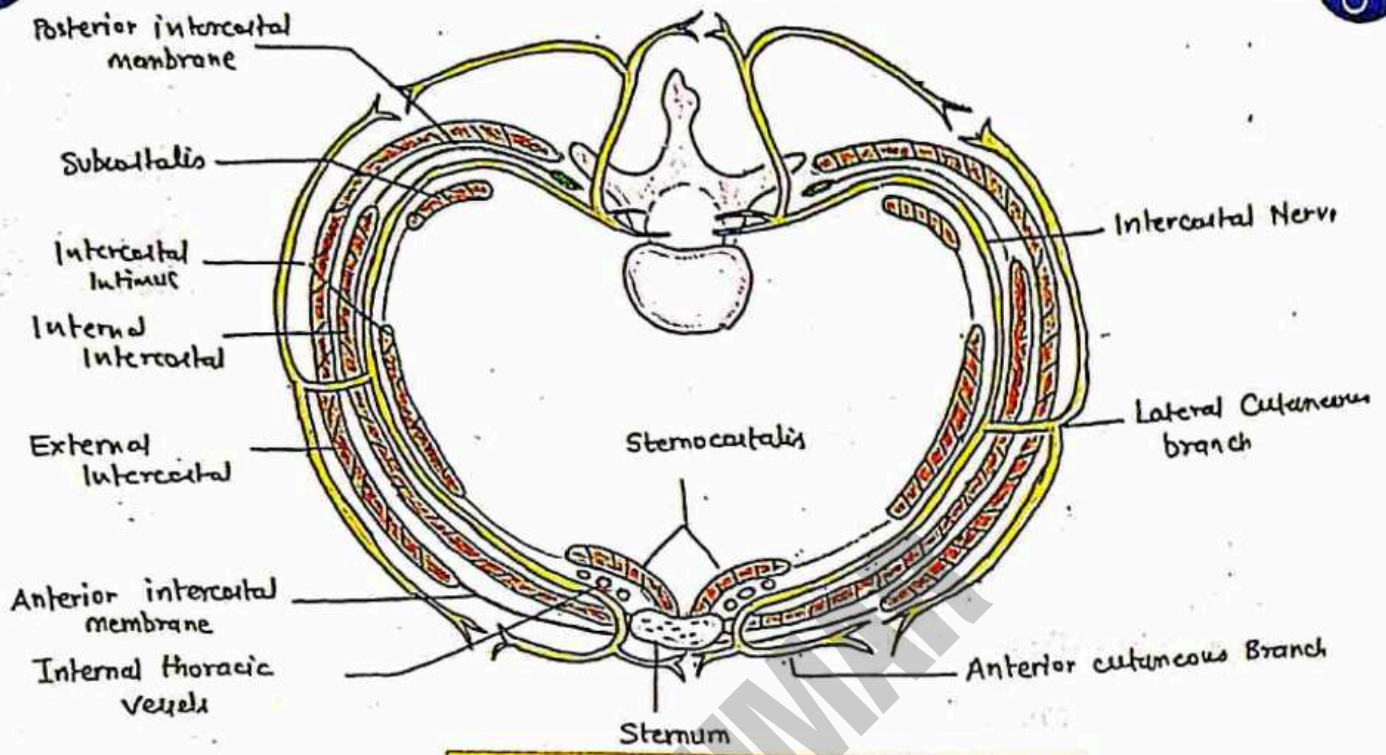
*ALL INTERCOSTAL NERVE - SUPPLY INTERCOSTAL MUSCLE

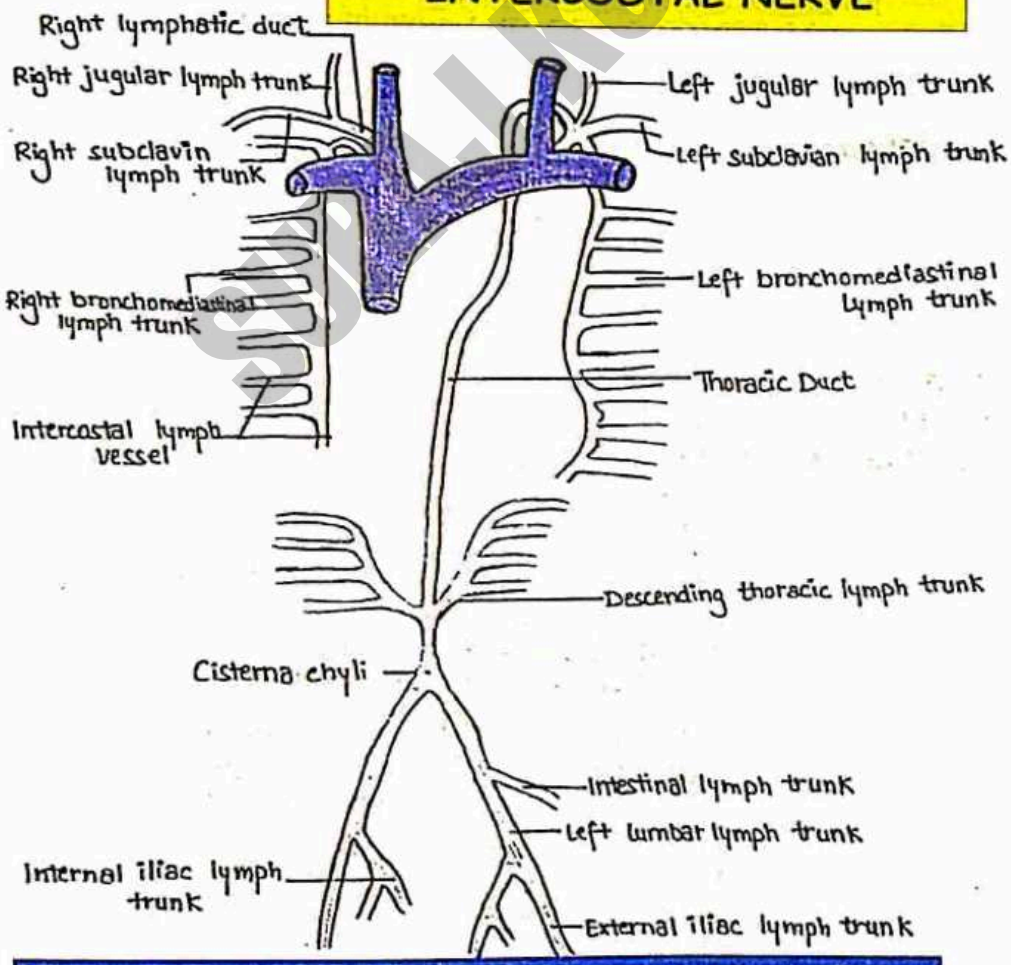
INTERCOSTAL NERVE :-
-INTERCOASTAL NERVE ARE ANT.PRIMARY RAMI OF THORACIC 1-12 SPINAL NERVE
-THE ANT.PRIMARY RAMI OF 12 THORACIC NERVE - FORM SUBCOASTAL NERVE.
-IN ADDITIONAL TO SUPPLY INTERCOASTAL SPACE -
-UPPER 2 INTERCOASTAL NERVE - ALSO SUPPLY UPPER LIMB MUSCLE
-LOWER 5 - SUPPLY ABDOMINAL WALL ALSO CALLED THORACO ABDOMINAL NERVE
-3-6 SUPPLY ONLY THORACIC WALL THEY ARE CALLED TYPICAL INTERCOASTAL NERVE.

*COURSE :- INTERCOASTAL NERVE RUN IN COASTAL GROOVE-→END NEAR STERNUM

*BRANCHES :- MUSCULAR BRANCH - NUMEROUS MUSCULAR BRANCH SUPPLY INTERCOASTAL MUSCLE - TRANSVERSUS THORACIS
-SENSORY BRANCH - MAIN BRANCH & COLLATERAL BRANCH ALSO SUPPLY PARIETIAL PLEURA & PERIOSTEUM OF RIB
-LAT.CUTANEOUS BRANCH ARISE-NEAR ANGLE OF RIB WHERE IT PAIR INTERCOASTAL MUSCLE & DISTRIBUTED TO THE SKIN.
-ANT.CUTANEOUS BRANCH - EMERGE ON THE SIDE OF STERNUM -SUPPLY OVERLYING SKIN.
-COMMUNICATING BRANCH - EACH NERVE IS CONNECTED TO THORACIC SYMPATHETIC GANGLION BY A DISTAL PLACE WHITE & PROXIMAL GREY RAMUS COMMUNICANS.



INTERCOSTAL NERVE



TRIBUTARIES OF THORACIC DUCT

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INTERCOASTAL SPACE

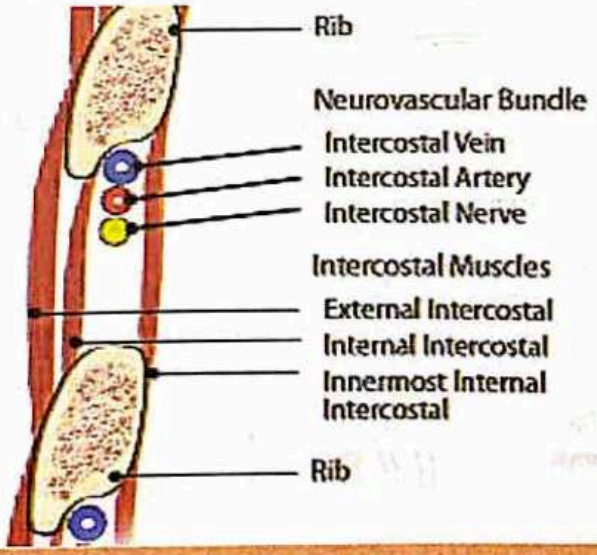
*GAP BETWEEN RIBS ARE CALLED INTERCOASTAL SPACE
- FILL BY INTERCOASTAL MUSCLE & CONTAIN INTERCOASTAL NERVE & VESSEL-
*9-INTERCOASTAL SPACE - ANTERIORLY
*11 INTERCOASTAL SPACE POSTERIORLY

INTERCOASTAL MUSCLE - 1.EXTERNAL INTERCOASTAL MUSCLE
2.INTERNAL INTERCOASTAL MUSCLE
3.TRANSVERSUS THORACIS - DIVIDED INTO 3 PARTS- SUBCOSTALIS, INTERCOSTALIS INTIMI, STERNOCOSTALIS

*EXTEND
1.EXTERNAL INTERCOASTAL - POST. - TUBERCLE OF RIB
- ANT. - COSTOCHONDRAL JUNCTION
*ANTERIORLY - BETWEEN COSTOCHONDRAL JUNCTION AND STERNUM
REPLACED BY EXTERNAL INTERCOASTAL MEMBRANE
*POSTERIORLY - END OF MUSCLE IS CONTINUOUS WITH POSTERIOR FIBRES
→SUPERIOR COSTO TRANSVERSE LIGAMENT
*DIRECTION OF FIBRE - DOWNWARD, FORWARD & MEDIALLY IN FRONT

2.INTERNAL INTERCOASTAL - ANT. - LATERAL BORDER OF STERNUM-->CONTINOUS
-->SUPERIOR COSTO LIGAMENT
- POST. - ANGLE OF RIB BEYOND POSTERIOR INTERCOASTAL MEMBRANE
*DIRECTION OF FIBRE - DOWNWARD, BACKWARD, LATERALLY

3.SUBCOSTALIS - POSTERIOR PART OF LOWER INTERCOASTAL SPACE
4. INTERCOSTALIS - MIDDLE 2/4 OF ALL INTERCOSTAL SPACE
5.STERNOCOSTALIS - ANTERIOR PART OF UPPER INTERCOSTAL SPACE



SECTION THROUGH INTERCOSTAL SPACE

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INLET OF THORAX

*NARROW UPPER END OF THORAX - CONTINUOUS WITH THE NECK, KIDNEY SHAPED.

*BOUNDARIES - ANT. - UPPER BORDER OF MANUBRIUM STERNI
-POST. - SUPERIOR SURFACE OF 1ST THORACIC VERTEBRA
-EACH SIDE ON 1ST RIB

*PLANE OF INLET IS DIRECTED DOWNWARDS AND FORWARD WITH AN OBLIQUITY OF 45 DEGREE.

*PARTITION AT INLET OF THORAX :-

1.PARTITION IN 2 HALVES----→RIGHT & LEFT --→WITH CLEFT BETWEEN
2.EACH HALF IS COVERED BY FASCIA - SIBSON FASCIA OR SUPRAPLEURAL MEMBRANE
3.MEMBRANE - TRIANGULAR IN SHAPE
-APEX ----→ATTACHED TO TIP OF TRANSVERSE PROCESS OF 7TH CERVICAL VERTEBRA
-BASE ----→INNER BORDER OF 1ST RIB
4.SIBSON FASCIA--→IS FLATTENED TENDON OF SCALENUS MINIMUS

STRUCTURE PASSING THROUGH THE INLET OF THORAX

1.VISCERA - TRACHEA
- OESOPHAGUS
- APICES OF LUNGS WITH PLEURA
- THYMUS

2.LARGE VESSELS - BRACHIOCEPHALIC ARTERY ON RIGHT SIDE
- LEFT COMMON CAROTID ARTERY -→ON LEFT SIDE
- LEFT SUBCLAVIAN ARTERY--→ON LEFT SIDE
- RIGHT & LEFT BROCHIOCEPHALIC VEIN

3.SMALLER VESSELS - RIGHT & LEFT INTERNAL THARACIC ARTERY
- RIGHT & LEFT SUPERIOR INTERCOSTAL ARTERIES
- RIGHT & LEFT FIRST POSTERIOR INTERCOSTAL VEINS
- INFERIOR THYROID VEINS

4.NERVES - RIGHT & LEFT PHRENIC NERVE, VAGUS NERVE, SYMPATHETIC TRUNKS

5.MUSCLES - STERNOHYOID
- STERNOTHYROID
- LONGUS COLLI

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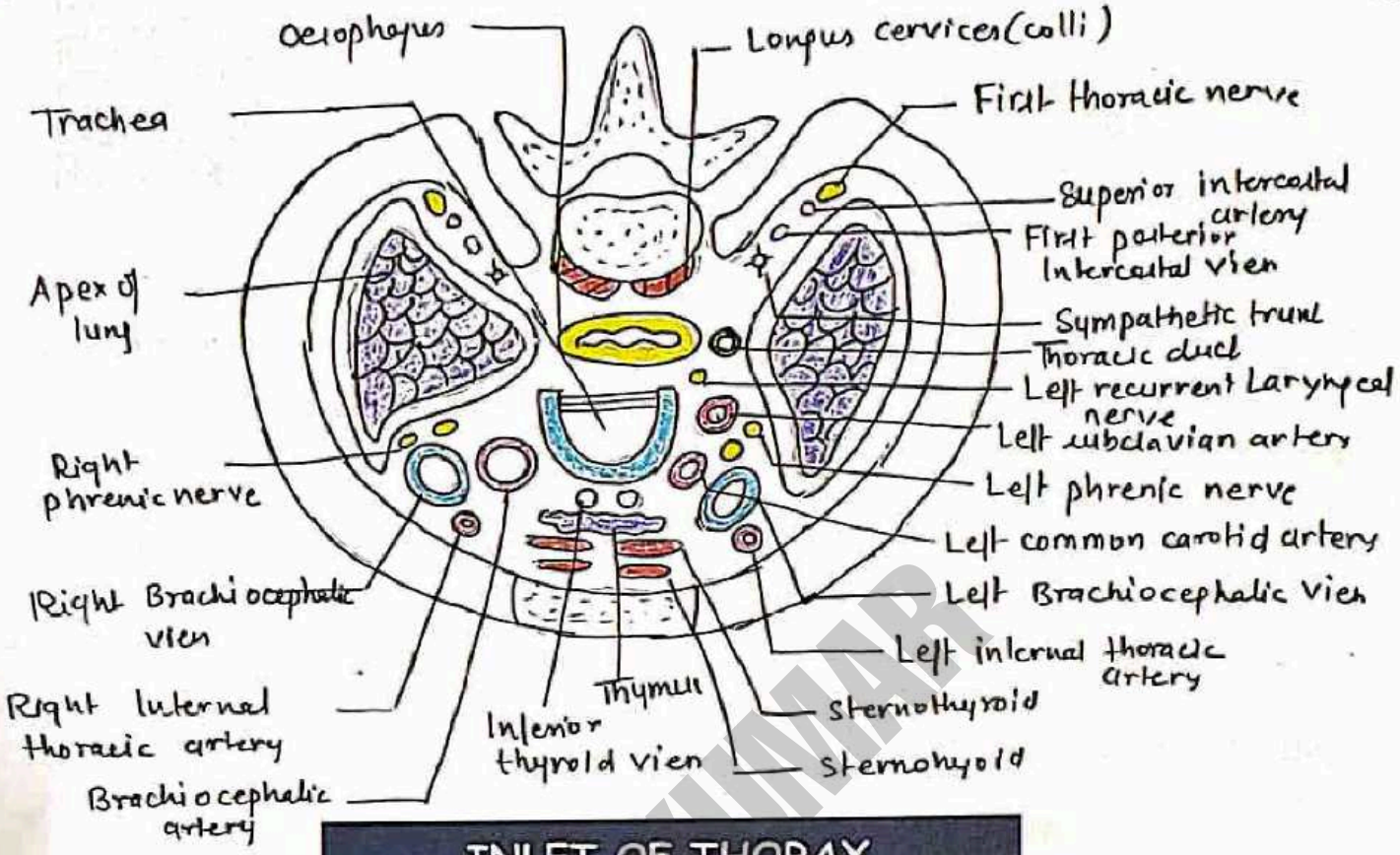
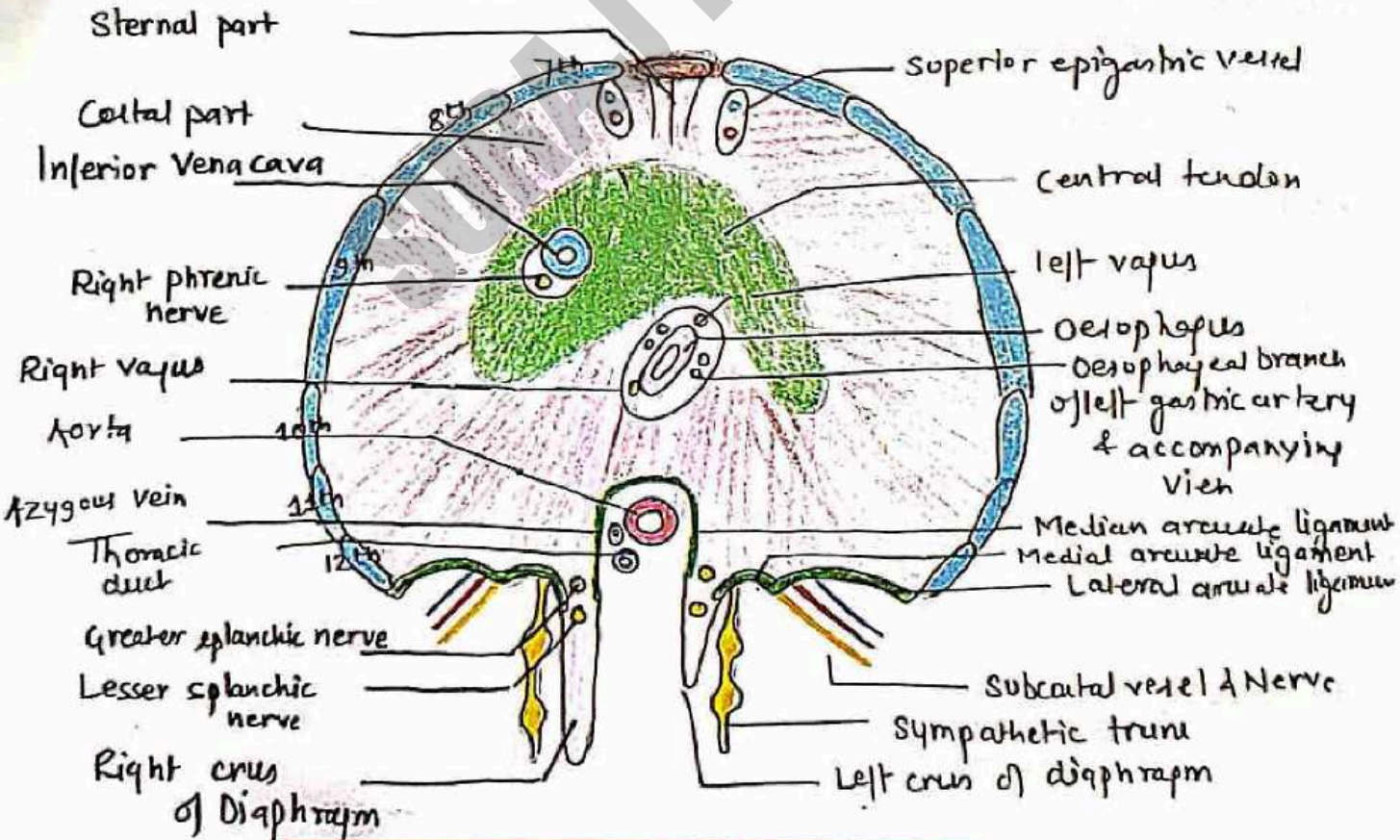


INLET OF THORAX



OUTLET OF THORAX

SUNIL KUMAR

STUDY TIPS FOR MAXIMUM MARKS

1.HOW TO WRITE IN UNIVERSITY EXAM AND HOW TO DRAW DIAGRAMS ALREADY MENTION IN UPPER LIMB (PAGE NUMBER - 25)

PREVIOUS YEAR UNIVERSITY QUESTIONS

*LONG QUESTIONS - 20 MARKS

1.DESCRIBE BRONCHOPULMONARY SEGMENTS WITH ITS APPLIED

2.DESCRIBE BLOOD SUPPLY OF HEART

3.DESCRIBE RIGHT ATRIUM UNDER FOLLOWING HEADINGS
1.EXTERNAL FEATURES 2.INTERNAL FEATURES 3.DEVELOPMENT

*SHORT NOTE - 5MARKS
1.ROOT OF LUNGS
2.RECESSES OF PLEURA
3.SUPERIOR MEDIA STINUM
4.INFERIOR MEDIA STINUM
5.AZYGUS VEIN
6.THORACIC DUCT
7.INTERCOASTAL NERVE
8.INTERCOASTAL SPACE
9.DRAW THE DIAGRAM OF INLET OF THORAX
10.DRAW THE DIAGRAM OF OUTLET OF THORAX

SUBAJ KUMAR

THORAX ANATOMY IS COMPLETED !!

WORK HARD TO HAVE A GOOD RESULT

ALL THE BEST FOR YOUR EXAMS

DR.JOHARI

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MCQ THORAX

1) Manubriosternal joint is a :
a synovial joint
b primary cartilaginous joint
c secondary cartilaginous joint
d syndesmosis

2) The inferior margin of pleura in the midaxillary line extends up to:
a 10th rib
b 9th rib
c 8th rib
d 7th rib

3) The pleura is supplied by all the following arteries except:
a intercostal arteries
b internal thoracic artery
c pulmonary artery
d bronchial arteries

4) The anterior most structure at the hilum of lung is:
a Pulmonary vein
b bronchial artery
c bronchus
d pulmonary artery

5) The structure that arches above the hilum of right lung is:
a superior venacava
b arch of aorta
c brachiocephalic vein
d azygous vein

6) Hypertrophy of following structures may compress oesophagus :
a left atrium
b right atrium
c left ventricle
d right ventricle

7) All the following are in right atrium except :
a azygous vein
b SVC
c IVC
d coronary sinus

8) Superior mediastinum shows all the following except:
a trachea
b ascending aorta
c arch of aorta
d vagus nerve

9) The venacaval opening foramen in the diaphragm lies at the level of:
a T8
b T12
c T10
d C7

10) The most superficial structure in the thoracic inlet is :
a vagus nerve
b thoracic duct
c superior venacava
d left subclavian artery

11) The most superficial structure in the thoracic inlet is :
a vagus nerve
b superior venacava
c right subclavian artery
d thoracic duct

12) which passes through the diaphragm with the oesophagus:
a azygous vein
b right vagus
c thoracic duct
d phrenic nerve

13) Which muscle is not used in forced expiration :
a transversus abdominis
b rectus abdominis
c diaphragm
d external oblique

14) Accessory muscle of inspiration includes all except:
a latissmus dorsi
b scalene muscle
c sternocleidomastoid
d quadratus lumborum

15) the posterior mediastinum contains all except :
a thoracic aorta
b oesophagus
c azygous vein
d sympathetic trunk

16) The oesophageal opening in the diaphragm transmits:
a azygous vein
b thoracic duct
c vagus nerve
d right phrenic nerve

17) Which doesn't drain into cardiac sinus :
a great cardiac vein
b anterior cardiac vein
c small cardiac vein
d oblique vein if rt atrium

18) Which muscle is not used in inspiration :
a transversus thoracis
b lattismus dorsi
c erector spinae
d pectoralis major

1)C 2)A 3)C 4)A 5)D 6)A 7)A 8)B 9)A 10)C 11)B 12)B 13)C 14)A 15)D 16)C 17)B 18)A

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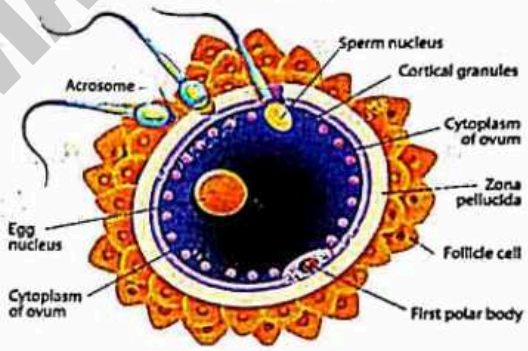
FIRST WEEK OF DEVELOPMENT

*In the first week - fertilization takes place in ampulla of fallopian tube
*zona pellucida - a membrane to prevent implantation and polyspermy.

The sperm binds to zona pellucida
↓
Triggers the release if acrosomal enzyme called - Acrosomal reaction
(Acrosin, acid phosphatase, hyaluronidase - acrosomal enzyme)
↓
The sperm penetrates the zona pellucida
↓
The cortical granules release its secretion called - Cortical reaction
↓
The cortical enzymes reach the zona
↓
Zona changes permeability of sperms called - zona reaction
↓
Fusion of male and female pronucleus
↓
Forms zygote
↓
undergoes 1st cleavage - 2 cells stage
↓
and cleavage - 4 cells stage
↓
3rd cleavage - 8 cells stage.



TIME AND SITE OF EVENTS

1.Fertilization- takes place within 12 to 24 hours of ovulation.
2.One day after fertilization- 2 cell stage.
3.Third day after fertilization- 8 to 16 cells stage [morula].
4.Fourth day after fertilization- 32 cells [advanced morula]--->Enters uterine cavity.

5. uterine fluid enters and advanced morula forms blastocyst.

6.At the time of formation - blastocyst contains 58 cells:
*53 cells in outer cell mass forms trophoblast - forms placenta.
*5 cells in inner cell mass forms embryoblast-forms embryo proper

7. Blastocyst is formed on 4 1/2 days after fertilization.
8. 5th day after fertilization zona pellucida disappears.
9. 6th to 7th after fertilization implantation occurs.

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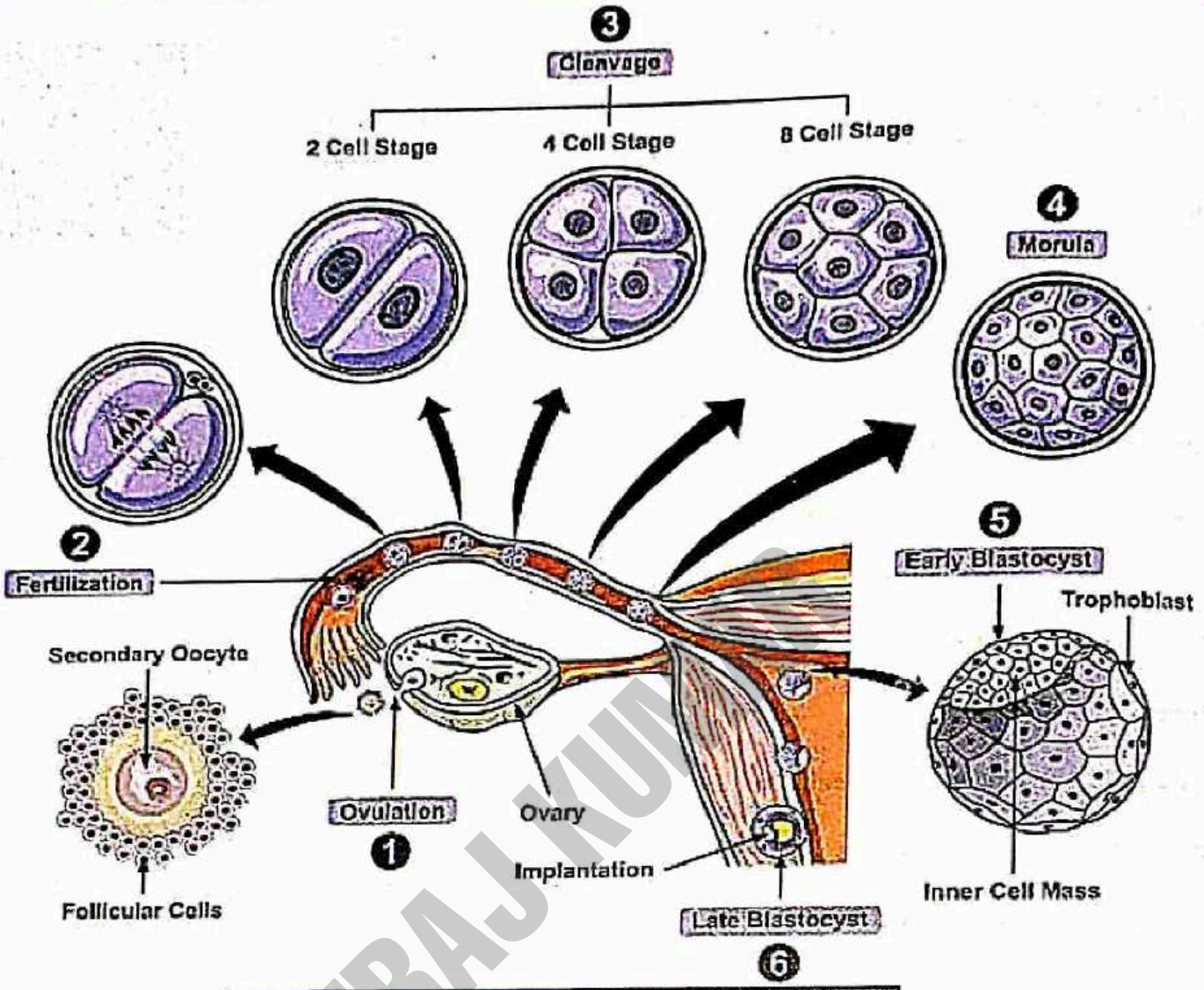
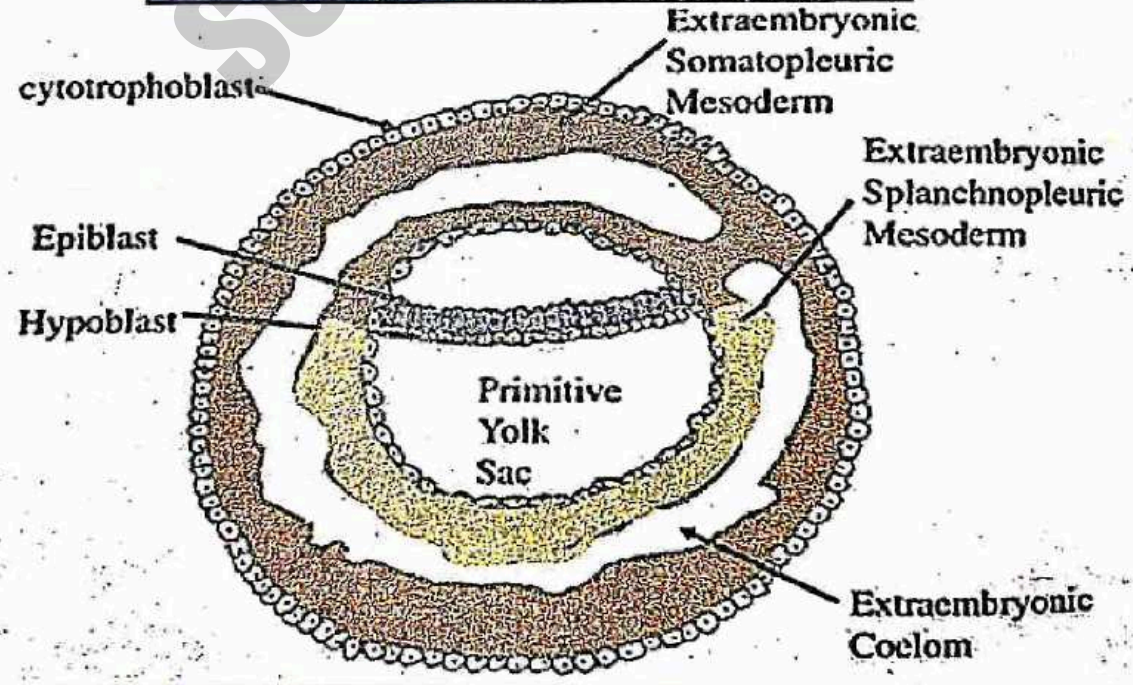


FIRST WEEK OF DEVELOPMENT



SECOND WEEK OF DEVELOPMENT

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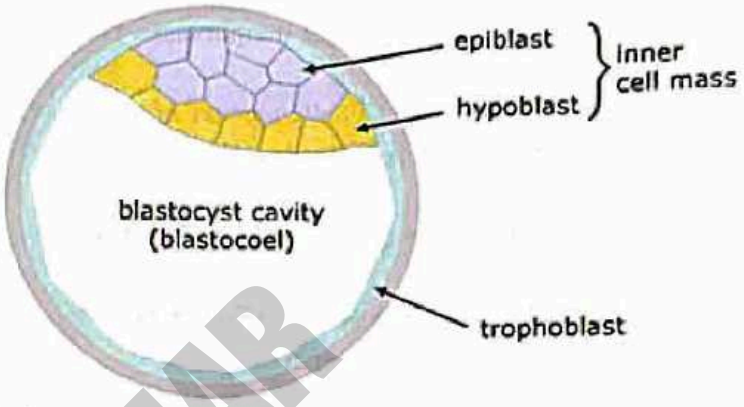
SECOND WEEK OF DEVELOPMENT

*Development of embryoblast and trophoblast :-

-The inner cell mass - forms embryoblast.
-The embryoblast forms two layers - Columnar cells & cuboidal cell

1.Columnar cells
↓
-Epiblast cells
↓
Formed on dorsal end
↓
Encloses amniotic cavity



2.Cuboidal cells -→ Hypoblast cell
-Hypoblast cells
↓
Formed on ventral end -→forms Exocoelom membrane
↓
It is flattened cell---→Encloses the primary yolk sac

*Formation of secondary yolk sac and umbilical cord :-

-The small cavity formed in the Eem joins to form a larger cavity.
-The undivided part of Eem (extra embryonic mesoderm) - cranial to amniotic cavity
↓
forms connecting stalk -→which becomes umbilical cord.

-The cavity in Eem divides it into:
↓
Splanchnopleuric layer of Eem- surrounding the yolk sac makes the yolk sac smaller in size
Called the secondary yolk sac.

*Summary of second week development: -

1.Two layers of inner cell mass - Epiblast
-Hypoblast.

2. Two layers of outer cell mass - Cytotrophoblast
-Syncytiotrophoblast.

3. Amnion and chorion are formed.
4.Two layers of eem somatic and visceral layer.
5.Yolk sac divided into primary and secondary yolk sac.

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THIRD WEEK OF DEVELOPMENT

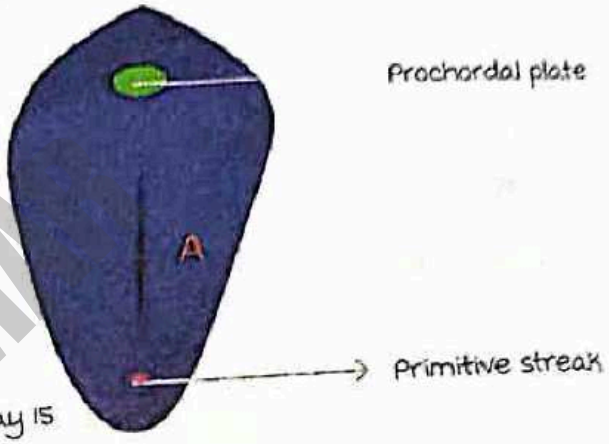
*GASTRULATION & NEURULATION OCCUR IN 3RD WEEK
- GASTRULATION - FORMATION OF GERM LAYER
- NEURULATION - FORMATION OF NEURAL TUBE

*GASTRULATION :-

*Prochordal plate: columnar hypoblast cell
-From Buccopharyngeal membrane- ruptures & forms oral cavity

1. Head end :-
-Primitive streak: Proliferation of epiblast.
-Primitive streak indicates the beginning of gastrulation.

2. Tail end :-
-End of and week/beginning of 3rd week



1.Ist germ layer to develop -→Endoderm
2. 2nd germ layer: mesoderm
↑
derived from primitive streak
↑
derived from epiblast

3.Last germ layer: Ectoderm
-The remains of epiblast after the formation of intraembryonic mesoderm.
-All germ layers derived from epiblast > yolk sac

NOTOCHORD DEVELOPMENT

*Introduction →midline structure
- Develop lying Cranial end Primitive streak & Caudal end prochordal plate

* Stage -

1. Cranial end primitive streak -→thick-→called Primitive knot

2. Depression appear-→center of primitive knot called primitive pit

3. Cell of primitive knot multiply-→passing between ectoderm & endoderm to prochordal plate called notochordal process

4. Cavity of promotive pot extend into -→ notochordal process called notochordal canal

5. Cell of notochordal canal -→intercalated in cell of endoderm and break down ,this stage form communication of amniatic cavity & yolk sac

6. Wall of canal flatted -→notochordal plate

7. Cell proliferation-→flat to curve -→notochord

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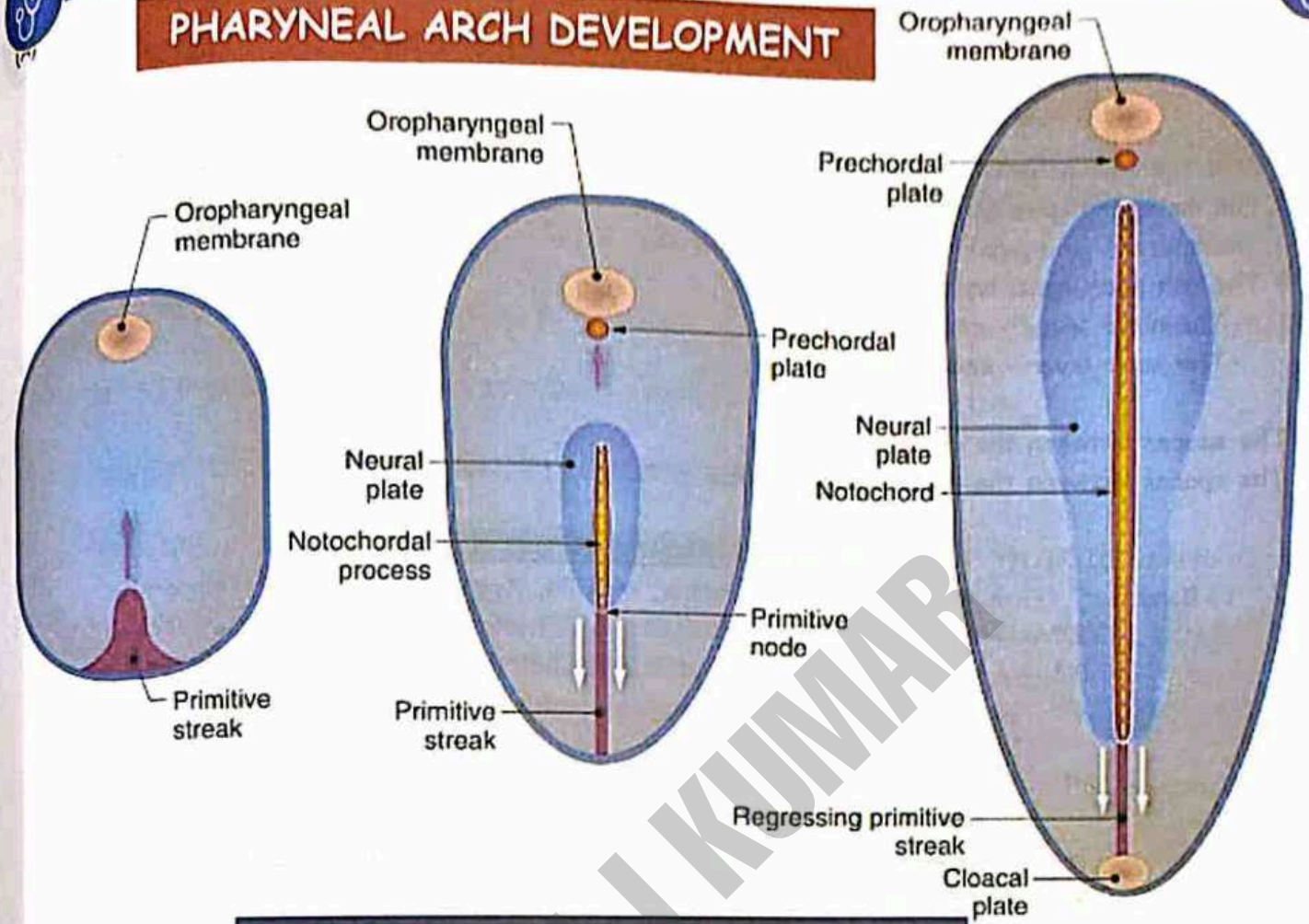
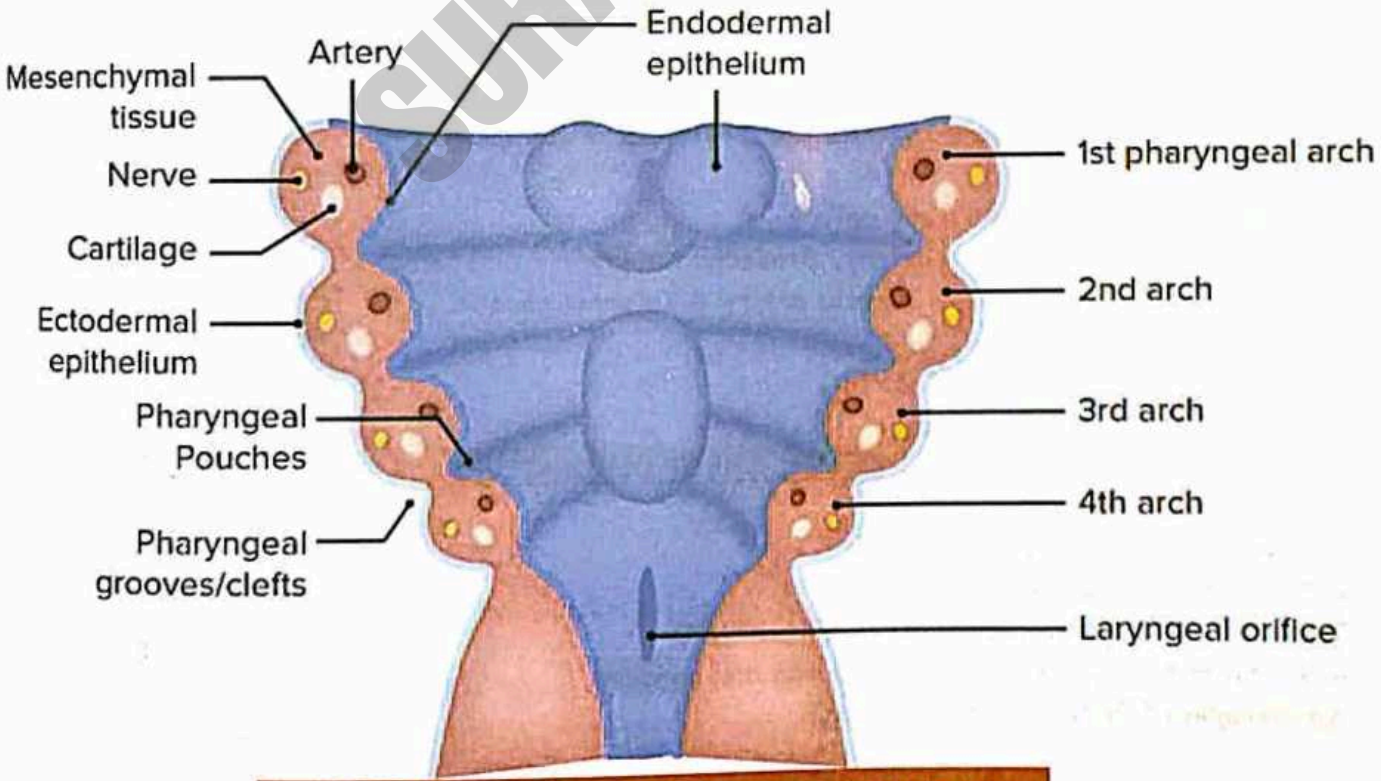
PHARYNEAL ARCH DEVELOPMENT



NOTOCHORD DEVELOPMENT



PHARYNEAL ARCH DEVELOPMENT

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PHARYNEAL ARCH DEVELOPMENT

1.It is the mesenchymal condensation.
2.The mesenchyme is derived from the NCC, paraxial & lateral mesoderm.
3.There are 5 pharyngeal arches - I, II, III, IV, V, VI.
4.The 5th pharyngeal arch disappears :-
-The outer layer - ectoderm
-The inner layer - endoderm

The spaces between the ectoderm - cleft.
The spaces between the endoderm - pouches

PHARYNGEAL ARCH DERIVATIVES

1. Bone - 1st ---→ Malleus, Incus, Mandible, Maxilla, Zygomatic, Palatine, Temporal
2nd ---→Stapes, Styloid, lesser cornu of hyoid, Upper body of hyoid
3rd ---→Greater cornu of hyoid & lower half of body of hyoid
4th ---→Thyroid cartilage, Arytenoid cartilage, cuneiform cartilage

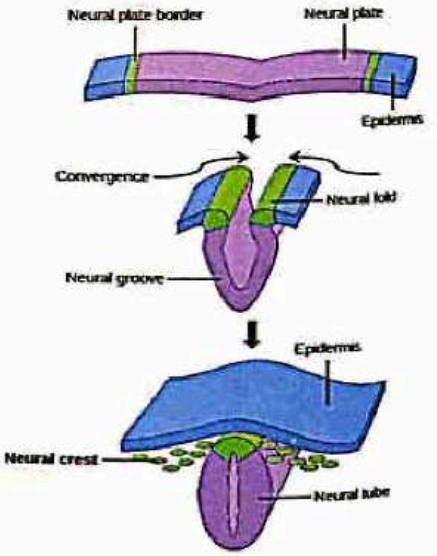
2.Muscle- 1st ---→Muscle of Mastication, Tensor tympani, Tensor plate, Anterior belly Digastric muscle
2nd --→Stapedius, Stylohyoid, Posterior belly of Digastric, Facial muscle
3rd---→Stylopharyngeus
4th --→larynx muscle

3.Nerve - 1st ----→ Trigeminal nerve - maxillary & mandibular
2nd ----→ Facial Nerve
3rd ----→ Glossopharyngeal nerve
4th ----→ Superior laryngeal branch of vagus

4.Artery- 1st-----→ Maxillary artery
2nd ----→ Hyoid artery, Stapedial artery
3rd----→ Common carotid artery & Internal carotid artery
4th ----→Arch of aorta

NEURAL CREST DERIVATIVES

1.Neuron out side CNS
2.Dorsal root of ganglion
3.Sensory ganglion of cranial nerve 5th , 7th ,10
4.Sympathetic & parasympathetic neuron and ganglion
5.Neuroglia of PNS---→ satellite cell



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THYROID DEVELOPMENT

1. THYROID GLAND--→ DEVELOP FROM --→ THYROGLOSSAL DUCT

2. PARAFOLLICULAR CELL --→DERIVED FROM -→CAUDAL PHARYNGEAL COMPLEX

3. MIDLINE OF MANDIBULAR ACHES-→SEPARATED BY MIDLINE SWELLING -→ TUBERCULUM IMPAR → BEHIND DEPRESSED SURFACE→ TO FORM THYROGLOSSAL DUCT

4. SITE OF ORIGIN OF DIVERTICULUM -→CALLED FORAMEN CAECUM

5. DIVERTICULUM GROW DOWN -→ MIDLINE INTO NECK

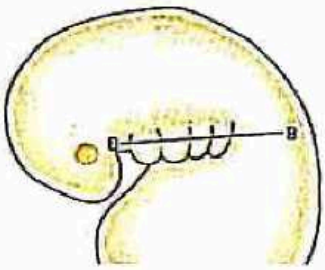
6. PROLIFERATION OF DIVERTICULUM CELL -→ FORM 2 LOBES OF THYROID GLAND

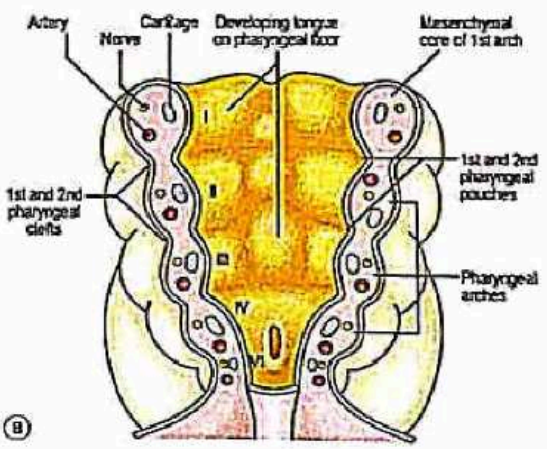
7. THYROID COME--→ INTIMATE RELATIONSHIP WITH CAUDAL PHARYNGEAL COMPLEX----→FUSE TO FORM→ PARAFOLLICULAR CELL

TONGUE DEVELOPMENT

*Development of tongue

1. Epithelium of the tongue : - Anterior 2/3rd- surface ectoderm.
   -Posterior 1/3rd- endoderm.
2. All the muscles are derived from myotome of occipital somites [Except palatoglossus CIV pharyngeal arch].
3. Nerve supply of the tongue
   -Anterior 2/3rd tongue : mandibular nerve-lingual nerve.
   -Posterior 1/3rd tongue : 9th nerve/glossopharyngeal nerve.
   -Posterior most part of tongue:10 nerve.





*On the floor of Ist arch there are three swellings:-

1. Two lingual swellings
2. One tuberculum impar
   ↓
   Forms anterior 2/3rd of tongue
3. Floor of II, III, IV arches - hypobranchial eminence
4. The III arch mesoderm of hypobranchial eminence forms
   ↓
   posterior 1/3rd of the tongue.
5. The IV arch mesoderm of hypobranchial eminence forms
   ↓
   Posterior most part of the tongue and epiglottis.
6. The II arch mesoderm does not contribute to tongue development
   ↓
   Because mesoderm of III arch overgrows 11 & fuse with anterior a/3rd of tongue

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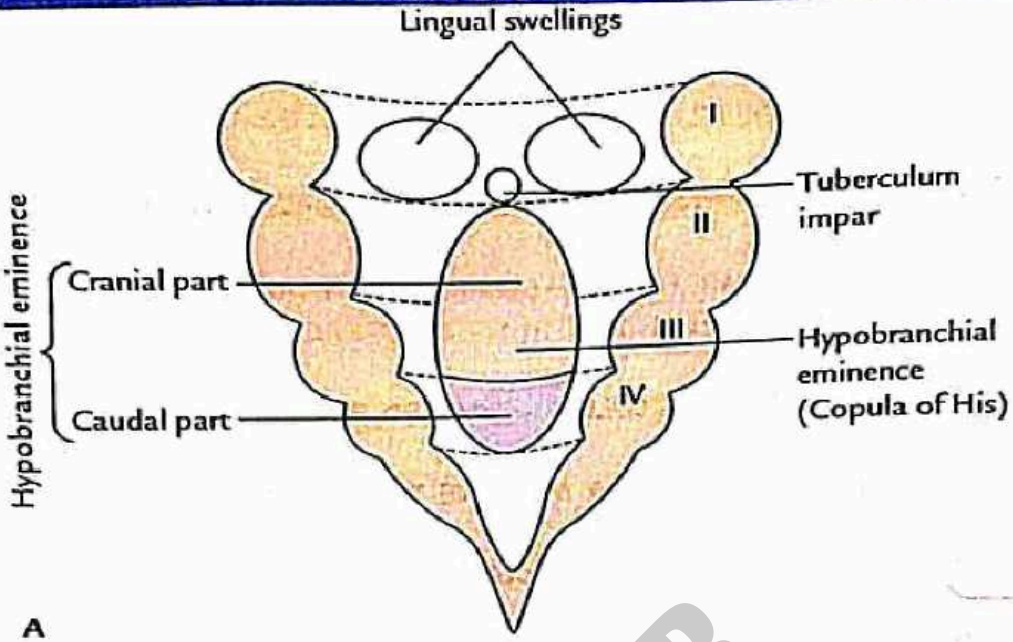

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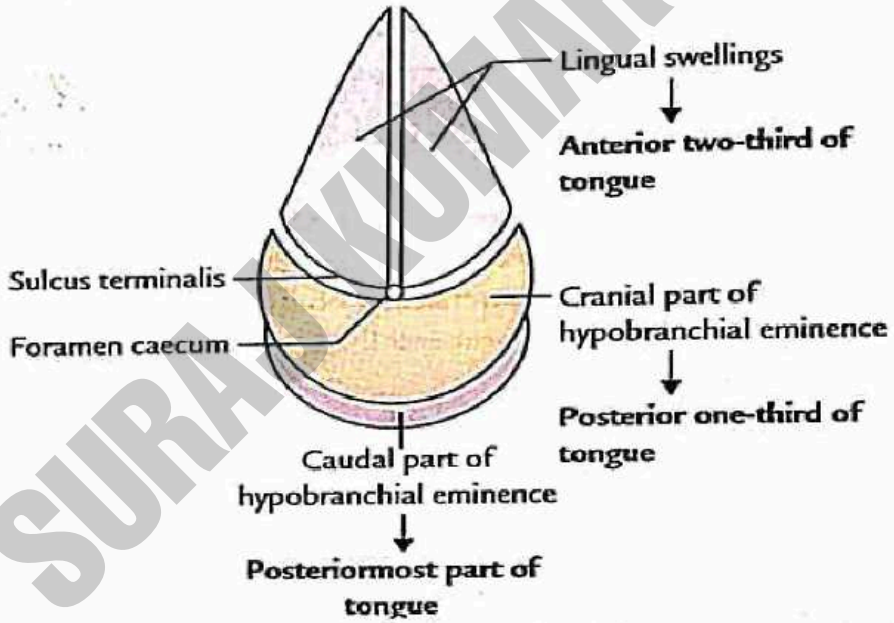

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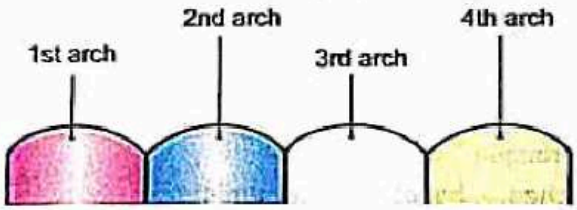

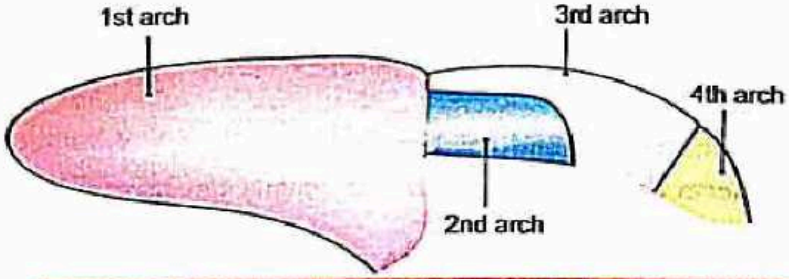

TONGUE OF DEVELOPMENT

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HEART TUBE FORMATION

1.Bulbous cordis :-
   a. Proximal 1/3 will form the rough part of the right ventricle
   b. Conus part will form infundibulum of both right ventricle and left ventricle
   c. Spiral septum [derived from neural crest cell] is developed in truncus arteriosus and divides it into aorta and pulmonary trunk

2.Primitive ventricle: develops into rough part of left ventricle

3.Bulbo ventricular sulcus: filled with septum which forms Interventricular septum

4.Primitive atrium: will form the rough part of both right and left atrium

5.Sinus venosus:
   a. Body absorbed into right atrium along with right horn
   b. Left horn obliterates and forms coronary sinus
   c. Right common cardinal vein will form superior vena cava
   d. Right vitelline vein will form inferior vena cava (cranial part)

*LOOPING OF HEART - Due to dextro looping atrial chamber goes above and behind, ventricle chamber move front and below

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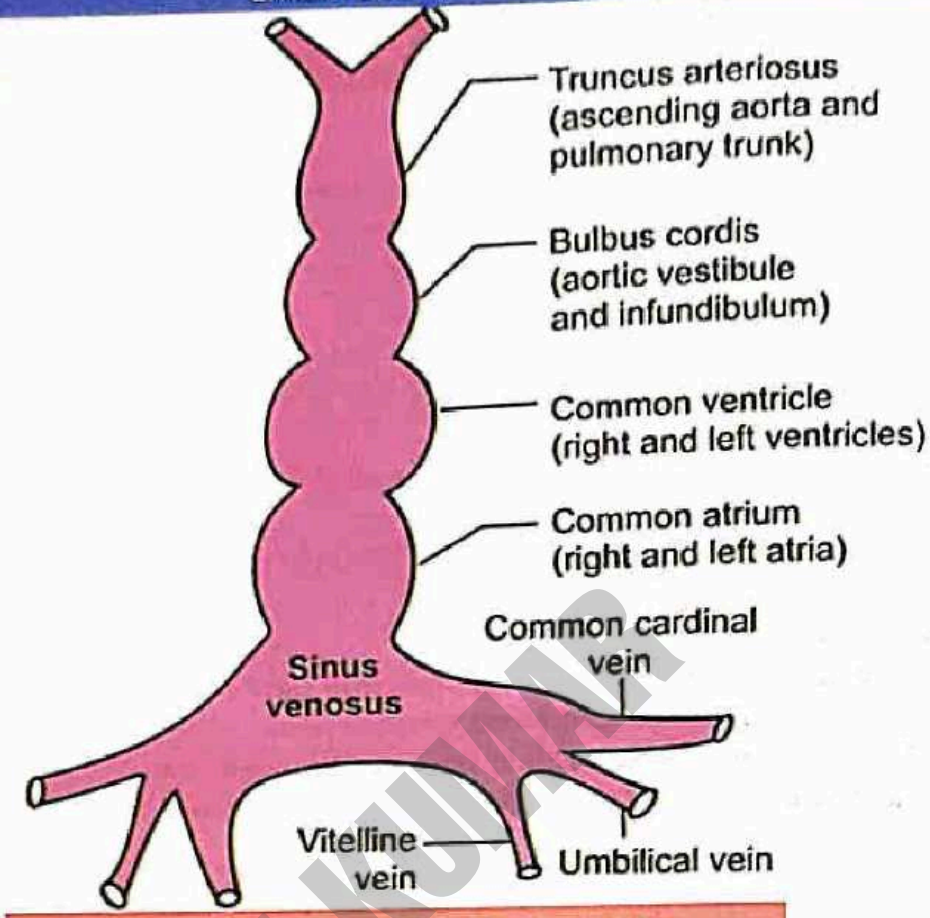
INTERVENTRICULAR SEPTUM FORMATION

1) Muscular part: from the muscular proliferation of Bulb ventricular cavity
2) Membranous part: derived from proliferation of Av cushion
3) Bulbar part: derived from spiral septum
   - Proximal 1/3 of bulbous cordis (RV) + Primitive ventricle (LV)
   ↓
   Common ventricular chamber

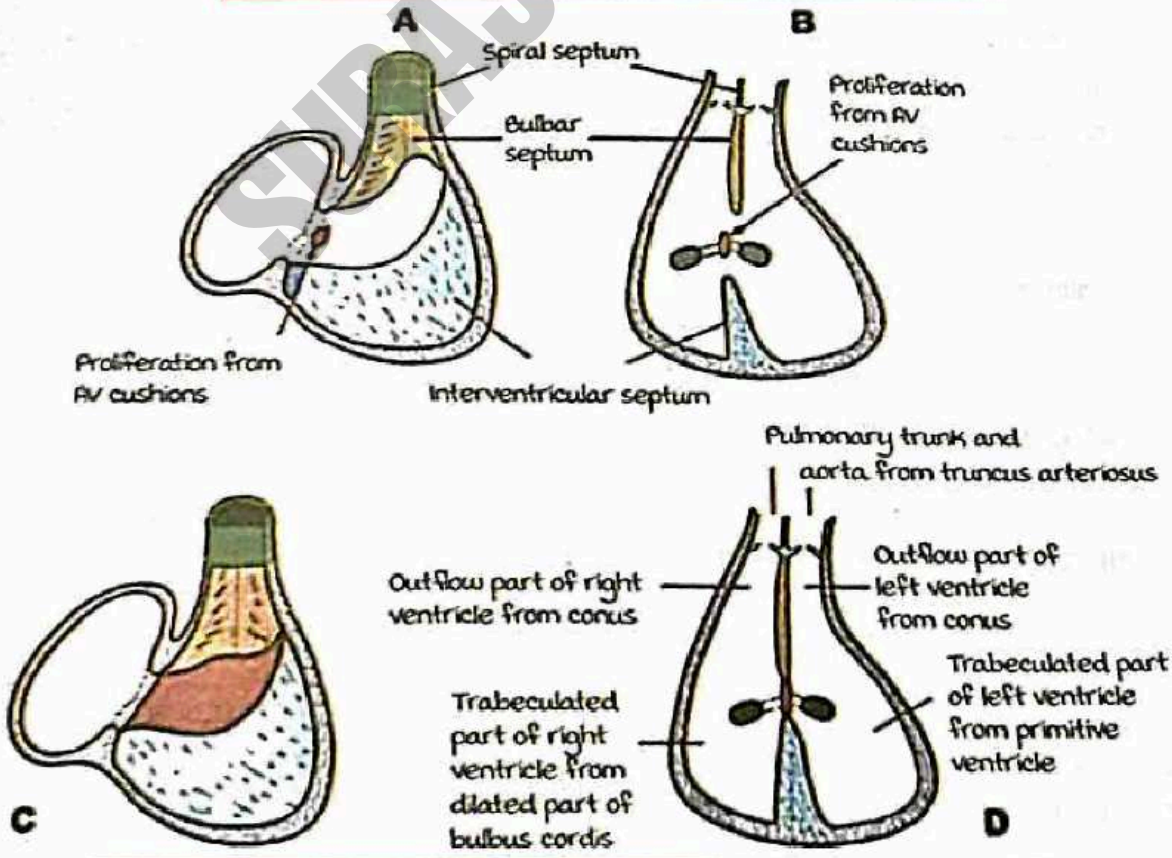
RIGHT ATRIUM DEVELOPMENT

*RIGHT ATRIUM -→ 3 SOURSE
1.RIGHT HALF OF PRIMITIVE ATRIUM - FROM ROUGH TRABECULATED PART -IN FRONT CRISTA TERMINALIS
2.ABORPTION OF RIGHT HORN OF SINUS VENOSUS - SMOOTH PART BEGIND CRISTA TERMINALIS
3.ABORPTION OF RIGHT HALF OF AV CANAL FORM- MOST VENTRAL -→SMOOTH PART

*ABSORPTION OF SINUS VENOUS - CORONARY SINUS & VENA CAVA ARE SEEN
*RIGHT VENOUS VALVE --→BECOME - CRISTA TERMINALIS, INFERIOR VENA CAVA
*LEFT VENOUS VALVE --→ FUSED WITH ATRIAL SEPTUM



HEART TUBE FORMATION



INTERVENTRICULAR SEPTUM FORMATION

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DEVELOPMENT OF FACE

*Face: Derived from - Frontonasal process
-medial & lateral process
-maxillary process
-mandibular process
-Lateral nasal process

1.Frontonasal process - forms forehead.
2.medial Nasal Process - fuse with each other
↓
Forms philtrum and primitive palate.
3. Lateral Nasal process - forms ala of the nose.
4.maxillary process - upper lip and cheek.
5.mandibular process - lower lip and chin.
6.The maxillary process fuses with MNP and LNP

DEVELOPMENT OF SOFT PALATE

*PALATE CONSIST OF 2 PART - PRIMARY PALATE & SECONDARY PALATE
*PRIMARY PALATE DEVELOPS FROM -→ FRONTONASAL PROCESS
*SECONDARY PALATE DEVELOPS FROM-→ MAXILLARY PROCESS

*DEVELOPMENT OF PALATE BEGINS IN - 6TH WEEK
*DEVELOPMENT OF PALATE END IN - 12TH WEEK

*PALATE DEVELOP IN 2 STAGES-
1.DEVELOPMENT OF PIRMARY PALATE
2.DEVELOPMENT OF SECONDARY PALATE

*DEVELOPMENT OF PRIMARY PALATE - FORM BY FUSION OF -→2 MEDAIL PROCESS OF FRONTONASAL PROCESS

*DEVELOPMENT OF SECONDARY PALATE - FORM BY FUSION OF 2 SHELVE LIKE OUTGROWTH CALLED PALATINE SHELVES - APPEAR IN 6TH WEEK

*INTIALLY - GROWTH - DOWNAWARD, MEDIALLY AND BELOW TONGUE

*7TH AND 8TH WEEK - ASSUME HORIZONTAL POSTION ABOVE TONGUE & FUSE TO FORM SECONDARY PALATE

*POSTERIOR PART OF THESE--→ FAIL RO OSSIFY AND FORM - SOFT PALATE

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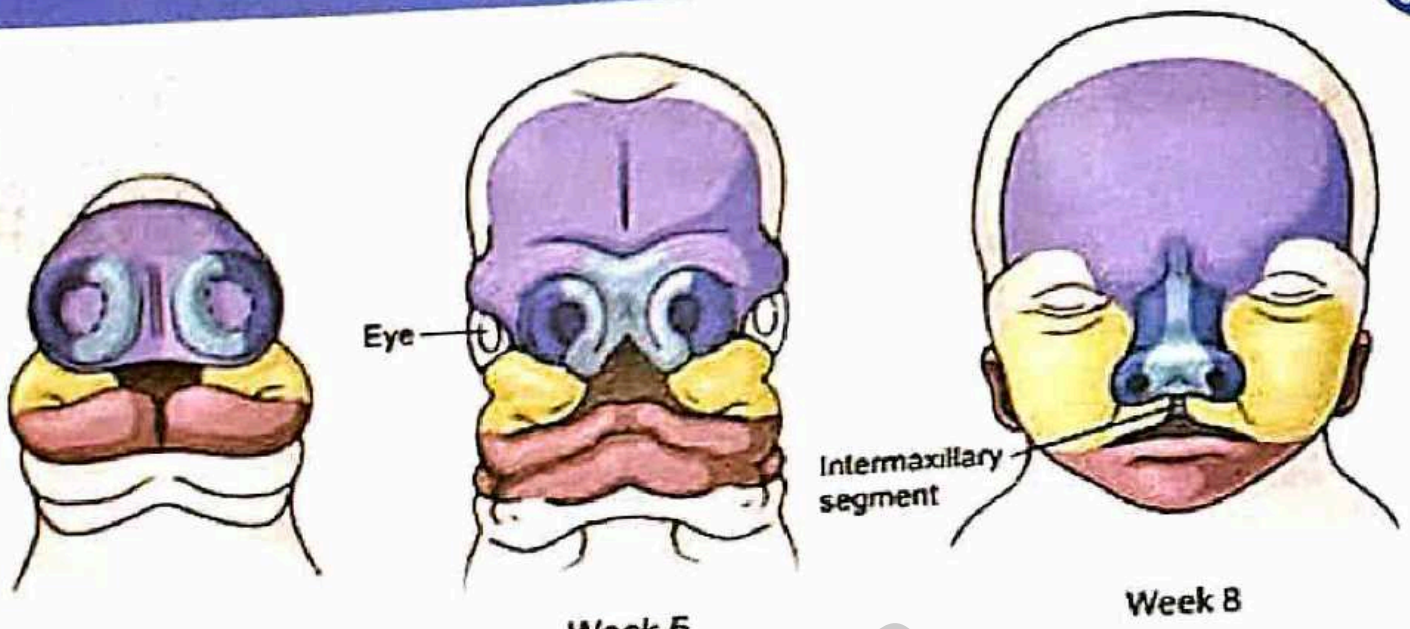
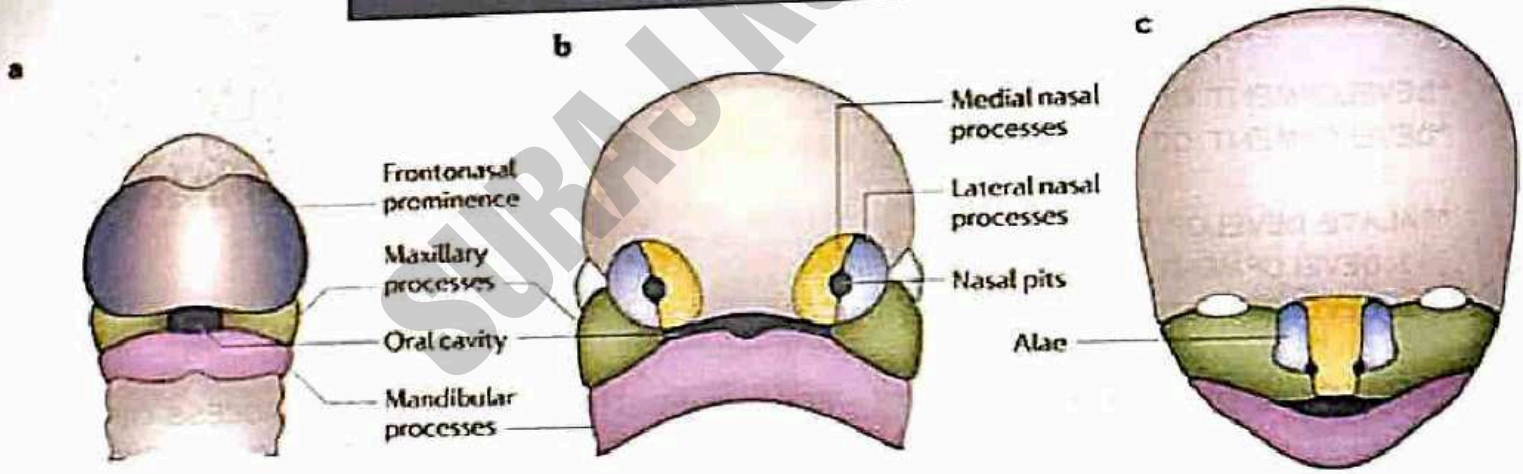
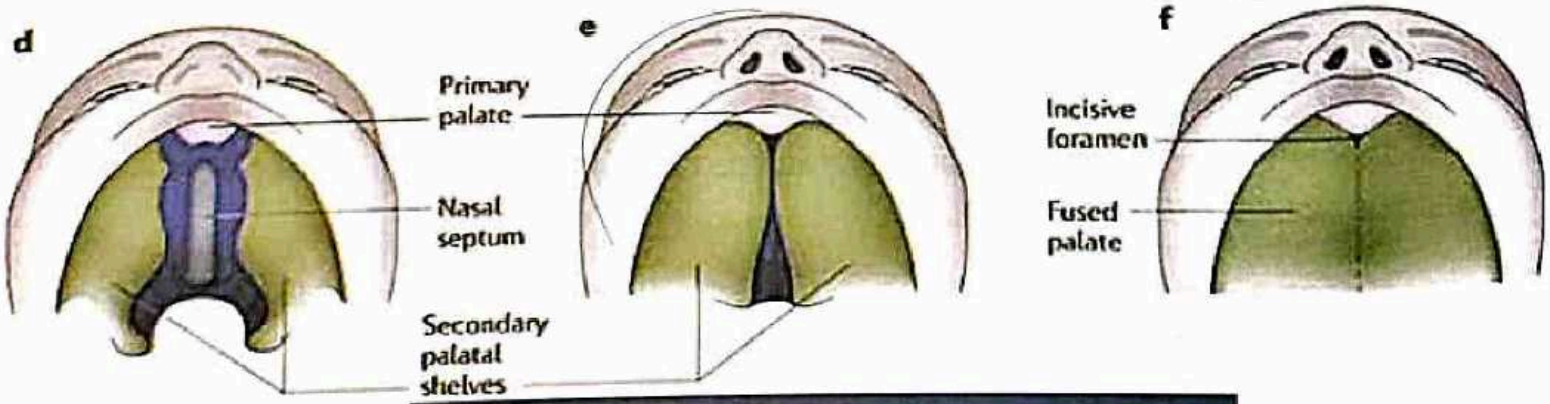


DEVELOPMENT OF FACE





DEVELOPMENT OF SOFT PALATE

SUBAJ KUMAR

DEVELOPMENT OF PANCREAS

*Pancreas Development from - Two buds : Dorsal pancreatic bud
Ventral pancreatic bud

*Ventral bud moves dorsally, lies below the dorsal bud in - 7th week

1.ventral bud - part of head of pancreas + uncinate process
2.Dorsal bud- remaining entire pancreas

*PANCREATIC DUCT -
1.Main pancreatic duct derived from distal part of dorsal bud & whole of ventral bud
2.Accessory Pancreatic Duct - derived from - proximal part of dorsal bud

*PARENCHYMA DEVELOPS - from - surrounding mesoderm

DEVELOPMENT OF DUODENUM

*Duodenum --->Derived from foregut + midgut
*Duodenum Supply - celiac trunk & superior mesenteric artery
*During rotation of stomach, duodenum comes to right side, forms a C - shaped loop
*Mesoduodenum suspends duodenum from Posterior abdominal wall.
↓
Disappears- duodenum becomes retroperitoneal.

* Duodenum retroperitoneal except the 1st proximal part of duodenum

DEVELOPMENT OF LIVER

*Development of liver, From a sources-liver bud and septum transversum
1.Liver bud- In foregut tube, Forms parenchyma of liver & biliary system
2.Septum transversum-Forms connective tissue of liver, capsule, sinusoids, Kupffer cells.

ROTATION OF MIDGUT

*During reduction, midgut rotates 3 times, each time by 90° in counter clockwise/anti-clockwise manner.

1.1st rotation: - Pre arterial segment - right side.
-Post arterial segment - left side.
2. 2ND rotation:-Pre arterial segment - elongates and enters abdominal cavity and rotates by 90°anti-clockwise.

*Pre arterial segment comes to lie behind the superior mesenteric artery.

3.3rd rotation: -Post arterial segment - enters abdominal cavity and rotates by 90°anti-clockwise.

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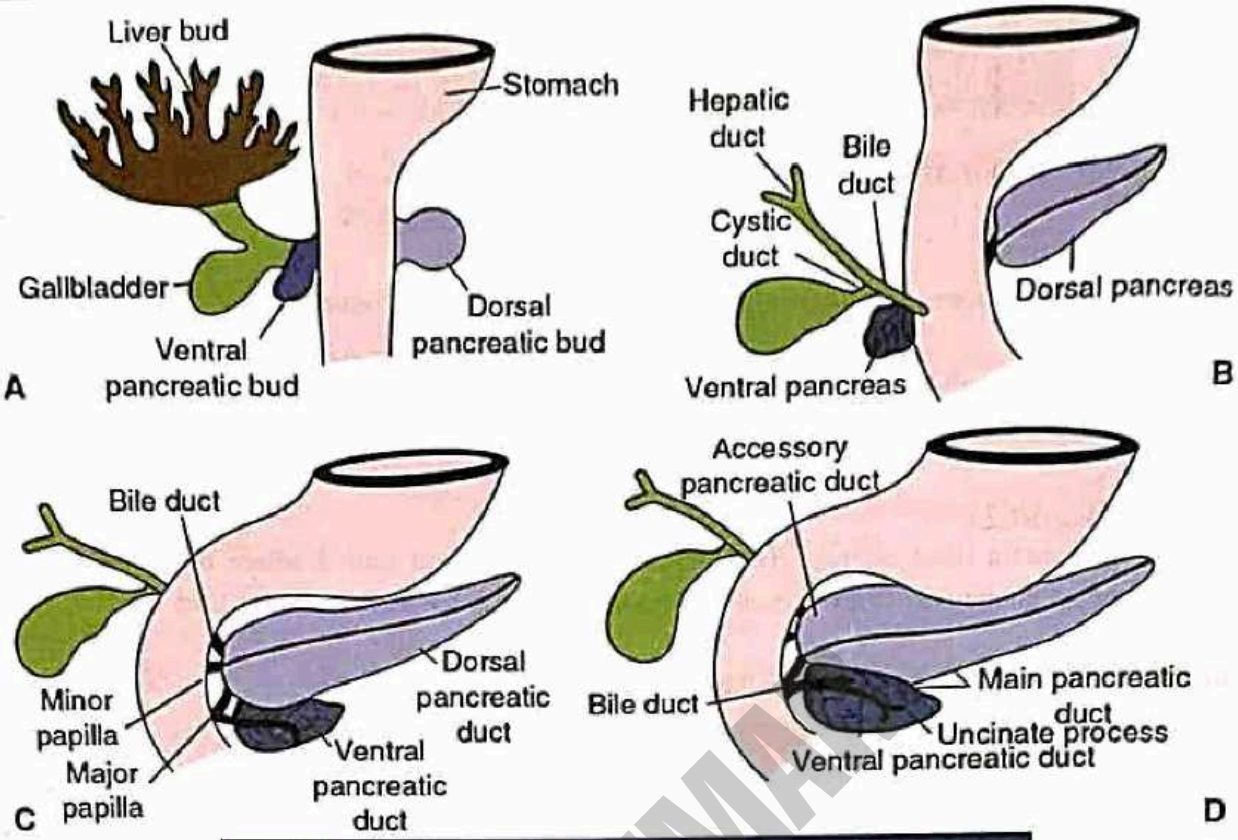
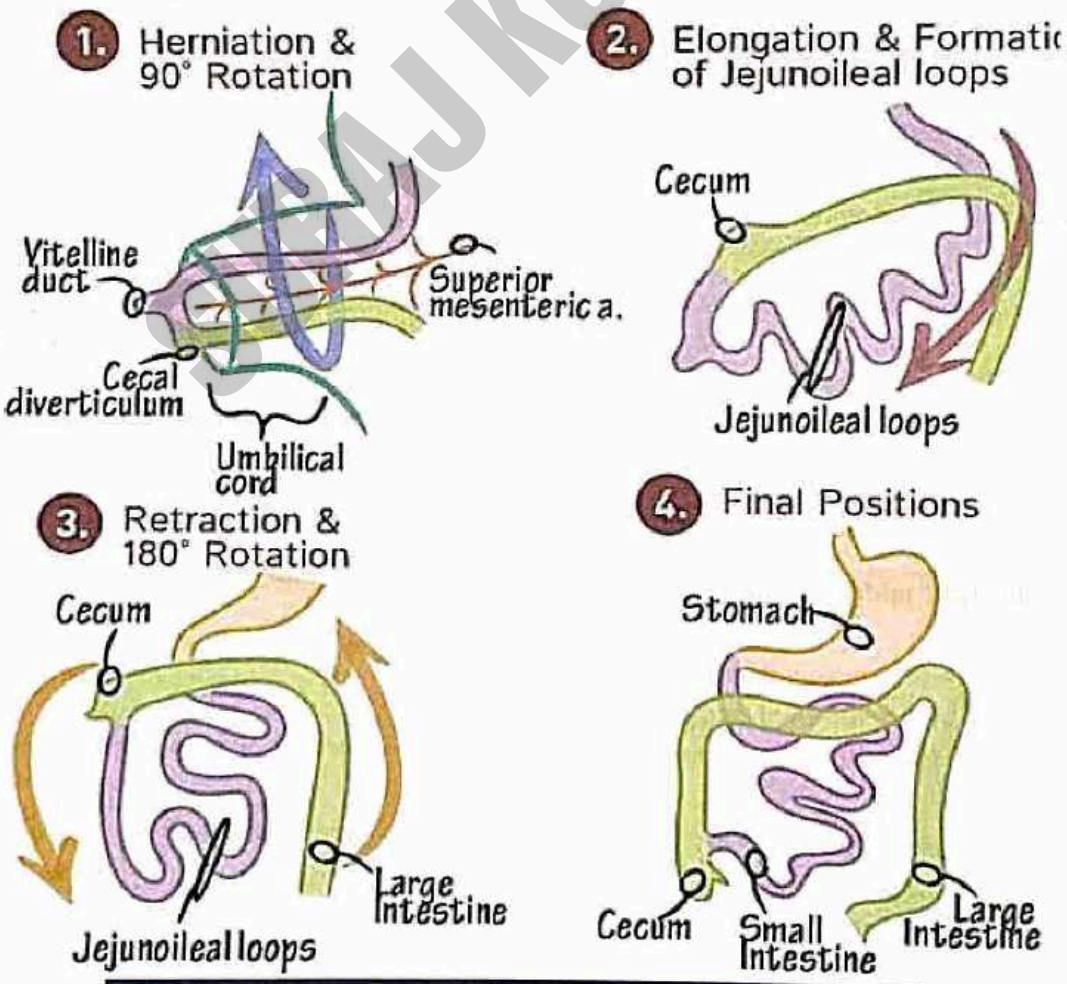


DEVELOPMENT OF PANCREAS



ROTATION OF MIDGUT

SIRAJ KUMAR

DEVELOPEMNT OF KIDNEY

*Nephrogenic cord - forms --->Kidney, mesonephric duct & Paramesonephric duct.

1. At 4th week - pronephros appears, functions as the kidney & disappears by end of 4th week.
2. By the end of 4th week - mesonephros appears, functions as interim Kidney up to a months.
3. By 5th week - metanephros appear and starts its function by 3rd month.
4. A diverticulum arises from the mesonephric duct called ureteric bud
   ↓
   which forms metanephric blastema.

*The kidneys are derived from two sources:-

1. Excreting component - derived from metanephros.
   - Metanephric blastema - contact with ampulla → under go differentiation form nephron

2. Collecting component - derived from ureteric bud.
   - uretic bud grow toward metanephric blastema → become ampulla
   -ampulla -→ form pelvis of kidney - then form minor & major calyx and collecting tubules

*The kidneys become functional from 12th week.

*Ascent of kidney :- Initial in sacral region -→ kidney ascend to lumbar region

*Rotation of kidney - Hilum - first anterior, later hilum come medially

DEVELOPEMNT OF HINDGUT

*Development of hindgut

1. Diverticulum arising from hindgut is called allantois.
2. Cloaca is a part of hindgut distal to allantois.
3. The urorectal septum - separates the cloaca and cloacal membrane.
4. The cloaca divides into - primitive urogenital sinus
   -Primitive rectum.
5. The cloacal membrane divides into - anal membrane & urogenital membrane.
6. The primitive rectum forms - rectum, Anal canal up to pectinate line

*Derivation of upper anal canal :

- Endoderm
  ↓
  Gut ---→ Hindgut -→ Cloaca -→ Primitive rectum -→ upper part of anal canal

*Derivation of lower part of anal canal :

- Ectoderm --→ Proctodeum --→lower part of anal canal

*The anal membrane is formed at the junction of upper and lower anal canal.
*It ruptures by the 9th week therefore, upper and lower anal canal communicate.
*The pectinate line is formed at the site of formation of anal membrane.

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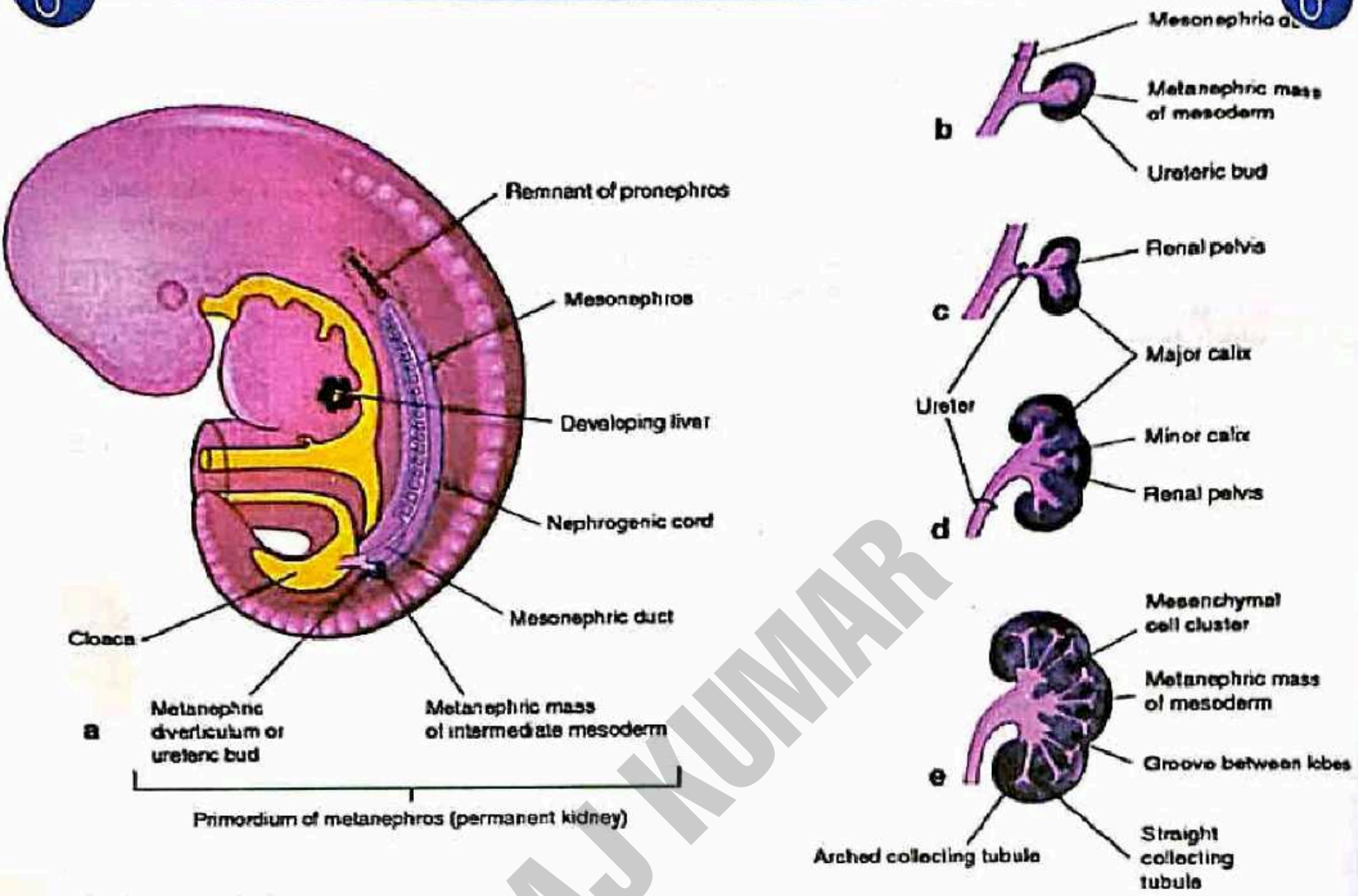
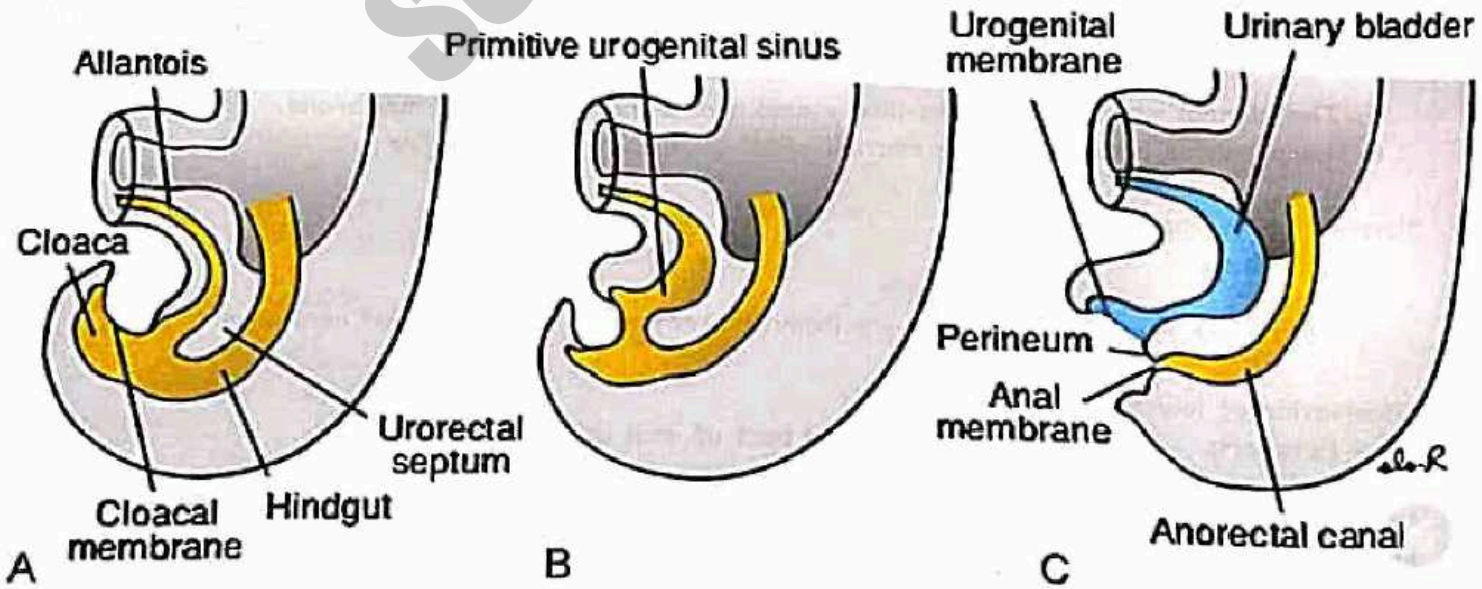


DEVELOPEMNT OF KIDNEY



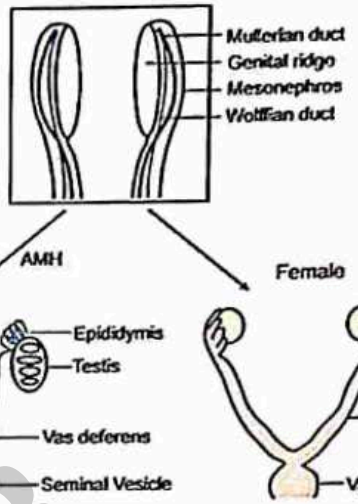
DEVELOPEMNT OF HINDGUT

SUMAJ KUMAR

DEVELOPEMNT OF GONADS

*Genotype that is xx/xy is determined at the time of fertilization itself

*Phenotypic differentiation will start only at 7th week
- until 6th week gonads are indifferent gonads
- Primordial germ cells reach the gonads at 5th week



*Gonads:
- Testis develops at 7th week
- Ovary develops at 8th - 10th week

*External genitalia:
- In males: 14th week
- In females: 11th week

*Sex determination can be done at 15th week (illegal to reveal)

Structures	Male derivative	Female derivative
1. Gonadal ridge	-Testis	-Ovary
2. Genital Tubercle	-Penis	-Clitoris
3. Genital swelling	-Scrotum	-Labia majora
4. Mesonephros	-Efferent tubules	-Epoophoron
5. Mesonephric duct	-Appendix of epididymis vas deferens Epididymis Ejaculatory duct Seminal vesicle	-Appendix of ovary
6. Paramesonephric duct	- Appendix of Testis	-Fallopian tube uterus vagina (upper part)

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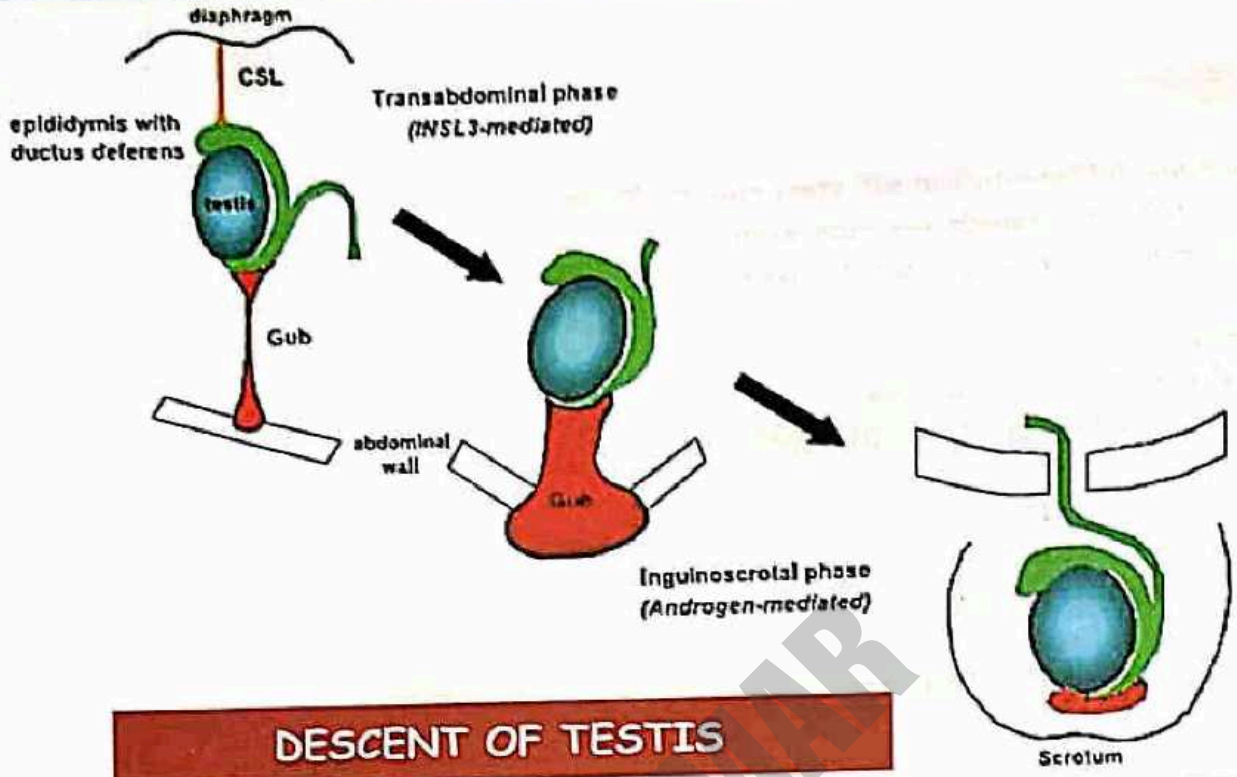
DESCENT OF TESTIS

1. Descend to reach scrotum.
2. Each testis begin to descent during second month of intrauterine life
3. Reach iliac fossa by 3RD month
   - Deep inguinal ring - 4th to 6th month
   - Inguinal canal 7th month
   - Superficial inguinal ring → 8th month
   - Bottom of scrotum by 9th month
4. Extension of peritoneal cavity - Called processus vaginalis-→Preceds the descent of tests into the scrotum
5. Processus vaginalis close above the testis
6. Doescent doesn't occur after one year of age



DESCENT OF TESTIS

STUDY TIPS FOR MAXIMUM MARKS

1) IN EMBRYOLOGY ONLY SHORT NOTES ARE ASKED

2) IN LONG QUESTION ,IT IS USUALLY ASKED AS SUB HEADING
(FOR EXAMPLE - DEVELOPMENT OF RIGHT ATRIUM)

3) ONLY WRITE MAIN CONTENT WITH DIAGRAMS, ABOUT 1/2 OR 1 PAGE

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PREVIOUS YEAR UNIVERSITY QUESTIONS

*SHORT NOTES - GENERAL EMBRYOLOGY
1.NOTOCORD DEVELOPMENT(VIMP)
2.PHARYNGEAL ARCHES DERIVATIVES
*SYSTEMIC EMBRYOLOGY -
1.DEVELOPMENT OF TONGUE
2. DEVELOPMENT OF THYROID
3. DEVELOPMENT OF SOFT PALATE
4. DEVELOPMENT OF PANCREAS
5. DEVELOPMENT OF DEUDENUM
6. DEVELOPMENT OF ANAL CANAL(HIND GUT)
7. DEVELOPMENT OF RIGHT ATRIUM
8. DEVELOPMENT OF INTERVENTRICULAR SEPTUM
9.ROTATION OF MID GUT
10.DESCENT OF TEST

GENERAL ANATOMY

EPIPHYSIS

- The ends and tips of a bone which ossify from secondary centre are called epiphysis.
- **Classification:**

According to number of epiphysis -

SIMPLE
- ends of long bones develop from many epiphysis, which fuse independently with shaft.

e.g. femur.

COMPOUND
- ends of bones develop from many centre which unite to form single epiphysis which in turn fuses with shaft.

e.g. humerus.

Based on function -

PRESSURE
- Articular
- takes part in weight transmission

e.g. head of Humerus.

TRACTION
- Non articular
- provide attachment to one or more tendon which exert traction on epiphysis.

e.g. Trochanter of femur.

ATAVISTIC
- phylogenitically independent bone which in man becomes fused to another.

e.g. Coracoid process of scapula.

ABERRANT
- not always present.

e.g. at head of first metacarpal.

COMPOUND
- in some bones 2-3 smaller epiphysis join to form compound epiphysis before joins a shaft.

e.g., upper end of humerus.

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BONES

- Bone is one-third connective tissue.
- It is impregnated which Calcium constituting remaining two-third.

- **Classification -**

According to Shape -

- LONG BONES - has elongated shaft (diaphysis) and two expanded ends (epiphysis) which are smooth & articular.
  - Typical long bones - two secondary epiphysis.
  - Miniature / short long bone - one epiphysis.
  - Modified long bones - no medullary cavity.

- **SHORT BONES** – Shape is usually cuboid / scaphoid.
- **FLAT BONES** – form boundaries of certain body cavities.
- **IRREGULAR BONES** – Hip bone.
- **PNEUMATIC BONES** – Irregular bone with large air space.
- **SESAMOID BONES** – Bony nodule embedded in tendon / joint capsules.
- **ACCESSORY BONES** – Not always present. May occur as ununited epiphysis developed from extra centre of ossification.

Developmental Classification

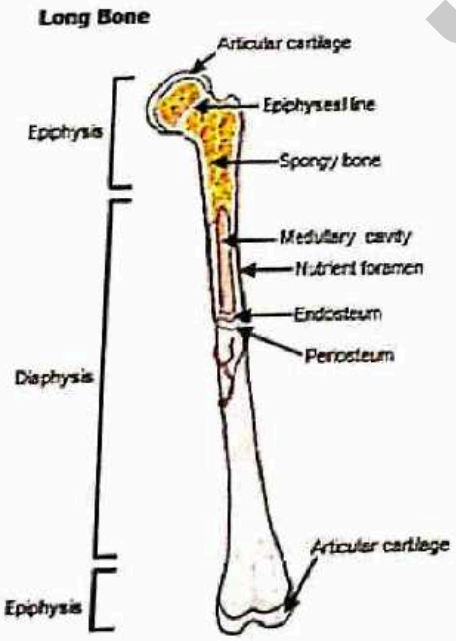
- **MEMBRANE (DERMAL) BONES** – Ossify in membrane.
- **CARTILAGINOUS BONES** – Ossify in cartilage.
- **MEMBRANO-CARTILAGINOUS BONES** – Ossify partly in membrane and partly in cartilage.
- **SOMATIC BONES** – Most bones are somatic.
- **VISCERAL BONES** – few and develop from pharyngeal arch.

Regional Classification

- **AXIAL SKELETON** – includes skull, vertebrae, thoracic cage.
- **APPENDICULAR SKELETON** – includes bone of limb

Structural Classification

- **MACROSCOPICALLY**
  - **COMPACT** – bone is dense, texture is ivory but extremely porus.
  - **CANCELLOUS** – Spony or trabaecular bone is open in texture and is made up of meshwork of trabeculae.
- **MICROSCOPICALLY**
  - **LAMELLAR** – most of mature bones are composed of thin plates of bony tissue called lamallae.
  - **WOVEN** – Seen in fetal bone, fracture repair or cancer bone.
  - **FIBROUS** – young foetal bone.
  - **DENTINE**, **CEMENT** – Occur in teeth.



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BLOOD SUPPLY of LONG BONE

- Arterial Supply is derived from -

NUTRIENT ARTERY -

enters cortex through nutrient foramen → divides into ascending and descending branch → at end, bend to form hairpin loop.

- Supplies metaphysis, inner 2/3rd cortex, medullary cavity.
- direction of nutrient foramen is away from growing end, i.e., towards elbow away from knee.
- any blood infection causes emboli can block nutrient artery at bend and cause osteomylitis.

PERIOSTEAL ARTERY -

- numerous beneath masculer and ligamentous attachments.
- enter Volkmann's canal to supply outer 1/3rd cortex.

EPIPHYSEAL ARTERY -

- derived from periarticular vascular arcades found on non-articula bony surface.

METAPHYSEAL ARTERY -

- derived from neighbowing systemic vessels.
- pass directly into metaphysis and reinforce metaphysial branch from primary nutrient artery.

LONG SHORT BONE

- Nutrient artery → enters middle of shaft → divide to form a plexus.
- major part of bone supplied by periosteal artery.

SHORT BONE

- Supplied by numerous periosteal arteries which enter their non-articular surface.

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VERTEBRA

- Body – Supplied by anterior posterior vessels
- Vertebral arch – Supplied by large vessels entering base of transverse process.

RIBS

- Nutrient artery – enter just beyond tubercle.
- Periosteal artery

VENOUS DRAINAGE –

- Veins are large and numerous – Cancellous bone.
- Accompany artery in Volkmann's canal – Compact bone.

CARTILAGE

- Cartilage is a connective tissue composed of cells and fibres embedded in a firm, gel-like matrix which is rich in a mucopolysaccharide.
- Classification → ELASTIC, HYALINE, FIBROCARTILAGE

→ **HYALINE** –
- located in articular cartilage of long bones, epiphyseal plates, nasal cartilage, thyroid, trachea, bronchi, costal cartilage.
- Bluish white in colour.
- Shiny or translucent.
- very thin fibres.
- flexible
- perichondrium is present.

Hyaline cartilage

→ **FIBROCARTILAGE** –
- located in intervertebral disc, Temporomandibular joint, sternoclavicular joint.
- Glistening white in colour.
- Opaque appearance.
- Numerous white fibres.
- perichondrium is absent

Fibrocartilage

→ **ELASTIC** –
- located in pinna, external auditory meatus, eustachian tube, epiglottis.
- yellowish in colour.
- Opaque appearance.
- numerous yellow fibres.
- Most flexible.
- perichondrium is present.

Elastic cartilage

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JOINT

- It is a junction between two or more cartilage.
- Device to permit movement.

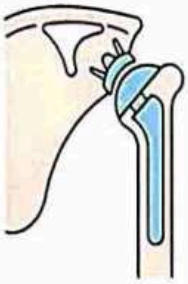
Classification –

Structural Classification-

- **FIBROUS JOINTS** – Immovable or permit slight movement
  - SUTURES – present only in skull.
  - SYNDESMOSIS – fibrous union b/w bones.
  - GOMPHOSIS – peg and socket junction between tooth and socket.
- **CARTILAGINOUS JOINTS** – bones are joined by plates of cartilage.
  - PRIMARY / SYNCHONDROSIS – temporary in nature as after a certain age cartilage plate is replaced by bone.
  - SECONDARY / SYMPHYSES – permanent and persist throughout life.
- **SYNOVIAL JOINTS** – most evolved, most mobile.
  - BALL AND SOCKET JOINT
  - SELLAR / SADDLE JOINT
  - CONDYLAR / BICONDYLAR JOINT
  - ELLIPSOID JOINT
  - HINGE JOINT
  - PIVOT JOINT
  - PLANE JOINT

ELBOW JOINT

KNEE JOINT



SHOULDER JOINT

HIP JOINT

Functional Classification

- SYNARTHROSES – Immovable.
- AMPHIARTHROSES – Slightly movable.
- DIARTHROSES – freely movable.

ANKLE JOINT

Regional Classification

- SKULL TYPE – Immovable.
- VERTEBRAL TYPE – Slightly movable.
- LIMB TYPE – freely movable.

According to number of Articulating Bones

- SIMPLE JOINT – when only two bones articulate.
- COMPOUND JOINT – more than two bones articulate in capsul
- COMPLEX JOINT – when joint cavity is divided by an intra-articular disc.

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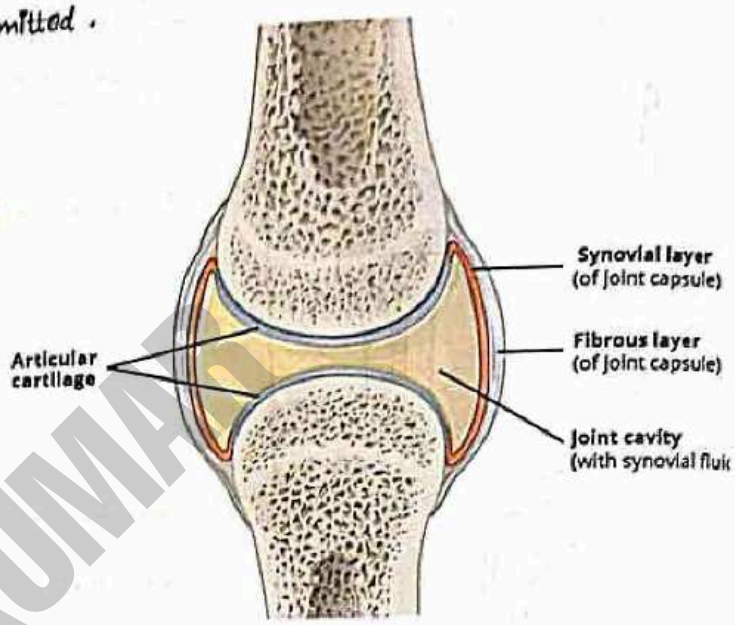
SYNOVIAL JOINT

- most evolved and most mobile.
- **Characters –**
  - Articular surface are covered with hyaline cartilage.
  - Synovial fluid is filled between articular surfaces.
  - Joint cavity is surrounded by articular capsule.
  - Varying degree of movement are always permitted.



- **Classification**

1. PLANE SYNOVIAL JOINT
- Articular surfaces are more or less flat.
- permit gliding movement in various directions

EXAMPLE –
- Intercarpal joints.
- Intertarsal joints.
- Carpometacarpal (except first).
- Tarsometatarsal
- Intermetatarsal.

2. HINGE JOINT
- Articular surface are pulley shaped.
- Movements are permitted in one plane.

EXAMPLE –
- Elbow joint.
- Ankle joint.

3. PIVOT JOINT
- Comparises a central bony pivot surrounded by osteoligamentous ring.
- Movement permitted in one plane.

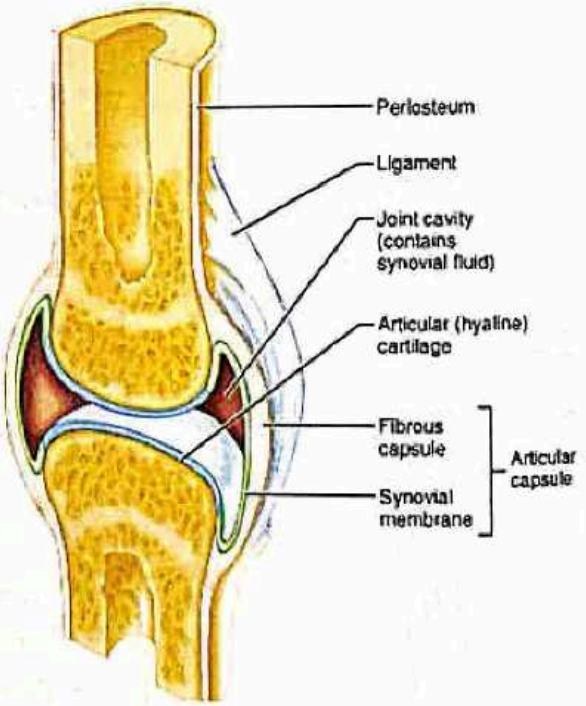
EXAMPLE –
- Median atlantoaxial joint.
- Superior and Inferior Radioulnar joint.

4. CONDYLAR JOINT
- Articular surface includes two distinct condyles.

EXAMPLE –
- Knee joint.
- TMJ.

5. ELLIPSOID JOINTS
- Articular surface include an oval.
- free movements are permitted.

EXAMPLES –
- wrist joint.
- Atlanto-axial joint.



6. SADDLE JOINT
- Articular surface are reciprocally concavoconvex.
- Movement are similar to those permitted by an ellipsoid joint.

EXAMPLE –
- Sternoclavicular joint.
- first carpometacarpal joint.

7. BALL-SOCKET JOINT
- Articular surface include a globular head filling into a cup-shaped socket.
- Movement occur around indefinite axis.

EXAMPLE –
- Shoulder joint.
- Hip joint.

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KARYOTYPING

- also called 'IDIOGRAM'.
- standard arrangement of photographed metaphase spread of chromosomes in decreasing length is called 'KARYOTYPING'.
- chromosomes are identified on the basis of :-
  - length of arm of chromosome.
  - position of centromere of chromosome.

Procedure -

- done by culture of lymphocyte in medium of phytohaemagglutinin (PHA).

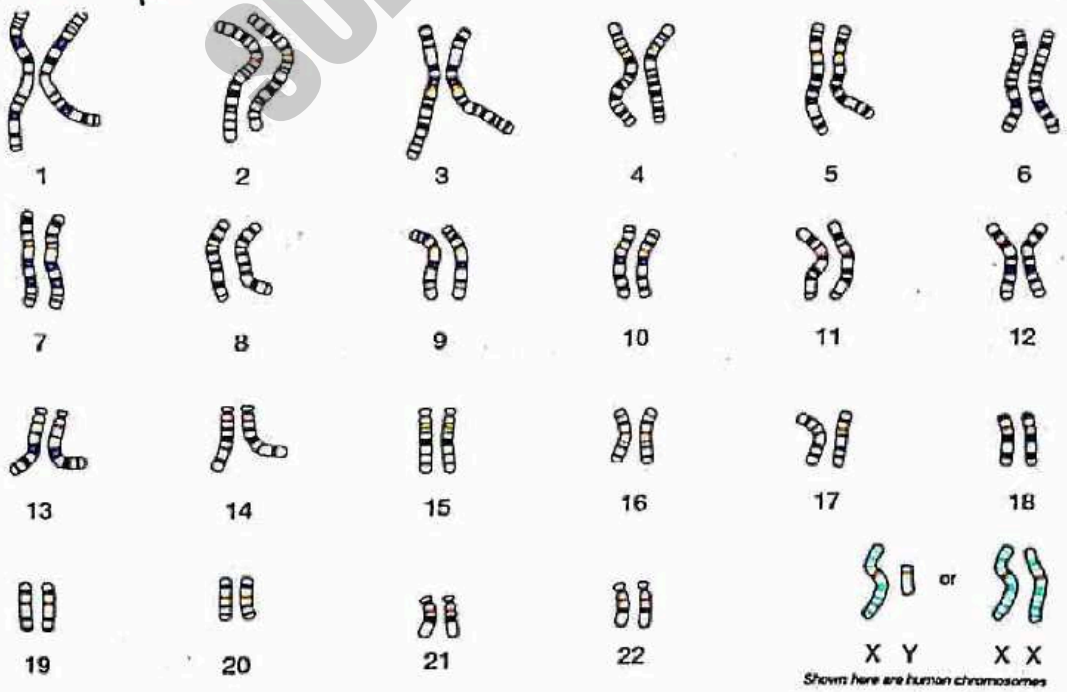
Cell division → Arrested at metaphase by action of cholchicine → Chromosome are spread, counted & photographed under microscope.

Image are cut & arranged in pair in decreasing order of length. → chromosomes are numbered 1-22 in decreasing order of length.

Karyotyping helps to -

1. Identify pattern of abnormal chromosome.
2. Determine the sex.

- the length of pair number 1 is long & 22 is short.
- Individual chromosomes can be recognized with quinacrine band and Giemsa band.



Shown here are human chromosomes

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DOWNS SYNDROME

- Most common congenital anomaly due to numerical abbreation first described by Dr. Down.
- There is trisomy of 21st chromosome, i.e., 4 XX or 47XY.
- Commonly seen in early primigravida or infants born to female suffering from viral infection during pregnancy.

Clinical features –
- Mentally retarded.
- poor growth & short stature.
- widely spaced great toe.

facial features
- moon face / short head / epicanthal fold / narrow palate / protruding tongue / flat nasal bridge / small ears.

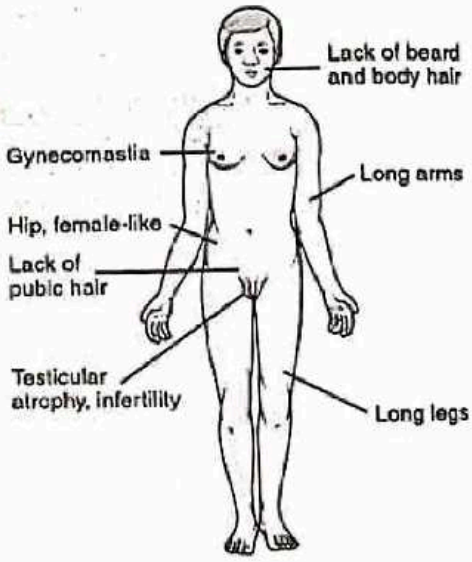
hand features
- short and broad hand / single characterstic crease called siminean crease.

Prenatal diagnosis –
- Aminocentesis
- Increased Inhibin A level.

- also called mongolism due to characterstic mongolian face of patient.

KLINEFELTER'S SYNDROME

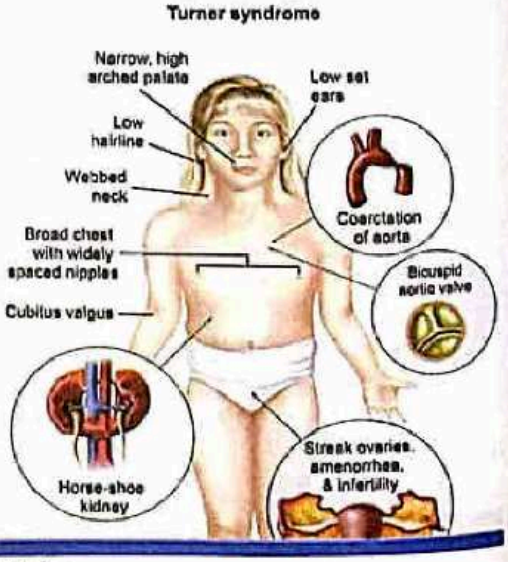
- It is a sex-chromosome abnormality with an extra X chromosome in male.
- Also called XXY syndrome.
- Karyotype is 47XXY.
- Barr body is seen in afflicted males.
- Occurs due to Meiosis II nondisjunction.



Clinical features –
- small degree of mental retardation.
- small testis leading to infertility.
- normal penis and scrotum.
- gynaecomastia
- scanty pubic and facial hairs.

TURNER'S SYNDROME

- sex chromosomal abnormality with partial or complete loss of X chromosome.
- Karyotype is 45X chromosome.
- No Barr body is seen in buccal smear as one chromosome is absent.



Clinical features:
- short stature.
- webbed neck.
- low set ears.
- normal / slightly retarded intelligence.
- Cubitus valgus, i.e., increased carrying angle at elbow.
- Amenorrhoea.
- widely placed nipples.
- Rudimentary ovary / streak gonads.

Treatment:
- Replacement therapy.
- IVF for sterile females.

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SKULL ANATOMY

*Study of Skull done from Outside or Inside

*Exterior View - Norma Verticalis (Superior aspect)
- -Norma Occipitalis (Posterior)
- -Norma frontalis (Anterior)
- - Norma lateralis (lateral)
- -Norma Basalis (Inferior)

NORMA VERTICALIS

→oval
→ Wider Posteriorly than anteriorly.
→ Bones 1. Frontal Bone --→Squamous Part
2. Parietal Bone -→ PAIRED
3. Occipital Bone → Squamous Part

→ Sutures 1. Coronal Suture - B/W Frontal and 2 Parietal Bonus
2. Sagittal Suture - B/W 2 Parietal Bones
3. lamdoid Suture - B/w occipital and 2 Parietal Bones

SCAN QR WITH ORBOT APP TO DIRECTLY OPEN VIDEO LINK

* Metopic Suture in 3%-8%. Cases Present in Median Plane meet

→ Other Features :-
1. Bregma→ Point Where Sagittal and coronal sutures
2. Parietal Eminence - Area of Maximum Convexity in Parietal Bone
3. Vertex → highest Point on skull
4. Parietal Foramen → Present near Sagittal Suture transmits emissary Vein
5. Obelion→ Point B/W 2 Parietal foramen

NORMA OCCIPITALIS

*Norma Occipitalis → Convex Upwards and sides and Flattened Below

*Bones - 1.Parietal Bones - Posterior Portions of Parietal Bone Superiorly One on each side
2.Occipital Bone - Upper Part of Occipital Bone Present inferiorly.
3.Mastoid Part of temporal Bone - Present inferolaterally on each side

*Sutures-lambdoid suture -Parietomastoid Suture

*Other features - tamda
- External Occipital protubrence
- Superior nuchal lines
- inferior nuchal lines
- highest nuchal lines (some Cases)

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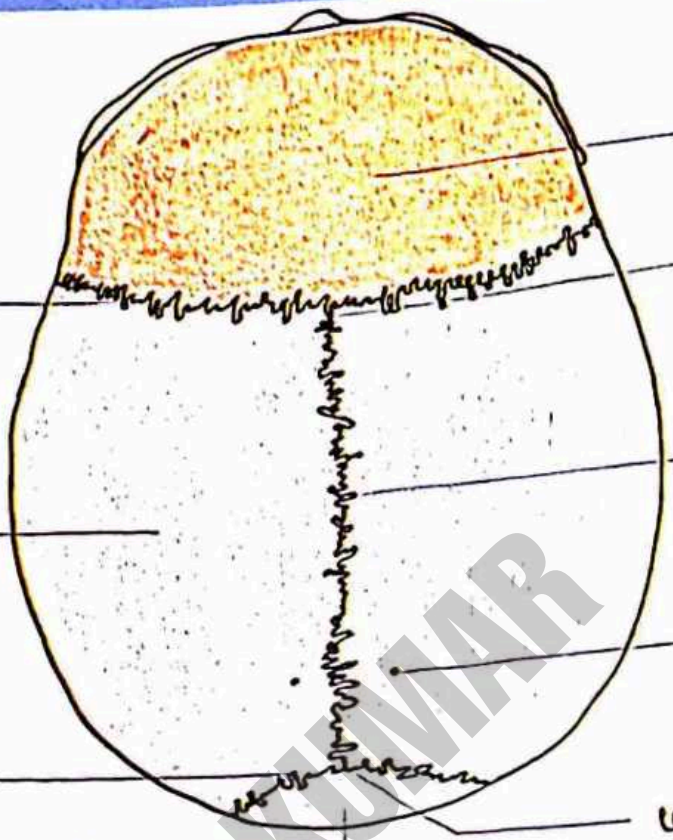
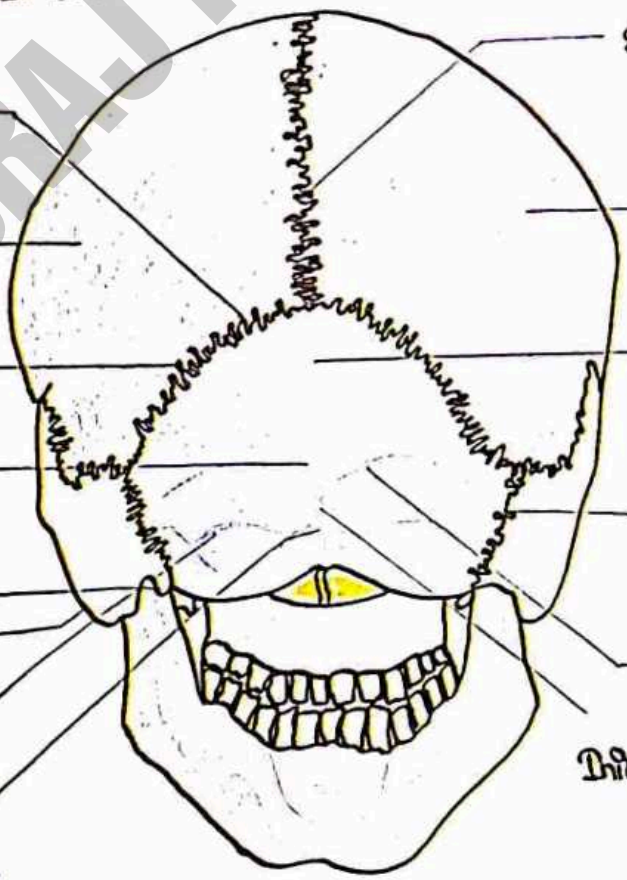
NORMA VERTICALIS



Frontal Bone
Bregma
Coronal suture
Sagittal suture
Parietal Bone
Parietal foramen
Lambdoid suture
Lamda
Occipital Bone



Sagittal suture
Suture Bone
Parietal Bone
Parietal Bone
Squamous part of occipital Bone
Lambdoid suture
External occipital protuberance
Occipitomastoid suture
Mastoid Notch
Mastoid process
Inferior Nuchal line
Superior Nuchal line
Inion
External occipital crest

NORMA OCCIPITALIS

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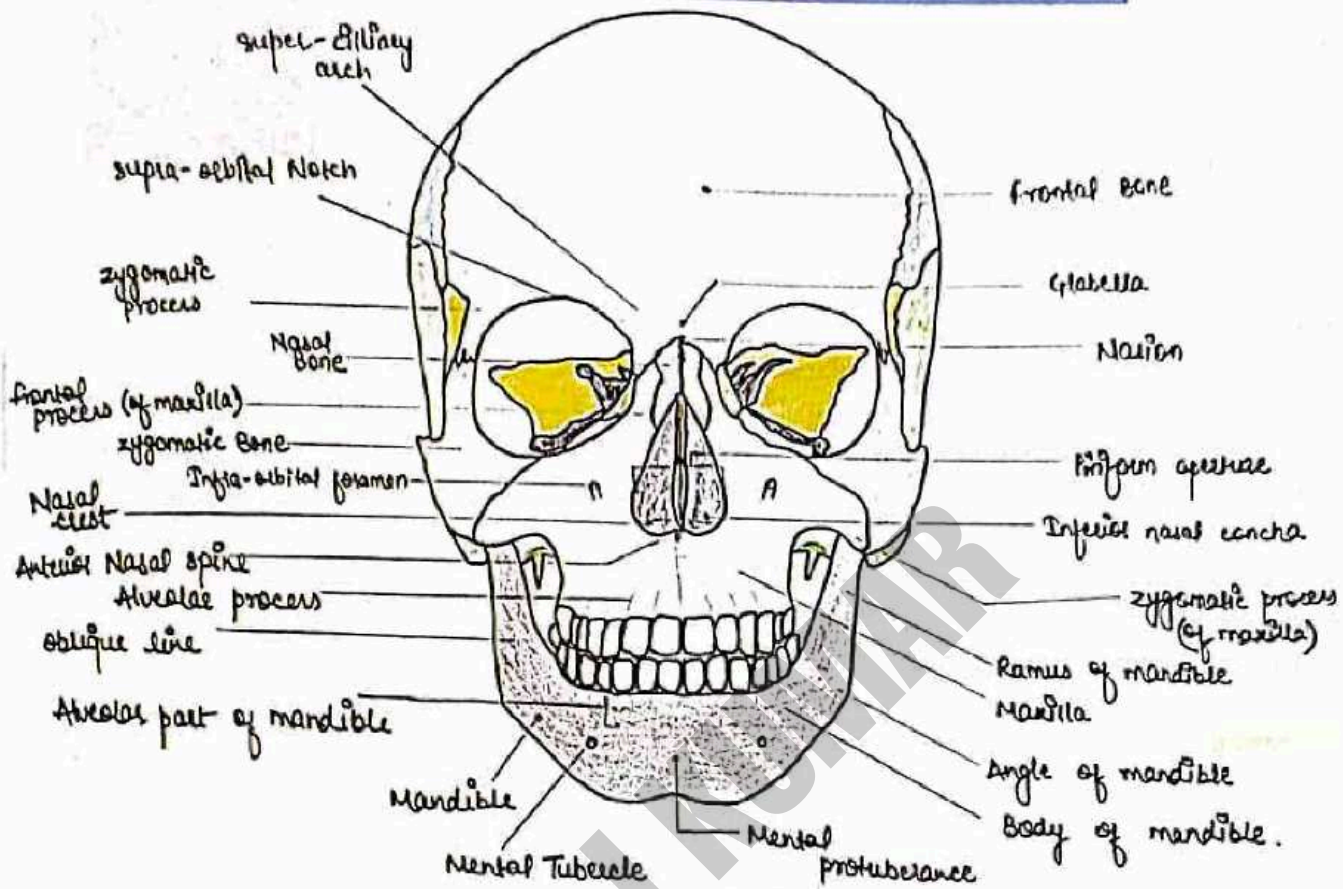
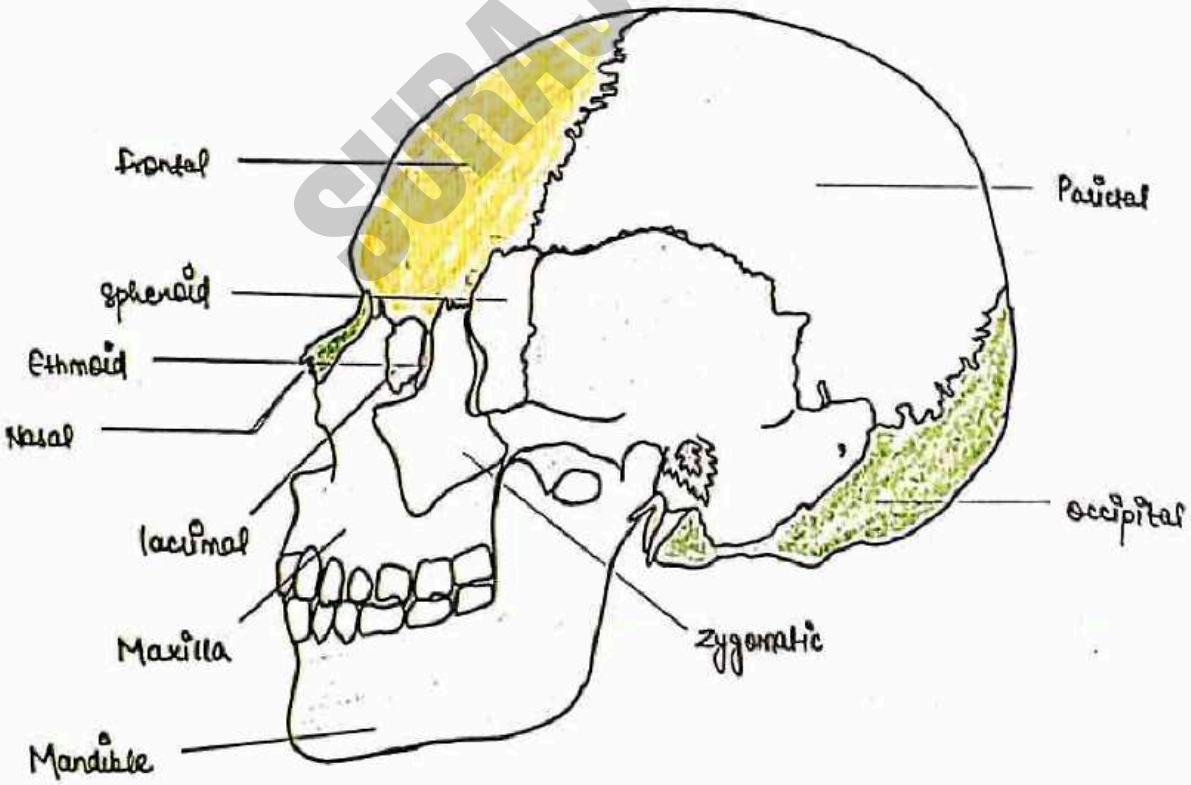
NORMA FRONTALIS





NORMA LATERALIS

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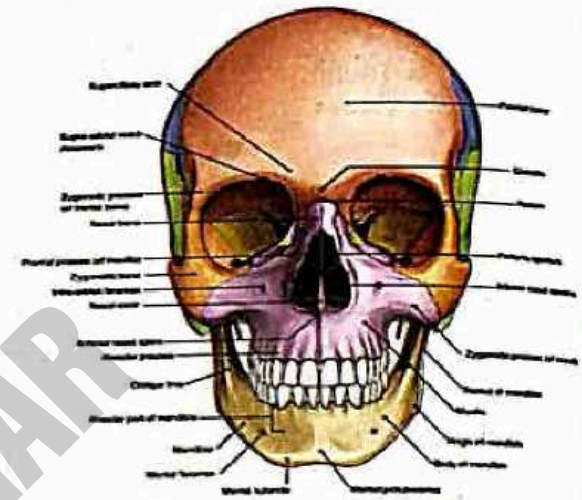
NORMA FRONTALIS

*Norma Frontalis -

*Bones - 1. Frontal Bone
2. Nasal Bone
3. Maxilla
4. Zygomatic Bone (Right and left)
5. Mandible



*Sutures - 1. Frontonasal 3. Frontomaxillary
2.Internasal 4. Zygomaticomaxillary

*Anterior aspect of skull Presents -

1. Frontal Region formed By Squamous Part of frontal Bone
2. Orbital openings
3. Prominences of the cheek formed By Zygomatic Bones
4. Bony External nose and Anterior Nasal Aperture.
5. Upper and lower Jaw Bearing teeth.

* foramen : 1. Supraorbital Foramen /Notch - Supraorbital nerve and Vessels
- Frontal diploic Vein.

2. Superior orbital Fissure - lacrimal artery and nerve
-Frontal nerve
-Trochlear nerve
-Abducent nerve
- Nasociliary nerve
*Superior and inferior division of oculomotor nerve
*Sympathetic Roots of Ciliary ganglions

3. Inferior orbital Fissure - Maxillary nerve
- Infraorbital Vessels
-Zygomatic nerve
-Vein Connecting inferior ophthalmic Vein with Pterygoid Venous Plexus.

4. Infraorbital Foramen - Infra Orbital nerve and artery → Continuation of maxillary artery
-Infraorbital vein

5. Zygomatico-Orbital foramen - Zygomatic Branch of maxillary nerve.

6. Mental Foramen.- Mental nerve and Vessels.

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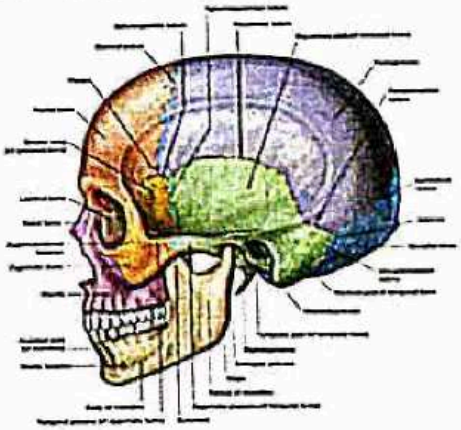
NORMA LATERALIS

*Bones - Frontal - Mandible - temporal
Parietal -Sphenoid - zygomatic
-Occipital - Maxilla - nasal

*Suture- -Coronal Suture
-Parietosquamosal suture - B/W Parietal and Squamous Part of temporal Bone
-lamdoid Suture

Other features -

1. Temporal line - Commences at the frontal Process of zygomatic Bone arches Upwards and Backwards across the Parietal Bone where it Splits in Superior and inferior Lines
2. Zygomatic arch → horizontal Bar of Bone form by temporal Process of zygomatic Bone and zygomatic Process of temporal Bone.
3. External auditory Meatus
4. Suprameatal triangle - Small depression posterosuperior to the External auditory meatus
5. Mastoid Process of temporal Bone
6. Anterior - Meeting Point of Parietomastoid, occipitomastoid and lambdoid Suture.
7. Styloid Process of temporal Bone
8. Temporal fossa and infratemporal Fossa
9. Pterion - Region in anterior Part of temporal fossa where 4 bones meet to form H Shaped structure
   1. Frontal
   2. Squamous temporal
   3. Parietal
   4. Greater Wing of sphenoid



***Foramen and Fissures of Norma lateralis**

1. Zygomatico temporal foramen - Zygomatico temporal artery and nerve Passes
2. Sphenopalatine foramen - Nasopalatine nerves & Sphenopalatine Vessels.
3. Pterygomaxillary fissure - 3rd part of Maxillary artery
   - Maxillary Nerve
4. Greater Palatine Canal - Palatine nerves and Vessels.

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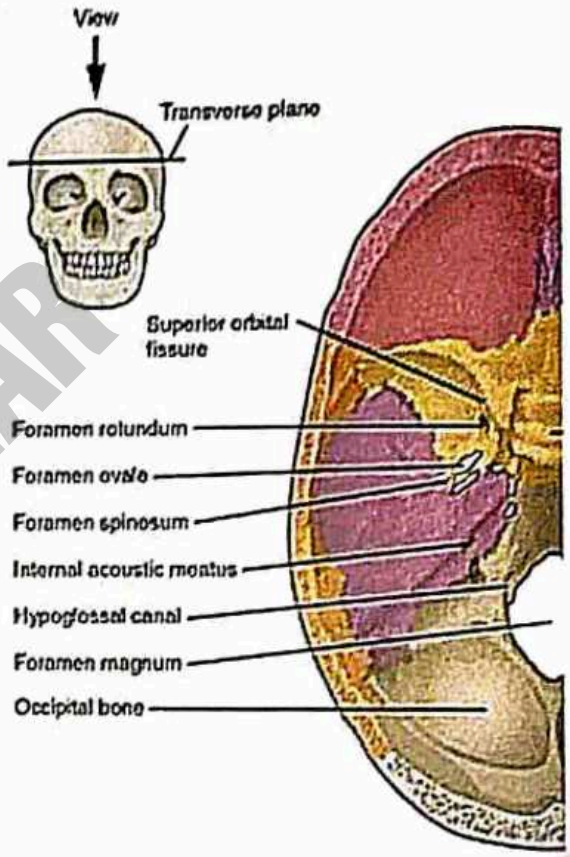
NORMA BASALIS

*Foramina and Structure Passing through : -

1. Incisive foramina - greater palatine Vessel
   -nasopalatine nerve
2. Greater palatine foramen- greater palatine nerves and Vessels
3. Lesser Palatine foramina.- Middle and Posterior palatine nerve
4. Foramen ovale - Mandibular nerve
   -Accessory middle meningeal artery
   -lesser Petrosal nerve
   -Emissary Vein
5. Foramen Rotendum - Maxillary nerve
   -Emissary Vein
6. Foramen Spinosum - Middle meningeal artery
   -Middle meningeal Vein
   -Nerve Spinous
7. Foramen Lacerum -Meningeal→
   Ascending pharyngeal Artery Branch
   -Emissary Vein
   -internal carotid artery
   -greater Petrosal nerve



8. Carotid canal - Infernal carotid artery and vein
   Emissary Vein.
9. Juglar Foramen - Anterior Part-Inferior Petrosal sinus
   -Middle part- 9th 10th 11th Cranial nerve
   -Posterior - Sigmoid sinus
10. Stylomastoid foramen - 7th Cranial nerve
    Stylomastoid artery
11. Foramen Magnum - Anterior Compartment
    Apical ligament of Dens
    Upper Band of Cruciform ligament of atlas
    Membrana tectoria
    - Posterior Compartment
    - Medulla oblangata
    -Two posterior spinal artery
    ·1x anterior Spinal artery
    · 2x Vertebral arteries
    - Sympathetic plexus
    -Spinal Root of two accessory
12. Hypoglossal Canal - 12th Cranial nerve
    - Meningeal Branch of ascending Pharyngeal artery

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CLAVICLE

• "Collar bone" • Medially - Sternum + 1st rib cartilage • Laterally - Acromion process

FEATURES
- Only long bone lying horizontally
- No medullary cavity
- Membrocartilaginous ossification
- Maybe pierced through cutaneous nerve "INTERMEDIATE SUPRACLAVICULAR NERVE"
- Subcutaneous throughout its extent
- Only long bone with 2 primary centres.
- First to ossify (8/w 5th-6th wk of IUL)
- Last to ossify (25 years)

PARTS 2 ends + Shaft

ENDS
- Medial / Sternal
  - Quadrilateral
  - with clavicular notch of manubrium sterni
- Lateral / Acromial
  - Flattened
  - with acromion process margin

SHAFT
- curved
- Medial 2/3rd round & convex forward
- Lateral 1/3rd flattened & concave forward
- Inferior surface Subclavian groove in middle third

LATERAL / ACROMIAL END
- Flattened above downward
- Oval facet articulate with medial margin of acromion and form 'Acromioclavicular Joint'
- Lateral end gives attachment to fibrous capsule of acromioclavicular joint.

MEDIAL / STERNAL END
- Saddle shaped articular surface with clavicular notch of manubrium sterni to form 'Sternoclavicular Joint'
- Gives attachment to
  - Fibrous capsule
  - Articular disc
  - Interclavicular ligament

SHAFT

LATERAL 1/3rd :- • Flattened • 2 surface and 2 borders

⇒ SURFACE
- SUPERIOR • Subcutaneous b/w attachment of deltoid and trapezius.
- INFERIOR
  1. Conoid Tubercle • Near posterior border at junction of lateral 1/4th and Medial 3/4th of clavicle
  2. Trapezoid ridge • Extend forward & laterally from conoid tubercle

  → Attachment to 'Coracoclavicular ligament'

⇒ BORDER
- ANTERIOR • Concave forward • Origin to deltoid muscle • Deltoid tubercle maybe present
- POSTERIOR • Convex forward • Insertion to trapezius muscle

MEDIAL 2/3rd :- • Cylindrical • Has 4 surface

⇒ SURFACE
- SUPERIOR • Origin of clavicular head of sternocleidomastoid muscle
- INFERIOR
  - Attachment of costoclavicular ligament to oval impression
  - Insertion of subclavius muscle into subclavian groove.
  - Attachment of clavipectoral fascia to margin of subclavian groove
  - Presence of nutrient foramen on late subclavian groove
- ANTERIOR
  - Convex forward
  - Origin to clavicular head of pectoralis major muscle.
- POSTERIOR
  - Concave backward • Origin to sternohyoid muscle.
  - Lateral part - forms anterior boundary of cervico-axillary canal.
    - Trunks of brachial plexus
    - 3rd part of subclavian artery.

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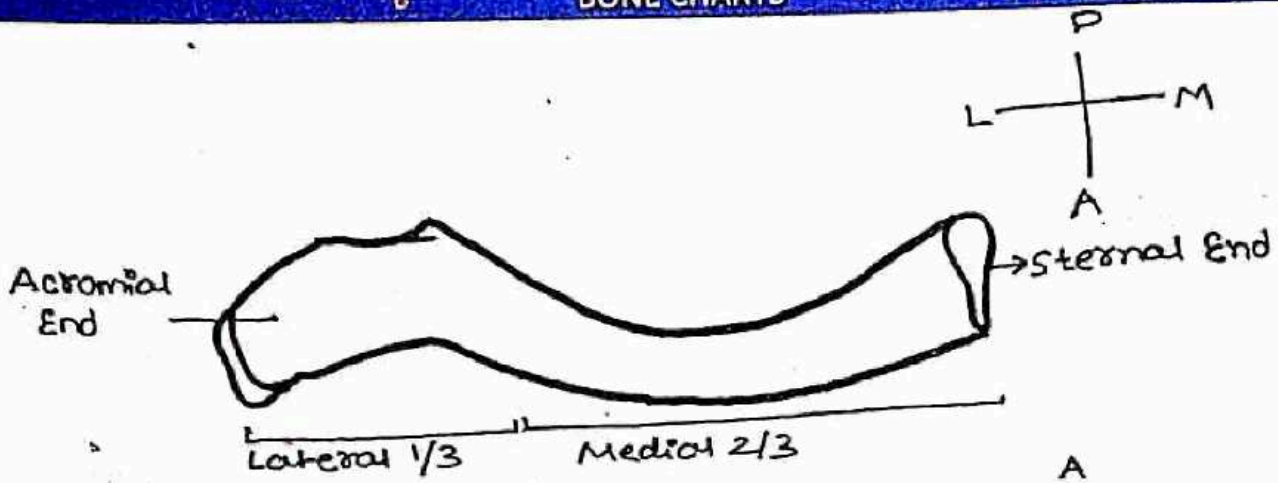
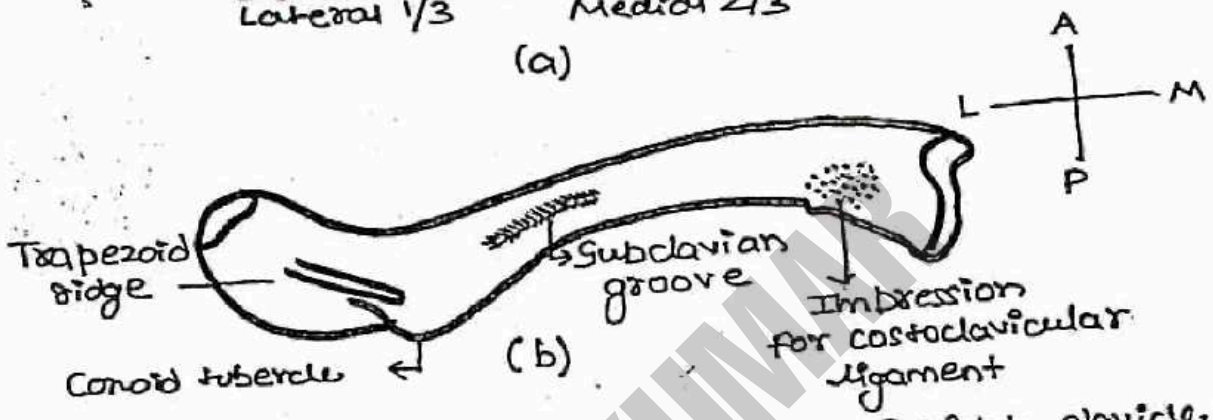
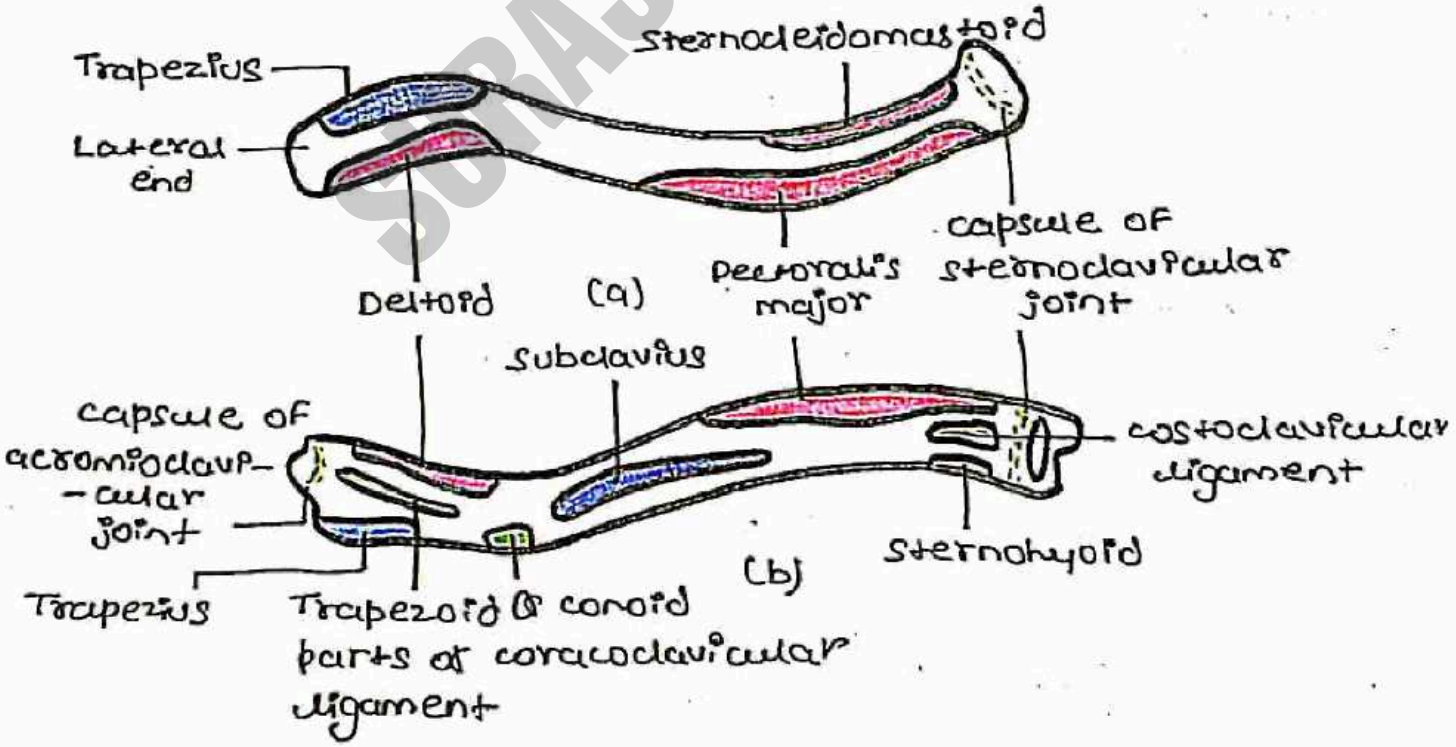


(a)



(b)

Figure showing general features of right clavicle
(a) Superior aspect (b) Inferior aspect



(a)

(b)

Figure showing attachments of right clavicle
(a) Superior aspect (b) Inferior aspect

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SCAPULA

: "Shoulder blade" • large, flattened, triangular bone • highly mobile • Upper part of posterolateral aspect of thorax, against 2nd - 7th ribs

PARTS 4 parts
- Body
- Processes spinous, acromion, coracoid

Body 2 surface + 3 borders + 3 angles

SURFACE

COSTAL SURFACE (Subscapular fossa)
- Concave & directed medially & forward
- 3 longitudinal ridge — attachment to subscapularis muscle — arising from medial 2/3rd except near subscapular bursa.

DORSAL SURFACE
- convex
- Presents spinous process (shelf-like)
  - supraspinatous fossa (1/3rd)
  - infraspinatus fossa (2/3rd)
- Spinoglenoid notch — suprascapular nerve & vessels
- Origin of supraspinatus and infraspinatous muscle from medial 2/3rd of supraspinatous fossa
- Origin of teres minor muscle from upper 2/3rd of dorsal surface of lateral border. Circumflex scapular artery interrupt its origin.
- Origin of teres major muscle from lower 1/3rd of dorsal surface of lateral border & inferior angle.
- Origin of latissimus dorsi muscle

BORDERS

SUPERIOR BORDER
- Shortest border
- Suprascapular notch on coracoid process — into suprascapular foramen by superior transverse ligament.
- Above the ligament ↳ suprascapular artery
  Below the ligament ↳ suprascapular nerve
- Origin of inferior belly of omohyoid near suprascapular notch

LATERAL BORDER
- Thickest border
- Presence of infraglenoid tubercle at upper end
- Origin of long head of triceps from this tubercle.

MEDIAL BORDER
- Thin and angled at root of spine
- Insertion of serratus anterior on costal surface.
- Insertion of levator scapulae on dorsal aspect.
- Insertion of rhomboideus minor on dorsal aspect
- Insertion of rhomboideus major on dorsal aspect.

ANGLES
- INFERIOR → lies over 7th rib or 7th intercostal space.
- SUPERIOR → Junction of superior and medial borders • lies over 2nd rib.
- LATERAL (Head of scapula)
  - → Glenoid cavity + head of humerus → 'Glenohumeral Joint' (Shoulder Joint)
  - → Glenoid labrum → margin of glenoid cavity to deepen its concavity.
  - → Capsule of shoulder joint → margin of glenoid cavity, proximal to attachment of glenoid labrum.
  - → Origin of long head of biceps brachii from supraglenoid tubercle. Origin is intracapsular.

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PROCESSES

SPINOUS PROCESS :- (Spine of Scapula)

- Junction of Upper 1/3rd and Lower 2/3rd.
- Upper Supraspinous fossa & Lower Infraspinous fossa.
- Has 2 surface & 3 borders.

- **SURFACE**
  - **SUPERIOR**
    - Lower boundary of supraspinous fossa
    - Origin to supraspinatus.
  - **INFERIOR**
    - Upper boundary of infraspinous fossa
    - Origin to infraspinatus.
- **BORDERS**
  - **ANTERIOR**
    - Attached to dorsal surface
  - **LATERAL**
    - Spinoglenoid notch — through which passes suprascapular nerve and vessels.
  - **POSTERIOR**
    - 'Crest of spine'
    - Insertion of trapezius on the upper lip of crest of spine.
    - Origin of posterior fibers of deltoid from lower border of crest of spine.

ACROMION PROCESS :- (Acromion)

- Projects forward almost at right angle from lateral end of spine & overhangs glenoid cavity.
- Superior surface — subcutaneous.
- Has tip, 2 borders, 2 surface.

- **SURFACE**
  - **SUPERIOR**
    - Rough & subcutaneous
    - Related to 'Subacromial bursa'.
  - **INFERIOR**
    - Smooth
- **BORDERS**
  - **MEDIAL**
    - Continues to form upper lip of crest of spine
    - Insertion to trapezius muscle
    - Circular facet articulate with lateral end of clavicle to form 'Acromioclavicular Joint'.
  - **LATERAL**
    - Continues to form lower lip of crest of spine
    - Origin to intermediate fibers of deltoid muscle.
- Attachment of coracoacromial ligament to the tip of acromion.
- Acromial angle — junction of lateral border of acromion and lateral border of crest of spine of scapula.

CORACOID PROCESS :-

- From upper part of head of scapula
- Bent sharply to project forward and slightly laterally.
- Origin of short head of biceps brachii and coracobrachialis from tip by a common tendon.
- Insertion of pectoralis minor muscle on medial border of upper surface.
- Attachment of coracoacromial ligament to its lateral border.
- Coracoclavicular ligament
  - conoid part (rhomboid ligament) — knuckle.
  - trapezoid part (rhomboid ligament) — ridge on superior aspect b/w pectoralis minor muscle and coracoacromial ligament.
- Coracohumeral ligament — root adjacent to glenoid cavity.

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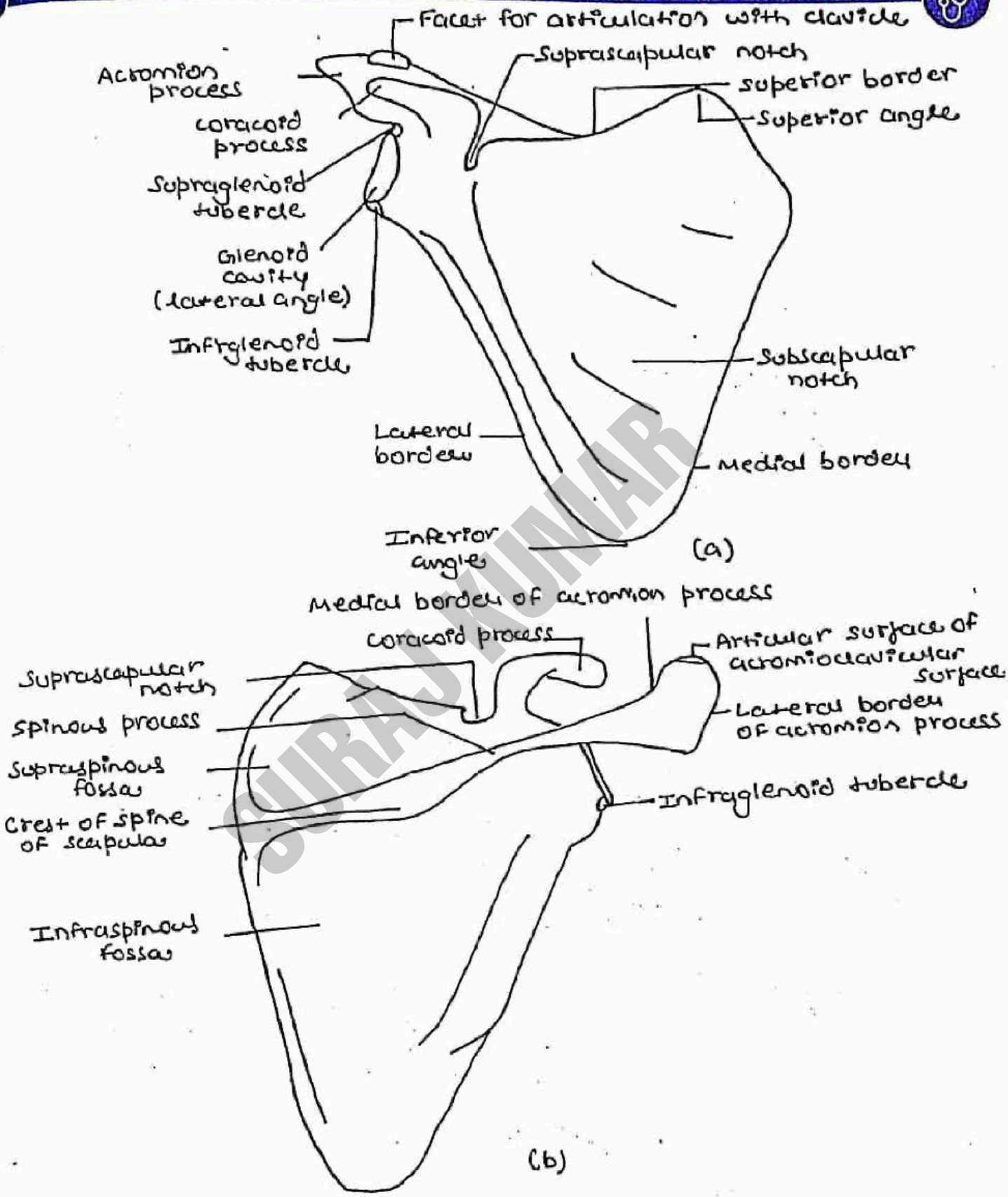

General Features of right scapula (a) Costal Surface (b) Dorsal surface

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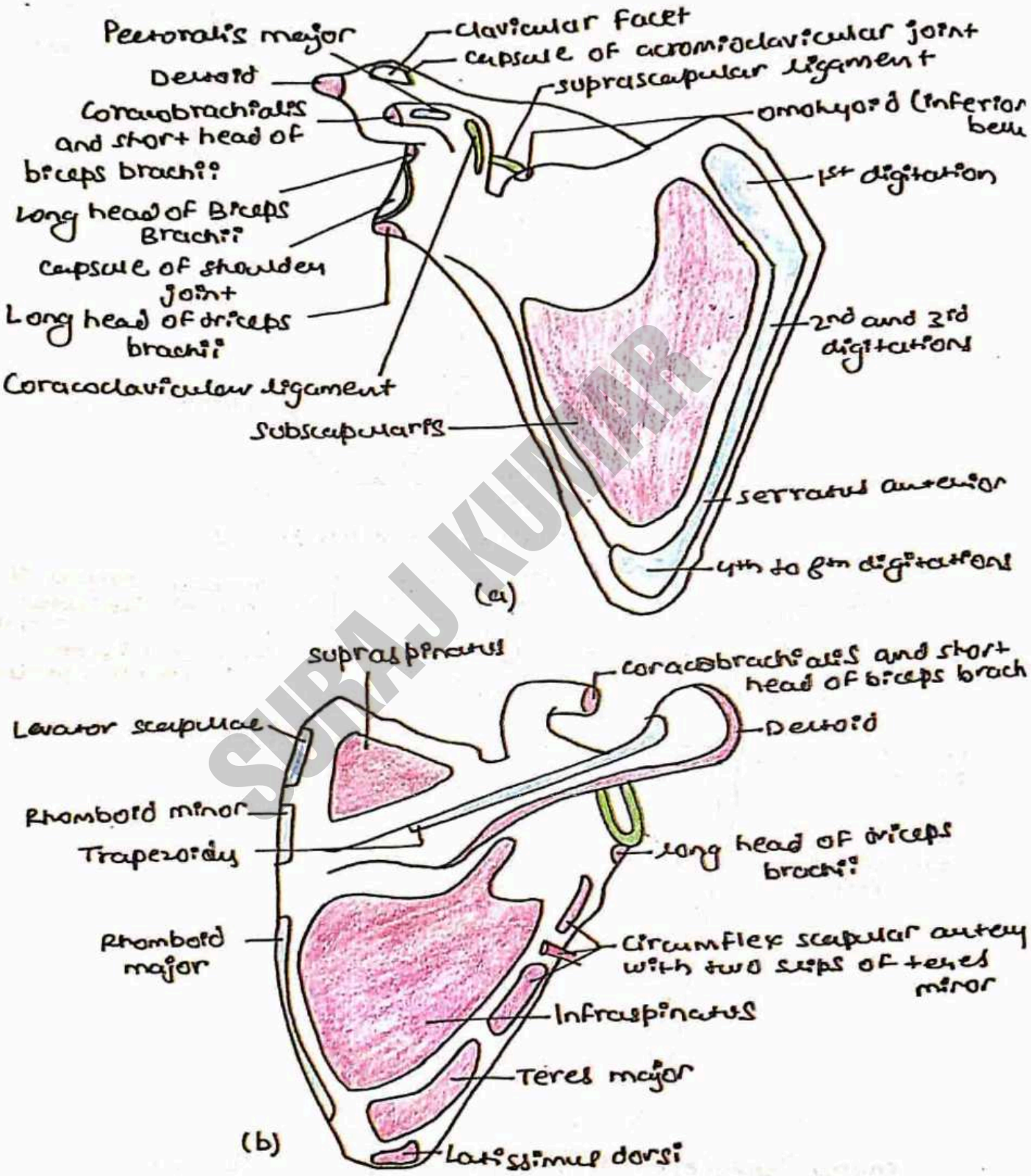

Attachments of right scapula (a) Coastal aspect (b) Dorsal aspect

SUBAJ KUMAR

HUMERUS

PARTS 2 ends + Shaft

UPPER END
1. Head : smooth rounded
   - forms glenohumeral joint
2. Neck
3. Greater tubercle
4. Lesser tubercle
5. Intertubercular sulcus.

SHAFT
- B/w upper & lower end
- Upper half - cylindrical
  Lower half - flattened
  - anteroposteriorly

LOWER END
1. Capitulum (lateral round convex projection)
2. Trochlea (medial pulley shaped)
3. Radial fossa (above capitulum)
4. Coronoid fossa (above trochlea)
5. Medial epicondyle
6. Lateral epicondyle
7. Olecranon fossa (deep hollow on posterior aspect above trochlea)

UPPER END

→ 1. HEAD :- . Smooth, round, 1/3rd of sphere . Covered by articular hyaline cartilage

→ 2. NECK
- → Anatomical :- . Provides attachment to capsular ligament of shoulder joint
  except, superiorly — for passage of tendon of long head of biceps brachii
  medially — capsule extends down for about 1-2 cm.
- → Morphological :- . B/w diaphysis and epiphysis . true junction of head with shaft
  . Represented by epiphyseal line
- → Surgical :- . Below greater and lesser tubercles / epiphyseal line
  . Related to axillary nerve and posterior & anterior circumflex humeral vessels.
  . One of sites of common humeral fractures.

→ 3. GREATER TUBERCLE :- . Most lateral part of proximal end
. Posterosuperior aspect — flattened facet like impressions
- upper - supraspinatus muscle
- middle - infraspinatus muscle
- lower - teres minor muscle

→ 4. LESSER TUBERCLE :- . Small elevation on front of upper end (just above surgical neck)
. Subscapularis muscle.

→ 5. INTERTUBERCULAR SULCUS :- (Bicipital Groove)
. Vertical groove
. Contains
1. long head of biceps in synovial sheath
2. Ascending branch of anterior circumflex humeral artery.

. Lateral lip of groove - Pectoralis major muscle
Floor of groove - Latissimus dorsi muscle
Medial lip of groove - Teres major muscle

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SHAFT

- Upper part – cylindrical
  lower part – triangular
- has 3 borders and 3 surface

SURFACE

- **Antero-lateral**
  - V-shaped deltoid tuberosity
  - Insertion to deltoid muscle.
- **Antero-medial**
  - Upper part – forms floor of intertubercular sulcus.
  - Nutrient foramen – about its middle and close to medial border – directing downward.
- **Posterior**
  - Upper 1/3rd – oblique ridge directing downward and laterally.
    - Origin to lateral head of triceps brachii.
  - Below & medial to oblique ridge – radial/spiral groove
    - Radial nerve & profunda brachii vessels.
  - Below spiral groove
    - origin to medial head of triceps brachii.

BORDERS

- **Anterior**
  - From lateral lip of intertubercular sulcus to anterior margin of deltoid tuberosity and ends in radial fossa
- **Medial**
  - Medial supracondylar ridge – medial intermuscular septum
  - Rough strip on middle – insertion to coracobrachialis muscle
  - Narrow area above medial epicondyle
    - origin to humeral head of pronator teres muscle.
- **Lateral**
  - Upper part – indistinct lower part – lateral supracondylar ridge
  - Above its middle – crossed by radial groove from behind.
  - lower part – lateral supracondylar ridge – lateral intermuscular septum.

LOWER END

- Capitulum articulate with head of radius
- Trochlea articulate with trochlear notch of ulna.
- Ulnar nerve – posterior surface of medial epicondyle.
- Anterior surface of medial epicondyle
  - common flexor origin of superficial flexors of forearm
- Anterolateral part of lateral epicondyle – common extensor origin
- Posterior surface of lateral epicondyle – Origin to anconeus muscle.
- Coronoid process of ulna – coronoid fossa (above trochlea) during full flexion
- Head of radius – radial fossa (above capitulum) during full flexion
- Olecranon process of ulna – olecranon fossa (posterior aspect above trochlea) during full extension
- Capsule of elbow joint is attached:
  Anteriorly: Above coronoid and radial fossae

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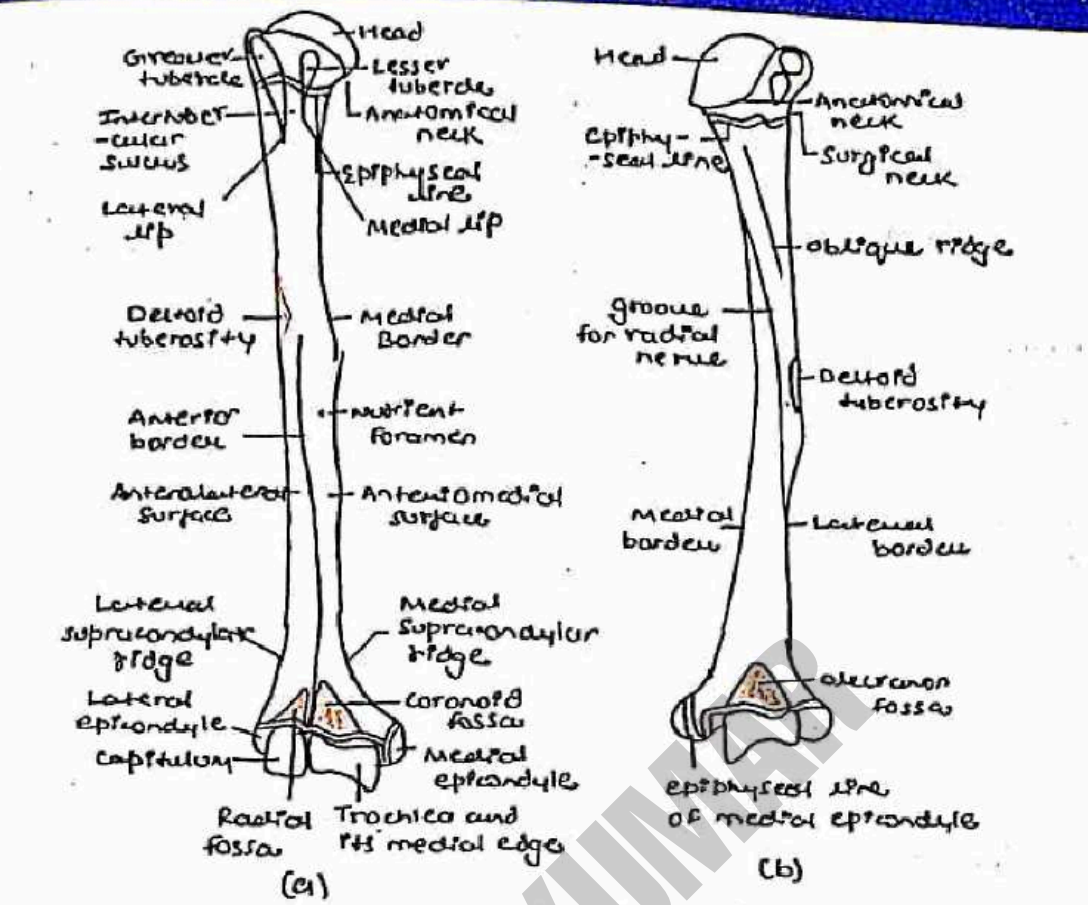

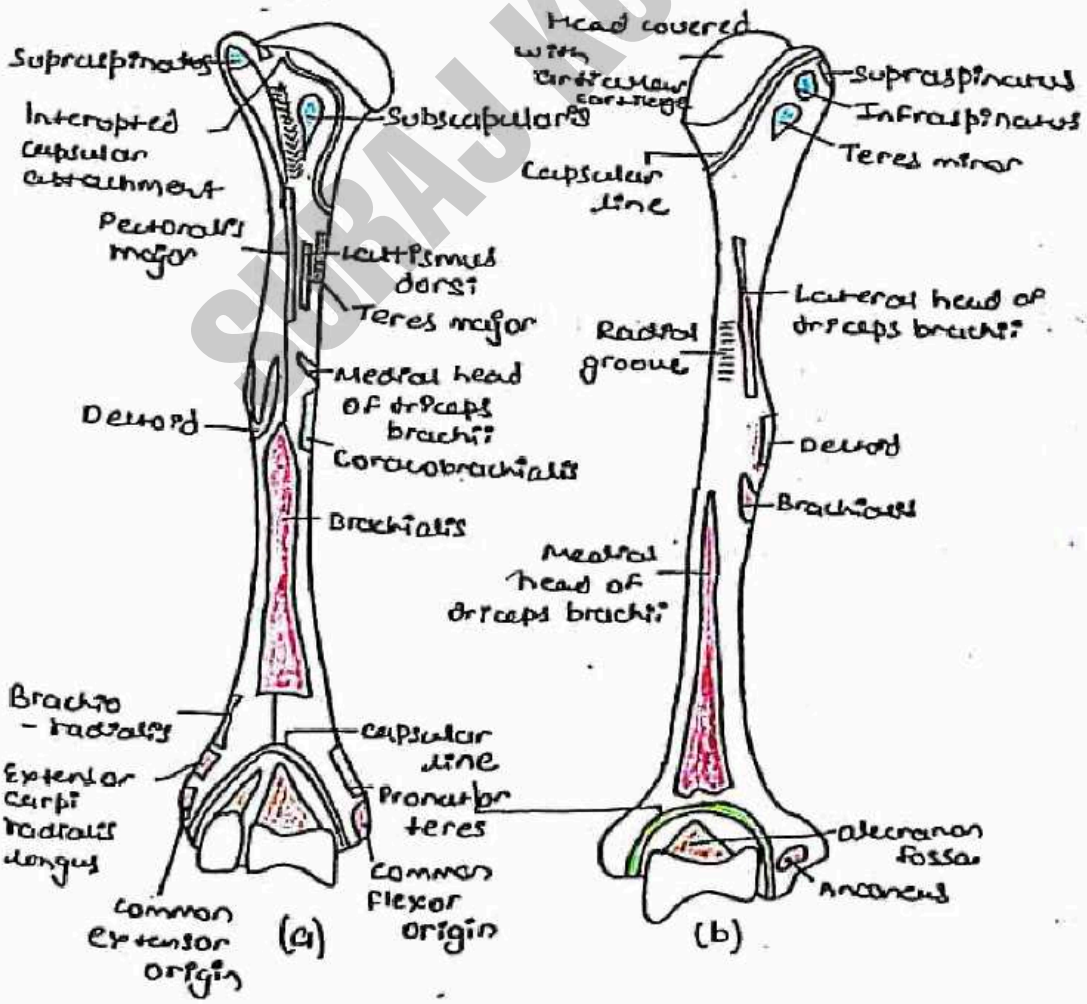

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Attachments of right humerus

(a) Anterior view (b) Posterior view

RADIUS

- Lateral bone of forearm
- Homologous to tibia

PARTS - 2ends + Shaft

UPPER END
- Head + Neck + Radial Tuberosity
- Head : - disc shaped
  - articulate with capitulum of humerus
- Neck : - constricted
- Radial Tuberosity : - just below medial part of neck

SHAFT
- Presents lateral convexity
- Sharpest interosseous border on medial side

LOWER END
- widest part
- 5 surfaces
- Lateral surface – styloid process
- Dorsal surface – Lister's tubercle

UPPER END
- HEAD :- · Humero-radial articulation · Annular ligament
- NECK :- · Quadrate ligament
- RADIAL TUBEROSITY :- · Insertion of tendon of biceps brachii

SHAFT - Has 3 borders & 3 surfaces.

SURFACE
- Anterior
  - Origin of Flexor policis longus — Upper 2/4th.
  - Insertion of Pronator quadratus — lower 1/4th.
  - Presence of nutrient foramen
    Nutrient artery — Anterior interosseous artery's branch.
- Posterior
  - Origin of Abductor policis Longus (APL) — middle 1/3rd.
  - Origin of Extensor policis brevis (EPB)
- Lateral
  - Insertion of Supinator — upper 1/3rd
  - Insertion of Pronator teres

BORDERS
- Anterior · Anterior oblique line — Origin of Flexor Digitorumsuperficialis (FDS)
- Posterior · Posterior oblique line
- Medial (Interosseous border) · Sharpest · Attachment of interrosseous membrane.

LOWER-END · Widest part with 5 surface

SURFACE
- Anterior · Palmar radio-carpal ligament of wrist joint
- Posterior · Dorsal tubercle of radius (Lister's tubercle) — for tendon of extensor policis longus.
  · Lateral to groove - ECRL & ECRB
  Medial to groove - Extensor policis longus — passes tendons of extensor digitorum and extensor indicis.
- Medial · Articular disc of inferior radio-ulnar joint
- Lateral · Tendons of Abductor pollicis longus & Extensor pollicis brevis.
  · Insertion of brachioradialis
- Inferior · Lateral Δr area — scaphoid
  Medial □r area — lateral part of lunate

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ULNA

- Medial bone of forearm
- Homologous to fibula.

PARTS - 2 ends + Shaft

UPPER END
- Trochlear notch b/w
  - Above : Olecranon process
  - Below : Coronoid process

SHAFT
- Lateral border (interosseous border) is sharp crest like.

LOWER END
- Has head & styloid process

UPPER END - Has 2 process & 2 notches.

PROCESS

OLECRANON - Has 5 surface

- **SURFACE**
  - **UPPER** - Insertion to triceps brachii - Capsular ligament of elbow jt
  - **ANTERIOR** - Upper part of trochlear notch.
  - **POSTERIOR** - Subcutaneous olecranon bursa
  - **MEDIAL** - Attachments to :-
    - Origin of Flexor carpi ulnaris (FCU)
    - Posterior & Oblique bands — Ulnar collateral ligament. (UCL)
  - **LATERAL**

CORONOID - Has 4 surface

- **SURFACE**
  - **SUPERIOR** - Lower part of trochlear notch
  - **ANTERIOR** - Ulnar tuberosity - Insertion of Brachialis muscle
    - Sublime tubercle :-
      - Ant band of UCL
      - Oblique band of UCL
      - Humero ulnar head of FDS
      - Pronator teres
      - Flexor pollicis longus
  - **MEDIAL** - Origin of Flexor digitorum Profundus.
  - **LATERAL** - Radial notch
    - Supinator fossa — Accommodates Radial tuberosity (during supination & pronation)
    - Origin to Supinator muscle

NOTCHES

- **TROCHLEAR** - Semilunar - With trochlea of humerus
  - Attachments to capsule of elbow joint
- **RADIAL** - Superior radio ulnar joint.

SHAFT - 3 surface + 3 borders

SURFACE

- **ANTERIOR**
  - Origin of Flexor digitorum profundus (Upper 3/4th)
  - Origin of Pronator quadratus (Lower 1/4th)
  - Presence of nutrient foramen
- **MEDIAL** - Origin of Flexor digitorum profundus (Upper 2/3rd)
- **POSTERIOR**
  - Insertion of Anconeus (above oblique line)
  - Below oblique line — larger medial, smaller lateral

BORDERS

- **LATERAL (interosseous)** - Sharpest - Attachment to interosseous membrane
- **ANTERIOR** - Origin to Flexor digitorum profundus
- **POSTERIOR** - Subcutaneous
  - Attachment by common aponeurosis
    1. Flexor digitorum profundus
    2. Flexor carpi ulnaris
    3. Extensor carpi ulnaris

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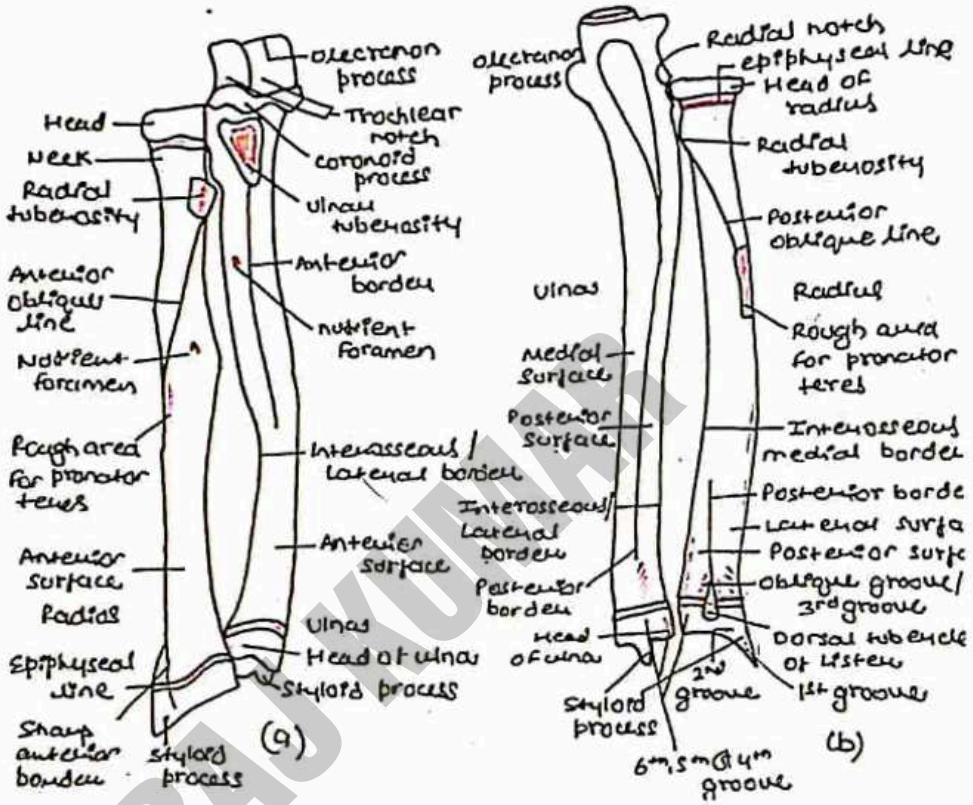
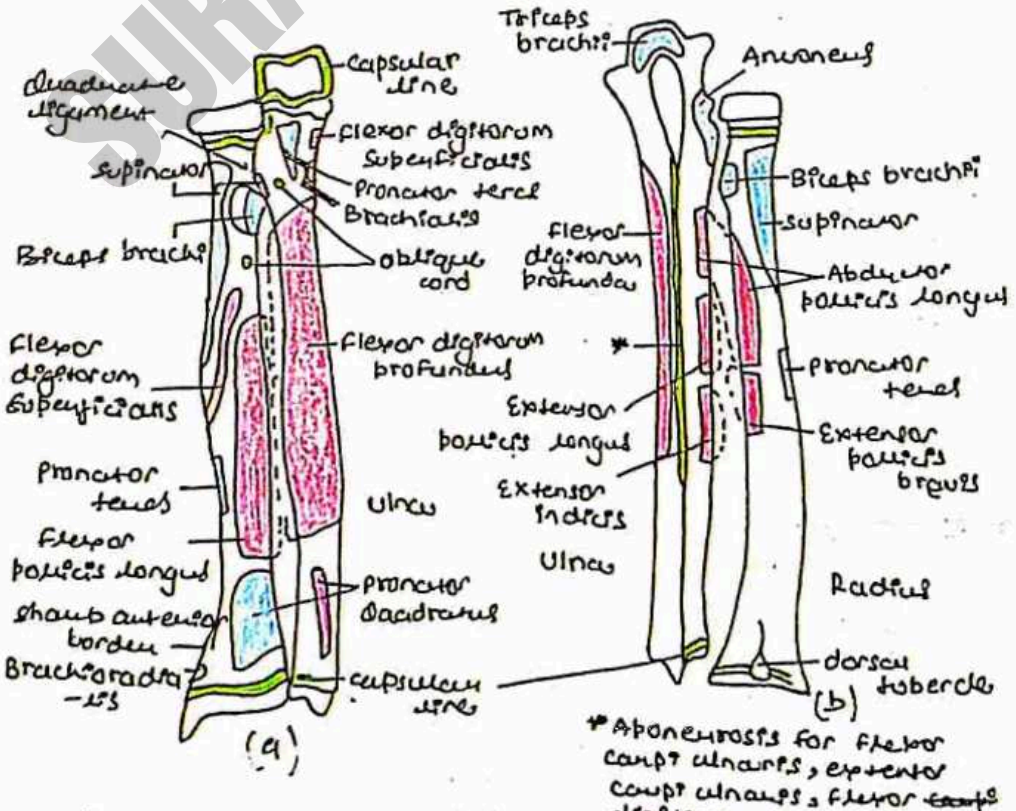
LOWER END · Head + Styloid process

- **HEAD** · Ulnar notch of radius — Inferior radio ulnar joint
- **STYLOID PROCESS**
  - Tip — medial collateral ligament of wrist joint
  - Apex — depression b/w head & base of process
  - Groove — tendon of extensor carpi ulnaris





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HIP BONE

. Also known as 'Coxae', 'Innominatum' · Fusion of Ilium, Ischium & Pubis

ILIUM

. 2 ends + 3 borders + 3 surfaces.

→ ENDS → Upper (Iliac crest)
Anterior superior iliac spine → Above : Inguinal ligament
Below : Sartorius
Posterior superior iliac spine → Sacrotuberous ligament

A. VENTRAL SEGMENT — Outer lip + intermediate area + Inner lip

1. Outer lip – tubercle of iliac crest
– Attachments to
· origin of tensor fascia latae
· insertion of external oblique muscle
· origin of latissimus dorsi
2. Intermediate area · Origin of internal oblique muscles.
3. Inner lip · Anterior 2/3rd : Transversus abdominis muscle
· Posterior 1/3rd : Quadratus lumborum

B. DORSAL SEGMENT
1. Outer sloping surface : Gluteus maximus muscle
2. Inner sloping surface : Erector spinae muscle

→ Lower

BORDERS → Anterior · Anterior inferior iliac spine
upper : straight head of rectus femoris
lower : iliofemoral ligament

→ Posterior · Posterior inferior iliac spine
Greater sciatic notch → foramina
Ischial spine
Lesser sciatic notch → foramina
· Structures passing from dorsal aspect of Ischial spine (Lateral → medial)
– Pudendal nerve
– Internal pudendal vessels
– Nerve to obturator internus.

→ Medial · Arcuate line

SURFACE → Iliac fossa · Origin of iliacus

→ Gluteal · 4 areas by 3 gluteal lines.
· Origin to 4 muscles
Gluteus maximus behind posterior gluteal line
Gluteus medius b/w anterior and posterior gluteal line
Gluteus minimus b/w anterior and inferior gluteal line
Rectus femoris b/w inferior gluteal line and upper margin of acetabulum.

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SACROPELVIC
- A. SACRAL PART
  1. Iliac tuberosity – attachments to ligaments
     a. iliolumbar
     b. dorsal sacroiliac
     c. interosseous sacroiliac
  2. Auricular surface – sacroiliac joint
- B. PELVIC PART – Pre auricular sulcus
  – Origin to obturator internus.

ISCHIUM

- Body – 2 ends + 3 borders + 3 surface & Ramus

→ BODY

ENDS → Upper
↳ Lower

- Ischial tuberosity
  → Upper quadrilateral area → upper lateral
  ↳ lower medial
  ↳ lower triangular area → outer
  ↳ inner
- Origin to 4 muscles.
  1. Semimembranosus
  2. Semitendinosus
  3. Semitendinosus + long head of biceps femoris
  4. Adductor magnus
- Medial margin → lower end of sacrotuberous ligament.

BORDERS → ANTERIOR – Attachment to obturator membrane

→ POSTERIOR – Ischial spine → attachments to
  – sacrospinous ligament
  – levator ani
  – coccygeus muscle
- Lesser sciatic notch
  → upper end : Gemellus superior
  ↳ lower end : Gemellus inferior

↳ LATERAL – tendon of obturator externus.

SURFACE → Femoral
- Obturator externus – along obturator foramen
- Quadratus femoris – along lateral border of ischial tuberosity.

→ Dorsal – Tendon of obturator internus + 2 gemelli — traverses the groove

↳ Pelvic – Origin to obturator internus.

→ RAMUS – Conjoint ischiopubic ramus – 2 borders + 2 surface

BORDERS → Upper – Attachment to Obturator membrane
↳ Lower

SURFACE → Outer – Linear attachment to 4 muscles
1. Obturator externus
2. Adductor magnus
3. Adductor brevis
4. Gracilis

↳ Inner – 2 ridges + 3 areas

- Ridges → Upper – Attachment to superior fascia of urogenital diaphragm
  Lower – Attachment to perineal membrane & falciform process of sacrotuberous ligament.

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- Area
  - Upper — origin to obturator internus
  - Middle — Attachment to sphincter urethrae & deep transversus perinei muscles.
  - Lower — Attachments to crus of penis, clitoris, ischiocavernosus & superficial transversus perinei muscles.

PUBIS

- 'Pubic bone / OS Pubis'.
- Body + superior ramus + inferior ramus.

→ BODY
- Pubic crest + Pubic tubercle + 3 surface.

, PUBIC CREST & PUBIC TUBERCLE

. SURFACE
- Anterior
  - Attachments to 4 Muscles
    1. Adductor longus
    2. Gracilis
    3. Adductor brevis
    4. Obturator externus
- Posterior (Pelvic) — Related to Urinary bladder.
- Medial (Symphyseal) — Symphysis pubis – 2° cartilaginous joint.

→ SUPERIOR RAMUS — 3 borders + 3 surfaces.

. BORDERS

→ Pectineal line (Posterior / superior)
- 'Pecten pubis'
- Attachments to
  - Medial
    - Lacunar ligament
    - inguinal ligament
    - conjoint tendon
    - Fascia transversalis
  - Lateral
    - Pectineal ligament (Ligament of cooper)
    - Pectineus muscle
    - Pectineus fascia
    - Psoas minor (if present)

→ Obturator crest (Anterior)

↳ Inferior — Attachment to obturator membrane leaving a gap for obturator canal.

SURFACE
- Pectineal — Origin of pectineus muscle.
- Pelvic — Featureless & covered by peritoneum.
- Obturator — Obturator nerve & vessels traverses the groove.

→ INFERIOR RAMUS — form conjoint ischiopubic ramus.

OBTURATOR FORAMEN
- Attachment of obturator membrane except the groove to convert it into obturator canal
- Obturator canal transmits obturator nerves & vessels.

ACETABULUM
- Shallow vinegar cup
- Acetabular notch → acetabular foramen due to transverse acetabular ligament
- Acetabular labrum (fibrocartilaginous rim) ↳ margin of acetabulum
- Acetabular fossa – filled with pad of fat
- Lunate surface – covered by hyaline cartilage

Articular – Lunate surface
Non articular area – Acetabular fossa

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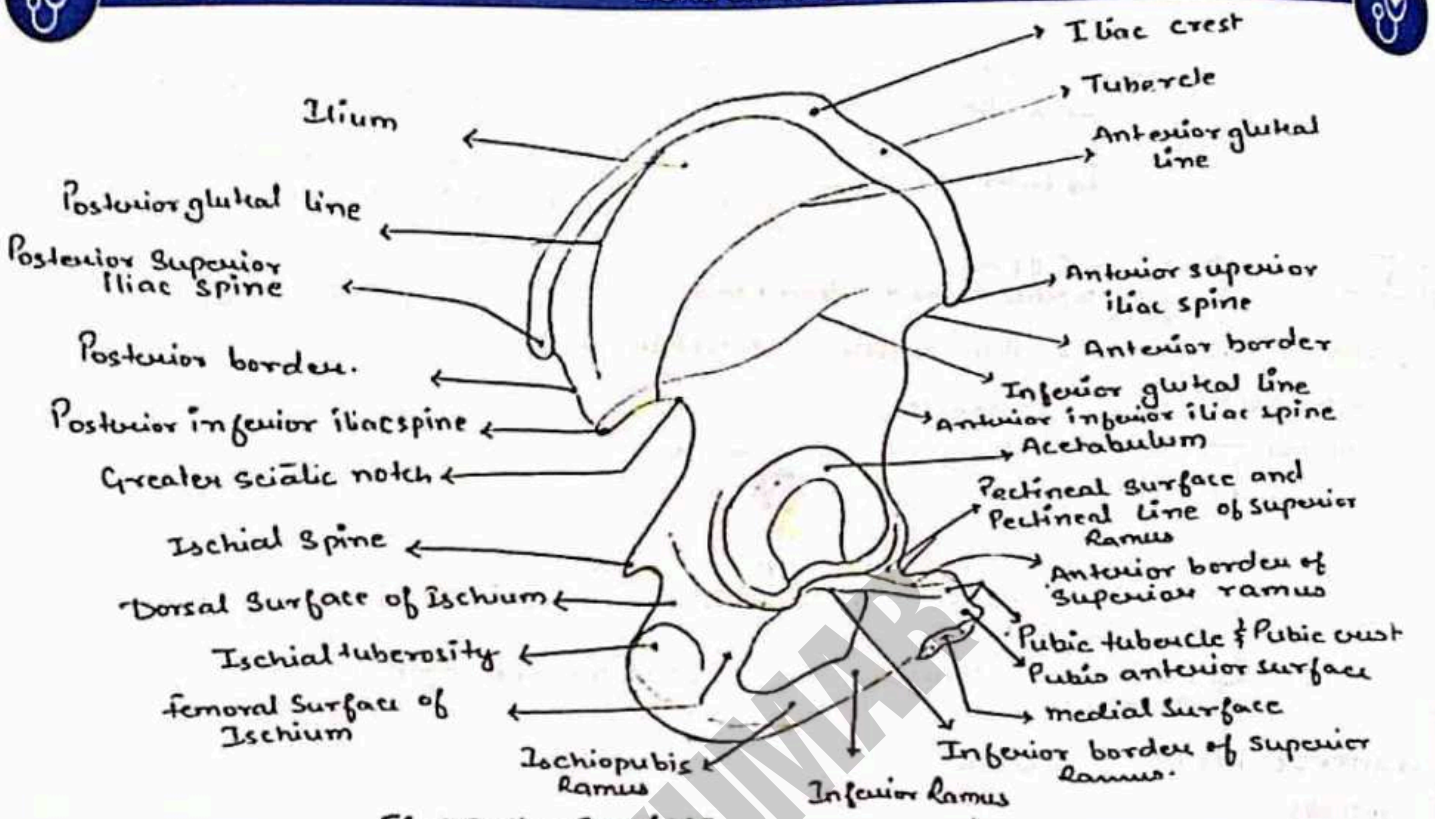

Fig: Outter Surface

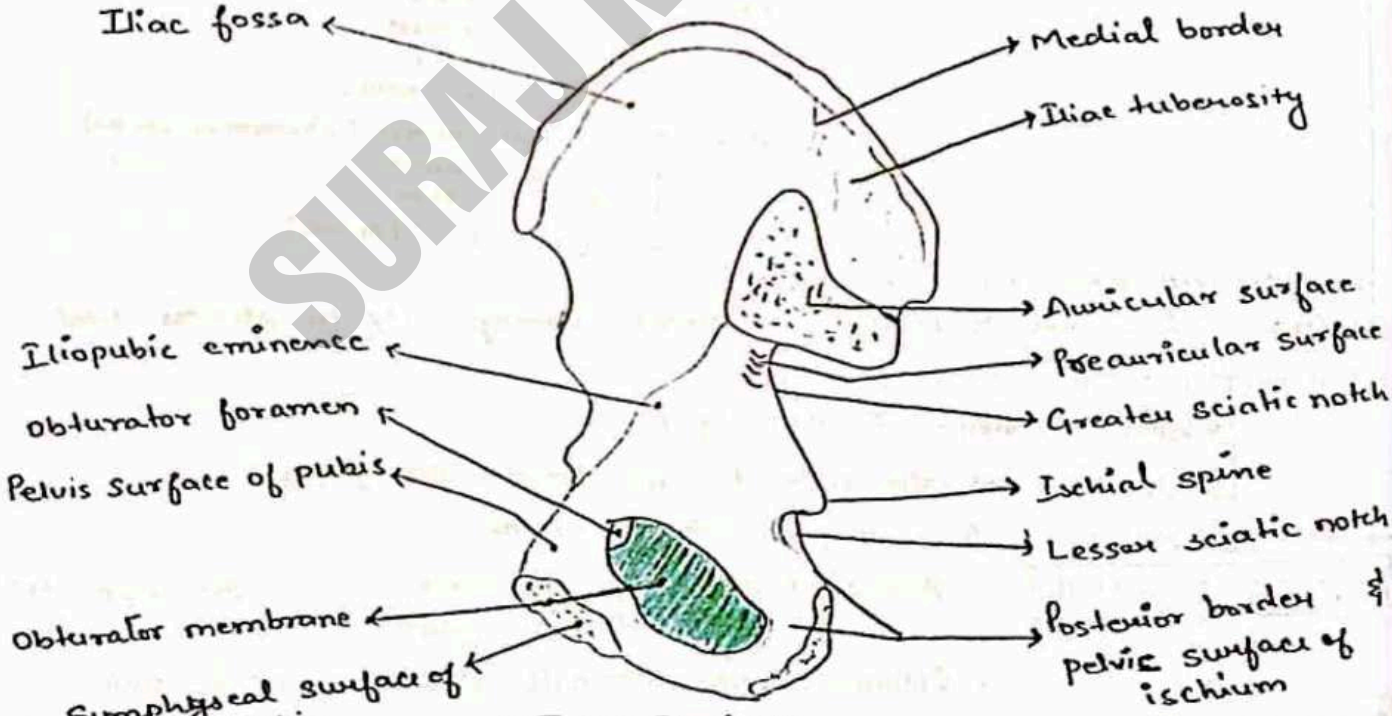

Fig: Inner Surface

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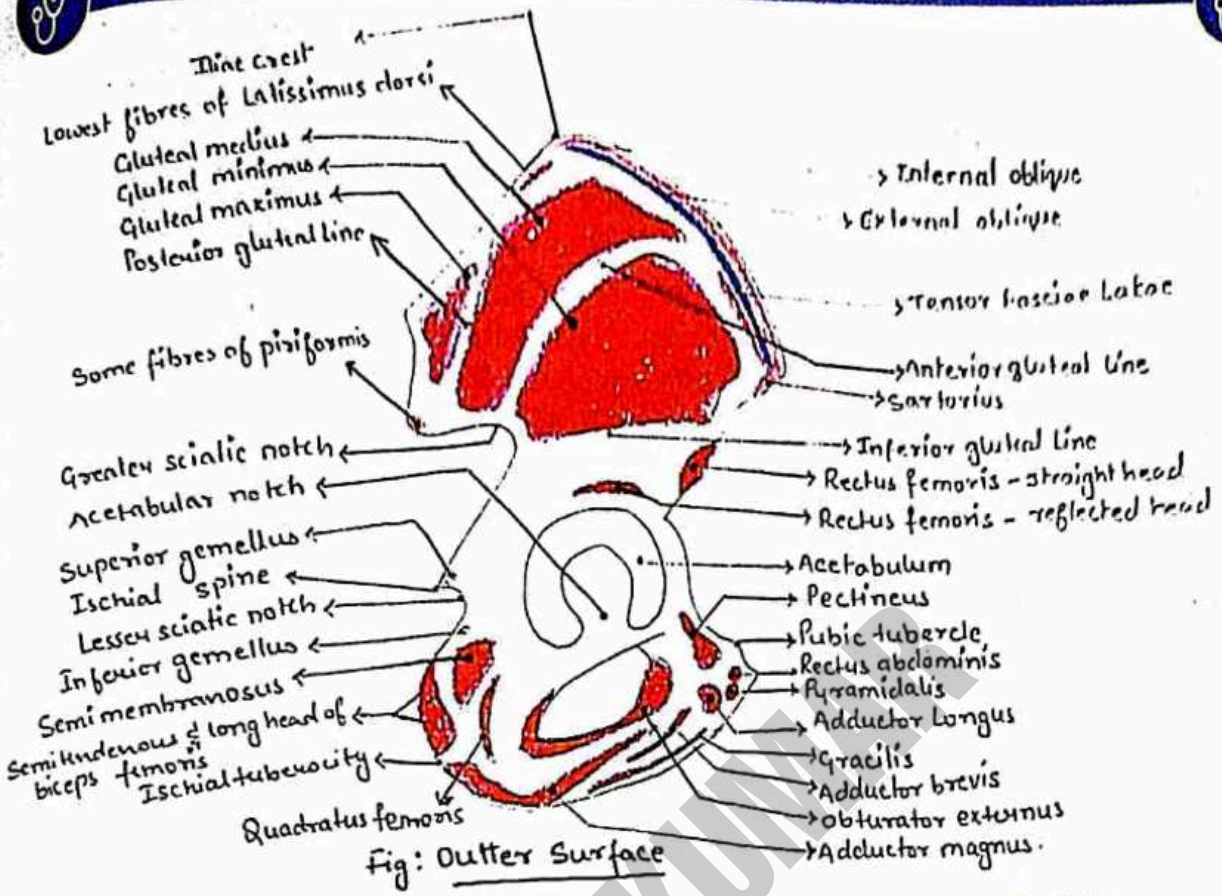

Fig: Outter Surface

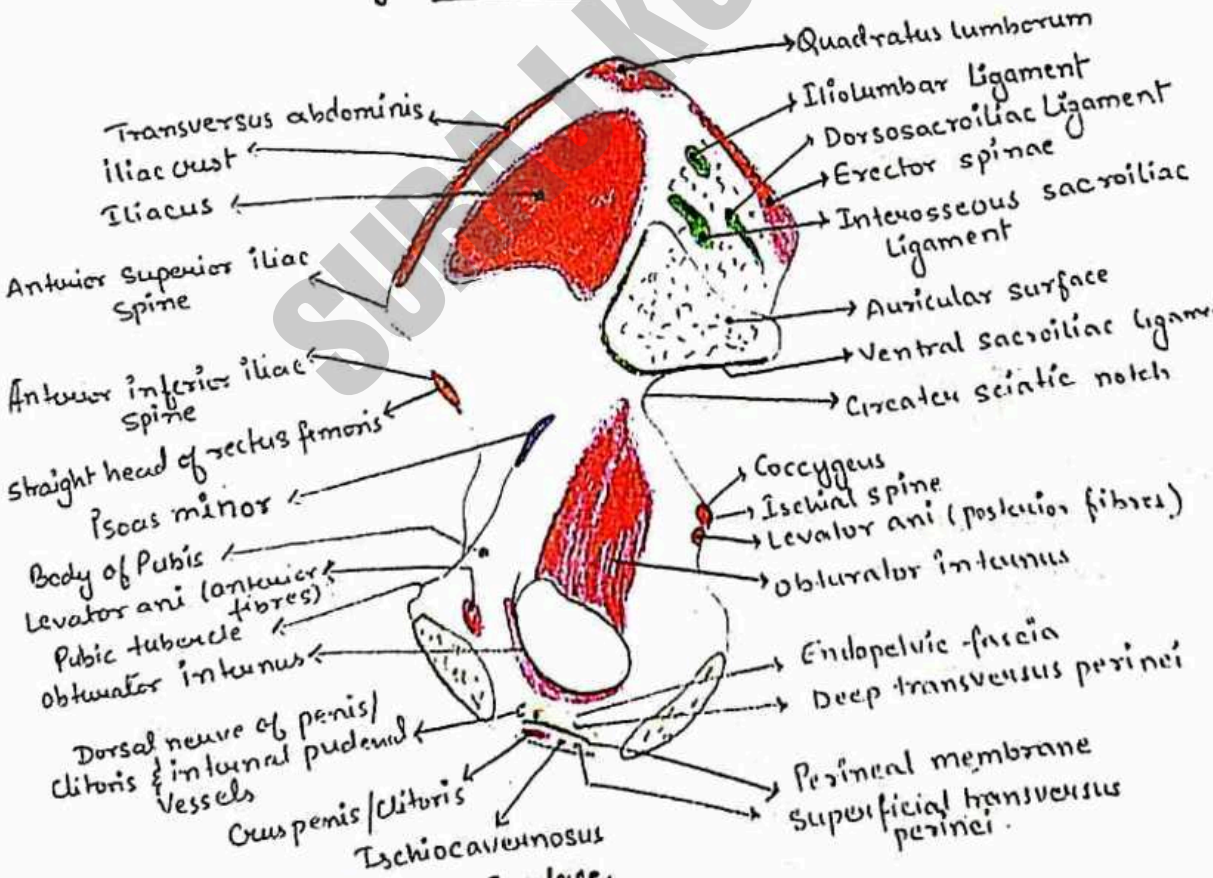

Fig: Inner Surface.

SUJIT KUMAR

⋗ FEMUR ⋖

- Longest & strongest

PARTS

UPPER END
- Head + Neck + Greater and Lesser trochanter

SHAFT
- Narrowest in middle

LOWER END
- Medial + Lateral condyles
- Intercondylar fossa

UPPER END

→ HEAD
- Acetabulam of hip bone — hip joint
- Fovea — ligamentum teres femoris.

→ NECK
- 2 borders and 2 surface

SURFACE → ANTERIOR
- Flat + no. of oblique bony ridges
- intertrochanteric line
- Completely intracapsular

★ INTERTROCHANTERIC LINE
- Spiral line
- Attachment to :
  1. Capsule of hip joint
  2. Iliofemoral ligament
  3. Upper end : Vastus lateralis
     Lower end : Vastus medialis

→ POSTERIOR
- intertrochanteric crest
- Medial half — intracapsular
- Tendon of obturator externus.

★ INTERTROCHANTERIC CREST
- Quadrate tubercle — Quadratus femoris

BORDER → UPPER
- Greater trochanter

→ LOWER
- Lesser trochanter.

→ GREATER TROCHANTER
- 1 border & 3 surface

BORDER → UPPER
- Piriformis

SURFACE → ANTERIOR
- Gluteus minimus - Lateral
- Trochanteric bursa of Gluteus minimus — Medial

→ MEDIAL
- Trochanteric fossa — obturator externus.
- Insertion of obturator internus + Gemellus superior and Gemellus inferior.

→ LATERAL
- Ridge — Gluteus medius muscle
- Ar areas — Anterior - Gluteus medius
  Posterior - Gluteus maximus

→ LESSER TROCHANTER
- Insertion to Psoas major
- Attachment of Iliacus.

SURA KUMAR

SHAFT

- It has 5 surface & 3 borders

SURFACE → ANTERIOR, MEDIAL, LATERAL, UPPER POSTERIOR, LOWER POSTERIOR (POPLITEAL SURFACE)

BORDERS → MEDIAL, LATERAL, POSTERIOR

LINEA ASPERA (Posterior border)

- Has inner lip + outer lip + intermediate area

UPPER END :-
- ⇒ Inner Medial Lip → spiral line
- ⇒ Outer lateral lip → Gluteal tuberosity
- ⇒ Posterior surface

LOWER END :-
- ⇒ Inner Medial lip → Medial Supracondylar line → Adductor tubercle
- ⇒ Outer lateral lip → Lateral supracondylar line
- ⇒ Popliteal surface

ATTACHMENTS → INTERMUSCULAR SEPTA
- Medial intermuscular septum – Medial lip
- Lateral intermuscular septum – Lateral lip

ATTACHMENTS → MUSCULAR
- Vastus intermedius
- Articularis genu
- Vastus lateralis
- Vastus medialis
- Gluteus maximus
- Adductor longus
- Pectineus
- Adductor brevis
- Adductor magnus
- Gastrocnemius
- Plantaris

LOWER END

MEDIAL CONDYLE
- Attachment to upper end of medial collateral ligament
- Adductor tubercle - ischial head of adductor magnus

INTERCONDYLAR FOSSA
- Medial wall – upper end of posterior cruciate ligament
- Lateral wall – upper end of anterior cruciate ligament [LAMP]

LATERAL CONDYLE
- Attachment to fibular collateral ligament of knee jt.
- Origin of lateral head of gastrocnemius.
- Attachment to popliteus

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Fig: Anterior aspect

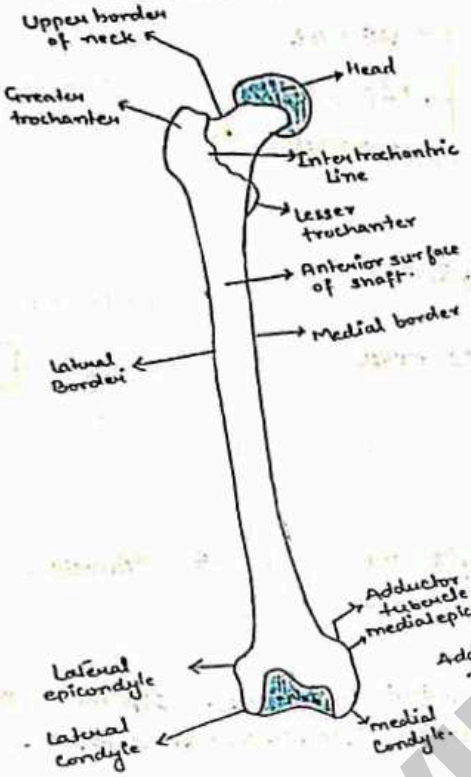
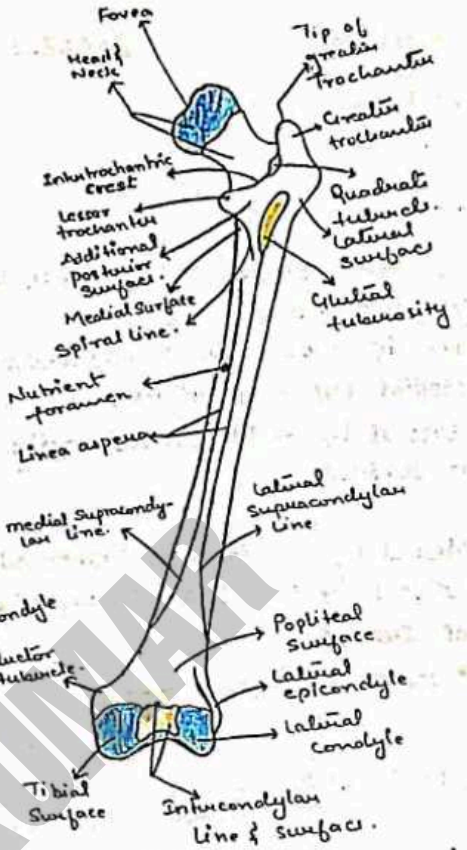
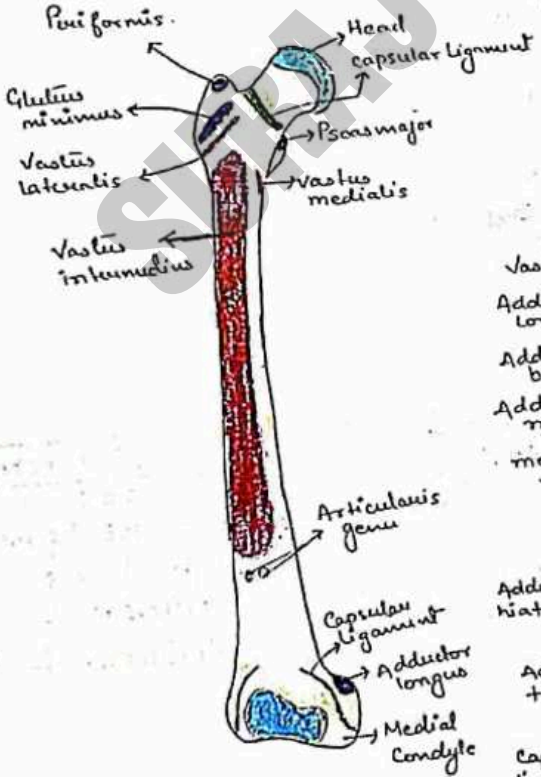


Fig: Posterior aspect





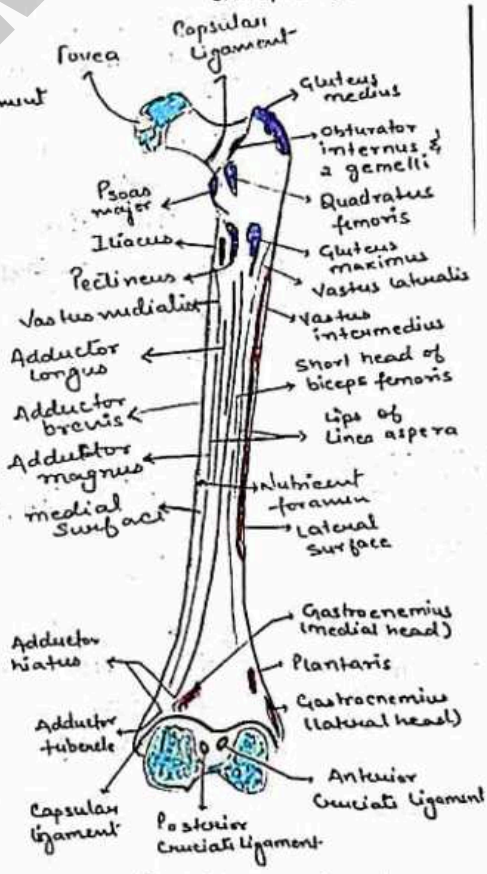


Fig: Anterior aspect

Fig: Posterior aspect

SURAJ KUMAR

FIBULA

- Homologous with ulna
- Pin-like bone

PARTS

UPPER END
- styloid process

SHAFT

LOWER END
- Malleolar fossa

UPPER END

→ HEAD
- Oval or circular articular facet
- Styloid process → fibular collateral ligament
- Sloping surface → insertion of biceps femoris

→ NECK
- Posterolateral → Common peroneal nerve
- Medial → Anterior tibial artery.

LOWER END

- 4 surface

SURFACE
- → ANTERIOR
  - Anterior talofibular ligament
  - Calcaneofibular ligament
- → POSTERIOR
  - tendons of peroneus brevis & longus.
- → MEDIAL
  - Malleolar fossa
  - Upper → posterior tibiofibular ligament
  - Lower → posterior talofibular ligament
- → LATERAL
  - Subcutaneous.

SHAFT

- 3 borders & 3 surface.

SURFACE
- → MEDIAL (Extensor)
  - Origin to extensor digitorum longus, extensor hallucis longus, peroneus tertius.
- → LATERAL (Peroneal)
  - Origin of Peroneus longus and brevis
- → POSTERIOR (Flexor)
  - Origin of tibialis posterior from medial concave part
  - Origin of soleus and flexor hallucis longus from lateral flattened part
  - Peroneal artery descends along medial crest
  - Presence of nutrient artery.

BORDERS
- → ANTERIOR
  - Attachment of superior extensor retinaculum and superior peroneal retinaculum
- → POSTERIOR
  - Attachment of posterior intermuscular septum.
- → MEDIAL (Interosseous)

SURAJ KUMAR

TIBIA

. Also known as 'Shinbone' . Second largest bone . Homologous to Radius

PARTS

UPPER END
. Medial & lateral condyles + tibial tuberosity

SHAFT
. 3 borders & 3 surface

LOWER END
. Medial malleolus

UPPER END . Has 2 condyles + intercondylar area + tibial tuberosity.

→ MEDIAL CONDYLE . 4 surface

SURFACE
- → SUPERIOR
  - . Articulate with femoral condyle
  - . Medial meniscus (fibrocartilaginous plate)
- → POSTERIOR
  - . Insertion of tendon of semimembranosus.
- → ANTERIOR
- → MEDIAL
  - . Vascular foramina
  - . Attachment to medial patellar retinaculum.

→ LATERAL CONDYLE . 4 surface

SURFACE
- → SUPERIOR
  - . Articullate with lateral femoral condyle
  - . Lateral meniscus
- → POSTERIOR
  - . Articular facet for head of fibula
  - . tendon of popliteus.
- → ANTERIOR
  - . Flat or facet – Iliotibial tract
- → LATERAL
  - . Along with anterior surface – attachment to lateral patellar retinaculum.

→ INTERCONDYLAR AREA
- . Intercondylar eminence → intercondylar tubercles
- . Attachments to
  1. Anterior horn of Medial meniscus
  2. Anterior cruciate ligament
  3. Anterior horn of Lateral meniscus
  4. Posterior horn of Medial meniscus
  5. Posterior horn of Lateral meniscus
  6. Posterior Cruciate ligament

→ TIBIAL TUBEROSITY
- . Upper smooth – ligamentum patellae
  Lower rough – subcutaneous infrapatellar bursa
- . Deep infrapatellar bursa.

SHAFT . has 3 borders & 3 surfaces

BORDERS
- → ANTERIOR
  - . Sharpest, subcutaneous, sinous
  - . Tibial tuberosity → Anterior border of medial malleolus.
- → MEDIAL
  - . Medial condyle → Posterior border of medial malleolus
- → LATERAL (Interosseous border)
  - . Lateral condyle → Anterior border of fibular notch.
  - . Attachment to interosseous membrane uniting tibia & fibula
  - . Attachment of interosseous inferior tibiofibular ligament.

SURAJ KUMAR

SURFACE
- MEDIAL
  - Almost entirely subcutaneous
  - Upper part — insertion of
    - S - Sartorius
    - G - Gracilis
    - S - Semitendinous
- LATERAL
  - Upper 2/3 — origin to tibialis anterior
  - Lower 1/3 — medial → lateral
    1. Tibialis anterior
    2. Extensor hallucis longus
    3. Anterior tibial artery
    4. Anterior tibial nerve / Deep peroneal nerve
    5. Extensor digitorum longus
    6. Peroneus tertius.
- POSTERIOR
  - Soleal line — from above downward
    1. Fascia covering popliteus
    2. Fascia covering soleus
    3. Soleus
    4. Deep transverse fascia
  - Ar area above soleal line — insertion to popliteus
  - Nutrient foramen — Nutrient artery branch of posterior tibial artery — largest nutrient artery of body.
  - Flexor digitorum longus — medial line below soleal line
  - Tibialis posterior — lateral area below soleal line.
  - Lower 1/4th - medial to lateral
    1. Tibialis posterior
    2. Flexor digitorum longus
    3. Posterior tibial artery
    4. Tibial nerve
    5. Flexor hallucis longus

LOWER END - 5 surface & Medial Malleolus.

SURFACES
- ANTERIOR
- POSTERIOR - Groove for tendon of tibialis posterior
- MEDIAL
- LATERAL - Fibular notch — anterior and posterior tibiofibular ligaments.
  upper part — interosseous tibiofibular ligament
- INFERIOR - Articulate with body of talus.

MEDIAL MALLEOLUS
- Attachment to deltoid ligament.
- Lateral surface — medial surface of body of talus
- Posterior surface — vertical groove for tendon of tibialis posterior.
- Medial margin — flexor retinaculum.

SURAJ KUMAR

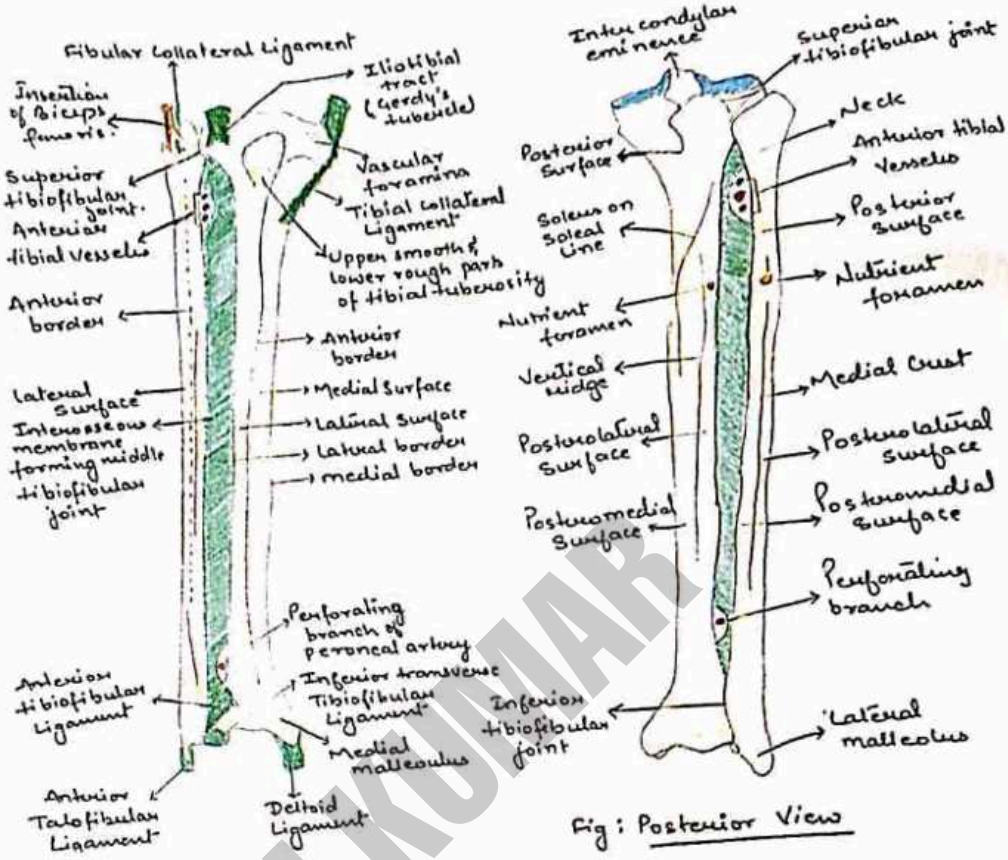


Fig: Posterior View

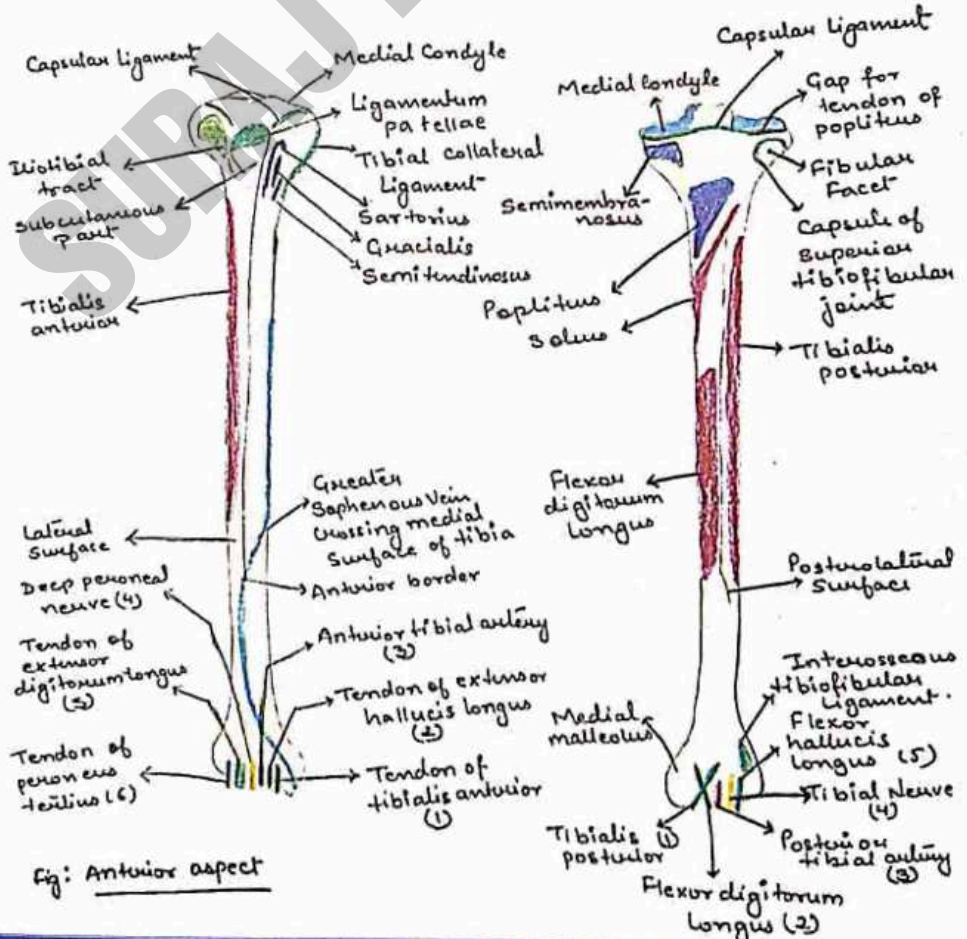
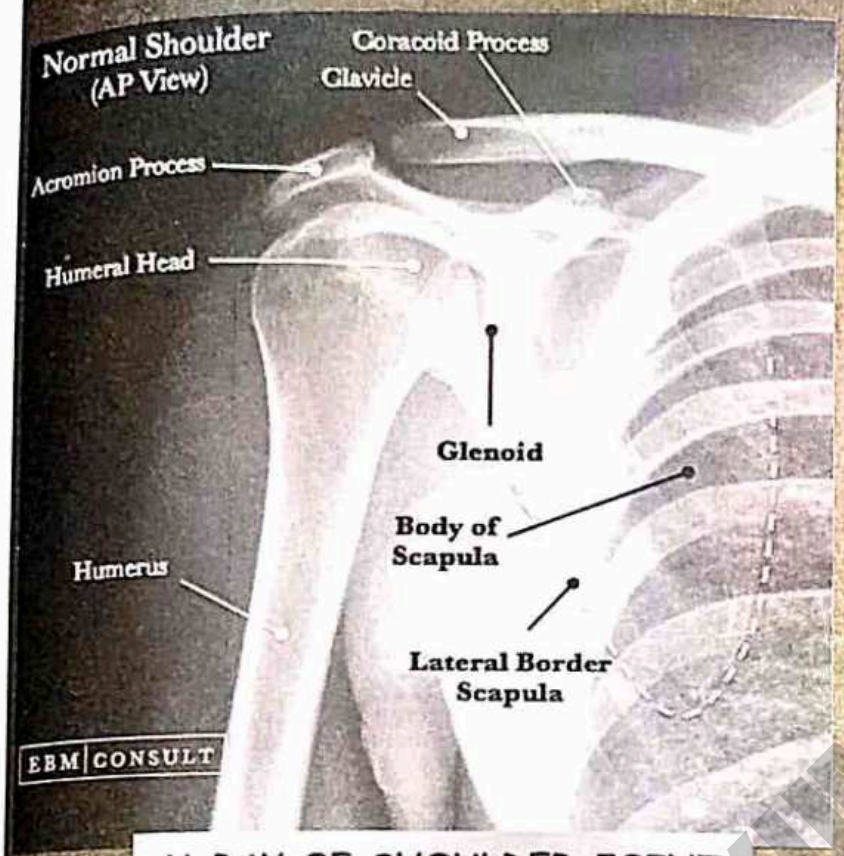


Fig: Anterior aspect

SUBAJ KUMAR

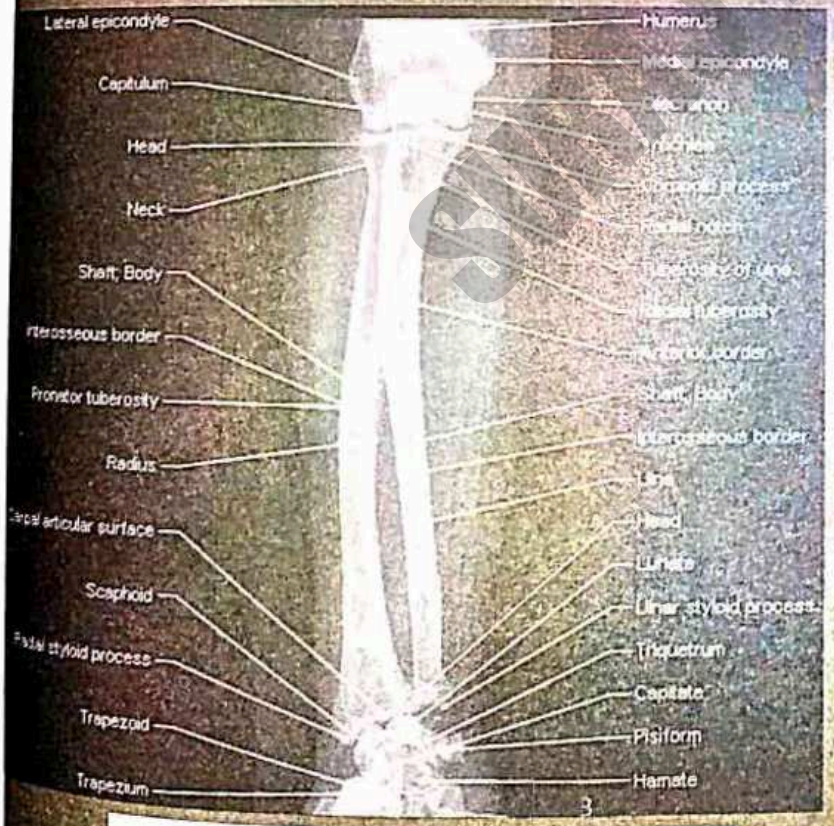
Fig: Po... YouTube

IMP X-RAY OF UPPER LIMB



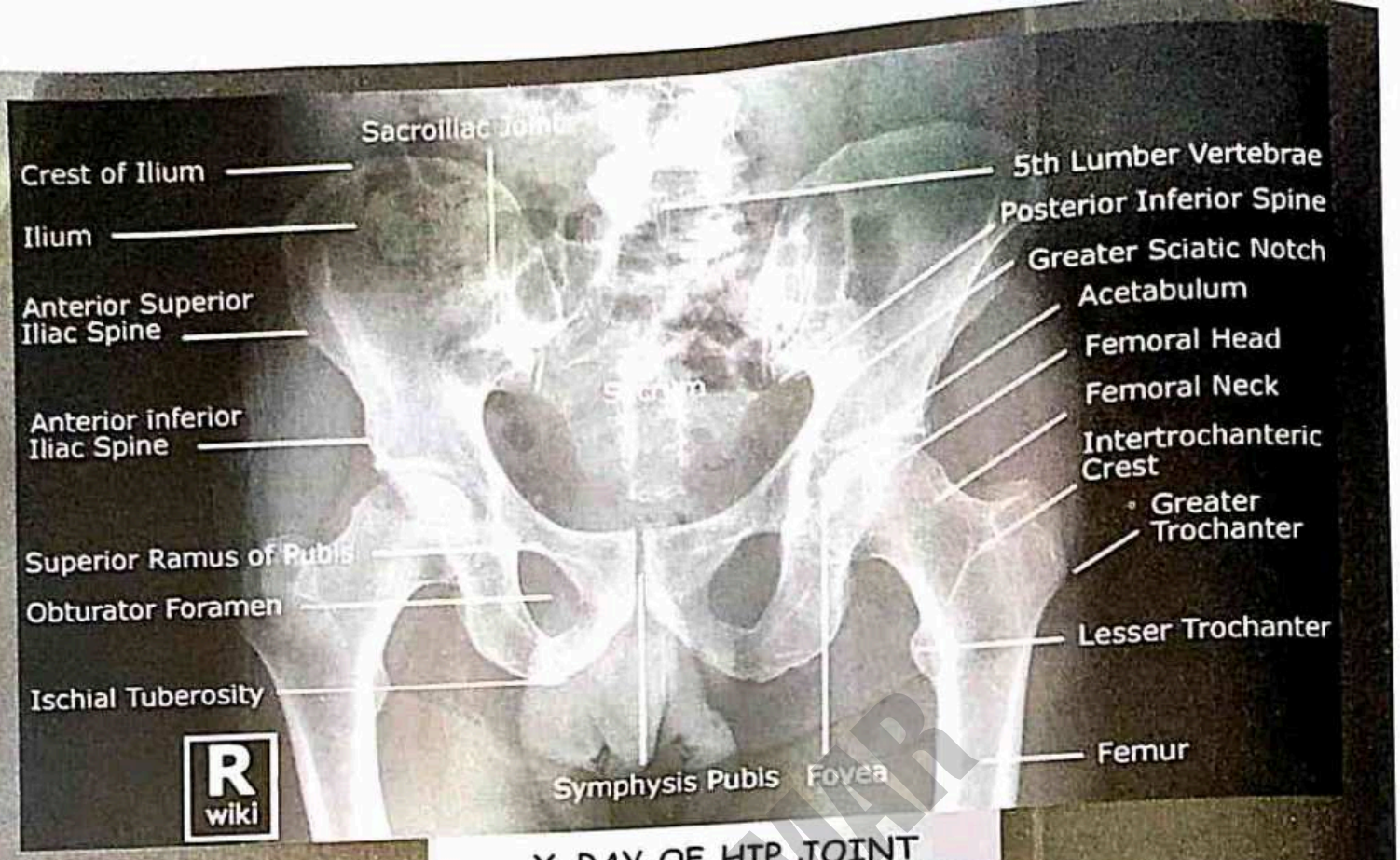
X-RAY OF SHOULDER JOINT

X-RAY OF HUMERUS BONE

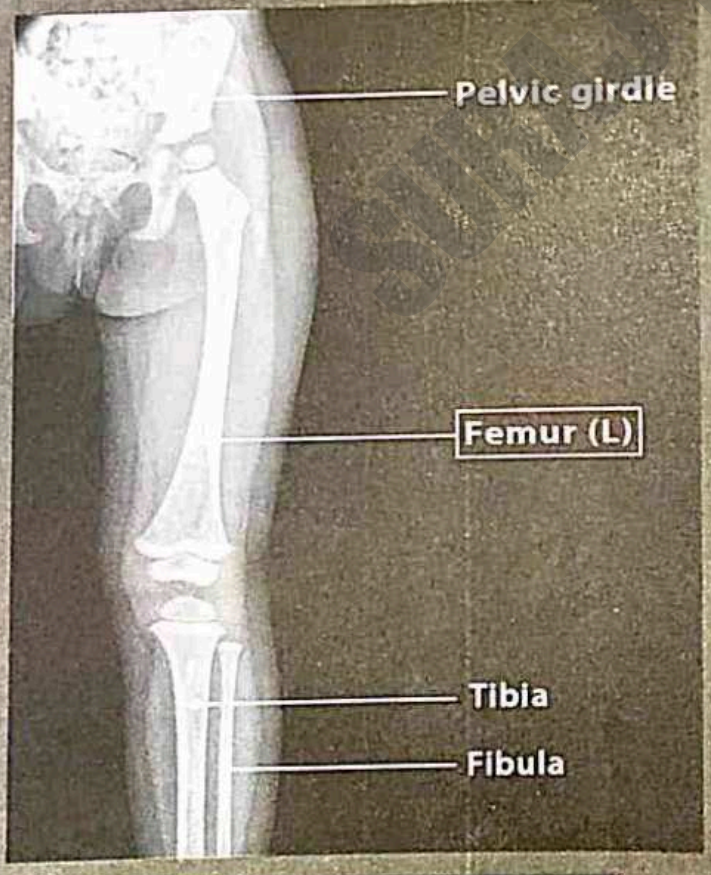


X-RAY OF FOREARM

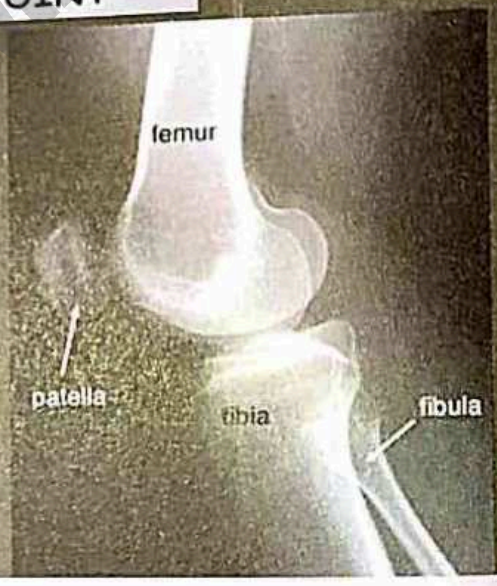
X-RAY OF HAND



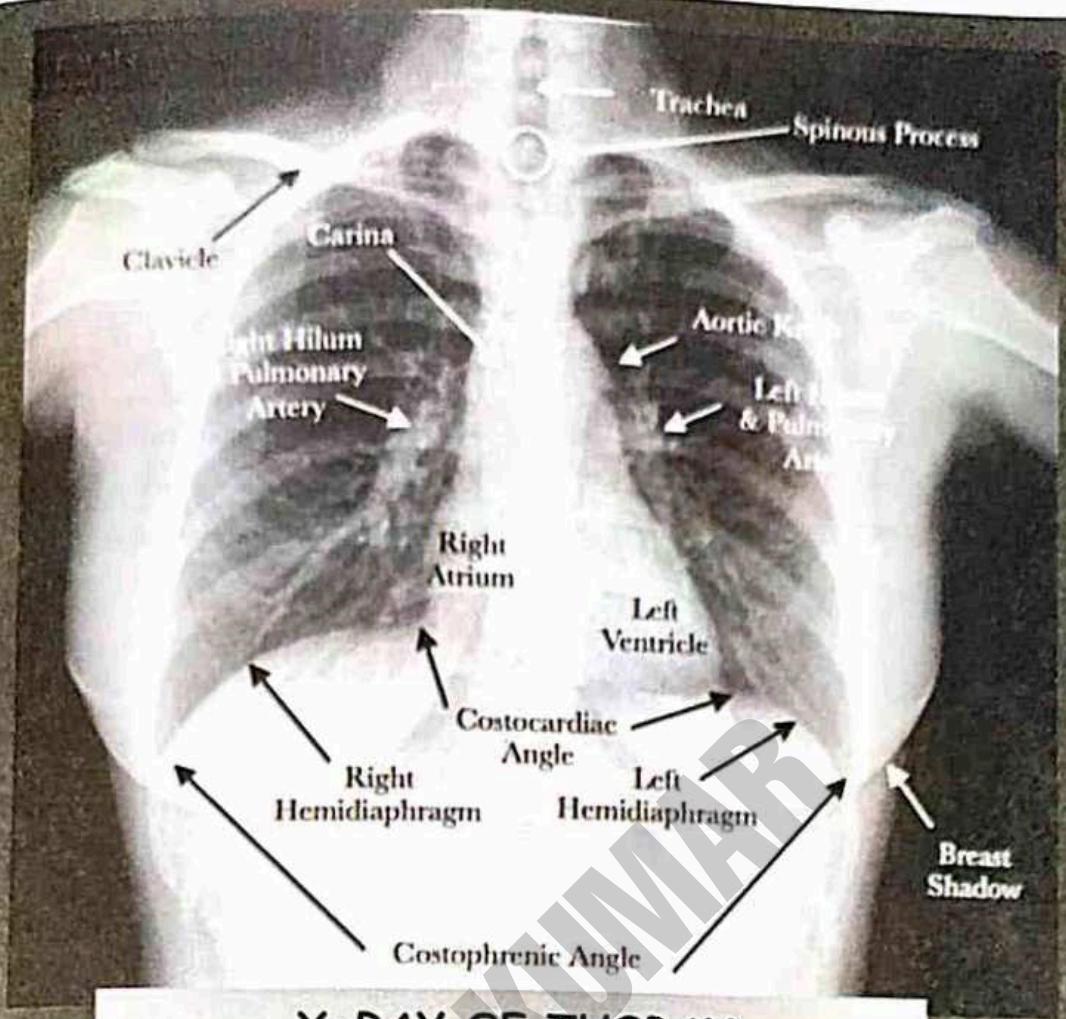
X-RAY OF HIP JOINT



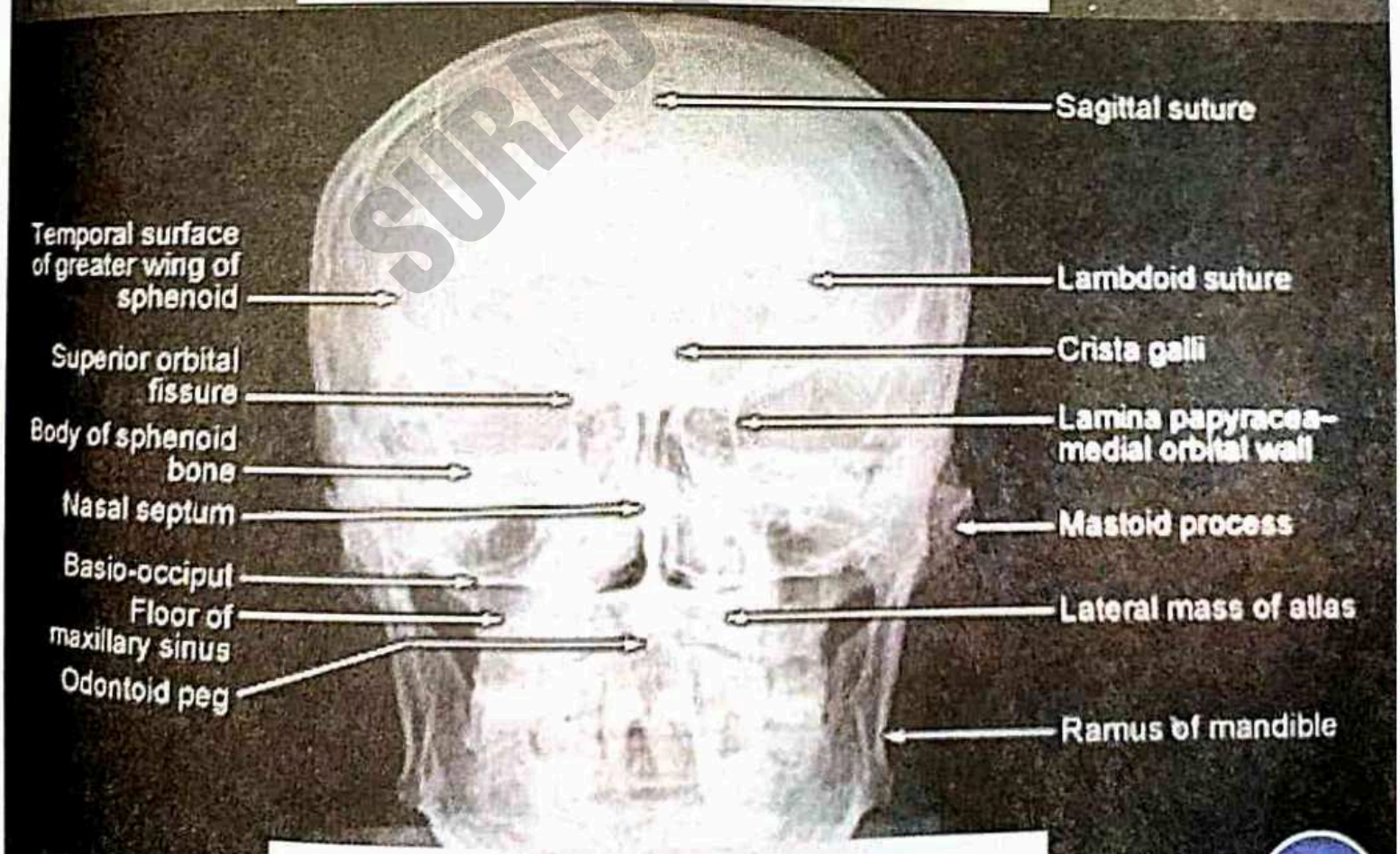
X-RAY OF LOWER LIMB



X-RAY OF KNEE JOINT



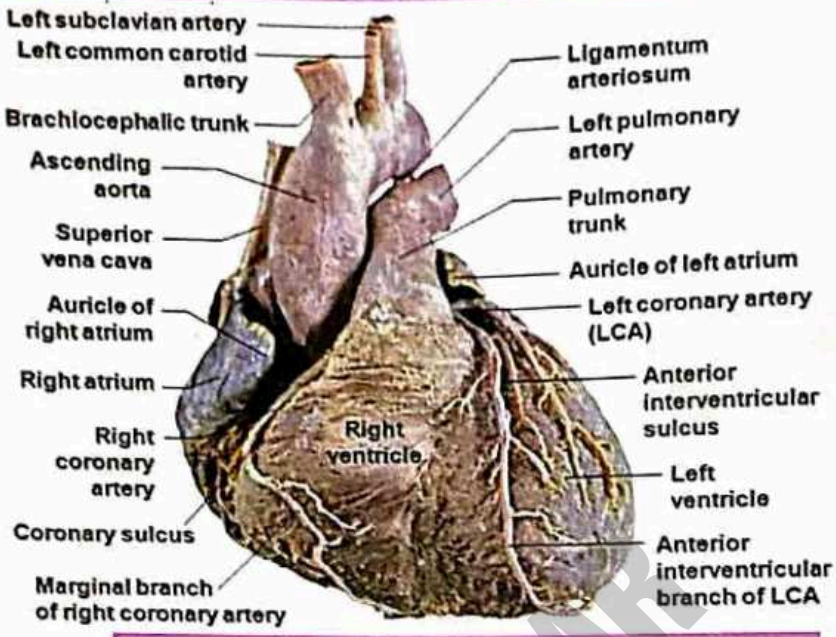
X-RAY OF THORAX



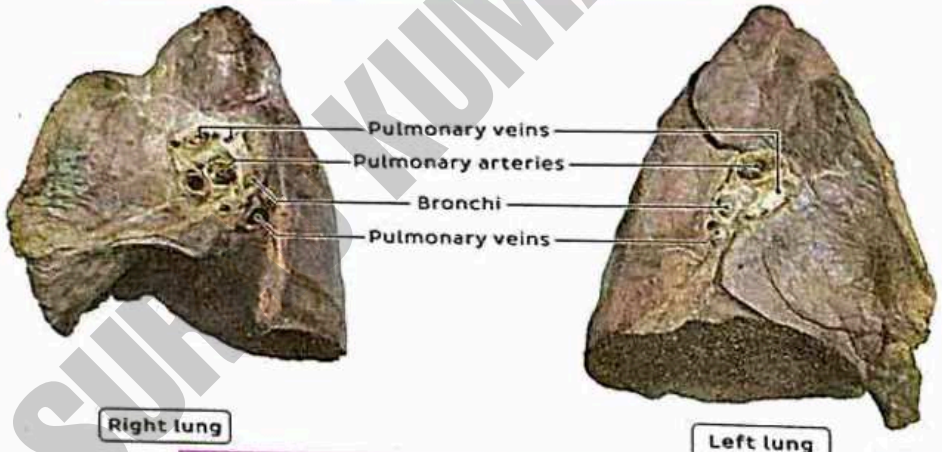
X-RAY OF SKULL

SURAJ KUMAR

IMP CADAVER ANATOMY IMAGES FOR VIVA



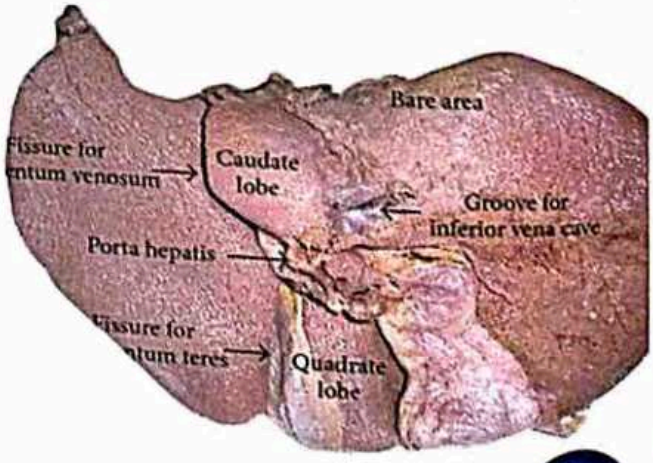
HEART CADAVER



LUNGS CADAVER

www.kenhub.com
KEN HUB

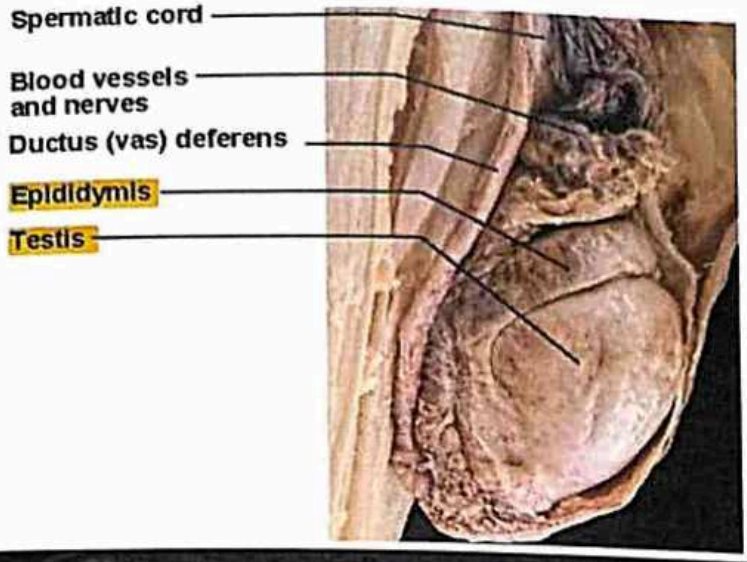
ABDOMEN CADAVER



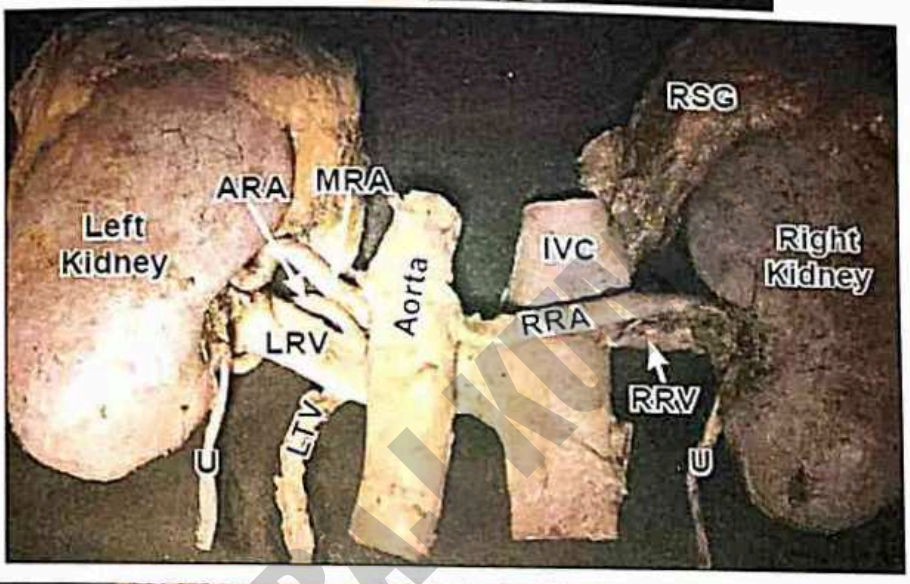
LIVER

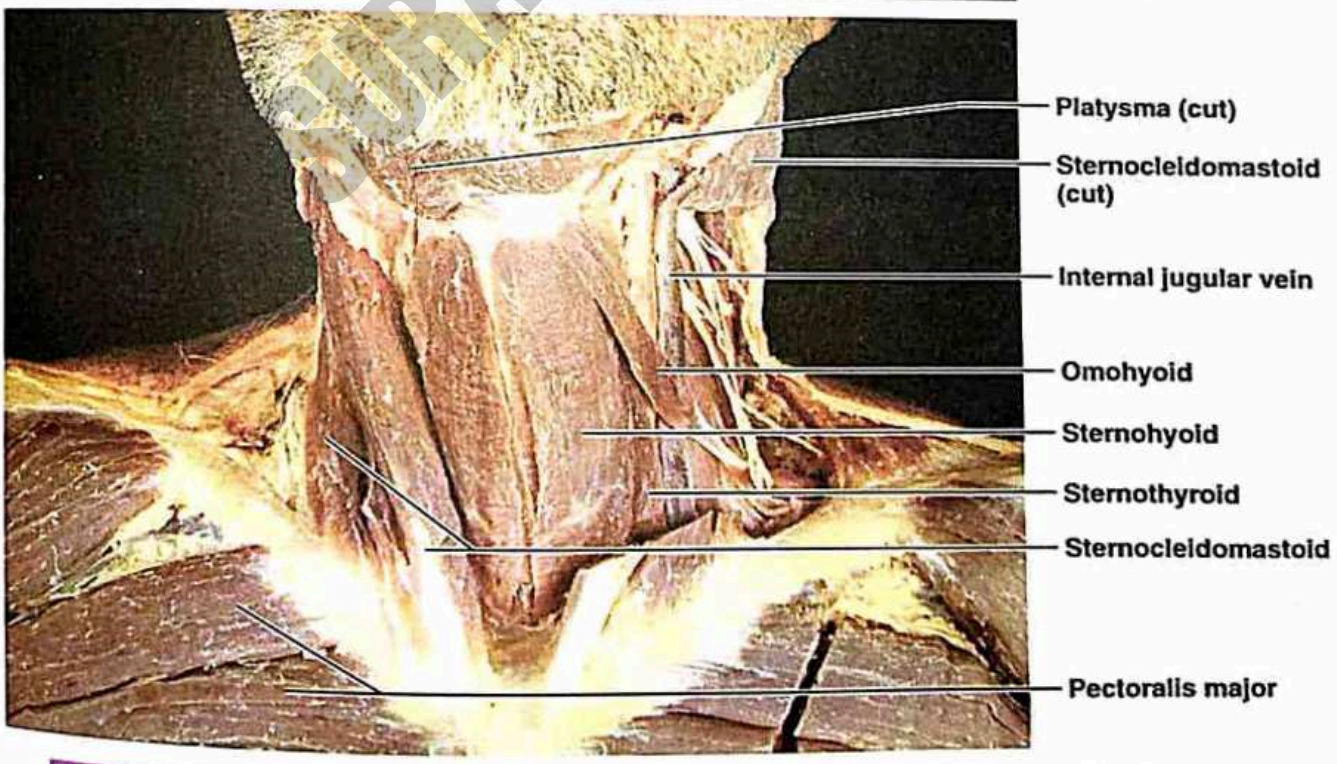
SURESH KUMAR

TESTIS



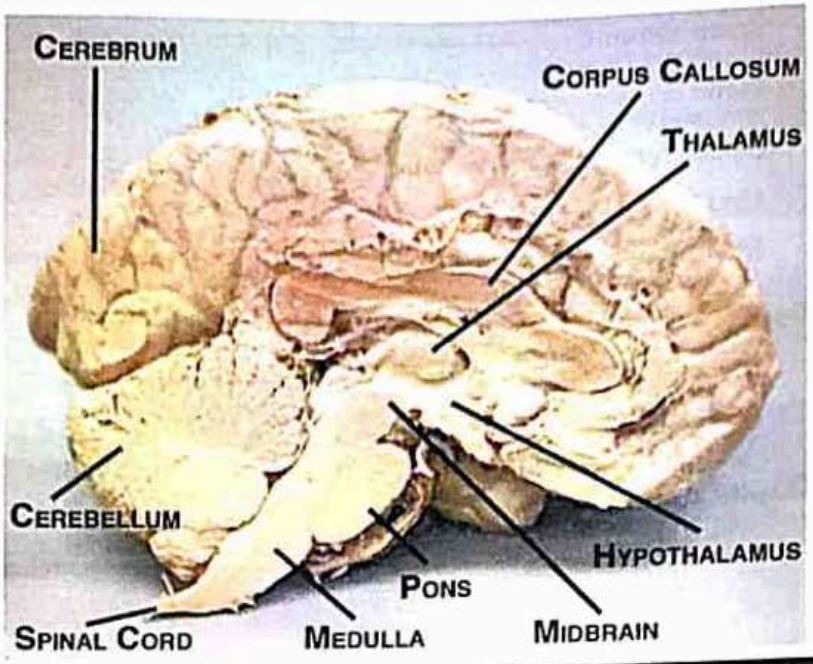
KIDNEY CADAVER



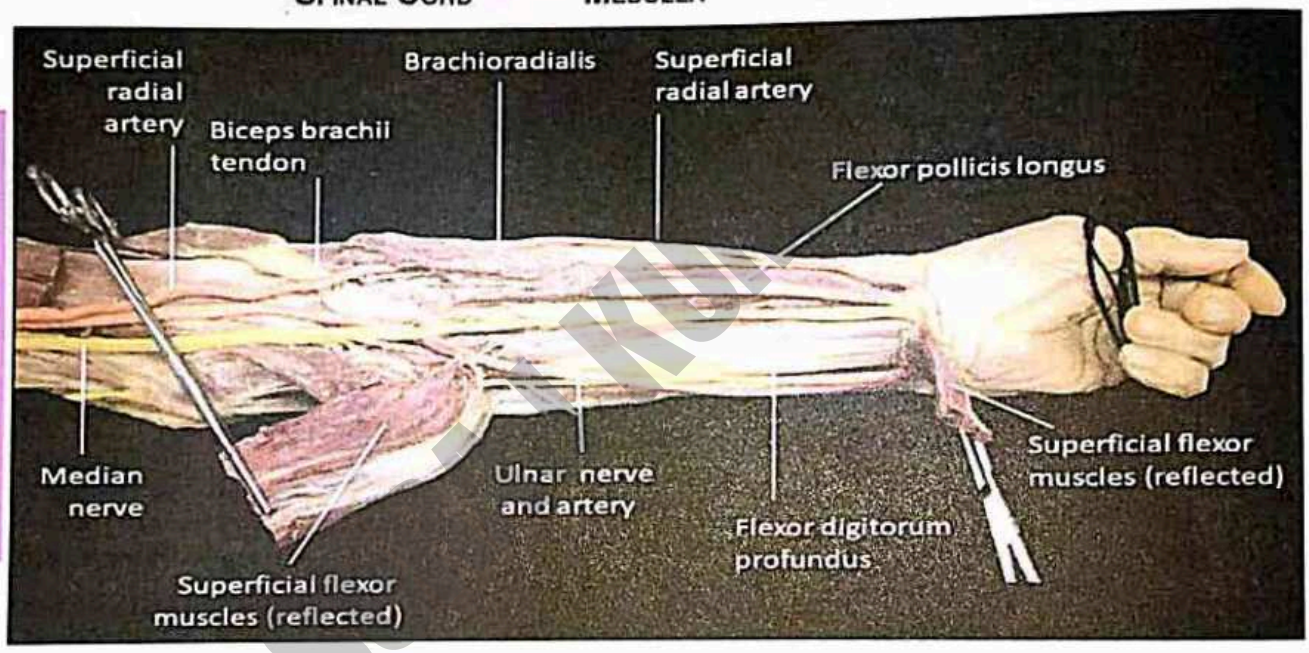


HEAD AND NECK CADAVER

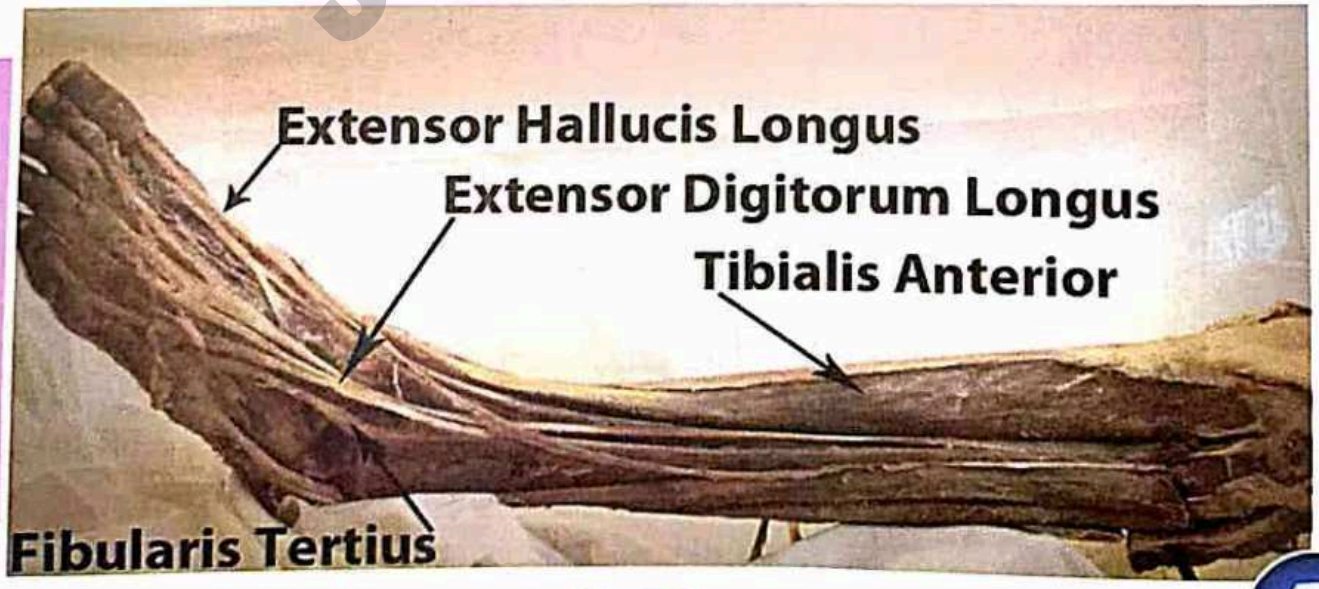
BRAIN CADAVER



UPPER LIMB

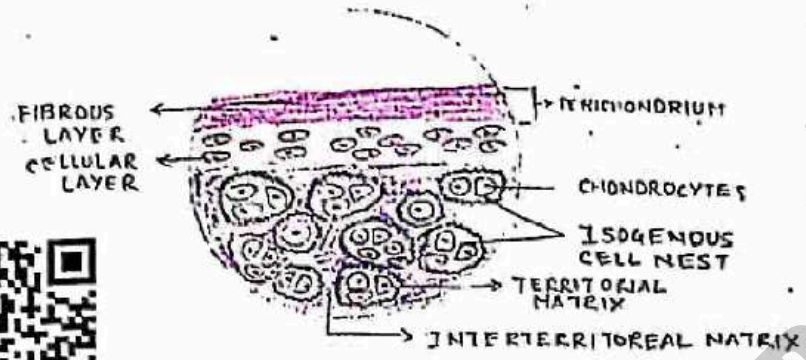


LOWER LIMB



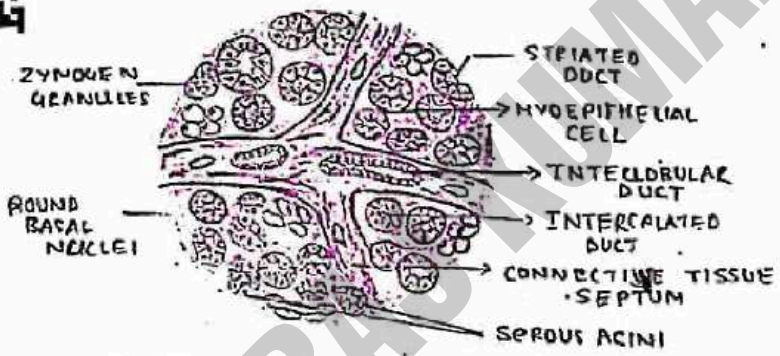
STUDY TIPS FOR MAXIMUM MARKS

1. ONLY ASK IN SHORT NOTES (DRAW A DIAGRAM OF HISTOLOGY)
2. DRAW HISTOLOGY SLIDE DIAGRAM IN ONE PAGE & WRITE 2-3 FEATURES
3. IMP. HISTOLOGY SLIDE DIAGRAMS ARE GIVEN BELOW



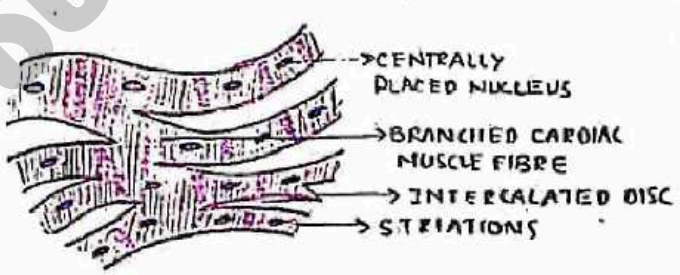
∵ HYALINE CARTILAGE

→ It has typical ground glass appearance with homogenous bluish matrix
→ chondrocytes lie within lacunae
→ It is surrounded by perichondrium which is two layered outer & inner
→ Matrix is basophilic



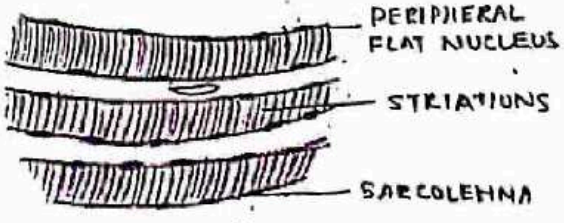
∵ SEROUS SALIVARY GLAND

→ Serous acini is small with narrow lumen
→ capsular connective tissue sends septa, which divide glands into lobes & lobules
→ Intralobular ducts are lined by cuboidal cells
→ Interlobular ducts, lined by stratified columnar cells



∵ CARDIAC MUSCLE

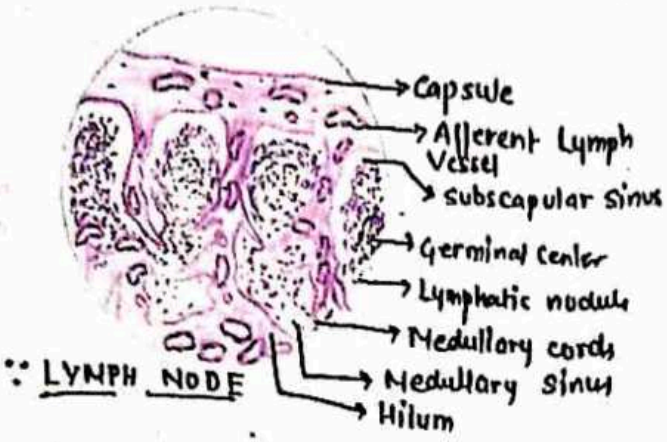
→ Found in Heart
→ Show cross-striations, less prominent than skeletal muscle
→ Made up of Cardiac myocytes
→ Has a centrally placed oval nucleus



∵ SKELETOL MUSCLE

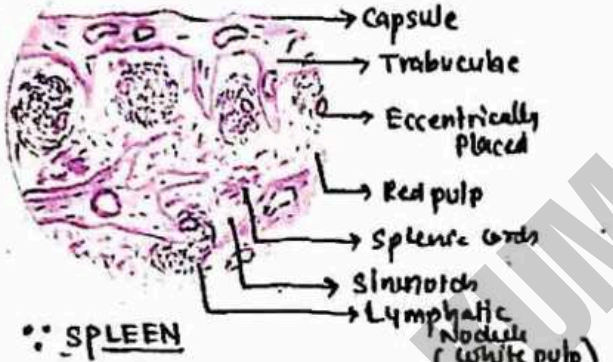
→ Sarcolemma contains numerous myofibrils
→ Striated appearence is due to regular arrays of actin & myosin
→ Each muscle fibre is variable ranging few mm to many cm.
→ Each muscle fibre is thou multinucleated flat oval

SURAJ KUMAR



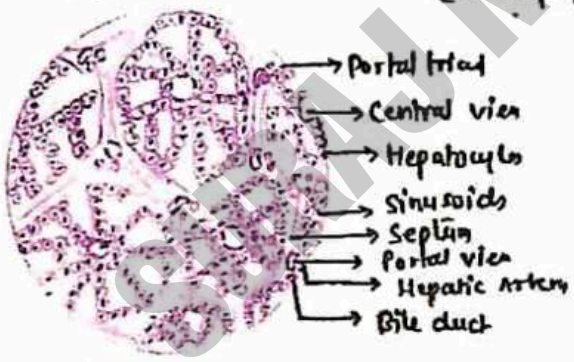
∵ LYNPH NODE

→ Lymph node has outer fibrous capsule
→ cortex contains lymphatic follicles
→ germinal center contain pale staining large, rapidly dividing B-lymphocytes
→ In medulla lymphocytes arranged like branching cords along the sinuses called "Medullary Cords"



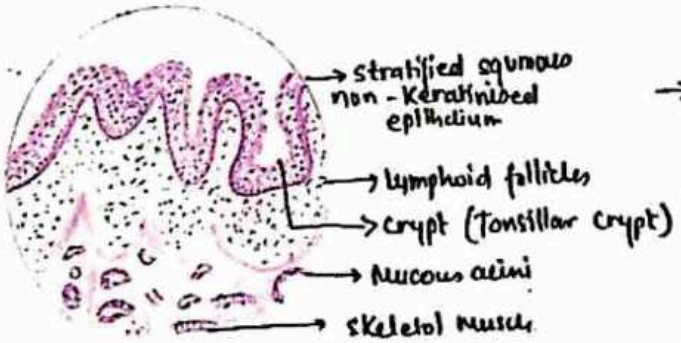
∵ SPLEEN

→ White pulp is aggregation of lymphocytes around arterial-periarterial lymphatic sheath.
→ Red pulp form major part of Spleen, it is made up of lymphocytes arranged like branching & anastomosing cords.
→ Spleen is largest hemolymphoid organ
→ Spleen has thick capsule



∵ LIVER

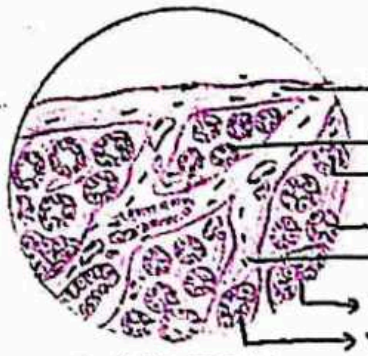
→ Hepatocytes are polyhedral in shape with round nuclei
→ Sinusoids is lined by epithelial cells
→ Portal triad consist of portal vein Hepatic artery & Bile duct
→ Center it contains central vein.



∵ Palatine Tonsil

→ Tonsils are large non-encapsulated mass of lymphoid tissue
→ Each lymph nodule has a germinal center where lymphocytes proliferates
→ Mucous acini is present
→ It is marked by stratified squamous non-keratinized epithelium.

SHAILENDRA KUMAR



∴ PANCREAS

→ Pancreatic acini are compo arranged
→ It has thin capsule, septa from capsule divides glan into lobules
→ Islets of Langerhans forms endocrine part of gland
→ Centeroacinar cells are th beginning of intercalated ducts.

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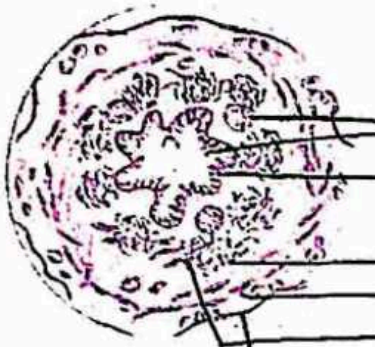

∴ APPENDIX

→ It has 4 layers - mucosa, submucosa, muscularis exter serosa.
→ Mucosa lined by simple columnar cells with microvi
→ Submucosa is made up of loose connective tissue
→ Serosa --> peritoneal coveri made of flat squamous cell

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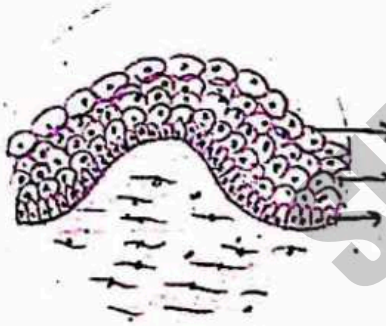

∴ UROTHELIUM

→ Highly specialized stratified epithelium
→ It form inner lining of urinar drainage system
→ It has property of expansion & contraction
→ Transitional / Urothelium is ma up of three types of cell layers: Basal, Intermediate & superficial

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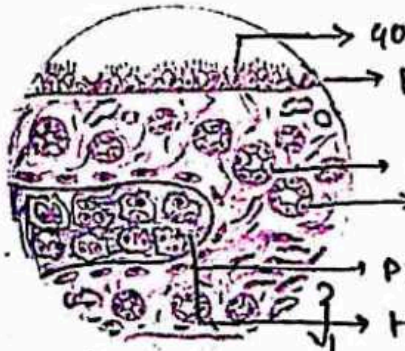

∴ RESPIRATORY EPITHELIUM

→ Mucosa is lined by pseudostrati ciliated columnar epithelium
→ Goblet cells & gland of submu secret mucus
→ Submucosa - made of CT with blood vessel & nerves
→ Adventitia - made of CT with blood vessel.

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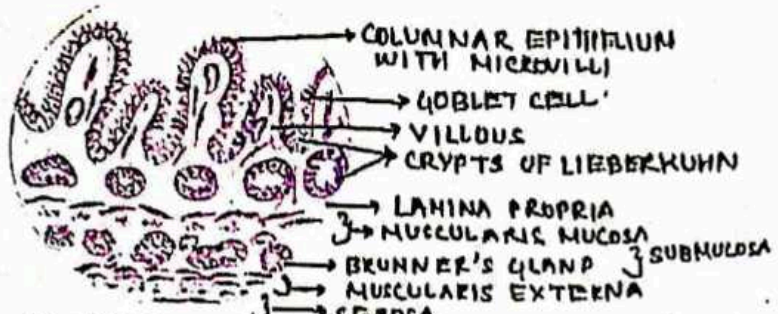
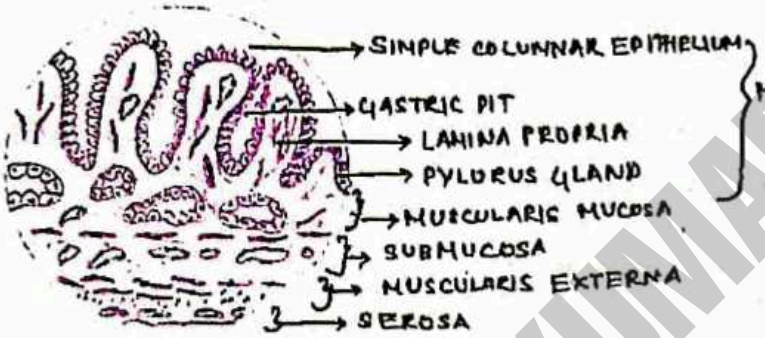


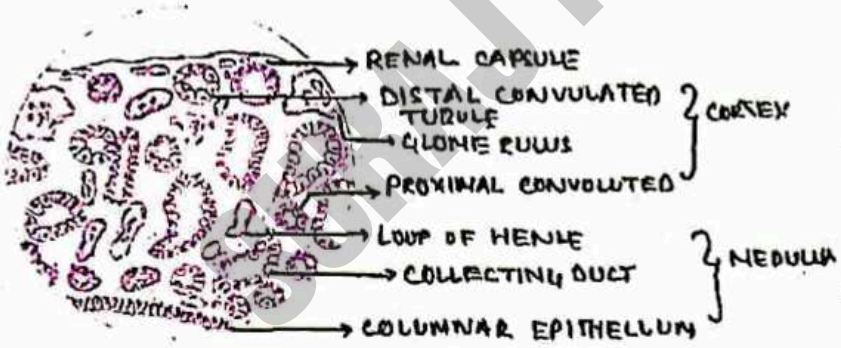
∴ DUODENUM

→ Mucosa - lined by simple columnar cells with microvilli
→ Mucosa thrown into folds called villi for increasing surface area
→ Lamina propria filled with intestinal glands - crypts of Lieberkühn
→ Brunner's gland is present in submucosa



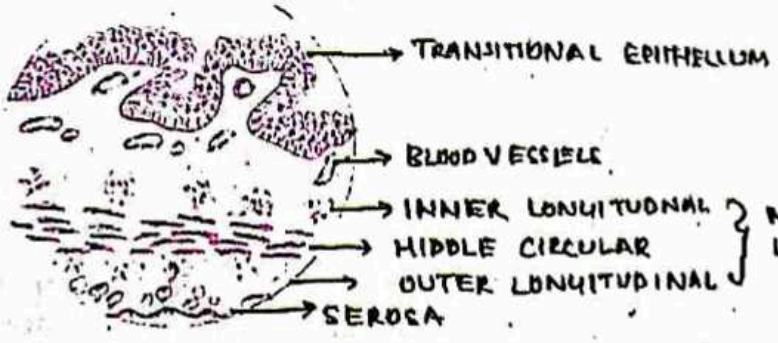
∴ STOMACH - PYLORUS

→ It has 4 layers - mucosa submucosa, muscularis & serosa
→ Lining epithelium is of simple columnar type
→ It has shows depression Kla gastric pits
→ Pyloric glands is filled in the lamina propria



∴ KIDNEY

→ outer cortex, inner medulla seen
→ Cortex contains renal corpuscules - Bowman's capsule with glomerular plexus.
→ Section of DCT - pale staining cuboidal cells, large lumen
→ Section of PCT - eosinophil columnar cells with brush border, narrow lumen



∴ URINARY BLADDER

→ Mucosa line by transit Epithelium
→ Empty bladder mucosa is thrown into folds. Distended bladder folds disappear
→ Thick muscle layer - Inner longitudinal, middle circular & outer longitudinal
→ Serosa lined by flat squ

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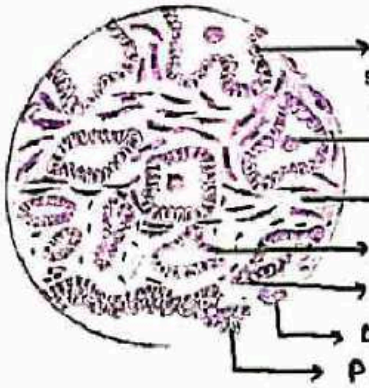
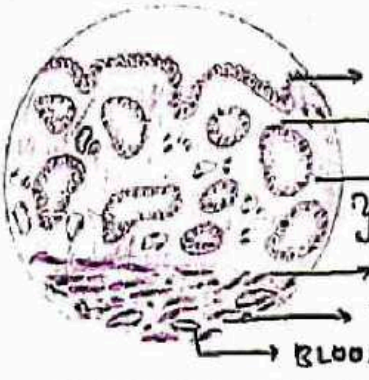


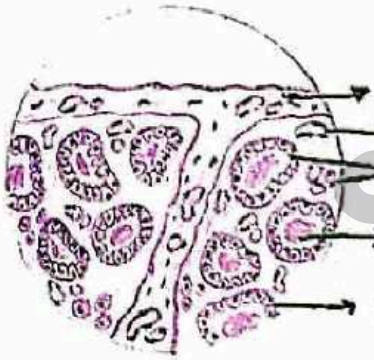
∴ PROSTATE GLAND

→ Tubuloalveolar glands separated by fibromuscular stroma
→ Gland lined by columnar epithelium
→ Lumen contains lamellated round masses of "corpora Amylacea"
→ Fibromuscular stroma is made up of collagen fibres & smooth muscles.



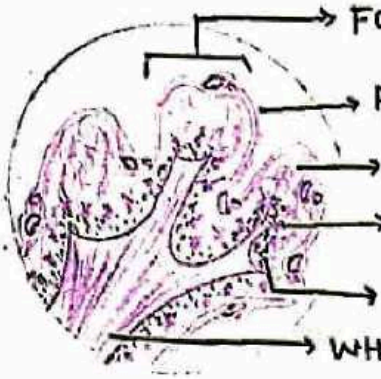
∴ UTERUS - PROLIFERATIVE PHASE

→ 3 layer - outer perimetrium middle myometrium & inner endometrium
→ Endometrium lined by columnar epithelium
→ Stroma is highly cellular contains simple tubular glands & spiral arteries
→ Myometrium is of smooth muscles.



∴ THYROID GLAND

→ Fibrous capsule surrounds the gland
→ Each lobule contain cords/cluster cells seperated by numerous sinusoids
→ Parafollicular cells are located b/w follicular cells & basement membrane.
→ Blood vessels are present



∴ CEREBELLUM

→ Cortex is three layered structure
→ Outer molecular layer
  - Basket cells
  - Stellate cells
  - Dendrite of purkinje cell
→ Middle purkinje cell is single layer
→ Inner granular cell layer made up of granule & Golgi cell & Synaptic glomeruli

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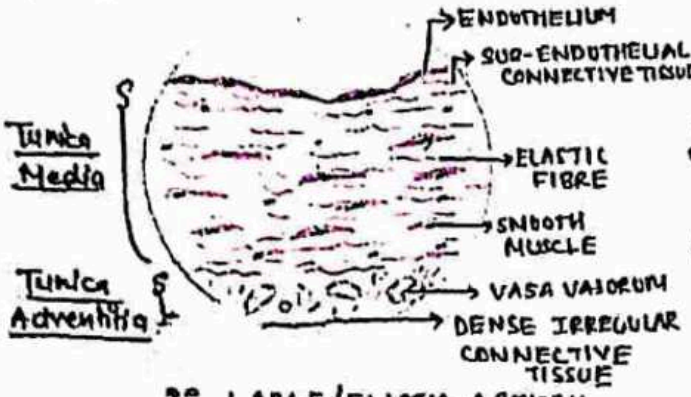
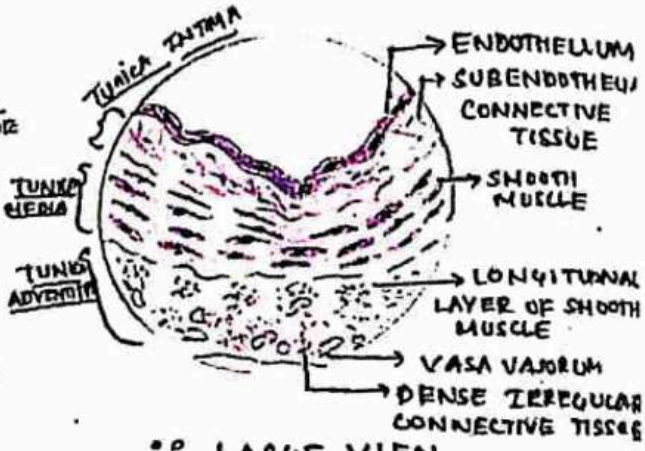


LARGE/ELASTIC ARTERY

→ Tunica intima contains endothelium flat squamous cells lines the lumen

→ Internal elastic lamina made of elastic fibres

→ Tunica media is thickest layer



LARGE VIEN

→ Tunica intima contains flat endothelium flat squamous cells lines the lumen

→ Internal elastic lamina is not well defined

→ Tunica media contain large amount of collagen fibers

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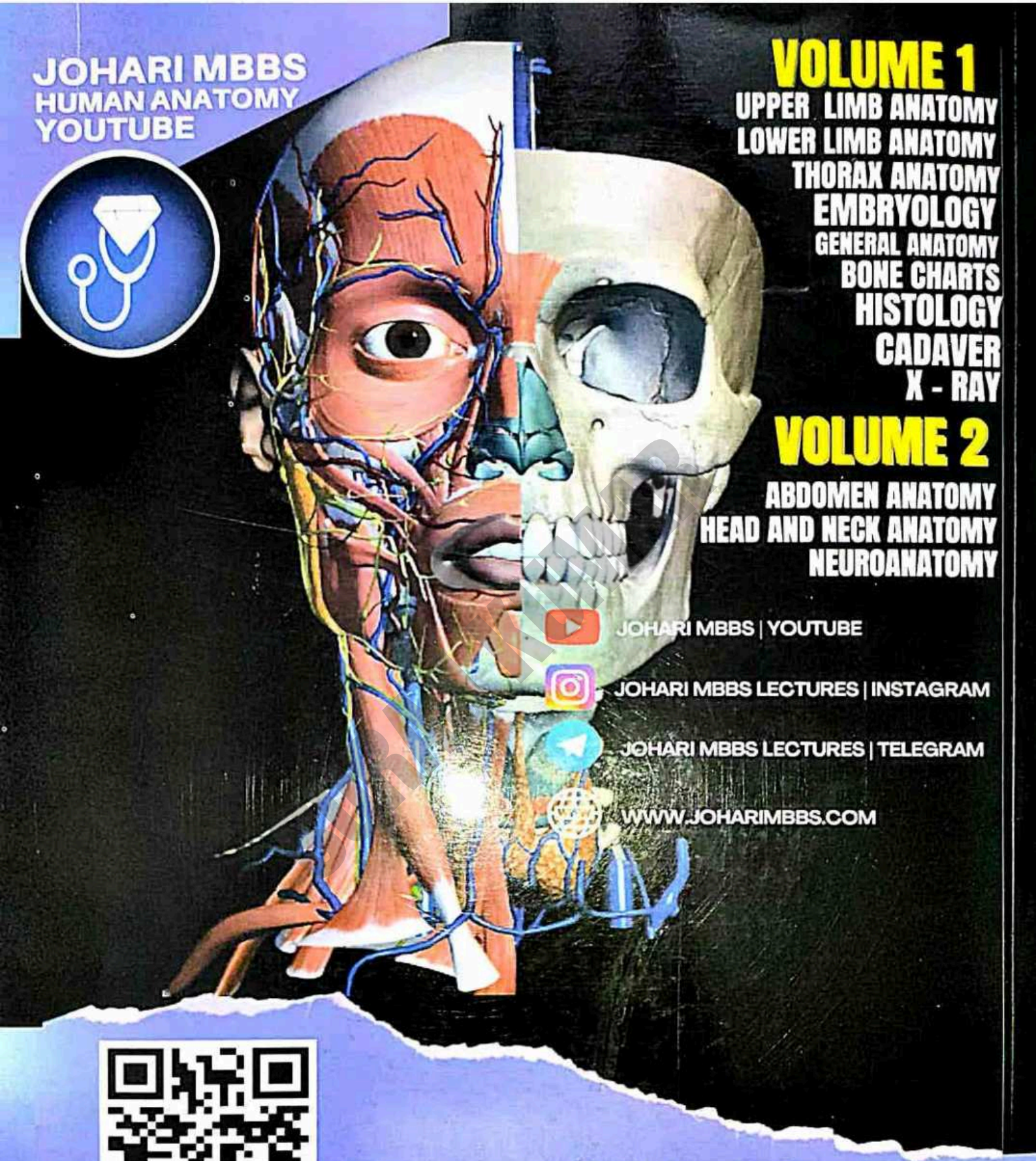

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ABDOMEN IMPS

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1) RECTUS SHEATH - 1
2) INGUINAL CANAL - 2
3) SPERMATIC CORD - 3
4) HERINA - 5
5) TESTIS - 6
6) STOMACH - 9
7) DUODENUM - 12
8) CAECUM - 15
9) VERMIFORM APPENDIX - 16
10) LIVER - 18
11) PANCREAS - 22
12) KIDNEY - 25
13) EXTRAHEPATIC BILIARY APPARATUS - 28
14) PORTAL VEIN - 30
15) ABDOMINAL AORTA - 31
16) GREATER OMENTUM - 33
17) LESSER OMENTUM - 33
18) EPIPLOIC FORAMEN - 34
19) ISCHIOANAL FOSSA - 36
20) URINARY BLADDER - 37
21) UTERUS - 40
22) ANAL CANAL - 43
23) DEEP PERONEAL SPACE - 44
24) PELVIC DIAPHRAGM - 46
25) LESSER SAC - 47
26) SPLEEN - 48

UNIVERITY QUESTION - 51
MCQ OF ABDOMEN - 52

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HEAD & NECK IMPS

FACE MUSCLE CHARTS - 53
SENSORY INNERVATION OF FACE - 54

1) SCALP - 55
2) LACRIMAL APPARATUS - 59
3) ANTERIOR TRIANGLE - 61
4) CAROTID TRIANGLE - 63
5) ANSA CERVICALIS - 65
6) POSTERIOR TRIANGLE - 67
7) PAROTID GLAND - 70
8) THYROID GLAND - 73
9) TMJ - 76
10) EXTERNAL CAROTID ARTERY - 79
11) FACIAL ARTERY - 80
12) MAXILLARY ARTERY - 82
13) MAXILLARY NERVE -83
14) DURAL VENOUS SINUS - 85
15) CAVERNOUS SINUS - 85
16) MUSCLE OF MASTIGATION - 88
17) SOFT PALATE - 90
18) PALATINE TONSIL - 90
19) TONGUE - 92
20) STYLOID APPARATUS - 96
21) EXTRAOCULAR MUSCLE - 97
22) LATERAL WALL OF NOSE - 99
23) MIDDLE EAR - 102
24) TYMPANIC MEMBRANE - 104
25) LARYNX - 105
26) SUB-MANDIBULAR GLAND - 107

UNIVERITY QUESTION - 108
MCQ OF ABDOMEN - 109

SURAJ KUMAR

NEUROANATOMY

1) ASCENDING & DESCENDING TRACT - 110
2) BRAIN STEM - MEDULLA - 111
3) BRAIN STEM - PONS - 112
4) BRAIN STEM - MIDBRAIN - 113
5) CRANIAL NERVE CHART - 114
   - 1ST OLFACTORY NERVE - 115
   - 2ND OPTIC NERVE - 115
   - 3RD OCULOMOTOR - 117
   - 4TH TROCHLEAR NERVE - 118
   - 5TH TRIGEMINAL NERVE - 119
   - 6HT ABDUCENT NERVE - 122
   - 7TH FACIAL NERVE - 123
   - 8TH VESTIBULOCOCHLEAR NERVE - 125
   - 9TH GLOSSOPHARUNEAL NERVE - 126
   - 10TH VAGUS NERVE - 128
   - 11TH ACCESSORY NERVE - 130
   - 12TH HYPOGLOSSAL NERVE - 132
6) SUB MANDIBULAR GANGLION - 135
7) OTIC GANGLION - 137
8) CILIARY GANGLION - 139
9) CEREBRUM - 141
10) CEREBELLUM - 142
11) LATERAL VENTRICLE - 145
12) THIRD VENTRICLE - 146
13) 4TH VENTRICLE - 147
14) THALAMUS - 149
15) CORPUS CALLOSUM - 150
16) WHITE MATTER OF CEREBRUM - 150
17) INTERNAL CAPSULE - 152
18) CIRCLE OF WILLIS - 153
19) BASAL NUCLEI - 155

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RECTUS SHEATH

SCAN QR WITH ORBOT APP TO DIRECTLY OPEN VIDEO LINK

1. IT IS AN APONEUROTIC SHEATH COVERING THE RECTUS ABDOMINIS.
2. IT HAS TWO PARTS - ANTERIOR & POSTERIOR

*FEATURES: -

ANT WALL - IT IS COMPLETE COVERING THE MS FROM END TO END.
- ITS COMPOSITION IS VARIABLE AS DESCRIBED BELOW.
-IT IS FIRMLY ADHERENT TO THE TENDINOUS INTERSECTIONS OF THE RECTUS MS.

POST WALL - IT IS INCOMPLETE, BEING DEFICIENT ABOVE THE COSTAL MARGIN & BELOW THE ARCUATE LINE.
- ITS COMPOSITION IS VARIBLE AS DESCRIBE BELOW.

*FORMATION: -

1.ABOVE THE COSTAL MARGIN -

ANT WALL- EXT OBLIQUE APONEUROSIS.
POST. WALL - IT IS DEFICIENT, THE RECTUS MS RESTS DIRECTLY ON 5TH 6TH & 7TH COSTAL CARTILAGE.

2 BETWEEN THE COSTAL MARGIN & THE ARCUATE LINE -

ANT WALL - EXT. OBLIQUE APONEUROSIS & ANT APONEUROSIS OF THE INT OBLIQUE.
POST WALL- POST LAMINA OF THE APONEUROSIS OF THE INTERNAL OBLIQUE & APONEUROSIS OF THE TRANSVERSUS MS.

3.BELOW THE ARCUATE LINE -

- ANT WALL- APONEUROSES OF ALL THE THREE FLAT MS OF THE ABDOMEN.APONEUROSES OF THE TRANSVERSUS & INTERNAL OBLIQUE ARE FUSED BUT THE EXTERNAL OBLIQUE APONEUROSIS REMAIN SEPARATE.
-POST WALL- IT IS DEFICIENT. --->RECTUS MS - FASCIA TRANSVERSALIS

*CONTENT -

1.MUSCLE - RECTUS ABDOMINIS & PYRAMIDALIS
2.ARTERY - SUP. EPIGASTRIC ARTERY. ---->ENTER SHEATH BY PASSING BETWEEN COSTAL & XIPHOID
-INF.EPIGASTRIC ARTERY----->ENTER SHEATH BY INFRONT OF ARCUTE LINE
3.VEIN - SUP. EPIGASTRIC VEIN & INF. EPIGASTRIC VEIN
4.NERVES - TERMINAL PART OF THE LOWER SIX THORACIC NERVES.

*FUNCTION - CHECK BOWING OF RECTUS MS. - DURING CONTRACTION.
- MAINTAIN STRENGTH OF ANT. ABDOMINAL WALL.

SUNIL KUMAR

INGUINAL CANAL

*OBLIQUE INTERMUSCULAR PASSAGE IN THE LOWER PART OF THE ANTERIOR ABDOMINAL WALL

*SITUATED --->JUST ABOVE MEDIAL HALF OF INGUINAL LIGAMENT

*DIRECTED DOWNWARD, FORWARD, MEDIAL.

*EXTEND - FROM DEEP INGUINAL RING TO SUPERFICIAL INGUINAL RING.

DEEP INGUINAL RING - OVAL OPENING IN FASICA TRANSVERSALIS
-SITUATED 1.2CM ABOVE THE MIDINGUINAL POINT.
-IMMIDIATELY---->LATERAL TO STEM OF INFERIOR EPIGASTRIC ARTERY

SUPERIOR INGUINAL RING - TRIANGULAR OPENING - IN EXTERNAL OBLIQUE APONEUROSIS
- IT IS LIKE AN OBTUSE ANGLE TRIANGLE
BASE -->FORM BY PUBIC CREST
TO MARGIN OF TRIANGULAR ---->LATERAL & MEDIAL REFERRED AS CRURA
BEYOND THE APEX OF TRIANGLE -->2 CRURA-> UNITED BY INERCRURAL FIBERS

*BOUNDARIS -

ANTERIOR WALL - WHOLE EXTEND - SKIN & SUPERFICIAL FASICA
- EXTERNAL OBLIQUE APONEUROSIS
- LATERAL 1/3RD - INTERNAL OBLIQUE MUSCLE

POSTERIOR WALL - WHOLE EXTEND - FASCIA TRANSVERSALIS
- EXTRAPERITONEAL TISSUE
- PARIETAL PERITONEUM
-MEDIAL 2/3RD - CONJOINT TENDON
- MEDIAL END BY THE REFLECTED PART OF INGUINAL LIGAMENT

ROOF - ARCHED FIBER - INTERNAL OBLIQUE
-TRANSVERSUS ABDOMINIS MUSCLE

FLOOR - GROOVED UPPER SURFACE OF INGUINAL LIGAMENT
- MEDIAL END BY LACUNAR LIGAMENT

SEX DIFFERENCE - INGUINAL CANAL IS LARGE IN MALE & SMALLER IN FEMALE

*STRUCTURES PASSING THROUGH THE CANAL -

1.SPERMATIC CORD IN MALES, ROUND LIGAMENT OF THE UTERUS IN FEMALE.
2.ILIOINGUINAL NERVE ENTERS THE CANAL THROUGH THE INTERVAL BETWEEN THE EXTERNAL & INTERNAL OBLIQUE MUSCLE & PASSES OUT THROUGH THE SUPERFICIAL INGUINAL RING.

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SPERMATIC CORD

*CONSTITUENT OF SPERMATIC CORD -

1. DUCTUS DEFERENS
2. TESTICULAR AND CREMASTERIC ARTERIES & DUCTUS DEFERENS ARTERY
3. PAMPINIFORM PLEXUS OF VEINS
4. LYMPH VESSEL
5. GRENITAL BRANCH-->OF GENITOFORMAL NERVE
   -PLEXUS OF SYMPATHETIC NERVES
6. REMAINS OF PROCESSUS VAGINALIS

*COVERING OF SPERMATIC CORD - FROM WITHIN OUTWARD FOLLOWS -

1. INTERNAL SPERMATIC FASCIA - DERIVED FROM-FASCIA TRANSVERSALIS
2. CREMASTRIC FASCIA - DERIVED FROM - INTERNAL OBLIQUE TRANSVERSUS ABDOMINIS MUSCLES
   -MADE UP OF - MUSCLE LOOPS -->CONSTITUTING CREMASTER MUSCLE
3. EXTERNAL SPERMATIC FASCIA - DERIVED FROM - EXTERNAL OBLIQUE APONEUROSIS

MECHANISM OF INGUINAL CANAL

*PRESENCE OF INGUINAL CANAL IS CAUSE OF WEAKNESS IN LOWER PART OF ANTERIOR ABDOMINAL WALL.
*WEAKNESS IS COMPENSATED BY FOLLOWING FACTORS.

1. OBLIQUITY OF INGUINAL CANAL -
   -2 INGUINAL RING DO NOT LIE OPPOSITE TO EACH OTHER
   -THEREFORE WHEN INTRA-ABDOMINAL PRESSURE RISES, THE ANT.& POST. WALL ARE APPROXIMATED.
   -THIS IS KNOWN AS FLAP VALVE MECHANISM
2. SUPERFICIAL RING GUARDED - FROM BEHIND - CONJOINT TENDON
   -REFLECTED PART OF INGUINAL LIGAMENT
3. DEEP INGUINAL RING GUARDED - FROM FRONT - INTERNAL OBLIQUE
4. SHUTTER MECHANISM OF INTERNAL OBLIQUE MUSCLE -
   - MUSCLE HAS TRIPLE RELATION TO INGUINAL CANAL.
   - IT FORM ANTERIOR, POSTERIOR WALL & ROOF OF CANAL
   - WHEN IT CONTRACTS THE ROOF IS APPROXIMATED TO FLOOR LIKE SHUTTER.
5. CONTRACTION OF CREMASTER/BALL VALVE MECHANISM - HELP SPERMATIC CORD TO PLUG SUPERFICIAL INGUINAL RING.
6. SPLIT VALVE MECHANISM - CONTRACTION OF EXTERNAL OBLIQUE RESULT IN APPROXIMATION OF 2 CRURA OF SUPRFICIAL INGUINAL RING.

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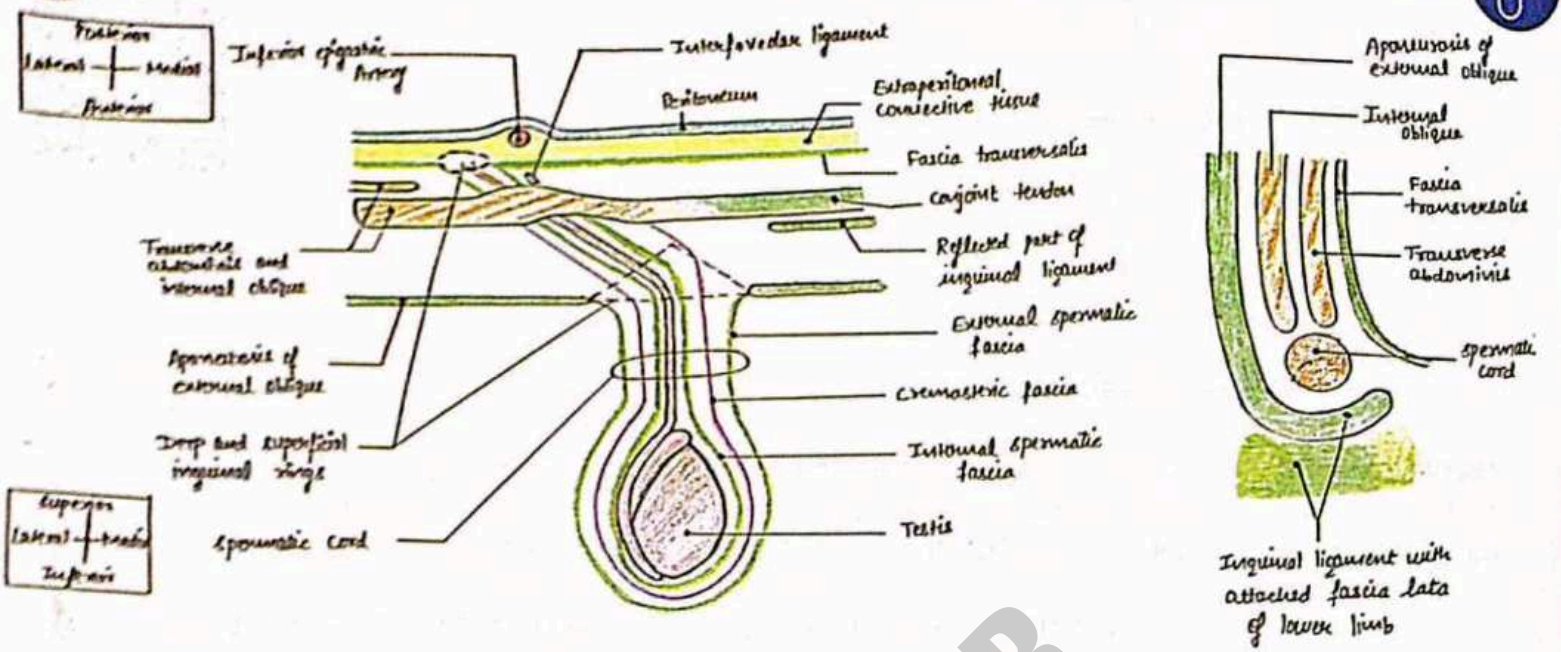
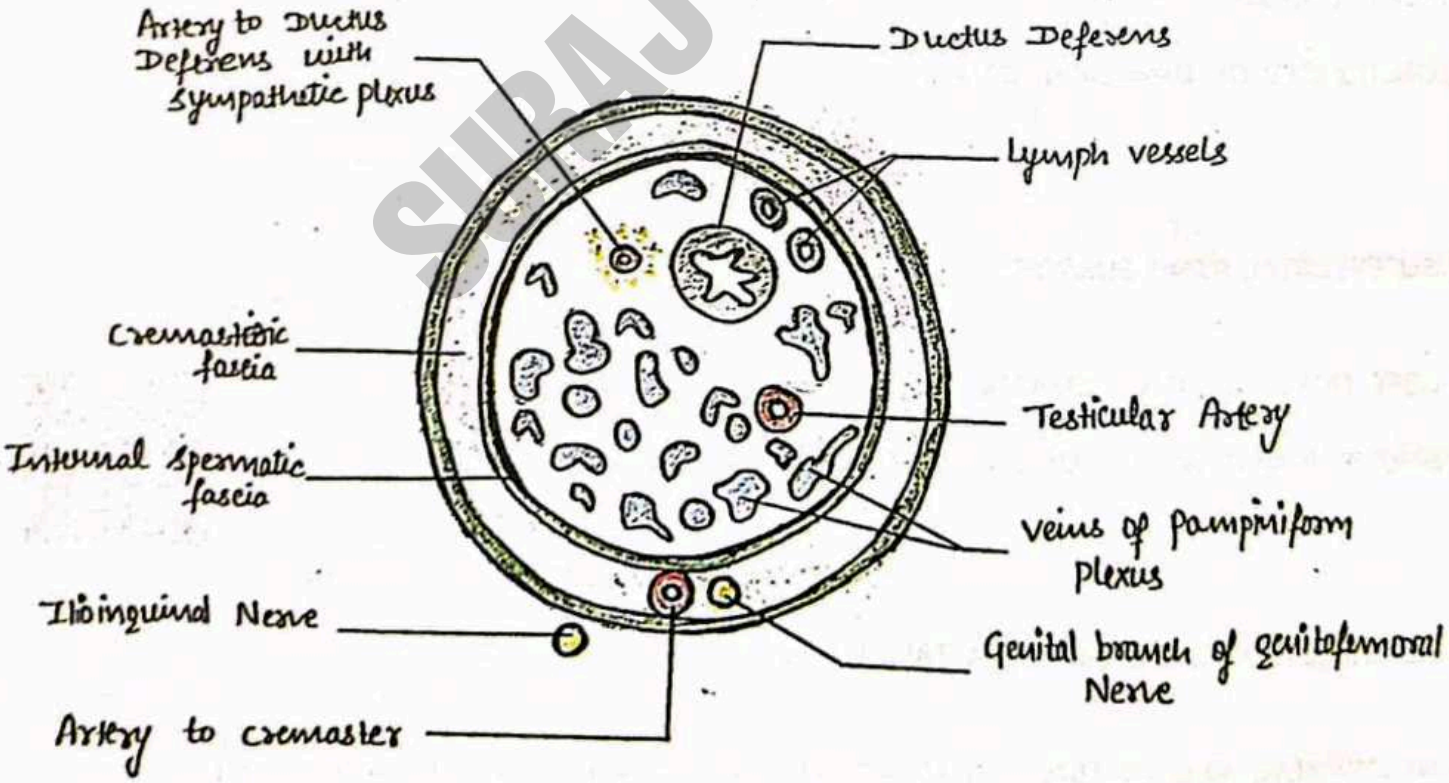


Boundaries of Inguinal canal ⓐ Anterior and posterior walls in a horizontal section and
ⓑ the roof and floor in a sagittal section.
Fascia lata attached to inguinal ligament.

INGUINAL CANAL



SPERMATIC CORD

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CLINICAL ANATOMY

HERNIA

*HERNIA IS A PROTRUSION OF ANY ABDOMINA CONTENT THROUGH ANY OF ITS WALL- THIS IS CALLED EXTERNAL HERNIA.
*INTESTINE PROTRUBES INTO THE NO ENTRY ZONE WITHIN ABDOMINALL CAVITY ITSELF-THIS CONDITION IS CALLED INTERNAL HERNIA.
*HERNIA CONSIST OF--→SAC, CONTENT, COVERING.

TYPE OF HERNIA - INTERNAL HERNIA & EXTERNAL HERNIA.

INGUINAL HERNIA - PROTUSION OF LOOP OF INTESTINE THROUGH INTESTINAL WALL IS CALLED INGUINAL HERNIA.

INDIRECT INGUINAL HERNIA - WHEN THE PROTUSION OCCUR THROUGH THE DEEP INGUINAL RING, INGUINAL CANAL, SUPERFICIAL INGUINAL RING INTO SCROTUM, IT IS CALLED INDIRECT HERNIA.

COVERING - EXTRAPERITONEAL TISSUE
-INTERNAL SPERMATIC FASCIA
-CREMASTRIC FASCIA
-EXTERNAL SPERMATIC FASCIA
-SKIN

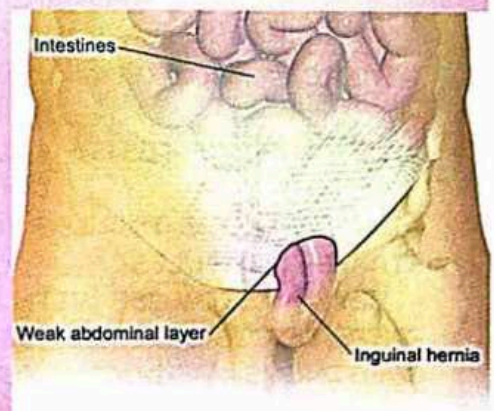
DIRECT HERNIA - WHEN THE PROTRUSION OCCUR THROUGH WEAK POSTERIOR WALLL OF INGUINAL CANAL OR TRIANGLE OF HESSELBACH THIS IS CALLED DIRECT HERNIA.

*HESSELBACH - BOUNDED BY INFERIOR EPIGASTRIC ARTERY, LATERAL BORDER OF RECTUS ABDOMINIS & INGUINAL LIGAMENT.
*THIS AREA IS DIVIDED INTO MEDIAL AND LATERAL PART BY PASSAGE OF OBLITERATED UMBILICAL ARTERY.

COVERING -

COVERINGS OF LATERAL DIRECT INGUINAL HERNIA -
1.EXTRAPERITONEAL TISSUE
2.FASCIA TRANSVERSALIS
3.CREMASTERIC FASCIA
4.EXTERNAL SPERMATIC FASCIA
5.SKIN

COVERINGS OF MEDIAL DIRECT INGUINAL HERNIA -
1.EXTRAPERITONEAL TISSUE
2.FASCIA TRANSVERSALIS
3.CONJOINT TENDON
4.EXTERNAL SPERMATIC FASCIA
5.SKIN



Inguinal Hernia

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TESTIS

*MALE GONAD - SUSPENDED IN THE SCROTUM BY SPERMATIC CORD.
*LIES OBLIQUETY, SO THAT UPPER POLE - FORWARD - LATERALLY.
*LEFT TETIS IS SLIGHTLY LOWER THAN RIGHT

*SHAPE & SIZE -
1.OVAL IN SHAPE - 3.75CM LONG, 2.CM BROAD, 1.8CM THICK FROM SIDE TO SIDE
2.WEIGHT ABOUT - 10 TO 15g

*EXTERNAL FEATURE -
1.TWO POLE - UPPER POLE & LOWER POLE ARE CONVEX & SMOOTH
2.TWO BORDER - ANTERIOR BORDER - CONVEX & SMOOTH
-FULLY COVERED - BY TUNICA UAGINALIS
- POSTERIOR BORDER - STRAIGHT
- PARTIALLY COVERED BY TUNICA VAGINALIS
- EPIDIDYMIS LIE - ALONG LATERAL PART OF POSTERIOR BORDER
3.TWO SURFACE - MEDIAL SURFACE & LATERAL SURFACE - CONVEX & SMOOTH

""MEDIAL SURFACE OF EPIDIDYMIS IS SEPERATED FROM TESTIS BY AN EXTENSION OF CAVITY OF TUNIICA VAGINALIS - THIS CALLED - SINUS OF EPIDIDYMIS""

*COVERINGS OF TESTIS -
*TESTIS IS COVERED BY LAYER OF SCROTUM
*IN ADDITION, ALSO COVERED BY 3 COATS - TUNICA VAGINALI - PARIETAL
- VISCERAL
- TUNICA ALBUGINEA
- TUNICA VASCULOSA

1.TUNICA VAGINALIS - REPRESENT LOWER PERSISTENT PORTION OF PROCESSUS VAGINALIS.
- COVER WHOLE TESTIS - EXCEPT FOR POSTERIOR BORDER
2.TUNICA ALBUGINEA - DENSE WHILE FIBROUS COAT COVERING THE TESTIS ALL AROUND.
- COVERED BY VISCERAL LAYER OF T.V EXCEPT - POSTERIORLY
WHERE TESTICULAR VESSEL & NERVE ENTER
-POSTERIOR BORDER OF TUNICA ALBUGINEA IS THICKENED
--->FORM INCOMPLETE VERTICAL SEPTUM - "MEDIASTINUM TESTIS"
-NUMEROUS SEPTA EXTEND FROM MEDIASTINUM
3.TUNICA VASCULOSA - INNER MOST - VASCULAR COAT

*STRUCTURE OF TESTIS - GLANDULAR PART - CONSIST OF 200 TO 300 LOBULES --> CONTAIN 2-3 SEMINIFEROUS TUBULES.
-SEMINIFEROUS TUBULES JOIN TOGETHER AT APICES OF LOBULES TO FORM 20-30 STRAIGHT TUBULES.
-THEY ANASTOMOSE WITH EACH OTHER - TO FORM RETE TESTIS.
-RETE TESTIS GIVE RISE 12-30 EFFERENT DUCTULES---->ENTER EPIDIDYMIS CONTINOUS--->DUCTUS DIFERENS.

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*ARTERIAL SUPPLY -

-TESTICULAR ARTERY----→BRANCH OF ABDOMINAL AORTA---→GIVEN OFF AT LEVEL OF L2.

----→DESCEND ON POSTERIOR ABDOMINAL WALL--→ TO REACH WHERE DEEP INGUINAL RING--→ENTER SPERMATIC CORD

-AT POSTERIOR BORDER OF TESTIS - IT DIVIDES INTO BRANCHES.

-SOME SMALL BRANCHES ---→ENTER POSTERIOR BORDER

-WHILE LARGE BRANCHES ---→MEDIAL & LATERAL ---→PIERCE TUNICA ALBUGINEA

*VEIN SUPPLY - VEIN - EMERGING FROM THE TESTIS FORM - PAMPINI FORM PLEXUS.

-ANTERIOR PART OF PLEXUS ARRANGED AROUND TESTICULAR ARTERY.

-MIDDLE PART OF PLEXUS ARRANGED TESTICULAR ARTERY-DUCTUS DEFERNS

-POSTERIOR PART IS - ISOLATED

-PLEXUS CONDENSES-→INTO 4 VEIN---→AT SUPERFICIAL INGUINAL RING

-→INTO 2 VEIN-→DEEP INGUINAL RING.

-ONE VEIN---→DRAIN INTO INFERIOR VENA CAVA ON RIGHT SIDE

-OTHER VEIN---→DRAIN INTO LEFT RENAL VEIN ON LEFT SIDE

*NERVE SUPPLY - SYM. - T10 OF SPINAL CORD

-NERVE ARE BOTH AFFERENT FOR TESTICULAR SENSATION & EFFERENT TO BLOOD VESSEL (VASOMOTOR).

CLINICAL ANATOMY

1.UNDESCENDED TESTIS OR CRYPTORCHIDISM : THE ORGAN MAY LIE IN THE LUMBER, ILIAC, INGUINAL, OR UPPER SCROTAL REGION. THE IMPORTANT FEATURES OF AN UNDESCENDED TESTIS ARE AS FOLLOWS.

A.THE TESTIS MAY COMPLETE ITS DESCENT AFTER BIRTH.

B.SPERMATOGENESIS MAY FAIL TO OCCUR IN IT.

C.A MALINGNANT TUMOUR IS MORE PRONE TO DEVELOP IN IT.

D.THE CONDITION CAN BE SURGICALLY CORRECTED.

EPTOPIC TESTIS :

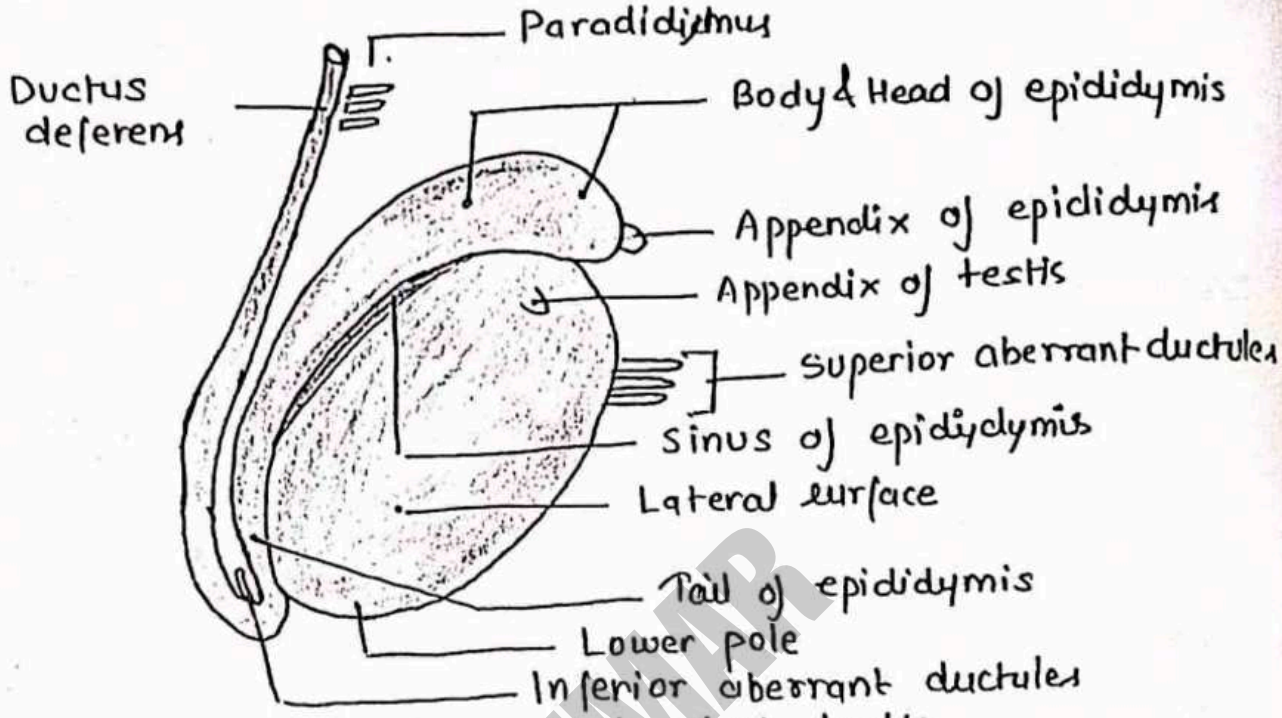
1.TESTIS MAY OCCUPY AN ABNORMAL POSITION DUE TO DEVIATION FROM THE NORMAL ROUTE OF DESCENT.

2.IT MAY BE UNDER THE SKIN OF THE LOWER PART OF THE ABDOMIN, UNDER THE SKIN OF THE FRONT OF THE THIGH, IN THE FEMORAL CANAL, UNDER THE SKIN OF THE PENIS & IN THE PERINEUM BEHIND THE SCROTUM.

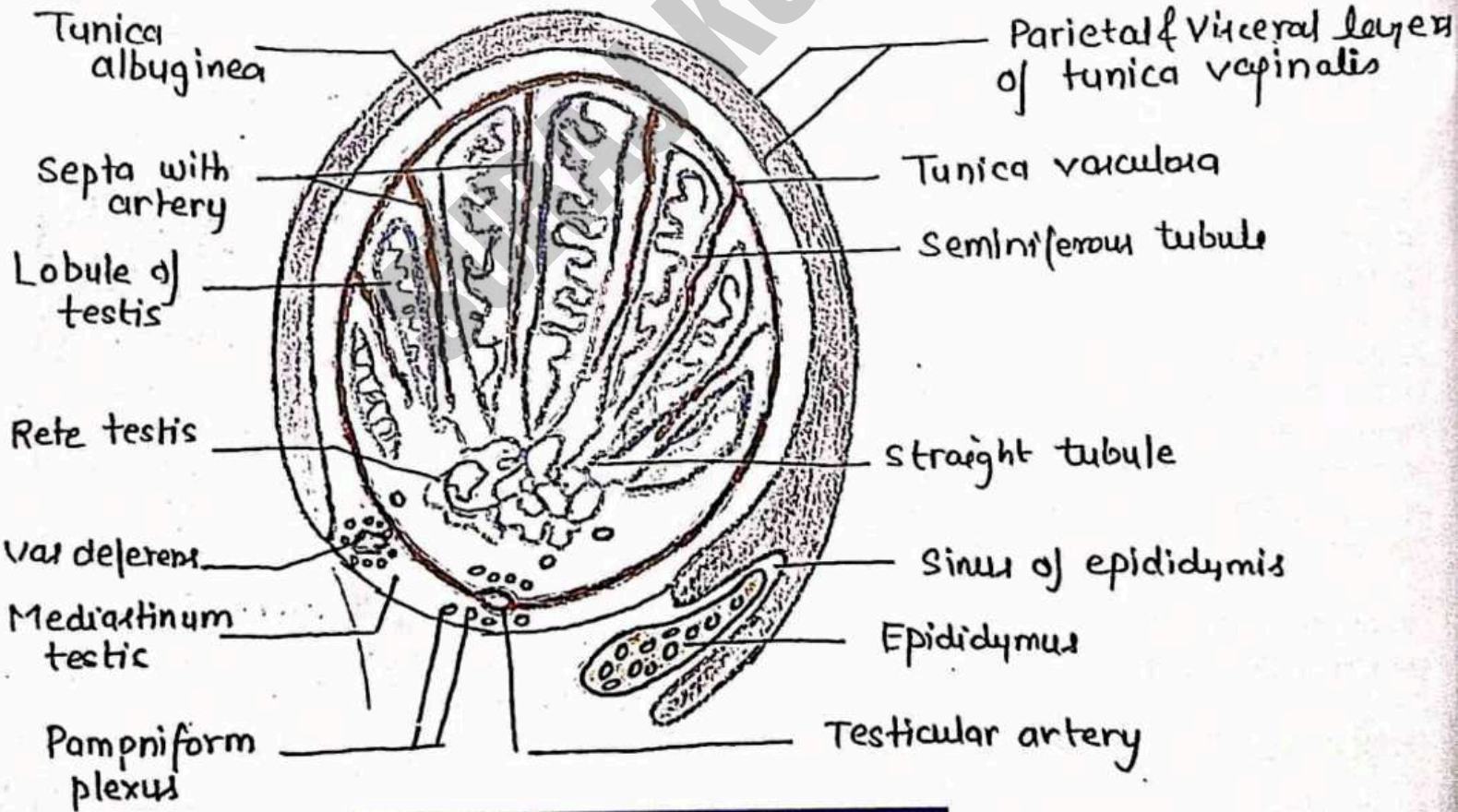
VARICOCOELE - IT PRODUCED BY DIALATATION OF PEMPINIFORM PLEXUS, IT IS USUALLYLEFT SIDE.

HERMAPHRODITISM - TRUE HERMAPHRODITISM - BOTH TESTIS & OVARY ARE PRESENT.

SUDEES KUMAR



∵ Lateral view of the right testis



TESTIS

STOMACH

*DEFINITION - A STOMACH IS--->MUSCULAR BAG--->FORMING WIDEST PART OF DIGESTIVE TUBE.
-ACT AS--> RESERVIOR OF FOOD->HELP IN DIGESTION OF CARBOHYDRATES, PROTIEN & FATS.

LOCATION - UPPER LEFT PART OF ABDOMEN.
-OCCUPYING - EPIGASTRIC, UMBILICAL & LEFT HYPOCHONDRIAC REGION.
SHAPE - IN NORMAL ACTIVE PERSON SHAPE IS "J-SHAPED"
SIZE & CAPACITY - ABOUT 25CM LONG
-CAPACITY - 30ML AT BIRTH
-1000ML AT PUBERTY
-2LITRES OR MORE IN ADULTS.

*EXTERNAL FEATURES -
-TWO ORIFICES - CARDIAC ORIFICE - IS JOINED BY - LOWER END OF OESOPHAGUS.
-BEHIND THE LEFT 7TH COASTAL CARTILAGE.
-2.5CM FROM STERNUM.
-AT LEVEL OF T11
-PYLORIC ORIFICE - OPEN INTO DEODENUM
-AT LEVEL OF L1
-ITS POSITION IS INDICATED ON THE SURFACE OF STOMACH: BY PREPYLORIC VEIN
-BY CIRCULAR GROOVE
- TWO CURVATURES - LESSER CURVATURE - CONCAVE
- FORM RIGHT BORDER OF STOMACH
- PROVIDES ATTACHMENT OF LESSER OMENTUM
- GREATER CURVATURE - CONVEX
- FORM LEFT BORDER OF STOMACH
- PROVIDES ATTACHMENT TO GREATER OMENTUM, GASTROSPLENIC LIGAMENT, GASTROPHRENIC LIGAMENT.

- TWO SURFACE - ANTERIOR - FACE - FORWARDS & UPWARDS
POSTERIOR - FACE - BACKWARD & DOWNWARD
-TWO PARTS - CARDIAC--->FURTHER SUB DIVIDED INTO--->FUNDUS & BODY
-PYLORIC - PYLORIC ANTRUM & PYLORIC CANAL.

*RELATION OF STOMACH -

-PERITONEAL RELATIONS - LINE BY PERITONEUM ON BOTH ITS SURFACES.
1.AT LESSER CURVATURE - LAYER OF PERITONEUM LINING ANTERIOR & POSTERIOR SURFACE MEET.
- AND BECOME CONTINOUS WITH LESSER OMENTUM
2.AT GREATER - TWO LAYER MEET --->TO FORM GASTROSPLENIC LIGAMENT.
-VISCERAL RELATION - ANTERIOR SURFACE - LIVER

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-DIAPHRAGM
-TRANSVERSE COLON
-LEFT PLEURA
-PERICARDIUM
-6-9 RIBS

- POSTERIOR SURFACE - DIAPHRAGM
- LEFT KIDNEY
- LEFT SUPRARENAL GLAND
- PANCREAS
- TRANSVERSE MESOCOLON
- SPLENIC FLEXURE OF COLON
- SPLENIC ARTERY

*BLOOD SUPPLY - STOMACH IS SUPPLIED ALONG-

1.LESSER CURVATURE BY - LEFT GASTRIC ARTERY-→BRANCH COELIAC TRUNK
- RIGHT GASTRIC ARTERY-→PROPER HEPATIC ARTERY

2.GREATER CURVATURE BY - RIGHT GASTROEPIPLOIC A. →GASTRODUODENAL ARTERY
- LEFT GASTROEPIPLOIC A.-→SPLENIC

3.FUNDUS - 5-7 SHORT GASTRIC ARTRIES -→BRANCH OF SPLENIC.

*VEIN - RIGHT & LEFT GASTRIC VEIN -→DRAIN INTO PORTAL VEIN
-RIGHT GASTRO EPIPLOIC END-→SUPERIOR MESENTERIC VEIN
-LEFT GASTROEPIPLOIC END VEIN -→TERMINATE IN SPLENIC VEIN &SHORT GASTRIC VEINS

*LYMPHATIC DRAINAGE - DIVIDED INTO 4X LYMPHATIC TERRITORIES -

1) AREA(A) - UPPER PART OF LEFT 1/3RD -→DRAIN INTO PANCREATICOSPLENIC NODE- TO REACH COELIAC NODES

2) AREA(B) - RIGHT 2/3RD ----→DRAIN INTO LEFT GASTRIC NODE - TO REACH COELIAC NODE

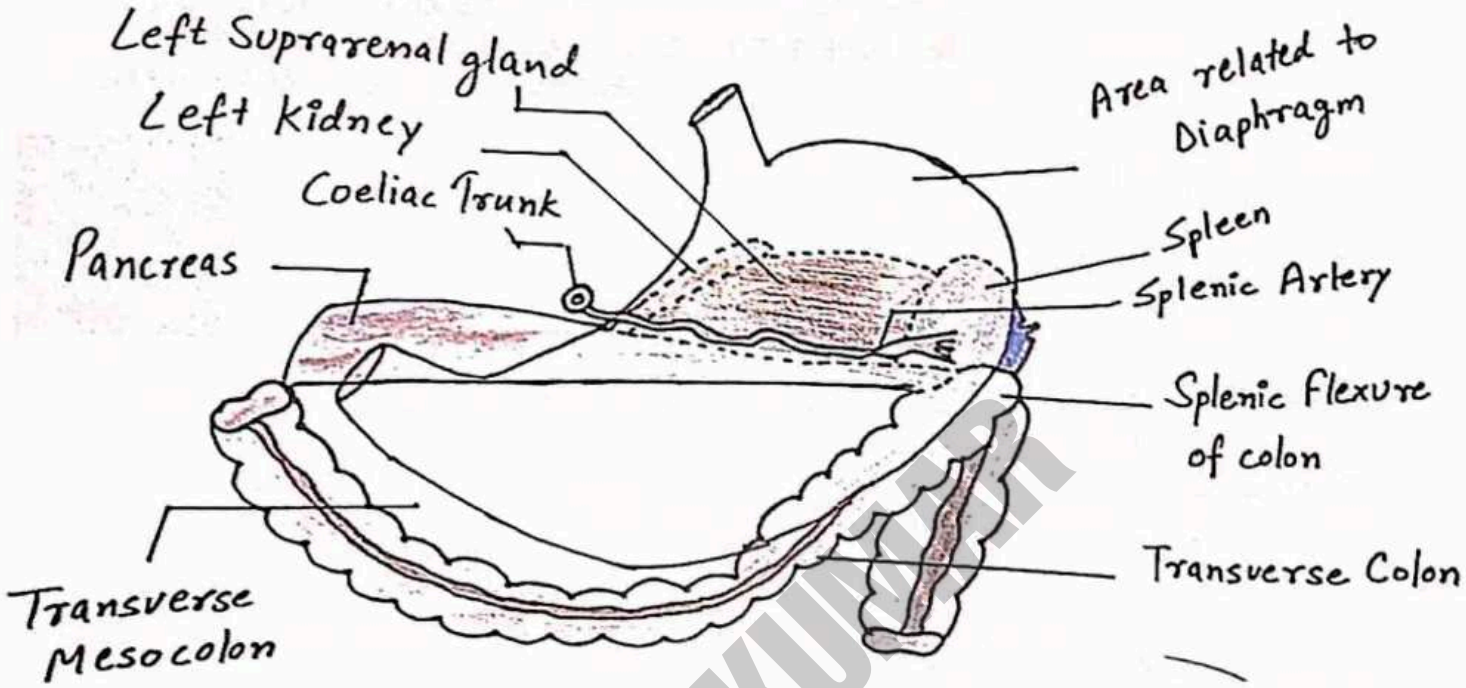
3) AREA(C) - LOWER PART OF LEFT 1/3RD -→RIGHT GASTROEPIPLOIC NODE-→LYING ALONG ARTERY OF SAME NODE
---→DRAIN INTO - SUBPYLORIC NODE, HEPATIC NODE, COELIAC NODE

4) AREA(D) - PYLORIC PART -→DRAIN INTO DIFFERENT DIRECTION-→INTO PYLORIC, HEPATIC, GASTRIC NODES-→TO REACH COELIAC NODES

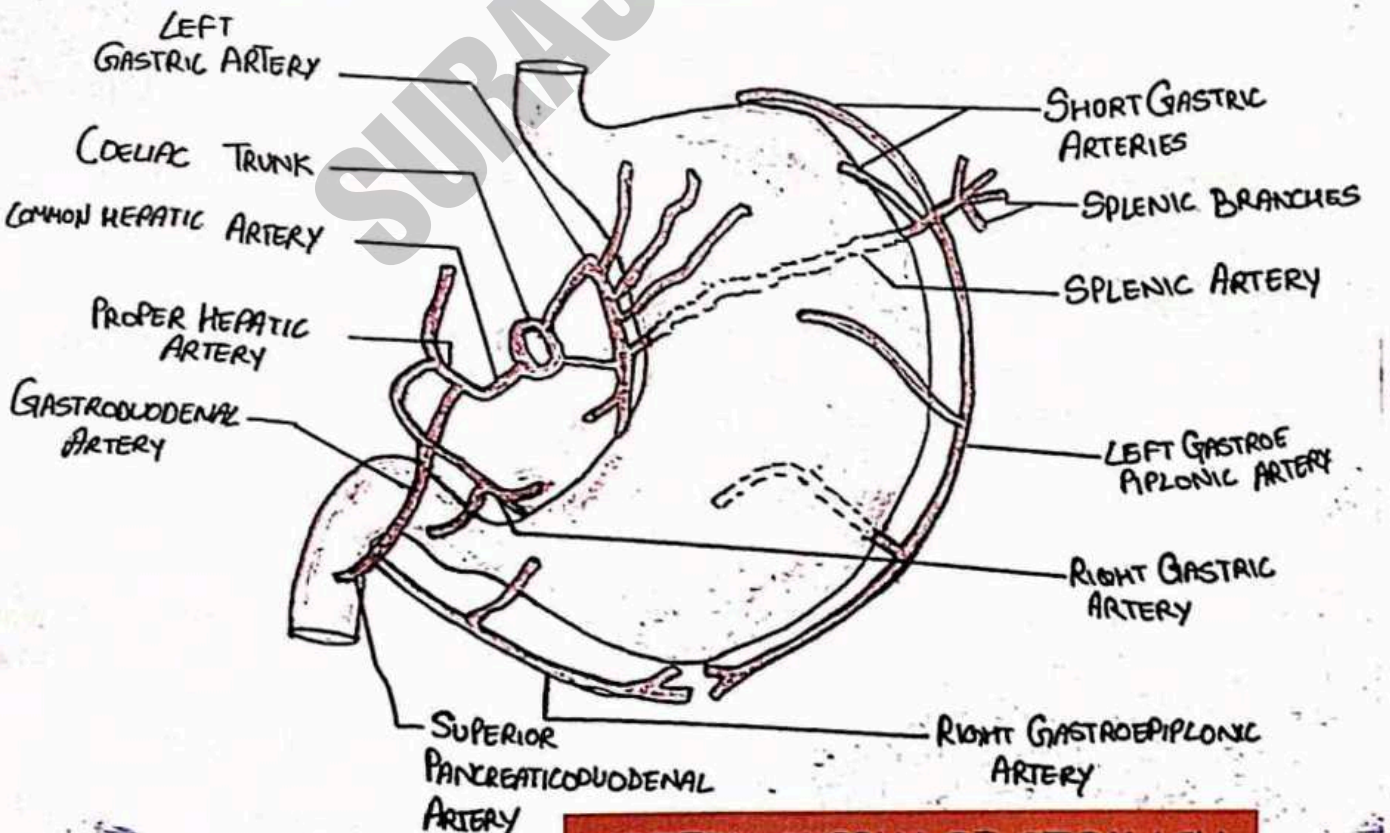
*LYMPH FROM ALL AREA OF STOMACH ULTIMATELY REACH - COELIAC NODE.

*NERVE SUPPLY - SYMPATHETIC NERVE -→DERIVED FROM T6-10SPINAL SEGMENT VIA GREATER SPLACHNIC NERVE, COELIAC & HEPATIC PLEXUS
-PARASYMPATHETIC NERVE - →DERIVED FROM VAGI THROUGH OESOPHAGEAL PLEXUS, GASTRIC NERVE.
-ANTERIOR GASTRIC NERVE -→CONTAIN MAINLY LEFT VEGAL FIBER
-POSTERIOR GASTRIC NERVE -→RIGHT VEGAL FIBERS

SUBAL KUMAR



POSTERIOR RELATION OF STOMATCH



ARTERY SUPPLY OF STOMACH

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DUODENUM

*DEFINITION & LOCATION - SHORT, WIDEST & MOST FIX PART OF SMALL INTESTINE
-EXTEND FROM PYLORUS TO DUODENOJEJUNAL FLEXURE
-LIE ABOVE THE LEVEL OF UMBILICUS - OPPOSITE FIRST, SECOND * THIRD LUMBAR VERTEBRA.

*PARTS - 25CM LONG & DIVIDED INTO 4x PART
1.FIRST OR SUPERIOR PART, 5CM LONG
2.SECOND OR DESCENDING PART, 7.5CM LONG
3.THIRD OR HORIZONTAL PART, 10CM LONG
4.FOURTH OR ASCENDING PART, 2.5CM LONG

1) FIRST PART - BEGIN AT PYLORUS TO MEET SECOND PART AT SUPERIOR DUODENAL FLEXURE.

*RELATION -
-PERITONEAL RELATION - PROXIMAL 2.5CM MOVABLE--->ATTACHMENT LESSER OMENTUM ABOVE
--->GREATER OMENTUM BELOW
-DISTAL 2.5CM FIXED-RETROPERITONEAL-COVER ONLY ANTERIORLY
-VISCERAL - ANTERIOR - QUADRATE LOBE OF LIVER
- GALL BLADDER
- POSTERIORLY - GASTRODUODENAL ARTERY
- BILE DUCT
- PARIETAL VEIN
- SUPERIORLY - EPIPLOIC FORAMEN
- INFERIORLY - HEAD & NECK OF PANCREAS

2) SECOND PART - BEGIN AT SUPERIOR DUODENAL FLEXURE TO REACH LOWER BORDER OF 3RD LUMBAR VERTEBRA.

*RELATION -
-PERITONEAL RELATION - RETROPERITONEAL - ANTERIOR SURFACE COVERED WITH PERITONEUM-EXCEPT-MIDDLE-->WHERE COLON RELATED
-VISCERAL - ANTERIORLY - RIGHT LOBE OF LIVER
- TRANSVERSE COLON
- ROOT OF THE TRANSVERSE MESOCOLON
- SMALL INTESTINE

3) THIRD PART - BEGIN AT INFERIOR DUONDENAL FLEXURE - AND BY JOINING 4TH PART IN FRONT OF ABDOMINAL AORTA.

RELATION -
-PERITONEAL RELATION - RETROPERITONEAL-ANTERIOR SURFACE IS COVERS BY PERITONEUM
-EXCEPT - MEDIAN PLANE - IT IS CROSSED BY SUPERIOR

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MESCENTERIC VESSEL

-VISCERAL RELATION - ANTERIORLY - SUPERIOR MESCENTERIC VESSEL
-ROOT OF MESCENTERY
- POSTERIORLY - RIGHT URETER
- RIGHT PSOAS MAJOR
- TESTICULAR OR OVARIAN VESSEL
- INFERIOR VENA CAVA
- ABDOMINAL AORTA
- SUPERFICIALY - HEAD OF PANCREAS
- INFERIORLY - COLIS OF JEJUNUM

4)FOURTH PART - RUN UPWARD UPTO UPPER BORDER OF 2ND LUMBAR VERTEBRA
- TO BECOME CONTINOUS WITH JEJUNUM AT DUODENOJEJUNAL FLEXURE

*RELATION -

-PERITONEAL RELATION - MOSTLY RETROPERITONEAL - COVERED WITH PERITONEUM ONLY ANTERIORLY.

-VISCERAL RELATION - ANTERIORLY - TRANSVERSE COLON
- TRANSVERSE MESOCOLON
- LESSER SAC
- STOMACH
- POSTERIORLY - LEFT SYMPATHETIC CHAIN
- LEFT RENAL ARTERY
- LEFT GONADAL ARTERY
- INFERIOR MESCENTRIC VEIN
- TO RIGHT - ATTACHMENT OF UPPER PART OF ROOT OF MESCENTERY
-TO LEFT - LEFT KIDNEY
- LEFT URETER

*ARTERIAL SUPPLY-DEVELOP PARTLY FROM--->FOREGUT - OPENING OF BILE DUCT INTO -->2ND PART -REPRESENT JUNCTION BETWEEN FOREGUT & MID GUT
- UP THE LEVEL OF OPENING -->SUPPLIED BY SUPERIOR PANCREATICODUODENAL ARTERY
-BELOW--> INFERIOR PANCREATICODUODENAL ARTERY

*VENOUS DRAINAGE - VEINS OF DUODENUM DRAIN INTO - SPLENIC SUPERIOR MESCENTRIC PORTAL VEN

*NERVE SUPPLY - SYMPATHETIC NERVE - T9-10 SPINAL
-PARASYMPATHETIC - FROM VAGUS PASS THROUGH COELIAC PLEXUS.

CLINICAL ANATOMY

1.PEPTIC ULCER CAN OCCUR IN THE SIDE OF PEPSIN & HCL ACID, NAMELY THE STOMACH AND FIRST PART OF DUODENUM.
2.GASTRIC ULCER OCCUR MAINLY ALONG THE LESSER CURVATURE.
3.CONGENITAL STENOSIS & OBSTRUCTION OF THE 2ND PART OF THE DUODENUM MAY OCCUR AT THE SITE OF THE OPENING OF THE BLUE DUCT.

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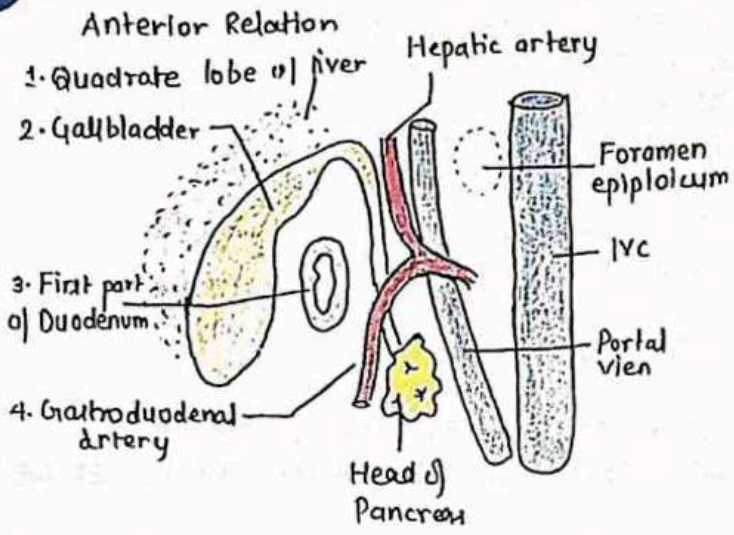
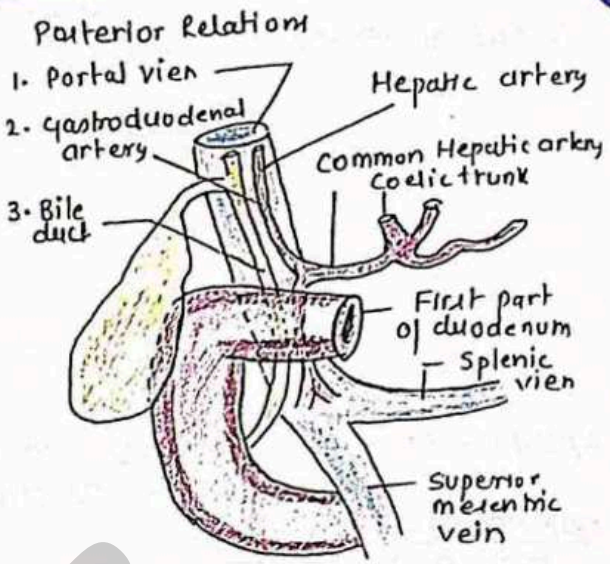
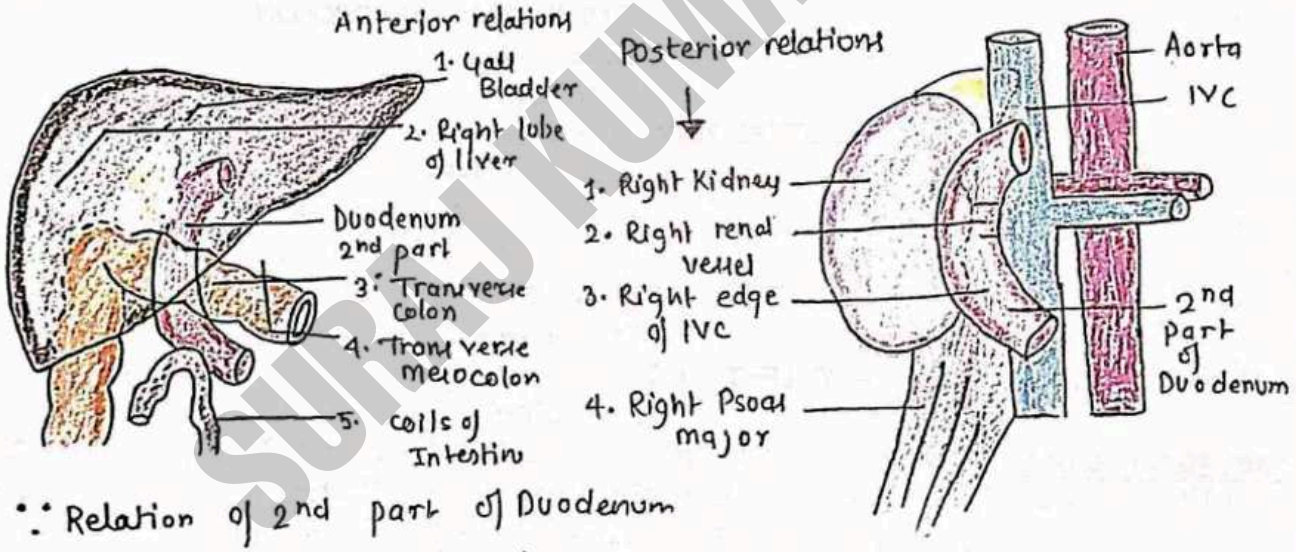
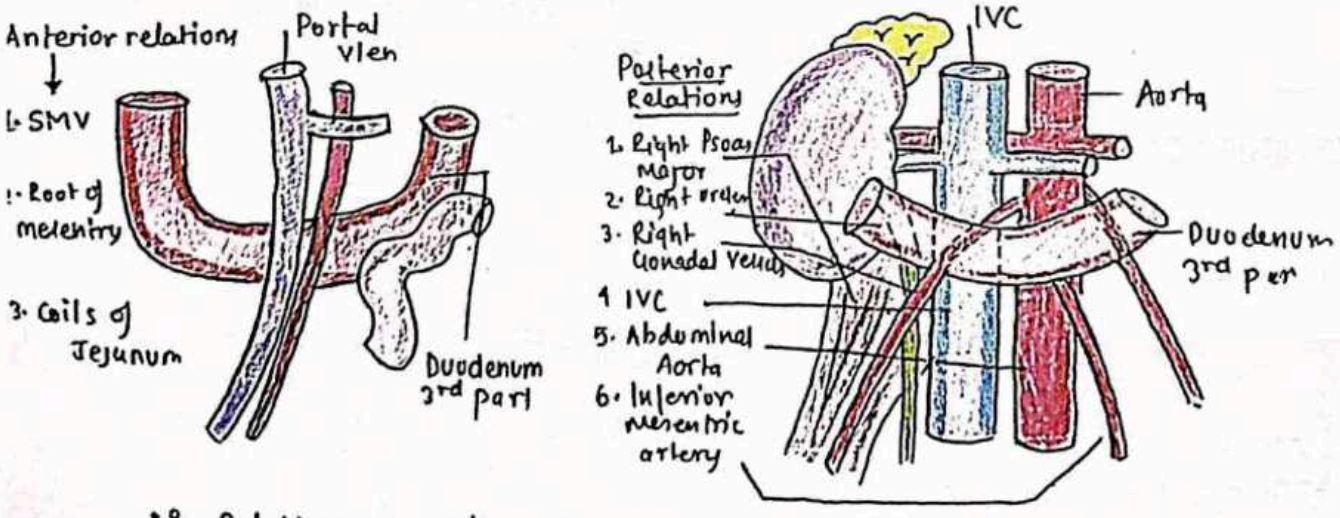




∴ Relation of First part of Duodenum



∴ Relation of 2nd part of Duodenum



∴ Relation of 3rd part of Duodenum

RELATION OF PARTS OF DUODENUM

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CAECUM

*FEATURES -

- LARGE BLIND SAC FORMING THE COMMENCEMENT OF LARGE INTESTINE
- SITUATED IN - RIGHT ILIAC FOSSA
- ABOVE THE LATERAL HALF OF INGUINAL LIGAMENT
- COMMUNICATION- SUPERIALLY - ASCENDING COLON
  MEDIALLY -CAECOCOLIC JUNCTION WITH ILEUM
  POSTEROMEDIALLY- APPEN DISC

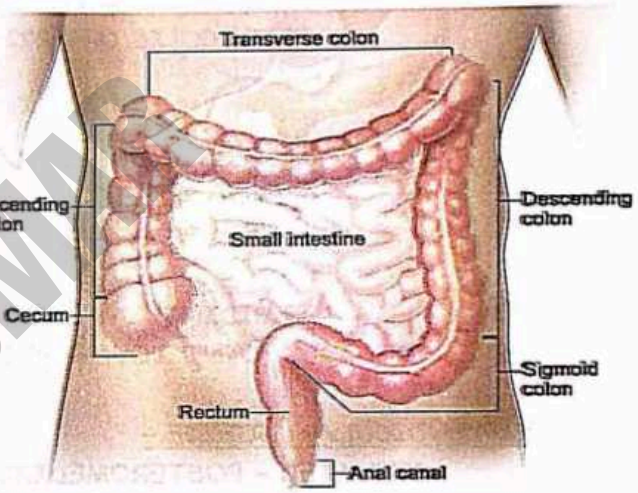
DIMENSION- 6CM LONG AND 7.5 CM BROAD (GREATER WIDTH THAN LENGTH)

*RELATION-

ANTERIOR - COIL OF INTESTINE
-ANTERIOR ABDOMINAL WALL
POSTERIOR - RIGHT PSOAS AND ILIACUS

NERVE-GENITOFEMORAL NERVE
- FEMORAL NERVE
- LATERAL CUTANEOUS NERVE OF THIGH

VESSEL- TESTICULAR OR OVARIAN



*ARTERIAL SUPPLY -
BRANCHES OF ILEOCOLIC ARTERY

*VEIN SUPPLY - SUPERIOR MESENTERIC VEIN

*NERVE - SYM: T11 TO L1
- PARA SYM:- VAGUS NERVE

*ILIOCAECAL VALVE- LOWER END OF ILEUM OPEN POSTEROMEDIAL - CAECOCOLIC JUNCTION
- ILEOCAECAL OPENING GUARDED, ILEOCAECAL VALUE
-TWO FRENULA - FORMED BY FUSION OF LIPS AT THE END OF APERTURE
-THESE ARE -→LEFT OR ANTERIOR FRENULA
--→ RIGHT OR POSTERIOR FRENULA
-VALVE IS ACTIVELY CLOSED BY SYMPATHETIC NERVE → CAUSE TONIC CONTRACTION OF ILEACAECAL SPHINCTER

*FUNCTION -
- PREVENT REFLEX. FROM CAECUM TO ILEUM
- REGULATE PASSAGE OF ILEAL CONTENTS INTO CAECUM

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VERMIFORM APPENDIX

*WORM LIKE DIVERTICULUM - ARISING FROM POSTEROMEIAL WALL OF CAECUM.
*ABOUT 2CM BELOW ILEOCAECAL ORIFICE.

*DIMENSIONS - LENGTH VARIES FROM 2 TO 20 CM WITH AN AVERAGE OF 9 CM
-LONGER IN CHILDREN THAN IN ADULT.
*POSITION - LIE IN RIGHT ILIAC FOSSA →BASE →IS FIXED
TIP → DIRECTION
-POSITION ARE OFTEN COMPARED TO HOUR HAND OF CLOCK.
-->PARACOLIC OR 11'O CLOCK POSITION - APPENDIX MAY PASS UPWARD
-->RETROCOLIC OR 12'O CLOCK POSITION - APPENDIX MAY LIE BEHIND CAECUM
- COMMONEST POSITION - 65%
--->SPLENIC OR 2'O CLOCK POSITION - APPENDIX POINT TOWARDS SPLEEN
--->PROMONTERIC OR 3'O CLOCK POSITION - HORIZONTALLY
---->PELVIC OR 4'O CLOCK POSITION - MAY DESCEND INTO PELVIS
-COMMON POSITION-30%
---->MIDINGUINAL OR 6'O CLOCK POSITION - BELOW THE CAECUM & MAY POINT TOWARDS INGUINAL LIGAMENT.

*APPENDICULAR ORIFICE -
-SITUATED - POSTEROMEDIAL OF CAECUM 2CM BELOW THE ILEOCAECAL ORIFICE
-OCCASIONALLY GUARDED BY - SEMILUNAR FOLD OF THE MUCOUS MEMBRANE, KNOWN AS 'VALVE OF GERLACH'
-ORIFICE IS MOVED ON SURFACE -→BY POINT SITUATED 2CM BELOW →TRANSTUBERCULAR & RIGHT LATERAL PLANES
-MC BURNEY'S POINT - SITE OF MAX.TENDERNESS IN APPENDICITIS
- POINT LIE AT JUNCTION OF LATERAL 1/3RD AND MEDIAL 2/3RD OF LINE JOINING THE UMBILICUS TO RIGHT ANTERIOR,SUPERIOR ILAC SPINE.

*PERITONEAL RELATION - APPENDIX IS SUSPENDED BY SMALL, TRIANGULAR FOLD OF PERITONEUM CALLED MESOAPPENDIX.

*BLOOD SUPPLY - APPENDICULAR ARTERY -->IS BRANCH OF LOWER DIVISION-→OF ILIOCOLIC ARTERY
-RUN BEHIND - TERMINAL PART OF ILEUM & ENTERS THE MESOAPPENDIX
-BLOOD DRAINED BY - APPENDICULAR VEIN
- ILEOCOLIC VEIN
- SUPERIOR MESCENTRIC VEIN--->TO PORTAL VEIN

*NERVE - SYMPATHETIC NERVE- T9 TO T10 SEGMENT THROUGH THE COELIAC PLEXUS
PARASYMPATHETIC NERVE--->DERIVED FROM VAGUS

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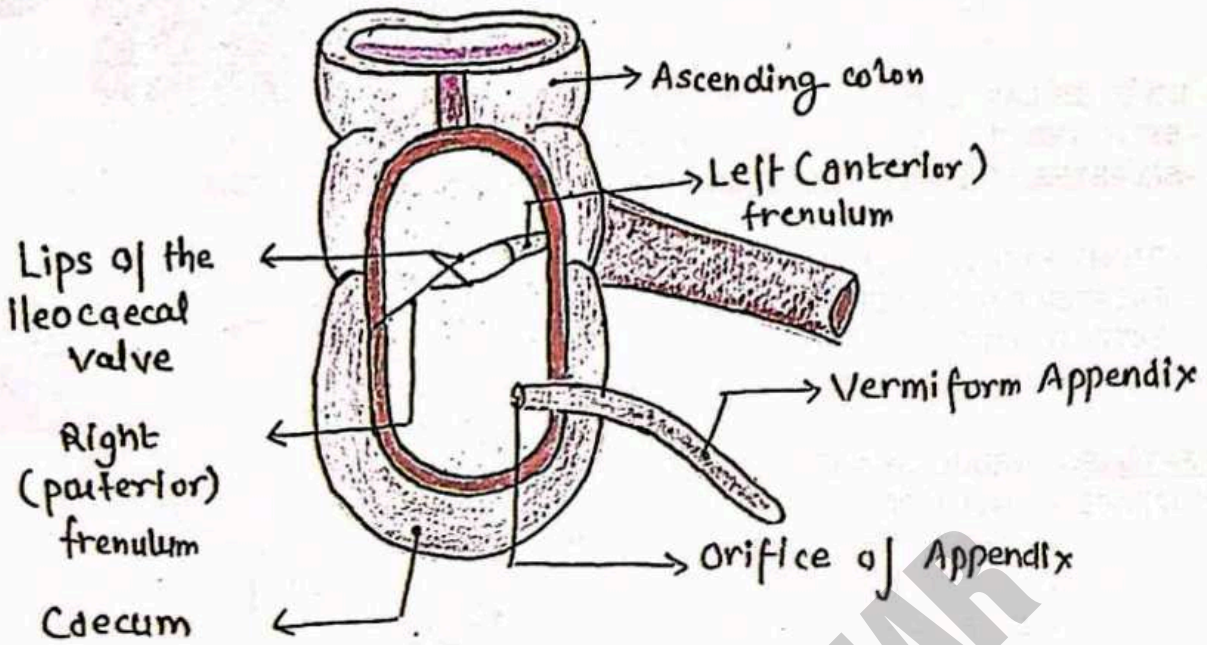
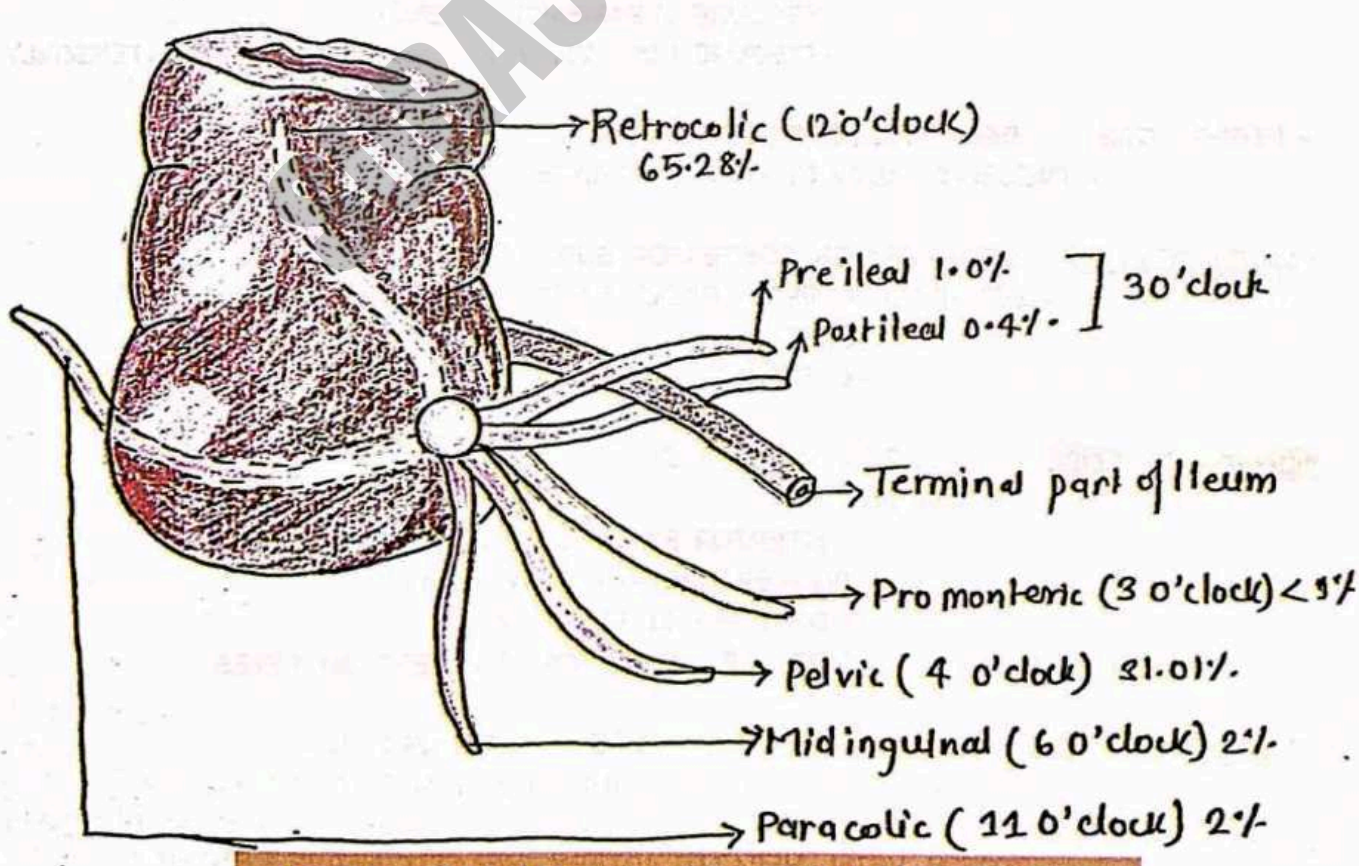


INTERIOR OF CAECUM



POSITION OF VERMIFORM APPENDIX

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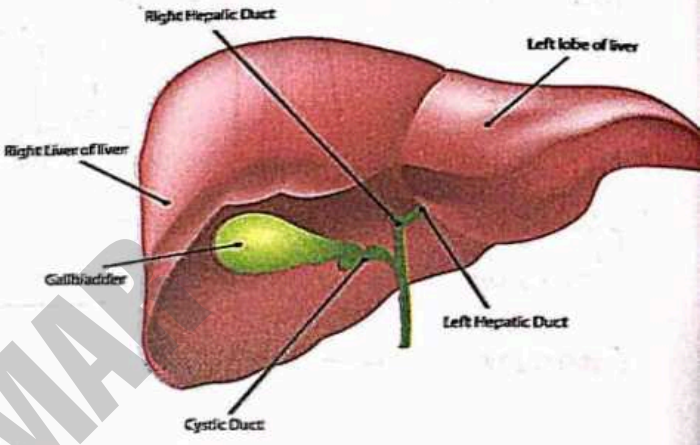
LIVER ANATOMY

*FEATURES:- LIVER IS LARGE AND SOLID GLAND
-SITUATED IN RIGHT UPPER QUADRANT OF ABDOMINAL CAVITY
-SECRETES BILE AND PERFORMS VARIOUS METABOLIC FUNCTION

*LOCATION:- -RIGHT HYPOCHONDRIUM
- GREATER PART IN EPIGASTRIUM
- EXTEND INTO LEFTHYPOCHONDRIUM



*EXTERNAL FEATURE:- WEDGE-SHAPED.
-FIVE SURFACE - ANTERIOR
- POSTERIOR
- SUPERIOR
- INFERIOR
- RIGHT

- ONE PROMINENT BORDER - INFERIOR BORDER
- SEPARATES ANTERIOR FROM INFERIOR SURFACE

- TWO LOBES - DIVIDED INTO- RIGHT & LEFT LOBES BY FALCIFORM LIGAMENT---→ ANTERIORLY & SUPERIORLY
-FISSURE LIGAMENTUM TERES - INFERIORLY
-FISSURE FOR LIGAMENTUM VENOSUM - POSTERIORLY

- RIGHT LOBE - LARGER THE LEFT LOBE
- PRESENT CAUDATE AND QUADRATE LOBES

*CAUDATE LOBE - SITUATED ON POSTERIOR SURFACE
- BOUNDED→RIGHT > GROOVE FOR THE INFERIOR VENA CAVA
-LEFT→ FISSURE FOR LIGAMENTUM VENOSUM
-INFERIORLY----→ PORTA HEPATIS.

*QUADRATE LOBE - SITUATED ON ---→ INFERIOR SURFACE
- BOUNDED ---→
ANTERIOR BY INFERIOR BORDER
POSTERIORLY→ PORTA HEPATIS
RIGHT - GALL BLADDER
LEFT - FISSURE FOR LIGAMENTUM TERES

*PORTA HEPATIS- DEEP, TRANSVERSE FISSURE ABOUT 5CM LONG
- SITUATED ON, INFERIOR SURFACE OF RIGHT LOBE OF LIVER
- LIE B/W CAUDATE LOBE ABOVE AND QUADRATE LO BE BELOW
- PORTAL VEIN, HEPATIC ARTERY, HEPATIC NERVE - ENTER

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*LIGAMENT :- LIGAMENT OF LIVER ARE 2X TYPE - FALSE & TRUE

*FALSE LIGAMENT - ARE ACTUALLY PERITONEAL FOLDS

1.FALCIFORM LIGAMENT - SICKLE-SHAPED FOLD OF PERITONEUM
-CONNECT THE LIVER TO UNDER SURFACE OF DIAPHRAGM AND ANTERIOR ABDOMINAL WALL.

2.CORONARY LIGAMENT -TRIANGLE FOLD OF PERITONEUM CONNECTING-BARE AREA TO DIAPHRAGM
- CONSIST OF TWO LAYER - UPPER AND LOWER

3.RIGHT TRIANGULAR LIGAMENT- SMALL TRIANGULAR FOLD OF PERITONEUM
-WHICH CONNECTING RIGHT LATERAL SURFACE OF LIVER TO DIAPHRAGM

4.LEFT TRIANGULAR LIGAMENT- SMALL TRIANGULAR FOLD OF PERITONEUM
-WHICH CONNECT UPPER SURFACE OF LEFT LOBE TO DIAPHRAGM

5.LESSER OMENTUM- FOLD OF PERITONEUM CONNECTING LESSER CURVATURE OF STOMACH ----->VISCERAL SURFACE OF LIVER
-HEPATIC ATTACHMENT IS -J SHAPE
- VERTICAL LIMB ATTACH, MARGIN OF FISSURE FOR LIGAMENT VENOSUM
- HORIZONTAL LIMB----> MARGIN OF PORTA HEPATIS.

*TRUE LIGAMENTS - THEY ARE ACTUALLY REMNANTS OF FETAL STRUCTURE

-LIGAMENTUM TERES HEPATIS- REMNANT OF LEFT UMBILICAL VEIN

-LIGAMENTUM VENOSUM - REMNANT OF DUCTUS VENOSUS
- WHICH IN FETAL LIFE CONNECT LEFT BRANCH OF PORTAL VEIN WITH INTERIOR VENA CAVA

*RELATION :-

*PERITONEAL RELATION - MOST OF LIVER IS COVERD BY PERITONEUM
- AREA NOT COVERED BY PERITONEUM ARE AS FOLLOWS -

1.TRIANGULAR BARE AREA - ON THE POSTERIOR SURFACE OF RIGHT LOBE
- LIMITED BY SUPERIOR & INFERIOR LAYER OF CORONARY LIGAMENT

2.GROOVE FOR INFERIOR VENA CAVA - POSTERIOR SURFACE OF RIGHT LOBE

3.FOSSA FOR GALLBLADDER - LIE ON INFERIOR SURFACE OF RIGHT LOBE

4.AREA OF ATTACHMENT OF LESSER OMENTUM

*VISCERAL RELATION -

1) ANTERIOR SURFACE - RELATED TO XIPHOID PROCESS

2) DIAPHRAGM ON EACH SIDE -->SEPARATE PLEURA- ABOVE LEVEL OF LINE DRAWN FROM.
-XIPHISTERNAL JOINT TO 10TH RIBS IN MID AXILLARY LINE
--->LUNGS - ABOVE LEVEL OF LINE FROM SAME JOINT TO 8TH RIB.

3) POSTERIOR SURFACE-->IS TRIANGULAR -BARE AREA->RIGHT SUPRARENAL

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-GROOVE FOR INFERIOR VENA CAVA--->UPPER PART OF VESSEL
-CAUDATE LOBE - CRURA OF DIAPHRAGM

-FISSURE FOR LIGAMENTUM VENOSUM - CONTAIN 2 LAYER OF LESSER OMENTUM

-POSTERIOR SURFACE OF LEFT LOBE-->OESOPHAGEAL IMPRESSION

4) INFERIOR SURFACE - INFERIOR SURFACE OF LEFT LOBE-->GASTRIC IMPRESSION
-FISSURE FOR LIGAMENTUM TERE - PASS INFERIOR TO PORTA HEPATITIS FROM BORDER
-QUADRATE LOBE->RELATED PYLORUS FIRST PART OF DUODENUM
-FOSSA FOR GALL BLADDER->LIE TO RIGHT OF QUADRATE LOBE
-->RIGHT OF THIS FOSSA-->BEAR COLIC IMPRESSION

5) RIGHT SURFACE - QUADRILATERAL & CONVEX
-RELATED TO DIAPHRAGM OPPOSITE 7TH TO 11TH RIBS IN MIDAXILLARY LINE
-DIAPHRAGM SEPARATE-->PLEURA UP TO 10TH RIB
-->LUNGS UPTO 8TH RIB

6) SUPERIOR SURFACE - QUADRILATERAL & CONCAVITY IN MIDDLE->IS CARDIAC IMPRESSION
-EACH SIDE - CONVEX-->TO FIT THE DOME OF DIAPHRAGM

*BLOOD SUPPLY -

*LIVER IS HIGHLY VASCULAR ORGAN
*RECEIVE BLOOD FROM 2x SOURCE->ARTERIAL BLOOD->IS SUPPLY BY ->HEPATIC ARTERY(20%)
----->VENOUS->IS SUPPLY BY -PORTAL VEIN(80%)

*VENOUS DRAINAGE - BY 3 LARGE HEPATIC VEIN - LEFT HEPATIC VEIN
-MIDDLE HEPATIC VEIN
-RIGHT HEPATIC VEIN
- 3 VEIN MAY ENTER IUC INDEPENDENTLY
- BUT LEFT & MIDDLE VEIN USUALLY JOIN, SO THAT ONLY 2 MAJOR VEIN JOINT THE IUC

*NERVE SUPPLY -
*SYMPATHETIC FIBRES-->DERIVED FROM COELIAC PLEXUS
*RUN ALONG LESSER OMENTUM & ENTER PORTA HEPATITIS

*PARASYMPATHETIC FIBRE ->DERIVED FROM HEPATIC BRANCH OF ANTERIOR VAGAL TRUNK
*REACH PORTA HEPATIC THROUGH LESSER OMENTUM

*HEPATIC SEGMENTS - STRUCTURAL UNIT OF LIVER -

- RIGHT LOBE - DIVIDED INTO ->ANTERIOR & POSTERIOR PART
- LEFT LOBE-DIVIDED INTO-->MEDIAL & LATERAL PARTS

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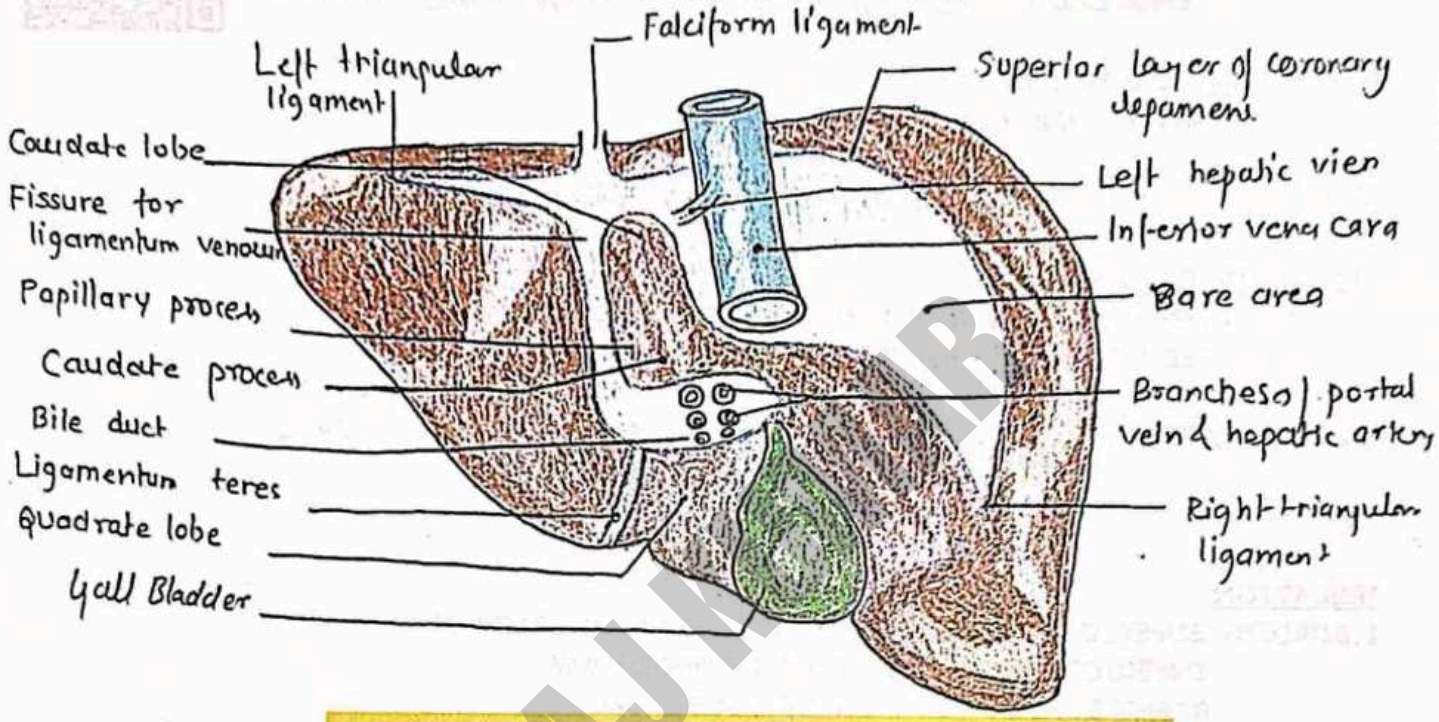
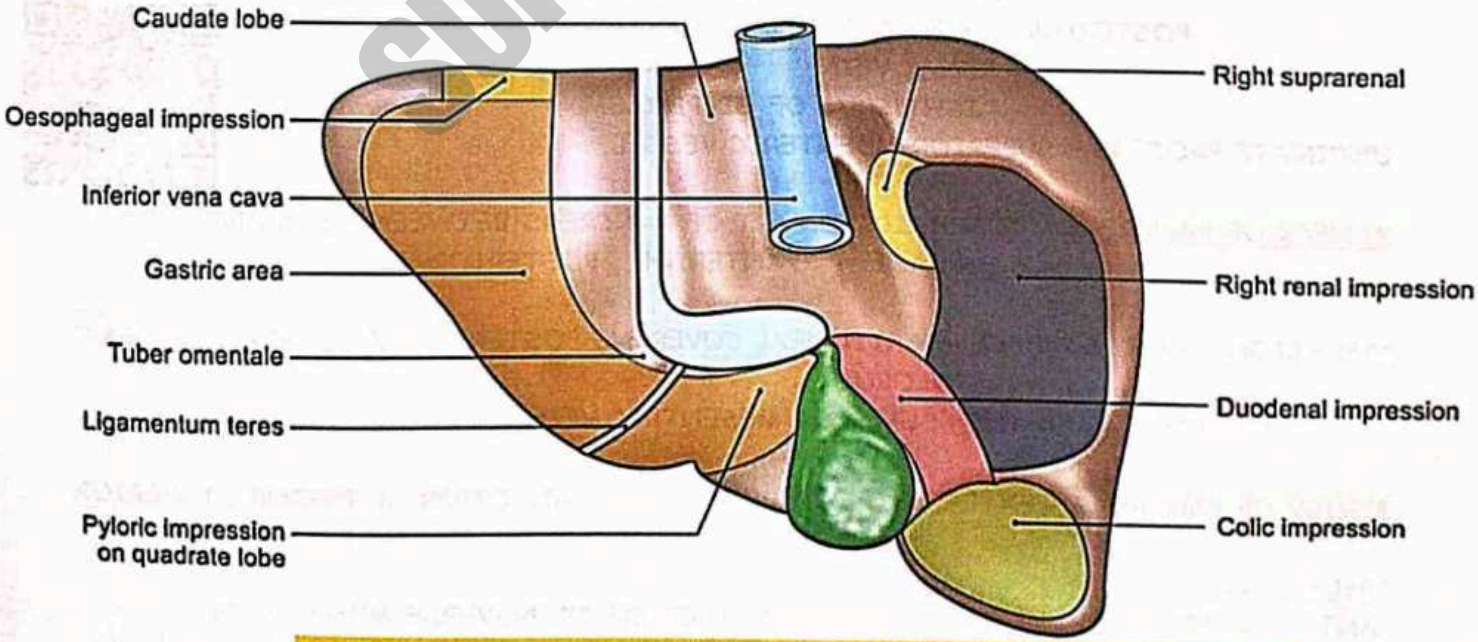
- EACH OF THESE PART FURTHUR DIVIDED INTO UPPER & LOWER PART
-VEIN DRAINING THE HEPATIC SEGMENTS ARE INTERSEGMENTAL
--->THEY DRAIN MORE THAN ONE SEGMENT

-FISSURE FOR LIGAMENTUM TERE & LIGAMENTUM VENOSUM-->MAKE JUNCTION BETWEEN MEDIAL & LATERAL SEGMENT OF LEFT LOBE.



LIVER SEEN FROM BEHIND & BELOW



RELATION OF INFERIOR SURFACE OF LIVER

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PANCREAS ANATOMY

*FEATURES- GLAND THAT IS PARTLY EXOCRINE AND PARTLY ENDOCRINE
- EXOCRINE - SECRETS - DIGESTIVE PANCREATIC JUICE
- ENDOCRINE - SECRETS - HORMONE E.G - INSULINE

*LOCATION - LIE IN POSTERIOR ADBOMINAL WALL
-AT 1ST AND 2ND LUMBAR VERTEBRA

*PANCREAS- DIVIDED INTO HEAD , NECK , BODY AND TAIL

1) HEAD OF PANCREAS :-
- ENLARGED FLATTENED RIGHT END OF PANCREAS
- SITUATED WITHIN C - SHAPE CURVE OF DUODENUM

*EXTERNAL FEATURE - 3 BORDER - SUPERIOR, INFERIOR, RIGHT LATERAL
- 2 SURFACE - ANTERIOR & POSTERIOR
- ONE PROCESS - UNCINATE PROCESS

*RELATION
1.BORDER- SUPERIOR - OVERLAPPED BY 1ST PART OF DUODENUM
INFERIOR BORDER - 3RD PART OF DUODENUM
RIGHT BORDER - 3RD PART OF DUODENUM

2.SURFACE - ANTERIOR - FRIST PART OF DUODENUM
- TRANSVERSE COLON & JEJUNUM
- POSTERIOR - INFERIOR VENA CAVA
- RENAL VEIN
- RIGHT CRUS OF DIAPHRAGM
UNCINATE PROCESS - SUPERIOR MSENTERIC VESSEL

2) NECK OF PANCREAS - CONSTRICTED PART OF PANCREAS BETWEEN HEAD AND BODY
- 2 SURFACE --->ANTERIOR , POSTERIOR

*RELATION - ANT. SURFACE - PERITONEAL COVERING POSTERIOR WALL OF LESSER SAC
-PYLORUS
-POST. SURFACE - SUPERIOR MESENTRIC VEIN

3)BODY OF PANCREAS - EXTERNAL FEATURE-3 BORDER-ANTERIOR, SUPERIOR , INFERIOR

*RELATION -
ANT. BORDER - PROVIDE ATTACHMENT TO ROOF OF TRANSVERSE MESOCOLON
SUP. BORDER - COELIAC TRUNK, HEPATIC ARTERY, SPLENIC ARTERY

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INF. BORDER - SUPERIOR MESENTRIC VESSEL

3.SRUFACE - ANT. SURFACE - COVERED BY PERITONEUM
- LESSER SAC & STOMACH

POST. SURFACE - DEVOID OF PERITONEUM
-AORTA
-LEFT SUPRARENAL GLAND & LEFT KIDNEY
-SPLENIC VEIN

INFERIOR SURFACE - COVERED BY PERITONEUM
-DUODENOJEJUNAL FLEXURE

4)TAIL OF PANCREAS

- LEFT END OF PANCREAS
- LIE IN LIENORENAL LIGAMENT

***DUCTS OF PANCREAS -2 DUCTS - MAIN & ACCESSARY PANCREATIC DUCT**

1) MAIN PANCREATIC DUCT - 3MM IN DIAMETER
-LIE NEAR POSTERIOR SURFACE OF PANCREASE
- BEGIN AT TAIL, RUN TOWARD RIGHT & BEND AT NECK DOWNWAR TO RIGHT IN THE HEAD
-WITH IN HEAD OF PANCREAS - PANCREATIC DUCT RELATE TO BILE DUCT
-2X DUCT ENTER THE WALL OF SECOND PART OF DUODENUM
-JOIN TO FORM HEPATO PANCREATIC AMPULLA OF VATER
-WHICH OPEN--→ AS MAJOR DUODENAL PAPILLA
-8 TO 10 CM DISTAL TO PYLORUS

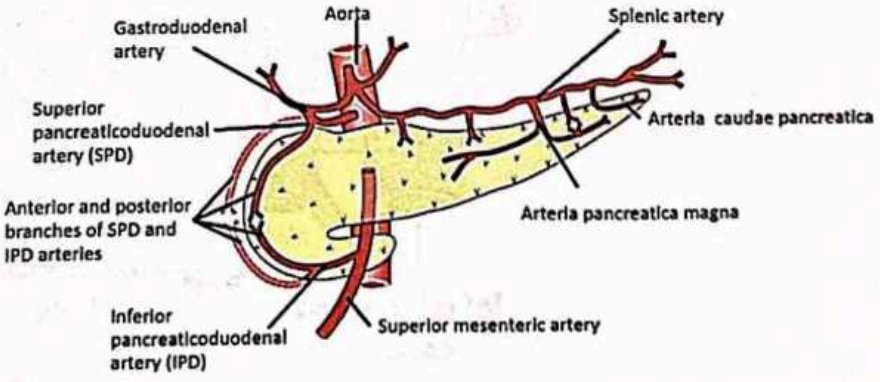
2) ACCESSARY PANCREATIC DUCT - BEGIN IN LOWER PART OF HEAD, CROSS FRONT OF MAIN DUCT
- OPEN INTO DUODENUM AT MINAR DUODENAL PAPILLA
-6 TO 18 CM DISTAL TO PYLORUS

***ARTERIAL SUPPLY:-**

-MAINLY BY- PANCREATIC BRANCHES OF SPLENIC ARTERY
- SUPERIOR PANCREATICODUODENAL ARTERY
- INFERIOR PANCREATICO DUODENAL ARTERY

***VENOUS DRAINAGE :-**

- SPLENIC VEIN
- SUPERIOR MESENTERIC VEIN
- PORTAL VEIN



*NERVE SUPPLY:-

-PARASYMPATHETIC NERVE
- VAGUS NETVE
-SYMPATHETIC NETUE -
SPLANCHNIC NERVE

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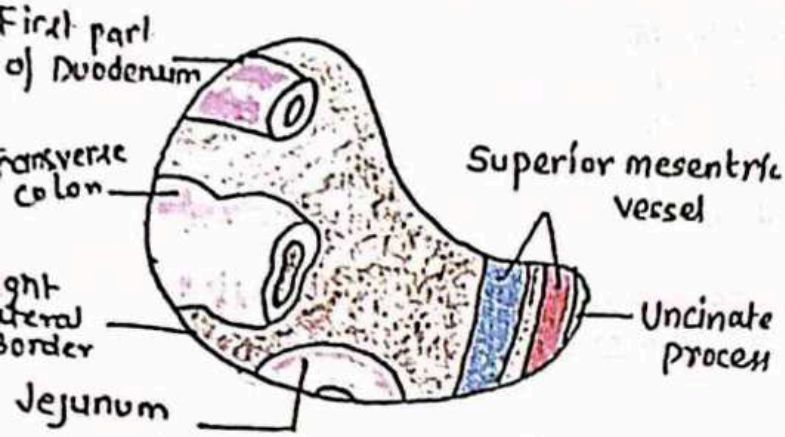
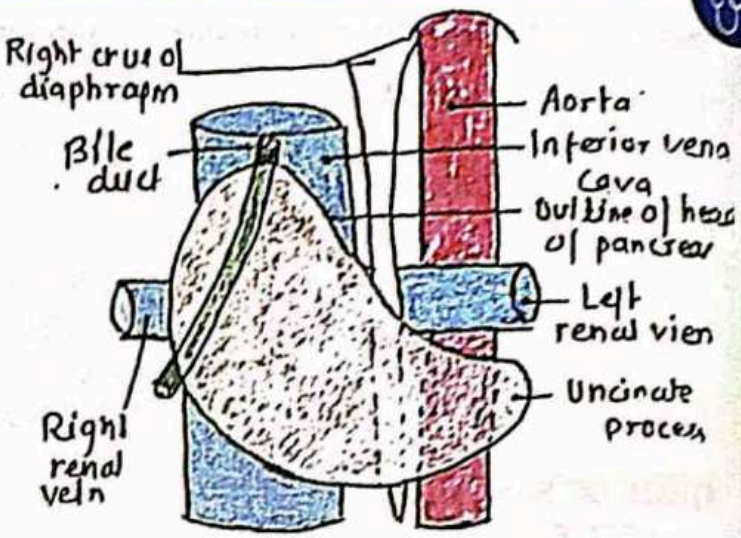
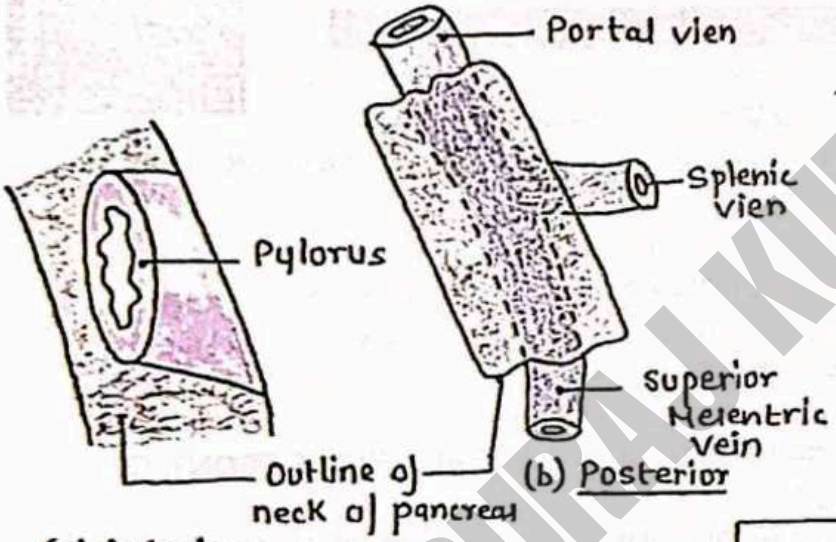
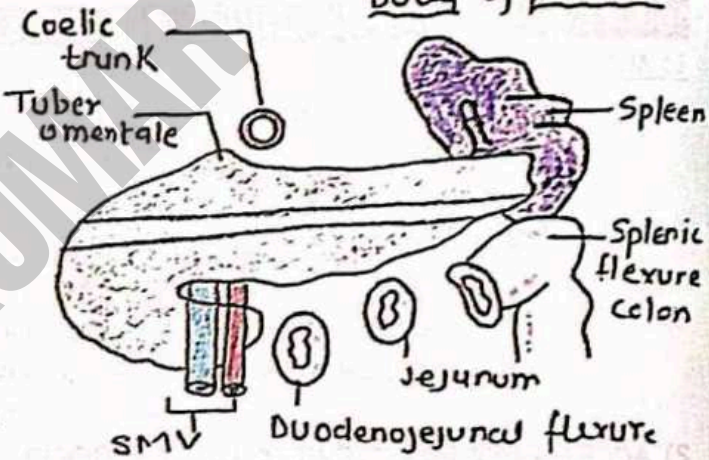


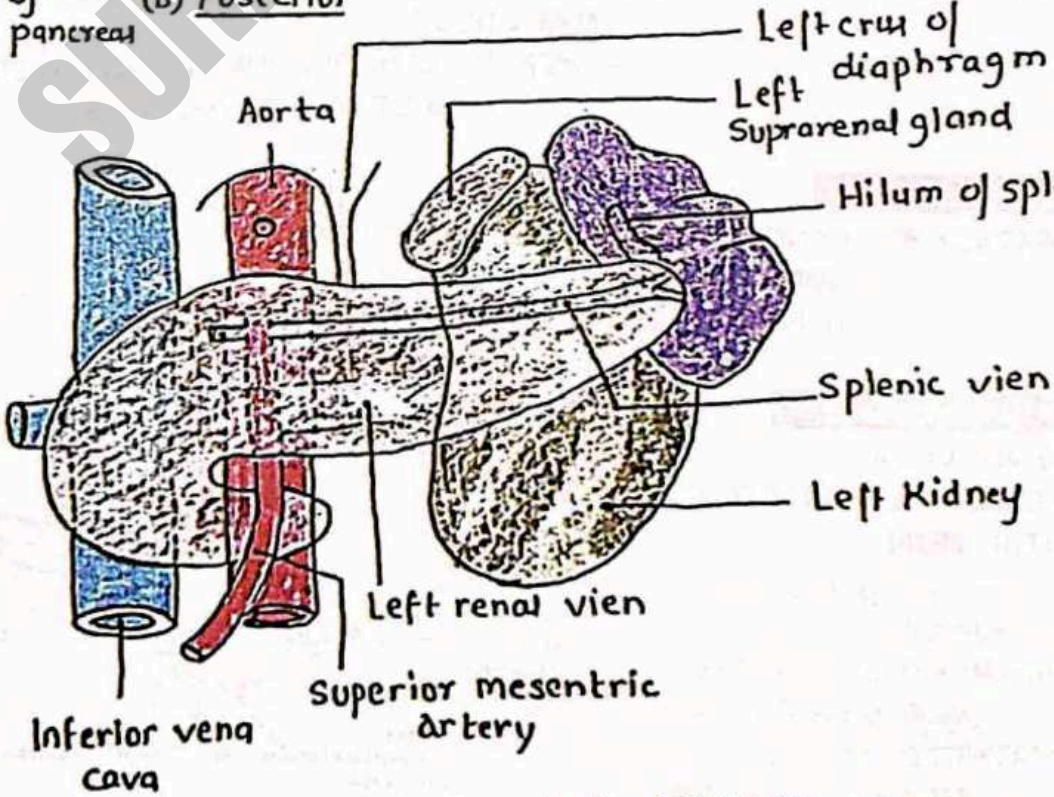


∵ Relation of the neck of pancreas



∵ Relation of anterior surface of body of pancreas





∵ Posterior relation of the body of the pancreas

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KIDNEY

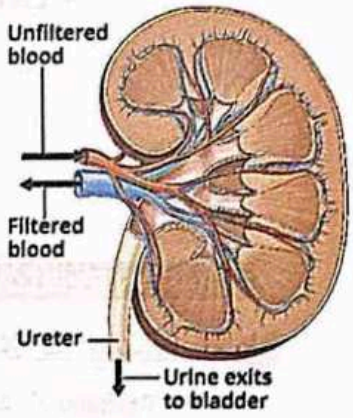
*DEFINITION - PAIR OF EXCRETORY ORGANS
-SITUATED ON POSTERIOR ABDOMINAL WALL
-EACH SIDE OF VERTEBRAL COLUMN BEHIND PERITONEUM

*LOCATION -
- OCCUPY EPIGASTRIC HYPOCHONDRIAC LUMBAR & UMBLICAL REGION
- EXTEND - UPPER BORDER OF T2 TO CENTER BODY OF L3
- RIGHT KIDNEY IS SLIGHTLY LOWER THAN LEFT

*SHAPE - EACH KIDNEY IS ABOUT - 11CM LONG 6CM BROAD 3CM THICK
-WEIGHT - 150g - IN MALE
-135g IN FEMALE

*EXTERNAL FEATURE - BEAN SHAPE
1 2 POLES-UPPER POLE- BROAD, CLOSE CONTACT WITH SUPRARENAL GLAND
2 2 SURFACE - ANTERIOR SURFACE - IRREGULAR
-POSTERIOR SURFACE - FLAT
3. 2 BORDER - LATERAL BORDER - CONVEX
-MEDIAL BORDER - CONCAVE, MIDDLE PART SHOW DEPRESSION - HILUM
4 HILUM - FOLLOWING STRUCTURE SEEN IN HILUM
-FROM ANTERIOR SIDE TO POSTERIOR -RENAL VEIN, RENAL ARTERY, RENAL PELVIS

RELATION OF KIDNEY - RETROPERITONEAL ORGAN
(PERITONEUM COVERED ANTERIOR ONLY)



RELATION COMMON TO 2 KIDNEY -
*UPPER POLE - SUPRARENAL GLAND
*MEDIAL BORDER - SUPRARENAL GLAND, URETER
*POSTERIOR SURFACE - DIAPHRAGM
-MEDIAL & LATERAL ARCUATE LIGAMENTS
-PSOAS MAJOR
-QUADRATUS LUMBORUM
-TRANSVERSUS ABDOMINIS
-SUBCOSTAL VESSELS
-SUBCOSTAL, ILIOHYPOGASTRIC & ILIOINGUINAL NERVES.

OTHER RELATION OF RIGHT KIDNEY -
*ANTERIOR RELATIONS - RIGHT SUPRARENAL GLAND
-LIVER
-SECOND PART OF DUODENUM
-HEPATIC FLEXURE OF COLON
-SMALL INTESTINE

OTHER RELATION TO LEFT KIDNEY -

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ANTERIOR RELATIONS - LEFT SUPRARENAL GLAND
-SPLEEN
-STOMACH
-PANCREAS
-SPLENIC VESSELS
-SPLENIC FLEXURE & DESCENDING COLON
-JEJUNUM

*CAPSULES OR COVERINGS OF KIDNEY -

1.FIBROUS CAPSULE - THIN MEMBRANE - CLOSELY INVEST THE KIDNEY
-NORMALLY IT CAN EASILY STRIPPED OFF
-BUT IN CERTAIN DISEASE IT BECOME ADHERENT & CANNOT STRIPPED.

2.PERINEPHRIC FAT - LAYER OF ADIPOSE TISSUE

3.RENAL FASCIA - FIBROAREOLAR SHEATH-→CONSIST OF 2 LAYER→ANTERIOR - FASCIA OF GEROTA.
→POSTERIOR - FASCIA OF ZUCRERKANDL'S

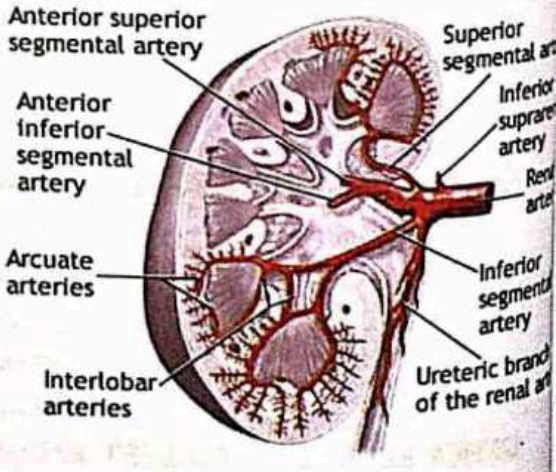
*EXTENSION - SUPERIORLY - 2 LAYER ENCLOSE SUPRARENAL GLAND
-THEN FUSE EACH OTHER & BECOME CONTINOUS AS DIAPHRAGMATIC FASCIA
- INFERIORLY - 2 LAYER REMAIN SEPARATE & ENCLOSE URETER
-ANTERIOR - LOST IN EXTRAPERITONEAL TISSUE OF ILIAC FOSSA
-POSTERIOR - BLEND WITH FASCIA ILIAC.
- LATERALLY - 2 LAYER UNITE FIRMLY & CONTINOUS WITH FASCIA TRANSVERSALIS
- MEDIALLY - ANTERIORLY LAYER PASSES IN FRONT OF KIDNEY AND MERGE WITH CONNECTIVE TISSUE SURROUND AORTA AND INFERIOR VENA CAVA
-POSTERIOR LAYER PASSES BEHIND KIDNEY AND ATTACH TO FASCIA COVERING QUADRATUS LUMBORUM AND PSOAS MAJOR

4) PARANEPHRIC FAT - LAYER OF FAT LYING OUTSIDE THE RENAL FASCIA

Blood Supply of Kidney :-

Abdominal aorta
↓
Renal artery
↓
5- segmental artery
↓
Each segmental artery
↓
Lobar artery
↓
interlobar artery
↓
arcuate artery
↓
Interlobula artery
↓
Afferent arteriole → Glomerulus → Efferent arteriole

inferior vena cava
↑
Renal vein
↑
5x segmental vein
↑
segmental vein
↑
Lobar vein
↑
interlobar vein
↑
Arcuate vein
↑
interlobular vein
↑
Peritubular Plexus
↑



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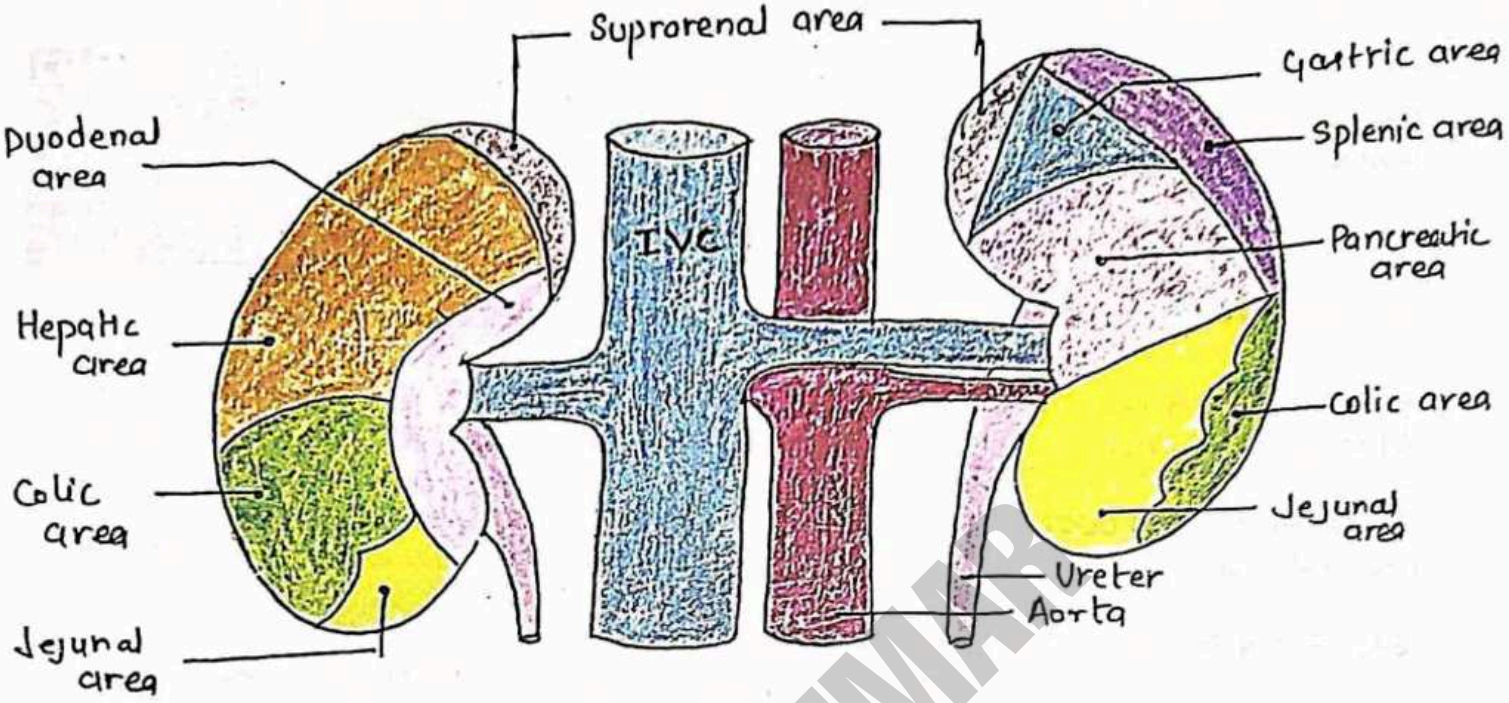
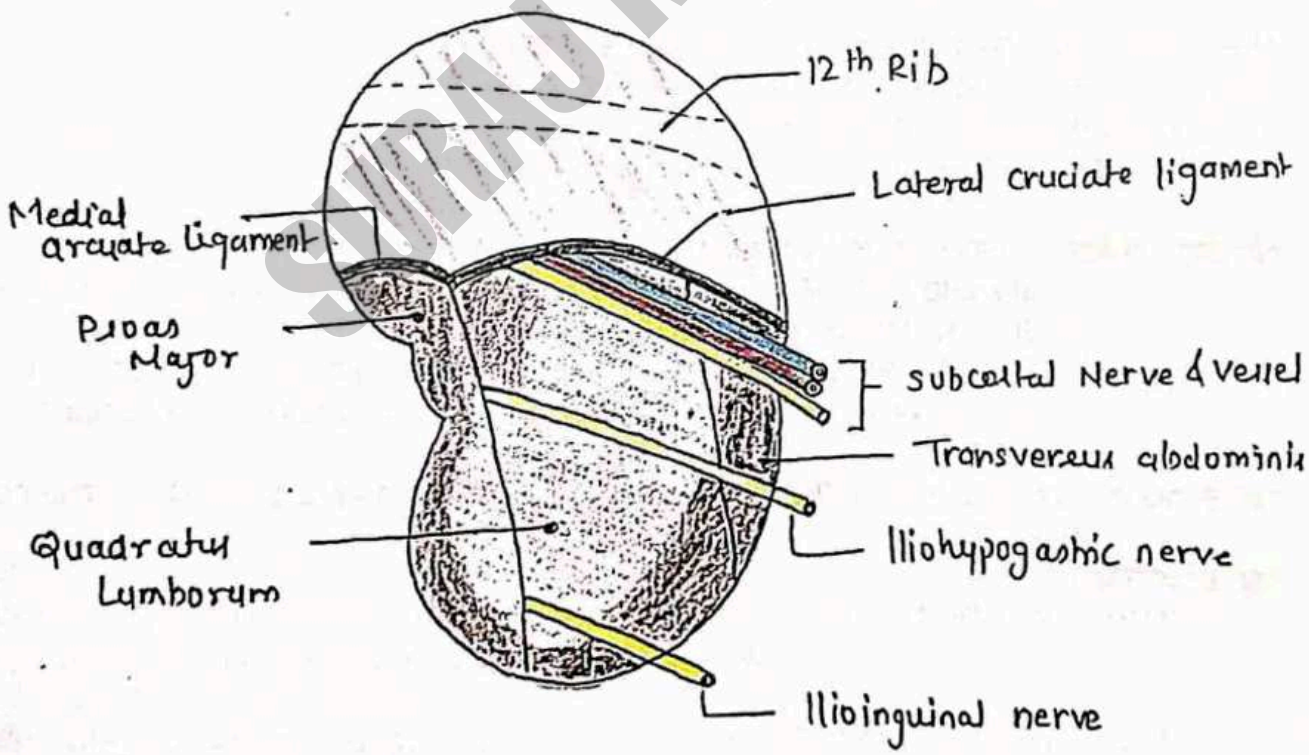


∴ Anterior relations of Kidneys



∴ Posterior relation of Kidney

KIDNEY

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EXTRAHEPATIC BILIARY APPARATUS

*EBA COLLECTS BILE FROM LIVER-->STORE IT IN GALL BLADDER-> TRANSMIT IT TO->SECOND PART OF DUODENUM.

*APPARATUS CONSIST OF - RIGHT & LEFT HEPATIC DUCT
-COMMON HEPATIC DUCT
-GALL BLADDER, CYSTIC DUCT & BILE DUCT

*RIGHT & LEFT HEPATIC DUCTS - EMERGE AT PORTA HEPATIS FROM THE RIGHT & LEFT LOBES OF THE LIVER.

*COMMON HEPATIC DUCT - FORMED BY UNION OF RIGHT & LEFT HEPATIC DUCT NEAR THE RIGHT END OF PORTA HEPATIS

*GALL BLADDER - PEAR SHAPED - RESERVOIR OF BILE
-SITUATED IN FOSSA →ON THE INFERIOR SURFACE OF RIGHT LOBE OF LIVER
-GALL BLADDER IS DIVIDED INTO - FUNDUS
-BODY
-NECK

1) FUNDUS - PROJECT BEYOND THE INFERIOR BORDER OF LIVER
2) BODY - LIE IN THE FOSSA FOR THE GALL BLADDER ON THE LIVER
3) NECK - NARROW UPPER END OF GALL BLADDER
-SITUATED NEAR THE RIGHT END OF PORTA HEPATIS

*CYSTIC DUCT - CYSTIC DUCT IS ABOUT 3-4CM LONG
- BEHIND THE NECK OF GALL BLADDER - RUN DOWNWARD, BACKWARD
- END BY JOINING THE COMMON HEPATIC
- MUCOUS MEMBRANE OF CYSTIC DUCT FORM FROM 5-12 CRECENTIC FOLD ---->ARRANGED SPIRALLY->CALLED SPIRAL VALVE OF HEISTER

*BILE DUCT -FORMED BY UNION OF CYSTIC & COMMON HEPATIC DUCT NEAR THE PORTA HEPATIS.

*SPHINCTER -

- SPHINTER CHOLEDOCHUS - TERMINAL PART OF BILE DUCT - SURROUND SMOOTH MUSCLE -THAT FORM SPHINCTER CHOLEDOCUS.

- SPHINCTER PANCREATICUS - ANOTHER LESS DEVELOPED SPHINCTER, AROUND TERMINAL PART OF PANCREATIC DUCT IS THE SPHINCTER PANCREATICUS.

- SPHINCTER AMPULLAE - THIS IS THIRD SPHINCTER SURROUNDS THE HEPATOPANCREATIC AMPULLA.

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PORTAL VEIN

*LARGE VEIN WHICH COLLECT BLOOD FROM - ABDOMINAL PART OF ALIMENTARY TRACT
-GALL BLADDER
- PANCREAS
- SPLEEN

*IN LIVER, THE PORTAL VEIN BREAK UP INTO SINUSOIDS - WHICH DRAINED BY - HEPATIV VEIN TO INFERIOR VENA CAVA.

*IT IS CALLED PORTAL VEIN BECAUSE - ITS MAIN TRIBUTARY, SUPERIOR MESCENTRIC VEIN, BEGIN IN ONE SET OF CAPILLARIES (IN THE GUT) & PORTAL ONE END IN ANOTHER SET OF CAPILLARIES IN LIVER.

*FORMATION - PORTAL VEIN ABOUT 8CM LONG.
- FORMED BY UNION OF - SUPERIOR MESCENTRIC VEIN BEHIND NECK OF PANCREAS & SPLENIC VEIN.

*COURSE - RUN UPWARD - FIRST BEHIND THE NECK OF PANCREAS
-NEXT BEHIND THE FIRST PART OF DUODENUM
-LASTLY - FREE MARGIN OF LESSER OMENTUM
-DIVIDED INTO -INFRADUODENAL
-RETRODUODENAL
-SUPRADUODENAL PART

-TERMINATION - VEIN END AT RIGHT END OF PORTA HEPATIS BY DIVIDING INTO-
→RIGHT & LEFT BRANCH

*RELATIONS -
- INFRADUODENAL PART - ANTERIORLY - NECK OF PANCREAS
- POSTERIORLY - INFERIOR VENA CAVA
- RETRODUODENAL PART - ANTERIORLY - FIRST PART OF DUODENUM
-BILE DUCT
-GASTRODUODENAL ARTERY
- POSTERIORLY -INTERIOR VENA CAVA
- SUPRADUODENAL PART - ANTERIORLY - HEPATIC ARTERY
-BILE DUCT
- POSTERIORLY - INFERIOR VENA CAVA

*BRANCHES →RIGHT BRANCH - SHORTER & WIDER
→LEFT BRANCH - LONGER & NARROWER

*TRIBUTARIES - PORTAL VEIN RECEIVE THE FOLLOWING VEIN :-
1.LEFT GASTRIC
2.RIGHT GASTRIC
3.SUPERIOR PANCREATICODUODENAL
4.CYSTIC VEIN
5.PARAUMBILICAL VEIN.

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Portosystemic Communications...

Position :-	Portal vein :-	Systemic vein
◦- Lower end of oesophagus	◦- Left gastric	◦- oesophageal vein
◦- Lower end of rectum	◦- Superior rectal	◦- Middle and inferior rectal vein
◦- Umbilicus	◦- Paraumbilical	Above - Superior epigastric lateral thoracic
		Below - Superior epigastric inferior epigastric
◦- Posterior abdominal wall	◦- Splenic	Sides - Posterior intercastal
◦- falciform ligament	◦- Paraumbilical	◦- Left renal vein
◦- Ligamentum venosum	◦- Left branch of portal	◦- Diaphragmatic
		◦- Inferior vena cava

CLINICAL ANATOMY

1.PORTAL PRESSURE : NORMAL PRESSURE IN THE PORTAL VEIN IS ABOUT 5-15MM HG IT IS USUALLY MEASURED BY SPLENIC PUNCTURE & RECORDING THE INTRASPLENIC PRESSURE.

2.PORTAL HYPERTENSION(PRESSURE ABOVE 40MM HG): IT CAN BE USED BY THE FOLLOWING :-

a.CIRRHOSIS OF LIVER, IN WHICH THE VASCULAR BED OF LIVER IS MARKEDLY OBLITERATED.

b.BANTI'S DISEASE

c.THROMBOSIS OF PORTAL VEIN.

3.THE EFECT OF PORTAL HYPERTENSION ARE AS FOLLOWS :-

1.CONGESTIVESPLENOMEGALY.

2.COLLATERAL CIRCULATION THROUGH THE PORTOSYSTEMIC COMMUNICATIONS.IT FORMS:-CAPUT MEDUSAE,OESOPHAGEAL VARICES,HAEMORRHOIDS IN ANAL CANAL.

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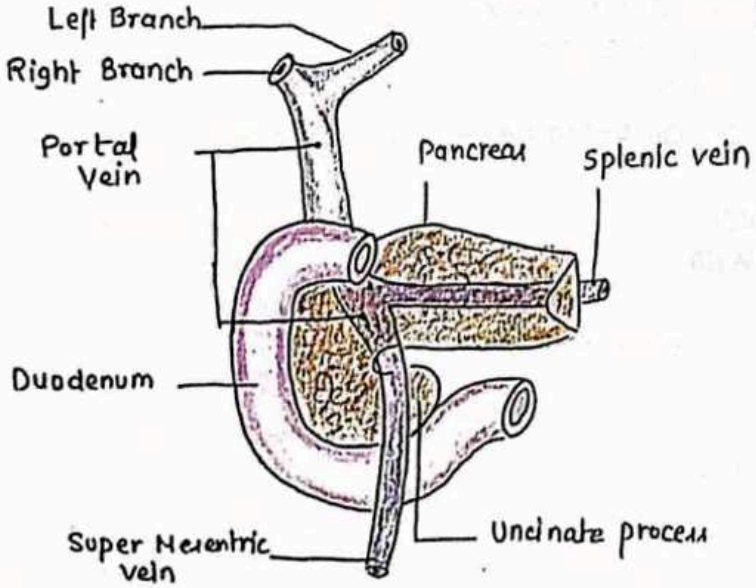
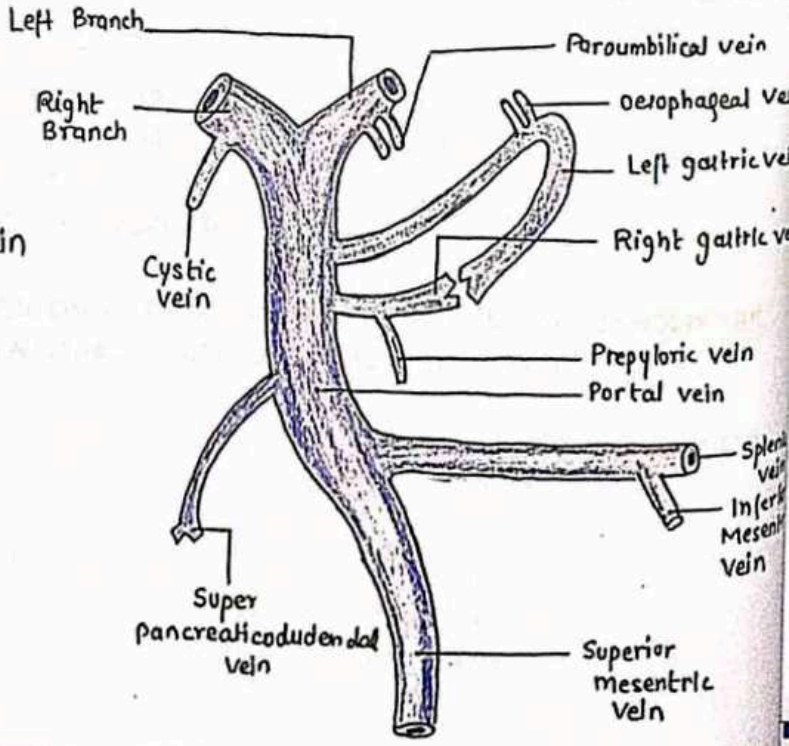


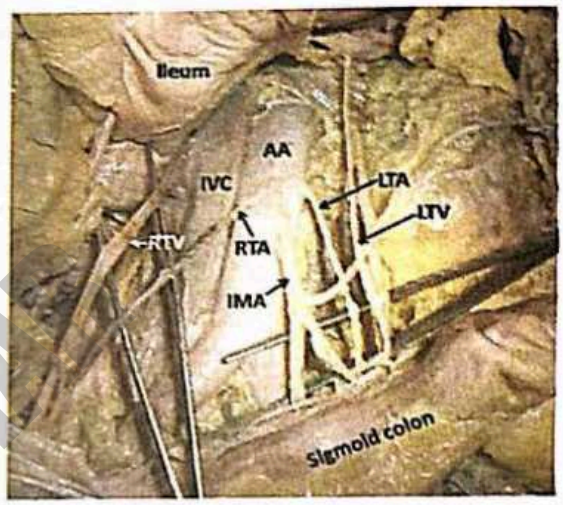


ABDOMINAL AORTA

***COURSE :-**

1. BEGIN AT THE AORTIC OPENING OF DIAPHRAGM
2. IT RUN DOWN WARD AND END IN FRONT OF LOWER PART OF BODY OF VERTEBRA L4.

***RELATIONS :-**



1 ANTERIORLY - COELIAC AND AORTIC PLEXUSES
- BODY OF PANCREAS
- THIRD PART OF DUODENUM

2.POSTERIORLY - BODY OF UPPER 4X LUMBAR VERTEBRA
- ANTERIOR LONGITUDINAL LIGAMENT

3.TO RIGHT SIDE - INFERIOR VENA CAVA,
- RIGHT CRUS OF DIAPHRAGM
- CISTERNA CHYLI

4.TO LEFT SIDE - LEFT CRUS OF DIAPHRAGM
- PANCREAS
- 4TH PART OF DUODENUM

***BRANCHES**

1) VENTRAL BRANCHES- 1. COELIAC TRUNK→ LEFT GASTRIC ARTERY
→ COMMON HEPATIC ARTERY
→ SPLENIC BRANCH

2.SUPERIOR MESENTERIC ARTERY → INFERIOR PANCREATICODUODENAL
→MIDDLE COLIC
→RIGHT COLIC
→ILEO COLIC
-12-15 JEJUNAL
→ ILEAL BRANCHES

3.INFERIOR MESENTERIC ARTERY --LEFT COLIC
→ SIGMOID ARTERY
→ SUPERIOR RECTAL

2) LATERAL BRANCHES - INFERIOR PHRENIC ARTERIES
- MIDDLE SUPRARENAL ARTERIES
- RENAL ARTERIES
- TESTICULAR OR OVARIAN ARTERIES

3) DORSAL BRANCHES - LUMBAR ARTERY
- MEDIAN SACRAL ARTERY

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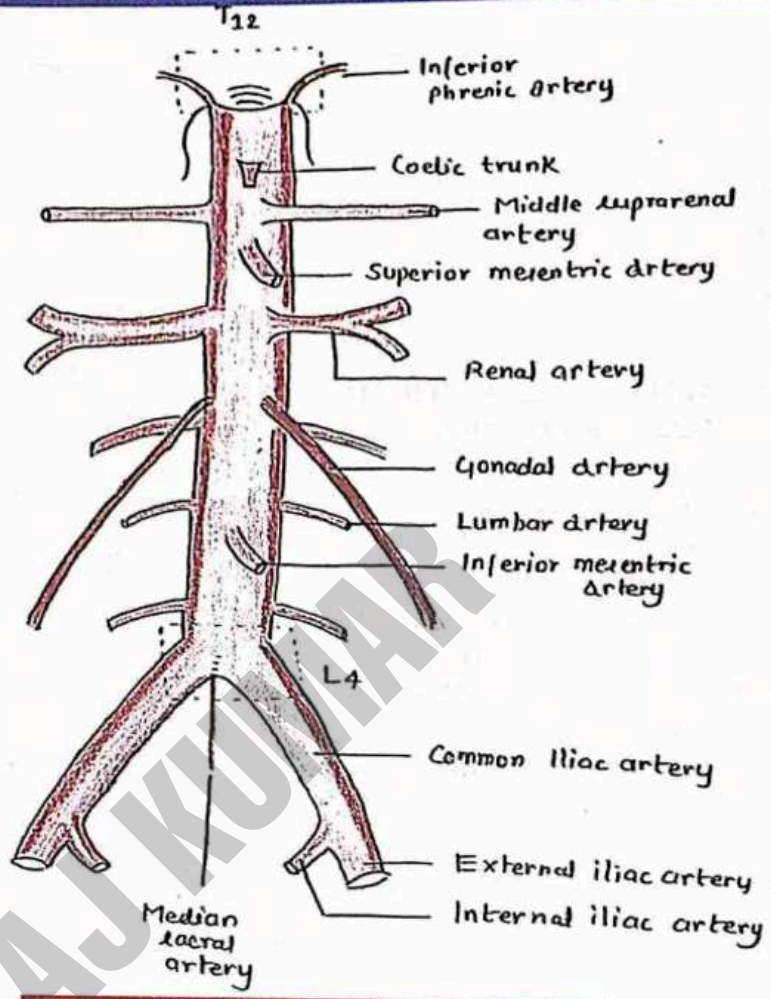
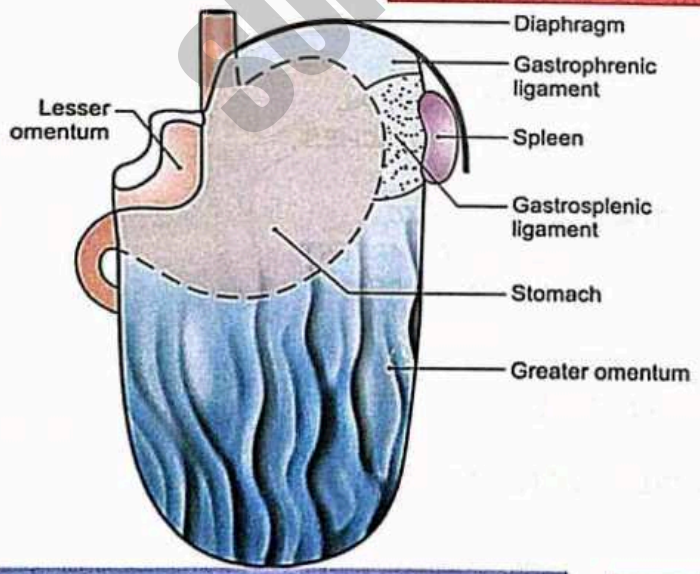
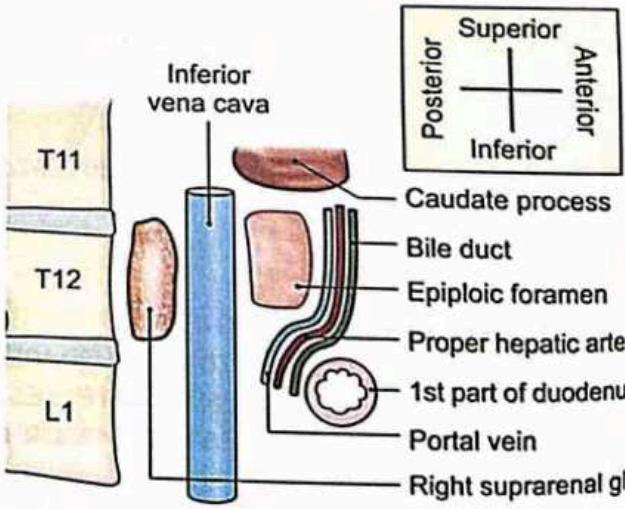


ABDOMINAL AORTA



GREATER OMENTUM



BOUNDARIES EPIPLOIC FORAMEN

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GREATER OMENTUM

*FOLD OF PERITONEUM - WHICH HANGS DOWN FROM-GREATER CURVATURE OF STOMACH.
*MADE UP OF FOUR LAYERS OF PERITONEUM- ALL WHICH ARE FUSED TOGETHER
*AND FORM THIN FENESTRATED MEMBRANE CONTAINING VARIABLE-QUANTITIES OF FAT
-SMALL ARTERIES & VEIN

*ATTACHMENT - ANTERIOR 2x LAYER DESCEND FROM GREATER CURVATURE OF STOMACH TO A, VARIABLE EXTEND.
- AND FOLD THEMSELVE TO FORM POSTEOIR TWO LAYER
- FOLDING OF OMENTUM IS SUCH THAT FIRST LAYER BECOME- FOURTH LAYER & 2ND LAYER-→ 3RD LAYER
- IN UPPER PART, FOURTH LAYER IS PARTIALLY FUSED TO ANTERIOR SURFACE OF TRANSVERSE COLON & TRANSVERSE MESOCOLON.

*CONTENT - RIGHT & LEFT GASTROPIPLOIC VESSELS
-FAT
*FUNCTIONS - STORE HOUSE OF FAT
- PROTECT PERITONEAL CAVITY AGAINST INFECTION BECAUSE OF PRESENCE OF MACRO PHAGES.
- COLLECTION OF MACROPHAGES FROM SMALL, DENSE PATCH KNOWN - AS MILKY SPOT.
- IT ALSO LIMIT SPREAD OF INFECTION BY MOVING TO SITE OF INFECTION & SEALING IT OFF.
- GREATER OMENTUM IS ALSO KNOWN AS POLICEMEN OF ABDOMIN.

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LESSER OMENTUM

1) FOLD OF PERITONEUM WHICH EXTEND-FROM- LESSER CURVATURE OF STOMACH TO FIRST 2CM OF DUODENUM TO THE LIVER
2) PORTION OF L.O B/W STOMACH AND LIVER- HEPATOGASTRIC LIGAMENT
3) PORTION OF L.O B/W DUODENUM AND LIVER- HEPATODUODENAL LIGAMENT

*ATTACHMENTS:-
1) INFERIORLY- LESSER CURVATURE OF STOMACH
- UPPER BARDER OF FIRST 2CM OF DUODENUM

2) SUPERIORLY - LIVER ATTACHMENT AS AN INVERTED 'L'
-VERTICAL LIMB OF L'- IS FISSURE FAR LIGAMENTUM VENOSUM
-HARIZONTAL LIMB - MARGIN OF PORTA HEPATIS.

*CONTENTS - PROPER HEPATIC ARTERY

- PORTAL VEIN
- BILE DUCT
- LYMPH NODES
- HEPATIC PLEXUS OF NERVE

- ALONG THE- LESSER CURVATURE OF STOMACH - CONTAIN - RIGHT & LEFT GASTRIC VESSELS.
-GASTRIC GROUP OF LYMPH NODE
-BRANCHES FROM GASTRIC NERVE

EPIPLOIC FORAMEN/OMENTAL AND WINSLOW FORAMEN

-VERTICAL SLIT-LIKE OPENING THROUGH WHICH - LESSER SAC COMMUNICATES WITH GREATER SAC.
-FORAMEN IS SITUATED BEHIND THE RIGHT FREE MARGIN OF LESSER OMENTUM AT LEVEL OF T12 VERTEBRA.

BOUNDARIES :-
1) ANTERIORLY: LESSER OMENTUM → CONTAINING - PORTAL VEIN
- PROPER HEPATIC ARTERY
- BILE DUCT

2) POSTERIORLY: - INFERIOR VENA CAVA
- RIGHT SUPRARENAL GLAND
- T12 VERTEBRA

3) SUPERIORLY - CAUDATE PROCESS OF LIVER
4) INFERIORLY - FIRST PART OF DUODENUM

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CLINICAL ANATOMY

CLINICAL ANATOMY OF ISCHIONAL FOSSA

1.PROLAPSE OF RECTUM : THE ISCHIONAL FAT ACTS AS A CUSHION LIKE SUPPORT TO THE RECTUM & ANAL CANAL. LOSS OF THIS FAT IN DEBILITATING DISEASES LIKE DIARRHOEA IN CHILDREN MAY RESULT IN PROLAPSE OF RECTUM.

2.BOTH THE PERIANAL & ISCHIOANAL ARE COMMON SIDE OF ABSCESSES.

3.ISCHIONAL HERNIA : THE OCCASIONAL GAP BETWEEN THE TENDINOUS ORIGIN OF LEVATOR ANI & THE OBTURATOR FASCIA IS KNOWN AS HIATUS OF SCHWALBE. RARELY, PELVIC ORGANS MAY HERNIATE THROUGH THIS GAP INTO THE ISCHIONAL FOSSA, RESULTING IN AN ISCHIONAL HERNIA.

ISCHIONAL FOSSA

-WEDGE-SHAPED FOSSA SITUATED - EACH SIDE ANAL CANAL BELOW→ PELVIC DIAPHRAGM.

- BASE - DOWNWARD, TOWARD SURFACE
- APEX - UPWARD

*BOUNDARIES:-

1) BASE- FORMED BY SKIN OF PERINEUM
2) APEX -FORMED BY - LINE WHERE- OBTURATOR FASCIA MEET INFERIOR FASCIA OF PELVIC DIAPHRAGM
3) ANTERIORLY- POSTERIOR BORDER OF PERINEAL MEMBRANE
4) POSTERIARLY - GLUTEUS MAXIMUS
   -SACROTUBEROUS LIGAMENT
5) LATERAL WALL - OBTURATOR INTERNUS
   -ISCHIAL TUBEROSITY
6) MEDIAL - LEVATOR ANI
   - EXTERNAL ANAL SPHINCTER

*SPACE AND CANNALS OF FOSSA:

1) PERIANAL SPACE - PERIANAL FOSSA FORM OF SEPTUM PASSES LATERALLY LONGITUDINAL COAT OF ANAL CANAL

↳ EXTEND-MEDIALLY FROM WHITE LINE OF HILTON
LATERALLY→ PUDENDAL CANAL
↳ SEPARATES SHALLOW SUBCUTANEOUS PERIANAL SPACE FROM DEEP ISCHIOANAL SPACE
↳ FAT IN TIGHTLY ARRANGED IN SMALL LOCULI FORMED BY → COMPLETE SEPTA

2) ISCHIOANAL SPACE - LARGE AND DEEP
- FAT- IN LOSSELY ARRANGED IN LARGE LOCULI FORMED- BY→INCOMPLETE DELICATE SEPTA.
-LUNATE FASCIA-ARCHES OVER THE ISCHIAL FAT

3) PUDENDAL CANNAL- FASCIAL CANNAL IN LATERAL WALL OF IF
- ENCLOSING - PUDENDAL NERVE
- INTERNAL PUDENDAL VESSEL

SUPERIOR SURFACE -> MALE - COMPLETELY COVERED BY PERITONEUM
->FEMALE - PERITONEUM COVERED GREATER POINT EXCEPT SMALL AREA NEAR POSTERIOR BORDER.

*CONTENTS OF ISCHIONAL FOSSA -

1.ISCHIOANAL PAD OF FAT
2.INFERIOR RECTAL NERVE & VESSEL

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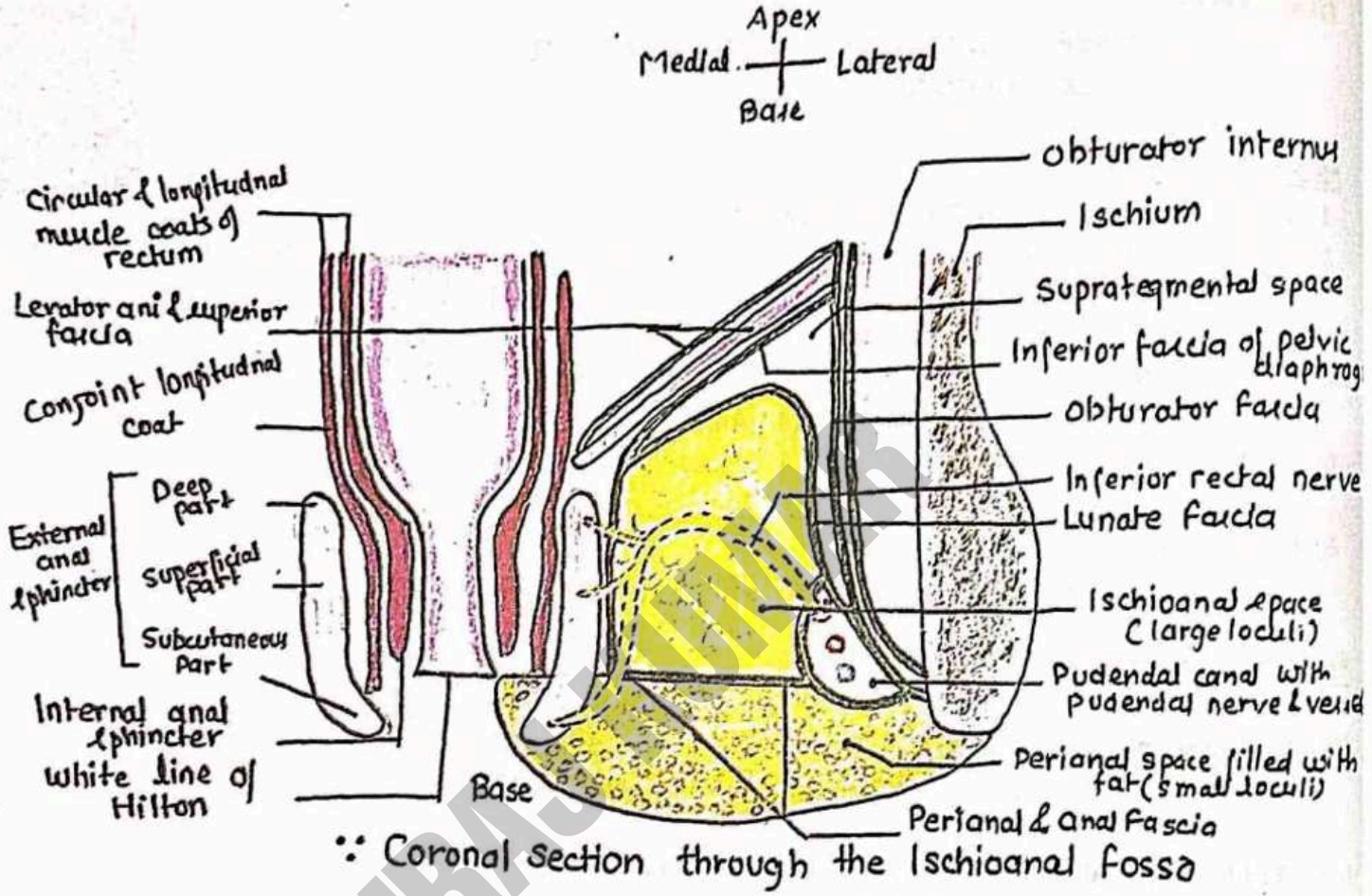
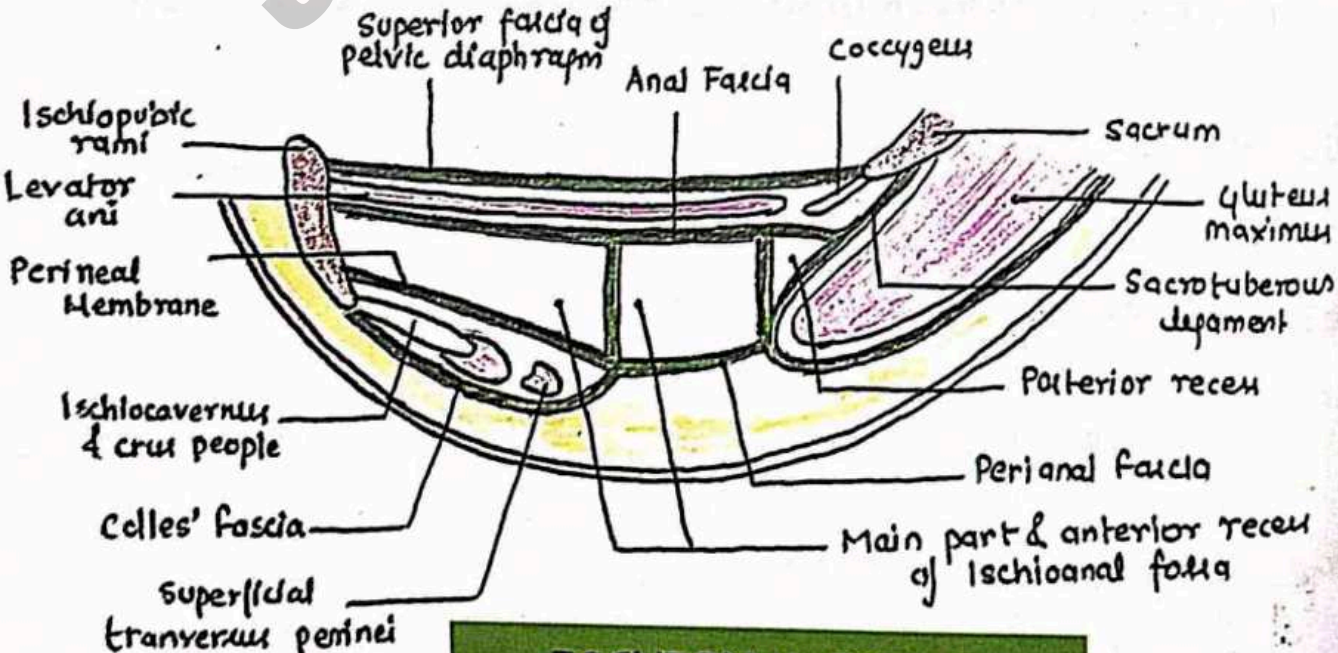
3.PUDENDAL CANAL
4.POSTERIOR SCROTAL OR POSTERIOR LABIAL NERVE & VESSEL
5.PERINEAL BRANCH OF 4TH SACRAL(S4)
6.PERFORATING CUTANEOUS BRANCHES OF NERVE (S2, S3)



∵ Coronal section through the Ischioanal fossa



ISCHIONAL FOSSA

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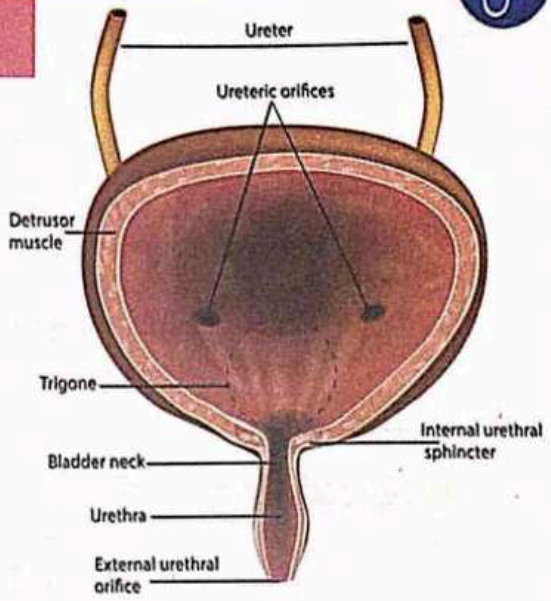
URINARY BLADDER



*FEATURE:-

- MUSCULAR RESERVOIR OF URINE
- LIE IN ANTERIOR PART OF PELVIC CAVITY
- SHAPE- EMPTY → TETRAHEDRAL
  FULL → OVOID

*EXTERNAL FEATURE :-

1) EMPTY BLADDER -APEX
   - BASE OR FUNDUS
   - NECK
   - 3 SURFACE -SUPERIOR
     -RIGHT AND LEFT INFEROLATERAL
   - 4 BORDER - 2 LATERAL , 1 ANTERIOR , 1 POSTERIOR

2) FULL BLADDER- APEX
   - NECK
   - TWO SUFACE - ANTERIOR OR POSTERIOR

*RELATION-

1) APEX - MEDIAN UMBILICAL LIGAMENT

2) BASE - FEMALE- UTERINE CERVIX
   -VAGINA

   - MALE- UPPER-SEPARATED FROM RECTUM BY RECTOVESICAL POUCH
     LOWER - SEMINAL VESICLE , VAS-DEFERENS

3) NECK - LOWER MOST FIXED PART OF BLADDER
   MALE- SMOOTH MUSCLE SURROUND BLADDER NECK AND PREPROSTATIC URETHRA→
   PREPROSTATIC → DEVOID OF PARASYMPATHETIC CHOLINERGIC NERVE

   FEMALE- RELATED TO PELVIC FASCIA → SURROUNDING THE UPPER PART OF URETHRA

4) SUPERIOR SURFACE -
MALE - COMPLETELY COVERED BY PERITONEUM

FEMALE - PERITONEUM COVER GREATER PART EXCEPT SMALL AREA NEAR POSTERIOR BORDER.

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5) INFEROLATERAL SURFACE - DEVOID OF PERITONEUM
- MALE- PUBIS
- PUBOPROSTATIC LIGAMENT
- RETROPUBIC FAT
- LEVATOR ANI
- OBTURATAN INTERNUS

-FEMALE- SAME EXCEPT - PUBOPROSTATIC LIGAMENT

*LIGAMENT:-

TRUE LIGAMENT - CONDENSATION OF PELVIC FASCIA AROUND NECK AND BASE OF BLADDER
- LATERAL TRUE LIGAMENT
- MEDIAL PUBOPROSTATIC LIGAMENT OR PUBOVESICAL LIGAMENT
- MEDIAN UMBILICAL LIGAMENT
- POSTERIOR LIGAMENT WITH VENOUS PLEXUS.

FALSE LIGAMENT- THEY ARE PERITONEAL FOLD
- MEDIAN UMBILICAL FOLD
- MEDIAL UMBILICAL FOLD
- LATERAL FALSE LIGAMENT
- POSTERION FALSE LIGAMENT

*ARTERIAL SUPPLY - MAIN- SUPERIOR, INFERIOR VESICAL ARTERY BRANCH OF INTERNAL ILIAC ARTERY
ADDITIONAL - OBTURATOR AND INFERIOR GLUTEAL ARTERIES SUPPLY

*VENOUS DRAINAGE - VESICAL VENOUS PLEXUS—->INTERNAL ILIAC VEIN

*NERVE -

1) PARASYMPATHETIC EFFERENT FIBRES - S2,3,4 → MOTOR TO DETRUSOR MUSCLE
2) SYMPATHETIC EFFERENT FIBRES - T11 TO L2 INHIBITORY TO DETRUSOR MUSCLE

3) SOMATIC PUDENDAL NERVE - SUPPLY-SPHINCTER URETHRAE - VOLUNTARY
4) SENSORY NERVE. PAIN SENSATION CARRIED BY - PARASYMPATHETIC NERVE.

CLINICAL ANATOMY

1.A DISTENDED BLADDER MAY BE RUPTURED BY INJURIES OF THE LOWER ABDOMINAL WALL.
2.CHRONIC OBSTRUCTION TO THE OUT FLOW OF URINE BY AN ENLARGED PROSTATE CAUSES HYPERTROPHY OF BLADDER LEADING TO TRABUCULATED BLADDER.
3.URINARY BLADDER IS ONE OF SITES FOR STONE FORMATION AS CONCENTRATED URINE LIES HERE.

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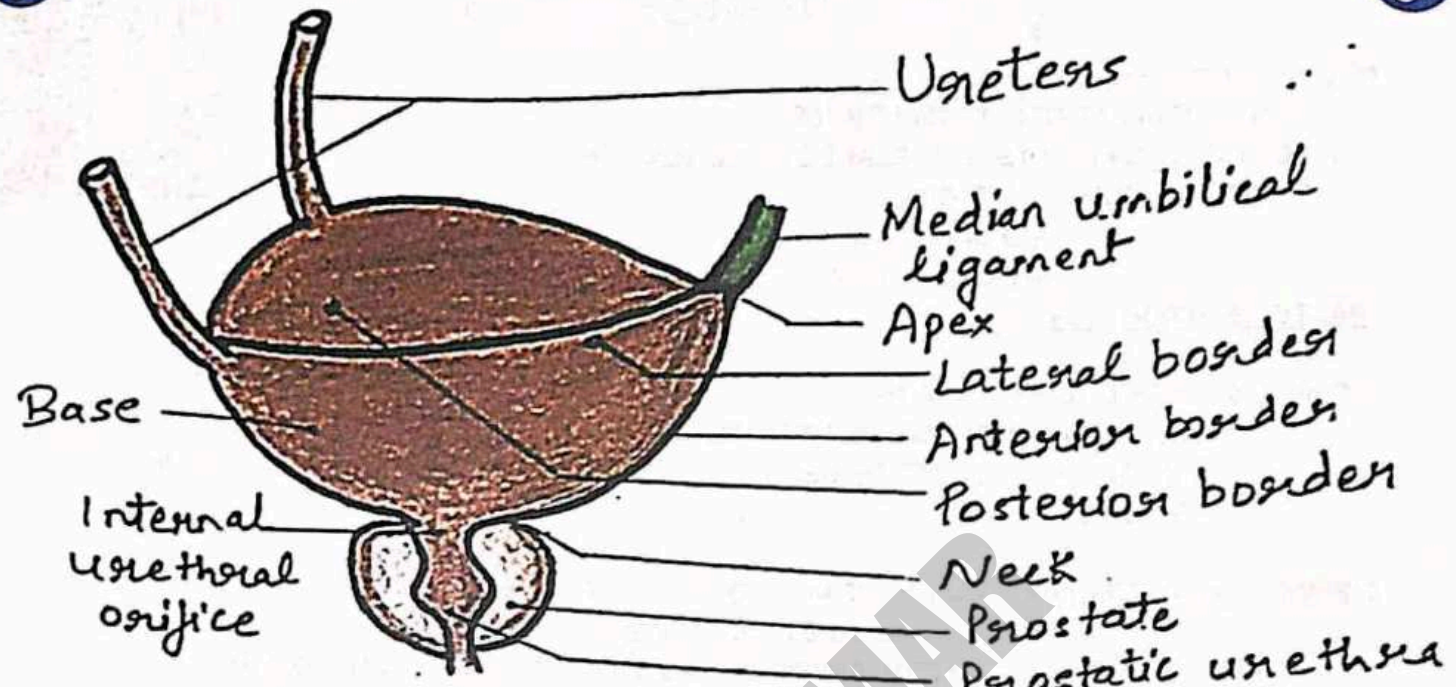
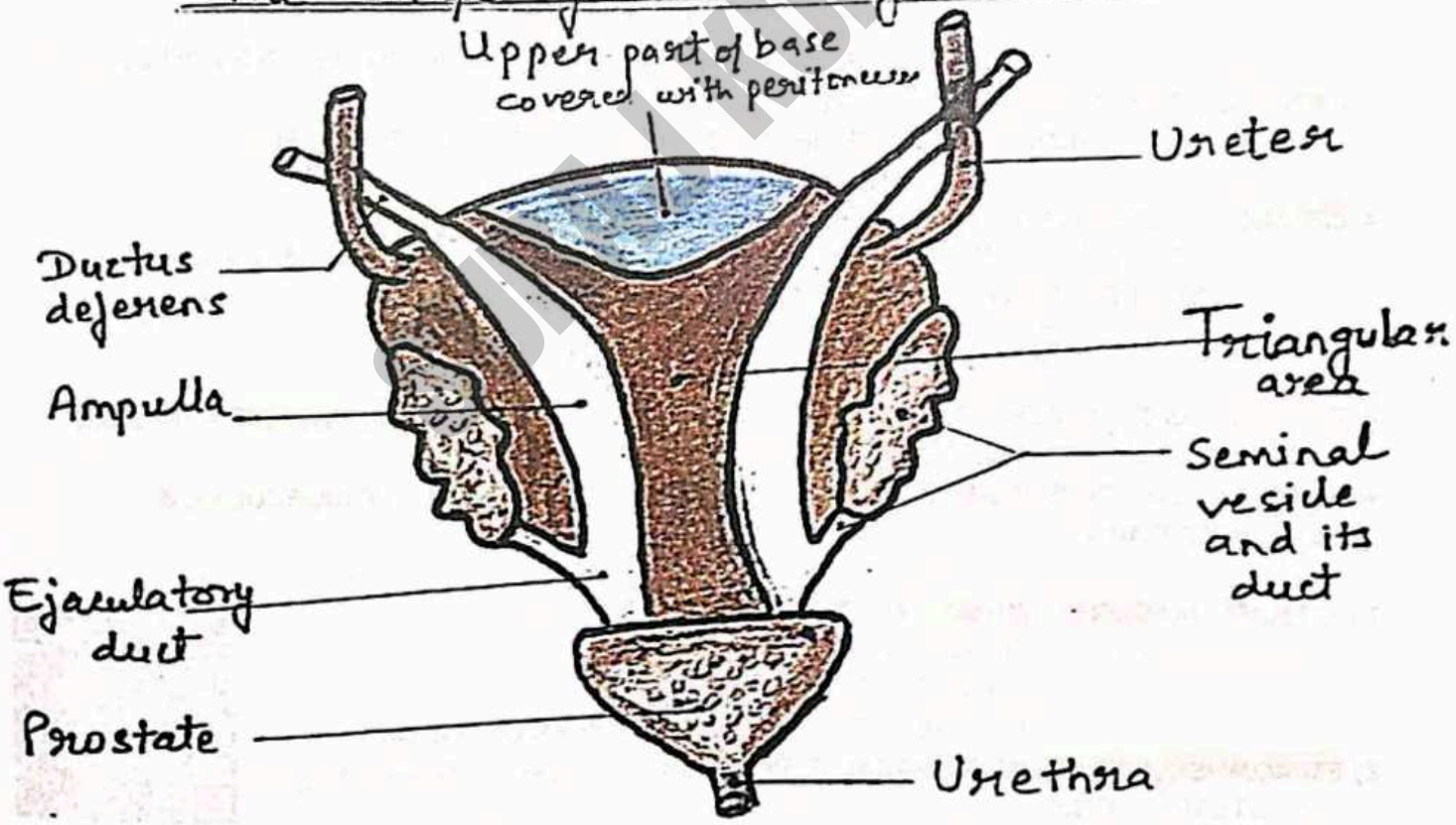


The Shape of the Urinary bladder



URINARY BLADDER

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UTERUS

*INTRODUCTION :-
- CHILD BEARING ORGAN IN FEMALE
- SITUATED IN PELVIS B/W BLADDER AND RECTUM
- PYRIFORM SHAPE - 7.5 CM LONG
  - 5 CM BROAD
  - 2.5CM THICK

*PARTS OF UTERUS :-

1.FUNDUS- FREE END OF UTERUS
- LIE ABOUT OPENING OF UTERINE TUBE
- FERTILISED OOCYTE IS USUALLY IMPLANTED IN POSTERIOR WALL OF FUNDUS

2.BODY- ANTERIOR PART - IS FLAT RELATED TO URINARY BLADDER
- COVERED BY PERITONEUM
- FORM SUPERIOR WALL OF UTEROVESICAL POUCH

- POSTERIOR SURFACE - IS CONVEX RELATED TO TERMINAL ILEUM
  → COVERED WITH PERITONEUM
  → FORM ANTERIOR WALL OF RECTOUTERINE POUCH

3.LATERAL BODY - ROUND AND CONVEX
- PROVIDE ATTACHMENT TO BROAD LIGAMENT OF UTERUS

4.CERVIX -THE CERVIX IS LOWER CYLINDRICAL PART OF UTERUS.
- LOWER PART OF CERVIX PROJECTS INTO ANTERIOR WALL OF VAGINA DIVIDED INTO SUPRA VAGINAL AND VAGINAL

SUPPORT OF UTERUS

- THE UTERUS IS A MOBILE ORGAN WHICH UNDERGOES EXTENSIVE CHANGES IN SIZE & SHAPE DURING THE REPRODUCTIVE PERIOD OF LIFE.
- IT IS SUPPORTED BY A NUMBER OF FACTORS WHICH ARE CHIEFLY MUSCULAR & FIBROMUSCULAR :-

1) PRIMARY SUPPORT:- MUSCULAR OR ACTIVE SUPPORT :-
- PELVIC DIAPHRAGM
- PERINEAL BODY
- DISTAL URETHRAL SPHINCTER MECHANISM

2) FIBROMUSCULAR OR MECHANICAL SUPPORT :-
1. UTERINE AXIS
2. PUBOCERVICAL LIGAMENT
3. TRANSVERSE CERVICAL LIGAMENT
4. UTERO SACRAL LIGAMENT
5. ROUND LIGAMENTS OF UTERUS

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3) SECONDARY SUPPORT :-

1. BROAD LIGAMENT
2. VESICOUTERINE POUCH & FOLD OF PERITONEUM
3. RECTOVAGINAL OR RECTOUTERINE POUCH & FOLD OF PERITONEUM

ROLE OF INDIVIDUED SUPPORTS:-

1. PELVIC DIAPHRAGM -
- SUPPORT PELVIC VISCERA & WRIST INCREASE IN INTRAABDOMINAL PRESSURE
- PUBOCOCCYGEUS- INSERT INTO PERINEAL BODY BETWEEN VAGINAL RECTUM
- TORN DURING PARTURITION THE SUPPORT OF UTERUS LOST.

2. PERINEAL BODY- FIBROMUSCULAR NODE TO WHICH TEN MUSCLES ARE ATTACHED.

THE MUSCLE ARE -
- 2 SUPERFICIAL TRANSVERSE PERINEI.
- 2 DEEP TRANSVERSE PERINEI.
- 2 PUBOCOCCYGEUS.
- 2 BULBOSPONGIOUS.
- 1 SINGLE SPHINCTER ANI EXTERNUS.
- LONGITUDINAL MS COATS OF ANAL CANAL.

3. URETHRAL SPHINCTER MECHANISM - UROGENITAL DIAPHRAGM DOES NOT EXIST IN URETHRA IS WITHIN THE WALL OF URETHRA.
- IN ADDITION COMPRESSOR URETHRAE & SPINCTER URETHRAE VAGINALIS INSERT INTO VAGINA SUPPORT UTERUS INDIRECTLY.

4) UTERINE AXIS -
ANTEVERTED POSITION OF UTERUS- PREVENT ORGAN SAGGING DOWN THROUGH VAGINA.

5) PUBOCERVICAL LIGAMENT - CONNECT CERVIX TO POST SURFACE OF PUBIS.
- DERIVED FROM ENDOPELVIC FASICA.

6 TRANSVERSE CERVICAL LIGAMENT - CONNECT LAT. ASPECT OF CERVIX TO LAT. PELVIC WALL
-DERIVED FROM ENDOPELVIC FASICA FAN SHAPE CONDENSATION

7. UTEROSACRAL LIGAMENT -CONNECT CERVIX TO PERIOSTEUM OF SACRUM.
-DERIVED FROM ENDOPELVIC FASICA.
-KEEP CERVIX BACKWARD AGAINST ROUND LIGAMENT.
-2 LIGAMENT COUPLE MAINTAIN UTERINE AXIS

8. ROUND LIGAMENT - 2 FIBROMUSCULAR BAND - 10 TO 12 CM LONG.

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EACH LIG. BEGIN INTO AT LAT. ANGLE OF UTERUS

↓

RUN FORWARD LATERALLY

↓

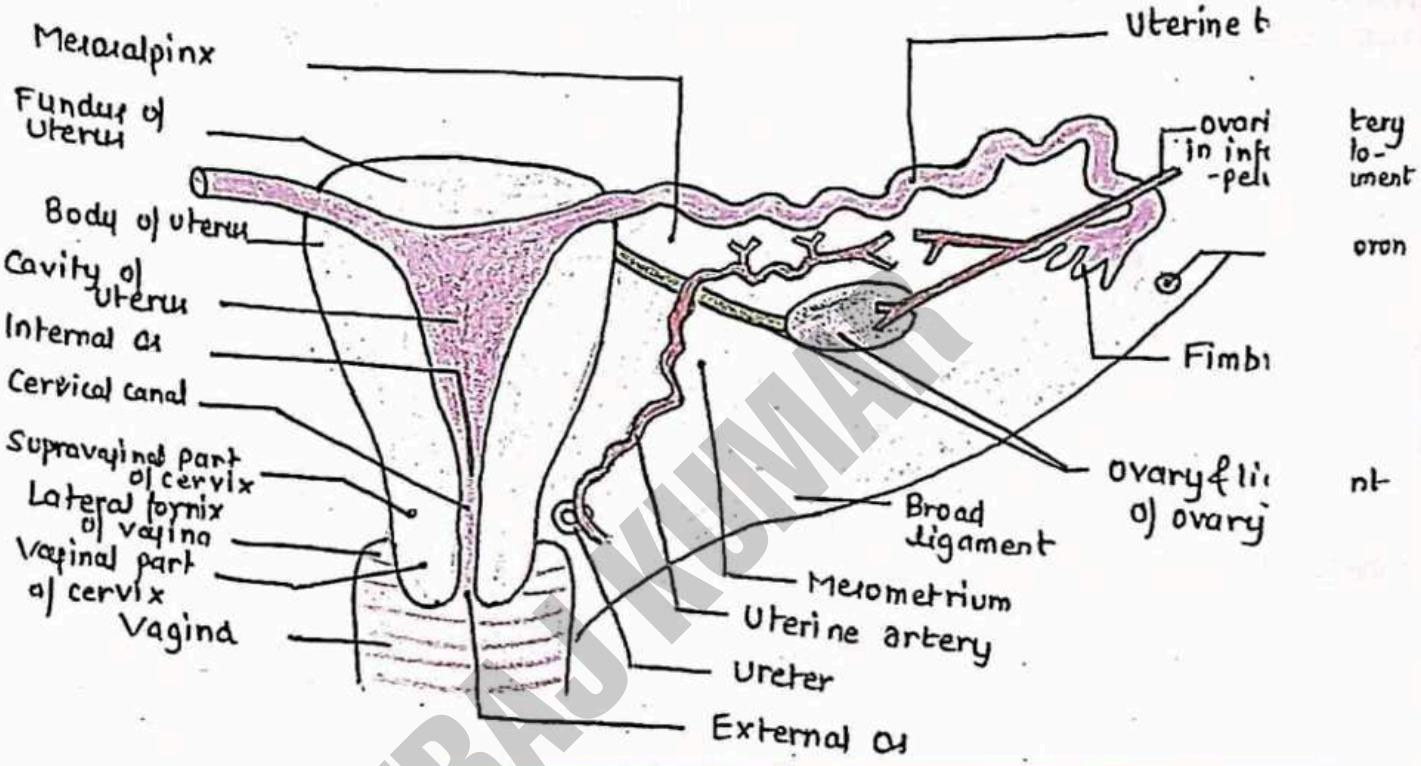
PASS DEEP INGUINAL RING

↓

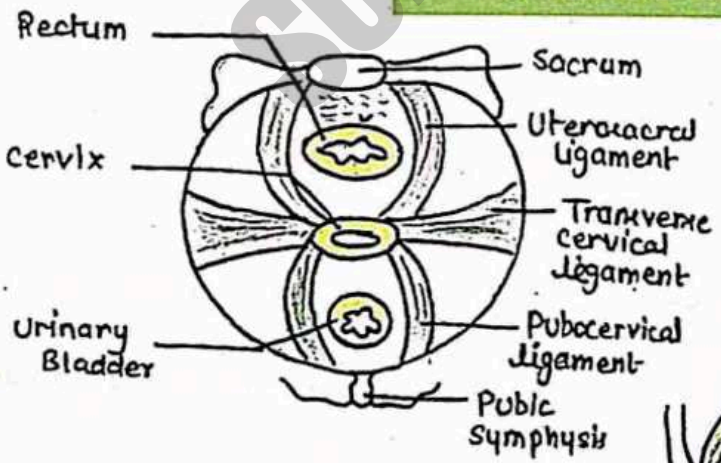
TRAVERSE THE INGUINAL CANAL

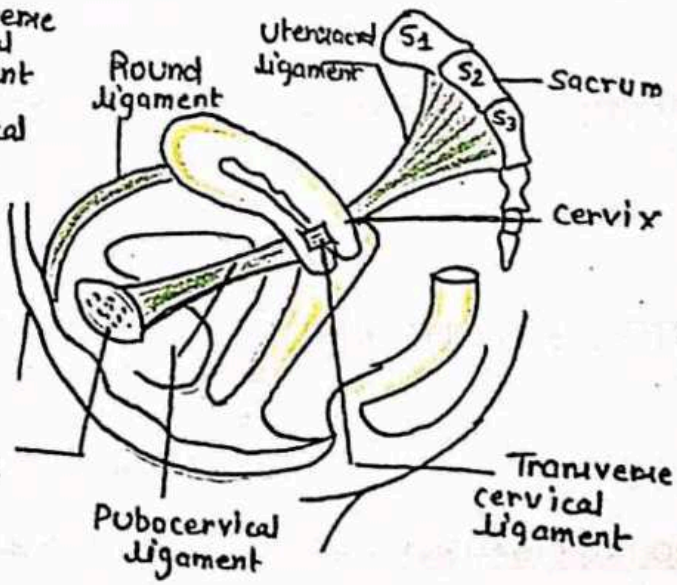
↓

MERGE WITH AEROLAR TISSUE TO LABIUM MAJUS.



UTERUS & RIGHT BROAD LIGAMENT





SUPPORT OF UTERUS

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UTERINE TUBES

*DEFINITION- TORTUOUS DUCT WHICH, CONVEY OOCYTE
-SITUATION IN UPPER FREE MARGIN OF BROAD LIGAMENT OF UTERUS
-DIMENSION- UTERINE TUBE IS ABOUT 10CM LONG.

*SUBDIVISION:

1) INFUNDIBULUM - LATERAL END OF UTERINE TUBE IS LIKE FUNNEL SHAPE
- BEAR NUMBER OF FRINGER LIKE PROJECTION - FIMBRIAE

2) AMPULLA - MEDIAL TO THE INFUNDIBULUM
-THIN WALLED DILATED AND TORTUOUS - 4MM DIAMETER

3) ISTHMUS - SUCCEEDS THE AMPULLA
4) INTRAMURAL - INTERSTITIAL PART OF THE TUBE IS ABOUT 1CM LONG

*COURSE:-

1) ISTHMUS - POSTEROLATERALLY IN HORIZONTAL PLANE
2) AMPULLA - ARCHES OVER THE OVARY
3)INFUNDIBULUM - PROJECT BEYOND THE FREE MARGIN OF BROAD LIGAMENT
4) UTERINE TUBE LIE - IN UPPER FREE MARGINE OF THE BROAD LIGAMENT

*BLOOD SUPPLY -

UTERINE ARTERY - SUPPLIES - MEDIAL 2/3 OF TUBE
OVARIAN ARTERY - SUPPLIES- LATERAL 1/3 OF TUBE

*NERVE SUPPLY:-

1) SYMPATHETIC NERVE- T10 TO T_{12} & L1-L2 SEGMENT FROM HYPOGASTRIC PLEXUSES
2) PARASYMPATHETIC NERVE- DERIVED FROM VAGUS - LATERAL HALF

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ANAL CANAL

*INTRODUCTION -

- ANAL CANAL IS TERMINAL PART→ GASTROINTESTINAL TRACK
- LIE - IN ANAL TRIANGLE B/W ISCHIOANAL FOSSA
- EXTEND FROM ANORECTAL JUNCTION TO ANUS
- SURROUNDING - OUTER VOLUNTARY & INNER-INVOLUNTARY SPHINCTER

*RELATION:-

1) ANTERIORLY - BOTH SEX - PERINEAL BODY
- IN MALE - MEMBRANOUS URETHRA AND BULB OF PENIS
- IN FEMALE-LOWER END OF VAGINA

2) POSTERIORLY - ANOCOCCYGEAL LIGAMENT.
- TIP OF COCCYX
3) LATERALLY - ISCHIANAL FOSSA

4) INFERIOR OF ANAL CANAL - DIVIDE INTO 3 PART - UPPER - 15MM
- MIDDLE - 15MM
- LOWER - 8MM

CHARACTERISTIC EPITHELIUM :-

1) UPPER MUCOUS PART:-
- LINE BY COLUMNAR EPITHELIUM (MUCOUS MEMBRANE]
- 6-10 VERTICAL RIDGE - ANAL COLUMNS
- LOWER END OF ANAL COLUMN UNITED BY SHORT TRANSVERSE FOLD CALLED ANAL VALVE
- ABOVE VALUE DEPRESSION- ANAL SINUS
- ANAL VALUE TOGETHER FORM TRANSVERSE LINE

2) MIDDLE PART:-
- NON- KERATINIZING SQUMOUSE EPITHELIUM (ALSO MUCOUS]
- ANAL COLUMNS ARE NOT PRESENT
- MUCOSA BLUISH APPEARANCE -BECAUSE DENSE VENOUS PLEXUS
-THIS REGION REFERRED TO PECTEN OR TRANSITIONAL ZONE
-LOWER LIMIT OF PECTIN OFTEN HAS WHITISH APPEARANCE ALSO REFERRED AS- "WHITE LINE OF HILTON"

3) LOWER PART:-
-LINE BY TRUE SKIN - CONTAINING SEBACEOUS GLAND (KERATINIZING SQUMOUSE EPITHELIUM)

DEEP PERINEAL SPACE

*DEFINTION - THIN SPACE - - -> UROGENITAL REGION
- SITUATED DEEP TO PERINEAL MEMBRANE

*BOUNDARIES - SUPERFICIAL - PERINEAL MEMBRANE
- ON EACH SIDE - ISCHIOPUBC RAMI

CONTENT-	MALE -	FEMALE-
1) TUBES -	PART OF URETHRA	- URETHRA & VAGINA
2) MUSCLES	- SPHINCTER URETHRA - DEEP TRANSVERSE PERINEI	- SPHINCTER URETHRA - COMPRESSOR URETHRA - SPHINCTER URETHROVAGINALIS
3) NERVE	- DORSAL NERVE OF PENIS - MUSCULAR BRANCHES FROM - PERINEAL NERVE	- DORSAL NERVE OF CLITORIS
4) VESSEL	- ARTERY OF PENIS - ARTERY TO BULB OF PENIS - URETHRA ARTERY - DEEP & DORSAL ATERIES OF PENIS	- ARTERY IF CLITORIS - ARTERY TO BULB OF VESTIBULE - URETHRAL ARTERY

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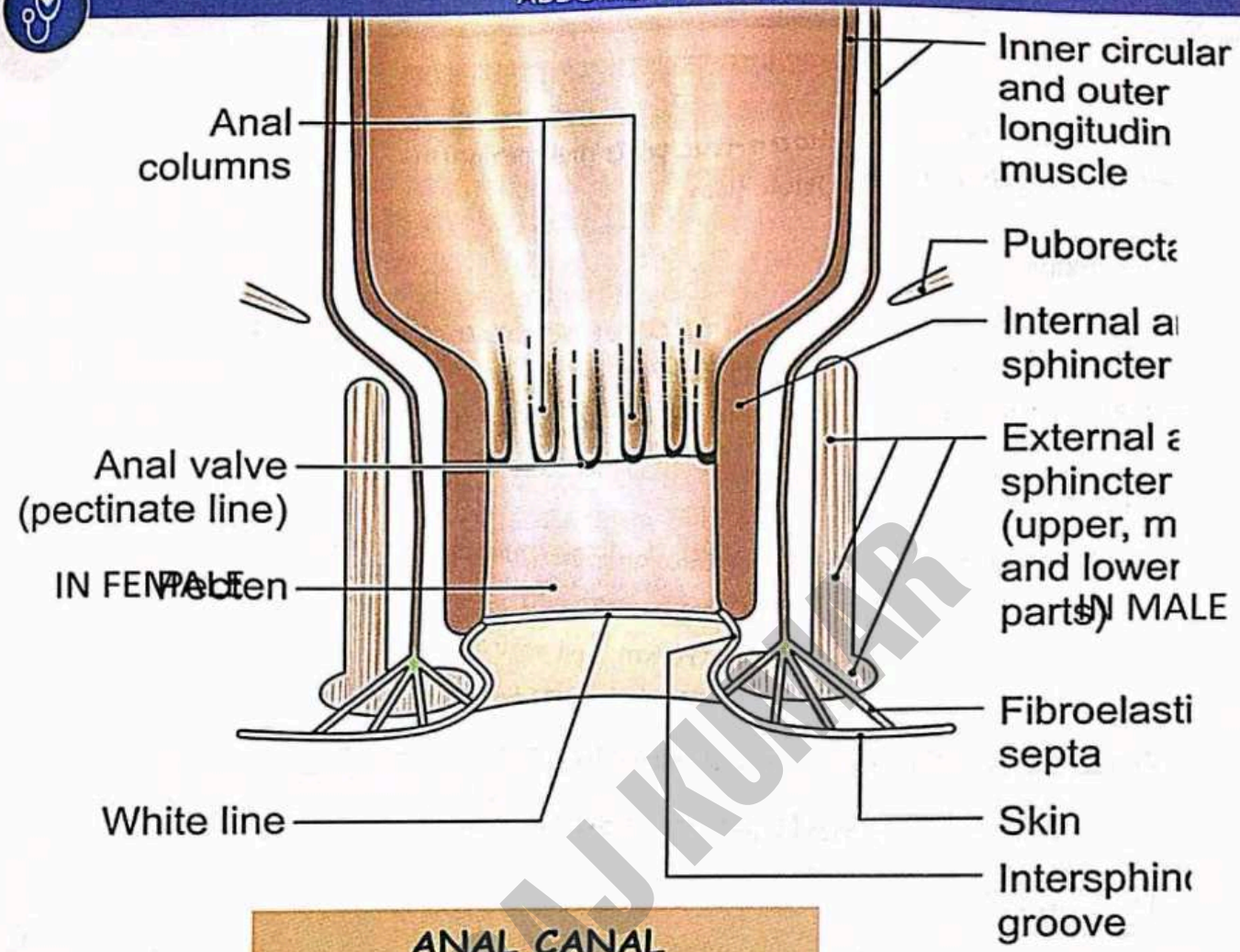
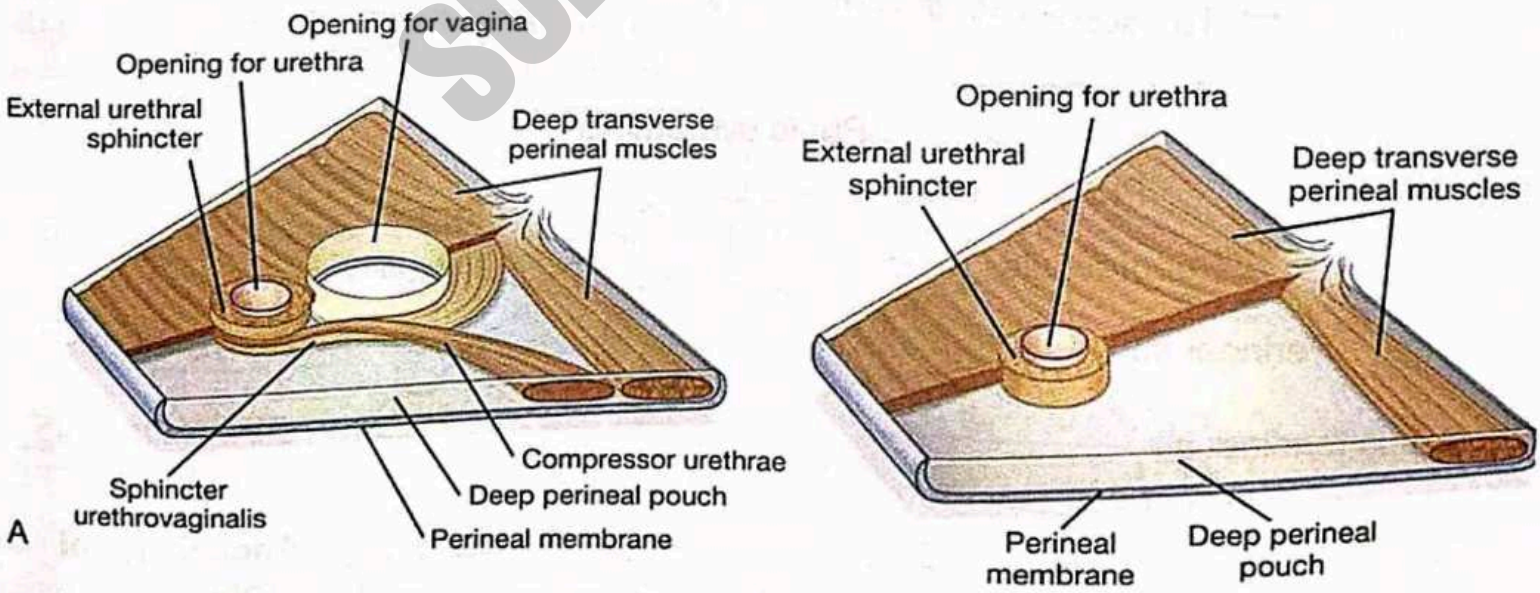


ANAL CANAL



DEEP PERINEAL SPACE

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PELVIC DIAPHRAGM

→ Muscular Partition between true pelvis and perineum.
→ form gutter shaped Pelvic floor.

Formation

→ formed by large levator Ani muscle and small coccygeus muscle on 2 sides & their covering fascia
→ Incomplete anteriorly to allow passage of urethra.

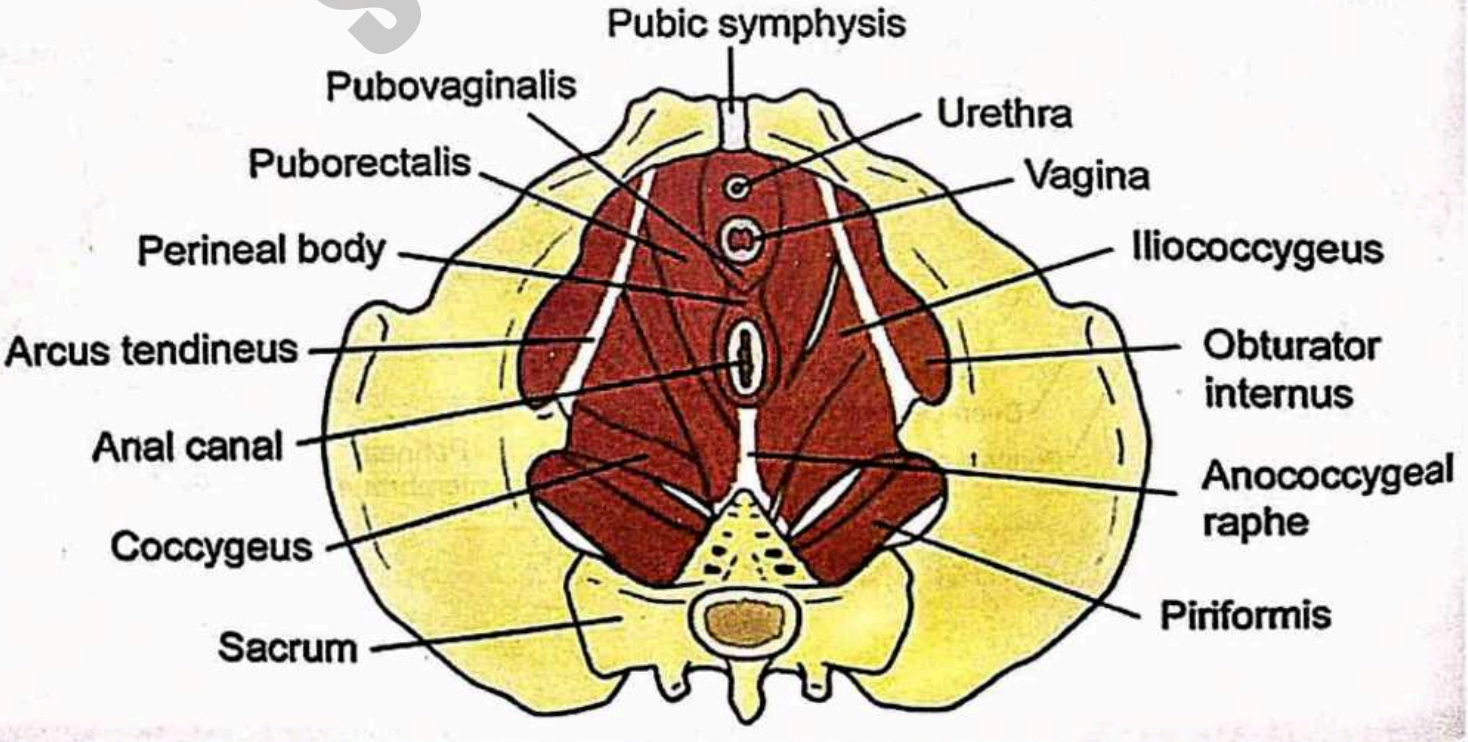
Openings

1) Hiatus Urogenitalis - Δ gap btw anterior fibres of levator Ani muscle
- transmit urethra in male & urethra and vagina in female.
- It closed from below by urogenital diaphragm.

2) Hiatus Rectalis - O(round) opening btw perineal body and anococcygeal raphe.
- provide passage to anorectal junction

Functions

→ provide support to pelvic viscera sphinter action on vagina & rectum.
→ Use abdominal pressure during defecation micturation parturition



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LESSER SAC

*IT IS LARGE RECESS OF PERITONRAL CAVITY BEHIND THE STOAMCH, THE OMENTUM AND CAUDATE LOBE OF LIVER

*CLOSE ALL ROUND

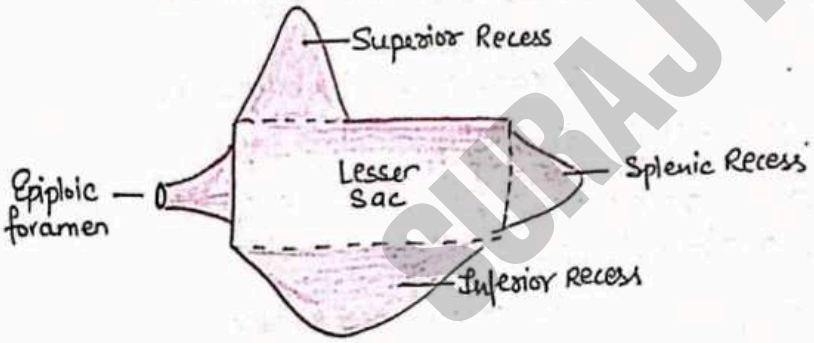
*ECXEPT IN THE UPPER PART OF ITS RIGHT BORDER -> WHERE IT COMMUNICATE WITH GREATER SAC ---> THROUGH --> EPIPLOIC FORAMEN

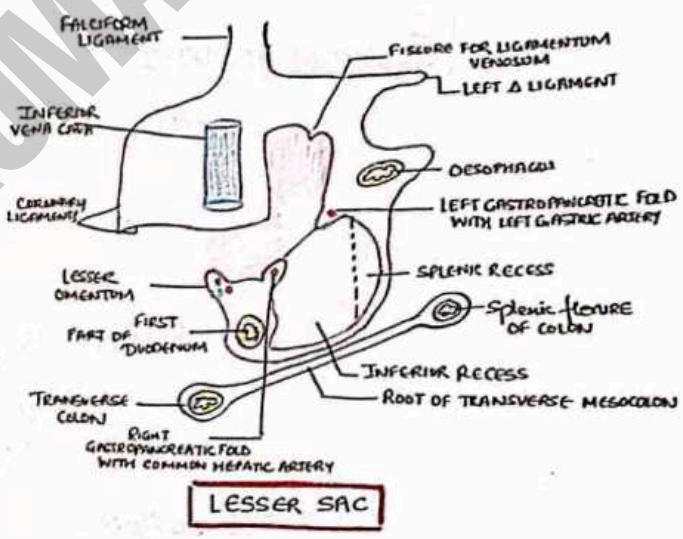
*BOUNDAIRES

1) ANTERIOR - PERITONEUM COVERING CAUDATE LOBE OF LIVER
   - POSTERIOR LAYER OF LESSER OEMNTUM
   - POSTERIOR SURFACE OF STOMACH
   - 2ND LAYER OF GREATER OMENTUM
2) POSTERIOR - 3RD LAYER OF GREATER OMENTUM
   - TRANSVERSE MESDODERM
   - UPPER LAYER OF TRANSVERSE MESOCOLON

SUBRAJ KUMAR

SUB DIVISION OF LESSER SAC



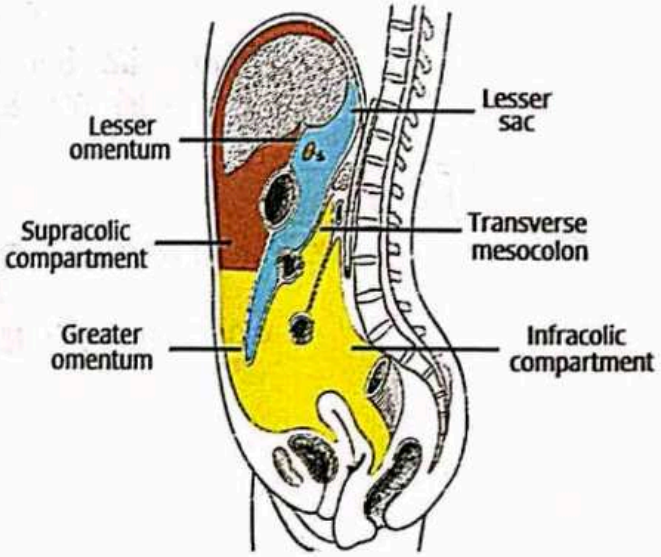


LESSER SAC

SUPERIOR RECESS - behind the lesser omentum and the liver

INFERIOR RECESS - behind the stomach and within greater omentum

SPLENIC RECESS - between the gastrosplenic and lienorenal ligament



SPLEEN

Introduction – ∘ lymphatic organ connected to Blood vascular system
∘ act as a filter for blood and plays an important role in immune response for body

Location – ∘ mainly in LEFT and partly in

∘ wedge shaped organ [tetrahedral in shape]

Dimensions – soft, highly vascular and dark purple in colour.
On average – 2.5 - thick
7.5 - Broad
12.5 - long

Position – lie obliquely along the long axis of 10th rib
Directed – downward, forward and laterally

External features –

∘ Two ends - Anterior or lateral end – expanded and is more like border
Directed - downward & forward
Posterior or medial end - round
Directed upward, backward

∘ Three Border - superior - characteristically notched near anterior end
inferior - rounded
intermediate - also rounded

∘ Two surface - Diaphragmatic surface - convex and smooth
visceral surface - concave and irregular

∘ Hilum - lie btw superior and intermediate borders
- pierced by branches and tributaries of splenic

Relations :–

★ Peritoneal relation - spleen is surrounded by peritoneum and suspended by ligament

∘ gastrosplenic ligament – extend hilum of spleen to greater curvature of stomach
contain short gastric vessels

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- Lienorenal ligament - extend from hilum of spleen to anterior surface of left kidney
  Contain - tail of pancreas
  Splenic vessels
  pancreaticosplenic lymph nodes
- Phrenicocolic ligament - not attached to spleen but support its anterior end

* Visceral relation - related to - fundus of stomach
  - anterior surface of left kidney
  - Splenic flexure of colon
  - tail of pancreas
  - gastric impression - fundus of stomach
    btw superior and intermediate border
  - Renal impression - Left kidney
    btw inferior and intermediate border
  - Pancreatic impression - tail of pancreas
    btw hilum and colic impression
  - Hilum - lie on inferomedial part of gastric impression
    - Transmits - splenic vessels & nerve
    - provide attachment to gastrosplenic & linorenal ligament
  - colic impression - splenic flexure of colon
    occupies Δ area of anterior border

* Diaphragmatic surface -
  - Related to diaphragm
  - which separated the spleen from pleura and lungs.

Arterial supply -
supplied by splenic artery

Venous supply -
- Drained by splenic vein
- Tributaries
  → Short gastric vein
  → left gastroepiploic vein
  → pancreatic vein
  → Inferior mesenteric vein

Nerve supply -
Sympathetic fibres - Coeliac plexus

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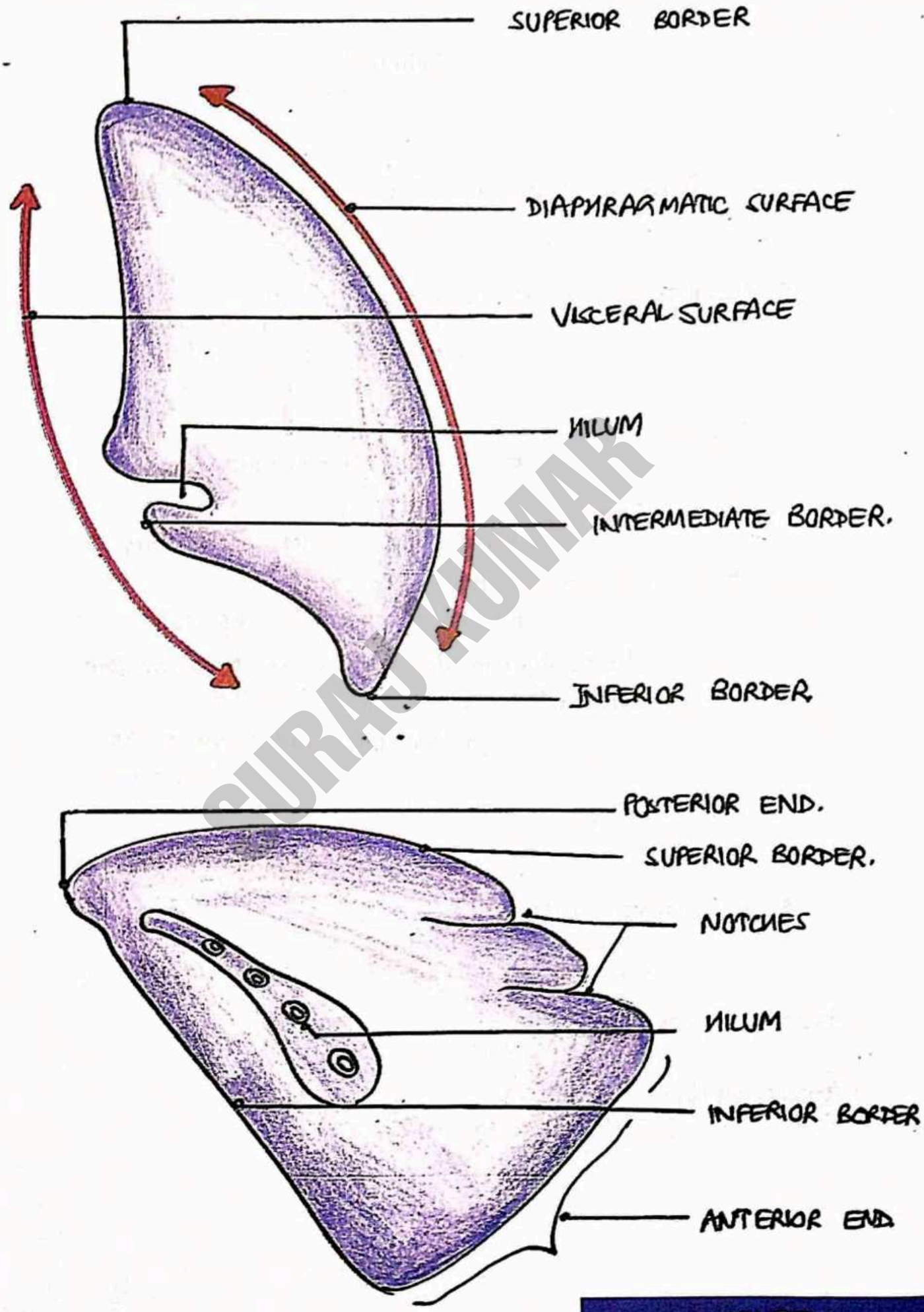

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SPLEEN

STUDY TIPS FOR MAXIMUM MARKS

1.HOW TO WRITE IN UNIVERSITY EXAM AND HOW TO DRAW DIAGRAMS ALREADY MENTION IN UPPER LIMB (PAGE NUMBER - 25 OF VOLUME - 1)

PREVIOUS YEAR UNIVERSITY QUESTIONS

*LONG QUESTION - 20MARKS

1.DESCRIBE INGUINAL CANAL UNDER FOLLOWING HEADINGS -
1.LOCATION & EXTEND 2.BOUNDARY & CONTENT 3.PROTECTIVE MECHANISM
4.APPLIED(HERNIA) 5.SPERMATIC CORD

2.DESCRIBE STOMACH UNDER FOLLOWING HEADINGS -
1.POSTERIOR RELATION 2.ARTERY SUPPLY 3.APPLIED ANATOMY

3.DESCRIBE VERMIFORM APPENDIX UNDER FOLLOWING HEADINGS -
1.POSITION 2.BLOOD SUPPLY & LYPHATIC DRAINAGE 3.APPLIED

4.DESCRIBE EXTERNAL FEATURE, RELATION, LIGAMENT, BLOOD SUPPLY & DEVELOPMENT OF URINARY BLADDER ?

5. DESCRIBE PARTS, RELATION, LIGAMENT, BLOOD SUPPLY & SUPPORT OF UTERUS ?

6.DESCRIBE SECOND PART OF DEUDENUM - UNDER FOLLOWING HEADING -
1.RELATION 2.DEVELOPMENT 3.APPLIED 4.ARTERY SUPPLY

*SHORT NOTES - 5MARKS

1.RECTUS SHEATH
2.DESCEND OF TESTIS
3.LIGAMENT OF LIVER & VISCERAL RELATION
4.PANCREAS
5.DEVELOPMENT OF KIDNEY
6.KIDNEY RELATION
7.EXTRA HAPATIS BILIARY APPARATUS
8.PORTAL VEIN & ITS CLINICAL ANATOMY
9.EPIPLOIC FORAMEN
10.GREATER & LESSER OENTUM
11.ISCHIONAL FOSSA
12.DEVELOPMENT OF ANAL CAN

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ABDOMEN ANATOMY IS COMPLETED !!

YOUR POSITIVE ACTION COMBINED WITHPOSITIVE THINKING RESULTS IN SUCCESS

ALL THE BEST FOR YOUR EXAMS

DR.JOHARI

THE COPYRIGHT ACT,1957

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MCQ ABDOMEN

1) Paralysis of which muscle causes drooping of upper eyelid (ptosis) :
a Occipito-frontalis
b Orbicularis oculi
c levator palpebrae superioris
d superior rectus

2) All of the following muscles have bony attachment except :
a procerus
b Orbicularis oculi
c Risorious
d platysma

3) Which one of the following is motor nerve of scalp:
a lesser occipital
b auriculo temporal
c posterior auricular
d supra orbital

4) Which of the following muscles is an abductor of vocal cords :
a posterior cricoarytenoid
b vocalis
c lateral cricoarytenoid
d cricothyroid

5) Which muscle is supplied by glossopharyngeal nerve
a stylohyoid
b stylopharyngeus
c salpingopharyngeus
d palatopharyngeus

6)Which muscle protrudes the tongue:
a genioglossus
b hyoglossus
c styloglossus
d palatoglossus

7) where does the nasolacrimal duct drains :
a superior meatus
b middle meatus
c inferior meatus
d sphenoethmoidal recess

8) Which muscle tenses the vocal cord
a Thyroarytenoid
b lateral cricoarytenoid
c posterior cricoarytenoid
d cricothyroid

9) Which of the following is present in the suprasternsl space :
a jugular venous arch
b external jugular vein
c supraclavivular nerve
d external jugular vein

10) All of the following pass through the superior orbital fissure except :
a ophthalmic artery
b ophthalmic vein
c occulomotor nerve
d trochlear nerve

11) Which of the following lies deep to the hyoglossus muscle
a Lingual nerve
b lingual artery
c submandibular gland
d submandibular ganglion

12) All of the following are intrinsic muscle of larynx excep
a cricothyroid
b thyrohyoid
c thyroepiglottic
d Thyroarytenoid

13) All of the following are supplied by hypoglossal nerve except:
a genioglossus
b styloglossus
c hyoglossus
d palatoglossus

14) The parotid duct opens into vestibule of mouth at whic level of molar tooth
a upper first
b lower first
c upper second
d lower second

15) Which nerve supplies the superior oblique muscle of t eye ball
a optic
b occulomotor
c trochlear
d abducens

16) Stylopharyngeus muscle recieves its motor innervatio from which nerve
a facial
b glossopharyngea
c vagus
d hypoglossal

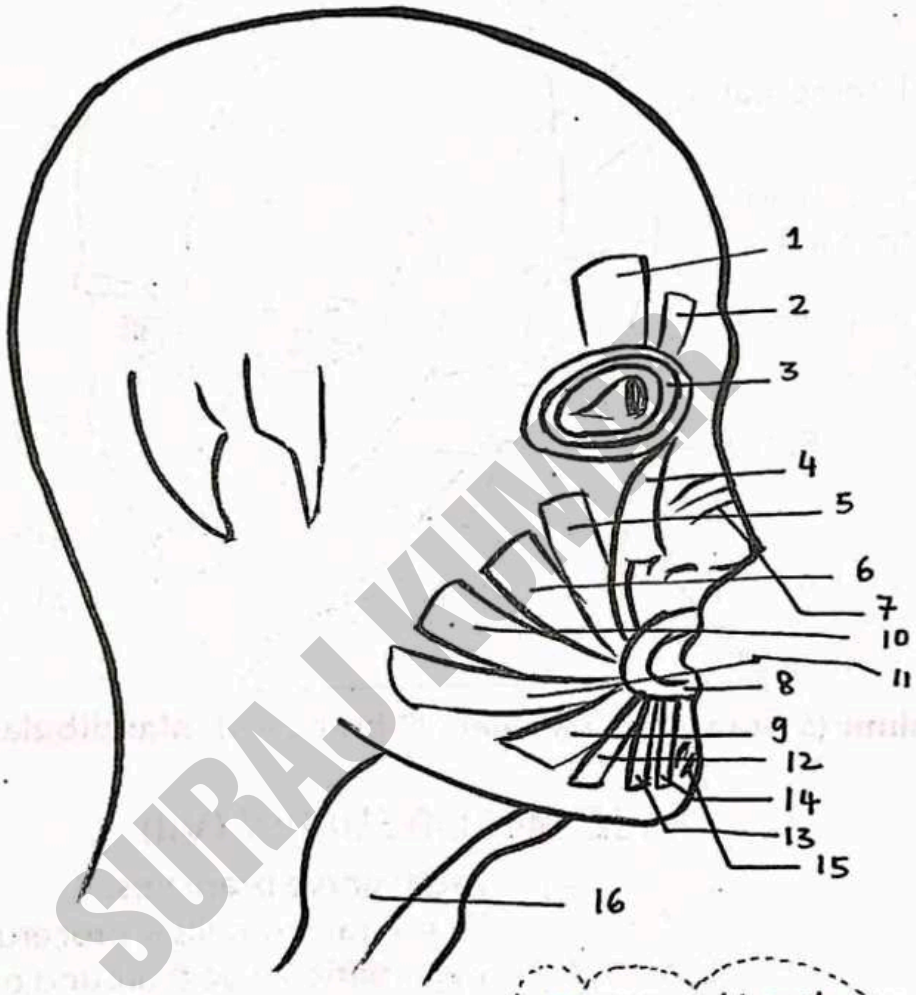
17) Which of the following layer of the scalp is known as dangerous layer
a skin
b connective tissue
c loose areolar tissue
d pericranium

18) The structure that does not pass through the parotid gland
a facial nerve
b external carotid artery
c retromandibular nerve
d auriculo temporal nerve

1)C 2)D 3)C 4)A 5)B 6)A 7)C 8)D 9)A 10)A 11)B 12)B 13)D 14)C 15)C 16)B 17)C 18)D

SUBHARIMBBS SURAJ KUMAR

FACIAL MUSCLES



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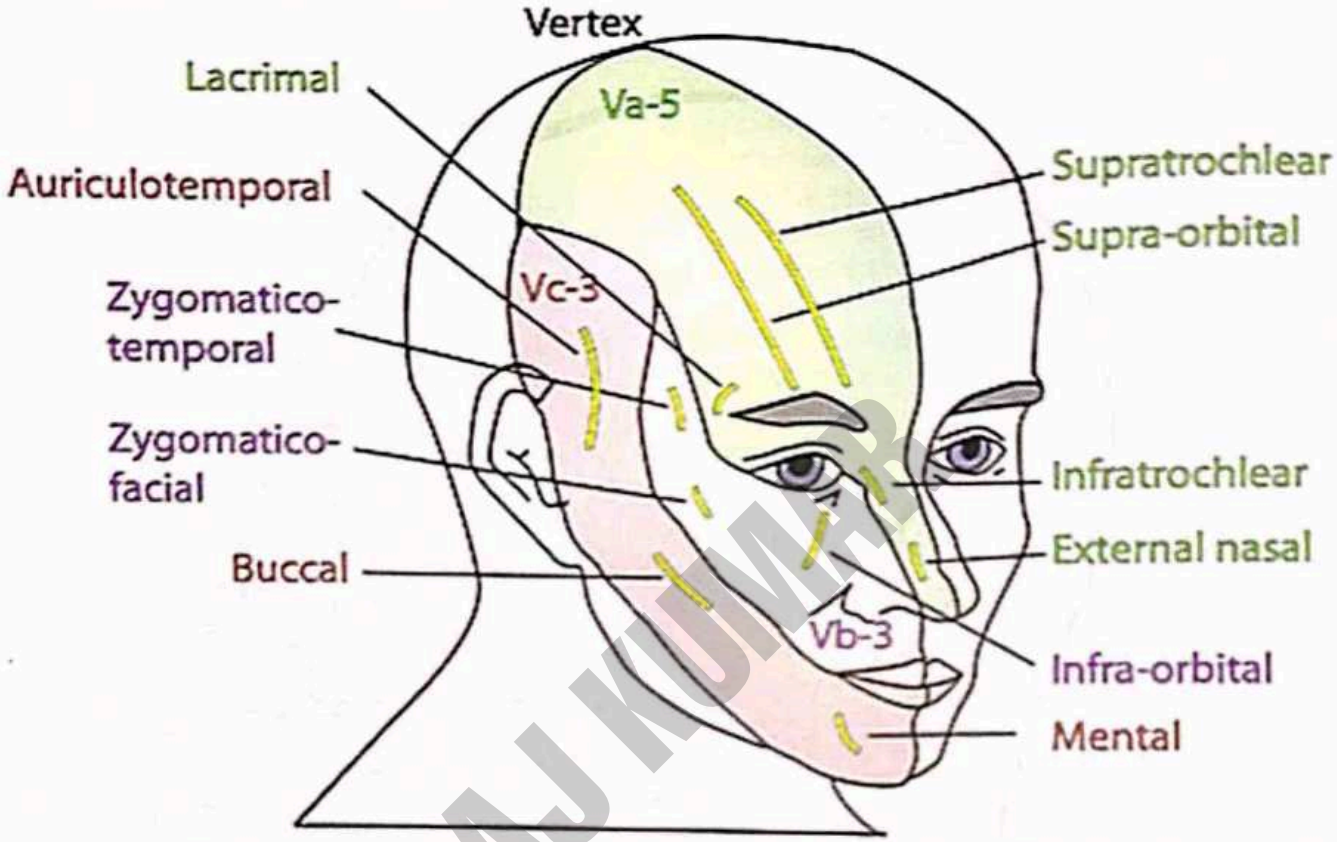
1 - Frontalis
2 - Corrugator supercilli
3 - Orbicularis oculi
4 - Levator labii superioris alaeque nasi
5 - Levator labii superioris
6 - Levator anguli oris
7 - Compressor naris
8 - Orbicularis oris
9 - Buccinator
10 - Zygomaticus minor
11 - Zygomaticor major
12 - Risorus
13 - Depressor labii oris
14 - Depressor labii inferioris
15 - Mentis
16 - Platysma

- Temporal branch
- Zygomatic branch
- Upper buccal branch
- Lower buccal branch
- Marginal Mandibular branch
- Cervical branch

SENSARY INERVATION OF FACE

SENSORY BRANCHES OF TRIGEMINAL N ON FACE



Ophthalmic (5 branches) Maxillary (3 branches) Mandibular (3 branches)

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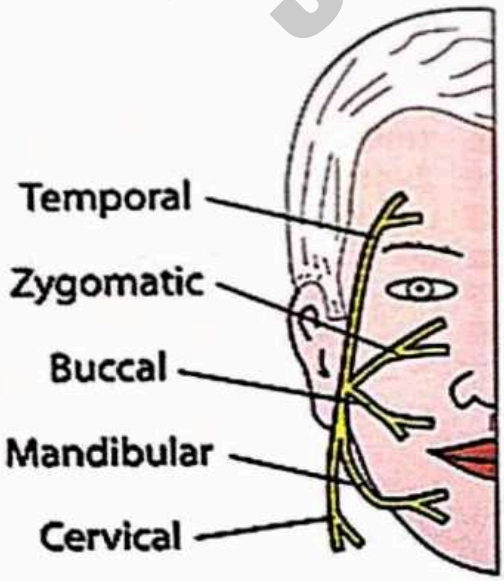
FACE: MOTOR SUPPLY (VII)



Facial nerve branches:
Temporal: frontalis & procerus
Zygomatic 1: eye & around orbit
Zygomatic 2: mid face & smile
Buccal: buccinator & upper lip
Mandibular: lower lip & orbicularis oris
Cervical: platysma
(note: proprioception is supplied by trigeminal)

Mnemonic:
Two
Zebras
Befriended
My
Cat

SCALP ANATOMY

SCAN QR
WITH QRBOT APP
TO DIRECTLY OPEN VIDEO LINK

*SOFT TISSUES COVERING - THE CRANIAL VAULT FROM SCALP

*EXTENT OF SCALP -
1) ANTERIORLY - SUPRAORBITAL MARGIN
2) POSTERIORLY - EXTERNAL OCCIPITAL PROTUBERANCE
- SUPERIOR NUCHAL LINES
3) EACH SIDE - SUPERIOR TEMPORAL LINES

-MADE UP OF FIVE LAYER - SKIN
- SUPERFICIAL FASCIA
- DEEP FASCIA OR GALEA APONEUROTICA
- LOOSE AREOLAR TISSUE
- PERICRANIUM

1) SKIN- THICK AND HAIRY
- IT HAS SWEET GLANDS AND SEBACEOUS GLANDS

2) SUPERFICIAL FASCIA - MORE FIBROUS AND DENSE IN CENTRE
- CONTAIN BLOOD VESSEL
- ITS BINDS THE SKIN TO SUBJACENT APONEUROSIS
- AND PROVIDE PROPER PASSAGE OF VESSEL AND NERVE OF SKIN

-OCCIPITOFRONTALIS MUSCLE:- HAS TWO BELLS OCCIPITALS AND FRONTALIS
- INSERTED-EPICRANIAL APRONEUROSIS

- OCCIPITALIS - SMALL AND SEPARATE
- ARISE - LATERAL 2/3 OF SUPERIOR NUCHAL LINE,
- SUPPLY BY - POSTERIOR AURICULAR BRANCH OF FACIAL NERVE

- FRONTALIS - LARGE AND PARTLY UNITED
- ARISE FROM- SKIN UPPER OF EYELID AND FORE HEAD
- SUPPLY BY- TEMPORAL BRANCH OF FACIAL NERVE

- TEMPARO PARIETALIS- PRESENT ON LATERAL SIDE
-ARISE FROM - TEMPORAL FASCIA AND FUSE WITH EPICRANIAL APONEUROSIS.
-SUPPLY BY TEMPORAL BRANCH OF FACIAL NERVE

3) EPICRANIAL APONEUROSIS-
- FREELY MOVABLE ON PERICRANIUM
- ATTACHMENT- ANTERIORLY - INSERTION-FRONTALIS
- POSTERIORLY - EXTERNAL OCCIPITAL PROTUBERANCE
- HIGHEST NUCHAL LINE B/W OCCIPITAL BELLIES
- SIDE - SUPERIOR TEMPORAL LINE
- EXPANSION - ZYGOMATIC ARCH

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5) LOOSE AREOLAR TISSUE

- EXTEND - ANTERIORLY-EYELIDS
- POSTERIORLY - SUPERIOR NUCHAL LINE.
- EACH SIDE - SUPERIOR TEMPORAL LINE
- GIVE PASSAGE TO EMISSARY VEIN WHICH CONNECT
- EXTRACRANIAL VEIN TO INTRACRANIAL VENOUS SINUSES

6) PERICRANIUM - LOOSELY ATTACHED TO SURFACE OF BONE
- BUT IS FIRMLY ADHERENT TO SUTURE.

*ARTERIAL SUPPLY - IN FRONT - SUPRATROCHLEAR-→BRANCH→INTERNAL CAROTID A.
- SUPRA ORBITAL-→BRANCH-→EXTERNAL CAROTID A.

- BEHIND - POSTERIOR AURICULAR-→EXTERNAL CAROTID ARTERY
- OCCIPITAL----→EXTERNAL CAROTID ARTERY

*VENOUS DRAINAGE -

-SUPRATROCHLEAR & SUPRA ORBITAL VEIN - JOIN EACH OTHER TO FORM ANGULAR VEIN--→ FACIAL VEIN

-SUPERFICIAL TEMPORAL VEIN - DESCEND IN FRONT OF TRAGUS--→TO ENTER PAROTID GLAND
-JOIN MAXILLARY VEIN -→FORM RETROMANDIBULAR VEIN
TERMINATED ANTERIOR & POSTERIOR DIVISION

-POSTERIOR AURICULAR VEIN - IT DESCEND BEHIND AURICLE &
-UNITES THE POSTERIOR DIVISION OF RETROMANDIBULAR VEIN--→FORM EXTERNAL JUGULAR VEIN

-OCCIPITAL VEIN - TERMINATE - SUBOCCIPITAL VENOUS PLEXUS

-VEIN OF SCALP COMMUNICATE WITH INTRACRANIAL DURAL VENOUS SINUSES THROUGH EMISSARY VEIN :

- EMISSARY VEIN - VEIN CONNECTING VEIN OUTSIDE THE CRANIUM WITH INTRACRANIAL VEIN BY PASSING THROUGH FORAMEN IN CRANIUM
- PARIETAL EMISSARY VEIN - COMMUNICATES SUPERIOR SAGGITAL SINUS
----→PASSES PARIETAL FORAMEN
- MASTOID EMISSARY VEIN - SIGMOID SINUS
-----→ PASSES MASTOID FORAMEN

*LYMPHATIC DRAINAGE -

- ANTERIOR PART OF SCALP--→ DRAIN PREAURICULAR OR SUPERFICIAL PAROTID
- POSTERIOR PART OF SCALP--→ POSTERIOR AURICULAR OR MASTOID
---→ OCCIPITAL LYMPH NODE

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*NERVE SUPPLY - SUPPLY BY 10 NERVE

-FRONT OF AURICLE - FIVE NERVE (4, SENSARY & 1 MOTOR)
-SUPRA ORBITAL
-SUPRATROCHLEAR NERVE
-ZYGOMATICOTEMPERAL NERVE
-TEMPORAL BRANCH OF FACIAL NERVE(MOTOR)
-AURICULOTEMPORAL NERVE

-BEHIND AURICLE - FIVE (4 SENSARY & 1 MOTOR)
-GREATER AURICULAR NERVE
-POSTERIOR AURICULAR NERVE(MOTOR)
-LESSER OCCIPITAL NERVE
-GREATER OCCIPITAL NERVE
-THIRD OCCIPITAL NERVE

CLINICAL ANATOMY

SCALP

1.THE LAYER OF LOOSE AREOLAR TISSUE IS KNOWN AS THE "DANGEROUS AREA OF THE SCALP" BECAUSE THE EMISSARY VEINS, WHICH COURSE HERE, MAY TRANSMIT INFECTION FROM THE SCALP TO THE CRANIAL VENOUS SINUSES.

2.BECAUSE OF THE ABUNDANCE OF SEBACEOUS GLANDS, THE SCALP IS A COMMON SITE FOR SEBACEOUS CYSTS.

3.COLLECTION OF BLOOD IN THE LAYER OF LOOSE CONNECTIVE TISSUE CAUSES GENERALISED SWELLING OF THE SCALP.

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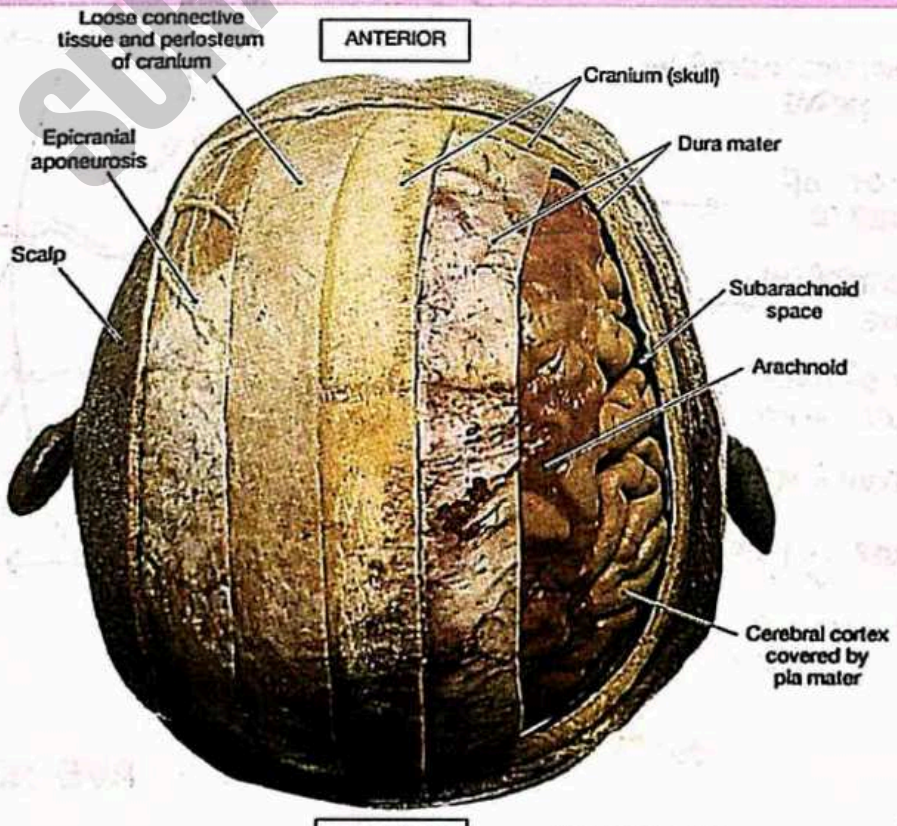
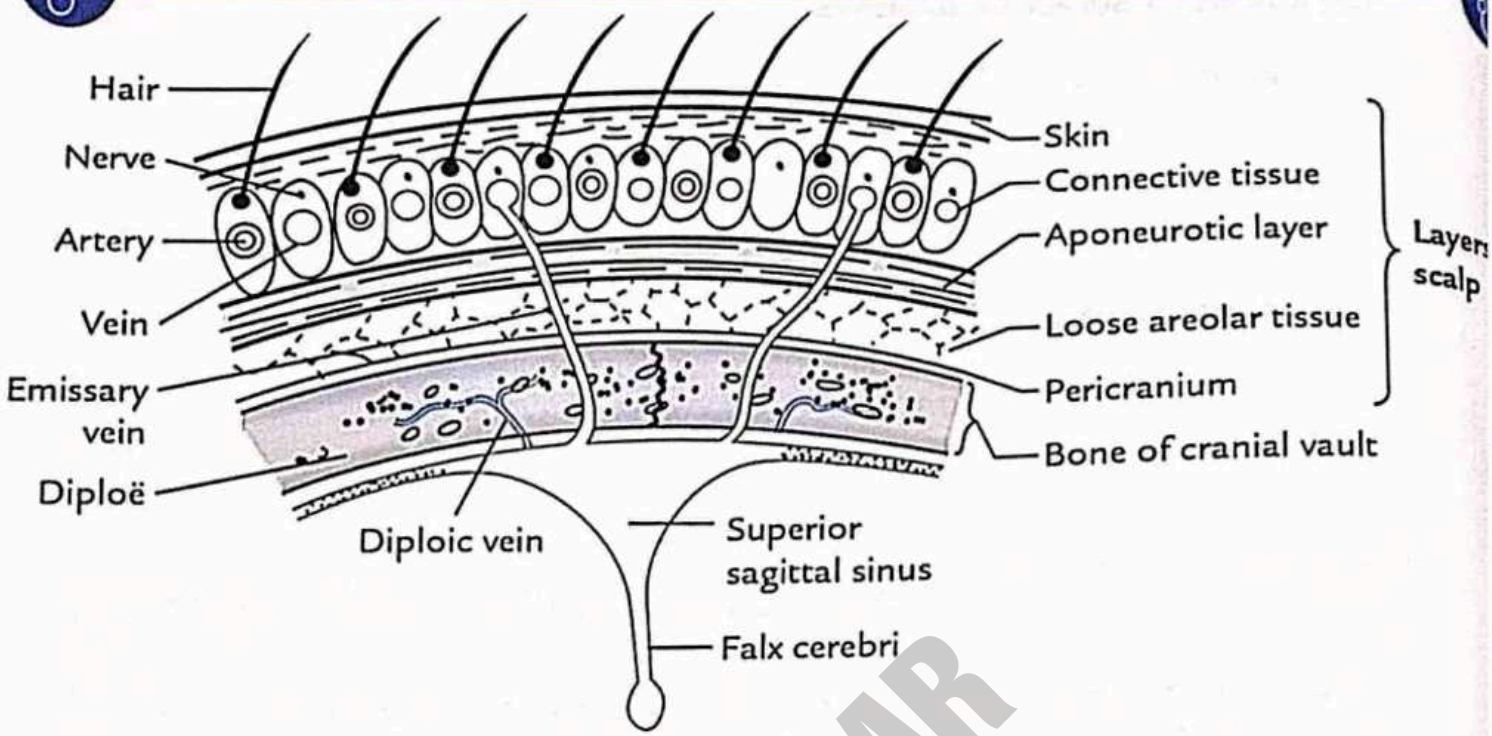
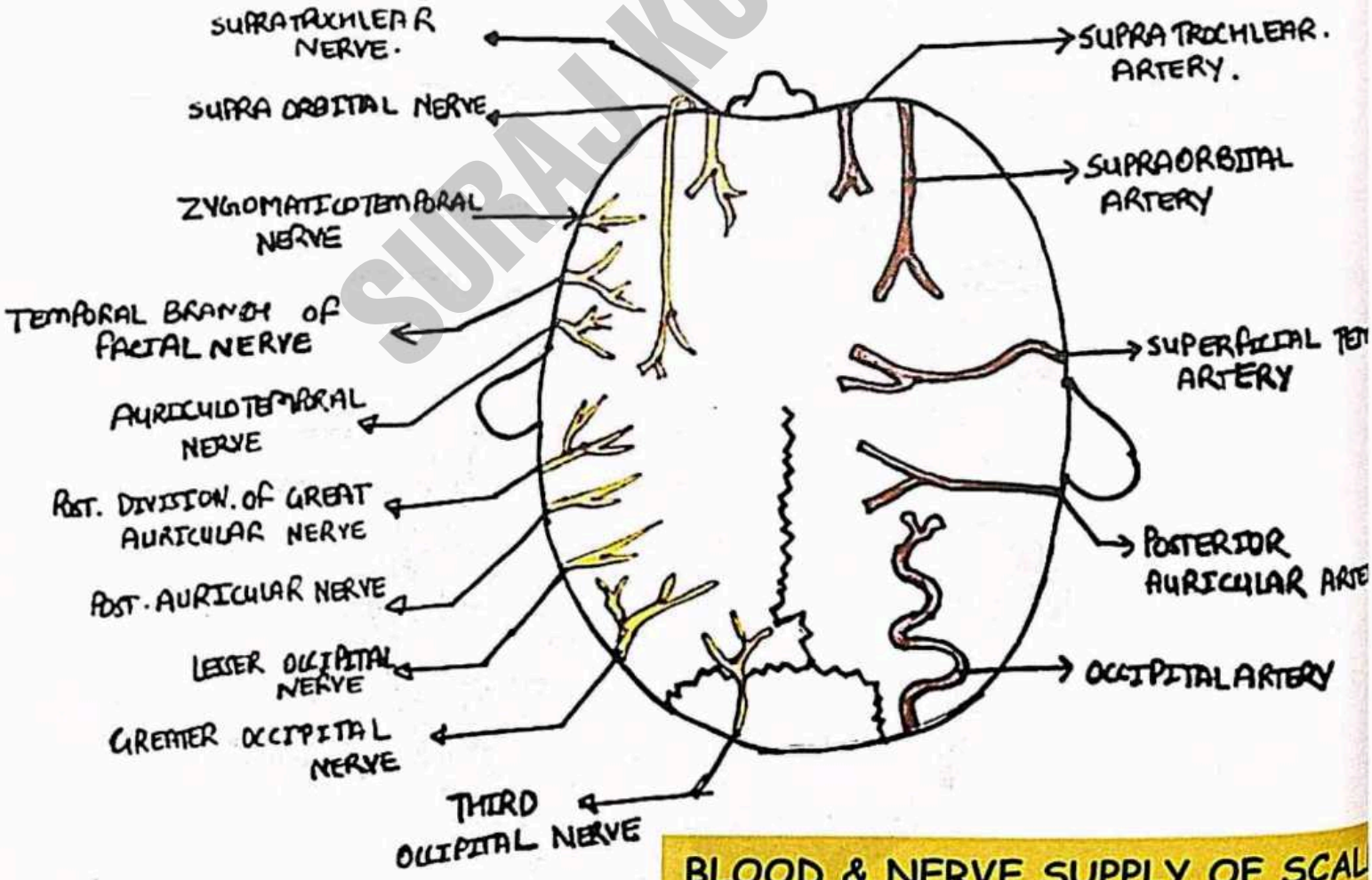




LAYER OF SCALP



BLOOD & NERVE SUPPLY OF SCAL

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LACRIMAL APPARATUS

*COMPONENTS - STRUCTURE CONCERNED WITH SECRETION AND DRAINAGE OF LACRIMAL TEAR FLUID CONSTITUTE THE LACRIMAL APPARATUS

- MADE - LACRIMAL GLAND AND DUCTS
- CONJUNCTIVAL SAC
- LACRIMAL PUNCTA AND LACRIMAL CANALICULI
- LACRIMAL SAC
- NASOLACRIMAL SAC

1) LACRIMAL GLAND - SEROUS GLAND
- SITUATED LACRIMAL FOSSA- ANTEROLATERAL OF ROOF OF BONY ORBIT
- "J" SHAPED GLAND - BEING INDENTED BY - TENDON-OF- LEVATOR PALPEBRAE SUPERIOR
  - ↳ ORBITAL PART - LARGER AND DEEPER
  - ↳ PALPEBRAL PART- SMALLER AND SUPERFICIAL
- DOZEN OF DUCTS PIERCE - CONJUNCTIVA OF UPPER LID AND OPEN INTO CONJUNCTIVAL SAC
- ARTERY - LACRIMAL BRANCH OF OPHTHALMIC ARTERY
- NERVE SUPPLY - LACRIMAL GLANDS NERVE
- LACRIMAL FUILD SECRETED —->LACRIMAL GLAND—->FLOW→ →CONJUNCTIVAL SAC-→ —> PERIODIC BLINKING —>SPREAD FLUID

2) CONJUNCTIVA SAC -
- CONJUNCTIVA LINING-DEEP SURFACE - OF EYELIDS → PALPEBRAL CONJUNCTIVA
  - ↳ FRONT OF EYEBALL →BULBAR CONJUNCTIVA
- POTENTIAL SPACE B/W PALPEBRAL AND BULBAR - CALLED CONJUNCTIVA SAC

3) LACRIMAL PUNCTA & CANALICULI -
- EACH LACRIMAL CANALICULI- BEGINS AT LACRIMAL PUNCTUM, 10MM LONG
- BOTH CANALICULI OPEN- LATERAL WALL OF LACRIMAL SAC
- BEHIND MEDIAL PALPEBRAL LIGAMENT

4) LACRIMAL SAC -
- MEMBRANOUS SAC 12MM LONG AND 5MM WIDE
- SITUATED IN LACRIMAL GROOVE
- BEHIND MEDIAL PALPEBRAL LIGAMENT
- UPPER - END IS BLIND
- LOWER END CONTINOUS WITH NASOLACRIMAL DUCT

5) NASOLACRIMAL DUCT:
- MEMBRANOUS PASSAGE - 18MM LONG
- BEGINS - LOWER END OF LACRIMAL SAC
- RUN DOWNWARD, BACKWARD - OPEN INTO- INFERIOR MEATUS OF NOSE

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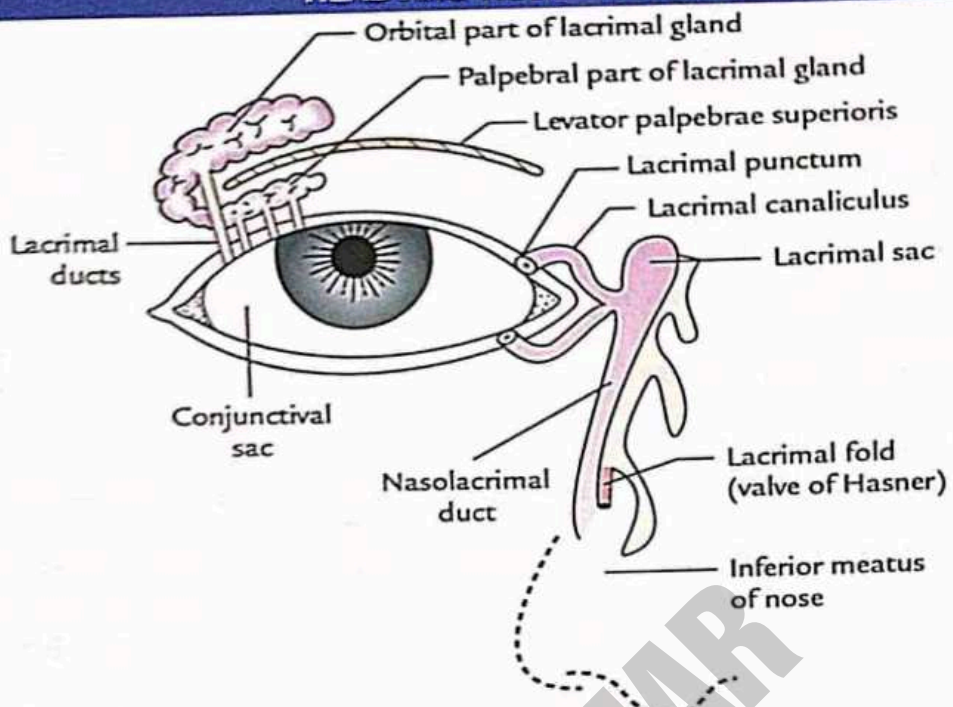
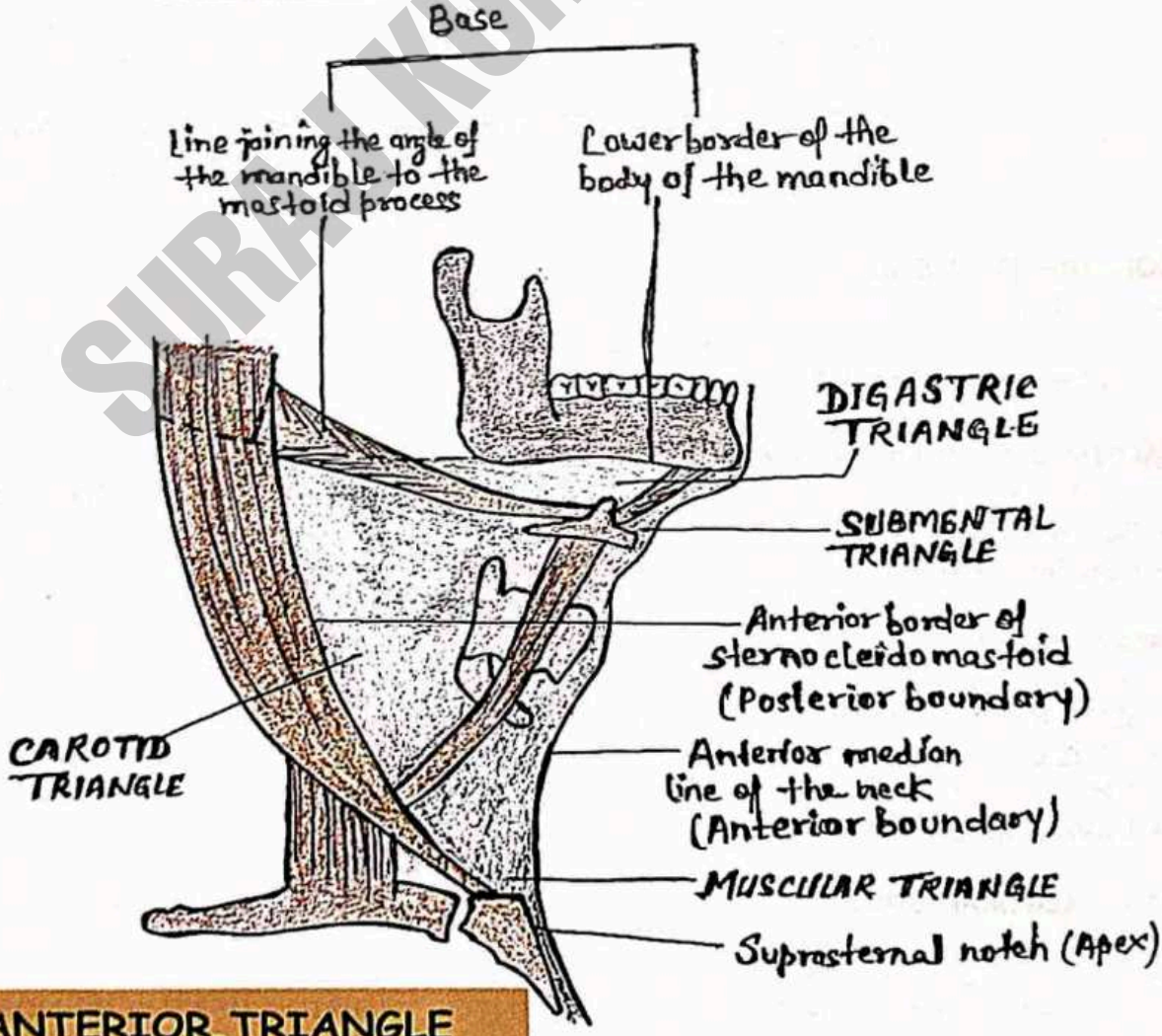


LACRIMAL APPARATUS



ANTERIOR TRIANGLE

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ANTERIOR TRIANGLE

*BOUNDARIES-

1) MEDIALLY-ANTERIOR MEDIAN PLANE OF NECK
2) LATERALLY-STERNOCLEIDOMASTOID
3) BASE -A LINE JOINING THE ANGLE OF MANDIBLE TO MASTOID PROCESS

*SUBDIVISION -
- SUB MENTAL TRIANGLE
- DIGASTRIC TRIANGLE
- CAROTID TRIANGLE
- MUSCULAR TRIANGLE

1) SUBMENTAL TRIANGLE -

*BOUNDARIES - EACH SIDE-ANTERIOR BELLY OF DIGASTRIC MUSCLE
- BASE - HYOID BONE
- FLOOR - RIGHT AND LEFT MYLOHYOID

*CONTENTS
- 4 SUBMENTAL LYMPH NODE
- SMALL SUB-MENTAL VEIN JOIN TO FORM ANTERIOR JUGULAR VEIN

2) DIGASTRIC TRIANGLE

*BOUNDARIES
- ANTERO-INFERIORLY - ANTERIOR BELLY OF DIGASTRIC
- POSTERO-INFERIORLY - POSTERIOR BETLY OF DIGASTRIC
  - STYLOHYOID
- BASE - BASE OF MANDIBLE
- ROOF - SKIN
  - SUPERFICIAL FASCIA→ PLATYSMA
    - CERVICAL BRANCH OF FACIAL NERVE
    - ANTERIOR CUTANEOUS NERVE
  - DEEP FASCIA
- FLOOR - ANTERIORLY - MYLOHYOID MUSDE
  - POSTERIORLY - HYOGLOSSUS MUSCLE

*CONTENTS -
- SALIVARY GLAND
- SUBMANDIBULAR LYMPH NODE
- FACIAL ARTERY AND VEIN
- SUB-MENTAL ARTERY
- HYPOGLOSSAL NERVE
- MYLOHYOID NERVE

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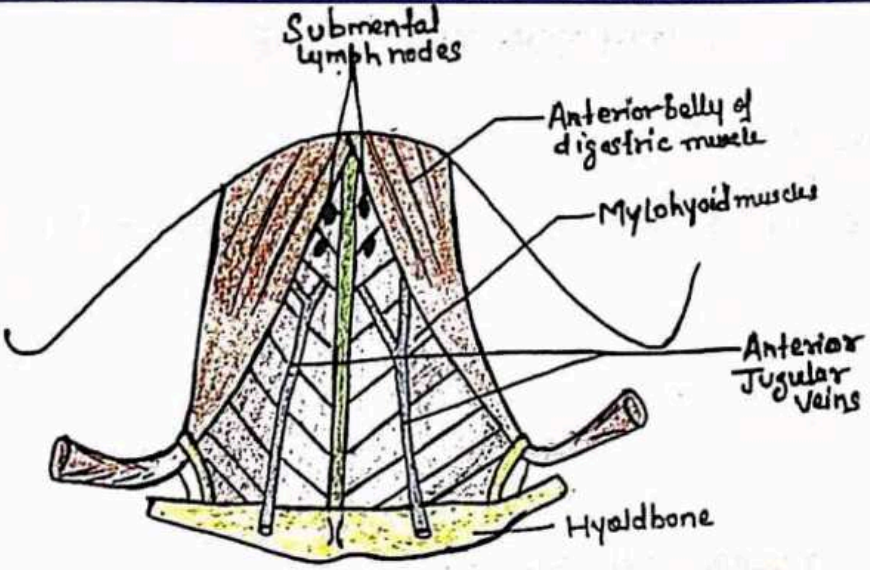
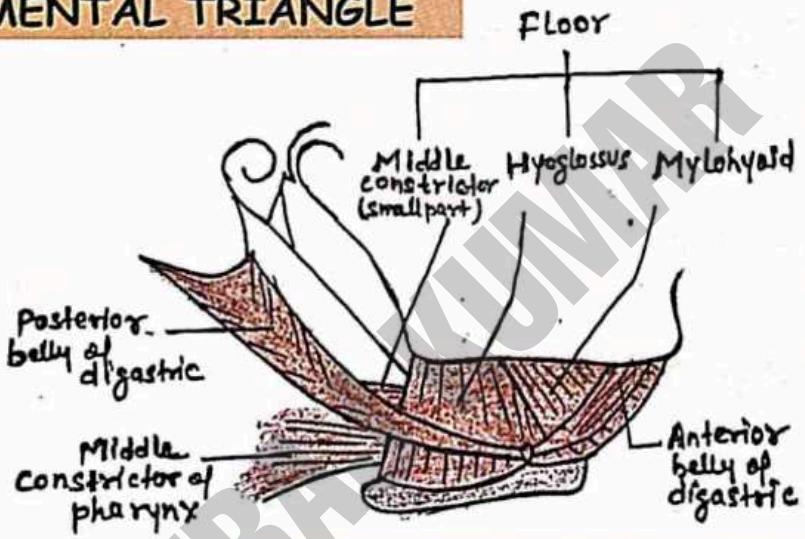
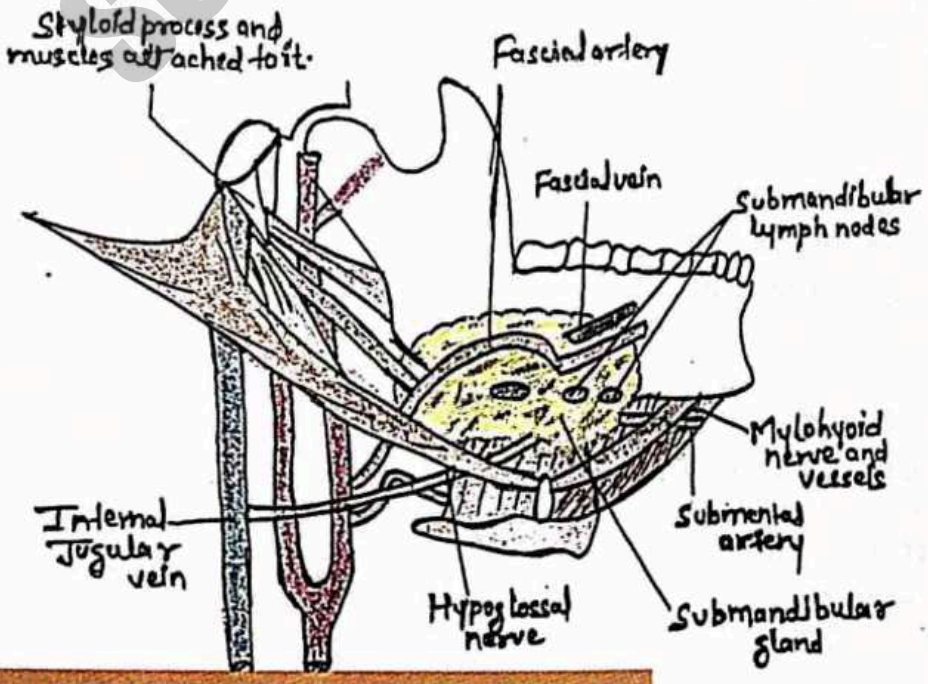


SUB MENTAL TRIANGLE



DIGASTRIC TRIANGLE



CONTENTS OF DIGASTRIC TRIANGLE

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CAROTID TRIANGLE

*INTRODUCTION - SUB-DIVISION OF -ANTERIOR TRIANGLE OF NECK

*BOUNDARIES -

- ANTEROSUPERIORLY - POSTERIOR BELLY OF DIGASTRIC MUSCLE
  - STYLOHYOID
- ANTERO INFERIORLY - SUPERIOR BELLY OF OMOHYOID
- ROOF - SKIN
  -SUPERFICIAL FASCIA CONTAIN - PLASTYSMA
  - CERVICAL BRANCH OF FACIAL NERVE
  -TRANSVERSE CUTANEOUS NERVE
  - DEEP CERVICAL FASCIA
- FLOOR - MIDDLE CONSTRICTOR OF PHARYNX
  - INFERIOR CONSTRICTOR OF PHARYNX
  - THYROHYOID MEMBRANE
- POSTERIORLY - ANTERIOR BORDER OF STERNOCLEIDOMASTOID

*CONTENTS -

*ARTERY

- COMMON CAROTID ARTERY
- INTERNAL CAROTID ARTERY
- EXTERNAL CAROTID ARTERY - SUPERIOR THYROID.
  - LINGUAL
  - FASCIAL
  - ASCENDING PHARYNGEAL ARTERY

*VEIN -

- INTERNAL JUGULAR VEIN
- COMMON FACIAL VEIN
- PHARYNGEAL VEIN
- LINGUAL VEIN

*NERVE-

- VAGUS NERVE
- SPINAL ACCESSARY NERVE
- SUPERIOR LARYNGEAL BRANCH - EXTERNAL LARYNGEAL NERVE
  - INTERNAL LARYNGEAL NERVE
- HYPOGLOSSAL NERVE
- SYMPATHETIC CHAIN NERVE

*LYMPH NODE - DEEP CERVICAL LYMPH NODE -
-JUGULODIGASTRIC NODE
- JUGULO-OMOHYOID NODE

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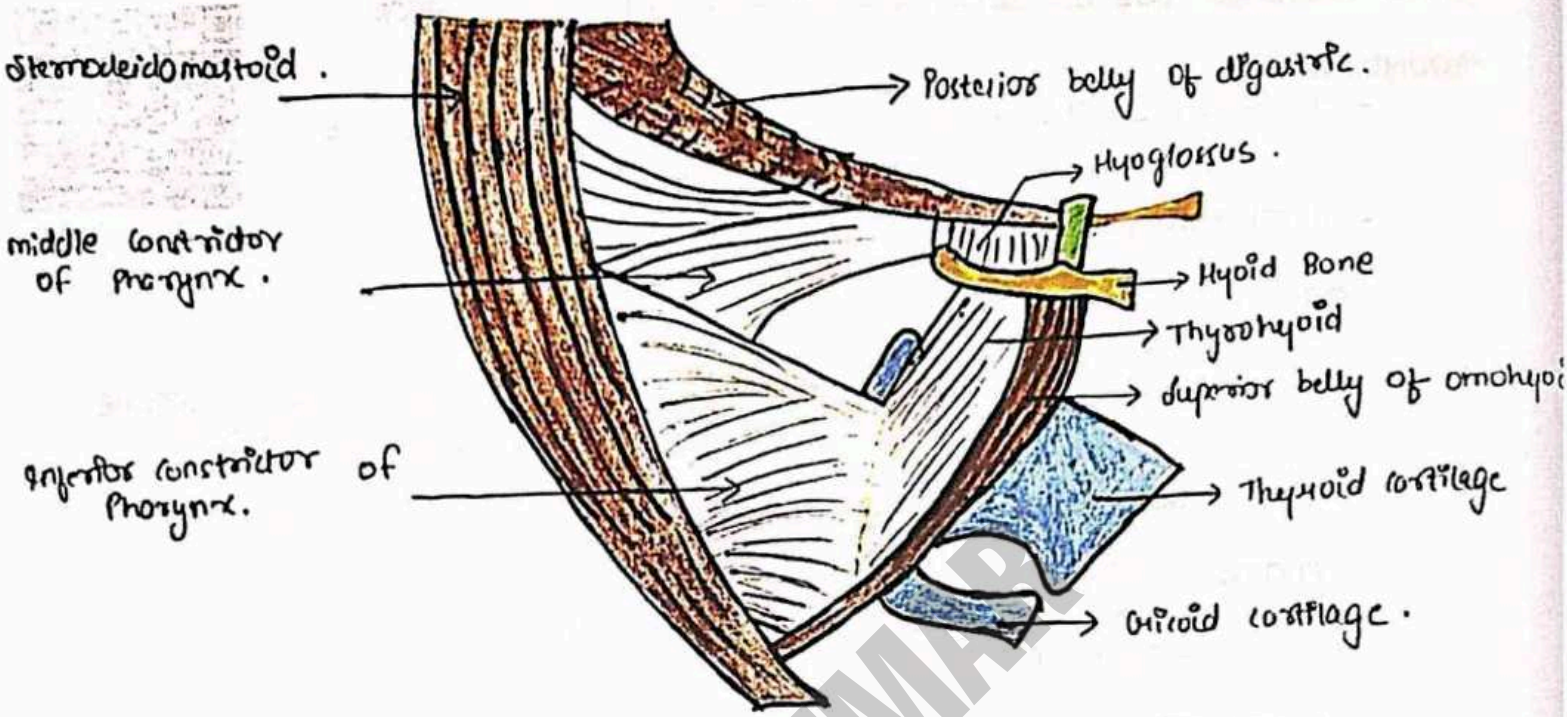
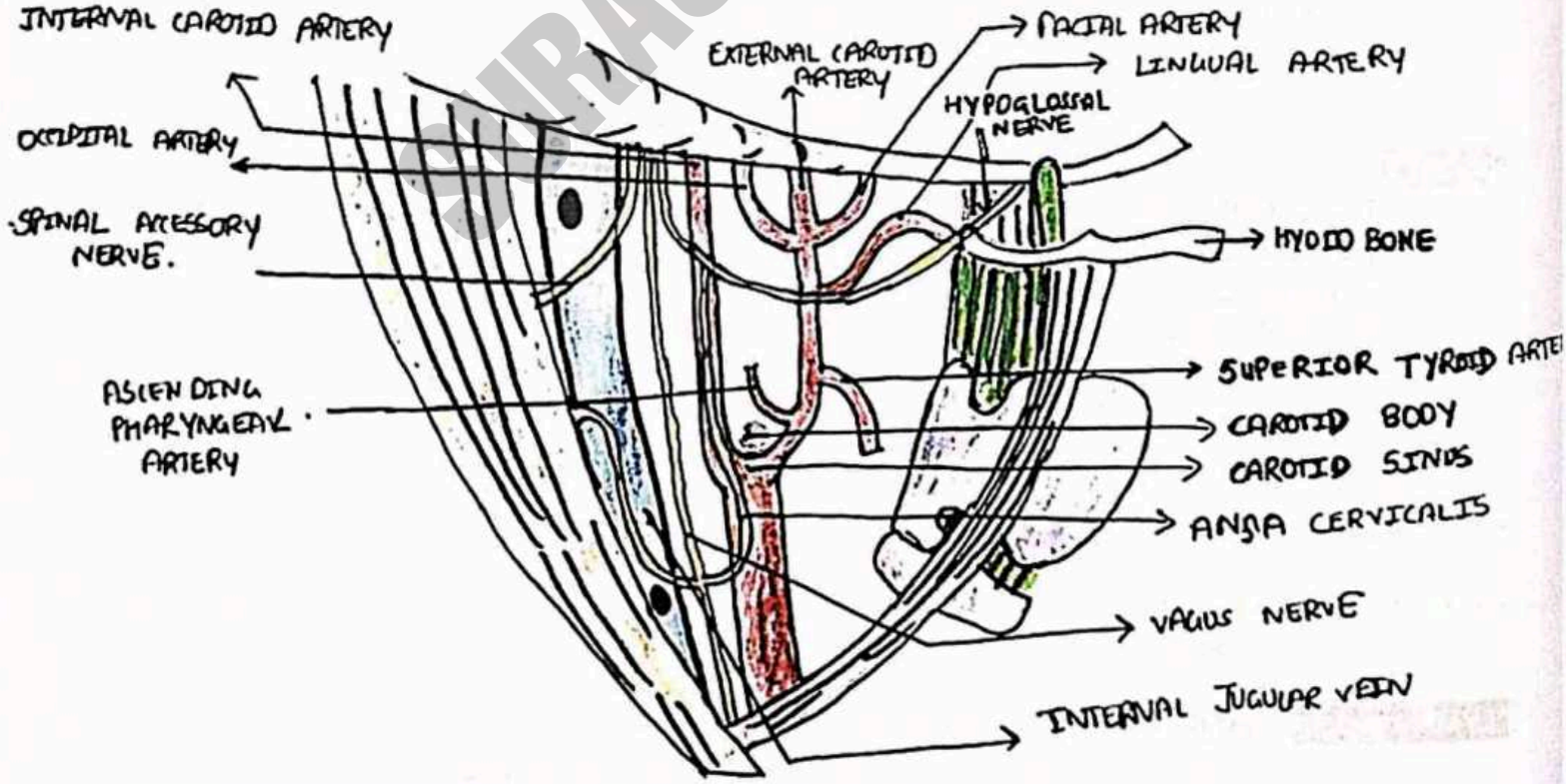


BOUNDARIES OF CAROTID TRIANGLE



CONTENTS OF CAROTID TRIANGLE

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CAROTID SHEATH

-CONDENSATION OF FIBROAREOLAR TISSUE - AROUND MAIN VESSEL OF NECK

*FORMATION: ANTERIOR - PRETRACHEAL FASCIA
POSTERIOR - PREVERTEBRAL FASCIA

*CONTENT-
- INTERNAL CAROTID ARTERIES
- INTERNAL JUGULAR VEIN
- VAGUS NERVE
- IN UPPER PART OF SHEATH - IX, XI, XII NERVE

*RELATION -
- ANTERIOR WALL - ANSA CERVICALIS LIE EMBEDDED
- POSTERIOR WALL - SYMPATHETIC CHAIN LIE BEHIND
- OVERLAPPED BY- STERNOCLEIDOMASTOID MUSCLE
- FUSED TO - DEEP CERVICAL FASCIA

ANSA CERVICALIS

*THIN NERVE LOOP (C1, C2 & C3)
*LIE EMBEDDED - IN ANTERIOR WALL OF CAROTID SHEATH
SUPPLIES - INFRAHYOID MUSCLES-→ THYROHYOID
→ OMOHYOID
→ STERNOHYOID & STERNOHYOID

*FORMATION - FORMED BY - SUPERIOR & INFERIOR ROOT

- SUPERIOR ROOT - FIBRES ARE DERIVED- FROM FIRST CERVICAL NERVE
- CONTINUATION OF DESCENDING BRANCH OF HYPOGLOSSAL NERVE.
- DESCEND OVER - INTERNAL COROTID ARTERY
- COMMON COROTID ARTERY

- INFERIOR ROOT - DERIVED FROM 2ND & 3RD CERVICAL SPINAL NERVE
- DESCENDS - >IT WIND ROUND - INTERNAL JUGULAR VEIN
- CONTINOUS - ANTEROINFERIORLY - TO JOIN SUPERIOR ROOT IN FRONT OF COMMON CAROTID ARTERY

DISTRIBUTION - SUPERIOR ROOT - SUPERIOR BELLY OF OMOHYOID
- INFERIOR ROOT - INFERIOR BELLY OF OMOHYOID
- ANSA CERVICALIS - STERNOTHYROID
- STERNOTHYROID

*THYROHYOID & GENIOHYOID SUPPLIED BY - SEPARATE BRANCHES FROM 1ST CERVICAL NERVE.

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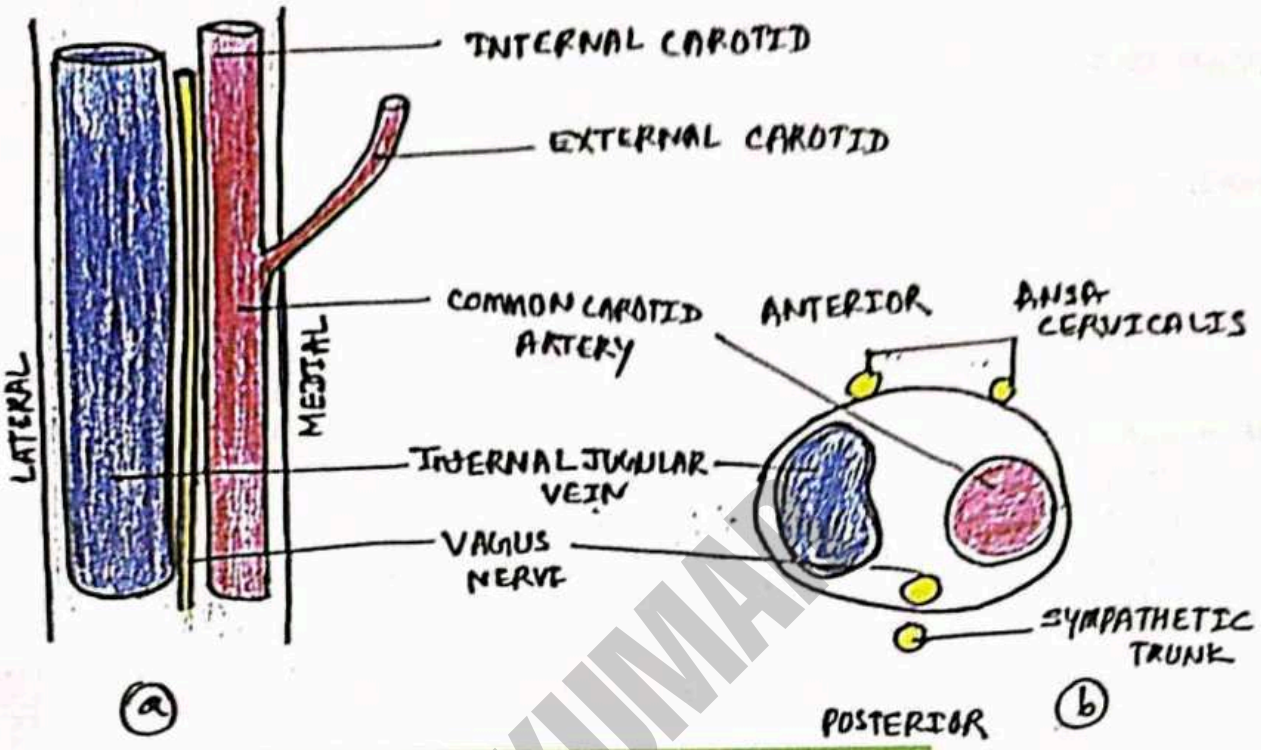
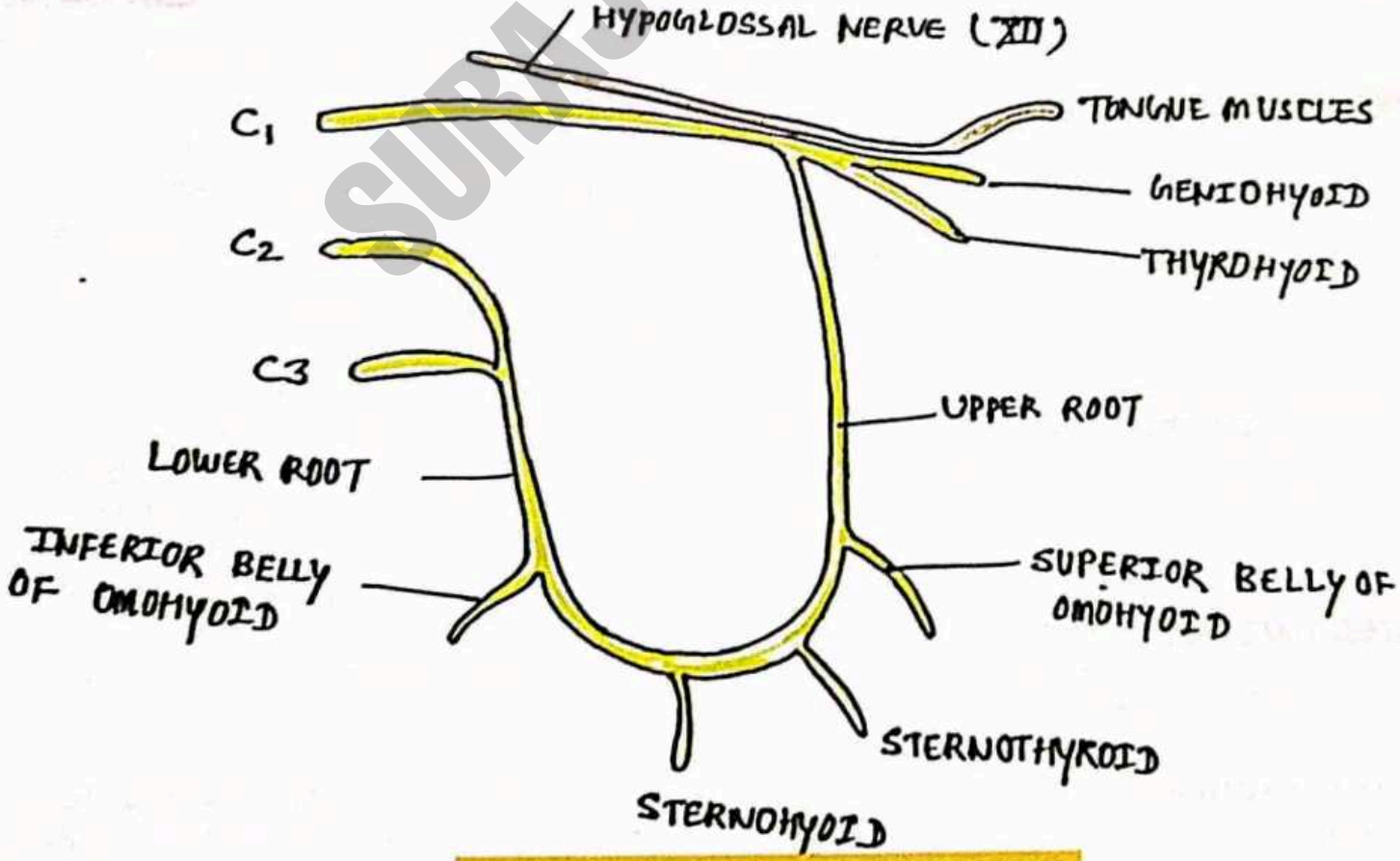


CAROTID SHEATH



ANSA CERVICALIS

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POSTERIOR TRIANGLE

*FEATURES -SPACE ON THE SIDE OF NECK
V- SITUATED BEHIND - STERNOCLEIDOMASTOID MUSCLE

*BOUNDARIES:-
*ANTERIOR - POSTERIOR BORDER OF STERNOCLEIDOMASTOID
*POSTERIOR - ANTERIOR BORDER OF TRAPEZIUS
*BASE - MIDDLE 1/3 OF CLAVICLE
*APEX - LIES ON SUPERIOR NUCHAL LINE
- WHERE TRAPEZIUS AND STERNOCLEIDOMASTOID MEET

*FLOOR - FORMED BY PREVERTEBRAL LAYER OF DEEP CERVICAL FASCIA
- COVERING FOLLOWING MUSCLE
- SPLENIUS CAPITIS
- LEVATOR SCAPULAE
- SCALENUS MEDIUS
- SEMISPINALIS CAPITIS

*ROOF - SKIN
- SUPERFICIAL FASCIA - INVESTING LAYER OF DEEP CERVICAL FASCIA
- CONTAINS - PLATYSMA
-EXTERNAL JUGULAR VEIN
-SUPRACLAVICULAR NERVE
-GREATER AURICULAR
-TRANSVERSE CUTANEOUS
-LESSER OCCIPITAL NERVE
-SUPRASCAPULAR ARTERY
-OCCIPITAL TRANSVERSE CERVICAL ARTERY
-LYMPH VESSEL

*DIVISION OF POSTERIOR TRIANGLE -
- SUB-DIVIDED BY- OMOHYOID MUSCLE
- LARGER UPPER PART - OCCIPITAL TRIANGLE
- SMALL LOWER PART- SUBCLAVIAN TRIANGLE

*CONTENTS -

OCCIPITAL TRIANGLE

1) NERVE
--> SPINAL ACCESSARY NERVE - 4X CUTANEOUS BRANCH
- TRANSVERSE CUTANEOUS NERVE
- SUPRACLAVICULAR NERVE
- GREAT AURICULAR NERVE
- LESSER OCCIPITAL NERVE
-->MUSCULAR BRANCH - 2 LEVATAR SCAPULAE
- 2 TRAPEZIUS
- NERVE TO RHOMBOID

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2) VESSELS - TRANSVERSE CERVICAL ARTERY AND VEIN
- OCCIPITAL ARTERY

3) LYMPH NODE - SUPRA-CLAVICULAR NODE
- OCCIPITAL NODE

SUBCLAVIAN TRIANGLE

1) NERVE -
- ROOT AND TRUNK BRACHIAL PLEXUS
- NERVE TO SUBCLAVIUS
- NERVE TO SERRATUS ANTERIOR
- SUPRASCAPULAR NERVE

2) VESSEL -
- 3RD PART OF SUBCLAVIAN ARTERY AND VEIN
- SUPRASCAPULAR ARTERY
- TRANSVERSE CERVICAL ARTERY
- EXTERNAL JUGULAR VEIN

CLINICAL ANATOMY

ANTTERIOR TRIANGLE

1.THE COMMON ANTERIOR MIDLINE SWELLINGS OF THE NECK ARE:

a.ENLARGED SUBMENTAL LYMPH NODES & SUBLINGUAL DERMOID IN THE SUBMENTAL REGION.
b.THYROGLOSSAL CYST & INFLAMED SUBHYOID BURSA JUST BELOW THE HYOID BONE.
c.GOITRE, CARCINOMA OF LARYNX & ENLARGED LYMPH NODES IN THE SUPRASTERNAL REGION.

POSTERIOR TRIANGLE

*THE MOST COMMON SWELLING IN THE POSTERIOR TRIANGLE IS DUE TO ENLARGEMENT OF THE SUPRACLAVICULAR LYMPH NODES.WHILE DOING BIOPSY OF THE LYMPH NODE, ONE MUST BE CAREFUL IN PRESERVING THE ACCESSORY WHICH MAY GET ENTANGLED AMONGST ENLARGED LYMPH NODES.

CAROTID TRIANGLE

*THE CAROTID SINUS IS RICHLY SUPPLIED BY NERVES.IN SOME PERSONS, THE SINUS MAY BE HYPERSENSITIVE.IN SUCH PERSONS, SUDDEN ROTATION OF THE HEAD MAY CAUSE SLOWING OF HEART.THIS CONDITION IS CALLED "CAROTID SINUS SYNDROME".

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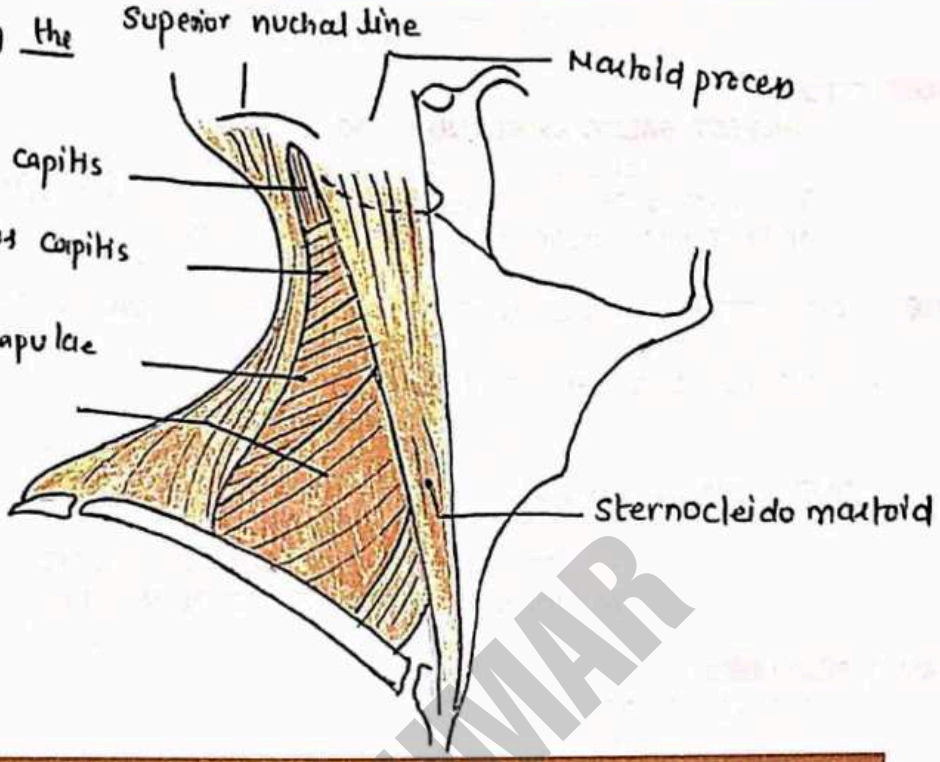
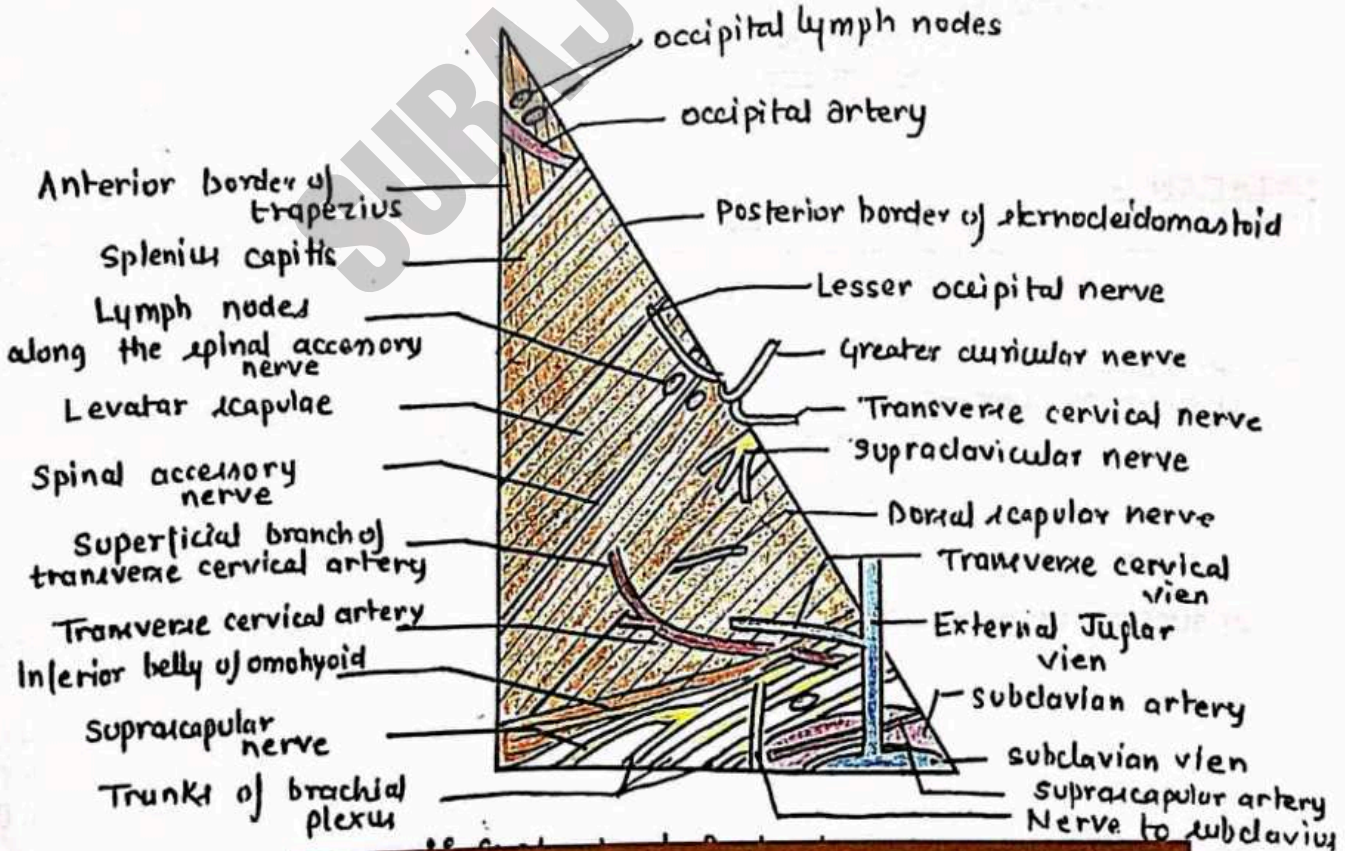
(*) Muscles forming the floor

1) Semispinalis capitis
2) Splenius capitis
3) Levator scapulae
4) Scalenus medius



BOUNDARIES OF POSTERIOR TRIANGLE



CONTENTS OF POSTERIOR TRIANGLE

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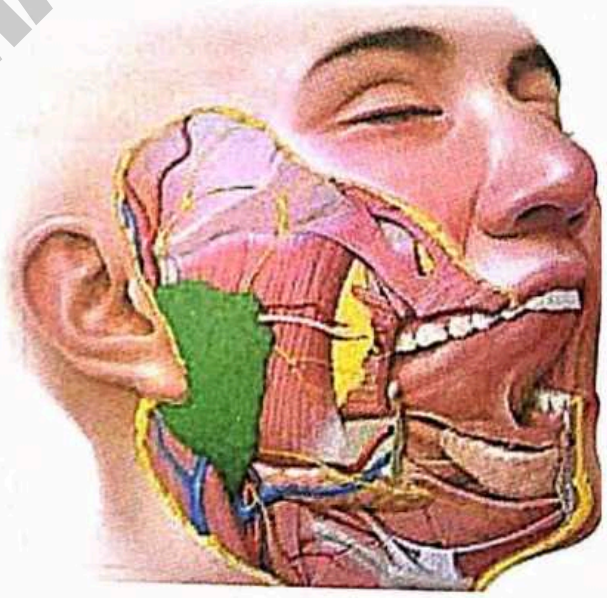
PAROTID GLAND

*INTRODUCTION -
- LARGEST SALIVARY GLAND - 25G
- SITUATED- BELOW EXTERNAL ACOUSTIC MEATUS.
- B/W RAMUS OF MANDIBLE AND STERNOCLEIDOMASTOID
- ANTERIORLY- OVERLAP - MASSETER MUSCLE

CAPSULE .- INVESTING LAYER OF DEEP CERVICAL FASCIA-FORM CAPSULE FOR GLAND
- SUPPLIED-BY AURICULAR NERVE
- FASCIA SPLITS - ENCLOSE GLAND
- SUPERIOR LAMINA- THICK & ADHERENT TO GLAND
  - ATTACHED ABOVE - ZYGOMATIC ARCH
- DEEP LAMINA - THIN
  - ATTACHED - TYMPANI PLATE
  - PORTION OF D.L EXTEND B/W STYLOID PROCESS AND MANDIBLE - IS THICKENED FORM STYLOMANDIBULAR LIGAMENT

*EXTERNAL FEATURES -
- 3 SIDE PYRAMID
- APEX - DOWNWARD
- 4 SURFACE - SUPERIOR
  - SUPERFICIAL
  - ANTEROMEDIAL
  - POSTEROMEDIAL
- -3 BORDER-
  - ANTERIOR
  - POSTERIOR
  - MEDIAL



*RELATION :-
1) APEX.- POSTERIOR BELLY DIGASTRIC
- CERVICAL BRANCH OF FACIAL NERVE
- 2 DIVISION OF RETROMANDIBULAR VEIN

2) SURFACE:-
1) SUPERIOR SURFACE-
- SUPERFICIAL TEMPORAL VESSEL
- AURICULO TEMPORAL NERVE
- EXTERNAL ACOUSTIC MEATUS
- POSTERIOR SURFACE OF TMJ

2) SUPERFICIAL SURFACE → SKIN
- SUPERFICIAL FASCIA
- PAROTID FASCIA
- DEEP PAROTID LYMPH NODE

3) ANTEROMEDIAL SURFACE -

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- RAMUS OF MANDIBLE
- MASSETER
- MEDIAL PTERYGOID
- LATERAL SURFACE OF TMJ

4) POSTEROMEDIAL SURFACE - MASTOID PROCESS
- STYLOID PROCESS

3) BORDER -

1) ANTERIOR BORDER - TERMINAL BRANCH OF FACIAL NERVE
- PAROTID DUCT
- TRANSVERSE FACIAL VESSEL

2) POSTERIOR BORDER SEPARATE — SUPERFICIAL SURFACE TO POSTEROMEDIAL SURFACE

3) MEDIAL BORDER SEPARATE — ANTEROMEDIAL TO POSTEROMEDIAL SURFACE

STRUCTURE WITHIN THE PEROTID GLAND

*ARTERIES - "ECA" ENTER POSTEROMEDIAL SURFACE
- MAXILLARY A. LEAVE —> ANTEROMEDIAL SURFACE
- TFA → ANTERIOR PART - SUPERIOR SURFACE

*VEIN - RETROMANDIBULAR VEIN → FORMED IN GLAND WITH BY STV + MV
- LOWER PART ANTERIOR, POSTERIOR DIVISION —>EMERGE, APEX

*NERVE - FACIAL NERVE - ENTER —> POSTEROMEDIAL SURFACE
- LEAVE —> ANTEROMEDIAL SURFACE

-BRANCH ->
TEMPORAL BRANCH
ZYGOMATIC BRANCH
BUCCAL BRANCH
MARGINAL MANDIBULAR NERVE
CERVICAL BRANCH

*PAROTID DUCT — EMERGS, MIDDLE OF ANTERIOR BORDER
- RUN FORWARD DOWNLOAD ON MASSETER
- PIERCE IN- BUCCINATOR MUSCLE
- FINALLY DUCT OPEN - VESTIBULE OF MOUTH

*BLOOD SUPPLY - ARTERY - EXTERNAL CAROTID ARTERY
VEIN SUPPLY - EJU AND IJU

*NERVE SUPPLY -
1) SENSORY NERVE - AURICULOTEMPORAL NERVE
2) SYMPATHETIC NERVE - VASOMOTAR - PLEXUS AROUND - MIDDLE MENINGEAL ARTERY

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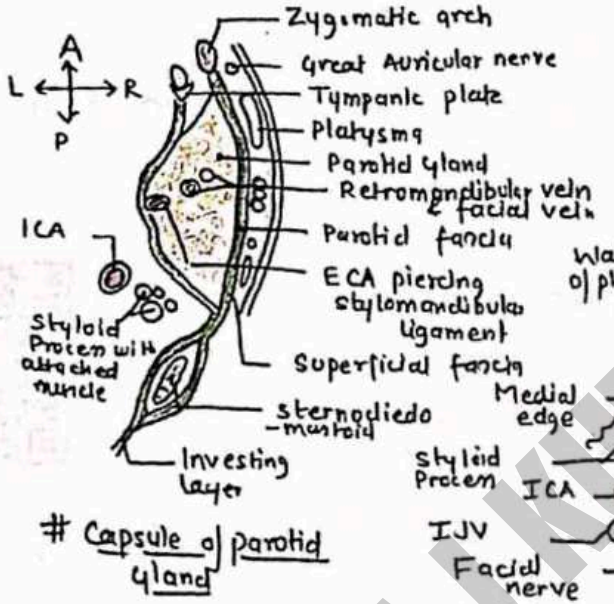
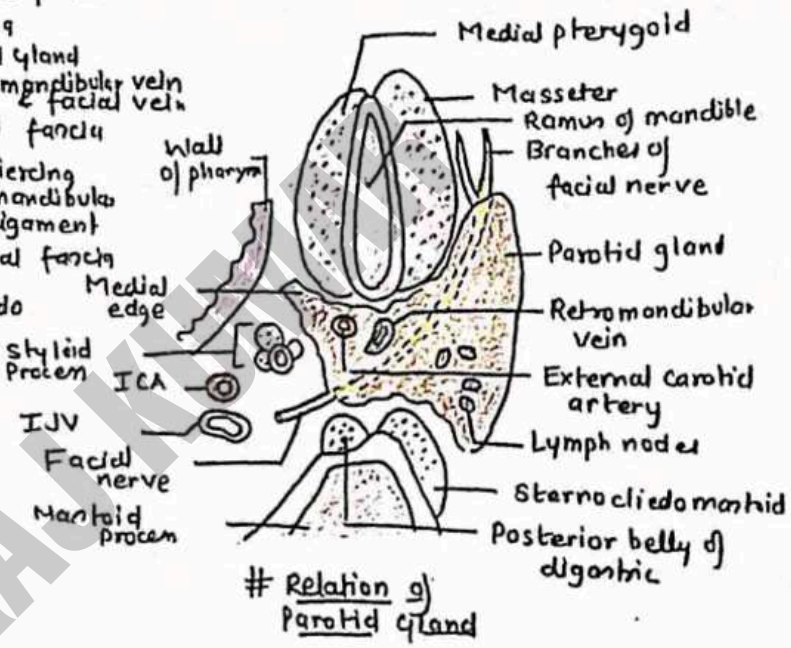
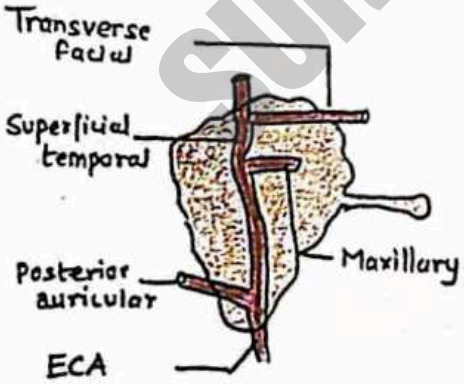
3) PARASYMPATHIC NERVE - SECRETO MOTOR
INFERIOR SALIVATORY NUCLEUS
GLOSSOPHARYNGEAL NERVE
TYMPANIC BRANCH
TYMPANIC NERVE
LESSER PETROSAL NERVE
OTIC GANGLION
POST-GANGLIONIC FIBRE THROUGH AURICULOTEMPORAL NERVE



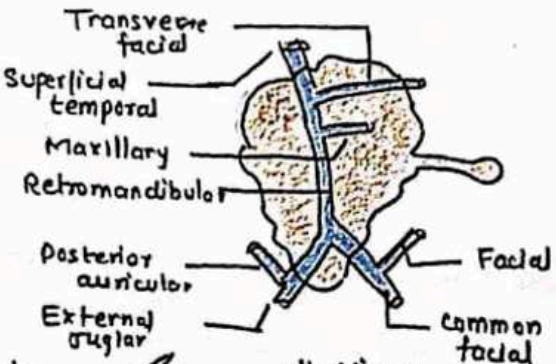
Capsule of parotid gland



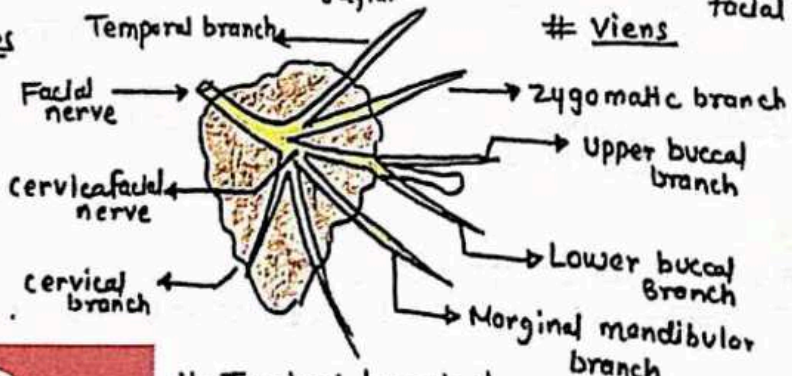
Relation of Parotid gland



Arteries



Viens



Terminal branch of

PAROTID GLAND

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THYROID GLAND

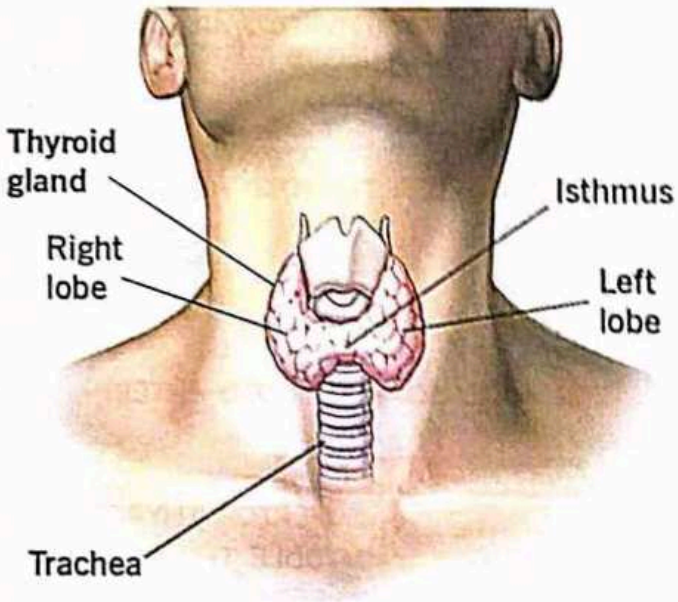
*THYROID GLAND - LARGEST ENDOCRINE GLAND
- LOWER PART OF FRONT AND SIDE OF NECK
- FUNCTION - BASAL METABOLIC RATE
- CALCIUM METABOLISM
- NATURAL IODINE SYNTHESIS

- CONSIST OF - 2 LOBES->RIGHT & LEFT LOBE-> JOINED TO EACH OTHER BY ISTHMUS
- SITUATION - GLAND LIES AGAINST - C5-C7, T1 VERTEBRAE
- EACH LOBE EXTEND→ MIDDLE OF THYROID CARTILAGE TO FIFTH TRACHEAL RING
- ISTHMUS EXTEND→ SECOND TO FOURTH TRACHEAL RING

*CAPSULE OF THYROID -
- TRUE CAPSULE → PERIPHERAL CONDENSATION OF CONNECTIVE TISSUE OF GLAND
- DENSE CAPILLARY PLEXUS IS PRESENT DEEP TO TRUE CAPSULE

- FALSE CAPSULE → DERIVED FROM - PRETRACHEAL LAYER OF DEEP CERVICAL FASCIA
- THIN IN POSTERIOR BORDER OF LOBE
- THICK INNER SURFACE OF GLAND TO FORM SUSPENSARY LIGAMENT (OF BERRY)

*PARTS AND RELATIONS:
1) LOBES - APEX
- BASE
- 3 SURFACES - LATERAL
- MEDIAL
- POSTEROLATERAL
- 2 BORDER - ANTERIOR
- POSTERIOR
2) ISTHMUS - 2 SURFACE - ANTERIOR
- POSTERIOR
- 2 BORDER - SUPERIOR
- INFERIOR



*THYROID GLAND RELATION : -
*LOBES:-
1) APEX- STERNOTHYROID MUSCLE
- SUPERIOR THYROID ARTERY
- EXTERNAL LARYNGEAL NERVE
2) BASE - INFERIOR THYROID ARTERY
- RECURRENT LARYNGEAL NERVE
3) SURFACE:-
1) LATERAL - STERNO THYROID
- STERNO HYOID
- SUPERIOR BELLY OF OMOHYOID
- STERNOCLEIDOMASTOID

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2) MEDIAL - 2X TUBE - TRACHEA
- OESOPHAGUS
-2 MUSCLE - INFERIOR CONSTRICTOR
-CRICOTHYROID
-2 NERVE - EXTERNAL RECURRENT LARYNGEAL NERVE

3) POSTEROLATERAL - CAROTID SHEATH

4) BORDER:-
ANTERIOR BORDER - THIN
- SUPERIOR THYROID ARTERY
POSTERIOR BORDER - THICK
- INFERIOR THYROID ARTERY
- PARATHYROID GLAND
- THORACIC DUCT

*ISTHMUS:-
1) ANTERIOR SURFACE - STERNO THYROID
- STERNO HYOID
2) POSTERIOR SURFACE- 2ND TO 4TH TRACHEAL RINGS
3) UPPER SURFACE- SUPERIOR THYROID ARTERIES
4) LOWER SURFACE - INFERIOR THYROID VEIN

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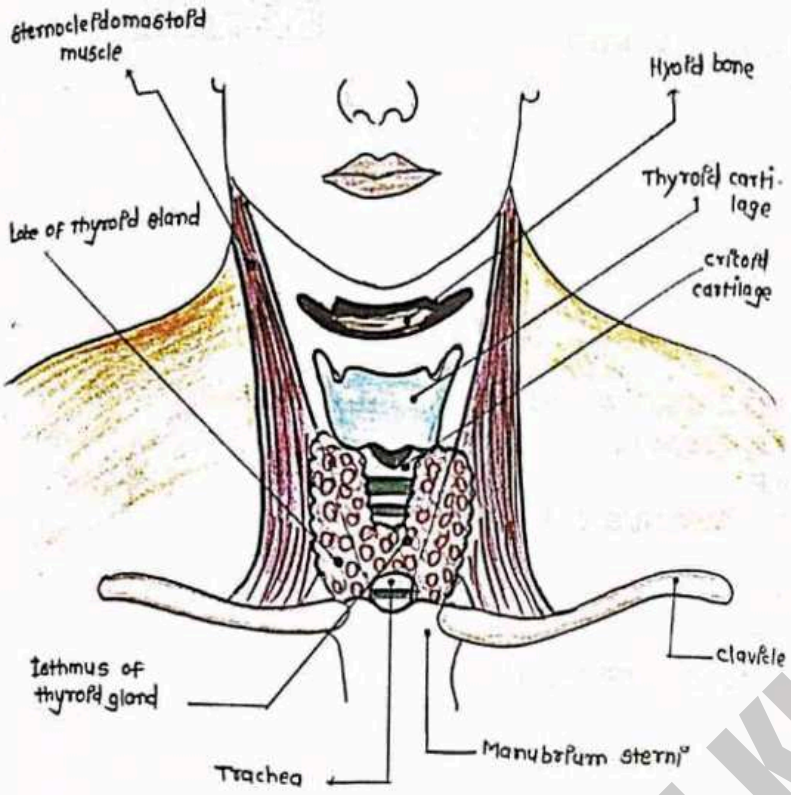
*ARTERIAL SUPPLY :-

1) SUPERIOR THYROID ARTERY - FIRST BRANCH OF ECA
- DOWNWARD, FORWARDS RUN
- RELATION-EXTERNAL LARYNGED NERVE
- APEX OF LOBE
2) INFERIOR THYROID ARTERY - BRANCH OF THYROCERVICAL TRUNK - SUB CLAVIAN ARTERY
- UPWARD, MEDIALLY RUN
- RELATION- RECURRENT LARYNGEAL NERVE
- BASE OF GLAND
3) ACCESSORY THYROID ARTERY - ARISE FROM TRACHEA AND OESOPHAGUS ARTERY

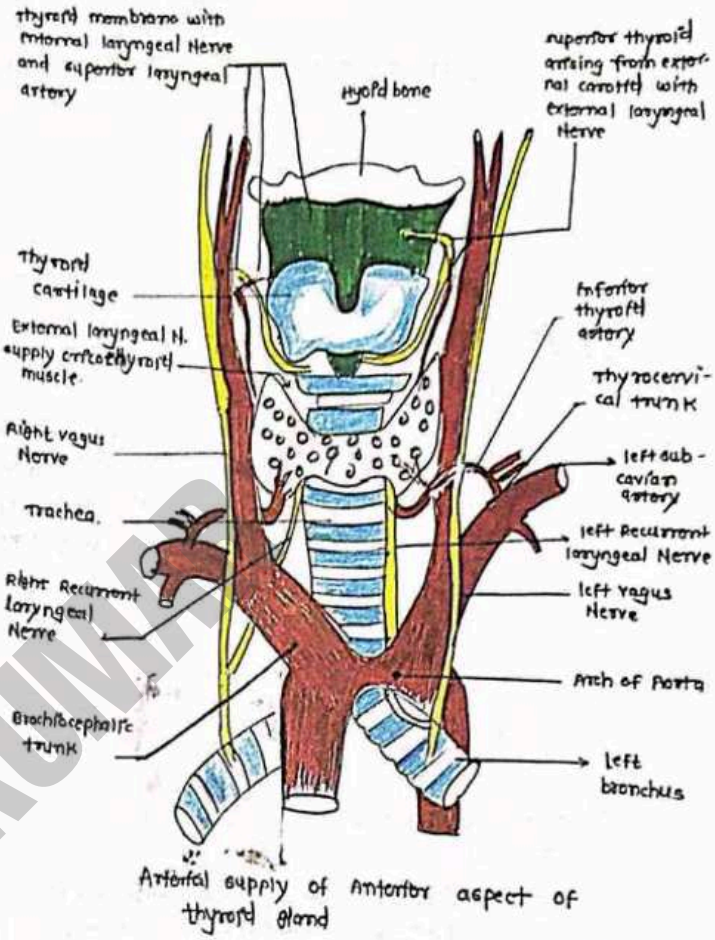
*VENOUS DRAINAGE -
- SUPERIOR THYROID VEIN - DRAIN - INTERNAL JUGULAR VEIN
- MIDDLE THYROID VEIN → INTERNAL JUGULAR VEIN
- INFERIOR THYROID VEIN - LEFT BRACHIOCEPHALIC VEIN

LYMPHATIC :- DEEP CERVICAL LYMPH NODES

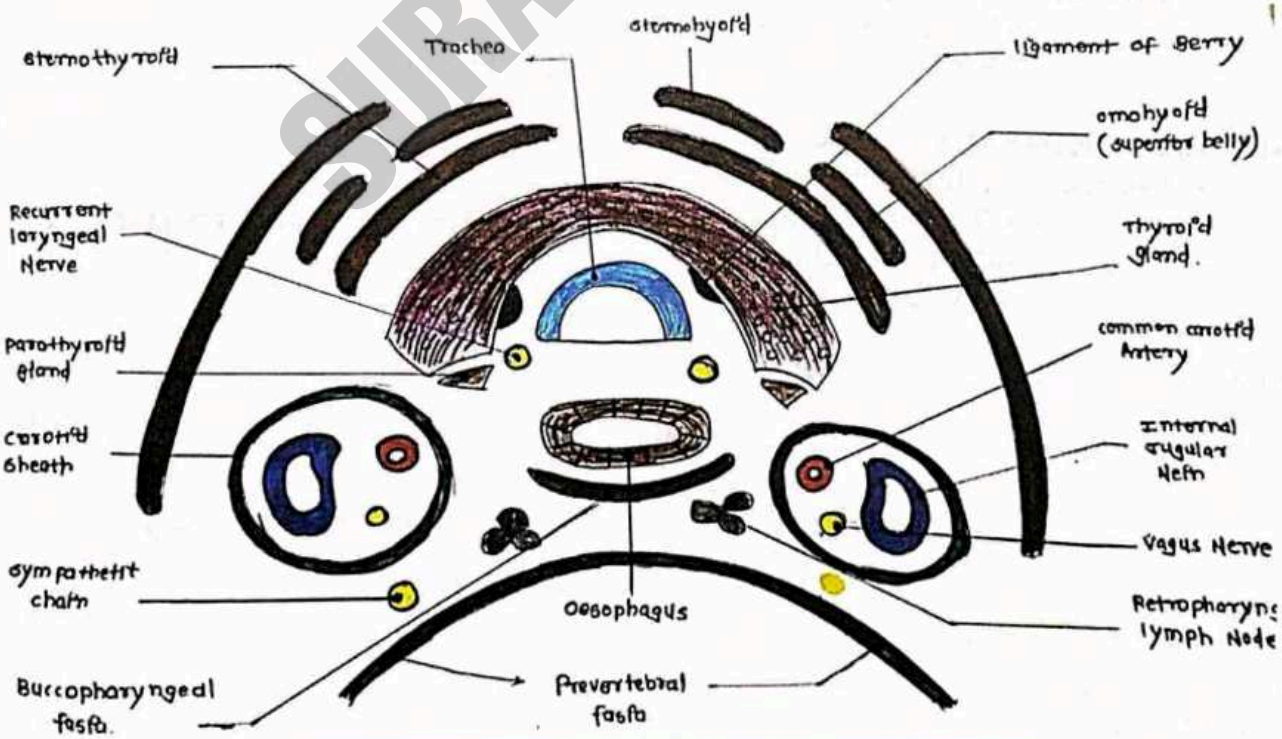
*NERVE - DERIVED FROM- MIDDLE CERVICAL GANGLION
PARTLY FROM - SUPERIOR AND INFERIOR CERVICAL GANGLION



Position of thyroid gland.



Arterial supply of Anterior aspect of thyroid gland



RELATION OF THYROID GLAND

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TEMPAROMANDIBULAR JOINT

TYPE OF JOINT: SYNOVIAL JOINT OF CONDYLAR VARIETY

ARTICULAR SURFACE - UPPER ARTICULAR SURFACE IS FORMED BY - TEMPORAL BONE
- ARTICULAR TUBERCLE
-MANDIBULAR FOSSA
-TYMPANIC PLATE

- INFERIOR ARTICULAR SURFACE → HEAD OF MANDIBLE

LIGAMENTS -

1) FIBROUS CAPSULE -
ATTACH - ABOVE - ARTICULAR TUBERCLE CIRCUMFERENCE OF MANDIBULAR FOSSA
- BEHIND - SQUAMOTYMPANIC FISSURE
- BELOW - NECK OF MANDIBLE
- SYNOVIAL MEMBRANE LINE - FIBROUS CAPSULE

2) LATERAL TEMPOROMANDIBULAR LIGAMENT -
ATTACH - ABOVE - ARTICULAR TUBERCLE
- BELOW - POSTEROLATERAL ASPECT OF NECK OF MANDIBLE
- STRENGTH— LATERAL PART - CAPSULE

3) SPHENOMANDIBULAR LIGAMENT -
ATTACH -
SUPERIORLY - SPINE OF SPHENOID
INFERIARLY - LINGUAL OF MANDIBULAR FORAMEN
-ACCESSARY LIGAMENT
- REMNANT - DORSAL PART OF MECKEL CARTILAGE

4) STYLOMANDIBULAR LIGAMENTS-
ATTACHED - ABOVE- STYLOID PROCESS
- BELOW - ANGLE AND POSTERIOR BORDER OF RAMUS OF MANDIBLE
- ACCESSARY LIGAMENT
- THICKEND - DEEP CERVICAL

5) ARTICULAR DISC-
- OVAL PREDOMINANTLY FIBROUS PLATE JOINT - DIVIDE JOINTS UPPER AND LOWER COMPARTMENT
- COMPOSED OF - ANTERIOR EXTENSION
ANTERIOR BAND - INTERMEDIATE- POSTERIOR BAND BILAMINAR REGION

- DISC REPRESENTS - DEGENERATED PRIMITIVE INSERTION OF LATERAL PTERYGOID
- PREVENTS FRICTION BETWEEN THE ARTICULATING SURFACES
- SHOCK ABSORPTION
- "PROPRIOCEPTIVE FIBRE" HELP REGULATE MOVEMENTS OF JOINT

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RELATION

1) LATERAL - SKIN
- SUPERFICIAL FASCIA
- PAROTID GLAND
- TEMPORAL BRANCH OF FACIAL NERVE

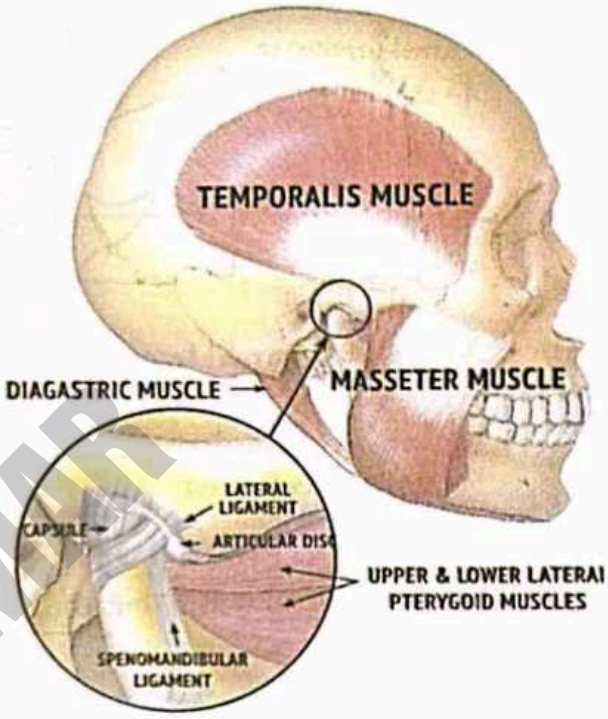
2) MEDIAL - TYMPANIC PLATE
- SPINE OF SPHENOID
- ANTERIOR-LATERAL PTERYGOID
- AURICULO TEMPORAL AND CHORDO-TYMPANI NERVE
- MIDDLE MENINGEAL ARTERY

3) ANTERIOR - LATERAL PTERYGOID
- MASSETERIC NERVE AND ARTERY

4) POSTERIOR - PAROTID GLAND
- SUPERFICIAL TEMPORAL VESSEL
- AURICULO-TEMPORAL NERVE

5) SUPERIOR- MIDDLE CRANIAL FOSSA

6) INFERIOR - MAXILLARY ARTERY



***BLOOD SUPPLY** - SUPERFICIAL TEMPORAL ARTERY AND VEIN
- MAXILLARY ARTERIES AND VEIN

***NERVE SUPPLY** - AURICULO-TEMPORAL NERVE
- MASSETERIC NERUE

***MOVEMENTS -**

1) DEPRESSION COPEN MOUTH) - LATERAL PTERYGOID
-DIGASTRIC AND GENIOHYOID
2) ELEVATION (CLASE MOUTH) → MASSETER AND MEDIAL PTERYGOID
3) PROTRUSION - MEDIAL & LATERAL PTERYGOID.
4) RETRUSION - TEMPORALIS & MASSETER
5) SIDE TO SIDE MOVEMENT DURING CHEWING

CLINICAL ANATOMY

DISLOCATION OF MANDIBLE : DURING EXCESSIVE OPENING OF THE MOUTH, THE HEAD OF THE MANDIBLE OF ONE OR BOTH SIDES MAY SLIP ANTERIORLY INTO THE INFRATEMPORAL FOSSA, AS A RESULT OF WHICH THERE IS INABILITY TO CLOSE THE MOUTH.

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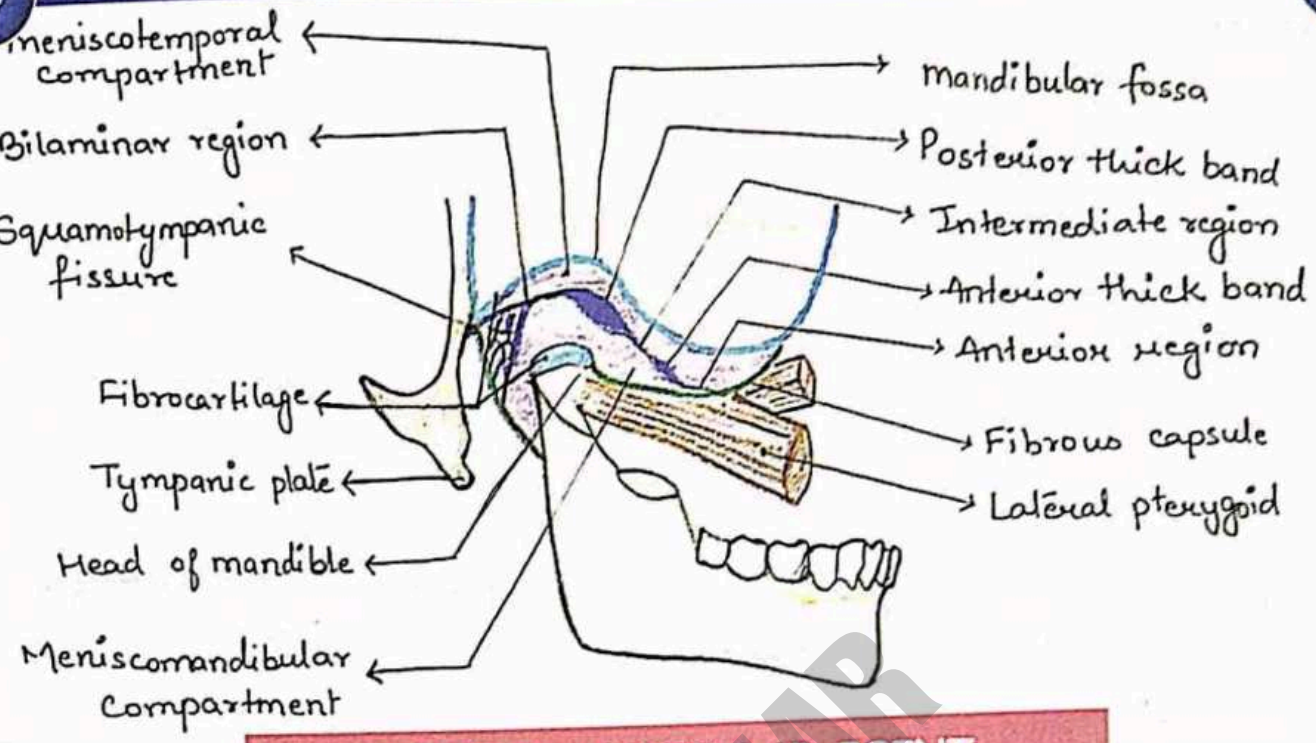
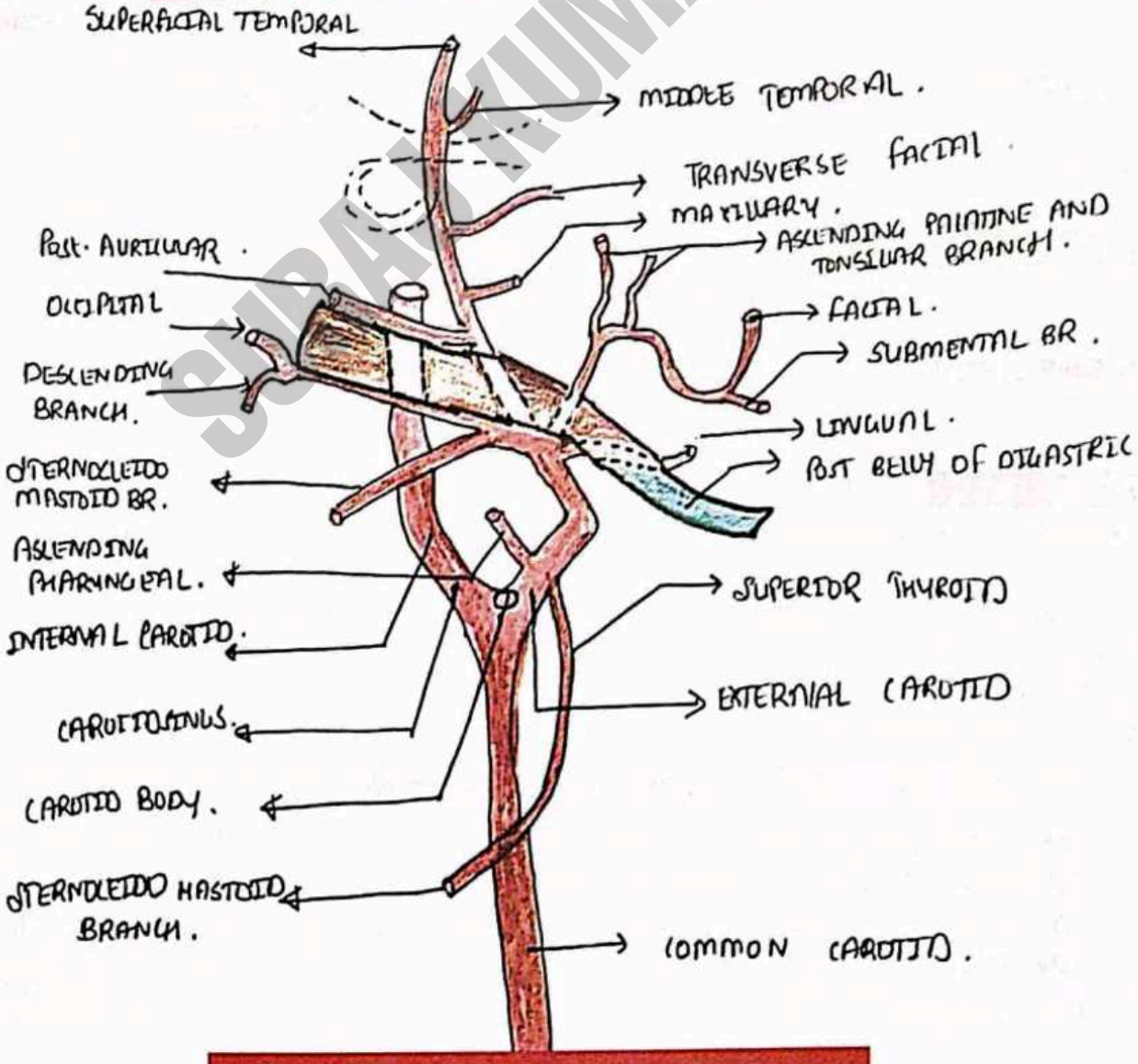


TEMEROMANDIBULAR JOINT



EXTERNAL CAROTID ARTERY

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EXTERNAL CAROTID ARTERY

*ECA IS TERMINAL BRANCH OF COMMON CAROTID ARTERY
*CHIEF ARTERY OF SUPPLY --→ FRONT OF NECK & FACE

*COURSE :-
*BEGIN - IN CAROTID TRIANGLE AT LEVEL OF UPPER BORDER OF THYROID CARTILAGE
- OPPOSITE TO C3 & C4
- RUN UPWARD , BACKWARD AN LATERALLY

*TERMINATE - BEHIND NECK OF MANDIBLE
- DIVIDING - MAXILLARY AND SUPERFICIAL TEMPORAL ARTERY
- SLIGHLY CURVED COURSE - LOWER ANTEROMEDIAL - TO ICA
- UPPER ANTEROLATERAL TO ICA
- IN CAROTID TRIANGLE - IS SUPERFICIAL
- LIE UPPER COVER OF ANTERIOR BORDER OF STERNOCLEIDOMASTOID MUSCLE
- ABOVE CAROTID - LIE - DEEP IN PAROTID GLAND
- RELATED - RETROMANDIBULAR VEIN
- & FACIAL NERVE

*BRANCHES - ANTERIOR - SUPERIOR THYROID A.
- LINGUAL A.
- FACIAL A
- POSTERIOR - OCCIPITAL
- POSTERIOR AUERICULAR A.
- MEDIAL - ASCENDING PHARYNGEAL A.
- TERMINAL - MAXILLARY ARTERY
- SUPERFICIAL TEMPORAL A.

1) SUPERIOR THYROID A. - BRANCH OF ECA - JUST BELOW GERATER CORNUA OF HYOID BONE
-RUN - DOWNWARD, FORWARD PARALLEL TO LATERAL LOBE OF THYROID
- SUPPLY - THYROID GLAND - TERMINAL BRANCH - SUPERIOR LARYNGEAL ARTERY

2) LINGUAL A - ARISE - OPPOSITE TIP OF GREATER CORNUA OF HYOID BONE
- BRANCH - SUPRAHYOID, DOSRAL LINGUAL, SUB- LINGUAL

3) FACIAL A.- SAME AS FACIAL NOTES

4) OCCIPITAL A - OPPOSITE ORIGIN OF FACIAL ARTERY
- CROSSED AT ITS ORIGIN HYPOGLOSSAL NERVE

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5) POSTERIOR ARTICULAR ARTERY - ABOVE POSTERIOR BELLY OF DIGASTRIC
- UPWARD BACKWARD→DEEP PAROTID GLAND & STYLOID PROCESS
- SUPPLY→SKIN-→OVER MASTOID PROCESS & BACK OF SCALP

6) ASCENDING PHARYNGEAL ARTERY - SMALLEST BRANCH
- ARISES→MEDIAL SIDE OF ECA
- VERTICALLY UPWARD-→BETWEEN SIDE WALL OF PHARYNX TONSIL & MEDIAL WALL OF MIDDLE EAR

7) MAXILLARY ARTERY - LARGEST BRANCH
- BEHIND NECK OF MANDIBLE UNDER PAROTID GLAND
- BELOW AURICULO-TEMPORAL NERVE

8) SUPERFICIAL TEMPORAL ARTERY - SMALLER TERMINAL BRANCH
- BEHIND NECK OF MANDIBLE
- DIVIDED→ANTERIOR & POSTERIOR BRANCH →SUPPLY TEMPLE & SCALP

FACIAL ARTERY

*CHIEF ARTERY OF FACE
*BRANCH -->OF EXTERNAL CAROTID ARTERY-->GIVE OFF IN CAROTID TRIANGLE-->JUST ABOVE GREATER CORNUA OF HYOID BONE.

*COURSE -

ENTER FACE→WINDING AROUND MANDIBLE-→AND PIERCING→DEEP CERVICAL FASCIA
-FIRST RUN UPWARD FORWARD→1.25CM LATERAL TO ANGLE OF MOUTH
-ASCENDS BY SIDE OF NOSE-→UP TO ANGLE OF EYE
-TERMINATE BY SUPPLYING LACRIMAL SAC →BY ANASTOMOSING WITH DORSAL NASAL BRANCH
-LIE BETWEEN SUPERFICIAL & DEEP MUSCLE OF FACE.

*BRANCHES - ANTERIOR BRANCH - INFERIOR LABIAL
- SUPERIOR LABIAL
- LATERAL NASAL

- POSTERIOR BRANCH - SMALL AND UNNAMED

*ANASTOMOSES - LARGE ANTERIOR→ANASTOMOSES SIMILAR BRANCH OF OPPOSITE SIDE
-SMALLER POSTERIOR-->TRANSVERSE FACIAL INFRA ORBITAL ARTERIES
-TERMINAL BRANCH OF FACIAL ARTERY-→OPHTHALAMIC ARTERY.

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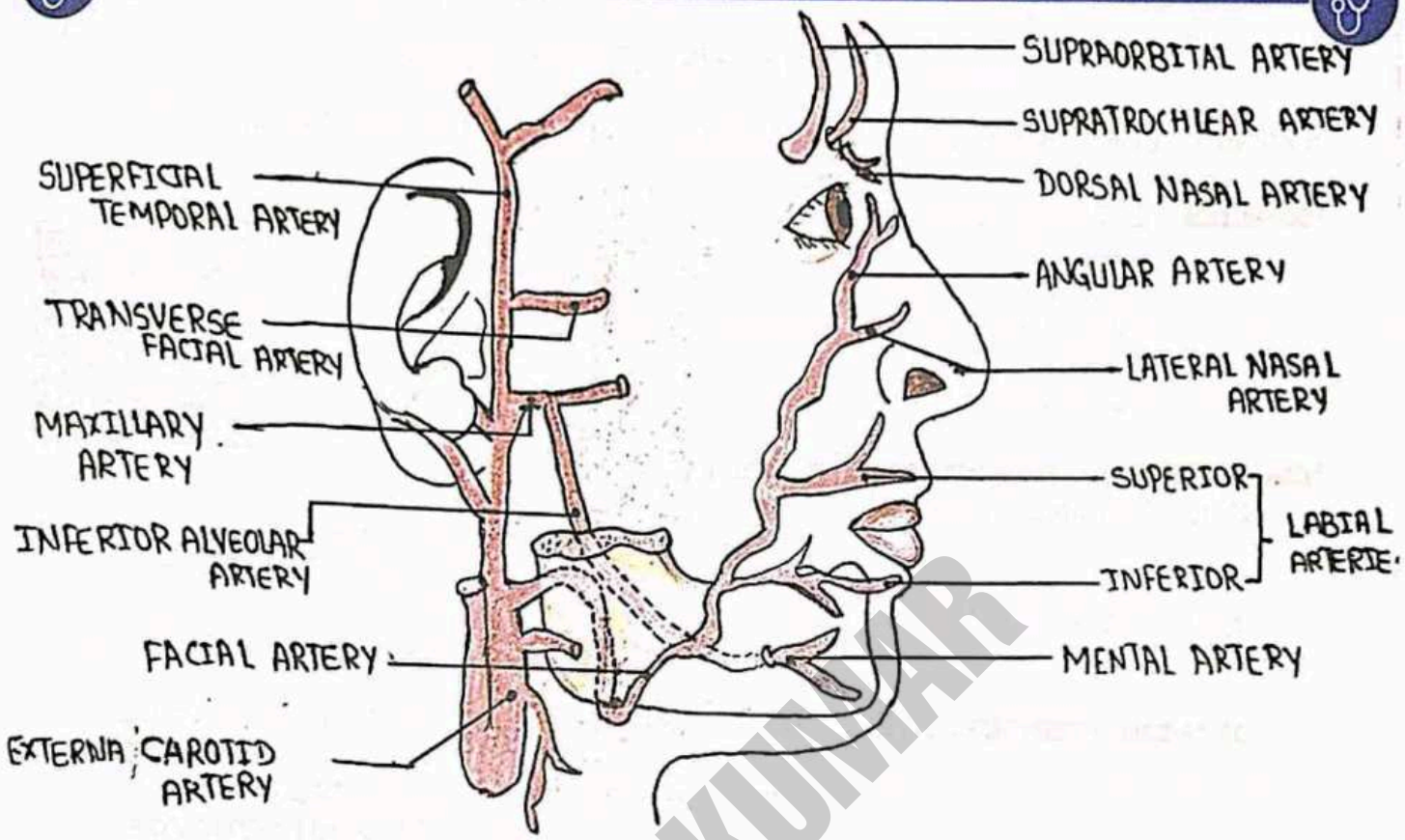
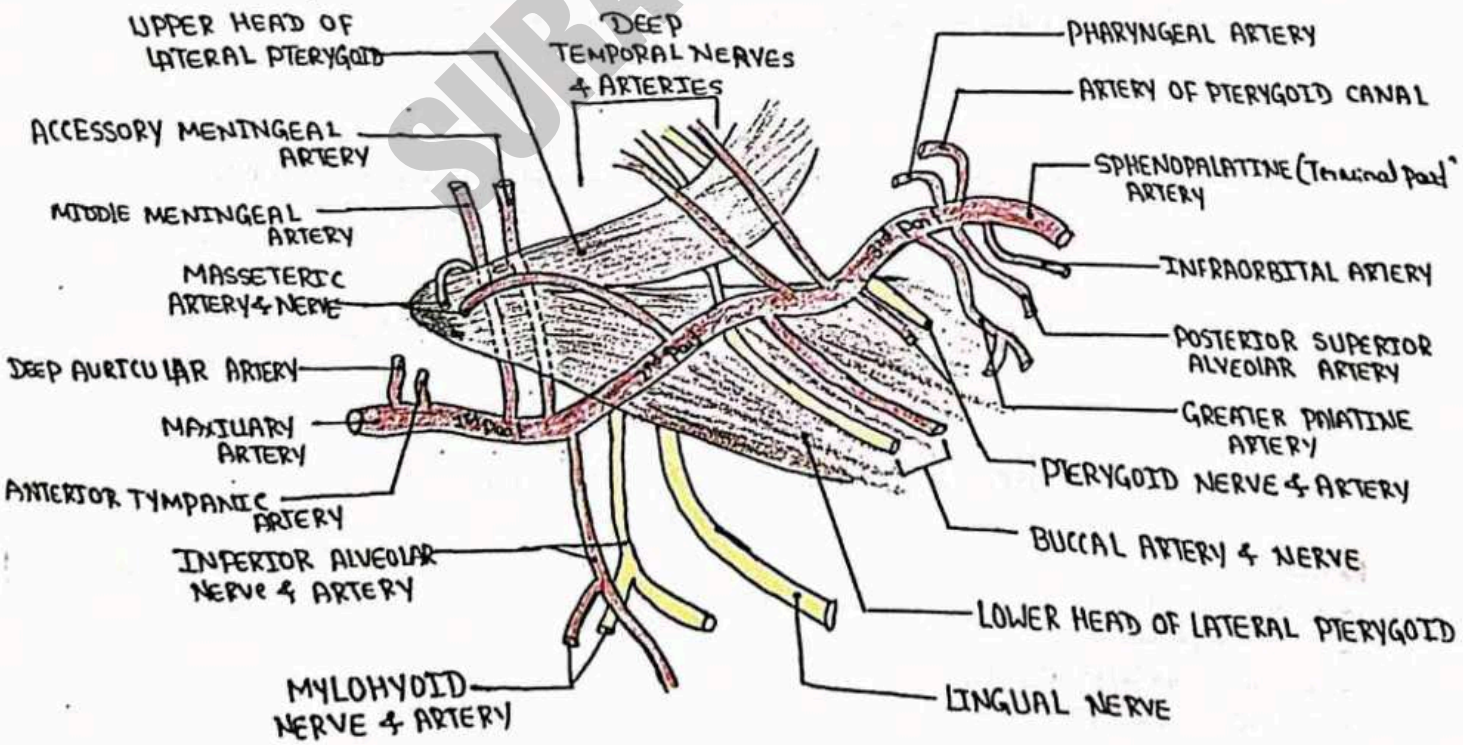


FACIAL ARTERY



MAXILLARY ARTERY

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MAXILLARY ARTERY

*LARGEST TERMINAL BRANCH OF ECA→GIVEN OFF BEHIND NECK OF MANDIBLE

*SUPPLIES - EXTERNAL & MIDDLE EAR
- DURA MATER
- MUSCLE OF TEMPORAL & INFRA TEMPORAL REGION
- NOSE & PARA-NASAL AIR SINUSES
- PALATE
- ROOF OF PHARYNX

*COURSE & RELATION - DIVIDED INTO 3x PARTS -

1) FIRST (MANDIBULAR) PART - RUN HORIZONTALLY FORWARDS
-BETWEEN NECK OF MANDIBLE & SPLENO MANDIBULAR LIGAMENT

2) SECOND (PTERYGOID) PART - RUN UPWARD & FORWARDS
-SUPERFICIAL TO LOWER HEAD OF LATERAL PTERYGOID

3) THIRD (PTERYGOPALATINE) PART - PASSES BETWEEN TWO HEAD OF LATERAL PTERYGOID
-THROUGH PTERYGOMAXILLARY FISSURE→TO
-→ENTER→PTERYGOPALATINE FOSSA

*BRANCHES -PART 1 1) DEEP AURICULAR ARTERY-→SUPPLY EXTERNAL ACOUSTIC MEATUS TYMPANIC MEMBRANE

2) ANTERIOR TYMPANIC→SUPPLY MIDDLE EAR MEDIAL SURFACE OF TYMPANIC MEMBRANE

3) MIDDLE MENINGEAL ARTERY→ENTER FORAMEN SPINOSUM
→REACH MIDDLE CRANIAL FOSSA
-→SUPPLY LARGE FRONTAL BRANCH & SMALL PARIETAL

4) ACCESSARY MENIGEAL ARTERY-→FORAMEN OVALE-→REACH MIDDLE CRANIAL FOSSA
-→SUPPLY INFRA TEMPORAL FOSSA

5) INFERIOR ALVEOLAR ARTERY -→MANDIBULAR FORAMEN
-→SUPPLY MANDIBLE & ROOT OF TOOTH

PART2
- 1) MASSETERIC
- 2) DEEP TEMPORAL
- 3) TO PTERYGOID MUSCLE
- 4) BUCCAL BRANCH

PART-3
1) POSTERIOR SUPERIOR ALVEOLAR ARTERY - ENTER - PTERYGOMAXILLARY FISSURE

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- SUPPLY - MOLAR AND PREMOLAR TEETH
- MAXILLARY AIR SINUS

2) INFRA-ORBITAL ARTERY - ENTER - PTERYGOMAXILLARY FISSURE ->INFERIOR - ORBITAL FISSURE --> ENTER ---> ORBITAL
-SUPPLY - ORBITAL BRANCHES
- MIDDLE SUPERIOR ALEVOLAR BRANCH
- ANTERIOR SUPERIOR ALVEOLAR BRANCH

3) GREATER PALATINE A.- ENTER - GREATER PALATINE FORAMEN ->GREATER PALATINE CANAL
-SUPPLY -> TERMINAL BRANCH ENTER NASAL CAVITY

4) PHARYNGEAL BRANCH - ENTER - PHARYNGEAL CANAL
- SUPPLY - NASOPHARYNX & AUDITORY TUBE

5) ANTERIOR ATERY OF PTERYGOID CANAL - PTERYGOID CANAL
- SUPPLY -> PHARYNX & TYMPANI CAVITY

6) SPHENOPALATINE ARTERY -> SPHENOPALATINE FORAMEN
- POSTEROLATERAL & POSTEROMEDIAL - NASAL BRANCH

MANDIBULAR NERVE

- LARGEST MIXED BRANCH OF TRIGEMINAL NERVE
- OTIC AND SUB-MANDIBULAR GANGLIA - ASSOCIATED WITH THIS NERVE

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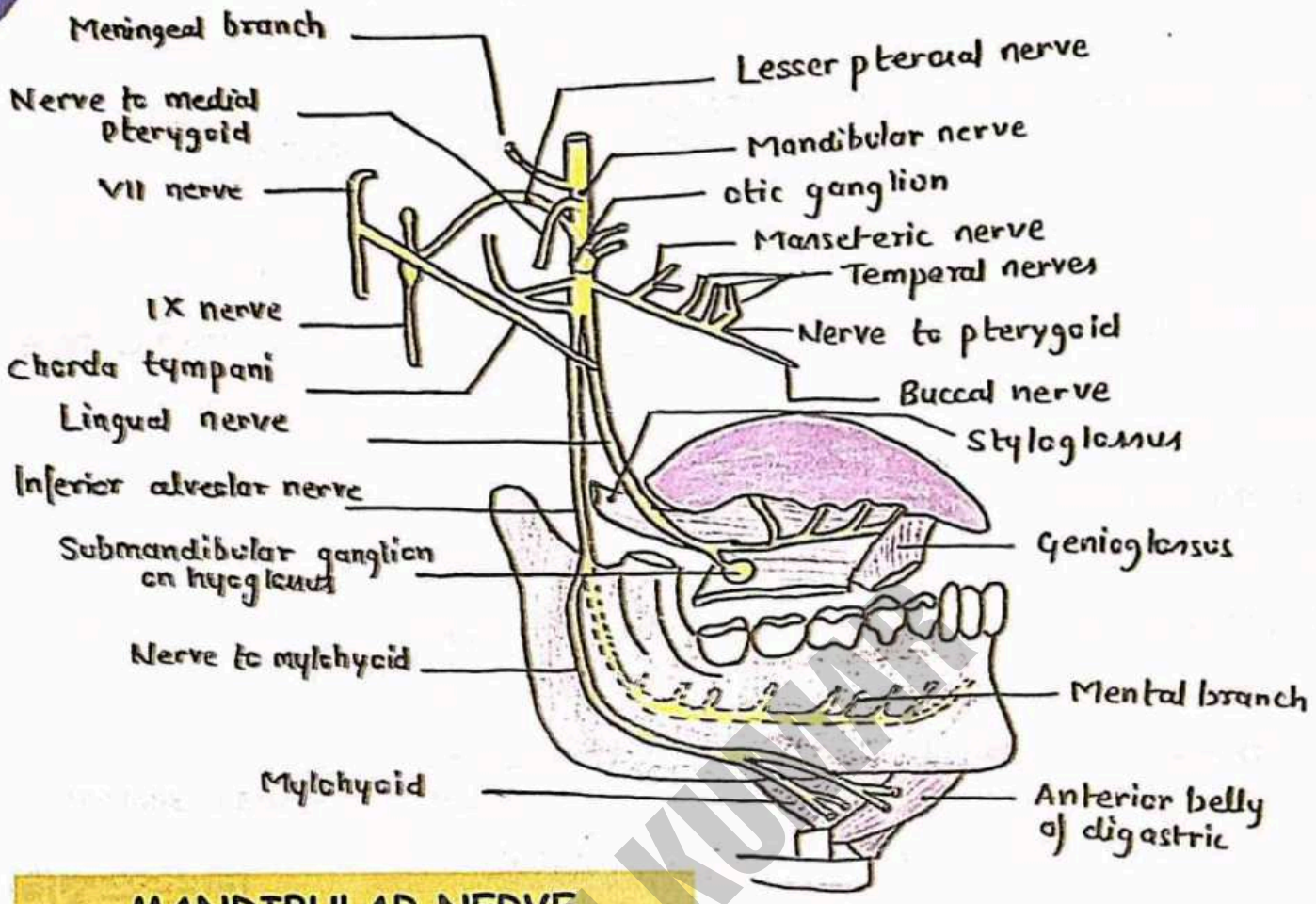
*COURSE-

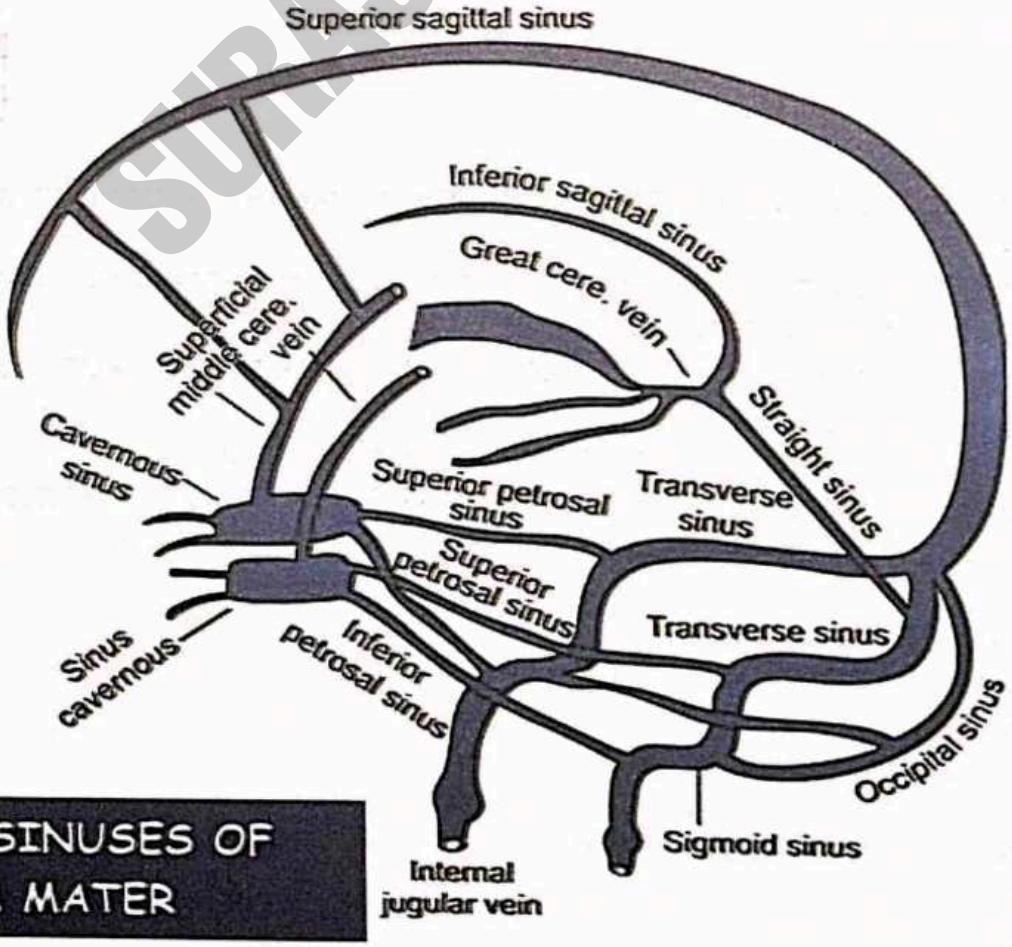
1. BEGIN IN MIDDLE CRANIAL FOSSA THROUGH, LARGE SENSORY ROOT & SMALL MOTOR ROOT
2. LARGE SENSARY ROOT ARISE FROM LATERAL PART OF TRIGEMINAL GANGLION LEAVE CRANIAL CAVITY FORAMEN OVALE
3. SMALL MOTOR ROOT LIE DEEP TO TRIGEMINAL GANGLION - LEAVE - CRANIAL CAVITY ----> FORAMEN OVALE
4. BOTH ROOT JOIN BELOW FORAMEN FORM - MAIN TRUNK
5. MAIN TRUNK IN INFRATEMPORAL FOSSA DEEP TO LATERAL PTERYGOID
6. MAIN TRUNK DIVIDED -> ANTERIOR TRUNK, POSTERIOR TRUNK

*BRANCHES :-

1.MAIN TRUNK - MENINGEAL BRANCH , NERVE TO MEDIAL PTERYGOID
2.ANTERIOR TRUNK - SENSORY BRANCH - BUCCAL NERVE
-MOTOR BRANCH -MASSETERIC, DEEP TEMPORAL NERVE
-NERVE TO LATERAL PTERYGOID
3.POSTERIOR TRUNK - AURICULO TEMPORAL NERVE ,LINGUAL, INFERIOR ALVEOLAR NERVE



MANDIBULAR NERVE



VENOUS SINUSES OF DURA MATER

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VENOUS SINUSES OF DURA MATER

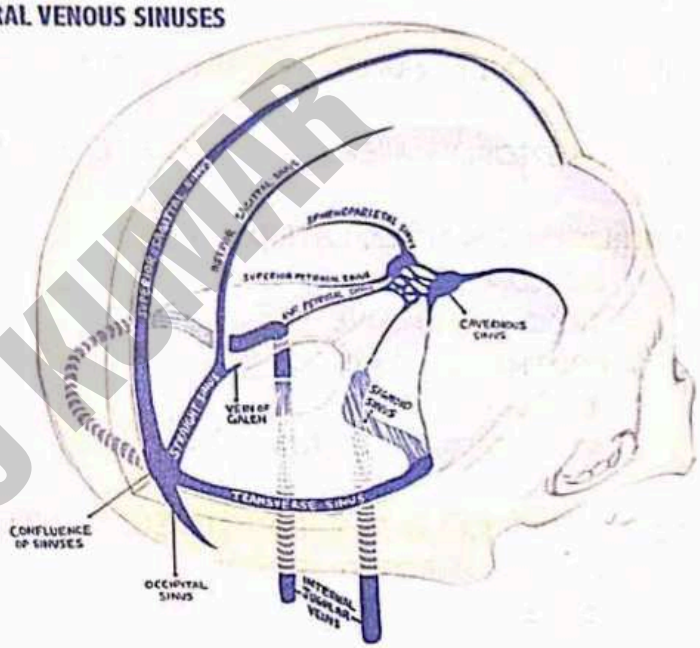
1) VENOUS SPACE IN - WALL WHICH ARE FORMED BY -DURA MATER
   I, NO MUSCLE IN THEIR WALL
   -THEY HAVE NO VALUE

2) VENOUS SINUSES RECEIVE → VENOUS BLOOD FROM BRAIN , MENINGS BONE OF SKULL
3) CRANIAL VENOUS SINUSES COMMUNICATED → WITH OUTSIDE SKULL THROUGH EMISSARY VEINS
4) COMMUNICATIONS HELP TO KEEP THE PRESSURE OF BLOOD CONSTANT
5) 23 VENOUS SINUSES --→ 8 PAIRED & 7 UNPAIRED

*PAIRED VENOUS SINUSES :-

- CAVERNOUS SINUS
- SUPERIOR PETROSAL SINUS
- INFERIOR PETROSAL SINUS
- TRANSVERSE SINUS
- SIGMOID SINUS
- SPHENO-PARIETAL SINUS
- PETRO-SQUAMOUS SINUS
- MIDDLE MENINGEAL SINUS

DURAL VENOUS SINUSES



*UNPAIRED VENOUS SINUSES:

- SUPERIOR SAGITAL SINUS
- INFERIOR SAGITAL SINUS
- STRAIGHT SINUS
- OCCIPITAL SINUS
- ANTERIOR INTER CAVERNOUS SINUS
- POSTERIOR INTER CAVERNOUS SINUS
- PLEXUS OF VEIN

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CAVERNOUS SINUS

- LARGE VENOUS SPACE- SITUATED IN MIDDLE CRANIAL FOSSA EITHER SIDE, BODY OF SPHENOID

* FLOOR & MEDIAL WALL → ENDOSTEAL DURA MATTER
* ROOF & LATERAL WALL → MENINGEAL DURA MATTER

- EXTEND - ANTERIORILY - SUPERIOR ORBITAL FISSURE
  - POSTERIORLY - APEX OF PETROUS TEMPORAL BONE

*RELATION -

1) SUPERIORLY- OPTIC TRACT
- OPTIC CHIASMA
- OLFACTORY TRACT
- INTERNAL CAROTID ARTERY

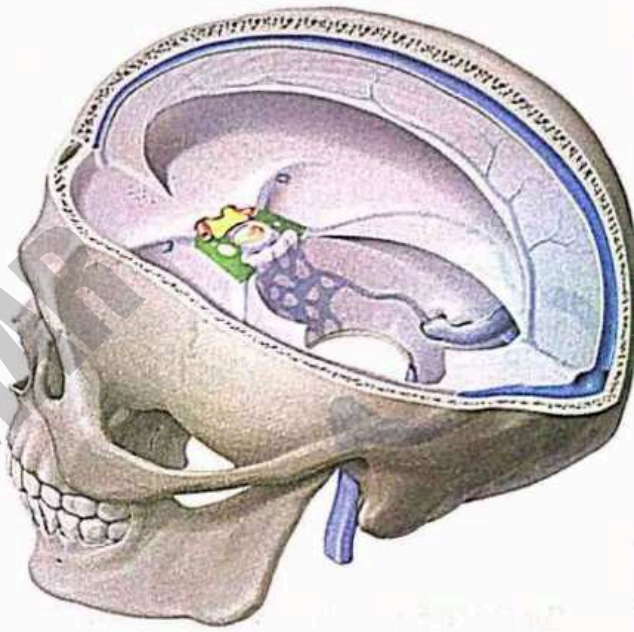
2) INFERIORLY - FORAMEN LACERUM
- GREATER WING OF SPHENOID BONE

3) MEDIALLY - HYPOPHYSIS CEREBRI
- SPHENOIDAL AIR SINUS

4) LATERALLY- TEMPORAL LOBE

5) ANTERIORLY- SUPERIOR ORBITAL FISSURE

6) POSTERIORLY- APEX OF PETROUS TEMPORAL



*STRUCTURE WITHIN LATERAL WALL:
- OCULOMOTOR NERVE
- TROCHLEAR NERVE
- OPHTHALMIC NERVE
- MAXILLARY NERVE
-TRIGEMINAL GANGLION

*STRUCTURE PASSING THROUGH MEDIAL ASPECT OF SINUS :
- INTERNAL CAROTID ARTERY
- ABDUCENT NERVER

*TRIBUTARIES OR INCOMING CHANNEL :
1) FROM ORBIT - SUPERIOR OPHTHALMIC VEIN
- INFERIOR OPHTHALMIC VEIN
- CENTRAL VEIN OF RETINA

2) FROM BRAIN - SUPERFICIAL MIDDLE CEREBRAL VEIN
- INFERIOR CEREBRAL VEIN

3) FROM MENINGES - SPHENO-PARIETAL SINUS
- MIDDLE MENINGEAL WEIN

*DRAINING CHANNEL -
-TRANSVERSE SINUS - THROUGH--→ SUPERIOR PETROSAL SINUS
- INTERNAL JUGULAR VEIN → INTERNAL CAROTID ARTERY
- PTERYGOID PLEXUS - EMISSARY VEIN
- FACIAL VEIN - SUPERIOR OPHTHALMIC VELN

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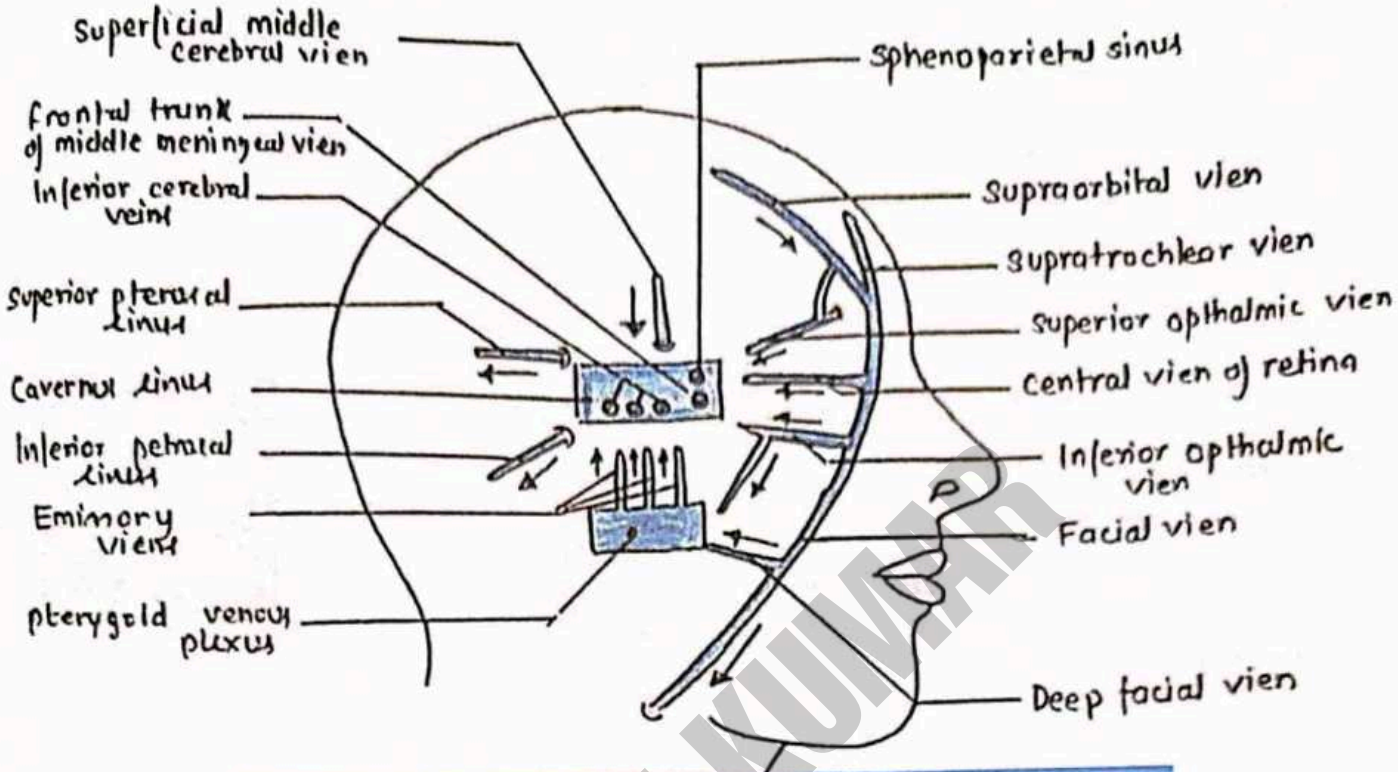
-RIGHT AND CAVERNOUS SINUSES → ANTERIOR & POSTERIOR -→ INTERCAVERNOUS SINUSES



VENOUS SINUSES OF DURA MATER

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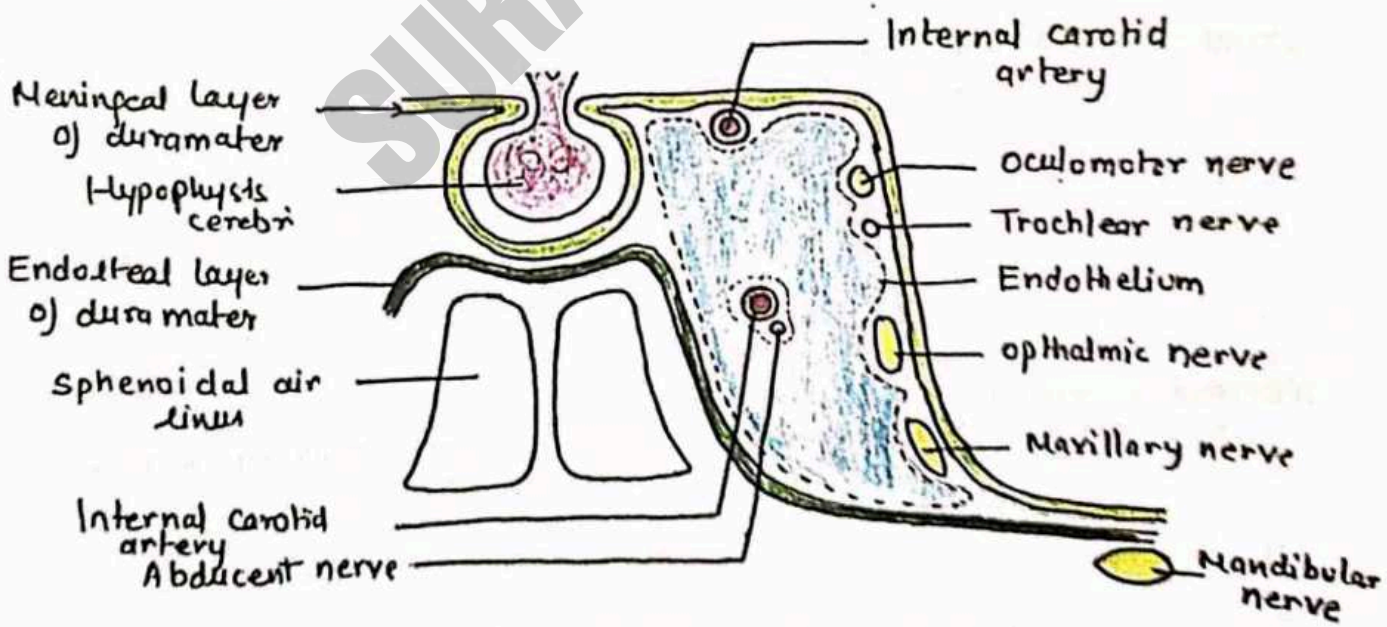

CAVERNOUS SINUS

CLINICAL ANATOMY

*THROMBOSIS OF THE CAVIRNUS SINUS : MAY BE CAUSED BY SEPSIS IN THE DANGEROUS AREA OF THE FACE, IN NASAL CAVITIES, & IN PARANASAL AIR SINUSES.THIS GIVES RISE TO THE FOLLOWING SYMPTOMS :

a.NERVOUS SYMPTOMS:SEVERE PAIN IN THE EYE & FOREHEAD IN THE AREA OF DISTRIBUTION OF OPHTHALMIC NERVE.

b.VENOUS SYMPTOMS : MARKED OEDEMA OF EYELIDS.

MUSCLE OF MASTICATION

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1) MASSETER - QUADRILATERAL

ORIGIN - SUPERFICIAL - ZYGOMATIC PROCESS OF MAXILLA
- ANTERIOR 1/3 ZYGOMATIC ARCH
- DEEP - ZYGOMATIC ARCH
- MIDDLE - 1/3 OF ZYGOMATIC ARCH

INSERTION - SUPERFICIAL - LOWER LATERAL SURFACE OF MANDIBLE
- DEEP - RAMUS OF MANDIBLE
- MIDDLE - CENTRAL PART OF RAMUS OF MANDIBLE

NERVE - MASSETER NERVE OF MANDIBULAR NERVE

ACTION - ELEVATION & PROTRUSION

2) TEMPORALIS - FAN-SHAPED

ORIGIN - TEMPORAL FOSSA EXCLUDING ZYGOMATIC BONE

INSERTION - DEEP SURFACE OF CORONOID PROCESS
- ANTERIOR BORDER OF RAMUL OF MANDIBLE

NERVE - DEEP TEMPORAL BRANCH OF M.N

ACTION -ELEVATION & PROTRUSION

3) LATERAL PTERYGOID:

ORIGIN- UPPER HEAD - INFRA-TEMPORAL SURFACE & INFRA-TEMPORAL CREST OF GREATER WING OF SPHENOID BONE
LOWER HEAD - LATERAL PTERYGOID PLATE

INSERTION- PTERYGOID FOVEA ON ANTERIOR OF NECK OF MANDIBLE

NERVE - ANTERIOR DIVISION OF MANDIBULAR NERVE

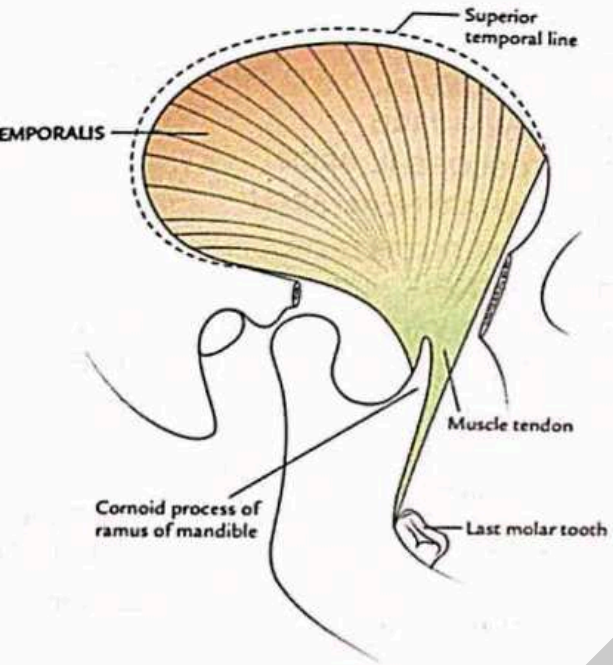
ACTION- DEPRESS & PROTRUSION

4) MEDIAL PTERYGOID - QUADRILATERAL

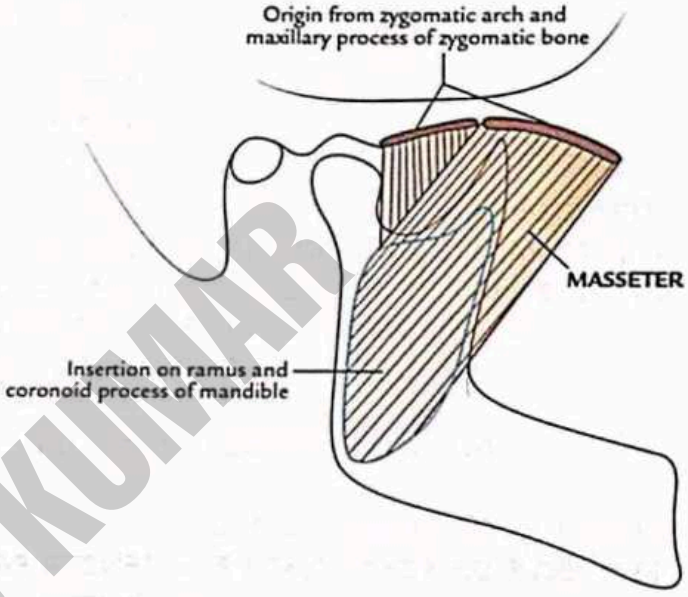
ORIGIN.- SUPERFICIAL HEAD - TUBEROSITY OF MAXILLA
DEEP HEAD - MEDIAL SURFACE OF - LATERAL PTERYGOID PLATE
INSERTION - MEDIAL SURFACE OF ANGLE OF ADJOINING RAMUS OF MANDIBLE

NERVE- NERVE TO MEDIAL PTERYGOID → MANDIBULAR NERVE

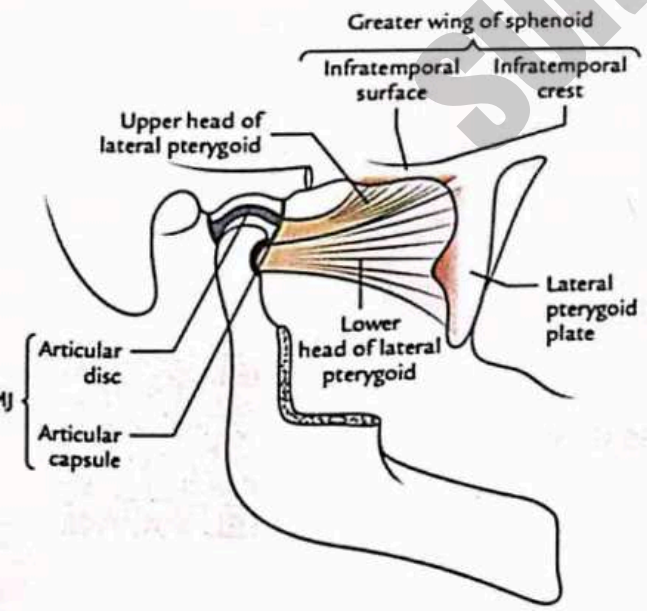
ACTION - ELEVATE & PROTRUDE



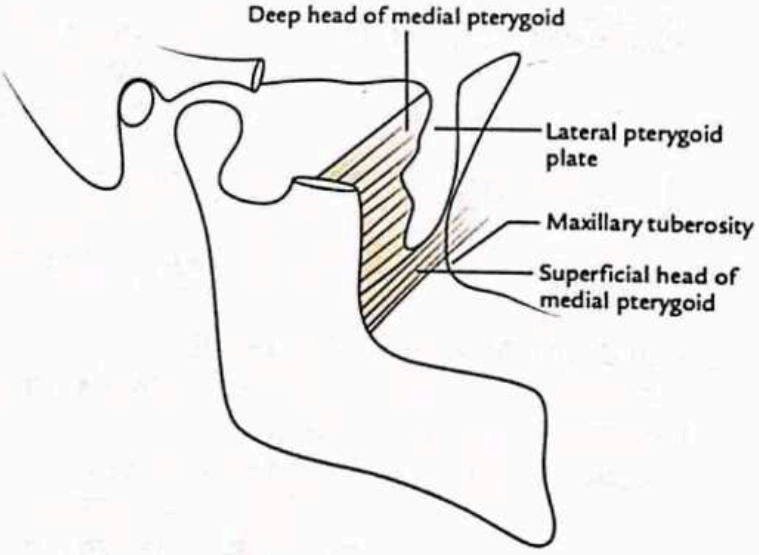
Origin and insertion of the temporalis muscle.



Origin and insertion of the masseter muscle.



Origin and insertion of the lateral pterygoid muscle.



Origin and insertion of the medial pterygoid

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SOFT PALATE

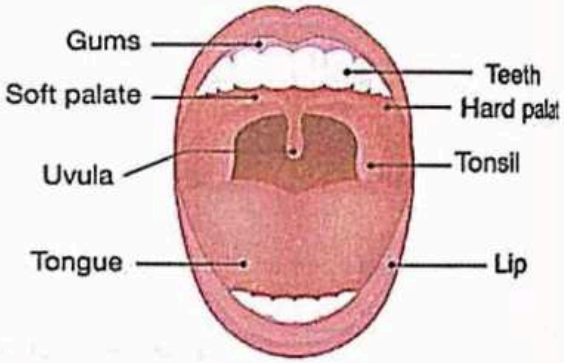
-MOVABLE, MUSCULAR FOLD, SUSPENDED FROM - POSTERIOR BORDER OF HARD PALATE

-2 SURFACE →ANTERIOR - CONCAVE
- POSTERIOR- CONVEX FLOOR OF NASAL CAVITY

-2 BORDER → SUPERIOR BORDER ATTACHED, POSTERIOR BORDER OF HARD PALATE
-INFERIOR BORDER - FREE
- CONICAL PROJECTION - "UVULA"
- FROM EACH SIDE BASE OF OVULA, TWO CURVED FOLD EXTEND - LATERALLY DOWNWARD
- ANTERIOR FOLD - PALATOGLOSSAL ARCH
- POSTERIOR FOLD-PALATOPHARYNEAL ARCH

STRUCTURE:-
- SOFT PALATE IS A FOLD OF MUCOUS MEMBRANE CONTAINING:-

1) PALATINE APONEUROSIS - FLATTENED TENDON OF TENSOR VELI PALATINI
- NEAR MEDIAN PLANE - APONEUROSIS SPLITS ENCLOSE - MUSCULUS UVALAE.
- SUPERIOR SURFACE CONTAIN --→ LEVATOR VELI PALATINI
-PALATOGLOSSUS
- INFERIOR SURFACE - PALATOGLOSSUS



2) NUMEROUS MUCOUS GLAND

3) MUSCLE OF SOFT PALATE 5 - TENSOR PALATI
- LEVATOR PALATI
- MUSCULAR UVULAE
- PALATOGLOSSUS
- PALATO PHARYNGEUS

*BLOOD SUPPLY: ARTERIES - GREATER PALATINE BRANCH MARTELY
- ASCENDING PALATINE FACIAL ARTERY
- PALATINE BRANCH OF ASCENDING PHARYNGEAL

*VEIN - TONSILLAR PLEXUSES OF VEIN

PALATINE TONSIL

- SECONDARY LYMPHOID TISSUE
- OCCUPIES- TONSILLAR FOSSA B/W PALATOGLOSSAL AND PALATOPHARYNGEAL ARCHES
- ALMOND-SHAPED - 2 SURFACE - MEDIAL
- LATERAL
- 2 BARDER - ANTERIOR
-POSTERIOR
- 2 POLES - UPPER AND LOWER

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1) SURFACE -

- MEDIAL SURFACE - COVERED BY STRATIFIED SQUAMOUS EPITHELIUM
  - SURFACE HAS - 12-15 CRYPTS
  - LARGEST OF THESE → INTRATONSILLAR CLEFT

- LATERAL SURFACE - COVERED BY SHEET OF FASCIA WHICH FORM HEMICAPSULE OF TONSILE
  - CAPSULE IS AN EXTENSION OF → PHARYNGOBASILAR FASCIA
  - TONSILLAR ARTERY - ENTER ---→ TONSIL
  - PARATONSILLAR VEIN → DESCEND → FROM PALATE IN LOOSE AREALOR TISSUE ON LATERAL SURFACE OF CAPSULE

2) BORDER → ANTERIOR BORDER- PALATOGLOSSAL ARCH
- POSTERIOR BORDER - PALATOPHARYNGEAL ARCH

3) POLE - UPPER POLE - SOFT PALATE
LOWER POLE - TONGUE

*BED OF TONSIL IS FORMED FROM WITHIN OUTWARDS BY :

- PHARYNGOBASILAR FASCIA
- SUPERIOR CONSTRICTOR AND PALATO PHARYNGEUS MUSCLE
- BUCCOPHARYNGEAL FASCIA
- GLOSSOPHARYNGEAL NERVES
- LOWER PART OF STYLOGLOSSUS

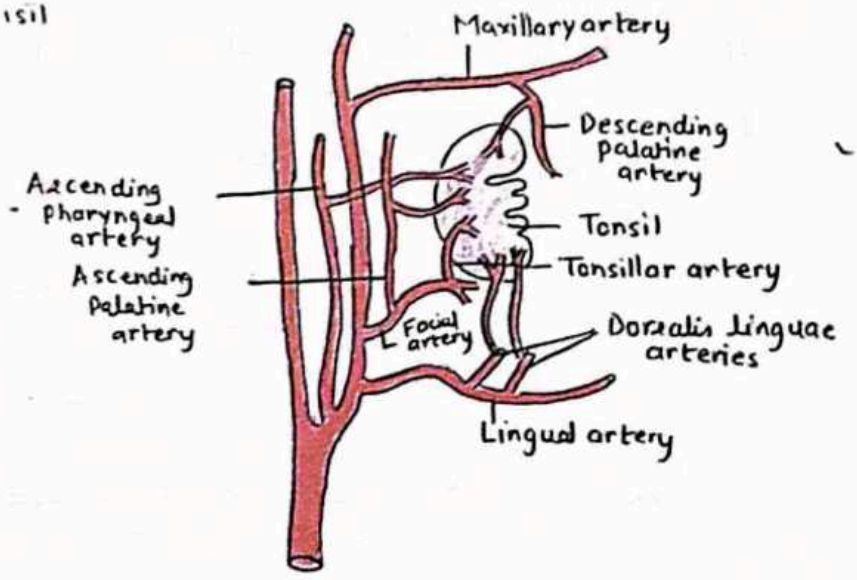
*BLOOD SUPPLY -

- MAIN-TONSILLAR BRANCH OF FACIAL ARTERY(FA)
- ASCENDING PALATINE - FA
- DORSAL LINGUAL BRACH - FA
- ASCENDING PHARYNGEAL BRANCH - ECA
- GREATER PALATINE - MAXILLARY A

*VEIN - 2-3 VEIN LEAVE LOWER PART OF TONSIL JOIN PALATINE,FACIAL VEIN

*NERVE

- GLOSSOPHARYNGEAL NERVE
- LESSER PALATINE NERWE



∴ Arteries supply of tonsil

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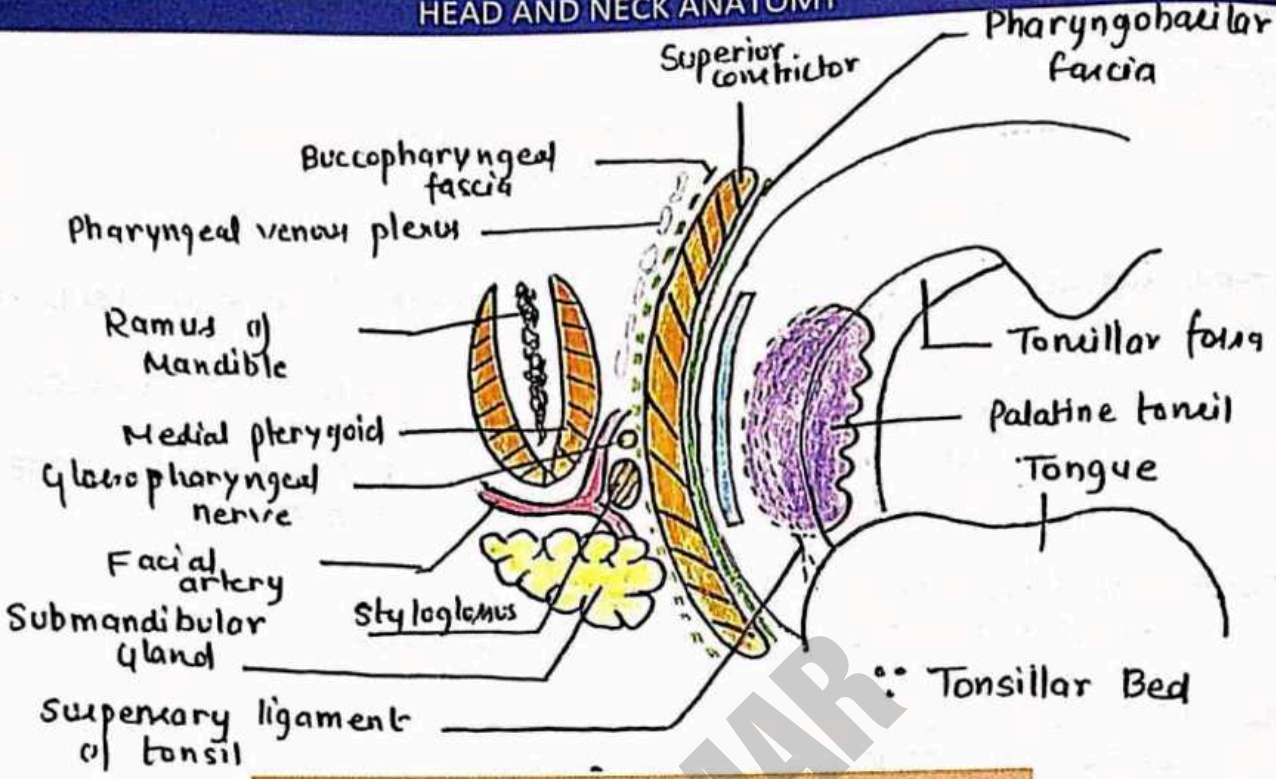

PALATINE TONSIL

TONGUE ANATOMY

*MUSCULAR ORGAN - SITUATED - FLOOR OF MOUTH
*FUNCTION- TASTE, SPEECH, CHEWING AND DEGLUTITION
*EXTERNAL FEATURES :- ROOT
- TIP
- BODY

SCAN QR WITH QRBOT APP TO DIRECTLY OPEN VIDEO LINK

1) ROOT - ATTACHED → STYLOID PROCESS & SOFT PALATE - ABOVE
- HYOID BONE - BELOW

2) TIP - ANTERIOR FREE END→ BEHIND UPPER INCISOR TEETH

3) BODY -
DORSUM - CONVEX IN ALL DIRECTION,
- ORAL PART - FLOOR OF MOUTH
- FREE MARGIN -IN CONTACT WITH GUMS AND TEETH
- SUPERIOR SURFACE - COVERED WITH PAPILLAE
- INFERIOR SURFACE - COVERED WITH SMOOTH MEMBRANE
- MEDIAN FOLD- FRENULUM
- LATERAL FOLD - PLICA FRENULUM

- PHARYNGEAL PART - LIE BEHIND PALABOGLOSSAL ARCHES
- BASE OF TONGUE

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- MUCOUS MEMBRANE - NO PAPILLAE BUT LYMPHOID FOLLICLE

- POSTERIOR MOST PART - CONNECTED TO EPIGLOTTIS
  - 3 FOLDS OF MUCOSA MEMBRANE
  - 2 LATERAL GLOSSOEPIGLOTTIC FOLD
  - MEDIAN GLOSSOEPIGLOTTIC FOLD

MUSCLE OF TONGUE

*MIDDLE FIBROUS SEPTUM → DIVIDES TONGUE INTO RIGHT & LEFT HALVES
*EACH HALF CONTAIN - 4 INTRINSIC MUSCLES
- 4 EXTRINSIC MUSCLES

*INTERINSIC MUSCLE - OCCUPY UPPER PART OF TONGUE
- ATTACHED- SUBMUCOUS FIBROUS LAYER

1) SUPERIOR LONGITUDINAL - ARISES FROM FIBROUS TISSUE DEEP TO MUCOUS MEMBRANE
   - ON DORSUM OF TONGUE
   - ACTION - SHORT TONGUE
     - MAKE DORSUM CANCAVE

2) INFERIOR LONGITUDINAL - ARISES FROM FIBROUS TISSUE → BENEATH MUCOUS MEMBRANE
   - ON TIP OF TONGUE
   - ACTION - SHORT TONGUE
     - MAKE DORSUM CONVEX

3) TRANSVERSE
   - LIE ON EITHER SIDE OF MID LINE
   - ORIGIN- FIBROUS MEDIAL SEPTUM TO SUBMUCOUS FIBROUS TISSUE
   - MAKE TONGUE NARROW

4) VERTICAL - FOUND AT BORDER OF ANTERIOR PART OF TONGUE
   - ACTION-FLATTED

*EXTRINSIC MUSCLE : -

ACTION

1) GENIOGLOSSUS → PROTRUDES
2) HYOGLOSSUS → DEPRESSES
3) STYLO GLOSSUS → RETRACTS
4) PALATOGLOSSUS → ELEVATES

PALATOGLOSSUS - ORIGIN - ORAL SURFACE OF PALATINE APONEUROSIS
-INSERTION- SIDE OF TONGUE
-NERVE - PHARYNGEAL PLEXUS

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HYOGLOSSUS - ORIGIN - LATERAL PART OF HYOID BONE
- INSERTION- SIDE OF TONGUE
- NERVE- REST ALL BE MUSCLE HYPOGLOSSAL NERVE

STYLOGLOSSUS - ORIGIN - TIP OF STYLOID PROCESS
- INSERTION- SIDE OF TONGUE

GENIOGLOSSUS - ORIGIN - UPPER GENITAL TUBERCLE OF MANDIBLE
- INSERTION - UPPER FIBRES INTO TIP OF TONGUE

*BLOOD & NERVE SUPPLY OF TONGUE

*ARTERY → DERIVED FROM → LINGUAL ARTERY BRANCH - ECA
-TIP ALSO SUPPLY BY - TONSILLAR ARTERY - FA

*VENOUS DRAINAGE - DEEP LINGUAL VEIN - LARGEST AND PRINCIPAL VEIN
- INFERIOR SURFACE OF TONGUE
- END IN INTERNAL JUGULAR VEIN THROUGH LINGUAL VEIN

*LYMPHATIC DRAINAGE
- SUBMENTAL NODE (TIP) -----> FINALLY DRAIN OMOHYOID NODE
- SUB-MANDI BULAR NODE (RIGHT AND LEFT HALVES) -----> FINALLY DRAIN OMOHYOID NODE
- DEEP-CERVICAL NODE (POSTERIO MOST) -----> FINALLY DRAIN OMOHYOID NODE

*NERVE SUPPLY :-
- MOTOR - ALL MUSCLE- HYPOGLOSSAL NERVE
EXCEPT - PALATOGLOSSUS - PHARYNGEAL PLEXUS
- SENSORY -ANTERIOR 2/3 - LINGUAL NERVE
- CHORDA - TYMPANI NERVE
POSTERIOR 1/3-GLOSSOPHARYNGEAL NERVE

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CLINICAL ANATOMY

TONGUE

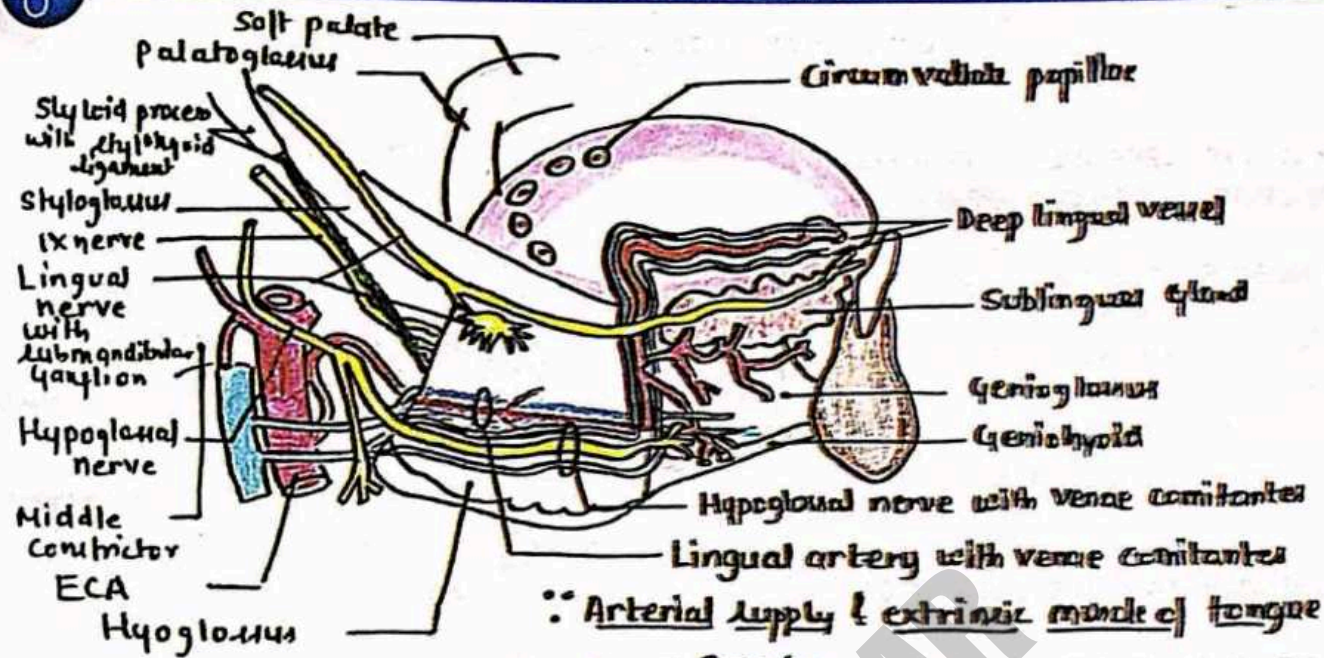
*GLOSSITIS IS USUALLY A PART OF GENERALIZED ULCERATION OF THE MOUTH CAVITY OR STOMATITIS.

*CARCINOMA OF THE TONGUE IS QUITE COMMON.THE AFFECTED SIDE OF THE TONGUE IS REMOVED SURGICALLY.

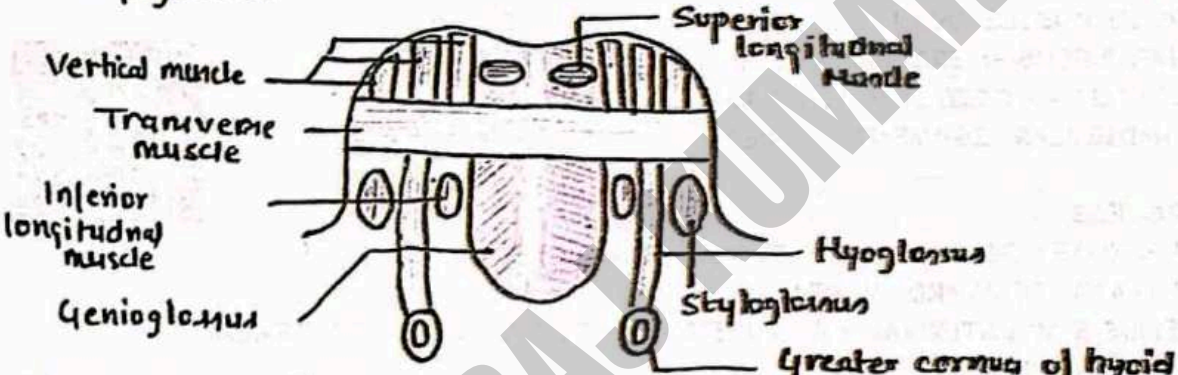
TONSILS

*THE TONSILS ARE LARGE IN CHILDREN.THEY RETROGRESS AFTER PUBERTY.

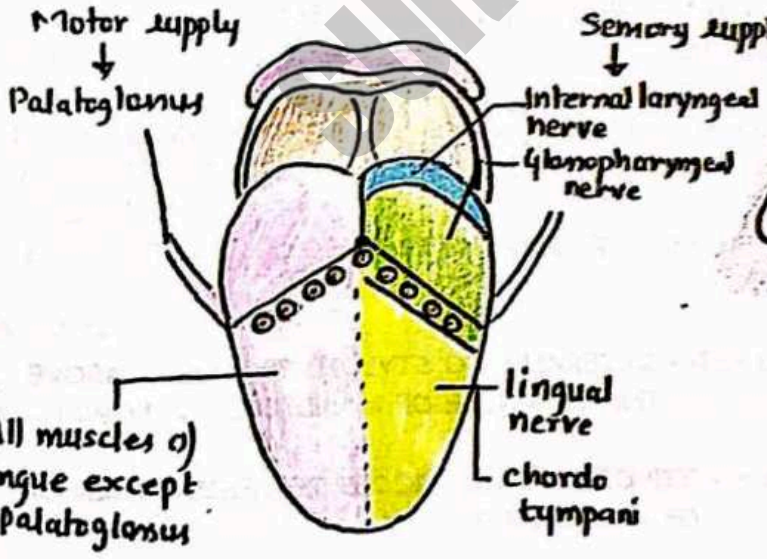
*TONSILLITIS MAY CAUSE REFERRED PAIN IN THE EAR AS GLOSSOPHARYNGEAL NERVE SUPPLIES BOTH THESE AREAS.



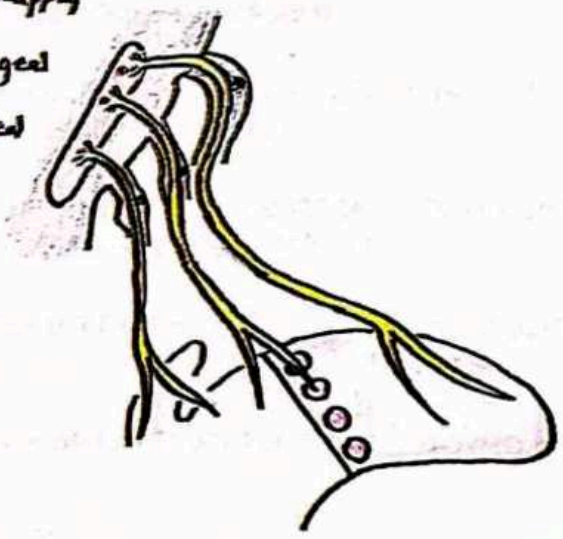
∵ Arterial supply & extrinsic muscle of tongue



∵ Intrinsic muscle



∵ Nerve supply of tongue



∵ Nerve carrying taste sensations

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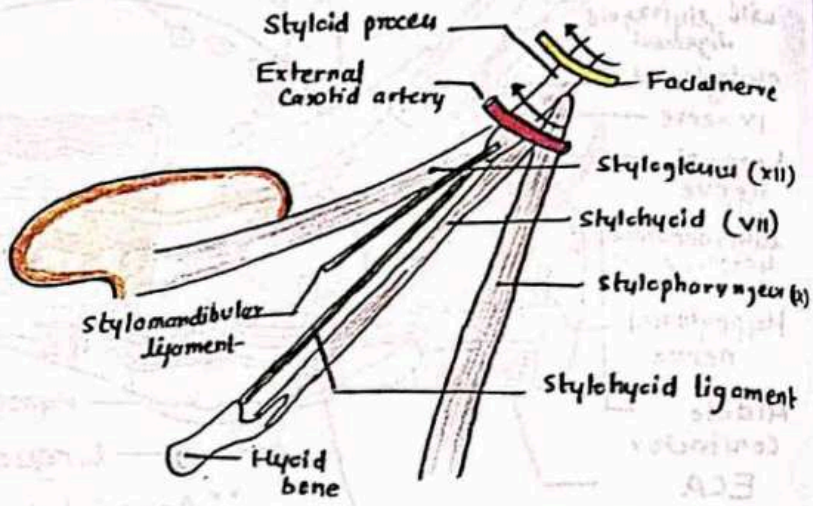
STYLOID APPARATUS

STYLOID PROCESS WITH ATTACHED STRUCTURE → STYLOID APPARATUS

- 3 MUSCLE - STYLOGLOSSUS
  - STYLOHYOID
  - STYLOPHARYNGEUS
- 2 LIGAMENTS - STYLOHYOID
  - STYLO MANDIBULAR



***APPARATUS IS OF DIVERSE ORIGIN :-**

- STYLOHYOID MUSCLE AND LIGAMENT -> 2RD BRANCHIAL ARCH
- STYLOPHARYNGEUS -> 3RD BRANCHIAL ARCH
- STYLOGLOSSUS -> OCCIPITAL MYOTOMES
- STYLOMANDIBULAR LIGAMENT -> DEEP FASCIA OF NECK

***STYLOID PROCESS :-**

- LONG POINTED BONY PROCESS OF TEMPORAL BONE
- DOWNWARD, FORWARD, MEDIALLY
- DESCENDS B/W EXTERNAL AND INTERNAL CAROTID ARTERY TO REACH SIDE OF PHARYNX

STYLOGLOSSUS - ARISES - ANTERIOR SURFACE OF STYLOID PROCESS
-INSERT SIDE OF TONGUE

STYLOPHARYNGEUS - ARISES - MEDIAL SURFACE OF BASE OF STYLOID PROCESS.
-INSERT - POSTERIOR BORDER OF THYROID CARTILAGE

STYLOHYOID - ARISES - POSTERIOR SURFACE OF STYLOID PROCESS
-INSERT - HYOID BONE

STYLOMANDIBULAR LIGAMENT ATTACHED - LATERALLY TO STYLOID PROCESS - ABOVE
-INSERT ANGLE OF MANDIBLE - BELOW

STYLOHYOID LIGAMENT EXTEND FROM - TIP OF STYLOID PROCES TO LESSER CORNUA OF HYOID BONE

***FEATURES -**

*EXTERNAL CAROTID ARTERY→ TIP OF STYLOID PROCESS
*FACIAL NERVE CROSSES -→ BASE OF STYLOID PROCES

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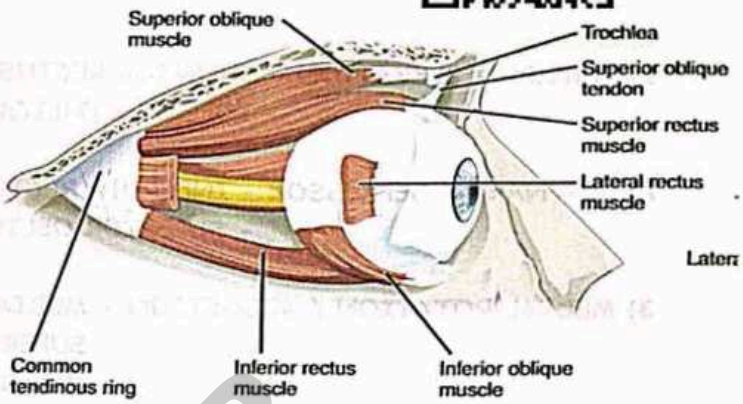
EXTRAOCULAR MUSCLE

*VOLUNTARY MUSCLE :-

4X RECT - SUPERIOR RECTUS
- INFERIOR RECTUS
- MEDIAL RECTUS
- LATERAL RECTUS

2X OBLIQUE - SUPERIOR OBLIQUE
- INFERIOR OBLIQUE

LEVATAR PALPEBRAE SUPERIORIS



Lateral view of the right eye

*INVOLUNTARY MUSCLE -

1) SUPERIOR TARSAL MUSCLE -DEEPER PORTION OF LPS
- INSERTED-UPPER MARGIN OF SUPERIOR TARSUS
- ELEVATE UPPER EYELID

2) INFERIOR TARSAL MUSCLE - EXTEND- FASCIAL SHEATH OF INFERIOR RECTUS
- DEPRESSES THE LOWER EYELID

3) ORBITALIS BRIDGE - THE INFERIOR ORBITAL FISSURE

VOLUNTARY MUSCLE -

*ORIGIN :-

-4 RECTI - COMMON ANNULAR TENDON - RING ATTACHED - SUPERIOR ORBITAL FISSURE
- LATERAL RECTUS ADDITIONAL ORIGIN -→ GREATER WING OF SPHENOID BONE
- SUPERIOR OBLIQUE ARISES --→ LESSER WING OF SPHENOID BONE
- INFERIOR OBLIQUE ARISES -→ ORBITAL SURFACE OF MAXILLA
- LEVATOR PALPEBRAE SUPERIOR -→ LESSER WING OF SPHENOID BONE

*INSERTION :-

- RECTI INSERTED → SCLERA (POSTERIOR TO LIMBUS]

- SUPERIOR OBLIQUE - PASSES THROUGH→ FIBRO-CARTILAGE PULLY - ATTACHED TO TROCHLEAR FOSSA OF FRONTAL BONE
-INSERTION - SCLERA BEHIND EQUATOR OF EYEBALL

- INFERIOR OBLIQUE MUSCLE INSERTED-→ CLOSE TO SO.

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NERVE - RECTUS X 3 = CN=3/OCCULOMOTAR NERVE
- SUPERIOR OBLIQUE = CN=4 / TROCHLEAR NERVE
- LATERAL RECTUS - CN=6 ABDUCENT NERVE

ACTION :-

1) UPWARD / ELEVATION - SUPERIOR RECTUS
- INFERIOR OBLIQUE

2) DOWNARD / DEPRESSOR - INFERIOR RECTUS
- SUPERIOR OBLIQUE

3) MEDIAL ROTATION / ADDUCTION - MEDIAL RECTUS
SUPERIOR RECTUS
INFERIOR RECTUS

4) LATERAL ROTATION / ABDUCTION - LATERAL RECTUS
SUPERIOR OBLIQUE
INFERIOR OBLIQUE

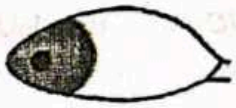
5) TORSION - SUPERIOR OBLIQUE
- SUPERIOR RECTUS

6) EXTORSION - INFERIOR OBLIQUE
- INFERIOR RECTUS

Supraduction

Infraduction

Adduction


Abduction

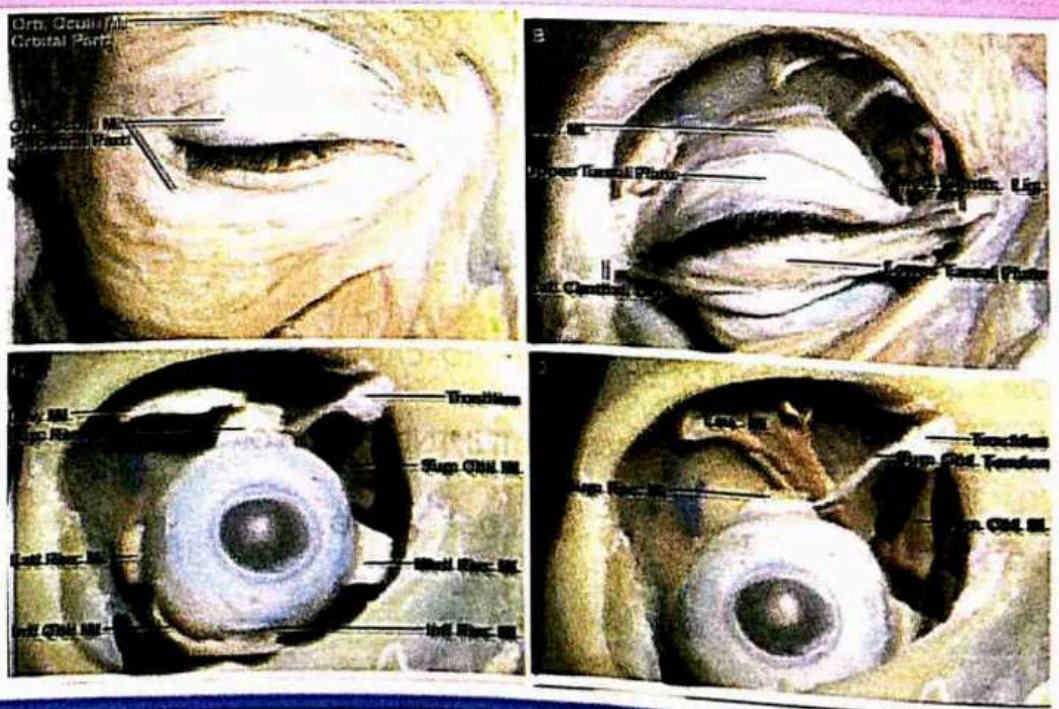
Intorsion

Extorsion

CLINICAL ANATOMY

*WEAKNESS OR PARALYSIS OF A MUSCLE CAUSES SQUINT OR STRABISMUS, WHICH MAY BE CONCOMITANT OR PARALITIC.

*MYSTAGMUS IS CHARACTERIZED BY INVOLUNTARY RYTHMICAL OSCILLATORY MOVEMENTS OF THE EYES.



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LATERAL WALL OF NOSE

*LATERAL WALL OF NOSE → IRREGULAR OWING → PRESENCE OF 3X SHELF LIKE BONE PROJECTION CALLED CONCHAE
*CONCHAE - INCREASE SURFACE AREA OF NOSE
- FOR EFFECTIVE AIR-CONDITIONING OF INSPIRED AIR

*SEPARATE THE NOSE - FROM ORBIT ABOUE
- MAXILLARY SINUS BELOW
- FROM LACRIMAL SAC AND NASOLACRIMAL DUCT FRONT

*SUB-DIVIDED INTO 3X PART - ANTERIOR - VESTIBULE
MIDDLE - ATRIUM OF MIDDLE MEATUS
POSTERIOR - MEATUSES

*SKELETON OF LATERAL WALL IS PARTLY BONY, PARTLY CARTILAGINOUS AND PARTLY MADE UP OF SOFT TISSUE
-BONY PART - NASAL
- FRONTAL PROCESS OF MAXILLA
- LACRIMAL
- LABYRINTH OF ETHMOID
- INFERIOR NASAL CONCHA
- PERPENDICULAR PLATE OF PALATINE BONE
- MEDIAL PTERYGOID PLATE

- CARTILAGINOUS PART - SUPERIOR NASAL CARTILAGE
- INFERIOR NASAL CARTILAGE
- 3-4 SMALL CARTILAGES OF ALA

- CUTICULAR LOWER PART -→ FORMED BY FIBROFATTY →COVERED WITH SKIN

CONCHAE AND MEATUSES

NASAL CONCHAE → CURVED BONE PROJECTION
- DIRECTED DOWNWARD AND MEDIALLY
- INFERIOR CONCHA - INDEPENDENT BONE
- MIDDLE CONCHA - MEDIAL SURFACE OF ETHMOIDAL LABYRINTH
- SUPERIOR CONCHA - MEDIAL SURFACE OF ETHMOIDAL LABYRINTH

MEATUSE OF NOSE - PASSAGE BENEATH THE OVERHANGING CONCHAE

1) INFERIOR MEATUS - LARGEST OF THREE MEATUSES

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- NASOLACRIMAL DUCT OPEN INTO JUNCTION OF POSTERIOR 2/3 & ANTERIOR 1/3

2) MIDDLE MEATUS - ETHMOIDAL BULLA - ROUND ELEVATION
- HIATUS SEMILUNARIS - DEEP SEMICIRCULAR SULCUS
- OPENING OF FRONTAL AIR SINUS → ANTERIOR PART OF HS
- OPENING OF ETHMOIDAL AIR SINUS → BEHIND F.A
- OPENING OF MAXILLARY AIR SINUS → POSTERIOR PART OF HS

3) SUPERIOR MEATUS - SHORTEST AND SHALLOWEST OF MEATUSES
- OPENING OF POSTERIOR ETHMOIDAL AIR SINUSES

SPHENO ETHMOIDAL RECESS - TRIANGLE FOSSA
- JUST ABOVE - SUPERIOR CONCHA
- RECEIVES - OPENING OF SPHENOIDAL AIR SINUS

***ARTERIAL SUPPLY -**
1) ANTERIOR POSTERIOR ETHMOIDAL A. - ANTEROSUPERIOR QUADRANT
2) SPHENOPALATINE A. - POSTEROSUPERIOR QUADRANT
3) FACIAL A. - ANTEROINFERIOR QUADRANT
4) GREATER PALATINE A. - POSTEROINFERIOR QUADRANT

***VENOUS DRAINAGE** - VEIN FROM A PLEXUS WHICH DRAINS -
- ANTERIORLY - FACIAL VEIN
- MIDDLE - PTERYGOID PLEXUS VEIN
- POSTERIORLY - PTERYGOID PLEXUS VEIN

***NERVE SUPPLY -**
1) ANTERIOR ETHMOID NERVE - ANTEROSUPERIOR QUADRANT
2) LATERAL POSTERIOR NASAL BRANCH NERVE - POSTEROSUPERIOR QUADRANT
3) ANTERIOR SUPERIOR ALVEOLAR NERVE - ANTEROINFERIOR QUADRANT
4) ANTERIOR PALATINE NERVE - POSTEROINFERIOR QUADRANT

CLINICAL ANATOMY

*HYPERTROPHY OF THE MUCOSA OVER THE INFERIOR NASAL CONCHA IS A COMMON FEATURE OF ALLERGIC RHINITIS, WHICH IS CHARACTERISED BY SNEEZING, NASAL BOLCKAGE AND EXCESSIVE WATERY DISCHARGE FROM NOSE.

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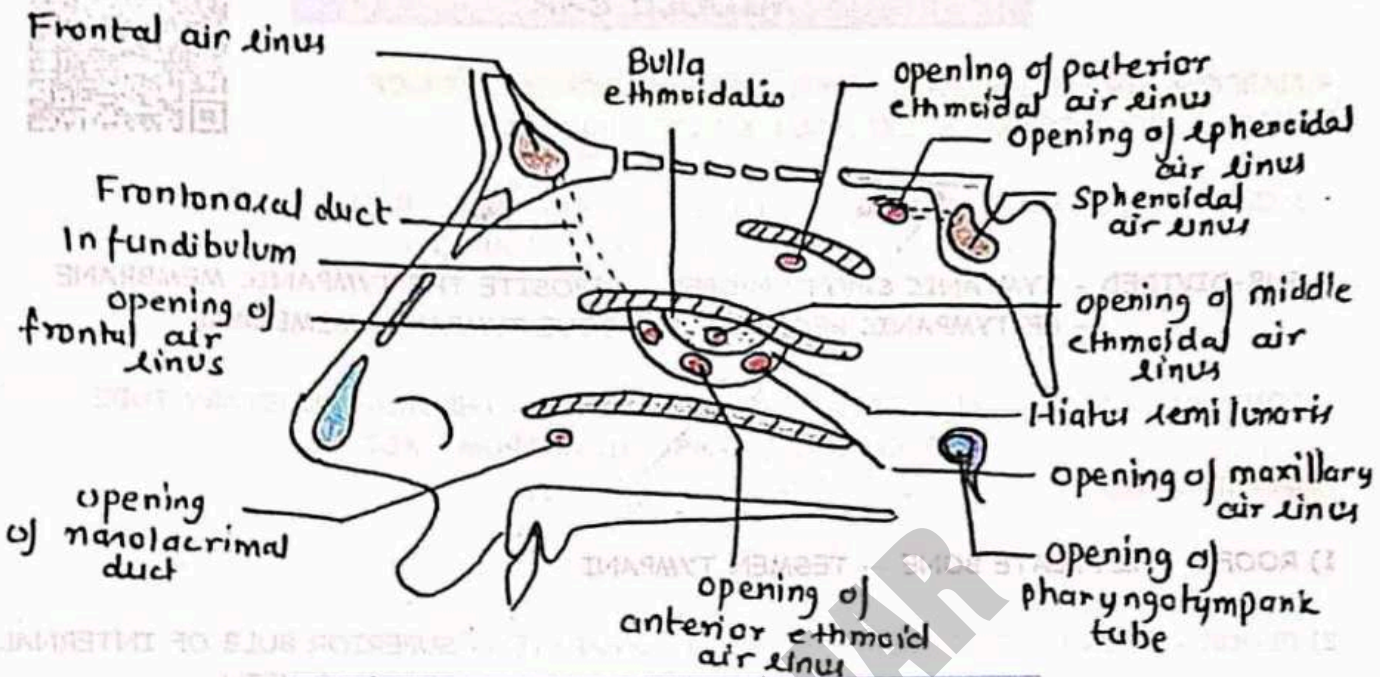
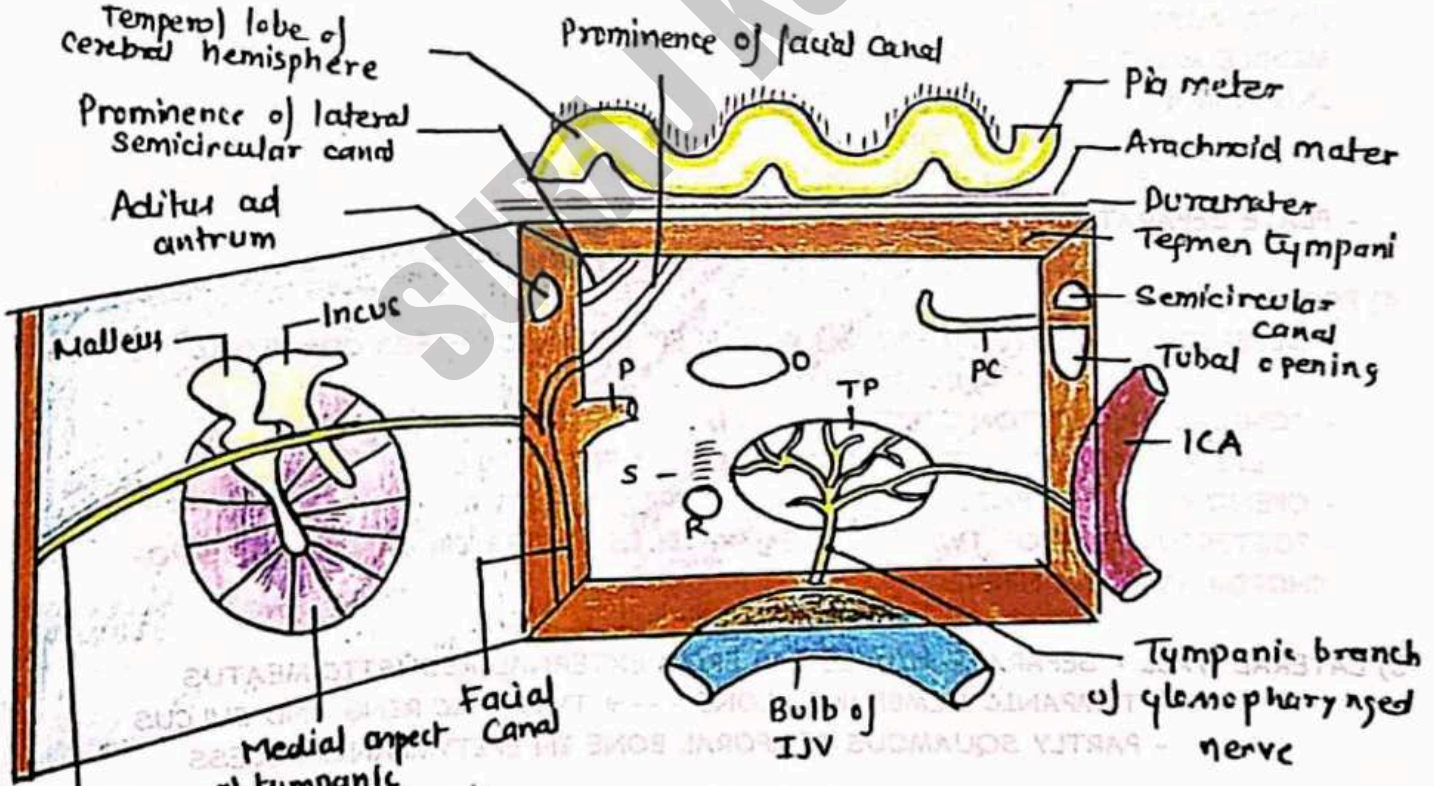


LATERAL WALL OF NOSE



Middle Ear

[P → Pyramid, O ⇒ oval window, PC ⇒ Procesus cochleariformis, PM ⇒ Promontory, R ⇒ Round window, S ⇒ Sinus tympani, TP ⇒ Tympanic Plexus]

MIDDLE EAR

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MIDDLE EAR

- NARROW AIR FILLED SPACE - SITUATED IN PETROUS PART OF TEMPORAL BONE BETWEEN EXTERNAL & INTERNAL EAR

- MIDDLE EAR SHAPED, LIKE CUBE - LATERAL AND MEDIAL WALL - LARGE
  - OTHER WALL - NARROW
- SUB-DIVIDED - TYMPANIC CAVITY PROPER - OPPOSITE THE TYMPANIC MEMBRANE
  - EPITYMPANIC RECESS - LIE ABOUE TYMPANIC MEMBRANE

- COMMUNICATION → ANTERIORLY - NASOPHARYNX - THROUGH AUDITARY TUBE
  POSTERIORLY - MASTOID ANTRUM - ADITUS

*BOUNDARIES -

1) ROOF - THIN PLATE BONE → TEGMEN TYMPANI

2) FLOOR - THIN PLATE BONE OF TEMPORAL SEPARATE → SUPERIOR BULB OF INTERNAL JUGULAR VEIN
   - GLOSSOPHARYNGEAL NERVE

3) ANTERIOR WALL -
   UPPER MOST - TENSOR TYMPANI
   MIDDLE MOST - AUDITORY TUBE
   INNER MOST - THIN PLATE OF BONE
   ↓ WHICH
   FORM POSTERIOR WALL OF CAROTID CANAL
   - PLATE SEPARATED INTERNAL CAROTID ARTERY

4) POSTERIOR WALL -
   - SUPERIORLY - ADITUS THROUGH WHICH EPITYMANIC RECESS COMMUNICATE TYMPANIC ANTRUM
   - CONICAL PROJECTION CALLED PYRAMID
     LIE - NEAR JUNCTION OF POSTERIOR & MEDIA WALL
   - OPENING -→ FOR PASSAGE OF TENDON OF" STAPEDIUS"
   - POSTERIOR EDGE OF TYMPANIC MEMBRANE IS POSTERIOR CANALICULUS FOR CHORDA TYMPANI NERVE

5) LATERAL WALL - SEPARATE MIDDLE EAR FROM EXTERNAL ACOUSTIC MEATUS
   - TYMPANIC MEMBRANE ALONG ---→ TYMPANIC RING AND SULCUS
   - PARTLY SQUAMOUS TEMPORAL BONE IN EPITYMPANIC RECESS

6) MEDIAL WALL - SEPARATE MIDDLE EAR FROM INTERNAL EAR
   - PROMONTARY - FIRST TURN OF COCHLEA

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-FENESTRA COCHLEAE - ROUND OPENING
- POSTERO INFERIOR → PROMONTORY

-FENESTRA VESTIBULI - OVAL OPENING
- POSTERO SUPERION → PROMONTORY
-SEMICIRCULAR CANAL → ABOVE FACIAL CANAL
-FACIAL CANAL - ABOVE FENESTRA VESTIBULI

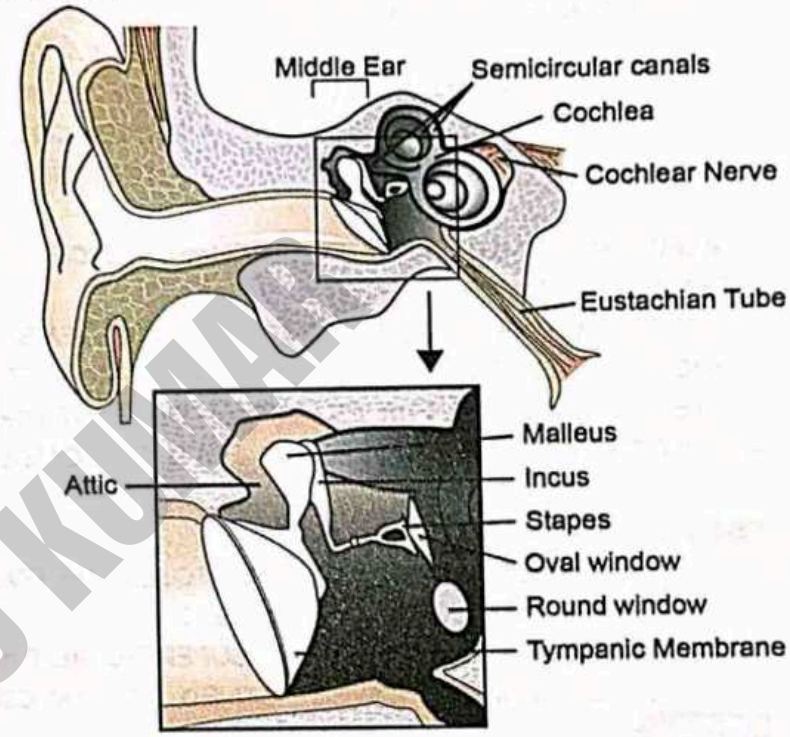
*CONTENTS :-
- 3X SMALL BONE' - MALLEUS
INCUS
STAPE

- LIGAMENT OF EAR OSSICLE

- 2X MUSCLE - TENSAR TYMPANI
- STAPEDIUS

- VESSELS SUPPLY AND DRAINING THE MIDDLE EAR

- NERVE - CHORDA TYMPANI
- TYMPANIC PIEXUS



*ARTERIAL SUPPLY -

- ANTERIOR TYMPANIC BRANCH - MAXILLARY ARTERY
- POSTERIOR TYMPANIC BRANCH POSTERIOR AURICULAR ARTERY
- PETROSAL AND SUPERIOR TYMPANIC BRANCH- MIDDLE MENINGEAL ARTERY
- BRANCHES OF ASCENDING PHARYNGEAL ARTERY
- TYMPANIC BRANCHES - ICA

*VENOUS DRAINAGE
- DRAIN --→ INTO SUPERIOR PETROSAL SINUS & PTERYGOID PLEXUS

*LYMPHATIC DRAINAGE
- PREAURICULAR & RETROPHARYNGEAL LYMPH NODE

*NERVE SUPPLY
- DERIVED FROM - TYMPANIC PLEXUS --→WHICH LIE OVER PROMOTONARY

- PLEXUS FORMED BY - 1) TYMPANIC BRANCH OF GLOSSOPHARYNGEAL NERVE
2) SUP. & INF. CAROTICOTYMPANIC NERVE-→ARISE FROM SYMPATHETIC PLEXUS

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TYMPANIC MEMBRANE

*THIN, TRANSLUCENT PARTITION BETWEEN EXTERNAL ACOUSTIC MEATUS AND MIDDLE EAR

*OVAL IN SHAPE

*PLACED - OBLIQUELY AT AN ANGLE OF 55 DEGREE WITH FLOOR OF MEATUS

*FACE - DOWNWARD, FORWARD AND LATERALLY

- 2 SURFACE ----→OUTER SURFACE - LINED BY THIN SKIN
  - CONCAVE
  ---→ INNER SURFACE - PROVIDE ATTACHMENT TO HANDLE OF MALLEUS
  - INNER CONVEX
  - POINT OF MAX. CONVEXITY LIE-→AT TIP OF HANDLE OF MALLEUS CALLED UMBO
- MEMBRANE THICKENED AT CIRCUMFERENCE →WHICH FIXED TYMPANIC SULCUS OF TEMPORAL PLATE
- SUPERIOR SULCUS DEFICIENT -→ HERE MEMBRANE ATTACHED TO →TYMPANIC NOTCH
- FROM END OF NOTCH - 2 BAND - ANT. & POST. MALLEOLAR FOLD PROLONGED
- GREATER PART OF T.M IS -→TIGHTLY STRETCHED ---→ PARS TENSA
- PART BETWEEEN 2 MALLEOLAR FOLD OS LOOSE ---→ PARS FLACCIDA

*STRUCTURE

- TYMPANIC MEMBRNE IS COMPOSED OF FOLLOWING 3 LAYER
- OUTER CUTICULAR LAYER - SKIN
- MIDDLE FIBROUS LAYER - SUPERFICIAL RADIATING FIBRES
- INNER MUCOUS LAYER - LINED BY LOW CILIATED COLUMNAR EPITHELIUM

*BLOOD-

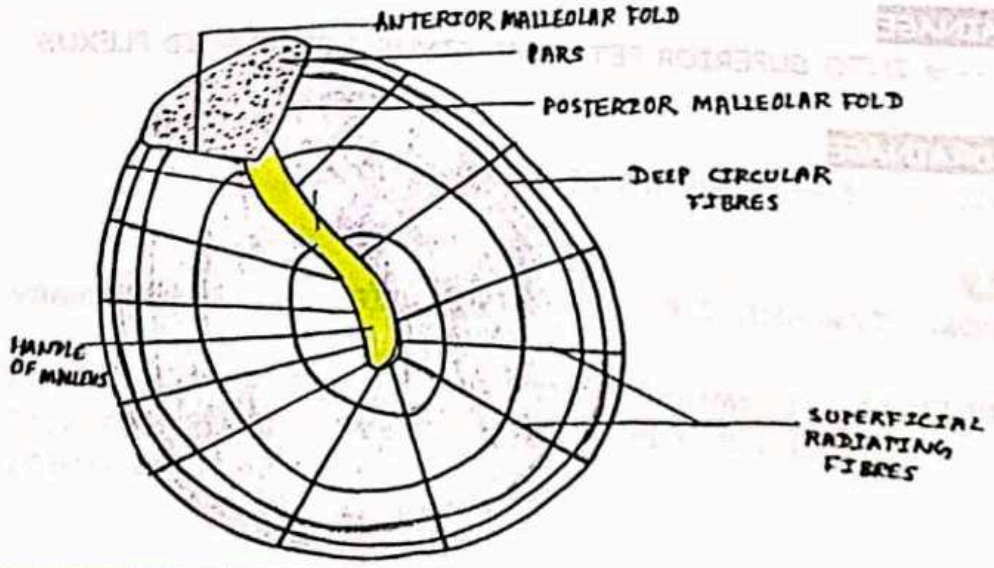
OUTER SYRFACE - DEEP AURICULAR A. BRANCH OF MAXILLARY ARTERY

INNER SURFACE - ANTERIOR TYMPANIC BRANCH - MAXILLARY A.
- POSTERIOR TYMPANI BRANCH - POSTERIOR AURICULAR A.

*NERVE-

OUTER SYRFACE - AURICLUOTEMPORAL NERVE

INNER SURFACE - TYMPANIC BRANCH →GLOSSOPHARYNGEAL NERVE



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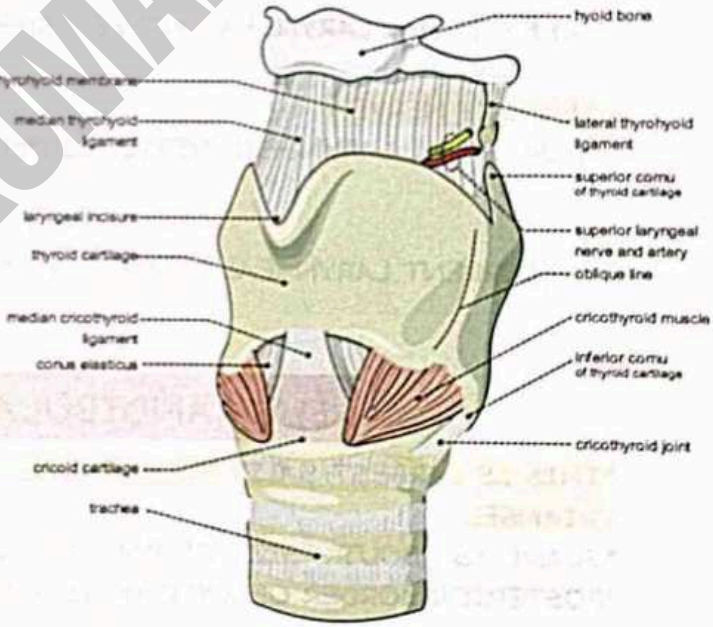
LARYNX ANATOMY

*LARYNX - IS ORGAN FOR PRODUTION OF VOICE OR PHONATION
*IT IS NALSO AN AIR PASSAGE AND ACT AS SPHINCTER AT INLET OF LOWER RESPIRATORY PASSAGES

*SITUATION-
LIE IN -→ ANTERIOR MIFLINE OF NECK
EXTENDING FROM - ROOT OF TONGUE TO TRACHEA
IN ADULT MALE - C3-C6 VERTEBRAE
IN ADULT FEMALE - C1 - C4

*SIZE - LENGTH -→ 44MM IN MALE & 36MM IN FEMALE

*CARTILAGES OF LARYNX -
-THE LARYNX CONTAINS NINE CARTILAGES, OF WHICH THREE ARE UNPAIRED AND THREE ARE PAIRED.



UNPAIRED CARTILAGES
1 THYROID (GREEK SHIELD-LIKE)
2 CRICOID (GREEK RING-LIKE)
3 EPIGLOTTIS (GREEK LEAF-LIKE)

PAIRED CARTILAGES
1 ARYTENOID (GREEK CUP-SHAPED)
2 CORNICULATE (LATIN HORN-SHAPED)
3 CUNEIFORM (LATIN WEDGE-SHAPED)

* INTRINSIC MUSCLES OF LARYNX
1. CRICOTHYROID
2. POSTERIOR CRICOARYTENOID TRIANGULAR
3. LATERAL CRICOARYTENOID
4. TRANSVERSE ARYTENOID
5,6. OBLIQUE ARYTENOID AND ARYEPIGLOTTICAN
7,8. THYROARYTENOID AND THYROEPIGLOTTIC
9. VOCALIS

*MUSCLES ACTING ON THE LARYNX

MOVEMENT	MUSCLE
1. ELEVATION OF LARYNX	- THYROHYOID, MYLOHYOID
2. DEPRESSION OF LARYNX	- STERNOTHYROID, STERNOHYOID
3. OPENING INLET OF LARYNX	- THYROEPIGLOTTIC
4. CLOSING INLET OF LARYNX	- ARYEPIGLOTTIC
5. ABDUCTOR OF VOCAL CORDS	- POSTERIOR CRICOARYTENOID ONLY
6. ADDUCTOR OF VOCAL CORDS	- LATERAL CRICOARYTENOID,

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TRANS-VERSE AND OBLIQUE ARYTENOIDS

7. TENSOR OF VOCAL CORDS AND MODULATION OF VOICE - CRICOTHYROID
8. RELAXOR OF VOCAL CORDS - THYROARYTENOID AND VOCALIS

*BLOOS SUPPLY

- UP TO VOCAL FOLD - BY SUPERIOR LARYNGEAL A.
  -VEIN - DRAIN BY SUPERIOR LARYNGEAL VEIN

- BELOW THE VOCAL FOLD - BY INFERIORLARYNGEAL A.
  - VEIN - DRAIN BY INFERIOR LARYNGEAL VEIN

*NERVE SUPPLY -

-MOTOR NERVE-

1) RECURRENT LARYNGEAL NERVE - SUPPLIS ALL INTRINSIC MUSCLE
   - EXCPET - CRICOTHYROID

2) EXTERNAL LARYNGEAL NERVE - SUPPLIES CRICOTHYROID MUSCLE

-SENSORY NERVE -

1) INTERNAL LARYNGEAL NERVE - SUPPLIES - MUCOUS MEMBRANE UP TO LEVEL OF VOVAL FOLD

2) RECURRENT LARYNGEAL NERVE - SUPPLIES - MUCOUS MEMBRANE BELOW LEVEL OF VOVAL FOLD

SUBMANDIBULAR SALIVARY GLAND

*THIS IS LARGEST SALIVARY GLAND, SITUATED IN ANTERIOR PART OF DIGASTRIC TRIANGEL

*GLAND IS ABOUT - SIZE OF WALNUT WEIGHING ABOUT - 15 TO 20G

*POSTERIOR BORDER OF MYLOHYOID - DIVIDE - INTO LARGER PART -SUPERFICIAL TO MUSCLE
- SMALL PART - DEEP TO MUSCLE

*COVERINGS -

1) GLAND PARTIALLY ENCLOSED BETWEEN TWO LAYERS OF DEEP CERVICAL FASCIA
2) SUPERFICIAL LAYER OF FASCIA - COVER - INFERIOR SURFACE OF SURFACE & ATTACHED TO BASE OF MANDIBLE
3) DEEP LAYER - COVER - MEDIAL SURFACE OF GLAND & SUPERIORLY TO MYLOHYOID LINE OF MANDIBLE

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*SUPERFICIAL PART -

- THIS PART OF GLAND FILLS - DIGASTRIC TRIANGLE
- EXTENDS - SUPERIORLY DEEP TO MANDIBLE UP TO MYLOHYOID LINE

*DEEP PART -

- THIS PART IS SMALL IN SIZE
- LIE DEEP TO MYLOHYOID

*BLOOS SUPPLY & LYMPHATIC DRAINAGE :-

*ARTERY -FACIAL ARTERY

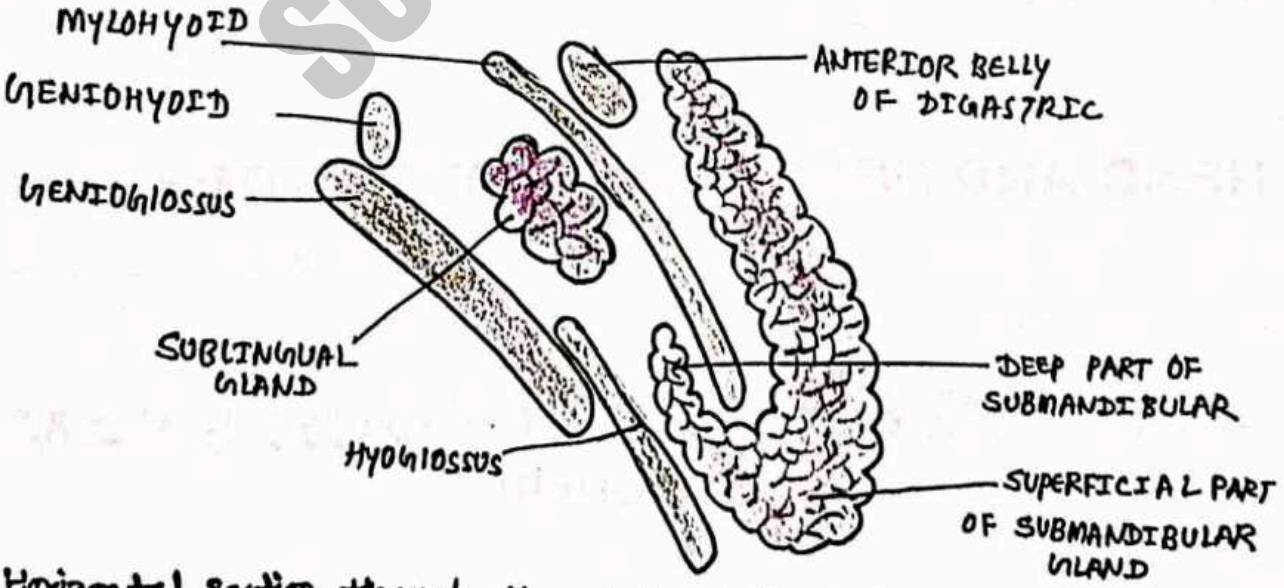
*VEIN DRAIN - FACIAL VEIN

*NERVE SUPPLY -

1) SENSORY FIBRES - LINGUAL NERVE

2) VASOMOTOR SYMPATHETIC FIBRES FROM - PLEXUS ON FACIAL ARTERY

3) SECRETOMOTOR FIBRES - SUPERIOR SALIVATORY NUCLEUS
↓
7TH CRANIAL NERVE
↓
CHORDA TYMPANIC BRANCH
↓
JOINS LINGUAL NERVE
↓
SUSPENDS SUBMANDIBULAR GANGLION
↓
SUBMANDIBULAR G



Horizontal section through the submandibular region showing the location of the submandibular and sublingual glands.

SURAJ KUMAR

PREVIOUS YEAR UNIVERSITY QUESTIONS

*LONG QUESTIONS - 20MARKS

1. CLASSIFY DURAL VENOUS SINUS & DESCRIBE CAVERNOUS SINUS IN DETAIL
2. DESCRIBE MIDDLE EAR UNDER FOLLOWING HEADINGS -
   1.LOCATION 2.CONTENT 3.BOUNDARY & NERVE SUPPLY 4.APPLIED
3. DESCRIBE TONGUE UNDER THE FOLLOWING HEADINGS -
   1.LOCATION & PARTS 2.NERVE SUPPLY & LYMPHATIC DRAINAGE 3.DEVELOPMENT
4. DESCRIBE ANTERIOR TRIANGLE & CAROTID TRIANGLE
5. DESCRIBE PALATINE TONSIL UNDER FOLLOWING HEADINGS -
   1.EXTERNAL FEATURES 2.BLOOD SUPPLY 3.HISTOLOGY
6. DESCRIBE THYROID GLAND OR PAROTID GLAND UNDER FOLLOWING HEADINGS -
   1.LOCATION 2.EXTERNAL FEATURE & RELATION 3.BLOOD & NERVE SUPPLY
7. DESCRIBE TEMPEROMADIBULAR JOINT

*SHORT QUESTIONS - 5MARKS

1. SCALP
2. LACRIMAL APPARATUS
3. POSTERIOR TRIANGLE
4. EXTERNAL CAROTID ARTERY
5. MAXILLARY ARTERY & MAXILLARY NERVE
6. MUSCLE MUSTIGATION
7. DEVELOPMENT OF SOF PALATES
8. HISTOLOGY DIAGRAM OF PALATINE TONSIL
9. EXTRA OCULAR MUSCLE
10. LARYNX
11 OTIC GANGLION
12. SUBMANDICULAR GLAND
13. LATERAL WALL OF NOSE

SURAJ KUMAR

HEAD AND NECK ANATOMY IS COMPLETED !!

IT ALWAYS SEEMS IMPOSSIBLE UNTIL IT'S DONE

ALL THE BEST FOR YOUR EXAMS

DR.JOHARI

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MCQ HEAD AND NECK

1) Paralysis of which muscle causes drooping of upper eyelid (ptosis) :
a Occipito-frontalis
b Orbicularis oculi
c levator palpebrae superioris
d superior rectus

2) Which one of the following is motor nerve of scalp:
a lesser occipital
b auriculo temporal
c posterior auricular
d supra orbital

3) Which of the following muscles is an abductor of vocal cords :
a posterior cricoarytenoid
b vocalis
c lateral cricoarytenoid
d cricothyroid

4) Which muscle is supplied by glossopharyngeal nerve
a stylohyoid
b stylopharyngeus
c salpingopharyngeus
d palatopharyngeus

5) Which muscle protrudes the tongue:
a genioglossus
b hyoglossus
c styloglossus
d palatoglossus

6) where does the nasolacrimal duct drains :
a superior meatus
b middle meatus
c inferior meatus
d sphenoethmoidal recess

7) Which muscle tenses the vocal cord
a Thyroarytenoid
b lateral cricoarytenoid
c posterior cricoarytenoid
d cricothyroid

8)All of the following pass through the superior orbital fissure except :
a ophthalmic artery
b ophthalmic vein
c occulomotor nerve
d trochlear nerve

9) Which of the following lies deep to the hyoglossus muscle
a Lingual nerve
b lingual artery
c submandibular gland
d submandibular ganglion

10) All of the following are intrinsic muscle of larynx except
a cricothyroid
b thyrohyoid
c thyroepiglottic
d Thyroarytenoid

11) All of the following are supplied by hypoglossal nerve except:
a genioglossus
b styloglossus
c hyoglossus
d palatoglossus

12) The parotid duct opens into vestibule of mouth at which level of molar tooth
a upper first
b lower first
c upper second
d lower second

13) Which nerve supplies the superior oblique muscle of the eye ball
a optic
b occulomotor
c trochlear
d abducens

14)stylopharyngeus muscle recieves its motor innervations from which nerve
a facial
b glossopharyngea
c vagus
d hypoglossal

15) Which of the following layer of the scalp is known as dangerous layer
a skin
b connective tissue
c loose areolar tissue
d pericranium

16) The structure that does not pass through the parotid gland is:
a facial nerve
b external carotid artery
c retromandibular nerve
d auriculo temporal nerve

17) All the following nerves traverses carotid triangle except
a glossopharyngeal
b vagus
c spinal part of acessory
d hypoglossal

18) Muscle responsible for depression of mandible
a masseter
b temporalis
c medial pterygoid
d lateral pterygoid

1)C 2)C 3)A 4)B 5)A 6)C 7)D 8)A 9)B 10)B 11)D 12)C 13)C 14)B 15)C 16)D 17)A 18)D

SURAJ KUMAR BBS

NEUROANATOMY BASICS

NEURON - PARTS OF NEURON:
- CELL BODY
- AXON

NUCLEUS - AGGREGATION OF CELL BODIES IN CENTRAL NERVOUS SYSTEM (CNS)
-DERIVED FROM NEUROECTODERM

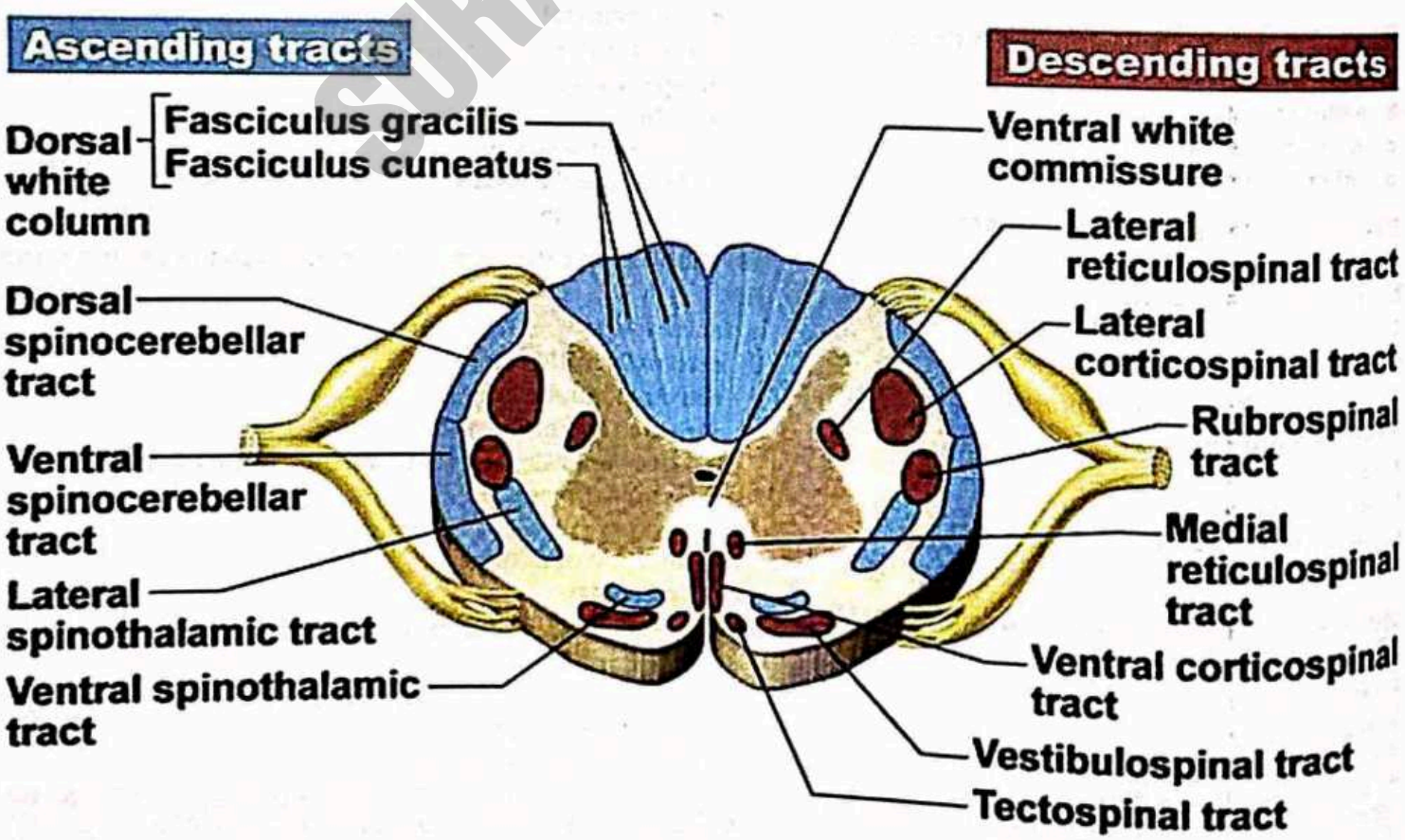
GANGLION - AGGREGATION OF CELL BODIES IN PERIPHERAL NERVOUS SYSTEM (PNS)
-DERIVED FROM NEURAL CREST CELLS

*THE NUCLEUS IS ALSO CALLED GREY MATTER.
*THE AXONS ARE ALSO CALLED WHITE MATTER/FASCICULUS/LEMNISCUS.
*THE COLLECTION OF AXONS IN CNS IS CALLED THE TRACT.
*THE COLLECTION OF AXONS IN PNS IS CALLED THE NERVE.

ASCENDING & DESCENDING TRACT

1) ASCENDING TRACTS ARE SENSORY PATHWAYS THAT BEGIN AT THE SPINAL CORD AND STRETCH ALL THE WAY UP TO THE CEREBRAL CORTEX.

2) DESCENDING TRACTS ARE THE PATHWAYS BY WHICH MOTOR SIGNALS ARE SENT FROM THE BRAIN TO THE SPINAL CORD. THEY ARE ALSO CALLED MOTOR TRACTS, DUE TO THEIR INVOLVEMENT IN MOVEMENT COORDINATION.



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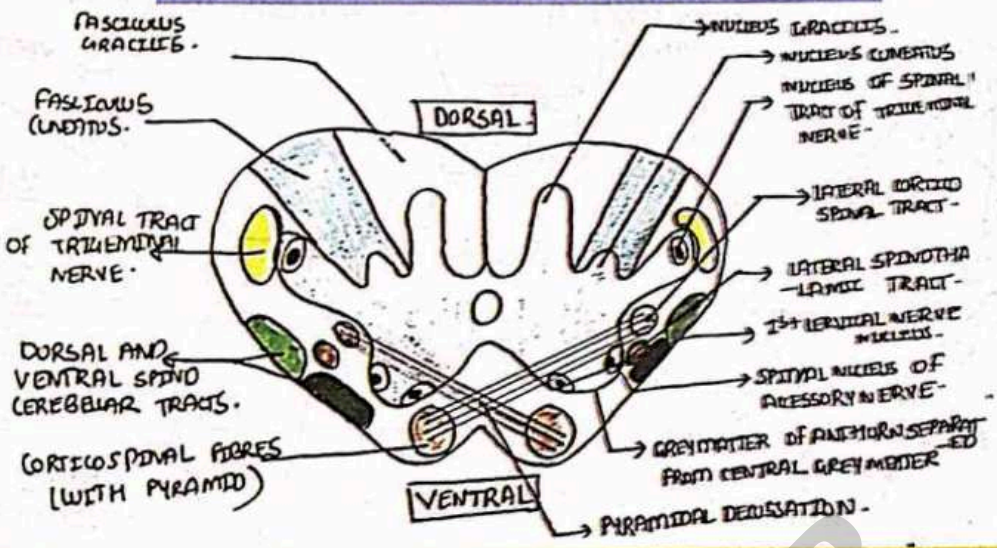
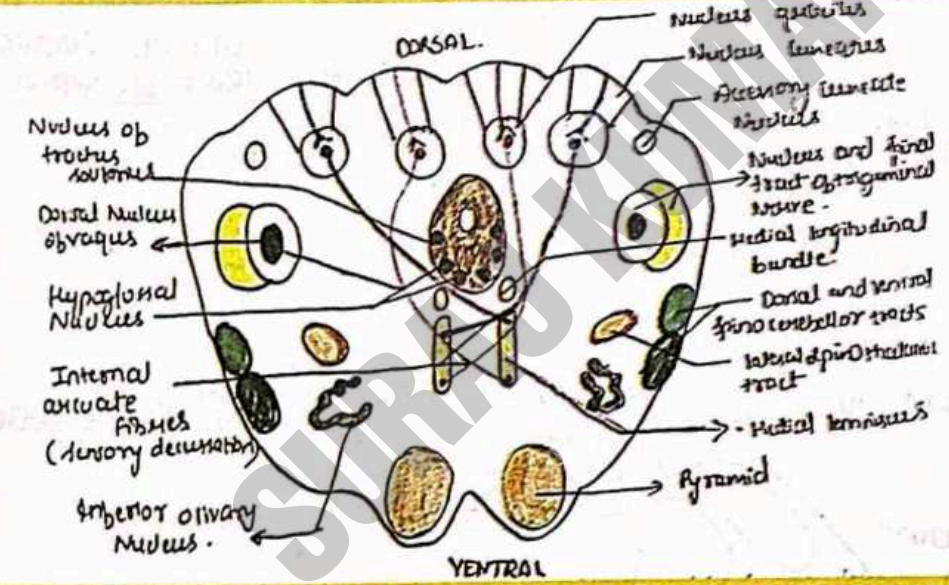
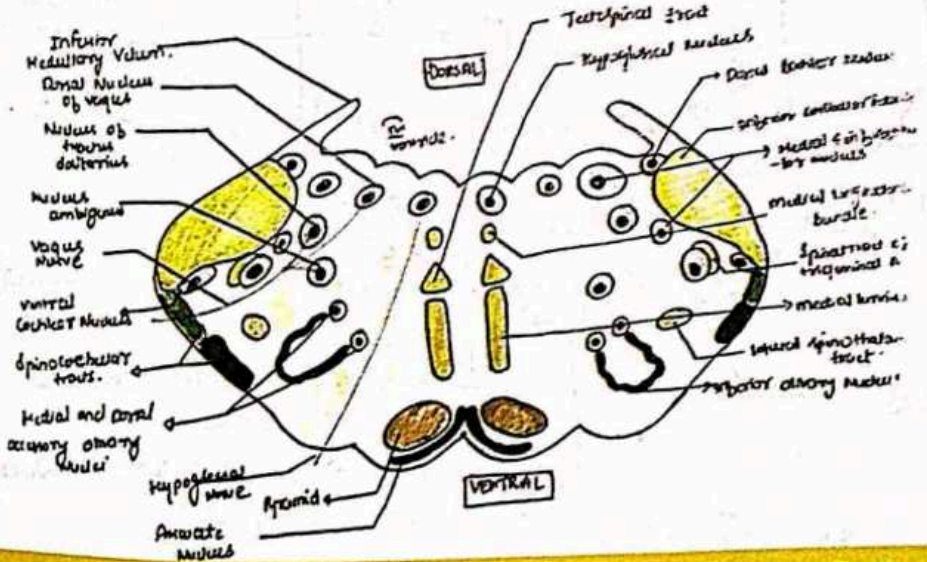
BRAIN STEM



TS OF MEDULLA AT LEVEL OF PYRAMIDAL DECUSSATION



TS OF MEDULLA AT LEVEL OF SENSORY DECUSSATION



TS OF MEDULLA AT LEVEL OF OLIVARY NUCLEUS

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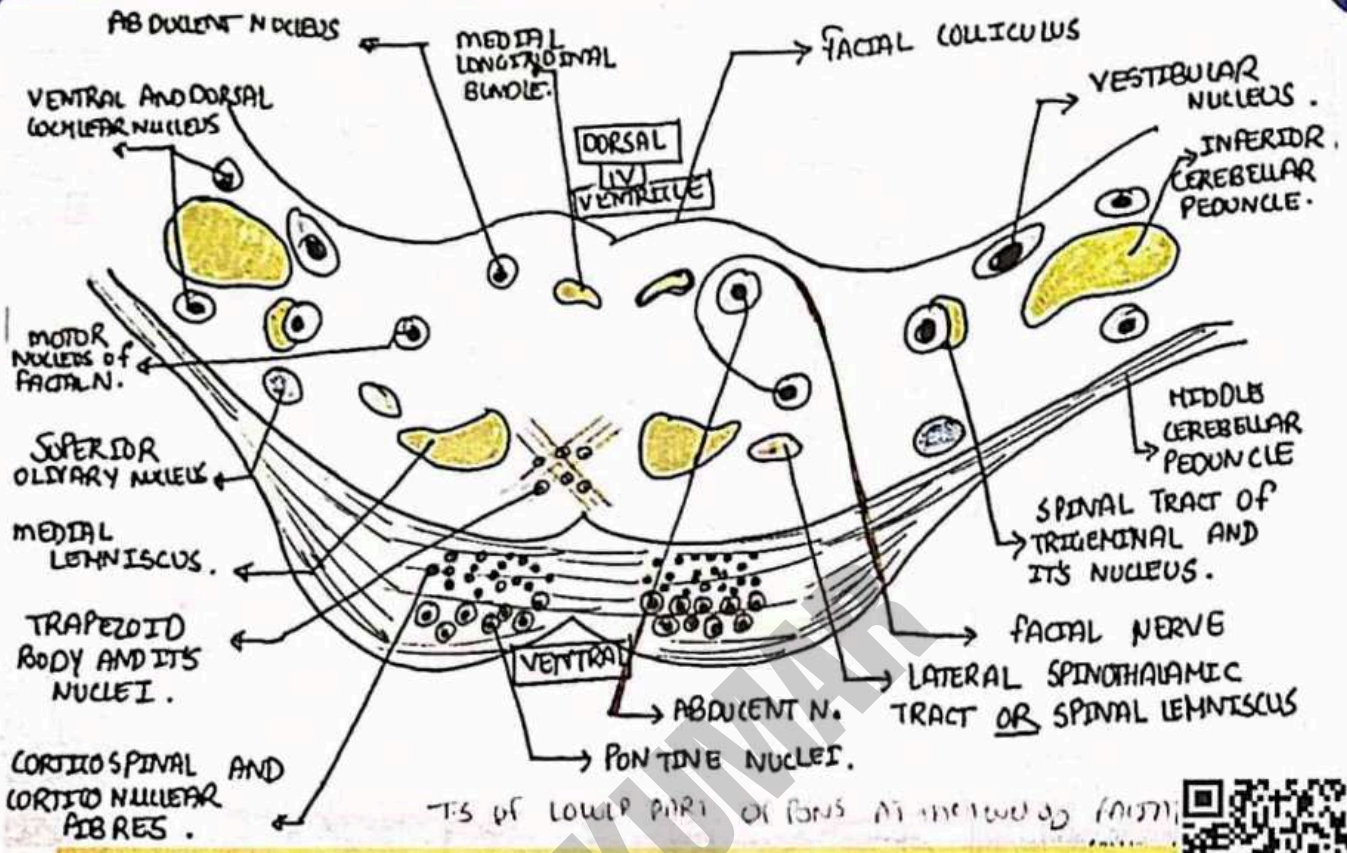
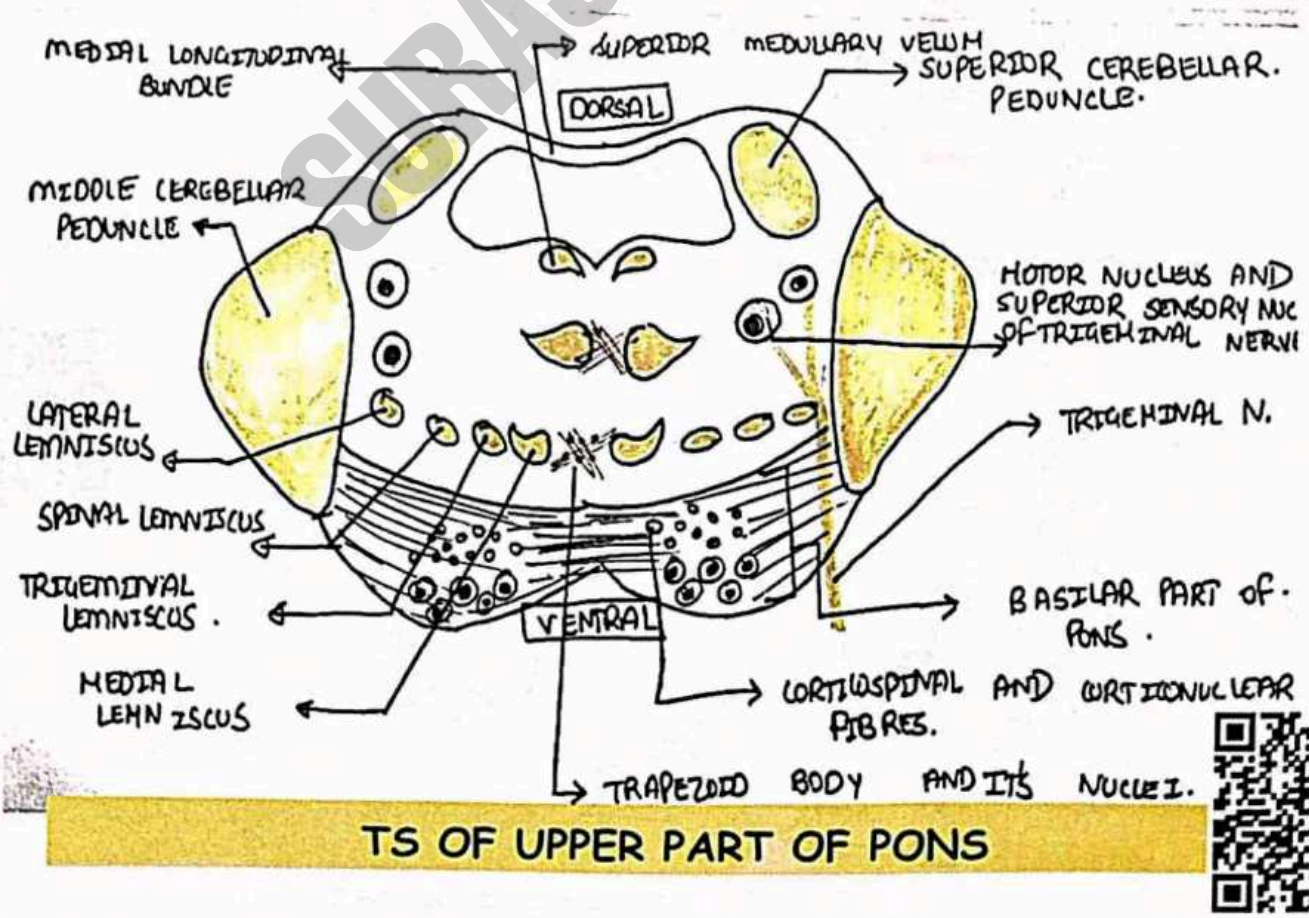


TS OF OF PONS AT FACIAL COLLICULUS



TS OF UPPER PART OF PONS

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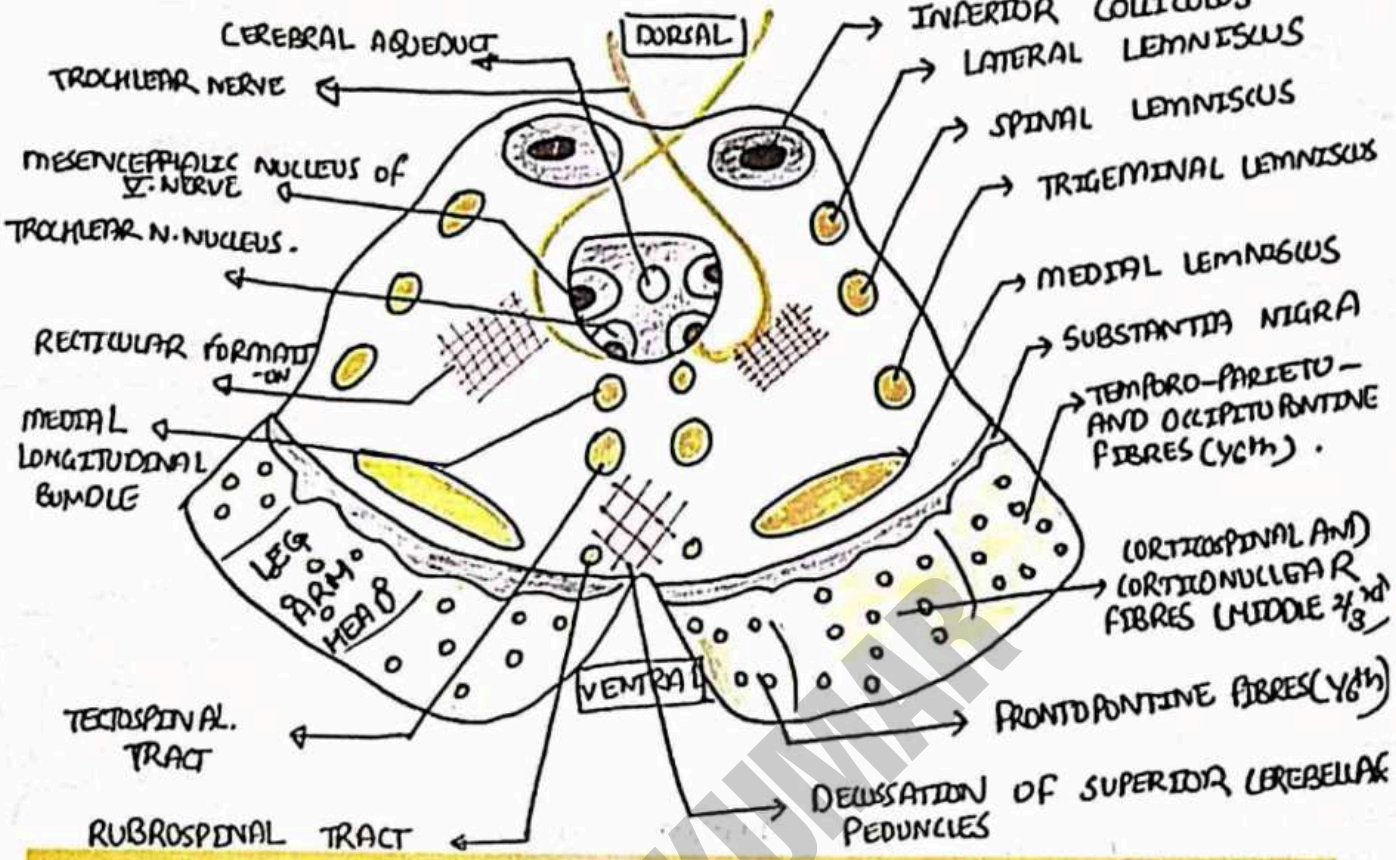
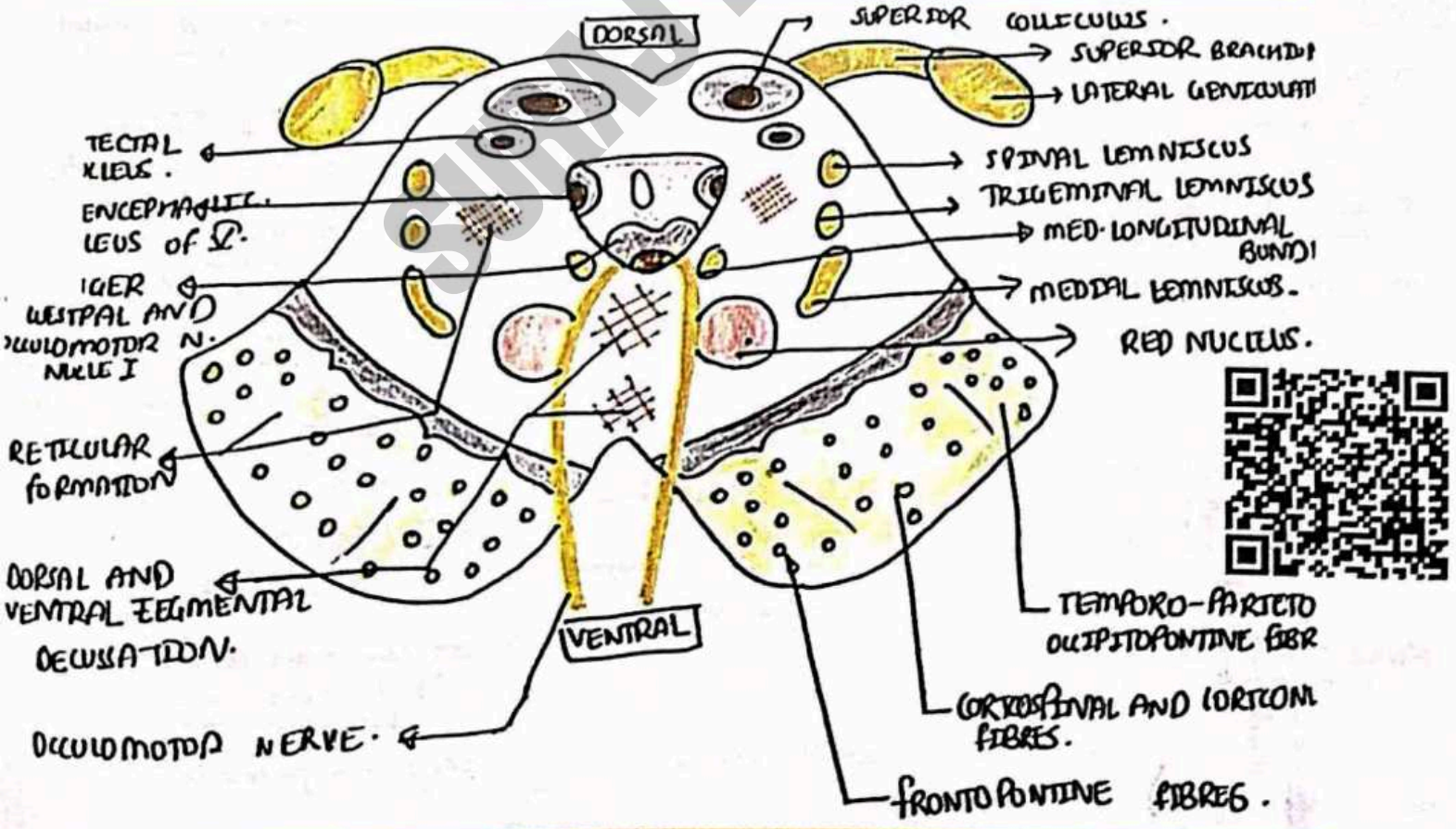


TS OF MIDBRAIN AT LEVEL OF INFERIOR COLLICLUS



TS OF MIDBRAIN AT LEVEL OF SUPERIOR COLLICLUS

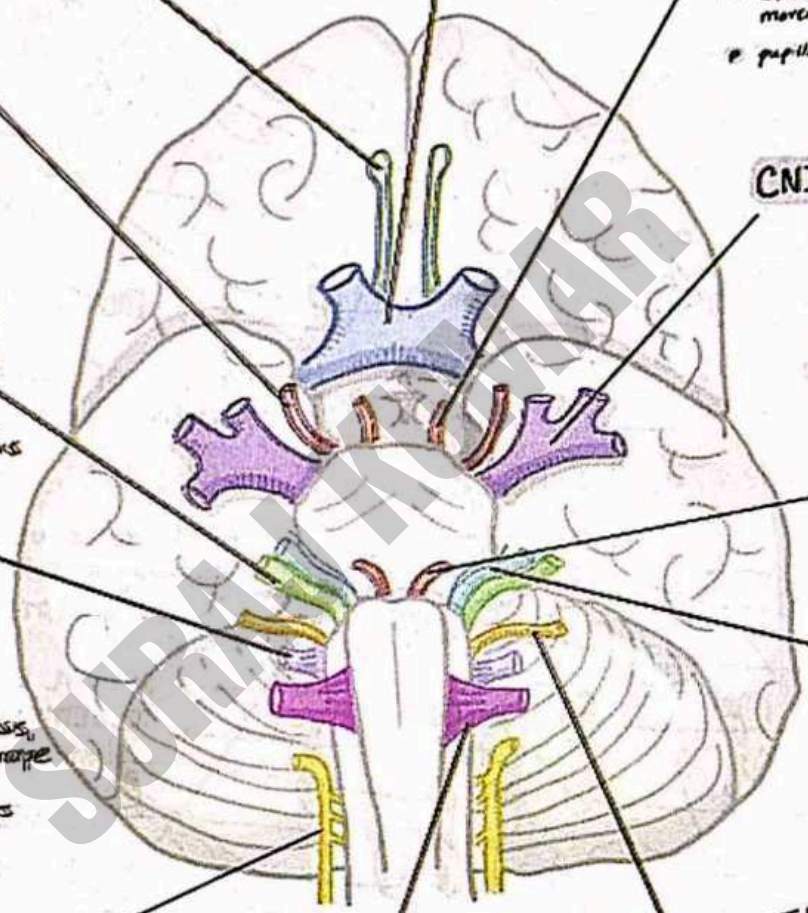
SUNIL KUMAR

CRANIAL NERVES

MAI STEWART

ANATOMY & FUNCTIONS

FUNCTIONS
— Legend —
- S Sensory
- M Motor
- P Parasympathetic
- SM Sympathetic

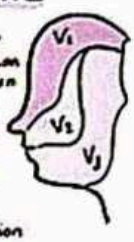


CN I: OLFACTORY
S smell

CN II: OPTIC
S sight

CN III: OCULOMOTOR
M eyeball movement: MR, IR, SR, IO
movement of upper eyelid: LP
P pupillary light reflex

CN IV: TROCHLEAR
M eyeball movement: superior oblique

CN V: TRIGEMINAL


V_1: OPTHALMIC
S facial sensation
SM pupillary dilation
V_2: MAXILLARY
S facial sensation
V_3: MANDIBULAR
M chewing
S facial sensation

CN VIII: VESTIBULO-COCHLEAR
S Hearing: cochlear
Balance: vestibular apparatus

CN VI: ABDUCENS
M eyeball movement
-lateral rectus

CN X: VAGUS
M swallowing, elevation of palate, hoarseness, coughing, gag reflex
P heart rate, GIT peristalsis, secretion, vasoconstriction
SM peripheral chemoreceptors -hyperglycemia

CN VII: FACIAL
S taste: anterior 2/3 of tongue
M facial expression
eyelid closing
stapedius muscle: dampens sound
P lacrimation
salivation – submandibular + sublingual glands

CN XI: ACCESSORY
M head turning: SCM
shrugging: trapezius

CN XII: HYPOGLOSSAL
M most muscles of tongue (except palatoglossus)
→ tongue movement

CN IX: GLOSSO-PHARYNGEAL
S sensation from posterior 1/3 of tongue, tonsils, middle ear, pharynx
taste from posterior 1/3 of tongue
M Stylopharyngeus muscle - elevates pharynx + larynx
P salivation – parotid gland

FUNCTION MNEMONIC
S-sensory M-motor B-both

Some Say Marry Money But My Brother
Cranial Nerve: I II III IV V XI XII

Says Big Brains Matter More
VIII IX X XI XII

SURAJ KUMAR

YouTube

OLFACTORY NERVE

- 1st cranial nerve

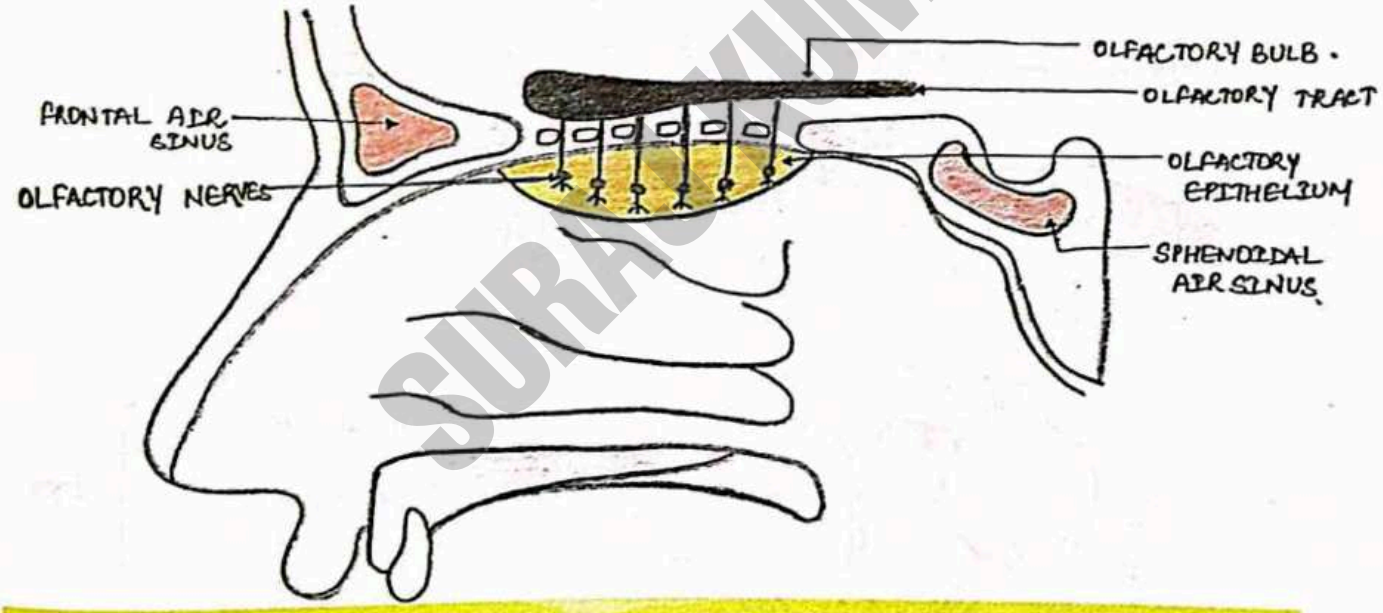
NEURONS

- Receptors & 1st neuron — olfactory cells (in olfactory par of nasal mucosa)
  olfactory neurons (central processes of olfactory cells
- 2nd neuron — located in primary olfactory cortex
- 3rd neuron — located in primary olfactory cortex
- 4th neuron — fibers goes to secondary olfactory cortex

COURSE

. Fibers from primary receptor neurons pass through → foramina in cribrif plate of ethmoid
↓ enter
Anterior cranial fo ← terminate ← olfactory bulb

. Nerves distributes — olfactory region of nasal cavity.



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OLFACTORY NERVE

OPTIC NERVE

- 2nd cranial nerve
- "Nerve of sight"

COURSE

- Fibers arise from ganglionic cells of retina of eyeball converge ↓ towards optic disc pierce → . outer layer of retina
  . choroid
  . sclera
  leaves → eyeball

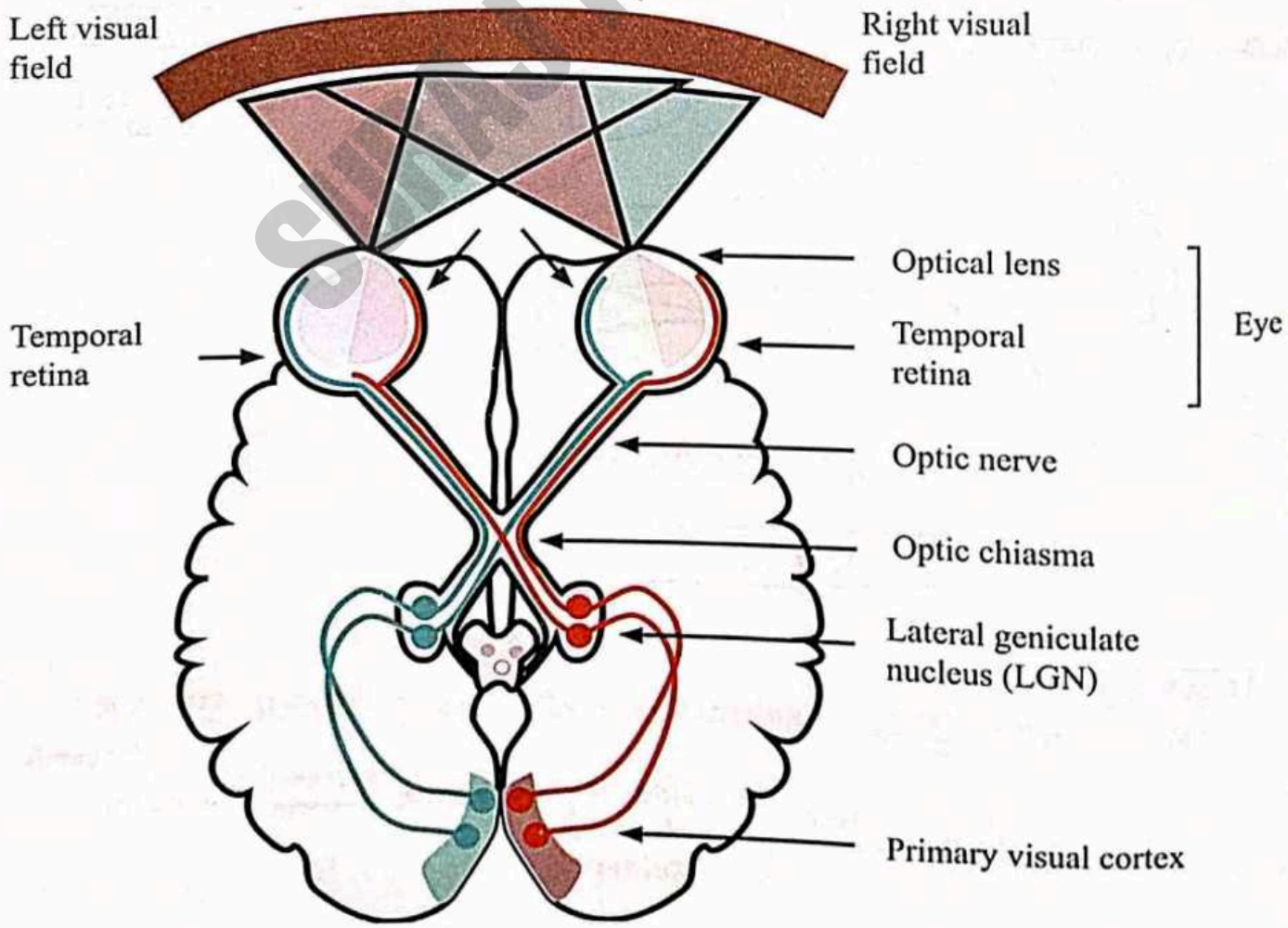
- Fibers unite & form → **OPTIC NERVE** runs → posteromedially
  ↓ through
  posterior half of orbit
  ↓ enters
  middle cranial fossa, through optic canal.
  ↓ unites to form
  OPTIC CHIASMA

- Optic nerve divided into →
  - intra-orbital part
  - canalicular part
  - intra-cranial part

- Enclosed by 3 meninges of brain

RELATIONS

- At apex ⟶ Recti muscles, ciliary ganglion
- Superior ⟶ ophthalmic artery, nasociliary nerve, superior ophthalmic vein
- Inferior ⟶ nerve to medial rectus.



SUNIL KUMAR

OCCULOMOTOR NERVE

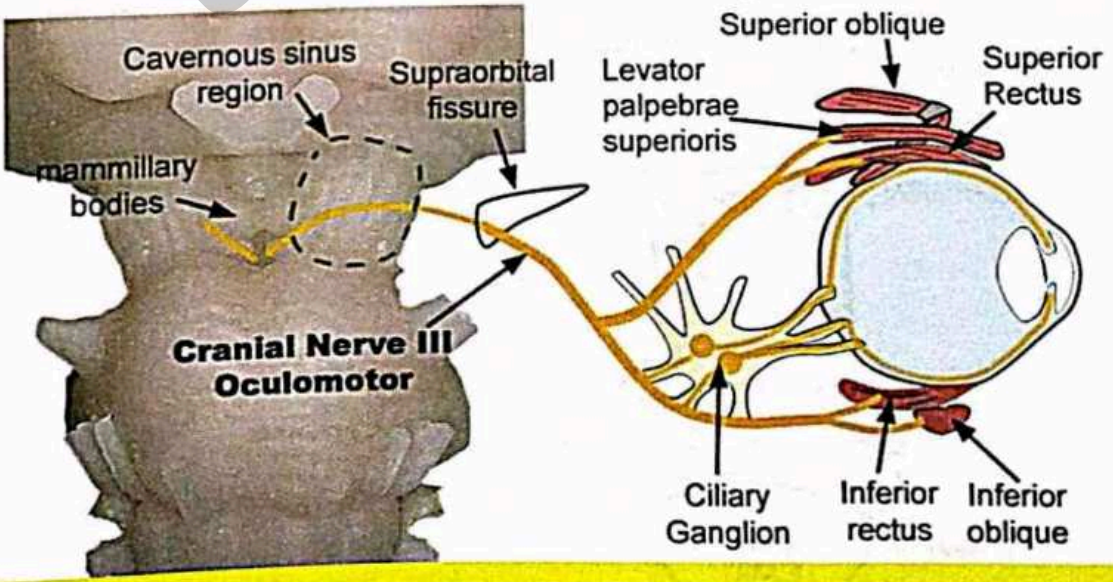
- 3rd cranial nerve
- Distributed to extraocular & intraocula muscles.

NUCLEUS
- Occulomotor nucleus

COURSE

Intraneural Course
- Nerve pass → ventrally through → tegmentum, red nucleus & substantia nigra
- Nerve attached → occulomotor sulcus on medial side of crus cerebri
- Nerve passes between → Superior cerebellar & posterior cerebellar arteries
  - runs forward in interpeduncular cistern, lateral side to posterior communicating artery
  - reach → cavernous sinus
  - descends → lateral wall of sinus, & lies above trochlear nerve
  - Anterior part of sinus → nerve divides → Upper division, Lower division
- Divisions enter → orbit through → middle part of superior orbital fissure

In orbit
- → Smaller upper division supplies → Superior rectus, Levator palpebrae superioris
- → larger lower division divides into → Medial rectus, Inferior rectus, Inferior oblique (↑ longest)
  - supplies → inferior oblique muscle.



COURSE OF TROCHLEAR NERVE

SURAJ KUMAR

TROCHLEAR NERVE

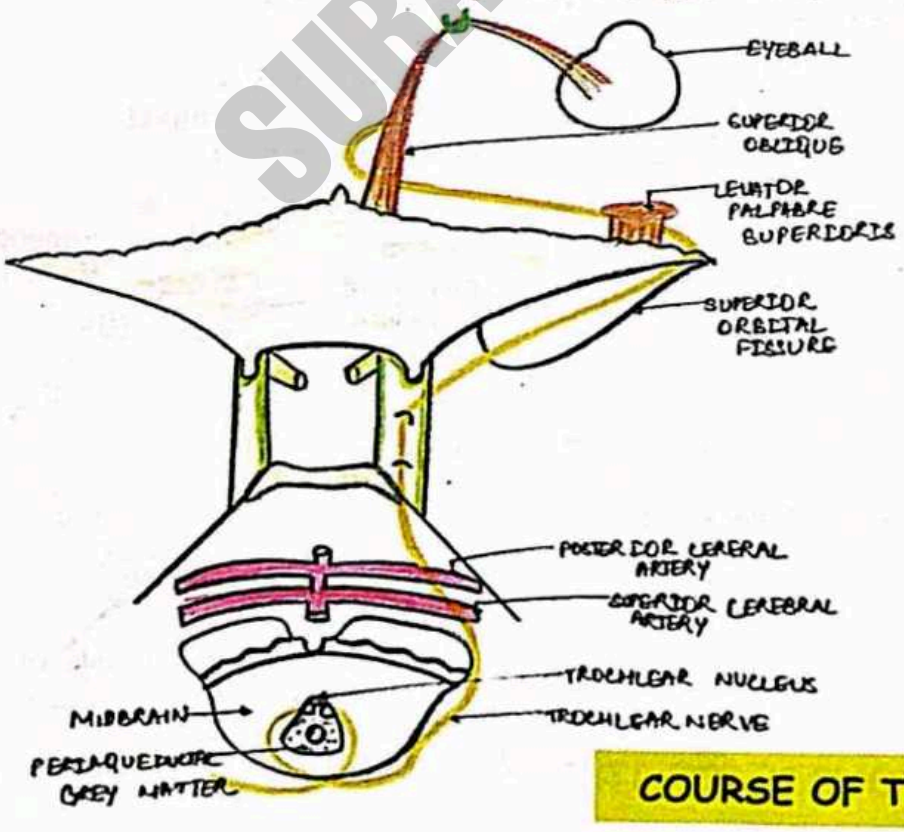
- 4th cranial nerve
- Supplies only superior oblique muscle of eyeball

NUCLEUS - Trochlear nucleus

COURSE

Intraneural Course

- Nerve runs dorsally round central grey matter → reach upper part of superior & anterior medullary velum
- Nerve attached each side – superior medullary velum.
  - ★ Only nerve to emerge on dorsal aspect of brainstem.
- Nerve enters cavernous sinus piercing posterior corner of roof → runs forward in lateral wall of cavernous sinus, between occulomotor & ophthalmic nerves
  - ★ It has longest intracranial course.
- Nerve enters orbit through lateral part of superior orbital fissure
- In orbit runs medially, above origin of levator palpebrae superioris → supply superior oblique muscle on its orbital surface



COURSE OF TROCHLEAR NERVE

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TRIGEMINAL NERVE

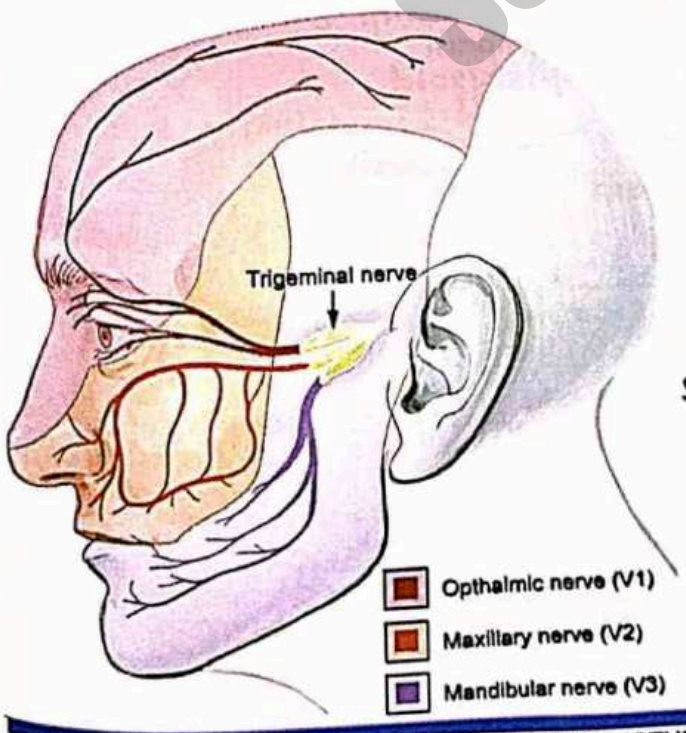
- 5th cranial nerve · Largest · Nerve of 1st brachial arch
- Consists of — Ophthalmic, Maxillary, Mandibular

NUCLEUS

1. General somatic afferent column
   a. Spinal nucleus of V nerve
   b. Principal sensory nucleus of V nerve
   c. Mesencephalic nucleus
2. Branchial efferent column

COURSE

- Trigeminal Nerve
  - larger lateral sensory root → runs forward & lateral → enter middle cranial fa → Trigeminal (Gasserian) ganglia in shallow fossa of petrous temporal bon
    - Ophthalmic Nerve → enters lateral wall of cavernous sinus, below trochlear nerve → branches: nasociliary, Lacrimal, frontal → enter orbit → through superior orbital fissure
    - Maxillary Nerve → enters lateral wall of cavernous sinus, lowest position → enter pterygopalatine fossa → through foramen rotundum
    - Mandibular Nerve → enter infratempora fossa → throu foramen ovale
  - smaller medial motor root → runs forward & laterally → enters infratemporal fossa, through foramen ovale → joins Mandibular nerve



SUBAJ KUMAR

BRANCHES

OPHTHALMIC NERVE (SENSORY)

- FRONTAL
  - Supratrochlear → Upper eyelid, Conjunctiva, lower part of forehead
  - Supra orbital → Upper eyelid, frontal air sinus, forehead, scalp
- NASOCILIARY
  - long ciliary → eyeball
  - Ciliary ganglion
  - Ethmoid
    - Anterior
    - posterior
  - Infratrochlear → Both eyelid, lacrimal sac
- LACRIMAL → lacrimal gland

MAXILLARY NERVE (SENSORY)

- MIDDLE CRANIAL FOSSA — Meningeal branch
- PTERYGOPALATINE FOSSA
  - Ganglionic branch
  - Zygomatic
    - → Zygomaticotemporal
    - → Zygomaticofacial
  - Posterior superior alveolar
- INFRAORBITAL CANAL
  - Middle superior alveolar
  - Anterior superior alveolar
- FACE — Infraorbital
  - → Palpebral
  - → Labial
  - → Nasal

MANDIBULAR NERVE (SENSORY + MOTOR)

- TRUNK
  - Meningeal
  - Medial pterygoid
    - → tensor veli palatini
    - → tensor tympani
    - → medial pterygoid
- ANTERIOR DIVISION
  - Deep temporal
  - Lateral pterygoid
  - Masseteric
  - Buccal
- POSTERIOR DIVISION
  - Auriculotemporal
  - Lingual
  - Inferior alveolar

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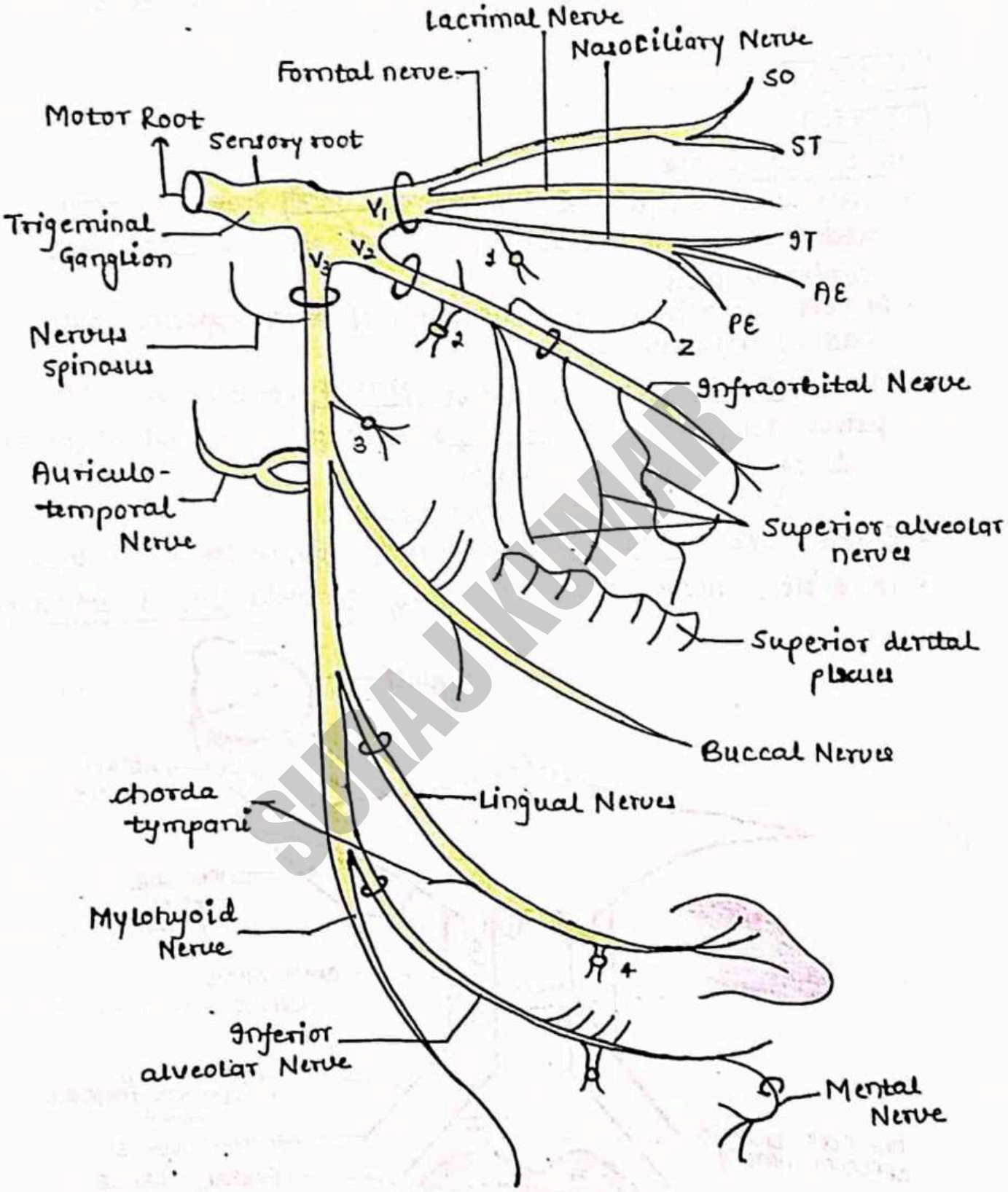

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BRANCHES OF TRIGEMINAL NERVE

ABDUCENT NERVE

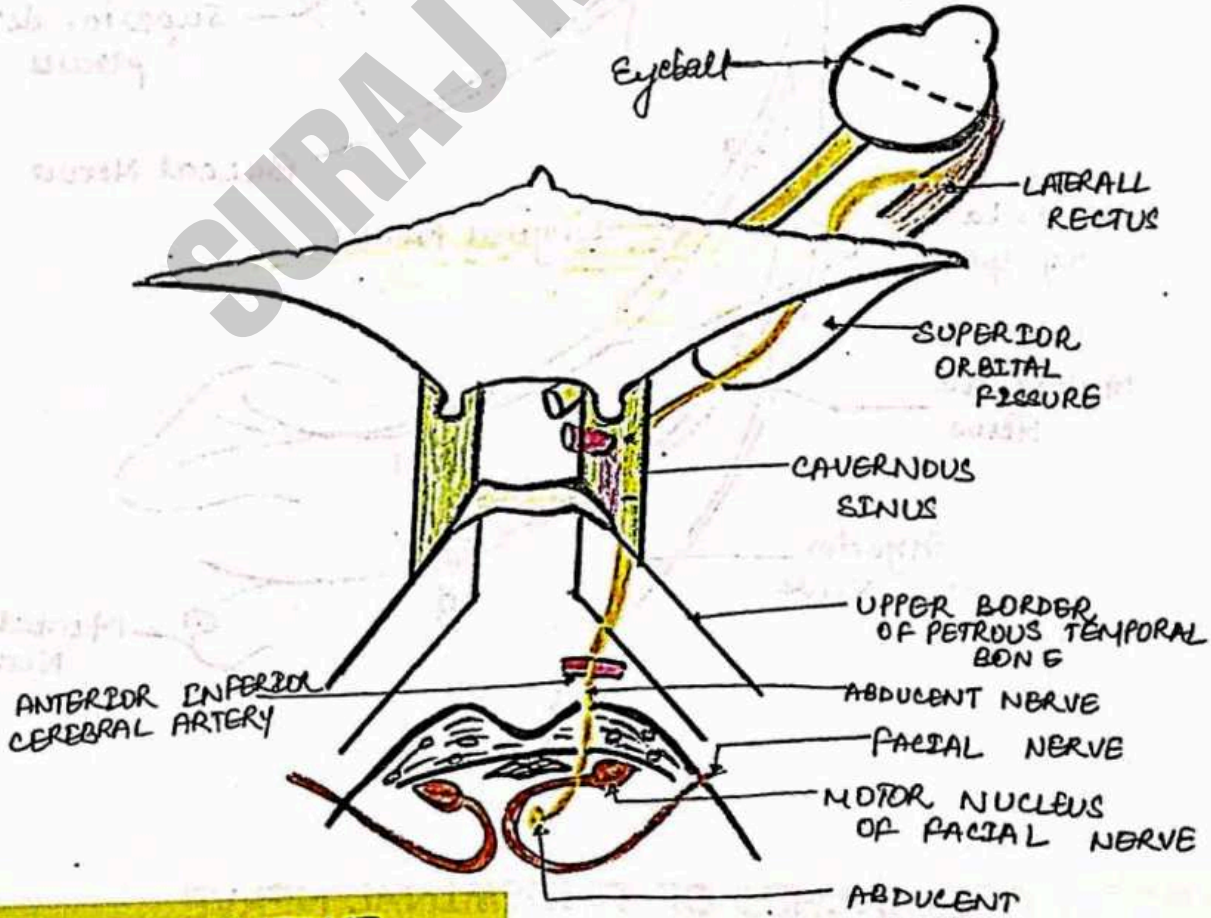
- It is the 6th cranial nerve and supplies lateral rectus muscle of eyeball.

NUCLEUS Abducent nucleus.

COURSE

Intraneural Course

- Fibers run ventrally & downward through trapezoid body, medial lemniscus & basilar part of pons to reach lower border of pons.
- Nerves attached to lower end of pons, opposite upper end of pyramid of medulla
- Nerve enters cavernous sinus piercing posterior wall of petrous temporal bone passing beneath petrosphenoidal ligament & bends sharply forwards.
  - It has long intracranial course
- Enters orbit through middle part of superior orbital fissure.
- In orbit, nerve ends by supplying only Lateral rectus musc



ABDUCENT NERVE

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FACIAL NERVE

- FACIAL NERVE IS NERVE OF SECOND BRANCHIAL ARCH

*NUCLEI : FIBRES OF NERVE ARE CONNECTED TO 4X NUCLEI→ SITUATED IN LOWER PONS
- MOTOR NUCLEUS
- SUPERIOR SALIVATORY NUCLEUS OR PARASYMPATHETIC
- LACRIMATORY NUCLEI - PARA SYMPATHETIC
- NUCLEUS OF TRACTUS SOLITARIUS

*COURSE :-
- ATTACHED TO BRAINSTEM BY 2 ROOTS, MOTOR AND SENSORY
- TWO ROOTS OF FACIAL NERVE ARE ATTACHED TO LATERAL PART OF PONS
- TWO ROOTS RUN LATERALLY AND FORWARD WITH, 8TH CRANIAL NERVE TO REACH---→INTERNAL ACOUSTIC MEATUS

*IN MEATUS → 2X ROOTS, SENSORY AND MOTOR FUSE
- 1ST PART --→ DIRECTLY LATERALLY ABOVE VESTIBULE
- 2RD PART ---→ RELATED TO MEDIAL WALL ABOVE - PROMONTORY.
- 3RD PART→ DIRECTLY VERTICALLY DOWNWARD BEHIND PROMONTORY

-FACIAL NERVE LEAVE SKULL BY PASSING THROUGH - STYLOMASTOID FORAMEN

*EXTRACRANIAL COURSE :-

- FACIAL NERVE CROSSES THE -> LATERAL SIDE OF BASE OF STYLOID PROCESS
- IT ENTERS - POSTERMEDIAL SURFACE OF PAROTID GLAND
- RUNS FORWARD THROUGH THE GLAND CROSSING -→ RETROMANDIBULAR VEIN & ECA
- BEHIND THE NECK OF MANDIBLE - IT DIVIDE INTO 5X TERMINAL BRANCHES EMERGE ALONG ANTERIAR BORDER OF PAROTID GLAND

*BRANCHES AND DISTRIBUTION-

1 WITHIN THE FACIAL CANAL: A. GREATER PETROSAL NERVE
B. THE NERVE TO THE STAPEDIUS
C. THE CHORDATYMPANI

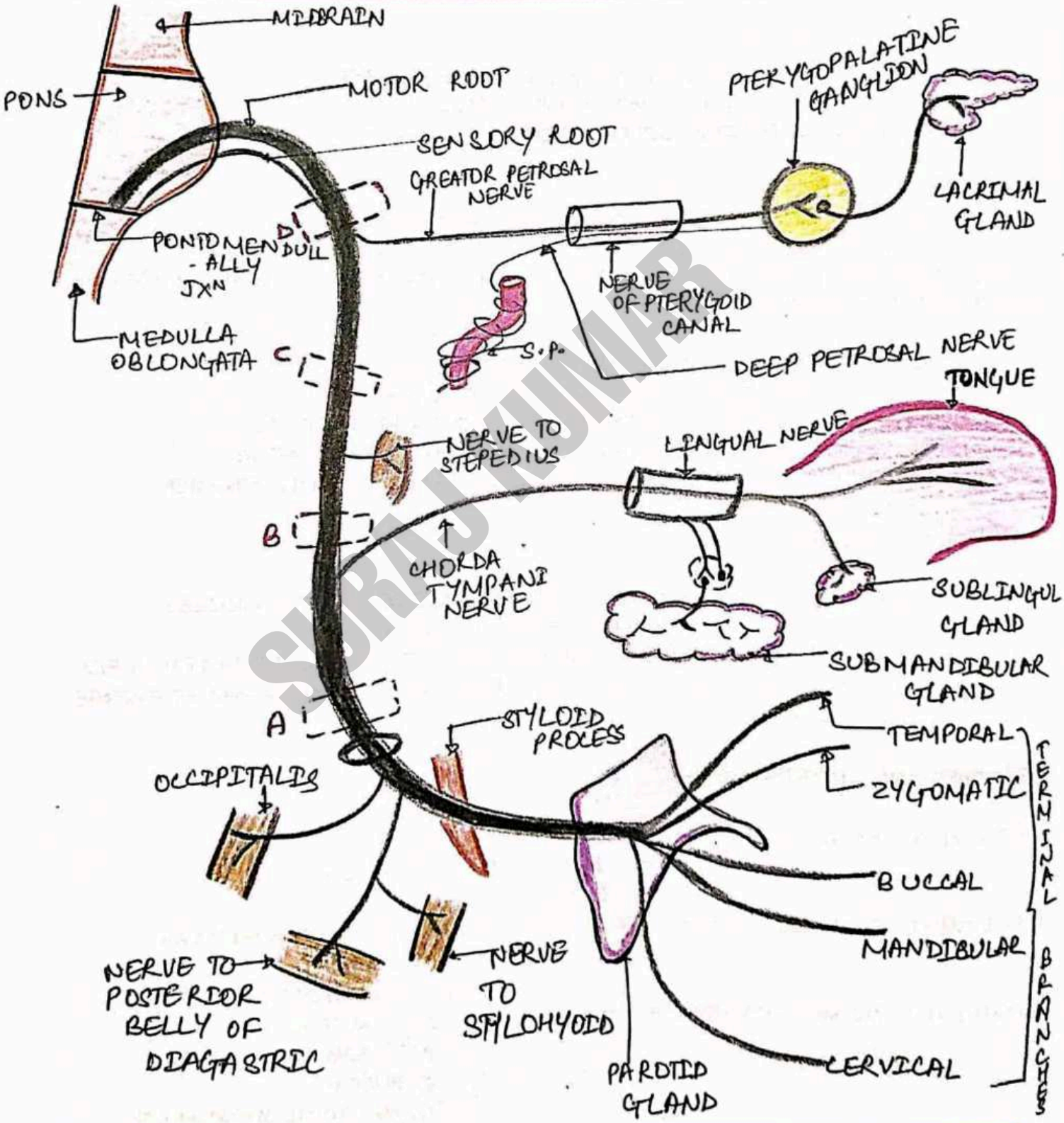
2 AT ITS EXIT FROM THE STYLOMASTOID FORAMEN: A. POSTERIOR AURICULAR
B. DIGASTRIC
C. STYLOHYOID.

3 TERMINAL BRANCHES WITHIN THE PAROTID GLAND: A. TEMPORAL
B. ZYGOMATIC
C. BUCCAL
D. MARGINAL MANDIBULAR
E. CERVICAL.

SURAJ KUMAR

CLINICAL ANATOMY

*BELLS PALSY : SUDDEN PARALYSIS OF NERVE AT THE STYLOMASTOID FORAMEN RESULT IN ASYMMETRY OF CORNER OF MONTH , INABILITY TO CLOSE THE EYE . OSS OF WRINKLING OF SKIN OF FOREHEAD ON THE SAME SIDE



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COURSE OF FACIAL NERVE

VESTIBULO-COCHLEAR NERVE

- It is the 8th cranial nerve.
- It comprises of — hearing & vestibular parts.

PATHWAY OF HEARING

1st neuron — in spiral ganglion, bipolar
- Peripheral → organ of Corti
  Central → Cochlear nerve
- Terminates in dorsal & ventral cochlear nuclei.

2nd neuron — in dorsal & ventral cochlear nuclei

3rd neuron — in superior olivary nucleus
- Axons form lateral lemniscus terminates inferior colliculus.

4th neuron — in inferior colliculus
- Axon pass through inferior brachium — reach medial geniculate body.

5th neuron — in medial geniculate body
- Axon form auditory radiation, passes through sublentiform part of internal capsule, to reach auditory area in temporal lobe

VESTIBULAR PATHWAY

- Receptors are maculae of saccule & utricle (static balance), & crista of ampullaris of semicircular ducts (kinetic balance)
- Fibres of crista of posterior semicircular canal lie → foramen singulare
- Most of fibres from maculae of utricle & saccule lie → in inferior vestibular area.
- Impulses can influence movements of — eyes, head, neck, trunk

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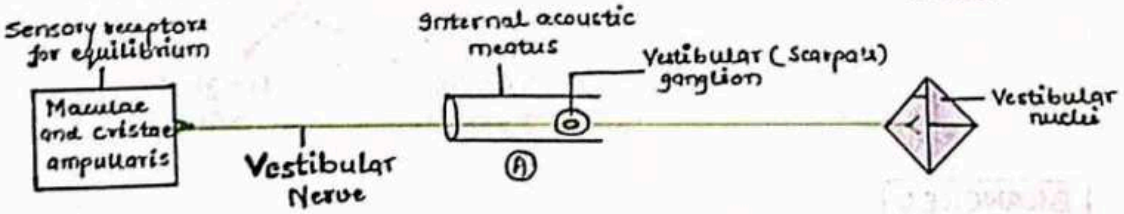

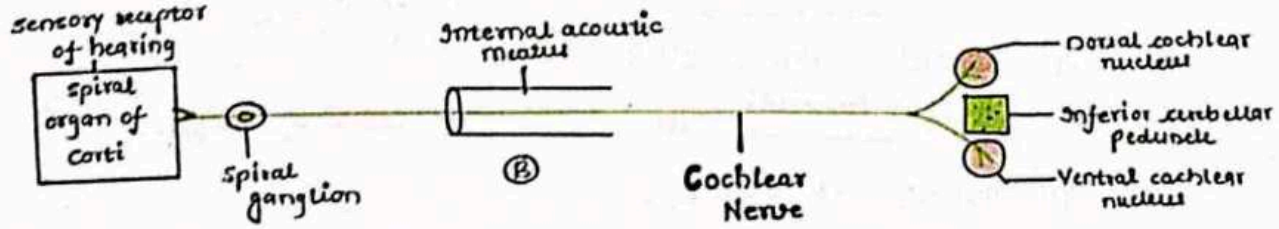

Ⓐ Vestibular Nerve Ⓑ Cochlear Nerve

GLOSSOPHARYNGEAL NERVE

- It is the 9th cranial nerve and nerve of third branchial arch.

NUCLEI The 3 nuclei in upper part of medulla are :—

(1) Nucleus Ambigus
(2) Inferior salivatory Nucleus
(3) Nucleus of tractus solitarius

COURSE

Intraneural Course

- Fibers pass forward & laterally, between olivary nucleus & inferior cerebellar peduncle, through reticular formation of medulla.
- At base of brain — Just above rootlets of Vagus nerve, nerve is attached by 3 to 4 filaments to upper part of posterolateral sulcus of medulla.

Intracranial course

- Filaments unite → forms single trunk → passes forward & laterally → JUGULAR FORAMEN
- Nerve is lodged in deep groove → leads to → COCHLEAR CANALICULUS
- Nerve is separated from vagus & Accessory nerves, by inferior petrosal sinus.

Extracranial Course Nerve descends

- Between Internal Jugular vein & Internal carotid Artery (deep to styloid process & muscles attached to it
- then turns forwards — winding stylopharyngeus & supplies pharynx to give pharyngeal branches
- Enters submandibular region — by passing deep to hyoglossus → breaks into → tonsillar branch, lingular branch.

BRANCHES

(1) Tympanic branch (Jacobson's nerve) → gives off → lesser petrosal nerve
(2) Carotid nerve (nerve of Herring) → supplies → carotid sinus & carotid body.
(3) Pharyngeal branches
(4) Muscular branches → supply → stylopharyngeus

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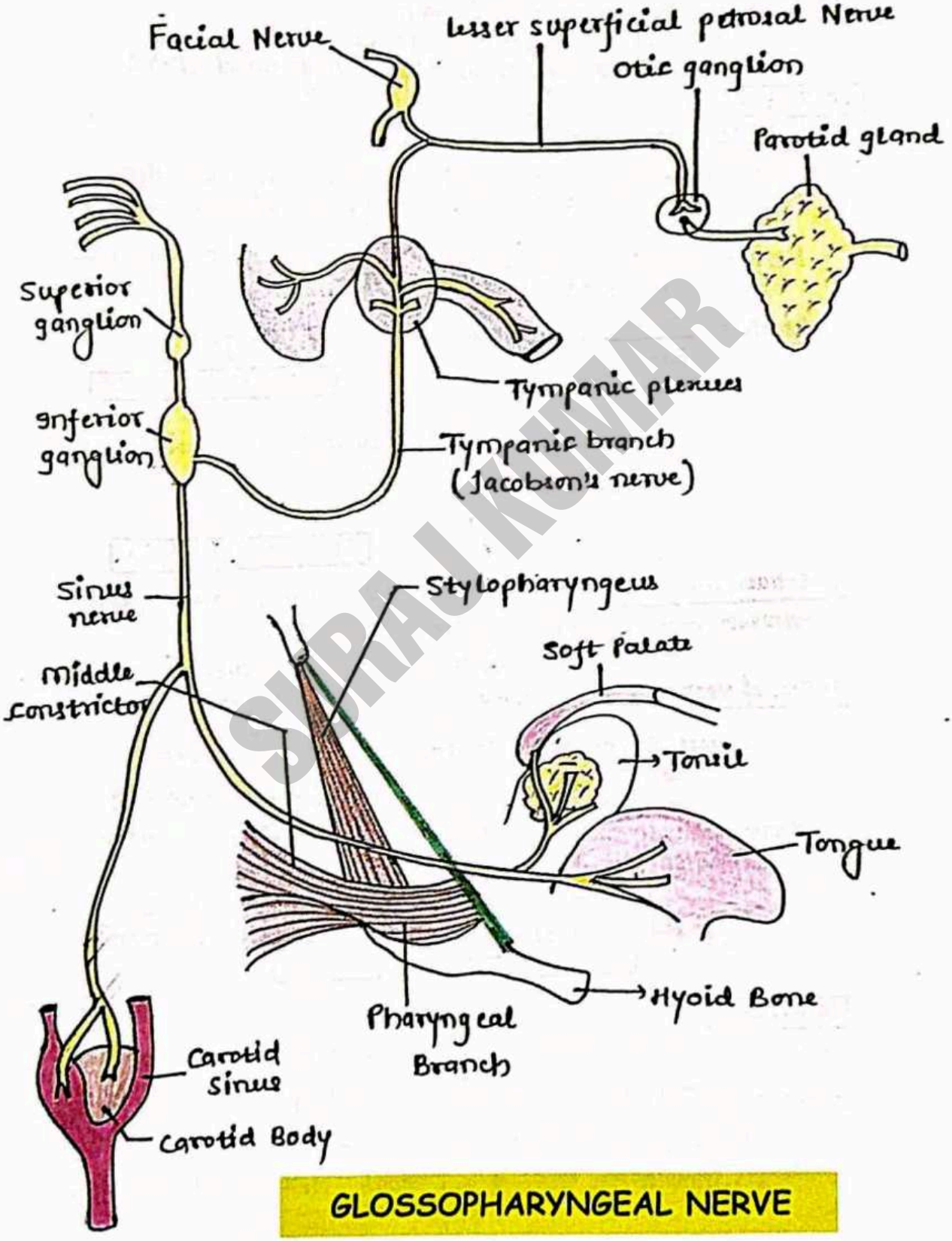
(5) Tonsillar branches <u>supply</u> → tonsil, fauces & palate

(6) Lingual branches <u>supply</u> → posterior 1/3 tongue, vallate papillae



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GLOSSOPHARYNGEAL NERVE

VAGUS NERVE

- 10th cranial nerve
- Named - so for its extensive (vague) course

NUCLEI

(1) Nucleus Ambigus
(2) Dorsal nucleus of vagus
(3) Nucleus of tractus solitarius
(4) Nucleus of spinal tract of trigeminal nerve

COURSE

In Intracranial Course

- Fibers run forward & laterally, between olivary nucleus & inferior cerebellar peduncle, through reticular formation of medulla.
- Nerve attached by about 10 rootlets to posterolateral sulcus of medulla.
- Rootlets → (unites & forms) → large trunk → passes laterally across → **JUGULAR TUBERCLE** with Glossopharyngeal & Accessory nerves → reaches → **JUGULAR FORAMEN**

In Extracranial Course

- Nerve descends within Carotid sheath, between
  - lateral : Internal Jugular vein
  - medial : Common carotid Arteries

At Root of Neck

- Right Vagus → crosses 1st part of subclavian artery → then inclinig medially, behind brachiocephalic vessels → reach Right side of trachea → **THORAX**
- Left Vagus → passing between Left common carotid & Left subclavian arteries behind Internal Jugular & Brachiocephalic veins → **THORAX**

BRANCHES

(1) Meningeal branch → supply → dura mater of posterior cranial fossa

(2) Auricular branch (Arnold's nerve or Alderman's nerve) → supply → floor of meatus & tympanic membrane
- It's stimulation → ear cough, vomiting, cardiac arrest

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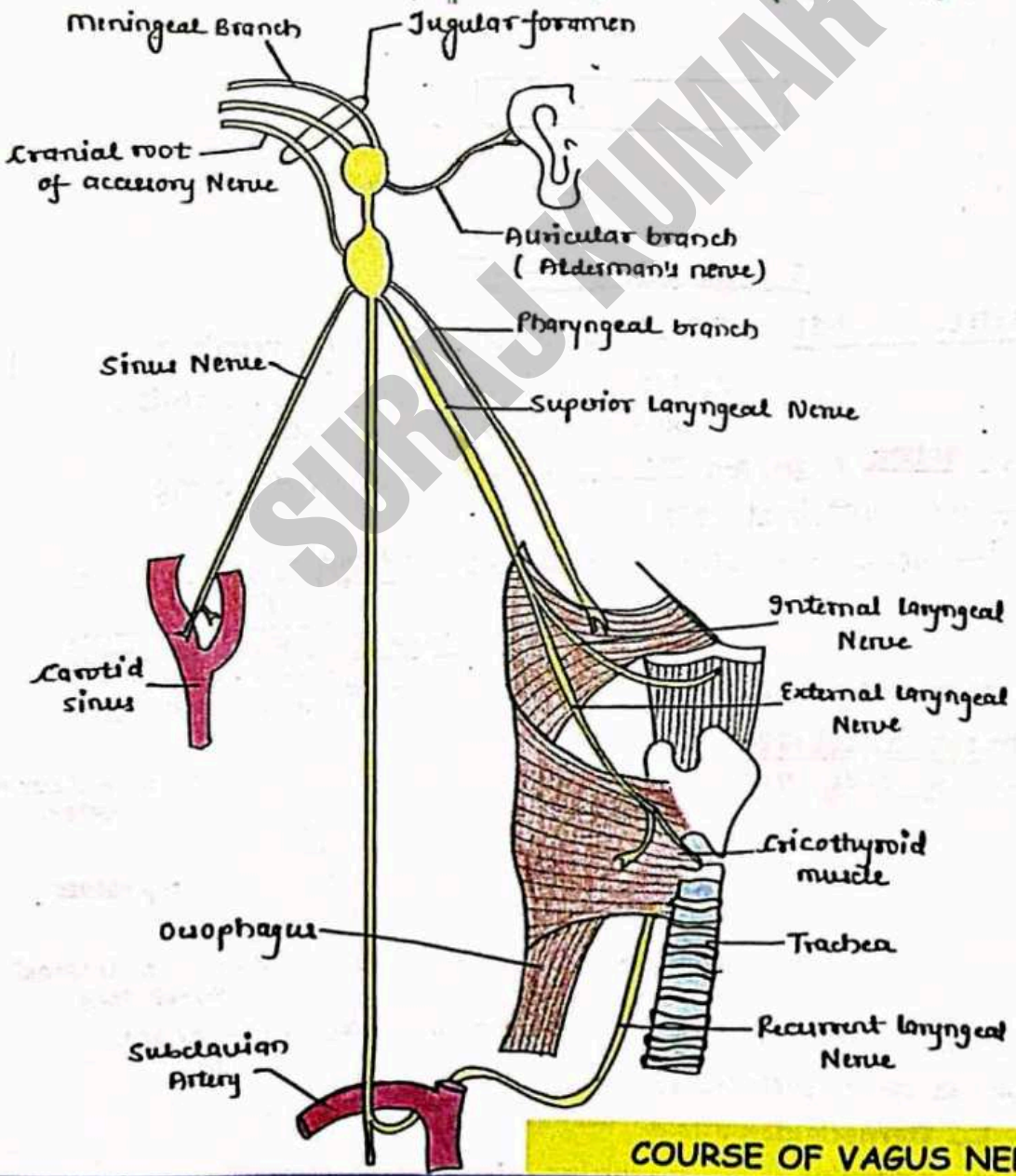
(3) Pharyngeal branch supplies
- → all muscles of pharynx, except – Stylopharyngeus ↑ supplied by Glossopharyngeal nerve
- → all muscles of soft palate, except – tensor palati ↑ supplied by mandibular nerve

(4) Branches to carotid body

(5) Superior laryngeal nerve
- Motor – external laryngeal nerve
- Sensory – internal laryngeal nerve

(6) Superior & inferior cervical cardiac branches

(7) Recurrent laryngeal nerve (nerve of 6th arch).



COURSE OF VAGUS NERVE

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ACCESSORY NERVE

- It is the 11th cranial nerve & has 2 roots → cranial, → spinal

NUCLEI

CRANIAL ROOT — Nucleus Ambiguus
SPINAL ROOT — Long Spinal nucleus

COURSE

CRANIAL ROOT

- Emerges in form of 4 to 5 rootlets — attached to posterolateral sulcus of medulla.
  ↓ joins to form
  single trunk
- Runs laterally with Glossopharyngeal, Vagus & Spinal Accessory
  ↓ crosses
  Jugular tubercle
  ↓ reaches
  Jugular foramen.
- Finally fuses with Vagus — at inferior ganglion, and distributes to muscles of palate, pharynx, larynx & heart.

SPINAL ROOT

In vertebral canal

- Filaments unite → form single trunk → ascends in front of dorsal nerve roots, behind ligamentum denticulatum.
- Nerve enters cranium through foramen magnum, lying behind vertebral artery.
  ← nerve runs upwards & laterally → crossing Jugular tubercle & reaches **Jugular foramen**
- Nerve leaves skull through middle part of Jugular foramen.

In extracranial Course

- Nerve descends vertically, between internal Jugular vein & internal carotid artery
  ↓ reaches between
  angle of mandible & mastoid process
  ↓ runs
  downward & backward — superficial to internal jugular vein, surrounded by lymph nodes.
- Nerve is then distributed to
  - Sternocleidomastoid — chin turning muscle
  - trapezius — shrugging muscle.

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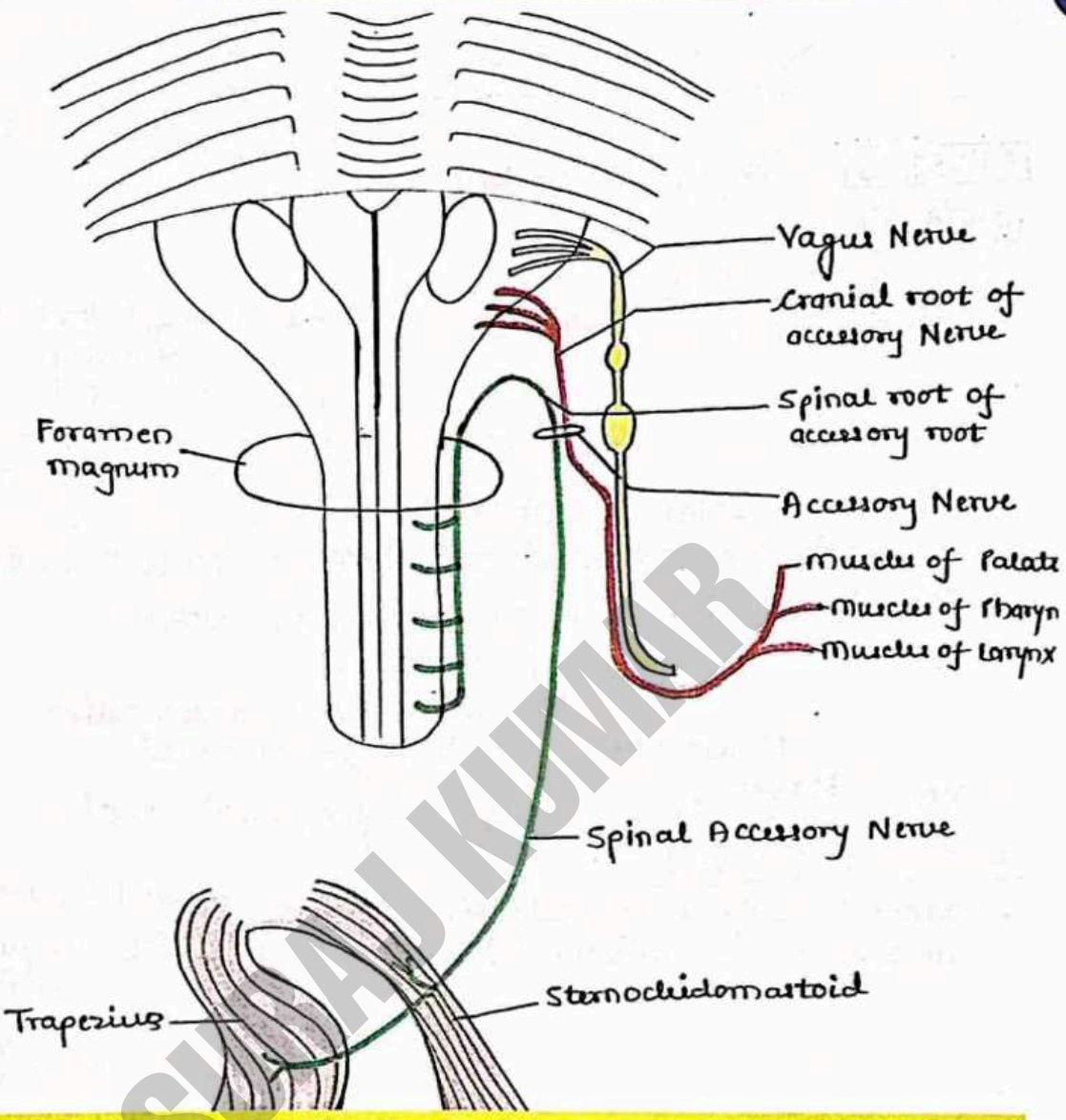

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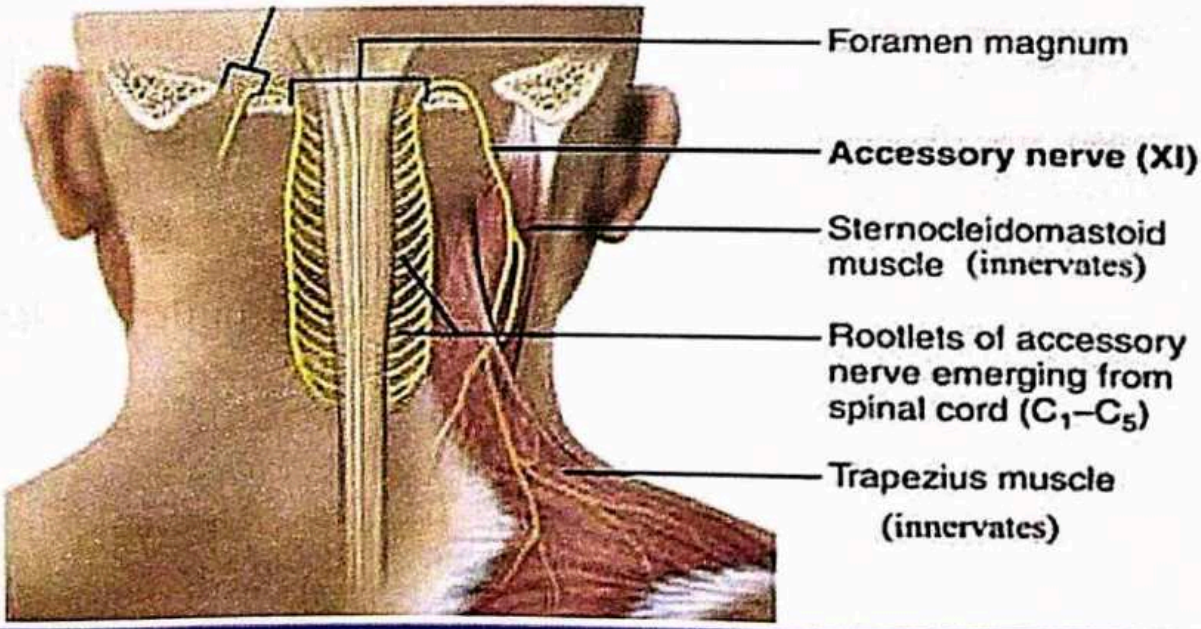
COURSE OF ACCESSORY NERVE



HYPOGLOSSAL NERVE

- It is the 12th nerve and supplies muscles of tongue

NUCLEUS · Hypoglossal nucleus

COURSE

Intraneural Course

- Fibers runs forwards – lateral → medial longitudinal bundle, medial leminiscus, pyramidal tract
  medial → reticular formation and olivary nucleus.
- Nerve attached to anterolateral sulcus of medulla, between pyramid and olive by 10 to 15 rootlets.
- Rootlets run laterally behind vertebral artery
  joins to form
  2 bundles — pierce dura mater separately near hypoglossal canal.
- Nerve leaves skull through hypoglossal canal

Extracranial Course

- Nerve lies deep to internal jugular vein → inclines b/w internal jugular vein & internal carotid artery crosses → Vagus (laterally in front
  ↓
  deep to parotid gland, styloid process, posterior belly of digastric
- At lower border of posterior belly of digastric – curves forward
  ↓ crosses
  Internal & External carotid Arteries & loop of lingual Arte ← passes deep to ← posterior belly of digastric again ← enters ← Submandibular region
- Nerves runs forward on genioglossus & hyoglossus, deep to submandibular gland & mylohyoid
  enters → tongue to supply — all intrinsic muscles & most extrinsic muscles.

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BRANCHES & DISTRIBUTION

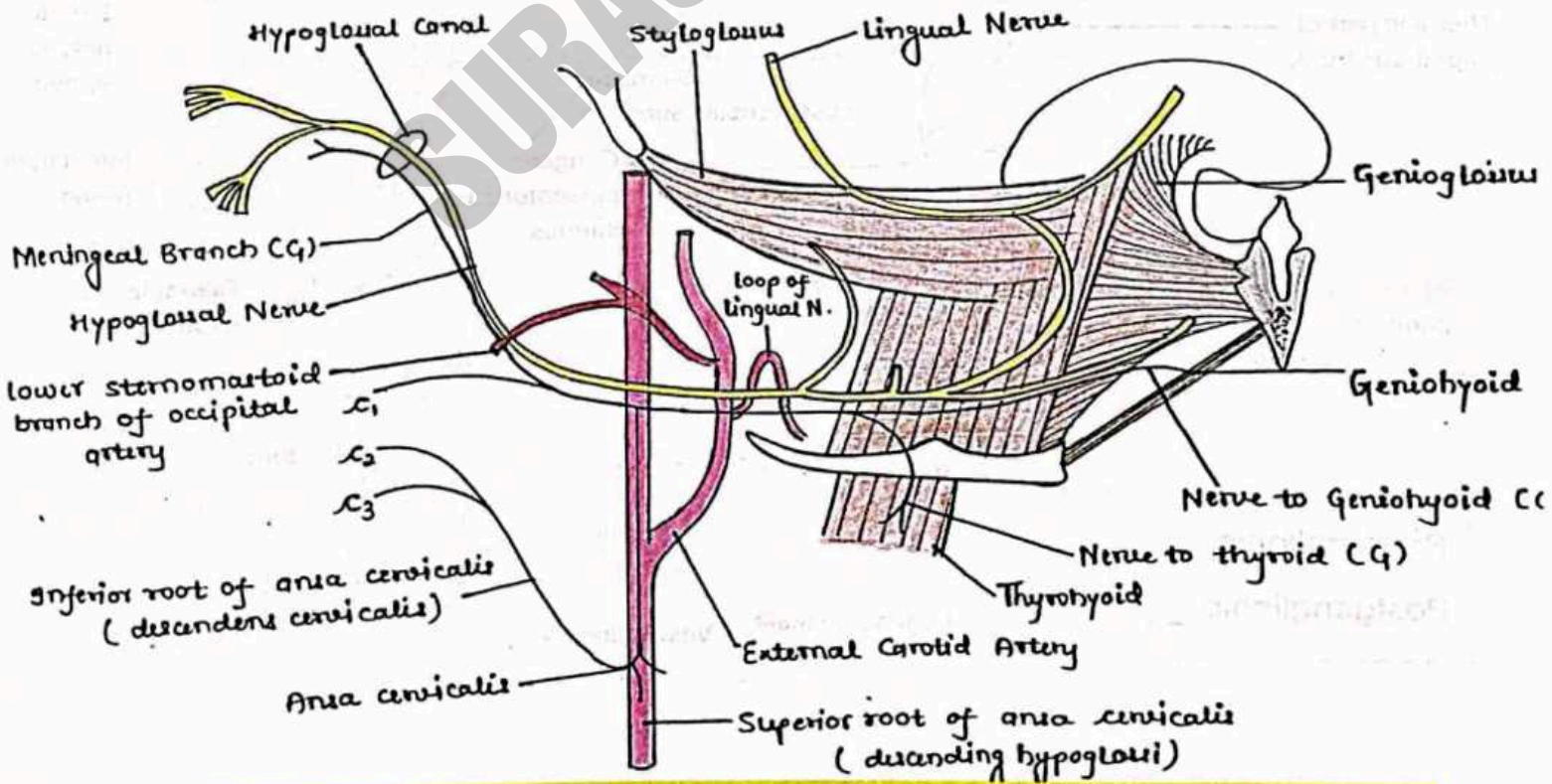
- **Branches containing fibres of hypoglossal nerve proper**
  - Supplies all intrinsic muscles
    - Superior longitudinal
    - Inferior longitudinal
    - Transverse & Vertical muscles

    &

    extrinsic muscles
    - Styloglossus
    - Genioglossus
    - Hyoglossus

  * **Exception** : Palatoglossus supplied by → Cranial Accessory Nerve

- **Branches of hypoglossal nerve containing fibers of nerve**
  - (1) Meningeal branch supplies → bones & meninges in anterior part of Posterior Cranial fossa
  - (2) Descending branch continues → Descendens hypoglossi in ansa cervicalis
  - (3) Branches also supplies → thyrohyoid muscle, Geniohyoid muscle.



COURSE OF HYPOGLOSSAL NERVE

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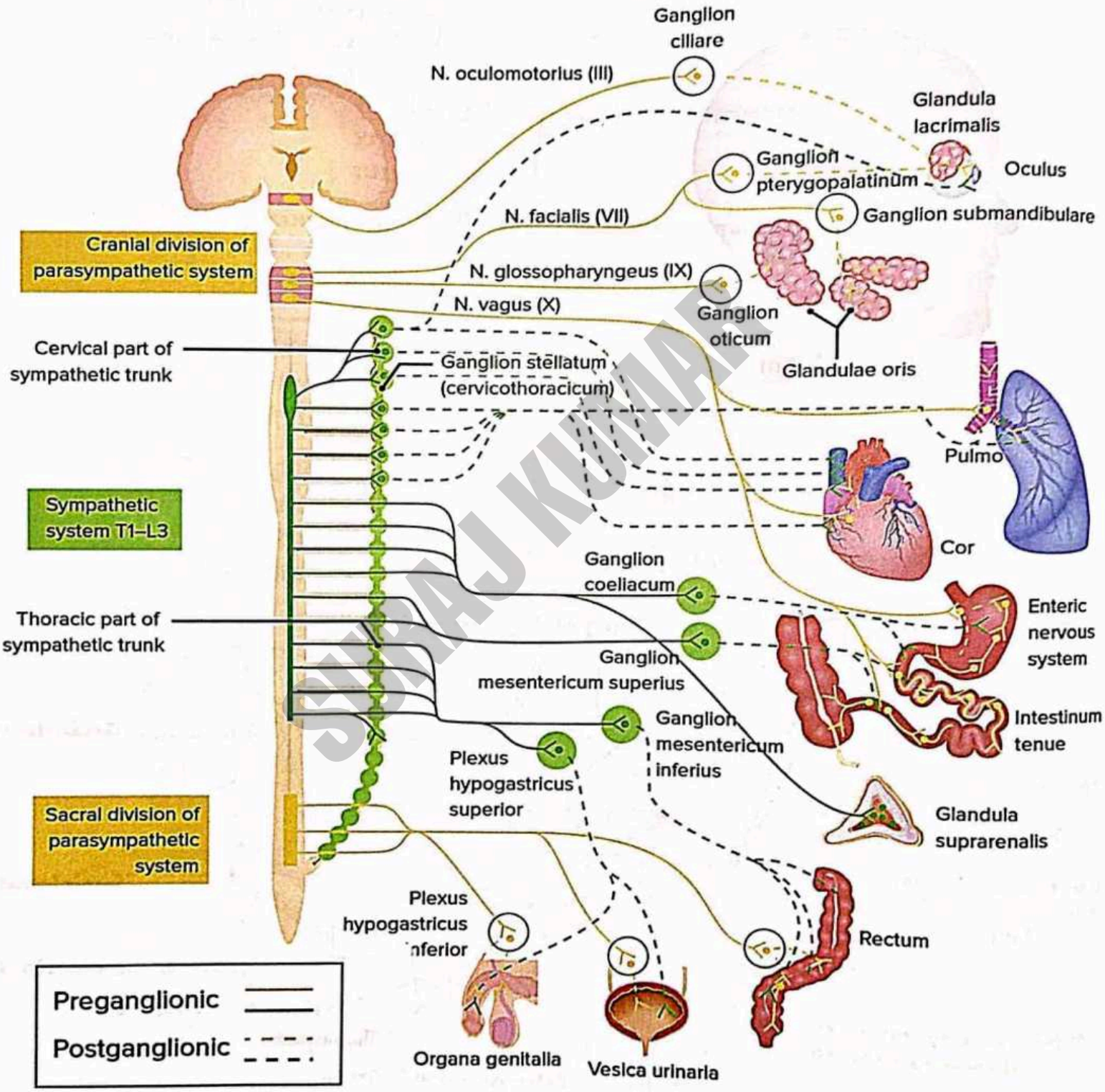
PARASYMPATHETIC & SYMPATHETIC GANGLIONS



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SUBMANDIBULAR ganglia

- It is a parasympathetic ganglion, which acts as a relay station for secretomotor fibres supplying submandibular and sub-lingual salivary glands.
- Topographically connected to :- Trigeminal Nerve
  Functionally connected to :- Facial Nerve

Location

→ Size → Pin-head

→ Situation → on outer surface of hyoglossal muscle.

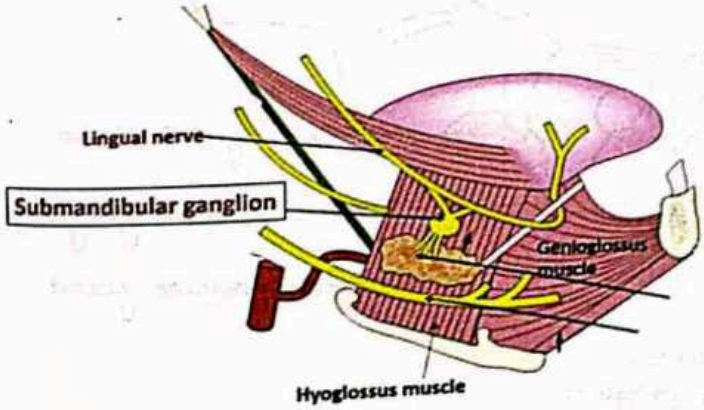
Relations

Above → Lingual Nerve
Below → Deep part of submandibular gland
Medial → Hyoglossus muscle
Lateral → Submandibular gland

Roots

→ Has 3 roots :-

1. Parasympathetic Root :- Superior Salivatory Nucleus
   ↓
   Nervus Intermedius
   ↓
   Facial Nerve
   ↓
   Chorda tympani Nerve
   ↓
   Joins Lingual Nerve
   ↓
   Submandibular Ganglion
   ↓ Relay
   Postganglionic Fibres
   ↓
   Submandibular and Sublingual glands



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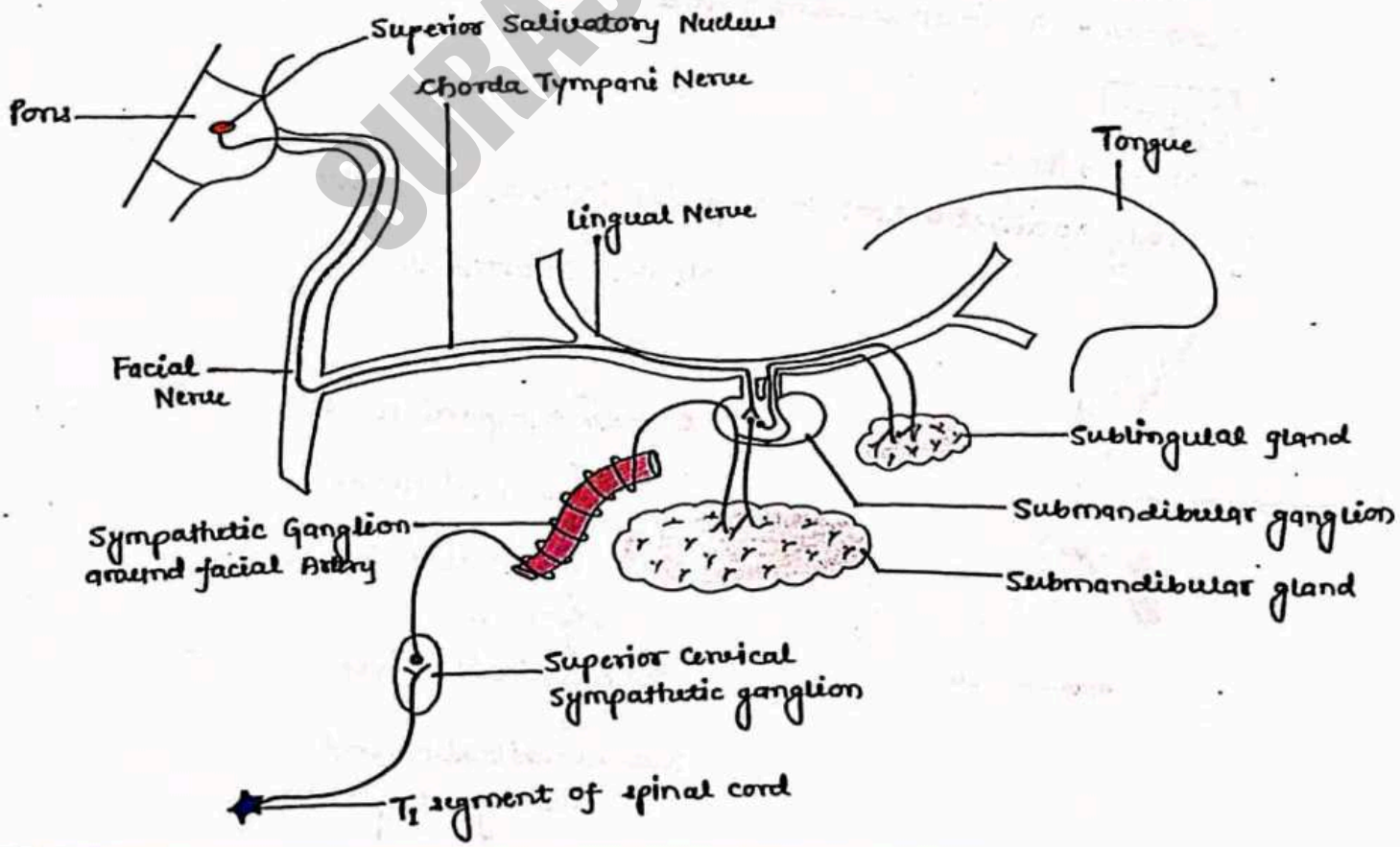
2. Sympathetic Root :- Sympathetic Plexus around facial Artery
↓
First thoracic spinal segment (T_1)
(pre ganglionic fibres)
↓
Superior cervical Sympathetic Ganglion
(post ganglionic fibres)
↓
Plexus around facial Artery
↓
Pass through ganglion without relay
↓
Supply Submandibular and Sublingual Salivary glands

3. Sensory Root → Derived from lingual Nerve

- **Branches**

→ 5 to 6 :- to supply submandibular gland
→ others :- join sublingual Nerve to supply sublingual & anterior lingual gland.



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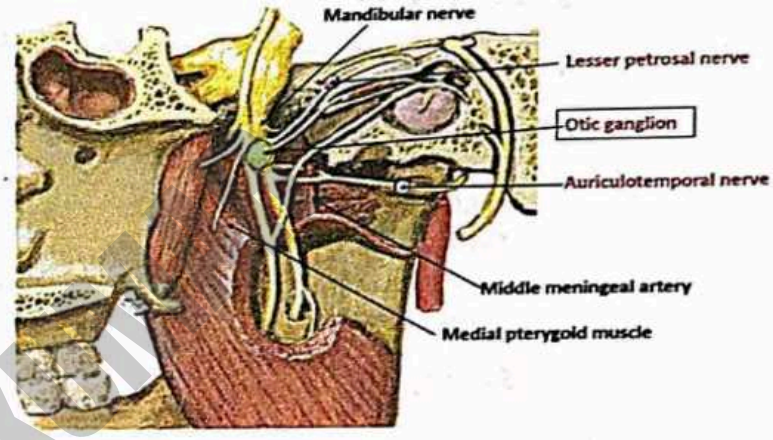
OTIC GANGLIA

- Small parasympathetic ganglion connected to mandibular div. of Trigeminal N.
- Provide relay station to secretomotor fibres to parotid gland.
- Topographically connected to :- Mandibular Nerve
- Functionally connected to :- Glossopharyngeal Nerve



- **Size**

→ Pin-head

- **Location**

→ Infratemporal fossa

- **Relation**

Lateral → Mandibular Nerve
Medial → Tensor Palati muscle
Posterior → Middle Meningeal Artery
Anterior → Medial Pterygoid Muscle

- **Roots**

1. Parasympathetic Root :- Inferior Salivatory Nucleus
↓
Glossopharyngeal N. (IX)
↓
Tympanic Br. of IX N. (Jacobson's N.)
↓
Tympanic Plexus
↓
Lesser Petrosal Nerve
↓
Otic Ganglion
↓ Relay
Postganglionic fibres through auriculo-temporal N.
↓
Supply Parotid gland

2. Sympathetic Root :- Sympathetic plexus around Middle Meningeal Artery
↓
T_1 & T_2 spinal segment
(pre ganglionic fibres)
↓
Superior cervical Sympathetic Ganglion

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↕

Plexus around Middle Meningeal Artery

↓

Pass through ganglion without relay

↓

Supply parotid gland (vasomotor in nature & cause thick salivary secretions)

3. Sensory Root :- From auriculotemporal Nerve

4. Somatic Motor :- Nerve to medial pterygoid

- **Branches**

Postganglionic Parasympathetic
Postganglionic Sympathetic
Sensory

} All supply parotid gland through auriculotemporal Nerve

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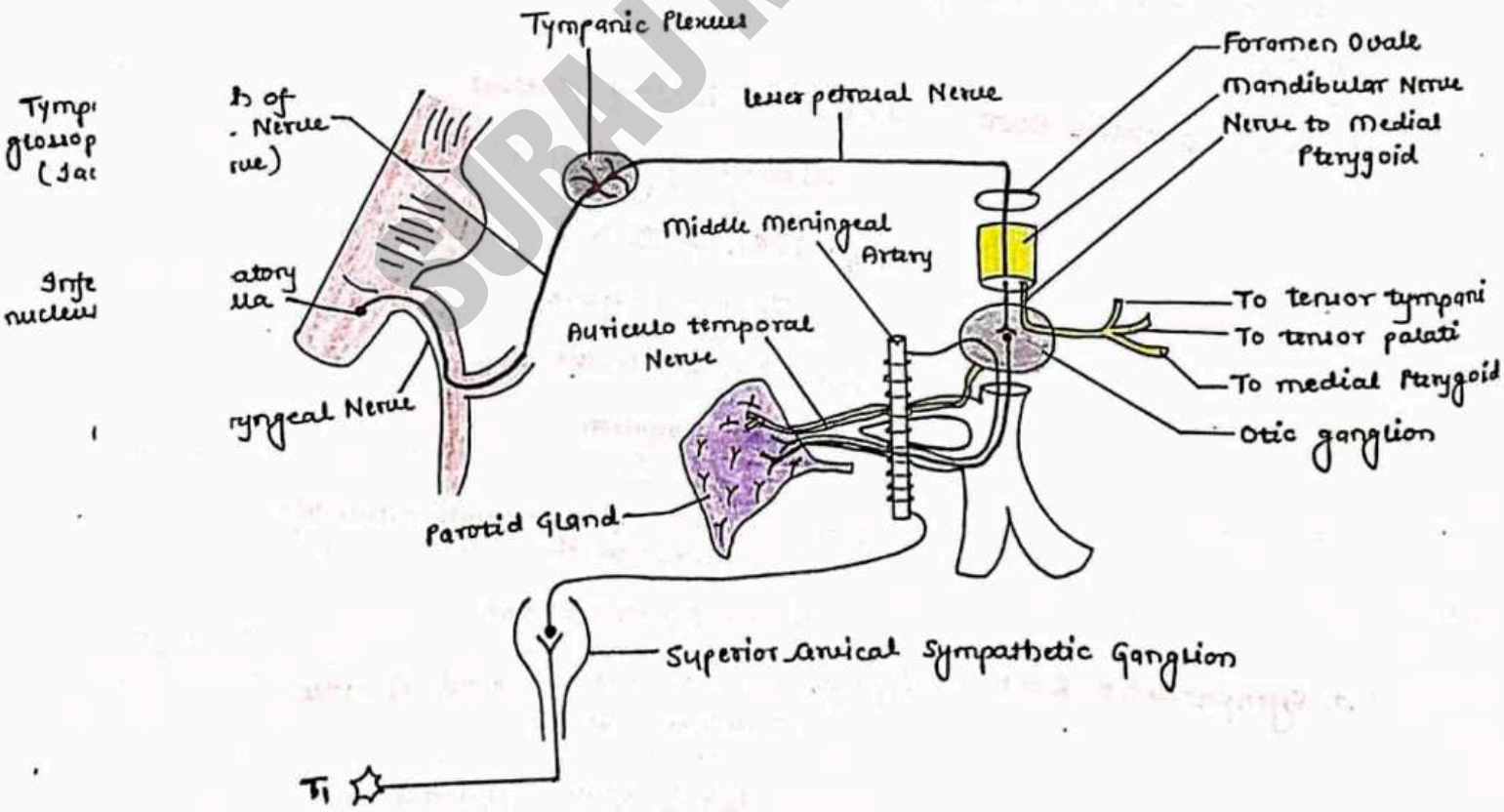

ROOTS & DISTRIBUTION OF OTIC GANAGLION

CILIARY GANGLIA

- Peripheral parasympathetic ganglion —connected with→ Nasociliary N.

- Topographically Connected to :- Nasociliary Nerve
- Functionally Connected to :- Oculomotor Nerve

- **Location**

→ Minute body lying near apex of orbit between optic and lateral rectus muscle.

- **Roots**

1. Parasympathetic Root → Edinger Westphal Nucleus
↓
III Nerve
↓
Nerve to Inferior oblique
↓
Branch to ciliary ganglion
↓
Short ciliary nerve supplies ciliaris & constrictor pupillae muscle.

2. Sympathetic Root :- Plexus around Internal Carotid Artery
↓
Superior Cervical Sympathetic Ganglion
(Post ganglionic fibres)
↓
Passes through ganglion without relay
↓
Short ciliary Nerve
↓
Supply dilator pupillae and Blood vessels of Eyeball

3. Sensory Root :- Derived from Nasociliary Nerve

- **Branches**

→ Branches are :- Short Ciliary Nerves (8-10)

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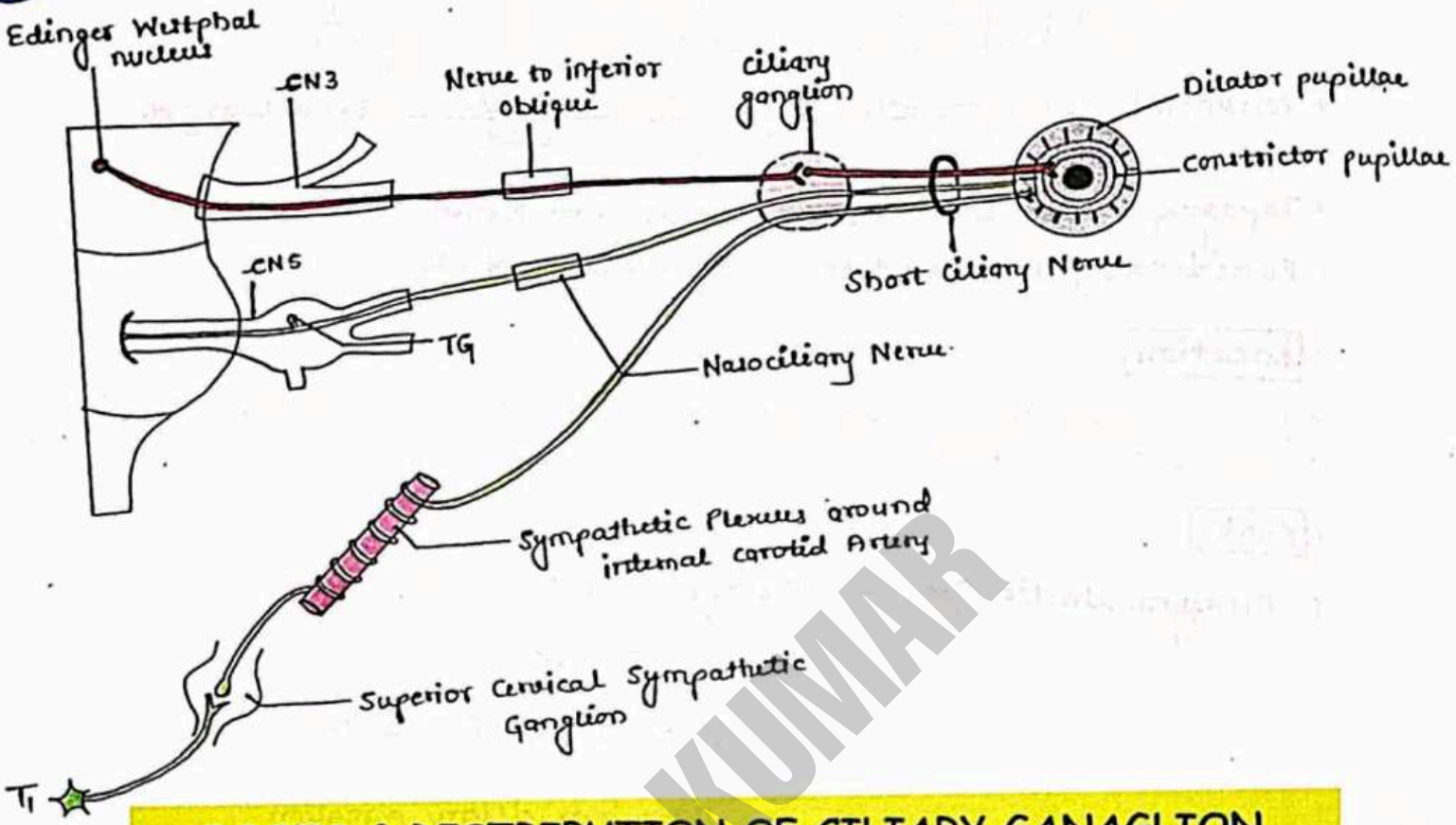
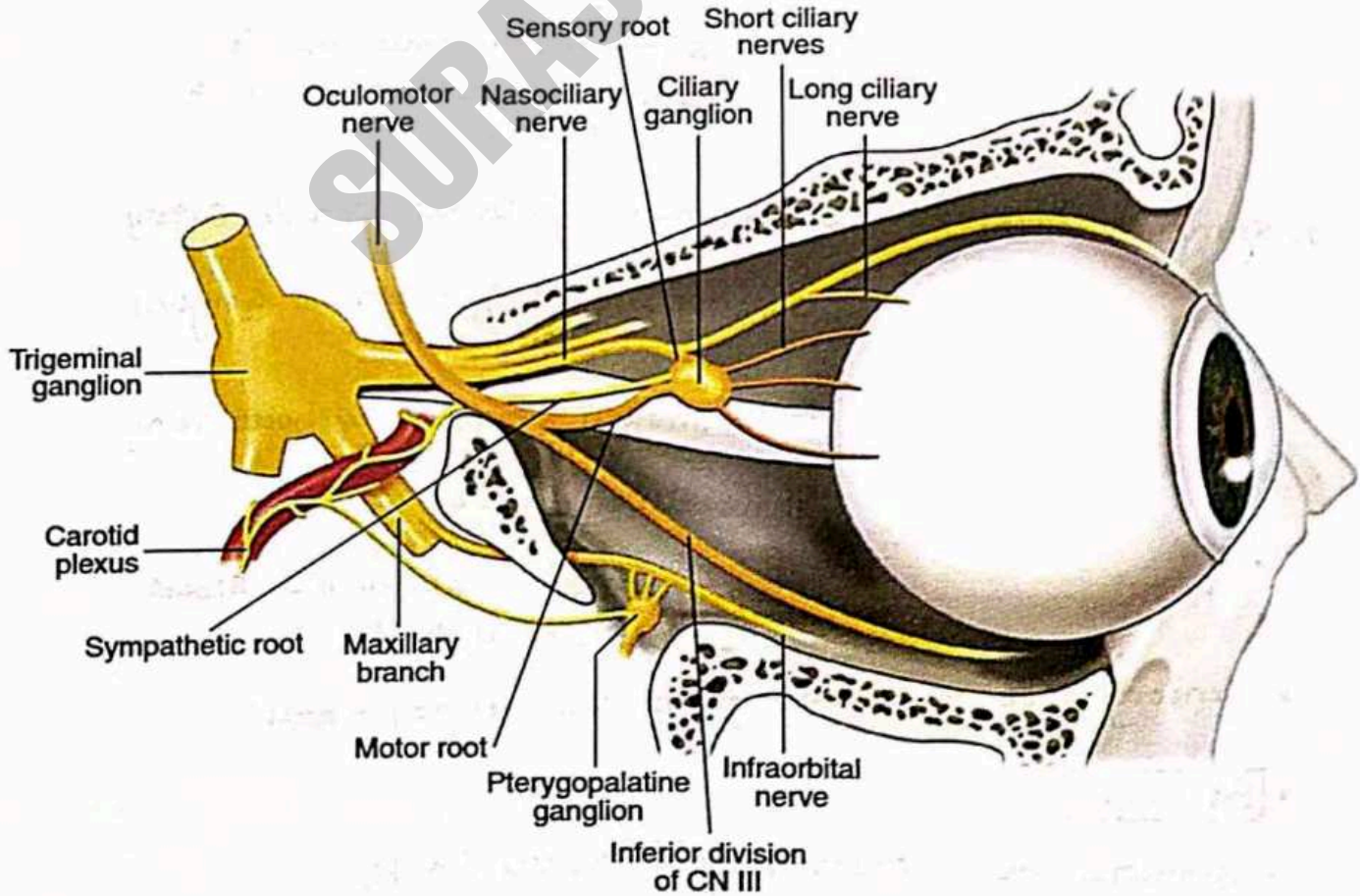


ROOTS & DISTRIBUTION OF CILIARY GANAGLION



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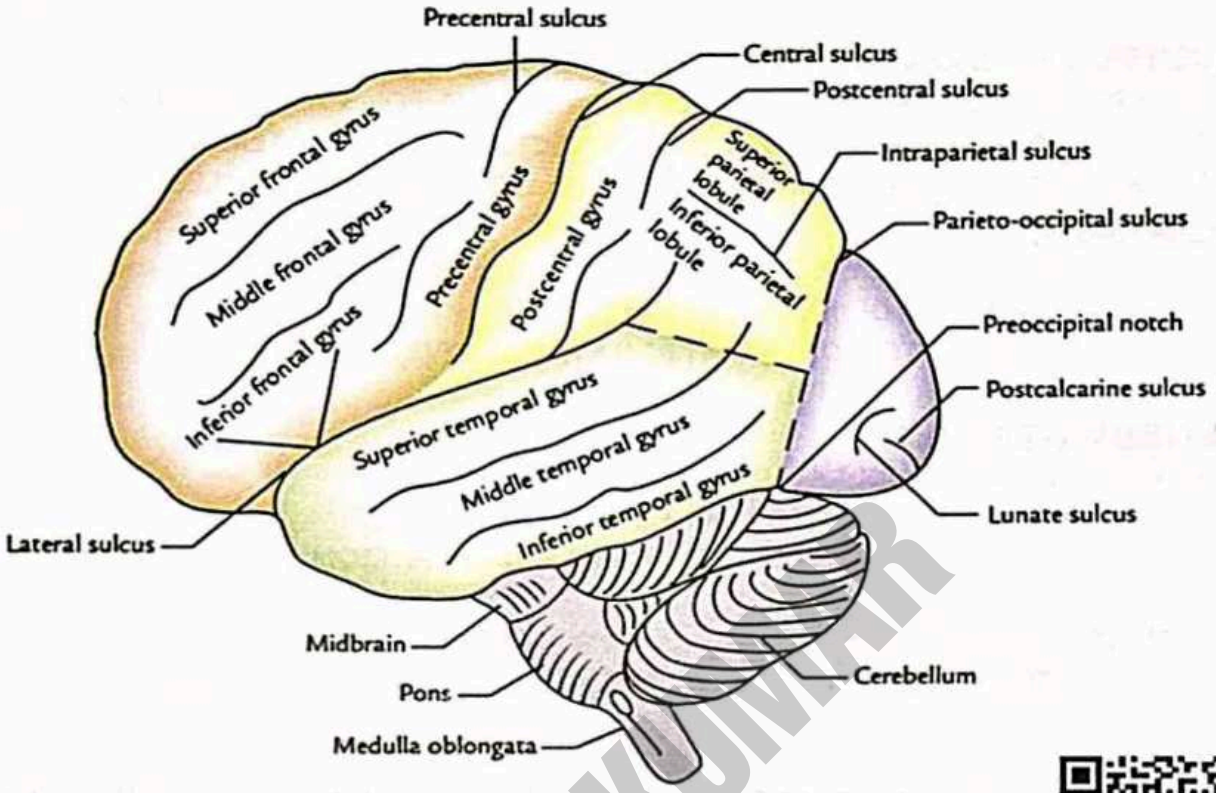
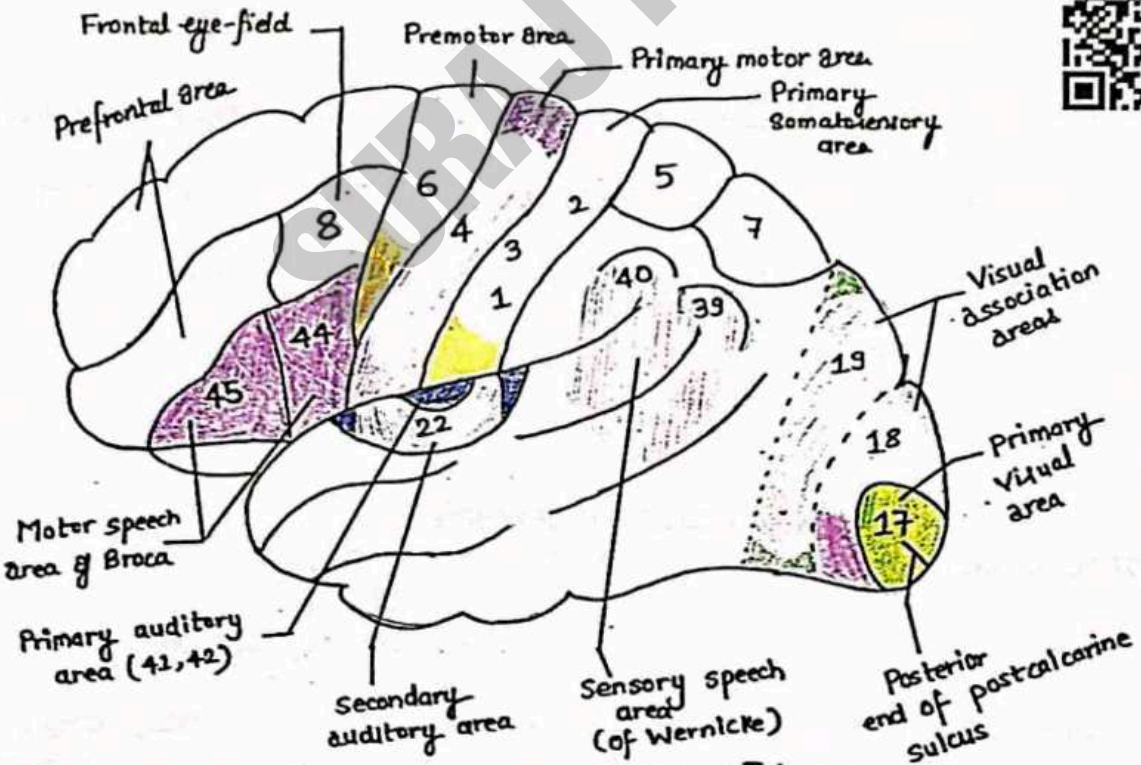
SUPEROLATERAL SURFACE OF SURFACE





[∵ Functional cortical areas on superolateral surface of cerebral hemisphere.]

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CEREBELLUM ANATOMY

- CEREBELLUM IS LARGEST PART OF HIND BRAIN
- LOCATED IN POSTORIOR CRANIAL FOSSA BEHIND PONS AND MEDULLA
- EACH HEMISPHERE OF CEREBELLUM CONNECTED TO BRAINSTEM BY 3 PAIR OF 19 LARGE FIBRE TRACT CALLED CEREBULLAR PEDUNCLES

FUNCTION:-

- MAINTANACE OF EQUILIBRIUM
- REGULATION OF MUSCLE TONE
- COORDINATION OF SOMATIC MATAR ACTIVITIES

***EXTERNAL FEATURES**

- PARTS - 3X PART - 2X LATERAL HEMISPHERICAL LOBES
  -2X CEREBELLAR HEMISPHERE
  -NARROW MEDIAN WARN LIKE PARTION - VERMIS

-SURFACE - SURPERIOR SURFACE
-INFERIOR

*NOTCHES :-
-ANTERIOR ASPECT OF CEREBELLUM MARKED BY WIDE SHALLOW ANTERIOR CEREBELLAR NOTCH
- POSTERIOR CEREBELLAR NOTCH IS, DEEP AND NARROW

*FISSURE. - HORIZONTAL FISSURE → ALONG LATERAL AND POSTERIAR MARGIN OF CEREBELLAM
*MARKS JUNCTION B/W - SUPERIOR AND INFERIOR SURFACE OF CEREBELLUM.

-POSTEROLATERAL SURFACE → INFERIOR SURFACE OF CEREBELLUM
- SEPARATE - FLOCCULONODULAR LOBE FROM REST OF CEREBELLUM

-V-SHAPED FISSURE PRIMA - DIVIDE CEREBELLUM INTO ANTERIOR AND POSTERIOR LOBES

SUB-DIVISION OF CEREBELLUM:

ANATOMICAL SUB-DIVISIONS:- INTO 3X LOBES - ANTERIOR
- POSTERIAR
- FLOCCULONODULAR

1) ANTERIOR LOBE- LIE ON SUPERIOR SURFACE ANTERIOR TO FISSURE PRIMA
2) POSTERIOR/MIDDLE LOBE:- BETWEEN FISSURA PRIMA AND POSTEROLATERAL
3) FLOCCULONODULAR LOBE :- SMALLEST OF ALL
- LIES ON INFERIOR SUFACE

MORPHOLOGICAL SUB-DIVISIONS :-

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*BASED ON PHYLOGENETIC AND FUNCTIONAL CRITERIA

1) ARCHICEREBULLUM (VESTIBULAR CEREBELLUM)

- OLDEST PART OF CEREBELLUM
- CONSIST OF FLOCCULO NODULAR AND LINGULA
- FUNCTION- MAINTANANCE OF EQUILIBRIUM, TONE

2) PALEO CEREBELLUM (SPINAL CERE BELLUM) -

- PHYLOGENETICALLY, IT IS NEXT PART OF CEREBELLUM TO APPEAR
- CONSIST OF ANTERIOR LOBE (EXCEPT LINGULA) AND PYRAMID AND UVALA OF INFERIOR VERMIS
- FUNCTION- CONCERNED WITH TONE, POSTURE AND CRUDE MOVEMENT OF LIMB

3) NEOCEREBELLUM (CEREBRAL CEREBALLUM) -

-PHYLOGENETICALLY-MOST RECENT PART OF CEREBELLUM TO DEVELOP
-MADE UP OF- MIDDLE LOBE, LARGE PART OF CEREBELLUM
-CHIEFLY CORTICO PONTOCEREBELLAR IN CONNECTION
-CONCERNED WITH SMOOTH PERFORMANCE OF SILKED VOLUNTARY MOVEMENT

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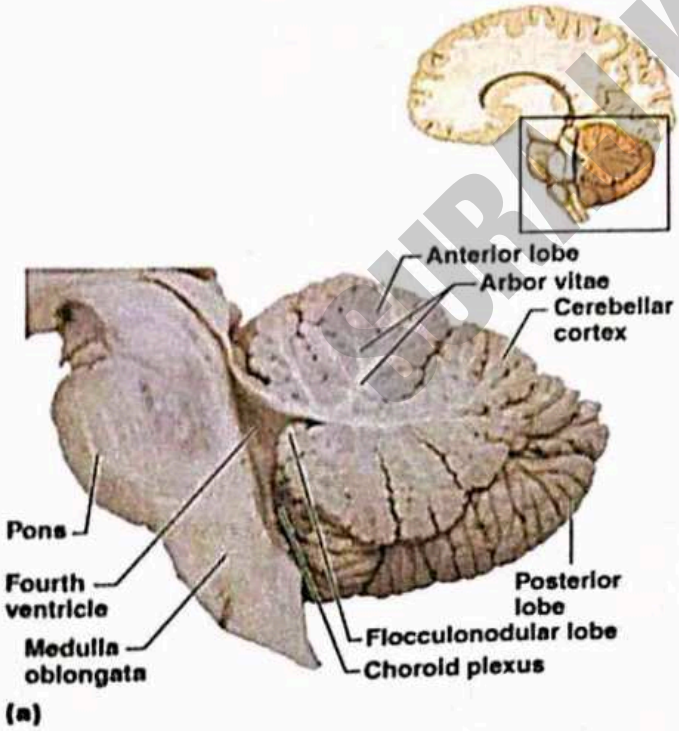

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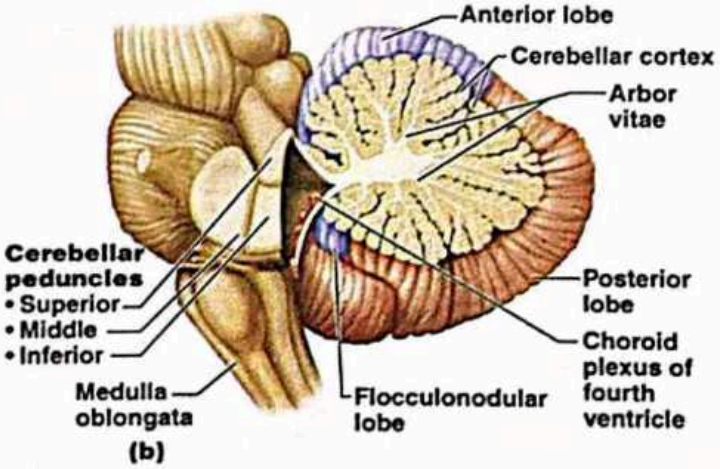

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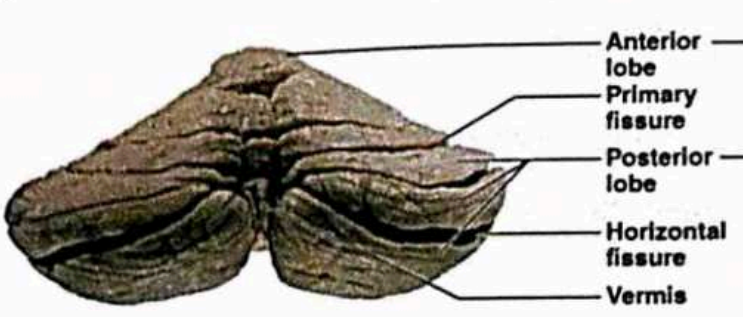

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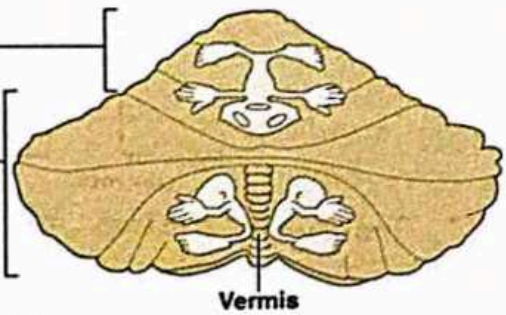

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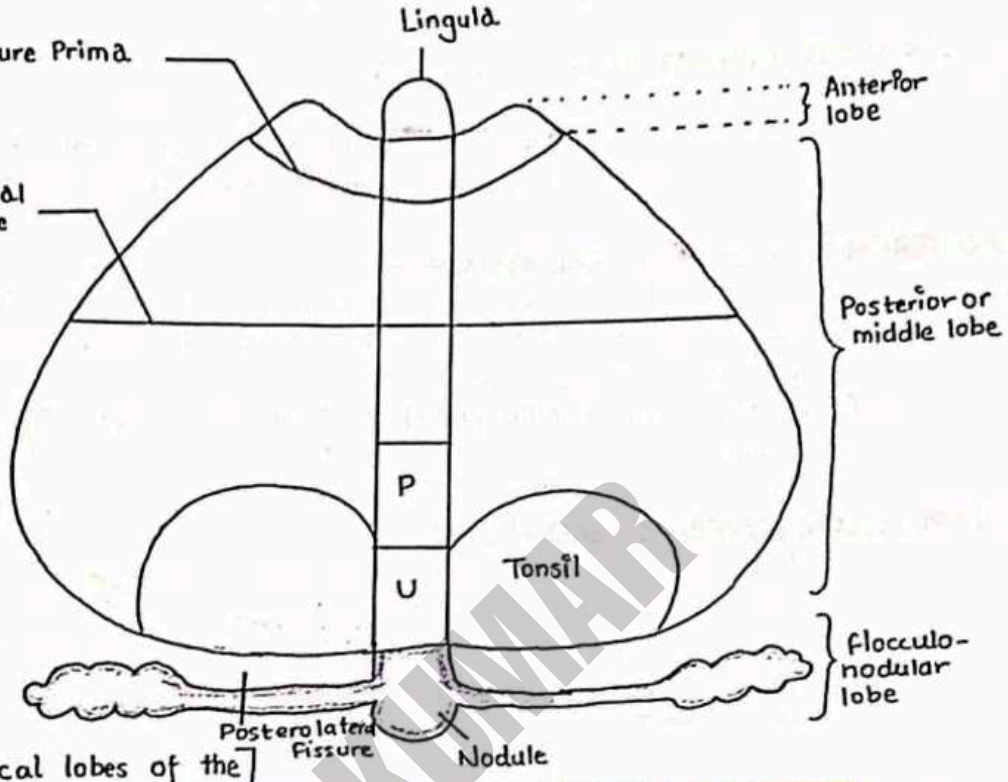
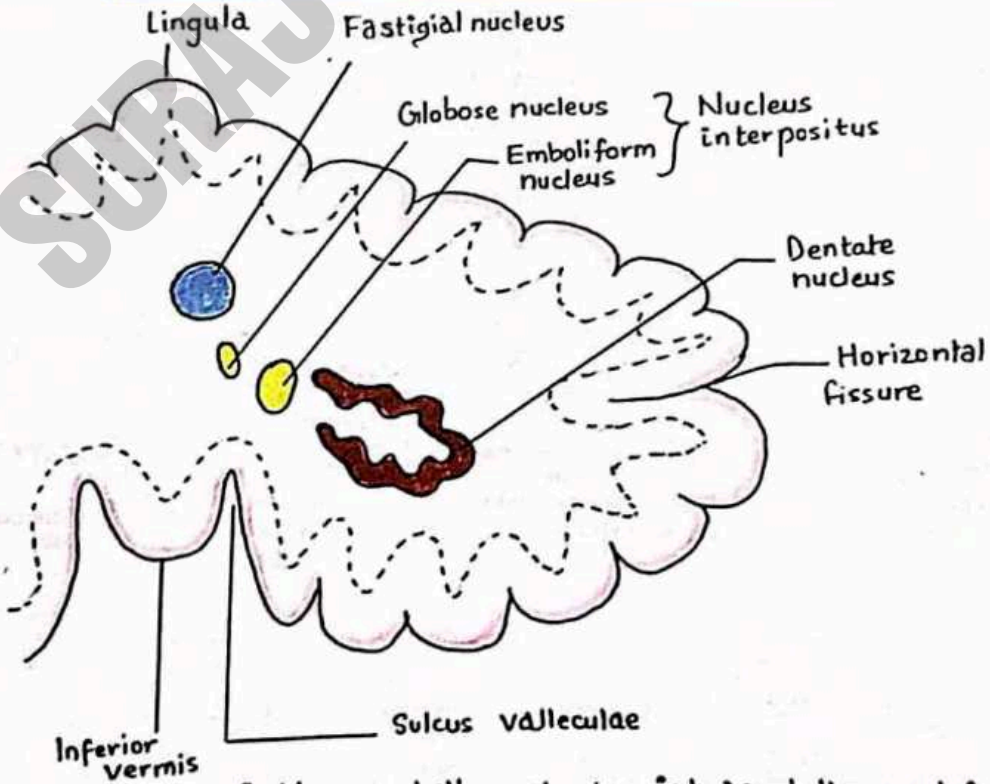


[∵ Anatomical lobes of the Cerebellum]

CEREBELLUM ANATOMY



∵ Transverse section of the cerebellum showing intracerebellar nuclei.

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LATERAL VENTRCILE

- TWO IRREGULAR CAVITIES IN CEREBRAL HEMISPHERE
- COMMUNICATION THIRD VENTRIDE THROUGH - INTERVENTRICULAR FORAMEN (FORAMEN OF MONRO]
- DIVIDE - CENTRAL PART AND 3 HORNS - ANT,POST,INF. HORN

*CENTRAL PART

-EXTEND INTERVENTRICULAR FORAMEN TO SPLENIUM OF CORPUS CALLOSUM

- BOUNDARIES :- ROOF - CORPUS CALLOSUM
  FLOOR - BODY OF CAUDATE NUCLEUS
  - THALAMOSTRIATE VEIN & THALAMUS
  MEDIAL WALL - SEPTUM PELLUCIDUM
  - BODY OF FORNIX

1) ANTERIOR HORN :-

- EXTEND - FRONT OF INTER VENTRICULAR FORAMEN EXTEND INTO FRONTAL LOBE

*BOUNDARIES - ANTERIOR→ POSTERIOR SURFACE OF GENU OF CORPUS CALLOSUM
ROSTRUM OF CORPUS CALLOSUM
- ROOF - CORPUS CALLOSUM
- FLOOR - CALLDATE NUCLEUS
- MEDIAL - SEPTUM PELLUCIDUM ,COLUMN OF FORNIX

2) POSTERIOR HORN

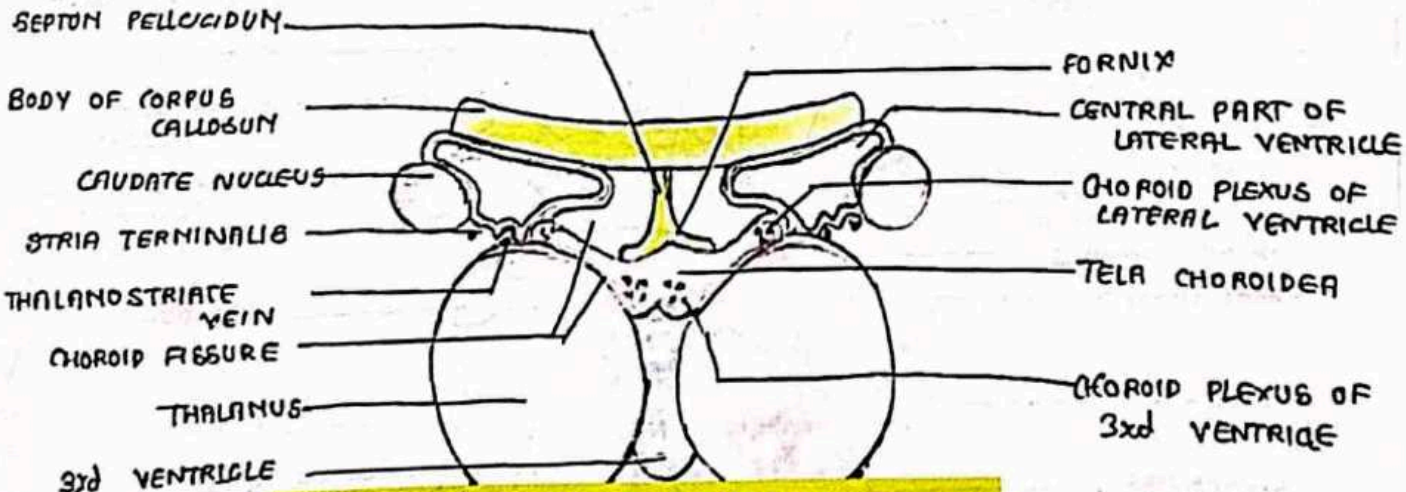
- LIE BEHIND SPLENIUM OF CARPUS CALLOSUM AND EXTEND INTO OCCIPITAL LOBES

*BOUNDARIES - FLOOR AND MEDIAL WALL → FORCEP MAJOR
- ROOF AND LATERAL WALL → TAPETUM OF OPTIC RADIATION

3)I NFERIOR HORNS

- BEGINS AT JUNCTION OF CENTRAL PART WITH POSTERIOR HORN INTO TEMPORAL LOBE

*BOUNDARIES - ROOF AND LATERAL WALL - TAIL OF CAUDATE NUCLEUS, AMYGDALOID
FLOOR → HIPPOCAMPUS



THIRD VENTRCILE

SURAJ KUMAR

THIRD VENTRCILE

-DIENCEPHALON CAVITY IS A -→ MEDIAN CLEFT B/W TWO THALAMI
-THIRD AND LATERAL VENTRICLE SECRETE CEREBRO-SPINAL FLUID

COMMUNICATIONS -

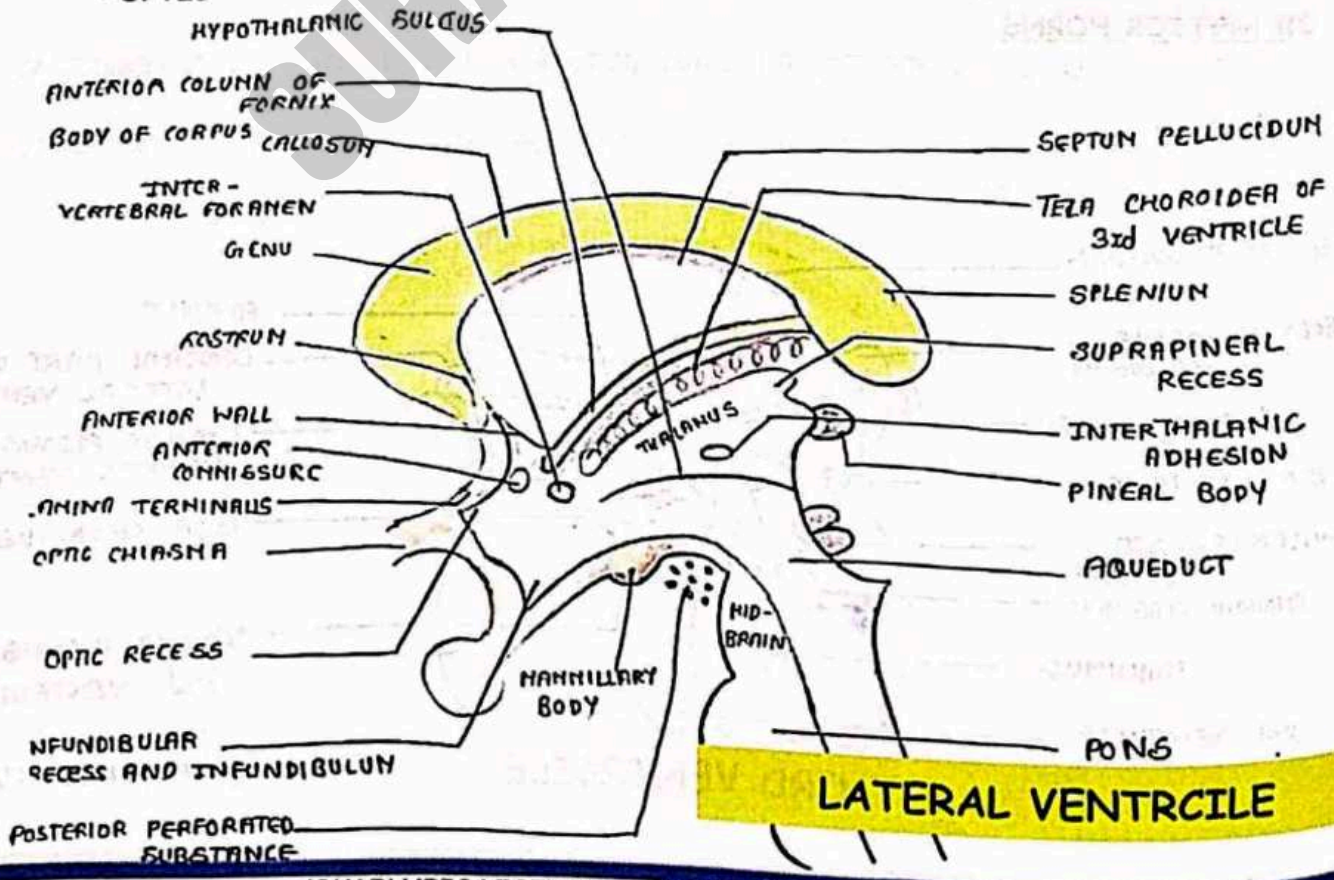
* ANTEROSUPERIORLY - LATERAL VENTRICLE THROUGH, FORAMEN OF MONRO.
* POSTEROINFERIORLY - FOURTH VENTRICLE THROUGH, CEREBRAL AQUEDUCT

BOUNDARIES

* ANTERIOR WALL - LAMINA TERMINALIS
  - ANTERIOR COMMISSURE
* POSTERIAR WALL - PINEAL BODY
  - POSTERIOR COMMISSURE
  - CEREBRAL AQUEDUCT
* ROOF - BODY OF FORNIX
  - TELA CHOROIDEA
* FLOOR - MAMMILLARY BODIES
  - INFUNDIBULUM (PITUITARY STALK)
  - OPTIC CHIASMA
  - TEGMENTUM OF MIDBRAIN
* LATERAL WALL - MEDIAL SURFACE OF THALAMUS
  - HYPOTHALAMUS

***RECESSES:** RECESSES ARE EXTENSIONS OF CAVITY

-SUPRAPINEAL
-PINEAL
-INFUNDIBULAR
-OPTIC



LATERAL VENTRCILE

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FOURTH VENTRICLE

-CAVITY OF HIND BRAIN

-TENT-SHAPED SPACE BETWEEN PONS AND UPPER PART OF MEDULLA OBLONGATA

*STRUCTURAL LAYERS: FLOOR IS LINED BY - EPENDYMA
- THIN LAYER OF NEUROGLIA
- LAYER OF GREY MATTER

*BOUNDARIES:-

1) SUPEROLATERALLY -> SUPERIOR CEREBELLAR PEDUNCLE

2) INFEROLATERALLY -> INFERIOR CEREBELLAR PEDUNCLE
- GRACILE TUBERCLES
- CUNEATE TUBERCLE

3) FLOOR-->ALSO CALLED RHOMBOID FOSSA
- FORMED BY - DORSAL SURFACE OF LOWER PONS
- DORSAL UPPER OF MEDULLA OBLONGATA

-PART - IT IS DIVISIBLE INTO
- UPPER TRIANGULAR PART FORMED BY → DORSAL SURFACE OF PONS
- LOWER TRIANGULAR PART FORMED BY → DORSAL SURFACE OF MEDULLA
- INTERMEDIATE PART →JUNCTION OF PONS AND MEDULLA
- PART IS MARKED BY TRANSVERSELY RUNNING FIBRE → STRIA MEDULLARIS

*FEATURE OF FLOOR -
- DORSAL MEDIAN SULCUS - DIVIDE FLOOR → INTO 2 SYMMETRICAL HALVES
- SULCUS LIMITANS - DIVIDE EACH HALF - MEDIAN & LATERAL EMINENCE
- SULCUS LIMITANS PRESENTS DEPRESSION- AT CRANIAL END CALLED - SUPERIOR FOVEA
-AT CAUDAL END CALLED - INFERIOR FOVEA
- MEDIAL EMINENCE → FACIAL COLLICULUS - DEEP TO F.C IS GENU OF FACIAL NERVE
- UPPERMOST PART OF SULCUS LIMITANS → BLUISH IN COLOUR → CALLED LOCUS COERULEUS
- DESCENDING FROM INFERIOR FOVEA SULCUS RUN OBLIQUE - TOWARD →MIDLINE DIVIDES MEDIAL EMINENCE INTO 2 TRIANGLE - HYPOGLOSSAL & VAGAL TRIANGLE

4) ROOF - DIAMOND SHAPE & DIVIDE INTO SUPERIOR , INFERIOR PART

*SUPERIOR/CRANIAL PART - FORM BY - SUPERIOR CEREBELLAR PEDUNCLES
- SUPERIOR MEDULLARY VELUM
-SCP EMERGING -> WHITE MATTER OF CEREBELLUM REACH, INFERIOR COLLICULI
-SMU MADE UP OF -> NERVOUS TISSUE B/W SCP
*INFERIOR / CAUDAL PART FORM BY - VENTRICULAR EPENDYMA AND
- DOUBLE FLOD OF PIA MATTER

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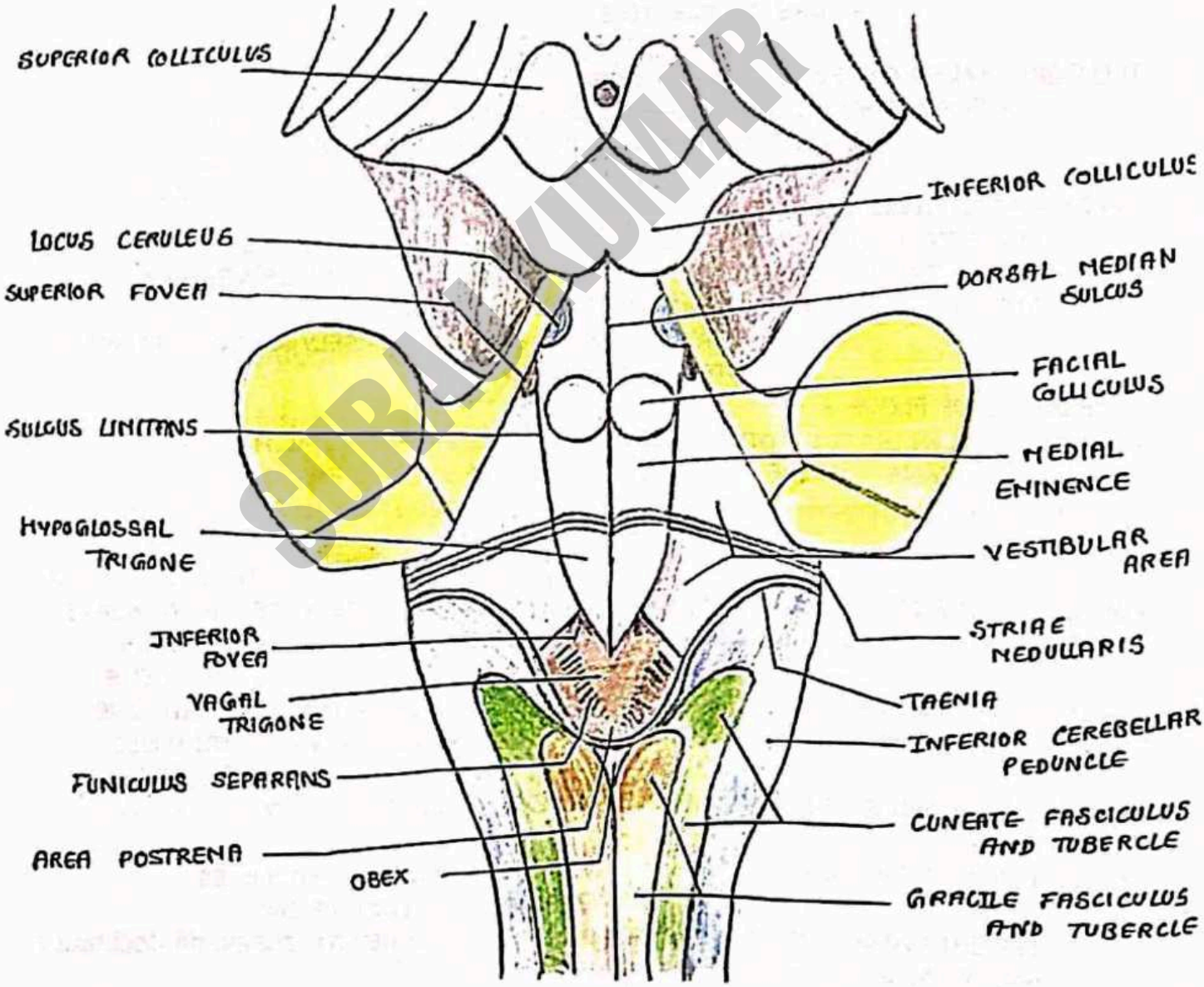
*COMMUNICATION: SUPERIORLY - CEREBRAL AQUEDUCT
INFERIORLY - CENTRAL CANAL

*OPENINGS IN ROOF :- 3X OPENING - 2X- LATERIAL - FORAMEN OF MAGENDIE
- 1X MEDIAN - FORAMINA OF LUSCHKA

*RECESSES OF FOURTH VENTRICLE:-
- 2X LATERAL RECESSES
- 1X MEDIAN DARSAL RECESSES
- 2X LATERAL DORSAL RECESSES



FOURTH VENTRICLE

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THALAMUS

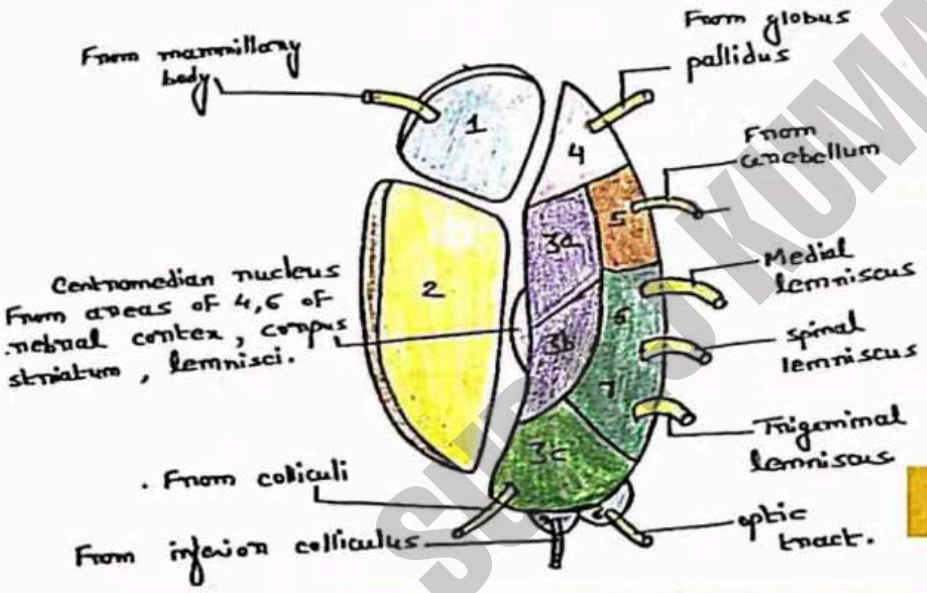
*THALAMUS-→LARGE MASS OF GREY MATTER
*4CM IN EACH - TRANSVERSE
- VERTICAL
- ANTERO-POSTERIOR DIAMETER

*SITUATED-→IN LATERAL WALL OF 3RD VENTRICLE & FLOOR OF CENTRAL PART OF →LATERAL VENTRICLE

STRUCTURE & NUCLEI OF THALAMUS -

1.WHITE MATTER-→EXTERNAL MEDULLARY LAMINA--→COVER LATERAL SURFACE
-INTERNAL MEDULLARY LAMINA--→DIVIDE THALAMUS--→IN 3x PARTS
-ANTERIOR
-MEDIAL
-LATERAL

2.GREY MATTER - DIVIDED INTO 7x NUCLEI



1. Anterior Nucleus.
2. Medial Nucleus.
3. Lateral nuclei
   a. Lateral dorsal.
   b. Lateral posterior.
   c. Pulvinar.

* Ventral Nuclei

4. Ventral Anterior
5. Ventral lateral
6. Ventral posterior lateral
7. Ventral posterior medial
8. Lateral geniculate body
9. Medial geniculate body

THALAMUS

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CORPUS CALLOSUM

- LARGEST COMMISSURE OF THE BRAIN.
- CONNECT 2 CEREBRAL HEMISPHERE.

PART OF BRAIN CONNECTED -
- CONNECTS ALL PART OF CEREBRAL CORTEX OF 2X SIDE EXCEPT, LOWER AND ANTERIOR PART OF TEMPORAL LOBES - WHICH CONNECT BY ANTERIOR COMMISSURE

*PART OF CARPUS CALLOSUM-
1) GENU - 4CM BEHIND THE FRONTAL POLE
RELATION - ANTERIORLY - ANTERIOR CEREBRAL ARTERIES
POSTERIORLY - ANTERIOR HORN OF LATERAL VENTRICLE

2) ROSTRUM - DIRECTED- DOWNWARD AND BACKWARD FROM GENU

- END BY JOINING → LAMINA TERMINALIS IN FRONT
ANTERIOR COMMISSURE
- RELATION - SUPERIOR TO ANTERIOR HORN

3) TRUNK - MIDDLE PART, BETWEEN GENU AND SPLENIUM
OVERLAPPED BY → CINGULATE GYRUS.
4)SPLENIUM -6CM FRONT OF OCCIPITAL POLE
- RELATION - INFERIOR SURFACE - TELA CHOROIDEA
- SUPERIOR SURFACE- INFERIOR SAGITTAL SINUS

*FIBRES OF CORPUS CALLOSUM-
*ROSTRUM CONNECT - ORBITAL SURFACE OF 2X FRONTAL LOBES
*FORCEPS MINOR CONNECT - 2X FRONTAL LOBES (MADE UP OF FIBRES OF GENU)
*FORCEPS MAJOR CONNECT - 2 X OCCIPITAL LOBES (MADE UP OF FIBRES OF SPLENIUM
*TAPETUM →(FORMED BY SOME FIBRES FROM TRUNK & SPLENIUM OF CORPUS)
IT'S FORM - ROOF AND LATERAL WALL OF POSTERIOR HORN
→ LATERAL WALL OF INFERIOR HORN

WHITE MATTER OF CEREBRUM

- WHITE MATTER OF CEREBRUM CONSIST OF MYELINATED FIBRES

- WHICH CONNECT VARIOUS PART OF CORTEX TO ONE ANOTHER

- FIBRES ARE CLASSIFIED INTO 3X - ASSOCIATION FIBRES (ARCUATE)
- COMMISSURAL FIBRES
- PROJECTION FIBRES
- CONNECT DIFFERENT CORTICAL AREA OF SAME HEMISPHERE TO ONE ANOTHER
- SUBDIVIDED INTO 2X GROUP-

1) SHORT ASSOCIATION FIBRES - CONNECT, ADJACENT GYRI
2) LONG ASSOCIATION FIBRES - CONNECT, MORE WIDELY SEPARATED GYRI
*UNCINATE FASCICULUS- CONNECT TEMPORAL LOBES TO MOTOR SPEECH AREA AND ORBITAL CORTEX

*CINGULUM -> CONNECT CINGULATE GYRUS TO PARAHIPPOCAMPAL GYRUS

*SUPERIOR LONGITUDINAL FASCICULUS - CONNECT -> FRONTAL LOBES TO OCCIPITAL AND TEMPORAL LOBES

*INFERIOR LONGITUDINAL FASCICULUS CONNECT -> OCCIPITAL LOBE TO TEMPORAL LOBES
3) COMMISSURE FIBRES - CONNECT - CORRESPONDING PARTS 2 HEMISPHERES
4) CORPUS CALLOSUM - CONNECTING CEREBRAL CORTEX OF TWO SIDES
5) ANTERIOR COMMISSURE - ARCHIPALLIA OF BOTH SIDES
6) POSTERIOR COMMISSURE - SUPERIOR COLLICULI
7) COMMISSURE OF FORNIX → CONNECTING THE CRURA OF FORNIX

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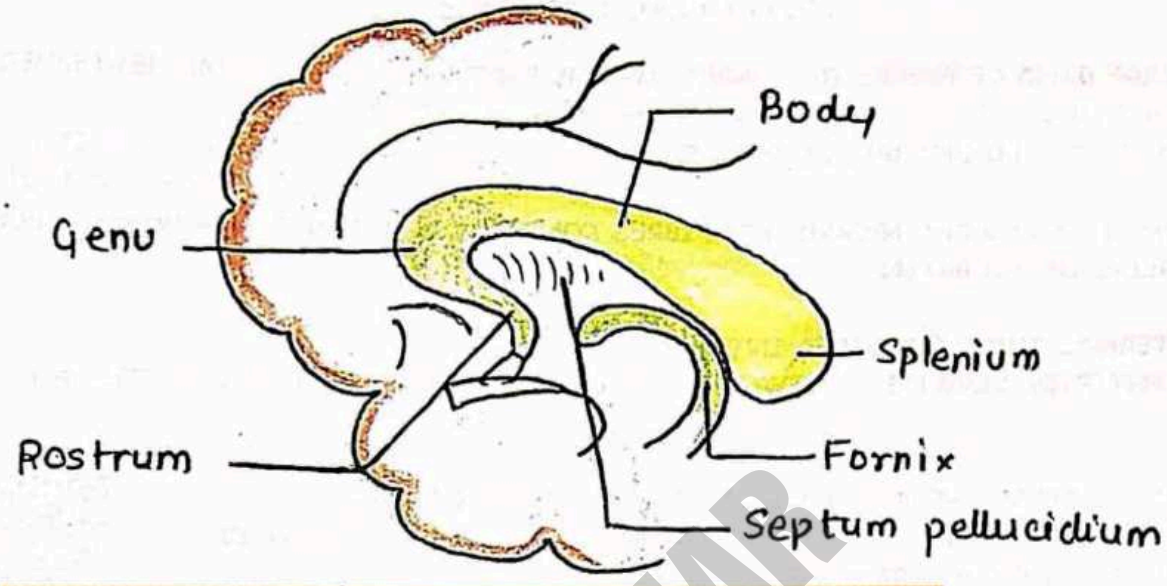
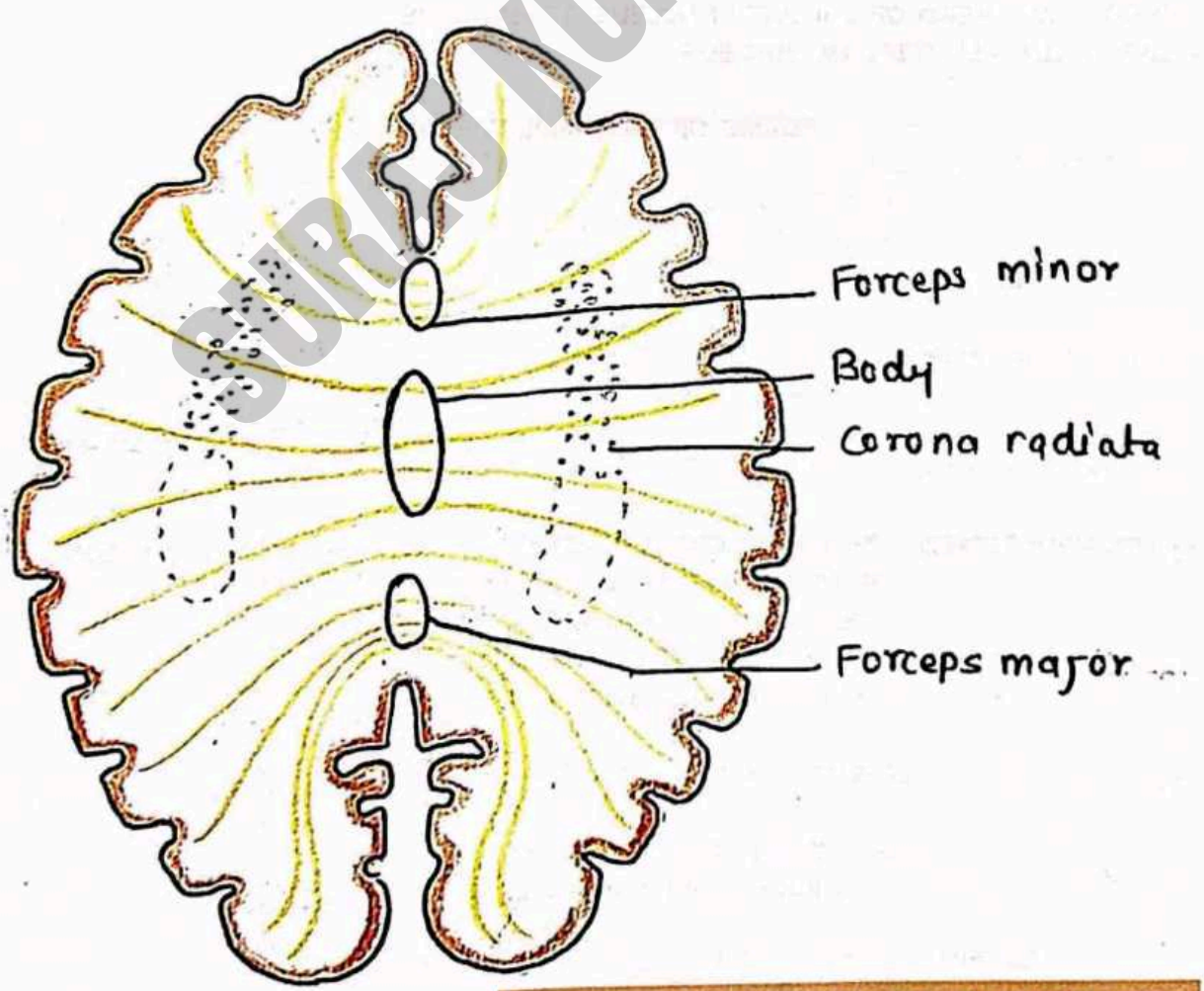


CORPUS CALLOSUM



WHITE MATTER OF CEREBRUM

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INTERNAL CAPSULE

-LARGE BAND OF FIBRES IN - INFEROMEDIAL PART OF EACH CEREBRAL HEMISPHER
-CONTAINS FIBRES GOING TO AND COMING FROM CEREBRAL CORTEX
-WHEN TRACED UPWARD, FIBRES OF CAPSULE DIVERGE AND CONTINUOUS WITH CORONA RADIATA

-WHEN TRACED DOWNWARD, ITS FIBRES CONVERGE AND CONTINUE- WITH - CRUS CEREBRI OF MIDBRAIN

INTERNAL CAPSULE DIVIDED INTO :-
1) ANTERIOR LIMB LIE - BETWEEN HEAD OF CAUDATE NUCLEUS AND LENTIFORM NUCLEUS

2) GENU BEND - BETWEEN ANTERIOR AND POSTERIOR LIMB
3) POSTERIOR LIMB → BETWEEN THALAMUS AND LENTIFORM NUDEI
4) SUB LENIFARM PART - LIE BELOW LENTI FARM NUDEUS
5) RETROLENTIFORM PART - LIE BEHIND LENTIFORM NUCLEUS

*RELATION :-
- MEDIALLY - HEAD OF CAUDATE NUCLEUS & THALAMUS
- LATERALLY - LENTIFORM NUCLEUS

FIBRES OF INTERNAL CAPSULE :-

1) MOTOR FIBRES - CORTICOPONTINE
FRONTOPONTINE
PARIETOPONTINE
TEMPOROPONTINE

2)-PYRAMIDAL FIBRES → CORTICONUCLEAR
→ CORTICOSPINAL

3) EXTRAPYRAMIDAL FIBRES - CORTICORUBRAL FIBRE & CORTICOSTRIATE

4) SENSORY FIBRES - THALAMOCORTICAL FIBRES FROM THALAMIC RADIATION
- ANTERIOR THALAMIC RADIATION
- SUPERIOR THALAMIC RADIATION
- INFERIOR THALAMIC RADIATION
- POSTERIOR THALAMIC RADIATION

*BLOOD SUPPLY - ANTERIOR LIMB - BRANCH OF ANTERIOR CEREBRAL
---→ ARTERY FROM RECCURENT BRANCH
- GENU - BRANCH OF INTERNAL CAROTID
- BRANCH OF POSTERIOR COMMUNICATING

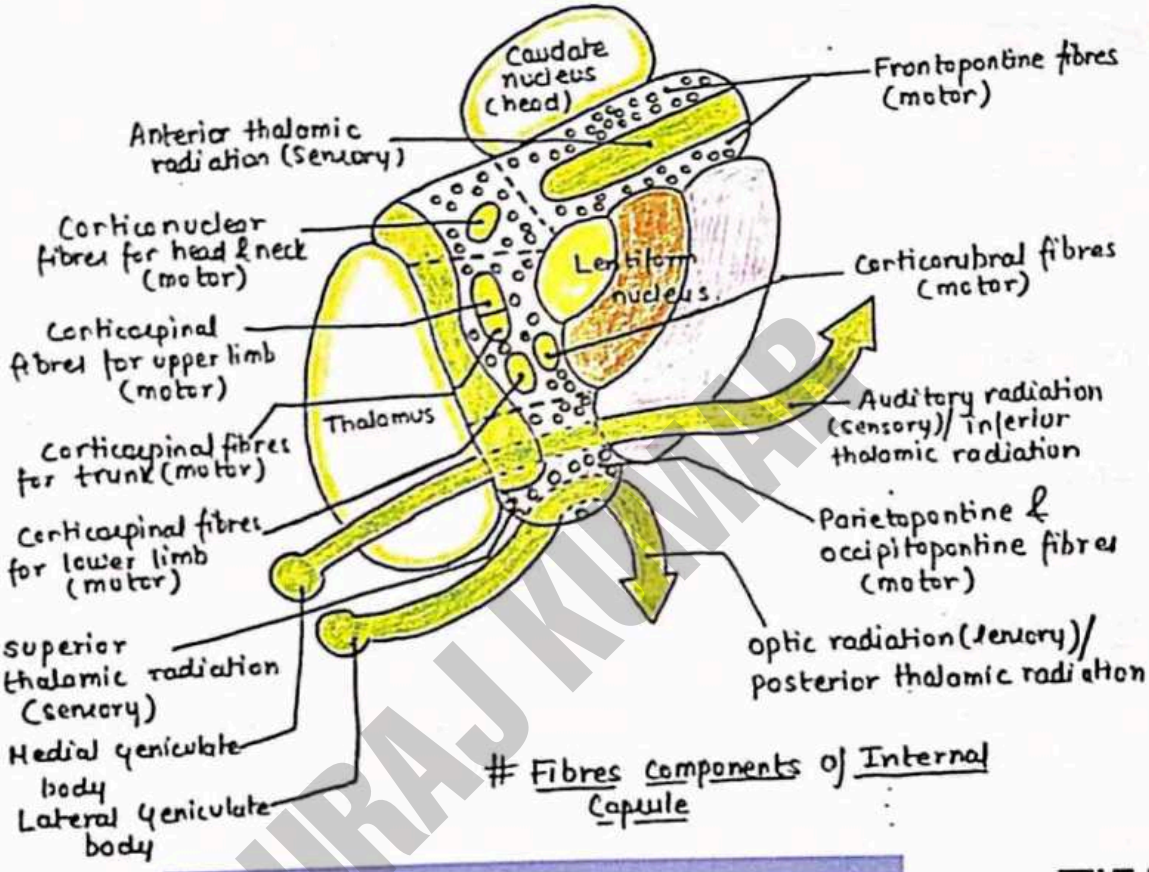
POSTERIOR LIMB → BRANCH OF ANTERIOR CHOROIDAL

SUNIL KUMAR

- BRANCH OF POSTERIOR COMMUNICATING
- BRANCH OF LATERAL AND MEDIAL STRIATE ARTERIES

SUBLENTIFORM → BRANCH OF ANTERIOR CHOROIDAL
-BRANCH OF POSTERIOR CEREBRAL

RETROLENTI FORM → BRANCH OF POSTERIOR CEREBRAL



Fibres components of Internal Capsule

CIRCLE OF WILLIS

*HEXAGONAL ARTERIAL CIRCLE
* SITUATED -->BASE OF BRAIN→IN INTERPEDUNCULAR FOSSA
* FORMED BY - ANTERIOR CEREBRAL-->BRANCH INTERNAL CAROTID
- TERMINAL PART OF INTERNAL CAROTID ARTERIES
- POSTERIOR CEREBRAL-→BRANCH BASILAR ARTERY

-TWO ANTERIOR CEREBRAL ARTERIES-->CONNECTED BY ANTERIOR COMMUNICATING ARTERY.

-INTERNAL CAROTID & POSTERIOR CEREBRAL ARTERY OF SAME SIDE-->UNITED BY POSTERIOR COMMUNICATING ARTERY.

*FORMATION - ANTERIORLY - ANTERIOR COMMUNICATED ARTERY
- ANTEROLATERALLY - ANTERIOR CEREBRAL ARTERY
- LATERALLY - INTERNAL CAROTID ARTERIES
- POSTEROLATERALLY - POSTERIOR COMMUNICATING ARTERY
- POSTERIORLY - POSTERIOR CEREBRAL ARTERIES

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*BRANCHES - CORTICAL BRANCHES-->SUPPLY CEREBRUM(ANASTOMOSE FREELY)
- CENTRAL BRANCHES-->THALAMUS
-CARPUS STRIATUM (DOESN'T ANASTOMOSE)
-INTERNAL CAPSULE
- CHOROIDAL BRANCH-->CHOROID PLEXUSES OF VENTRICLE

1.CORTICAL BRANCHES -->ARE - ANTERIOR CEREBRAL ARTERIES
- MIDDLE CEREBRAL ARTERIES
- POSTERIOR CEREBRAL ARTERIES

SUPPLY - CEREBRUM --- >SUPEROLATERAL SURFACE--->MAINLY BY MIDDLE CEREBRAL ARTERY
*EXCEPT - OCCIPITAL LOBE - PCA
- TEMPORAL LOBE - PCA

→ MEDIAL SURFACE → MAINLY BY ANTERIOR CEREBRAL ARTERY
*EXCEPT - OCCIPITAL LOBE------- >PCA
- TEMPORAL LOBE ------- >PCA
→ INFERIOR SURFACE----→ BY POSTERIOR CEREBRAL ARTERY
*EXCEPT - MEDIAL 1/3 ORBITAL LOBE-ACA
- LATERAL 2/3 TEMPORAL LOBE-MCA

2.CENTRAL BRANCHES - ANTEROMEDIAL---->ARISES ANTERIOR CEREBRAL ARTERY
---->ENTER MEDIAL MOST PART OF ANTERIOR PERFORATED SUBSTANCE
-ANTEROLATERAL-->ARISES MIDDLE CEREBRAL ARTERY
->SUPPLY - MEDIATE STRIATE → CAUDATE NUCLEUS
→ INTERNAL CAPSULE
- LATERAL STRIATE → ENTER INTERNAL CAPSULE
- POSTEROMEDIAL → ARISE POSTERIOR CEREBRAL ARTERY
-POSTEROLATERAL--->ARISES POSTERIOR CEREBRAL ARTERIES
3.CHOROIDAL BRANCH - ANTERIOR CHOROIDAL--->BRANCH INTERNAL CAROTID A.
----->SUPPLIES INFERIOR HORN OF LATERAL VENTRICLE

- POSTERIOR CHOROIDAL---->BRANCH POSTERIOR CEREBRAL
------->SUPPLIES REST PART OF CHOROID PLEXUS OF LATERAL VENTRICLE

CLINICAL ANATOMY

*THROMBOSIS OF THE PARACENTRAL ARTERY (TERMINAL CORTICAL BRANCH OF THE ANTERIOR CEREBRAL ARTERY) CAUSES CONTRALATERAL LOWER LIMB MONOPLEGIA.

*THROMBOSIS OF LATERAL STRIATE BRANCHES OF MIDDLE CEREBRAL ARTERY CAUSES MOTOR AND SENSORY LOSS TO MOST OF THE OPPOSITE SIDE OF BODY EXCEPT LOWER LIMB.

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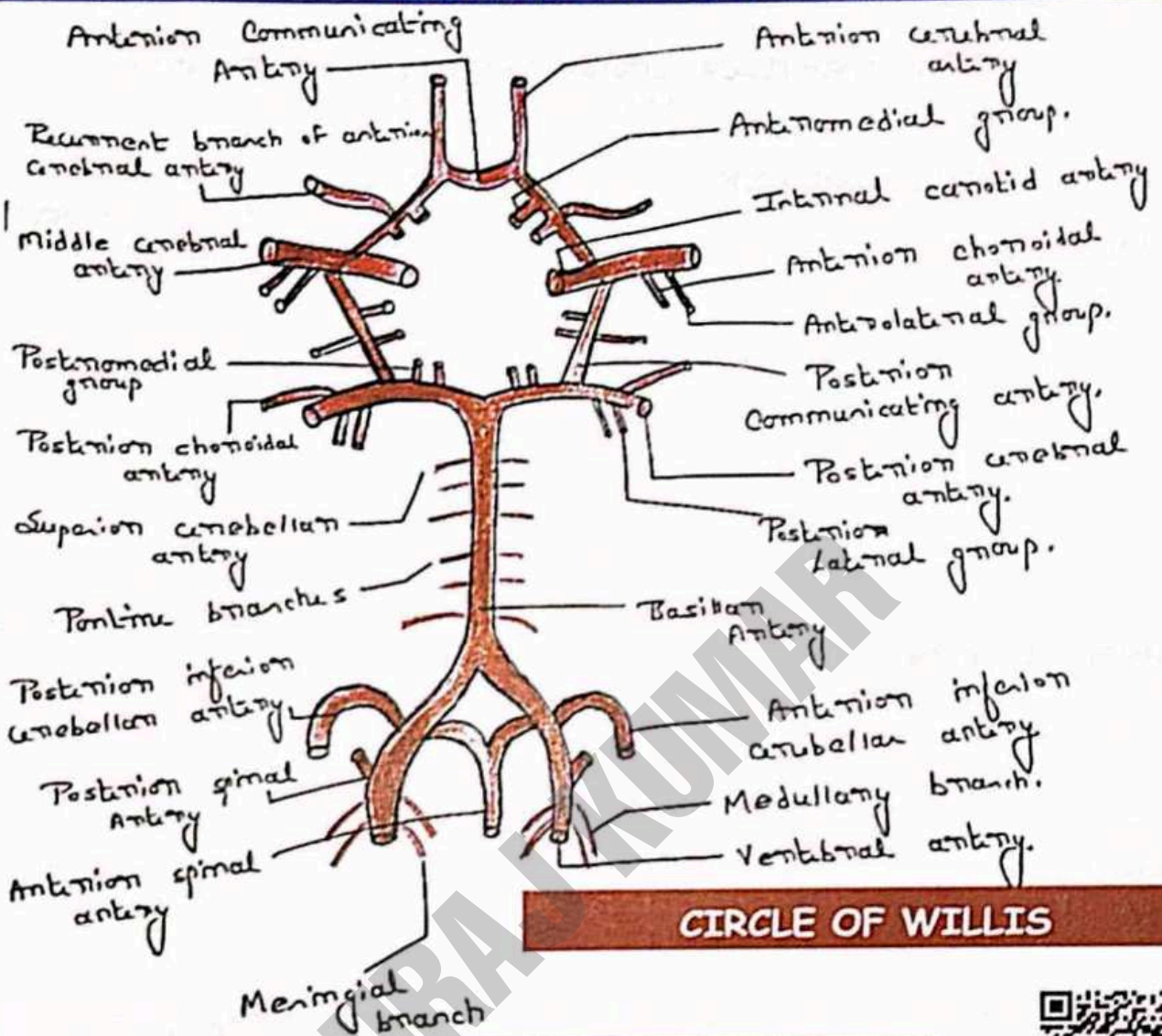
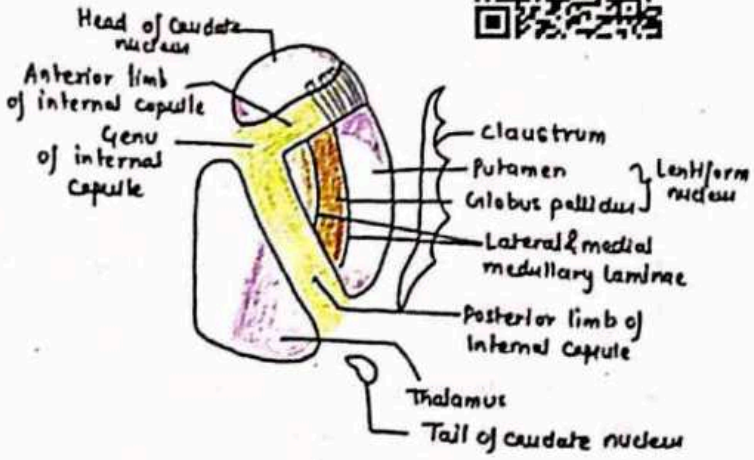


CIRCLE OF WILLIS

SUBA KUMAR

BASAL NUCLEI



- BASAL NUCLEI ARE SUBCORTICAL, INTRACEREBRAL MASSES OF GREY MATTER
- INCLUDES 1) CORPUS STRIATUM -
  - *CAUDATE NUCLEUS
  - *TENTIFORM NUCLEUS
    - GLOBUS
    - PALLIDUS
  2) AMYGDALOID BODY
  3) CLAUSTRUM

1) CARPUS STRIATUM :

*CAUDATE NUCLEUS - C-SHAPED NUDEUS - SURROUNDED - LATERAL VENTRICLE
- HEAD, BODY AND TAIL

- RELATION - HEAD - FORM FLOOR - ANTERIOR HORN OF LATERAL VENTRICLE
-MEDIAL WALL OF ANT. LIMB OF INTERNAL CAPSULE

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FIRST YEAR MBBS 2021
ANATOMY PAPER 1

Q1 LONG

1. DESCRIBE ARTERIAL SUPPLY AND VENOUS DRAINAGE OF HEART — ANS (1) – PAGE NO.75 V1
2. DESCRIBE PALATINE TONSIL UNDER THE FOLLOWING HEADINGS: GROSS FEATURES INCLUDING DEEP RELATIONS, HISTOLOGY AND APPLIED ANATOMY — ANS (2) – PAGE NO.90 V2
3. A 30 YEAR OLD MALE PRESENTED TO THE EMERGENCY DEPARTMENT AFTER A ROAD TRAFFIC ACCIDENT WITH COMPLAINING OF RIGHT ARM PAIN AND INABILITY TO MOVE HIS EXTREMITY. THE FOLLOWING MORNING, HE DEVELOPED CLAWING. NERVE CONDUCTION STUDIES ON THE PERIPHERAL NERVES OF HIS ARM CONFIRMED THE DIAGNOSIS OF INJURY TO THE ULNAR NERVE. KEEPING THIS SCENARIO IN MIND, ANSWER THE FOLLOWING QUESTIONS: ORIGIN AND ROOT VALUE OF THE NERVE, COURSE AND DISTRIBUTION, BRANCHES, EXPLAIN REASON FOR DEVELOPING THE CLAW HAND — ANS (3) – PAGE NO.13 V1

Q2 SHORT

1. CAROTID TRIANGLE — ANS (1) – PAGE NO.61 V2
2. FACIAL NERVE — ANS (2) – PAGE NO. V2
3. DESCRIBE MICROSCOPIC FEATURES OF TRACHEA — ANS (3) – PAGE NO.149 V1
4. WELL LABELLED DIAGRAM SHOWING THE TRANSVERSE SECTION OF UPPER PART OF PONS — ANS (4) – PAGE NO.112 V2
5. CUBITAL FOSSA — ANS (5) – PAGE NO.6 V1
6. CLASSIFICATION OF SYNOVIAL JOINT — ANS (6) – PAGE NO.110 V1
7. RESPIRATORY MOVEMENTS OF THE THORACIC WALL — ANS (7) – (EXTRA – PHYSIOLOGY)
8. DERIVATIVES OF FIRST PHARYNGEAL ARHC — ANS (8) – PAGE NO.93 V1
9. OPENINGS OF LATERAL WALL OF NOSE — ANS (9) – PAGE NO.99 V2
10. WRITE A NOTE ON OTIC GANGLION — ANS (10) – PAGE NO.137 V2

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ANATOMY PAPER 2

Q1 LONG

1. DESCRIBE INGUINAL CANAL UNDER THE FOLLOWING HEADS: BOUNDARIES, CONTENTS AND APPLIED ASPECTS — ANS (1) – PAGE NO.2 V2
2. DESCRIBE KNEE JOINT UNDER THE FOLLOWING HEADINGS: TYPE OF JOINT AND ARTICULAR SURFACE, CAPSULE AND LIGAMENTS, MOVEMENTS AND MUSCLES RESPONSIBLE AND APPLIED ANATOMY — ANS (2) – PAGE NO.52 V1
3. DESCRIBE DUODENUM UNDER THE FOLLOWING HEADINGS: PERITONEAL AND VISCERAL RELATIONS(FIRST, SECOND AND THIRD PART), HISTOLOGY, EXPLAIN ANATOMICAL BASIS OF DUODENAL CAP AND DUODENAL ULCER. — ANS (3) – PAGE NO.12 V2

Q2 SHORT

1. WRITE A NOTE ON VARIOUS POSITIONS OF APPENDIX — ANS (1) – PAGE NO.16 V2
2. LOCKING UNLOCKING MECHANISM OF KNEE JOINT — ANS (2) – PAGE NO.50 V1
3. ISCHIORECTAL FOSSA — ANS (3) – PAGE NO.36 V2
4. ADDUCTOR CANAL — ANS (4) – PAGE NO.37 V1
5. SUPPORT OF UTERUS — ANS (5) – PAGE NO.37 V2
6. GREAT SAPHENOUS VEIN — ANS (6) – PAGE NO.44 V1
7. HISTOLOGY OF OVARY — ANS (7) – PAGE NO.153 V1
8. PORTACAVAL ANASTOMOSIS (THREE IMPORTANT SITES) (DIAGRAM ONLY) AND MENTION THREE EFFECTS OF PORTAL HYPERTENSION — ANS (8) – PAGE NO.30 V2
9. DEVELOPMENT OF TESTIS — ANS (9) – PAGE NO.103 V1
10. MEDIAL LONGITUDINAL ARCH OF FOOT — ANS (10) – PAGE NO.40 V1

JOHARI KUMAR MBBS

MADHYA PRADESH MEDICAL SCIENCE UNIVERSITY
FIRST YEAR MBBS 2020
ANATOMY PAPER 1

Q1 LONG

1. DESCRIBE MEDIAN NERVE UNDER FOLLOWING HEADS: ROOT VALUE, COURSE AND RELATIONS, DISTRIBUTION AND CLINICAL ANATOMY — ANS (1) – PAGE NO.15 V1
2. DESCRIBE BRONCHOPULMONARY SEGMENTS WITH ITS APPLIED — ANS (2) – PAGE NO.66 V1
3. CLASSIFY THE DURAL VENOUS SINUSES AND DESCRIBE CAVERNOUS SINUS IN DETAIL — ANS (3) – PAGE NO.85 V2

Q2 SHORT

1. DEVELOPMENT OF PALATE AND ITS ANOMALIES — ANS (1) – PAGE NO.97V1
2. SCALP — ANS (2) – PAGE NO.55 V2
3. MIDDLE EAR CAVITY — ANS (3) – PAGE NO.102 V2
4. SUPINATION AND PRONATION OF FOREARM — ANS (4) – PAGE NO.24 V1
5. HISTOLOGY OF THYROID GLAND — ANS (5) – PAGE NO.149 V1
6. ROTATOR CUFF OF SHOULDER JOINT — ANS (6) – PAGE NO.17 V1
7. TYPICAL INTERCOSTAL SPACE — ANS (7) – PAGE NO.80 V1
8. CAROTID TRIANGLE — ANS (8) – PAGE NO.61 V2
9. LATERAL NASAL WAAL — ANS (9) – PAGE NO.99 V2
10. SECRETOMOTOR PATHWAY OF PAROTID GLAND — ANS (10) – PAGE NO.70 V2

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FIRST YEAR MBBS 2020
ANATOMY PAPER 2

Q1 LONG

1. DESCRIBE THE EXTERNAL FEATURES, RELATIONS, LIGAMENTS, BLOOD SUPPLY AND DEVELOPMENTAL ANATOMY OF URINARY BLADDER. ADD A NOTE ON ITS APPLIED ANATOMY. — ANS (1) – PAGE NO.37 V2
2. DESCRIBE THE ROOT VALUE, COURSE, RELATIONS, BRANCHES AND DISTRIBUTION AND APPLIED ANATOMY OF OBTURATOR NERVE — ANS (2) – PAGE NO.39 V1
3. DESCRIBE THE TYPE, LIGAMENTS, RELATION, MOVEMENTS AND MUSCLES PRODUCING THE MOVEMENTS AND APPLIED ANATOMY OF HIP JOINT — ANS (3) – PAGE NO.49 V1

Q2 SHORT

1. LESSER SAC — ANS (1) – PAGE NO.47 V1
2. HISTOLOGY OF SPLEEN — ANS (2) – PAGE NO.150 V1
3. POPLITEAL FOSSA — ANS (3) – PAGE NO.39 V1
4. SUPPORTS OF UTERUS — ANS (4) – PAGE NO.40 V1
5. DESCENT OF TESTIS — ANS (5) – PAGE NO.103 V1
6. BLOOD SUPPLY OF STOMACH — ANS (6) – PAGE NO.9 V2
7. ADDUCTOR CANAL — ANS (7) – PAGE NO.37 V1
8. MEDIAL LONGITUDINAL ARCH OF FOOT — ANS (8) – PAGE NO.43 V1
9. PORTOCAVAL ANASTOMOSIS — ANS (9) – PAGE NO.30 V2
10. PROSTATIC PART OF URETHRA — ANS (10) – (EXTRA – BDC)

JOHARI KUMAR MBBS

MADHYA PRADESH MEDICAL SCIENCE UNIVERSITY
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ANATOMY PAPER 1

Q1 LONG

1. DESCRIBE MIDDLE EAR UNDER FOLLOWING HEADINGS A) LOCATION B)CONTENTS C) BOUNDARIES D) NERVE SUPPLY E) APPLIED ANS (1) – PAGE NO.102 V2
2. DESCRIBE BRACHIAL PLEXUS WITH ITS APPLIED ANATOMY ANS (2) – PAGE NO.3 V1

Q2 SHORT

1. SENSORY INNERVATION OF FACE ANS (1) – PAGE NO.54 V2
2. HISTOLOGICAL FEATURES OF PAROTID GLAND ANS (2) – PAGE NO.70 V2
3. LATERAL PTERYGOID MUSCLE ANS (3) – PAGE NO.88 V2
4. DEVELOPMENT OF TONGUE ANS (4) – PAGE NO.73 V1
5. SYNAPSE ANS (5) – (EXTRA – PHYSIOLOGY)
6. TRANSVERSE SECTION OF OPEN PART OF MEDULLA ANS (6) – PAGE NO.111 V2
7. MAJOR OPENINGS OF THORACIC DIAPHRAGM ANS (7) – PAGE NO.84 V1
8. THORACIC DUCT ANS (8) – PAGE NO.79 V1
9. AXILLARY ARTERY ANS (9) – PAGE NO.5 V1
10. CORONARY SINUS ANS (10) – PAGE NO.75 V1

MADHYA PRADESH MEDICAL SCIENE UNIVERSITY
FIRST YEAR MBBS 2018
ANATOMY PAPER 1

Q1 LONG

1. DESCRIBE CUBITAL FOSSA ANS (1) – PAGE NO.6 V1
2. DESCRIBE SCALP WITH ITS APPLIED ASPECTS ANS (2) – PAGE NO.55 V2

Q2 SHORT

1. TASTE PATHWAY ANS (1) – PAGE NO.92 V2 (TONGUE NERVE)
2. ULNAR NERVE IN HAND ANS (2) – PAGE NO.13 V1
3. MEDIASTINAL SURFACE OF RIGHT LUNG ANS (3) – PAGE NO.62 V1
4. CLASSIFICATION SYNOVIAL JOINTS ANS (4) – PAGE NO.110 V1
5. RESPIRATORY EPITHELIUM ANS (5) – PAGE NO.151 V1
6. MORULA ANS (6) – (EXTRA)
7. BLOOD SUPPLY OF THYROID GLAND ANS (8) – PAGE NO.73 V2
8. ANATOMICAL SNUFF BOX ANS (8) – PAGE NO.19 V1
9. SECTION OF MEDULLA AT THE LEVEL OF MOTOR DECUSSATION ANS (9) – PAGE NO.111 V2
10. THORACIC DUCT ANS (10) – PAGE NO.79 V1

MADHYA PRADESH MEDICAL SCIENCE UNIVERSITY
FIRST YEAR MBBS 2017
ANATOMY PAPER 1

Q1 LONG

1. DESCRIBE BROCHOPULMONARY SEGMENTS WITH ITS APPLIED ASPECTS ANS (1) – PAGE NO.66 V1
2. DESCRIBE CAROTID TRIANGLE ANS (2) – PAGE NO.61 V2

Q2 SHORT

1. INTERIOR OF RIGHT ATRIUM ANS (1) – PAGE NO.73 V1
2. SECRETOMOTOR PATHWAY OF PAROTID GLAND ANS (2) – PAGE NO.70 V2
3. DIAGRAM OF FLOOR OF 4TH VENTRICLE OF BRAIN ANS (3) – PAGE NO.147 V2
4. MUSCULOCUTANEOUS NERVE IN ARM ANS (4) – (EXTRA)
5. SUPINATION AND PRONATION OF FOREARM ANS (5) – PAGE NO.24 V1
6. RELATIONS IN CORONAL SECTION OF CAVERNOUS SINUS ANS (6) – PAGE NO.85 V2
7. CIRCLE OF WILLIS ANS (7) – PAGE NO.153 V2
8. SPERMATOGENESIS ANS (8) – (EXTRA)
9. CLASSIFICATION OF SYNOVIAL JOINTS ANS (9) – PAGE NO.110 V1
10. UROTHELIUM ANS (10) – PAGE NO.153 V1

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MADHYA PRADESH MEDICAL SCIENCE UNIVERSITY
FIRST YEAR MBBS 2019
ANATOMY PAPER 2

Q1 LONG

1.DESCRIBE THE 2ND PART OF DUODENUM UNDER FOLLOWING HEADS: RELATIONS, DEVELOPMENT, ARTERIAL SUPPLY INTERIOR AND APPLIED ASPECT ANS (1) – PAGE NO.12 V2

2.DESCRIBE URINARY BLADDER UNDER FOLLOWING HEADS: RELATIONS, EXTERNAL FEATURES, SUPPORTS ARTERIAL AND NERVE SUPPLY AND APPLIED ASPECT ANS (2) – PAGE NO.37 V2

Q2 SHORT

1.DEFENSE MECHANISM OF INGUINAL CANAL ANS (1) – PAGE NO.2 V2
2.RECTOUTERINE POUCH ANS (2) – (EXTRA)
3.HISTOLOGY OF PANCREAS ANS (3) – PAGE NO.152 V1
4.KLINEFELTERS SYNDROME ANS (4) – PAGE NO.112 V1
5.PUDENDAL CANALANS (5) – PAGE NO.36 V2 (ISCHIOANAL FOSSA)
6.DEVELOPMENT OF KIDNEY ANS (6) – PAGE NO.81 V1
7.NOTOCHORD ANS (7) – PAGE NO.90 V1
8.MEDIAL LONGITUDINAL ARCH OF FOOT ANS (8) – PAGE NO.42 V1
9.TIBIALIS ANTERIOR MUSCLE
10.ADDUCTOR CANAL ANS (10) – PAGE NO.37 V1

MADHYA PRADESH MEDICAL SCIENCE UNIVERSITY
FIRST YEAR MBBS 2018
ANATOMY PAPER 2

Q1 LONG

1.DESCRIBE SPLEEN UNDER FOLLOWING HEADS: SITUATIONANS (1) – PAGE NO.48 V2 RELATIONS, EXTERNAL FEATURES AND APPLIED ASPECT

2.DESCRIBE UTERUS UNDER FOLLOWING HEADS: SITUATION, RELATIONS, SUPPORT, BLOOD SUPPLY AND APPLIED ASPECT ANS (2) – PAGE NO.40 V2

Q2 SHORT

1.BOUNDARIES AND CONTENTS OF POPLITEAL FOSSA ANS (1) – PAGE NO.39 V1
2.TURNER'S SYNDROME ANS (2) – PAGE NO.112 V1
3.PORTOCAVAL ANASTOMOSIS ANS (3) – PAGE NO.30 V2
4.RELATIONS AND BLOOD SUPPLY OF SUPRARENAL GLAND
5.DEVELOPMENT OF PANCREAS ANS (5) – PAGE NO.99 V1
6.MICROANATOMY OF DUODENUM ANS (6) – PAGE NO.151 V1
7.DEFINE INVERSION AND EVERSION MOVEMENT AND MENTION THE JOINT INVOLVED ANS (7) – PAGE NO.52 V1 (KNEE JOINT)
8.TIBIALIS POSTERIOR
9.STOMACH BED ANS (3) – PAGE NO.9 V2
10.DESCENT OF TESTIS AND ANOMALIES ANS (10) – PAGE NO.103 V1

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FIRST YEAR MBBS 2017
ANATOMY PAPER 2

Q1 LONG

1.DESCRIBE VERMIFORM APPENDIX UNDER THE FOLLOWING HEADINGS: POSITION, RELATIONS, ARTERIAL SUPPLY, LYMPHATIC DRAINAGE AND APPLIED ASPECT. ANS (1) – PAGE NO.16 V2

2.DESCRIBE VENOUS DRAINAGE OF LOWER LIMB AND DISCUSS ITS APPLIED ASPECT ANS (2) – PAGE NO.44 V2

Q2 SHORT

1.OBTURATOR NERVE- ORIGIN, COURSE AND DISTRIBUTION ANS (1) – PAGE NO.39 V1
2.PELVIC DIAPHRAGM ANS (2) – PAGE NO.46 V2
3.INGUINAL CANAL ANS (3) – PAGE NO.5 V2
4.EXTRAHEPATIC BILIARY APPARATUS ANS (4) – PAGE NO.28 V2
5.MICRO ANATOMY OF THE PROSTATE ANS (5) – PAGE NO.153 V1
6.FEMORAL ARTERY-ORIGIN, COURSE AND DISTRIBUTION ANS (6) – PAGE NO.35 V1
7.DEVELOPMENT OF THE ANAL CANAL ANS (7) – PAGE NO.81 V1
8.ISCHIORECTAL FOSSA ANS (8) – PAGE NO.36 V2
9.RECTO-UTERINE POUCH ANS (9) – (EXTRA)
10.FALLOPIAN TUBEANS (10) – PAGE NO.40 V2

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